```
      }
    }
  ],
  "reserveDateTime": "2017-05-04T18:22:12+00:00"
}
```

Async placement calculation

When you POST a reservation, Turbonomic runs a plan to calculate the optimal placement of the workloads. Depending on the size of the reservation, this can take a significant amount of time. The API calculates the placement asynchronously. If your script relies on the reservation, you should verify that the reservation has completed before moving on.

When you execute a reservation, the response to your POST shows that the calculation is in progress:

```
{
    "uuid": "_0dPrYTDlEeePgeXuo0RRbw",
    "displayName": "MyReservation",
    "count": 4,
    "status": "IN_PROGRESS",
    "reserveDateTime": "Thu May 04 18:22:12 UTC 2017",
    "expireDateTime": "Tue Oct 10 12:38:17 UTC 2017"
}
```

To examine the placement results, GET the reservation using the UUID that the API gives in the response.

Using templates

To create a reservation you specify a template that contains workload requirements. VM Templates specify the resources that will be available to the VM, including:

- VCPUs
- Virtual Memory
- Storage
- Network Throughput
- IOPS
- IO Throughput

Note that templates can be created by users, and Turbonomic also discovers templates that are created by the management services in your environment. For example, a hypervisor or a cloud service provider typically manages a number of its own templates — Turbonomic discovers these. You should never edit a discovered template.

To see whether a template is discovered, GET the template you're interested in and look for the `discovered` property. This will be `true` or `false`.

Turbonomic REST API endpoints

The REST API resources give you full access to the Turbonomic software. This is a complete API that exposes the full set of Turbonomic capabilities. In fact, the product user interface is implemented as a Turbonomic client that uses this API.

While the REST API resources are documented in the SwaggerUI, this section provides extra details for the more expressive resources in the API.

To access the Turbonomic REST SwaggerUI, open a web browser to:

`https://<Your_Turbonomic_URL>/apidoc`

Utility requests

In the Turbonomic API Guide, utility requests are the requests within an endpoint that do not return information about your environment, but provide benefits to other calls you may make. For example,

`https://10.10.10.10/api/v3/settings/` is a utility request that returns a list of all possible settings managers. This information can be used to find settings on your instance, but the `/settings/` request itself does not provide any instance-specific information.

Actions endpoint

Instead of responding to thresholds, Turbonomic analyzes the operating conditions and constantly recommends actions to keep the entire environment within the desired state. If you execute these actions (or let Turbonomic execute them for you), the environment will maintain operating conditions that assure performance while also ensuring the lowest possible cost thanks to efficient utilization of your resources.

Using the actions endpoint, you can:

- Get a complete or filtered list of actions in your environment
- Accept or reject actions
- Get statistics related to a specified action
- Get notifications related to a specified action

Actions utility requests

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

Getting the valid `actionMode` parameters for a specified entity and action

The actions endpoint contains a utility request that shows the possible values for the `actionMode` parameter for a specified entity and action type. These can be used to set or understand the possibilities for various action settings.

To specify an entity/action type, this request takes the following parameters:

- action_type
  - `START`:

    Start an entity, and add it to the Turbonomic market.
  - `MOVE`:

    Move an entity from one provider to another. For example, moving a VM between hosts, or a datastore between disk arrays.
  - `SUSPEND`:

    Suspend the entity to exclude its capacity from your environment. Suspend does not remove the entity from your environment, but the market does not include the entity in its analysis. Based on market conditions, analysis can recommend restarting suspended entities, which includes them in market analysis once again.
  - `ADD PROVIDER`:

    This is equivalent to the START action, but for storage entities.
  - `CHANGE`:

    Move a virtual machine's storage between datastores.
  - `PROVISION`:

    Add capacity to your environment.
  - `RECONFIGURE`:

    Analysis recommends this when a VM is not able to correct a misconfiguration via another type of action. For example, analysis can recommend a RECONFIGURE action when a placement policy requires a VM to move to a host on a particular cluster, but cross-cluster moves are not permitted.
  - `DELETE`:

This is equivalent to the SUSPEND action, but for storage entities.

  - RESERVE_ON_PM:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular host.

  - RESERVE_ON_DS:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular datastore.

  - RESIZE_FOR_EFFICIENCY:

    A resize that reduces the capacity of a commodity, such as vCPU, or vMEM.

  - RESIZE_FOR_PERFORMANCE:

    A resize that increases the capacity of a commodity, such as vCPU, or vMEM.

- entity_type
  - APPLICATION
  - APPLICATION_SERVER
  - BUSINESS_ACCOUNT
  - BUSINESS_UNIT
  - CHASSIS
  - CLOUD_SERVICE
  - CONTAINER
  - CONTAINER_POD
  - DATABASE
  - DATABASE_SERVER
  - DATACENTER
  - DISK_ARRAY
  - IO_MODULE
  - LOAD_BALANCER
  - LOGICAL_POOL
  - NETWORK
  - PHYSICAL_MACHINE
  - STORAGE
  - STORAGE_CONTROLLER
  - SWITCH
  - VIRTUAL_APPLICATION
  - VIRTUAL_DATACENTER
  - VIRTUAL_MACHINE

Example: `GET https://10.10.10.10/api/v3/actions/availablemodes?action_type=PROVISION&entity_type=PhysicalMachine`

Response: An array of strings, where each string is a valid actionMode for the combination provided. In this example, a `PROVISION` action and the `PhysicalMachine` entity. An empty response body indicates that the requested entity does not have an action of that type, usually for practical reasons. For example, the Network entity cannot have a `MOVE` action.

```
[
  "RECOMMEND",
  "DISABLED",
```

```
  "MANUAL",
  "AUTOMATIC"
]
```

Getting a list of entry points for actions

The actions endpoint contains a utility request that shows the entry points for other action-related requests. These can be used to find the UUID and details relating to a specific action.

Example: `GET https://10.10.10.10/api/v3/actions`

Response: An array of `LinkApiDTO` objects, where each object contains a URL to get all actions for a specified market, entity, or group.

```
{
  "links":[
    {
      "rel":"Market actions",
      "href":"https://10.10.10.10/api/v3/markets/{uuid}/actions?ascending=true"
    },
    {
      "rel":"Entity actions",
      "href":"https://10.10.10.10/api/v3/entities/{uuid}/actions?ascending=false"
    },
    {
      "rel":"Group actions",
      "href":"https://10.10.10.10/api/v3/groups/{uuid}/actions?ascending=true"
    }
  ]
}
```

Actions requests

Action requests handle acceptance, rejection, filtering, and exploration of actions on your Turbonomic appliance.

If one or more of the entities in the scope of your request no longer exists in the topology, the ActionApiDTO for such entities will contain minimal information about the entities.

Getting a specified action

Gets the action details, including the creation time and target entity information, for a specified action UUID.

Example: `GET https://10.10.10.10/api/v3/actions/_6QsKkaNWEeiYWYj--U8W5g/details`

Response: The full `ActionApiDTO` for the requested action:

```
{
  "links":[
    {
      "rel":"self",
      "href":"https://10.10.10.10/api/v3/actions/_6QsKkaNWEeiYWYj--U8W5g"
    },
    {
      "rel":"self",
      "href":"https://10.10.10.10/api/v3/actions/2218532354592"
    }
  ],
  "uuid":"_6QsKkaNWEeiYWYj--U8W5g",
  "createTime":"2018-08-19T07:40:57-04:00",
  "actionType":"MOVE",
  "actionState":"PENDING_ACCEPT",
  "actionMode":"MANUAL",
  "details":"Move VirtualMachine EAP-10.203 from dc17-host-01.mycorp.com to dc17-
host-02.mycorp.com",
  "importance":7.4150376,
```

```
  "target":{
    "uuid":"421d8db4-3fc6-440b-0575-11383c020c6c",
    "displayName":"EAP-10.203",
    "className":"VirtualMachine",
    "aspects":{
      "virtualMachineAspect":{
        "os":"CentOS 4/5/6/7 (64-bit)",
        "connectedNetworks":[
          {
            "uuid":"34f9f5364cd843aa1fe99900621ee3b51a7dc6e5",
            "displayName":"VM Network"
          }
        ],
        "numVCPUs":2,
        "ebsOptimized":false
      }
    },
    "environmentType":"ONPREM"
  },
  "currentEntity":{
    "uuid":"34313836-3333-5553-4537-33364e385146",
    "displayName":"dc17-host-01.mycorp.com",
    "className":"PhysicalMachine",
    "environmentType":"ONPREM"
  },
  "newEntity":{
    "uuid":"34313836-3333-5553-4537-33394e43424e",
    "displayName":"dc17-host-02.mycorp.com",
    "className":"PhysicalMachine",
    "environmentType":"ONPREM"
  },
  "risk":{
    "uuid":"_CizZoKN2EeiYWYj--U8W5g",
    "subCategory":"Efficiency Improvement",
    "description":"dc17-host-01.mycorp.com can be suspended for efficiency",
    "severity":"MINOR",
    "importance":0
  },
  "actionID":2218532354592
}
```


Getting a filtered list of actions

Gets a list of actions, given a scope UUID and filtered by an ActionApiInputDTO using one or more of the following criteria:

- actionModeList
  - `DISABLED`:

    Do not recommend or perform the action. When you disable an action, the user interface will no longer display actions of that type.
  - `RECOMMENDED`:

    Recommend the action so a user can execute it via the given hypervisor or by other external means
  - `MANUAL`:

    Recommend the action, and provide the option to execute that action through the Turbonomic user interface or an /actions API request.
  - `AUTOMATED`:

    You have directed Turbonomic to execute the action automatically.
  - `COLLECTION` **(UNUSED)**:

    Legacy action mode, no longer used.
- actionStateList
  - `DISABLED`:

    An action whose policy has been disabled.
  - `RECOMMEND`:

An action that cannot be automated due to policy settings or must be executed outside the system, such as the purchase of hardware.

  - `PENDING_ACCEPT`:

    When the action mode is set to manual, a recommended action that has not been accepted or cleared.

  - `ACCEPTED`:

    An action that has been accepted, but is not yet in progress.

  - `IN_PROGRESS`:

    An action that is being executed.

  - `SUCCEEDED`:

    An action that was accepted and completed successfully.

  - `FAILED`:

    An action that was accepted and did not complete successfully.

  - `REJECTED`:

    When the action mode is set to manual, an action that has been rejected by the user.

  - `CLEARED`:

    When the action mode is set to manual, an action that is no longer recommended by the market.

  - `ACCOUNTING`:

    For cloud entities, an action to resize in order to use a different instance template.

  - `QUEUED`:

    When more than 10 actions are to be executed for a single target, Turbonomic will place the 11th and all subsequent actions into a QUEUED state, and they will execute as the previous actions complete.

- actionTypeList

  - `START`:

    Start an entity, and add it to the Turbonomic market.

  - `MOVE`:

    Move an entity from one provider to another. For example, moving a VM between hosts, or a datastore between disk arrays.

  - `SUSPEND`:

    Suspend the entity to exclude its capacity from your environment. Suspend does not remove the entity from your environment, but the market does not include the entity in its analysis. Based on market conditions, analysis can recommend restarting suspended entities, which includes them in market analysis once again.

  - `ADD PROVIDER`:

    This is equivalent to the START action, but for storage entities.

  - `CHANGE`:

    Move a virtual machine's storage between datastores.

  - `PROVISION`:

    Add capacity to your environment.

  - `RECONFIGURE`:

    Analysis recommends this when a VM is not able to correct a misconfiguration via another type of action. For example, analysis can recommend a RECONFIGURE action when a placement policy requires a VM to move to a host on a particular cluster, but cross-cluster moves are not permitted.

  - `DELETE`:

    This is equivalent to the SUSPEND action, but for storage entities.
  - `RESERVE_ON_PM`:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular host.
  - `RESERVE_ON_DS`:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular datastore.
  - `RESIZE_FOR_EFFICIENCY`:

    A resize that reduces the capacity of a commodity, such as vCPU, or vMEM.
  - `RESIZE_FOR_PERFORMANCE`:

    A resize that increases the capacity of a commodity, such as vCPU, or vMEM.
- cleared

  Default: `false`.

  When `true`, actions that were removed from the action list before being executed will be returned.
- costType
  - `Saving`

    An action that will decrease cloud spending.
  - `Investment`

    An action that will increase cloud spending.
- cumulative

  Default: `false`.

  When `true`, shows the savings for each action in a scope as a cumulative total, rather than showing the savings for individual actions.
- environmentType
  - `ONPREM`:

    Returns only actions related to entities that are part of your on-prem environment.
  - `CLOUD`:

    Returns only actions related to entities that are part of your cloud environment.
- groupBy

  You may group the actions returned by the following criteria: `[actionModes, actionStates, actionTypes, risk, riskSeverity, riskSubCategory]`
- hasReservedInstance

  Default: `false`.

  When `true`, only actions that are related to reserved instances will be returned.
- relatedEntityTypes

  Returns only actions related to entities of these types within the scope. For example, if your scope is a virtual datacenter and you specify VirtualMachine as a related entity type, the request will include only those actions related to VMs in the specified virtual datacenter.

  Choose from the following entity types: `[ Datacenter, PhysicalMachine, VirtualMachine, Storage, Application, Chassis, DiskArray, IOModule, StorageControl, Switch, VirtualDataCenter, VPod, DPod, Container, Database, DatabaseServerContainer, LogicalPool ]`

- riskSeverityList

  Returns only actions of the given severities: `[ Unknown, Normal, Minor, Major, Critical ]`
- riskSubCategoryList
- returns only actions of the given subcategories: `[ Performance Assurance, Efficiency Improvement, Prevention, Compliance ]`

The following commodities may be returned as the related commodity for an action:

Access
ActionPermit
ActiveSessions
ApplicationCommodity
Ballooning
Biclique
BufferCommodity
BurstBalance
CPU
CPUAllocation
CPUProvisioned
ClusterCommodity
CollectionTime
Connection
Cooling
Coupon
CrossCloudMoveSvc
CrossClusterMoveSvc
DBCacheHitRate
DBMem
DISK_ARRAY_ACCESS
DSPMAccessCommodity
DataCenterCommodity
DatastoreCommodity
DrsSegmentationCommodity
Extent
Flow
FlowAllocation
HACommodity
HOST_LUN_ACCESS
Heap
HotStorage
IOThroughput
ImageCPU
ImageMem
ImageStorage
InstanceDiskSize
InstanceDiskType
LICENSE_ACCESS
LicenseCommodity
Mem
MemAllocation
MemProvisioned
Move

NetThroughput
NetworkCommodity
NetworkInterfaceCount
NetworkPolicy
NumDisk
NumberConsumers
NumberConsumersPM
NumberConsumersStorage
PORT_CHANNEL
PoolCPU
PoolMem
PoolStorage
Power
ProcessingUnits
Q1VCPU
Q2VCPU
Q3VCPU
Q4VCPU
Q5VCPU
Q6VCPU
Q7VCPU
Q8VCPU
Q16VCPU
Q32VCPU
Q64VCPU
QNVCPU
ResponseTime
RightSizeDown
RightSizeSVC
RightSizeUp
SLACommodity
SameClusterMoveSvc
SegmentationCommodity
ServiceLevelCluster
SoftwareLicenseCommodity
Space
Storage
StorageAccess
StorageAllocation
StorageAmount
StorageClusterCommodity
StorageLatency
StorageProvisioned
Swapping
TemplateAccess
TenancyAccess
Threads
Transaction
TransactionLog
Unknown
VAppAccess

VCPU
VCPULimitQuota
VCPURequest
VCPURequestQuota
VDCCommodity
VMPMAccessCommodity
VMem
VMemLimitQuota
VMemRequest
VMemRequestQuota
VStorage
Zone

Example: `POST https://10.10.10.10/api/v3/actions`

Example input: In this example, a list of actions that meet the following criteria will be returned and grouped by the `actionMode` of the action:

- Is in state `PENDING_ACCEPT`
- Is part of an on-prem environment
- Is an action for a virtual machine
- Is an action generated by the Turbonomic real-time market

```
{
  "actionInput":{
    "actionStateList":[
      "PENDING_ACCEPT"
    ],
    "environmentType":"ONPREM",
    "groupBy":[
      "actionModes"
    ]
  },
  "relatedType":"VirtualMachine",
  "scopes":[
    "Market"
  ]
}
```

Response: A list of `ActionApiDTO`s that meet the specified criteria.

For information about filtering actions within a specified time period, see *Filtering Actions by Start and End Time* in "Actions endpoint tips" on page 1313.

Accepting or rejecting a specified action

This request accepts or rejects an action with the specified UUID, and takes the following parameters:

- action_UUID (Required)

  The UUID of the action.

- accept (Required)

  When `true`, accepts the action. When `false`, rejects the action.

Example: `POST https://10.10.10.10/api/v3/actions/_QJCzNd3JEeij-LI4LMEROw?accept=false`

Response: `true`, if the action was accepted.

Actions endpoint tips

GroupBy field criteria

When using the POST requests to query actions and action statistics, there is a `groupBy` parameter that you may use to group statistics. The valid values for this parameter differ by entity type. Here is the complete list of valid options per entity:

Note:

When sending an inputDTO using this field, only the value is needed.

Example:`"groupBy": "vmsByPMName"`

```
{
  "VirtualMachine":{
    "criteria":[
      {
        "filterType":"vmsByName"
      },
      {
        "filterType":"vmsByPMName"
      },
      {
        "filterType":"vmsByStorage"
      },
      {
        "filterType":"vmsByNetwork"
      },
      {
        "filterType":"vmsByApplication"
      },
      {
        "filterType":"vmsByDatabaseServer"
      },
      {
        "filterType":"vmsByDatabaseServerVersion"
      },
      {
        "filterType":"vmsByDC"
      },
      {
        "filterType":"vmsByVDC"
      },
      {
        "filterType":"vmsByDCnested"
      },
      {
        "filterType":"vmsByNumCPUs"
      },
      {
        "filterType":"vmsByMem"
      },
      {
        "filterType":"vmsByGuestName"
      },
      {
        "filterType":"vmsByAltName"
      },
      {
        "filterType":"vmsByClusterName"
      },
      {
        "filterType":"vmsByDiskArrayName"
      },
      {
        "filterType":"vmsByLogicalPoolName"
      },
      {
        "filterType":"vmsByProfileName"
      },
      {
        "filterType":"vmsByTag"
      },
      {
        "filterType":"vmsByState"
      },
```

```
      {
        "filterType":"vmsByBusinessAccountUuid"
      },
      {
        "filterType":"vmsByResourceGroupUuid"
      }
    ]
  },
  "VirtualDataCenter":{
    "criteria":[
      {
        "filterType":"vdcsByName"
      },
      {
        "filterType":"vdcsByVDCName"
      },
      {
        "filterType":"vdcsByTag"
      },
      {
        "filterType":"vdcsByState"
      }
    ]
  },
  "PhysicalMachine":{
    "criteria":[
      {
        "filterType":"pmsByName"
      },
      {
        "filterType":"pmsByStorage"
      },
      {
        "filterType":"pmsByNetwork"
      },
      {
        "filterType":"pmsBySwitch"
      },
      {
        "filterType":"pmsByNumVms"
      },
      {
        "filterType":"pmsByDC"
      },
      {
        "filterType":"pmsByMem"
      },
      {
        "filterType":"pmsByNumCPUs"
      },
      {
        "filterType":"pmsByVendorName"
      },
      {
        "filterType":"pmsByCPUModel"
      },
      {
        "filterType":"pmsByModel"
      },
      {
        "filterType":"pmsByTimezone"
      },
      {
        "filterType":"pmsByClusterName"
      },
      {
        "filterType":"pmsByTag"
      },
      {
        "filterType":"pmsByState"
      }
    ]
  },
  "Storage":{
    "criteria":[
      {
        "filterType":"storageByName"
      },
      {
        "filterType":"storageByTag"
      },
      {
```

```
        "filterType":"storageByVMs"
      },
      {
        "filterType":"storageByDC"
      },
      {
        "filterType":"storageByPMCluster"
      },
      {
        "filterType":"storageByState"
      }
    ]
  },
  "Application":{
    "criteria":[
      {
        "filterType":"appsByName"
      },
      {
        "filterType":"appsByTag"
      }
    ]
  },
  "ApplicationComponent":{
    "criteria":[
      {
        "filterType":"appSrvsByName"
      }
    ]
  },
  "WebServer":{
    "criteria":[
      {
        "filterType":"wbSrvsByName"
      }
    ]
  },
  "BusinessApplication":{
    "criteria":[
      {
        "filterType":"busAppsByName"
      }
    ]
  },
  "Database":{
    "criteria":[
      {
        "filterType":"databaseByName"
      },
      {
        "filterType":"databaseByTag"
      },
      {
        "filterType":"databaseByBusinessAccountUuid"
      },
      {
        "filterType":"databaseByResourceGroupUuid"
      }
    ]
  },
  "DatabaseServer":{
    "criteria":[
      {
        "filterType":"databaseServerByName"
      },
      {
        "filterType":"databaseServerByTag"
      },
      {
        "filterType":"databaseServerByBusinessAccountUuid"
      },
      {
        "filterType":"databaseServerByEngine"
      },
      {
        "filterType":"databaseServerByEdition"
      },
      {
        "filterType":"databaseServerByVersion"
      }
    ]
  },
```

```
"VirtualApplication":{
  "criteria":[
    {
      "filterType":"vappsByName"
    }
  ]
},
"Cluster":{
  "criteria":[
    {
      "filterType":"clustersByName"
    },
    {
      "filterType":"clustersByTag"
    }
  ]
},
"DataCenter":{
  "criteria":[
    {
      "filterType":"datacentersByName"
    },
    {
      "filterType":"datacentersByTag"
    }
  ]
},
"Group":{
  "criteria":[
    {
      "filterType":"groupsByName"
    },
    {
      "filterType":"groupsByTargetName"
    }
  ]
},
"StorageCluster":{
  "criteria":[
    {
      "filterType":"storageClustersByName"
    }
  ]
},
"DiskArray":{
  "criteria":[
    {
      "filterType":"diskarrayByName"
    }
  ]
},
"Zone":{
  "criteria":[
    {
      "filterType":"zonsByName"
    }
  ]
},
"Region":{
  "criteria":[
    {
      "filterType":"regsByName"
    }
  ]
},
"Network":{
  "criteria":[
    {
      "filterType":"netsByName"
    }
  ]
},
"LoadBalancer":{
  "criteria":[
    {
      "filterType":"lbsByName"
    }
  ]
},
"Chassis":{
  "criteria":[
    {
```

```
        "filterType":"chasByName"
      }
    ]
  },
  "IOModule":{
    "criteria":[
      {
        "filterType":"ioModuleByName"
      }
    ]
  },
  "StorageController":{
    "criteria":[
      {
        "filterType":"storagecontrollerByName"
      }
    ]
  },
  "DPod":{
    "criteria":[
      {
        "filterType":"dpodByName"
      }
    ]
  },
  "VPod":{
    "criteria":[
      {
        "filterType":"vpodByName"
      }
    ]
  },
  "LogicalPool":{
    "criteria":[
      {
        "filterType":"logicalPoolByName"
      }
    ]
  },
  "Switch":{
    "criteria":[
      {
        "filterType":"switchByName"
      }
    ]
  },
  "Container":{
    "criteria":[
      {
        "filterType":"containersByName"
      },
      {
        "filterType":"containersByVMName"
      }
    ]
  },
  "ContainerPod":{
    "criteria":[
      {
        "filterType":"containerpodsByName"
      },
      {
        "filterType":"containerpodsByVMName"
      }
    ]
  },
  "ServiceEntity":{
    "criteria":[
      {
        "filterType":"workloadByBusinessAccountUuid"
      },
      {
        "filterType":"workloadByResourceGroupUuid"
      }
    ]
  },
  "Workload":{
    "criteria":[
      {
        "filterType":"workloadByBusinessAccountUuid"
      },
      {
```

```
        "filterType":"workloadByResourceGroupUuid"
      }
    ]
  },
  "ResourceGroup":{
    "criteria":[
      {
        "filterType":"resourceGroupByName"
      },
      {
        "filterType":"resourceGroupByUuid"
      },
      {
        "filterType":"resourceGroupByBusinessAccountUuid"
      },
      {
        "filterType":"resourceGroupByTag"
      }
    ]
  },
  "BusinessAccount":{
    "criteria":[
      {
        "filterType":"businessAccountByName"
      },
      {
        "filterType":"businessAccountByUuid"
      },
      {
        "filterType":"subBusinessAccountOfUuid"
      },
      {
        "filterType":"businessAccountValidationStatus"
      },
      {
        "filterType":"businessAccountTargetName"
      },
      {
        "filterType":"businessAccountCloudProvider"
      }
    ]
  },
  "BillingFamily":{
    "criteria":[
      {
        "filterType":"billingFamilyByName"
      }
    ]
  },
  "ViewPod":{
    "criteria":[
      {
        "filterType":"viewPodByName"
      }
    ]
  },
  "DesktopPool":{
    "criteria":[
      {
        "filterType":"desktopPoolByName"
      },
      {
        "filterType":"desktopPoolByViewPod"
      }
    ]
  },
  "BusinessUser":{
    "criteria":[
      {
        "filterType":"businessUserByName"
      },
      {
        "filterType":"businessUserByDesktopPool"
      }
    ]
  }
}
```

Filtering actions by time

When you make a GET request to the /actions endpoint, you can filter the list of returned actions based on a specified start time, end time, or both.

- The startTime parameter excludes any actions that were executed (succeeded or failed) before a specified time.
- The endTime parameter excludes any actions with a createTime value that is after a specified time.

When you define both these parameters, the response includes all current and historical actions that were pending or executed during the specified time period.

For example, the following GET request to the markets/Market/actions endpoint returns all actions that were created, pending, or executed between 6:00 and 11:00 on May 8, 2024.

Example: GET https://10.10.10.10/api/v3/markets/Market/actions

Example input:

```
{ "startTime": "2024-05-08T06:00:19Z",
  "endTime": "2024-05-08T011:00:19Z",
  ...
}
```

Although the response does not include any actions that were created after the endTime, it might include actions that were created before the startTime, if they are still pending as of the startTime. Therefore, this filtering includes all actions that are either pending or executed during the specified time period, regardless of when those actions were created.

Admin endpoint

The admin endpoint is a collection of requests that enable you to perform various administrative functions. To use the admin endpoint, you must be logged into Turbonomic as an administrator user.

Using the admin endpoint, you can:

- Export diagnostics locally, and send them to Support
- Get health check results to see if there are issues related to the monitoring and analysis of your environment
- Get or set proxy settings
- Get or set logging levels for various components of your environment
- Get the current product version information and check for updates
- Load existing configuration files for use in your environment

Admin requests

Exporting diagnostics

If you are experiencing problems with Turbonomic, your support representative might request that you export diagnostic data. You can export the data and then send it to the support representative as requested.

Example: POST https://10.10.10.10/api/v3/admin/exports

Response: true if successful, false if an error occurs while Turbonomic creates the export. The export file is saved in the /tmp directory on your instance with a file name that begins with bkp-.

Exporting audit log

Audit logs record a series of user and system events that provide insight into application activity. These logs capture actions such as login, logout, account management, policy changes, and authorization-

related events. If you are experiencing problems with Turbonomic, your support representative might request that you export audit logs. You can export the data and then send it to the support representative as requested.

Example: `GET https://10.10.10.10/api/v3/admin/auditlogs`

Response: A zip file named `auditlog.zip` upon successful response. Extract the zip file to view one or more audit log files containing a chronological record of activities performed within the application.

Getting health check results

Get health check results regularly to see whether issues exist that are related to the monitoring and analysis of your environment.

Example: `GET https://10.10.10.10/api/v3/admin/health`

Response: HealthCategoryResponseDTO with the health check results.


```
[
  {
    "healthCategory": "TARGET",
    "categoryDisplayName": "Targets",
    "categoryHealthState": "CRITICAL",
    "responseItems": [
      {
        "subcategory": "VALIDATION",
        "healthState": "MAJOR",
        "numberOfItems": 2,
        "recommendations": []
      },
      {
        "subcategory": "DISCOVERY",
        "healthState": "CRITICAL",
        "numberOfItems": 6,
        "recommendations": [
          {
            "errorType": "INTERNAL_PROBE_ERROR",
            "description": "{Description}"
          }
        ]
      },
      {
        "subcategory": "DISCOVERY",
        "healthState": "NORMAL",
        "numberOfItems": 23,
        "recommendations": []
      }
    ]
  },
  {
    "healthCategory": "ACTION",
    "categoryDisplayName": "Actions",
    "categoryHealthState": "NORMAL",
    "responseItems": [
      {
        "subcategory": "ANALYSIS",
        "healthState": "NORMAL",
        "numberOfItems": 0,
        "recommendations": []
      }
    ]
  },
  {
    "healthCategory": "COMPONENT",
    "categoryDisplayName": "Components",
    "categoryHealthState": "NORMAL",
    "responseItems": [
      {
        "subcategory": "CONTAINER PODS",
        "healthState": "NORMAL",
        "numberOfItems": 45,
        "recommendations": [
          {
            "errorType": "INFO",
            "description": "Pods that are in a running state."
          }
        ]
```

```
      }
    ]
  }
]
```

The response includes the following health check categories (`healthCategory`):

- `TARGET` – see whether issues exist with the targets you added, such as validation or discovery failures.
- `ACTION` – see whether issues exist with analyzing your environment, such as analysis timeouts.
- `COMPONENT` – see whether issues exist with Turbonomic components, such as container pods that are not running.

Ideally, each `healthCategory` shows `"categoryHealthState": "NORMAL"`. If you see a different value, review the issues and recommendations in the `responseItems` key.

Use the following parameters to filter the results:

- `category`

  Get results for a specific `healthCategory`.
- `health_state`

  Get results with a specific `categoryHealthState`. For example, choose `CRITICAL` to see issues that require immediate attention.

Getting and setting a proxy

Gets the current state of the proxy and proxy details (if enabled).

Example: `GET https://10.10.10.10/api/v3/admin/httpproxy`

Response: A `HttpProxyDTO` describing details of the proxy. If no proxy is configured, this request returns `"isProxyEnabled": false`.

```
{
  "isProxyEnabled":true,
  "proxyHost":"10.10.111.1",
  "userName":"UserName",
  "password":"PW"
}
```

To set the proxy, pass an `HttpProxyDTO`, making sure to include `"isProxyEnabled": true` if you want to activate the proxy immediately.

Example: `POST https://10.10.10.10/api/v3/admin/httpproxy`

Example httpProxyDTO:

```
{
  "isProxyEnabled":true,
  "proxyHost":"10.10.111.1",
  "portNumber":"33128",
  "userName":"ProxyUser",
  "password":"Password"
}
```

Response: `true` if successful, `false` if an error occurs while Turbonomic is setting the proxy.

Getting and setting logging levels by component

You can set the level of logging for different components of the Turbonomic platform. The default logging level for all components is `INFO`. Setting more verbose logging levels can greatly increase the disk space that is required to store the log files. You normally change these settings only while you're working with a Turbonomic support representative.

The available logging levels are (in order of least to most verbose): `INFO, WARN, DEBUG,` and `TRACE`.

Example: `GET https://10.10.10.10/api/v3/admin/logginglevels`

Response: A `LoggingApiDTO` with details about your current logging level settings:

```
    {
  "componentLoggingLevel":{
    "Extension":"INFO",
    "Analysis":"INFO",
    "Discovery":"INFO",
    "API":"INFO",
    "Presentation":"INFO",
    "Abstraction":"INFO",
    "Monitoring":"INFO"
  }
}
```

To set the logging level for a specific component, pass a partial LoggingApiDTO containing the component and logging level.

Example: `POST https://10.10.10.10/api/v3/admin/logginglevels`

Example LoggingApiDTO:

```
    {
  "componentLoggingLevel":{
    "Extension":"WARN"
  }
}
```

Response: A successful response returns your input in the response body.

Getting the current version

Gets the version of each Turbonomic package that your instance is running. Set the `check_for_updates` parameter to `true` to also check for updates that might be available.

This request takes the following parameters:

```
check_for_updates
```

Check for available updates before returning version information. Default: `false`.

Example: `GET https://10.10.10.10/api/v3/admin/versions?check_for_updates=true`

Response: A `ProductVersionDTO` describing details about the current version and available updates. If you are running the latest version of Turbonomic, the updates value reads `"No Updates Available\n"`

```
{
    "versionInfo": "Turbonomic Operations Manager 6.1.7 (Build 20180813223950000)\nvmt-
platform-6.1.7-20180813223950000.i586
    ...,
    "updates": "Available Packages\nvmt-bundle.i586  6.2.0-20180817180112000 vmturbo\nvmt-
config.x86_64
    ...,
    "marketVersion": 2
}
```

Loading external configuration files

To help with diagnosis of some issues, a support representative might want you to load configuration files into your Turbonomic appliance. Use this request only when you are working with a support

representative, who can guide you through the process. The support representative specifies what kind of configuration file you are loading, and the contents of that file, by specifying the required parameters.

Example: `POST https://10.10.10.10/api/v3/admin/configfiles?config_type=TOPOLOGY&topology=abc`

Response: `true` if successful, `false` if an error occurs while you are uploading the configuration.

Authorization endpoint

You can use the authorization endpoint to create and manage OAuth 2.0 clients. You can then use these clients to authenticate applications to Turbonomic APIs.

Using the authorization endpoint, you can:

- Create OAuth 2.0 clients
- List OAuth 2.0 clients
- Delete OAuth 2.0 clients

Authorization requests

Use authorization requests to create, manage, and authenticate OAuth 2.0 clients.

Prerequisites

- Turbonomic version 8.12.4 or later
- `t8c-operator` version `42.57` or later
- `ADMINISTRATOR` or `SITE_ADMIN` role

  Note: `ADMINISTRATOR` role is required to create an OAuth 2.0 client with the `ADMINISTRATOR` role (that is, a client with the same privileges as administrator. The `SITE_ADMIN` role is not sufficient in this case.

To interact with OAuth 2.0 APIs, you must be authenticated with the Turbonomic instance. For more information, see "Authenticating with the API" on page 1568.

Authenticating calls to Turbonomic APIs

To authenticate calls to Turbonomic APIs, create an OAuth 2.0 client and then exchange the client credentials for an access token.

You can also try these calls by using the Turbonomic swagger UI. For more information, see Turbonomic REST API Swagger Documentation.

Currently, Turbonomic supports only the OAuth 2.0 client credentials flow.

Creating an OAuth 2.0 client

1. Construct a `POST https://{your_instance_address}/api/v3/authorization/oauth2/clients` request with the following payload.

```
{
  "clientName": "<client_name>",
  "grantTypes": [
    "client_credentials"
  ],
  "clientAuthenticationMethods": [
    "client_secret_post"
  ],
  "scopes": [
    "role:<ROLE_NAME>"
  ],
  "tokenSettings": {
    "accessToken": {
      "ttlSeconds": 700
```

```
    }
  }
}
```

2. Specify the following values.
   - `clientName`: Choose a name for your client.
   - `role`: Specify a Turbonomic user role. Use uppercase letters only. Any access tokens that you generate for this client must have the same role. The following roles are available:
     - `ADMINISTRATOR` (only if the logged-in user is an administrator)

       Users with this role can use all Turbonomic features and modify settings to configure the Turbonomic installation. For Turbonomic instances hosted in the public cloud, this role is limited to the Turbonomic representative that manages the instances.
     - `SITE_ADMIN`

       Users with this role can use all Turbonomic features and modify site-specific settings to configure the Turbonomic installation. Users can also administer Groups, Policies, Workflows, Templates, Billing/Costs, and Target Configuration, but not Email, Licenses, Updates, and Maintenance. Users can create other user accounts, except accounts with the Administrator role.
     - `AUTOMATOR`

       Users with this role can use all Turbonomic features (including Plan, Park, and Place), but cannot configure the Turbonomic installation or create policies.
     - `DEPLOYER`

       Users with this role can view all Turbonomic charts and data, use Place to reserve workloads, and create placement policies and templates. However, users cannot run plans or execute any recommended actions.
     - `PARKER`

       Users with this role can park workloads manually or automatically (using parking schedules), manage parking schedules, and view all Turbonomic charts and data. However, users cannot create policies and templates, place workloads, run plans, or execute any recommended actions.
     - `PARKER_AUTOMATOR`

       Users with this role can park workloads manually or automatically (using parking schedules) and view all Turbonomic charts and data. However, users cannot manage parking schedules, create policies and templates, place workloads, run plans, or execute any recommended actions.
     - `ADVISOR`

       Users with this role can view all Turbonomic charts and data, and run plans. However, users cannot use Place to reserve workloads, create policies, or execute any recommended actions.
     - `OBSERVER`

       Users with this role can view the environment, including the Home Page and Dashboards. Users can also use Search to set a scope to the session. For scope, only VM groups and Resource Groups are supported.
     - `OPERATIONAL_OBSERVER`

       Users with this role can view the environment, including the Home Page, Dashboards, Groups, and Policies. Users can also use Search to set a scope to the session.
     - `SHARED_ADVISOR`

       Users with this role are scoped users. They can view the Home Page and Dashboards, but only see VMs and Applications. Users cannot execute Turbonomic actions.
     - `SHARED_OBSERVER`

Users with this role are scoped users. They can view the Home Page, custom dashboards, and application maps, but can see only VMs and Applications. They cannot view Executive Dashboards or execute Turbonomic actions. Shared Observer is the most restricted user role.

- REPORT_EDITOR

  Users with this role can create, edit, and delete reports. Due to limits to the reporting license, only one user per instance is allowed to have this role (by default, the local **administrator** user). To assign this role to another user, you must first remove it from the current user. Be sure that the new user is *not* a scoped user.

3. (Optional) Specify the following parameters:
   - `clientAuthenticationMethods`: This value determines whether the client credentials must be supplied in a basic auth header or in the request body when you request an access token for the client. Specify one or both of the following values:
     - `client_secret_basic`: the default, which requires the credentials to be supplied as part of a basic auth header
     - `client_secret_post`: the credentials must be supplied in the request body
   - `tokenSettings.accessToken.ttlSeconds`: This value determines the lifespan of access tokens that the OAuth 2.0 authorization server issues. The default value is `600` seconds.
4. Execute the request.

   The response is similar to the following example.

```
{
  "clientName": "my-client",
  "clientId": "My_ID",
  "grantTypes": [
    "client_secret"
  ],
  "clientAuthenticationMethods": [
    "client_secret_post"
  ],
  "scopes": [
    "role:SITE_ADMIN"
  ],
  "audience": [
    "turbonomic"
  ],
  "createdAt": 1702042743,
  "tokenSettings": {
    "accessToken": {
      "ttlseconds": 700
    }
  },
  "clientSecret": "{secret}"
}
```

5. Copy the `clientId` and `clientSecret` values.

   Important: The `clientSecret` cannot be recovered if lost.

Exchanging the client credentials for an access token

1. Construct a request to the `/oauth2/token` endpoint. This endpoint is a standard OAuth 2.0 API. You can use any library that supports OAuth 2.0 to facilitate this transaction programmatically. For more information, see the documentation for your chosen library. The following steps describe how to construct the call manually.
   - If your client uses `client_secret_basic` as the authentication method, include the client credentials in a basic `auth` header, not in the request body.
     a. Construct a POST `https://{your_instance_address}:443/oauth2/token` request with the following payload.

```
grant_type=client_credentials&scope=role:<ROLE_NAME>
```

b. For the `role` parameter, specify the role that you configured in a previous step.

- If your client uses `client_secret_post` as the authentication method, include the client credentials in the request body.

  a. Construct a `POST https://{your_instance_address}:443/oauth2/token` request.

  ```
  grant_type=client_credentials&scope=role:<ROLE_NAME>&client_id=<client_ID>&client_secret
  =client_secret>
  ```

  b. Specify the following values:

  - `scope`: Specify the role that you configured in a previous step.
  - `client_id`: Specify the client ID that you copied in a previous step.
  - `client_secret`: Specify the client secret that you copied in a previous step.

2. Execute the request.

   The response is similar to the following example.

   ```
   {
     "access_token": "{token}",
     "scope": "role:SITE_ADMIN",
     "token_type": "Bearer",
     "expires_in": 599
   }
   ```

3. Copy the `access_token` value.
4. Use the access token to access Turbonomic APIs.

   Include the authorization token in a bearer authorization header of a request to a Turbonomic API that is available to your role, such as the `GET licenses` API. For example, you can set the token value as a variable in your terminal:

   ```
   token={token}
   ```

   When you make a request to a Turbonomic API, include the variable in a bearer authorization header, as shown in the following curl example:

   ```
   curl -H "Authorization: Bearer ${token}" https://$host:443/vmturbo/rest/licenses
   ```

Listing OAuth 2.0 clients

List the OAuth 2.0 clients in your Turbonomic instance.

Example: `GET https://10.10.10.10/api/v3/authorization/oauth2/clients`

Response: returns a list of OAuth2ClientApiDTOs for your instance, similar to the following example.

```
{
  "clientName": "my-client",
  "clientId": "My_ID",
  "grantTypes": [
    "client_secret"
  ],
  "clientAuthenticationMethods": [
    "client_secret_post"
  ],
  "scopes": [
    "role:SITE_ADMIN"
  ],
  "audience": [
    "turbonomic"
  ],
  "createdAt": 1702042743,
  "tokenSettings": {
    "accessToken": {
      "ttlSeconds": 700
    }
}
```

Deleting an OAuth 2.0 client

Delete an OAuth 2.0 client from your Turbonomic instance.

Example: `DELETE https://10.10.10.10/api/v3/authorization/oauth2/clients/[clientId]`

A successful `DELETE` request does not return a response body. You can confirm that the request succeeded by running the same request again. The response to that request is `404 NOT_FOUND` because the target client was already deleted.

Any tokens that were issued for the deleted client are immediately invalidated and can no longer be used to access Turbonomic APIs.

Troubleshooting OAuth 2.0 clients

If you cannot access the `authorization/oauth2/clients/` API, make sure that you satisfy the prerequisites described in this topic.

If the token you get from the OAuth 2.0 API is not working, check the following values:

- Make sure that the token is not expired. The default lifespan of a token is 10 minutes. If the token expires, you must retrieve a new one.
- Ensure that you specified a scope in your request. If you don't include a scope in your request, the token is not valid.

Businessunits endpoint

Business units are container objects used for modeling accounts and subscriptions in Public Cloud targets, and for cloud budgets and price adjustements.

Using the businessunits endpoint of the Turbonomic API, you can:

- For a specified business unit, get a list of:
  - All actions relating to the entities in the business unit
  - Cloud services and pricing models
  - All entities belonging to the business unit
  - Price adjustments
  - Any related (parent, sibling, or child) business units
  - Statistics relating to entities in the business unit
  - Templates available
- Get the supply chain for a specified business unit
- Create a business unit
- Edit a business unit
- Edit a price adjustment
- Delete a business unit

Business units may be one of three types:

- BUDGET

  A business unit that has a budget, and may contain children.
- DISCOUNT

  A business unit that has a discount, and may contain children.
- DISCOVERED

  A business unit that has been discovered by the Turbonomic instance. These business units should not be edited or deleted, except on the discovered target.

Businessunits requests

Getting business units

Gets a list of all business units created on or discovered by your Turbonomic instance. This request takes the following parameters:

- type

  The type of business unit to return. `[ BUDGET, DISCOUNT, DISCOVERED ]`

- cloud_type

  The cloud provider whose business units to return. `[ AWS, AZURE, GCP ]`

- has_parent

  Whether to return only business units that have parent business units. Default: `False`. When `true`, all relevant business units will be returned irrespective of parent business units.

Example: `GET https://10.10.10.10/api/v3/businessunits?type=DISCOVERED&cloud_type=AWS`

Response: A list of `BusinessUnitApiDTO`s that meet the selected criteria. In this case, discovered business units that are part of an AWS environment.


```
[
  {
    "uuid": "323871187550",
    "displayName": "Product Trust",
    "className": "BusinessAccount",
    "environmentType": "CLOUD",
    "master": false,
    "hasRelatedTarget": true,
    "budget": {
      "value": 0
    },
    "businessUnitType": "DISCOVERED",
    "cloudType": "AWS",
    "targets": [
      {
        "uuid": "_6glgQEEhEemQ645gSx8RYw",
        "displayName": "PT-AWS",
        "type": "AWS"
      }
    ],
    "costPrice": 0,
    "severity": "Critical",
    "membersCount": 80,
    "memberType": "Workload",
    "related": {
      "rateCard": null,
      "priceAdjustment": null
    },
    "resourceGroupsCount": 0
  },
  {
    "uuid": "001844731978",
    "displayName": "Advanced Engineering",
    "className": "BusinessAccount",
    "environmentType": "CLOUD",
    "master": false,
    "hasRelatedTarget": false,
    "budget": {
      "value": 0
    },
    "businessUnitType": "DISCOVERED",
    "cloudType": "AWS",
    "costPrice": 0,
    "severity": "Normal",
    "membersCount": 0,
    "memberType": "Workload",
    "related": {
      "rateCard": null,
      "priceAdjustment": null
    },
```

```
    "resourceGroupsCount": 0
  },
  {
    "uuid": "192821421245",
    "displayName": "Development",
    "className": "BusinessAccount",
    "environmentType": "CLOUD",
    "master": true,
    "hasRelatedTarget": true,
    "budget": {
      "value": 0
    },
    "businessUnitType": "DISCOVERED",
    "cloudType": "AWS",
    "childrenBusinessUnits": [
      "631949720430",
      "001844731978",
      "323871187550"
    ],
    "targets": [
      {
        "uuid": "_cBVj0MvWEeiFJej-UwMSGg",
        "displayName": "engineering.aws.amazon.com",
        "type": "AWS"
      }
    ],
    "costPrice": 0,
    "severity": "Critical",
    "membersCount": 33,
    "memberType": "Workload",
    "related": {
      "rateCard": null,
      "priceAdjustment": null
    },
    "resourceGroupsCount": 0
  },
  {
    "uuid": "631949720430",
    "displayName": "Turbonomic SaaS",
    "className": "BusinessAccount",
    "environmentType": "CLOUD",
    "master": false,
    "hasRelatedTarget": false,
    "budget": {
      "value": 0
    },
    "businessUnitType": "DISCOVERED",
    "cloudType": "AWS",
    "costPrice": 0,
    "severity": "Normal",
    "membersCount": 0,
    "memberType": "Workload",
    "related": {
      "rateCard": null,
      "priceAdjustment": null
    },
    "resourceGroupsCount": 0
  }
]
```


Deleting a business unit

To delete a business unit, pass the UUID of the business unit in the request. Note, you cannot delete DISCOVERED business units.

Example: `DELETE https://10.10.10.10/api/v3/businessunits/49d50ac3f8adbfe649d53e72c42dd9428d1d3835`

Response: A response of 200 indicates successful deletion

Getting cloud service and pricing model pairs for a specified business unit

Gets the cloud service name and pricing model for each cloud service.

Example: `GET https://10.10.10.10/api/v3/businessunits/192821421245/cloudservices`

Response: A list of `CloudServicePricingModelApiDTO`s representing each cloud service available to the business unit:


```
[
  {
    "uuid": "aws::631949720430::CS::AWSCloudTrail",
    "displayName": "AWS CloudTrail",
    "pricingModel": "ON_DEMAND"
  },
  {
    "uuid": "aws::631949720430::CS::AmazonCloudWatch",
    "displayName": "AWS CloudWatch",
    "pricingModel": "ON_DEMAND"
  },
  {
    "uuid": "aws::192821421245::CS::AWSDeveloperSupport",
    "displayName": "AWS Developer Support",
    "pricingModel": "ON_DEMAND"
  },
  {
    "uuid": "aws::001844731978::CS::AmazonDynamoDB",
    "displayName": "AWS DynamoDB",
    "pricingModel": "ON_DEMAND"
  },
  {
    "uuid": "aws::631949720430::CS::AmazonEC2",
    "displayName": "AWS EC2",
    "pricingModel": "ON_DEMAND"
  },
  {
    "uuid": "aws::001844731978::CS::AmazonEKS",
    "displayName": "AWS EKS",
    "pricingModel": "ON_DEMAND"
  },
  ...
]
```


Getting price adjustments related to a specified business unit

Gets any price adjustments affecting the specified business unit.

Example: `GET https://10.10.10.10/api/v3/businessunits/192821421245/priceadjustments`

Response: A list of `BusinessUnitPriceAdjustmentApiDTO`s showing the corrected prices after adjustment:


```
[
  {
    "servicePriceAdjustments": [
      {
        "uuid": "aws::192821421245::CS::AmazonCloudWatch",
        "displayName": "AWS CloudWatch",
        "pricingModel": "ON_DEMAND"
      },
      {
        "uuid": "aws::192821421245::CS::AWSDeveloperSupport",
        "displayName": "AWS Developer Support",
        "pricingModel": "ON_DEMAND"
      },
      {
        "uuid": "aws::192821421245::CS::AmazonDynamoDB",
        "displayName": "AWS DynamoDB",
        "pricingModel": "ON_DEMAND"
      },
      {
        "uuid": "aws::192821421245::CS::AmazonEC2",
        "displayName": "AWS EC2",
        "pricingModel": "ON_DEMAND",
        "templateDiscounts": [
          {
            "uuid": "aws::VMPROFILE::g3.16xlarge",
            "displayName": "g3.16xlarge",
            "family": "g3",
```

```
            "pricesPerDatacenter": [
              {
                "uuid": "aws::ap-southeast-1::DC::ap-southeast-1",
                "displayName": "aws-Asia Pacific (Singapore)",
                "price": 6.68
              },
              {
                "uuid": "aws::us-west-2::DC::us-west-2",
                "displayName": "aws-US West (Oregon)",
                "price": 4.56
              },
              {
                "uuid": "aws::ca-central-1::DC::ca-central-1",
                "displayName": "aws-Canada (Central)",
                "price": 5.664
              },
              {
                "uuid": "aws::eu-west-1::DC::eu-west-1",
                "displayName": "aws-EU (Ireland)",
                "price": 4.84
              },
              {
                "uuid": "aws::eu-west-3::DC::eu-west-3",
                "displayName": "aws-EU (Paris)",
                "price": 0
              },
              {
                "uuid": "aws::ap-northeast-1::DC::ap-northeast-1",
                "displayName": "aws-Asia Pacific (Tokyo)",
                "price": 6.32
              },
              {
                "uuid": "aws::eu-central-1::DC::eu-central-1",
                "displayName": "aws-EU (Frankfurt)",
                "price": 5.7
              },
              {
                "uuid": "aws::ap-south-1::DC::ap-south-1",
                "displayName": "aws-Asia Pacific (Mumbai)",
                "price": 0
              },
              {
                "uuid": "aws::ap-northeast-2::DC::ap-northeast-2",
                "displayName": "aws-Asia Pacific (Seoul)",
                "price": 0
              },
              {
                "uuid": "aws::us-east-1::DC::us-east-1",
                "displayName": "aws-US East (N. Virginia)",
                "price": 4.56
              },
              {
                "uuid": "aws::us-west-1::DC::us-west-1",
                "displayName": "aws-US West (N. California)",
                "price": 6.136
              },
              {
                "uuid": "aws::sa-east-1::DC::sa-east-1",
                "displayName": "aws-South America (Sao Paulo)",
                "price": 0
              },
              {
                "uuid": "aws::ap-southeast-2::DC::ap-southeast-2",
                "displayName": "aws-Asia Pacific (Sydney)",
                "price": 7.016
              },
              {
                "uuid": "aws::us-east-2::DC::us-east-2",
                "displayName": "aws-US East (Ohio)",
                "price": 4.56
              },
              {
                "uuid": "aws::eu-north-1::DC::eu-north-1",
                "displayName": "aws-EU (Stockholm)",
                "price": 0
              },
              {
                "uuid": "aws::eu-west-2::DC::eu-west-2",
                "displayName": "aws-EU (London)",
                "price": 0
              }
            ]
```

```
          },
          {
            "uuid": "aws::VMPROFILE::p3.8xlarge",
            "displayName": "p3.8xlarge",
            "family": "p3",
            "pricesPerDatacenter": [
              {
                "uuid": "aws::ap-southeast-1::DC::ap-southeast-1",
                "displayName": "aws-Asia Pacific (Singapore)",
                "price": 16.936
              },
              {
                "uuid": "aws::us-west-2::DC::us-west-2",
                "displayName": "aws-US West (Oregon)",
                "price": 12.24
              },
              {
                "uuid": "aws::ca-central-1::DC::ca-central-1",
                "displayName": "aws-Canada (Central)",
                "price": 13.464
              },
              {
                "uuid": "aws::eu-west-1::DC::eu-west-1",
                "displayName": "aws-EU (Ireland)",
                "price": 13.22
              },
              {
                "uuid": "aws::eu-west-3::DC::eu-west-3",
                "displayName": "aws-EU (Paris)",
                "price": 0
              },
              {
                "uuid": "aws::ap-northeast-1::DC::ap-northeast-1",
                "displayName": "aws-Asia Pacific (Tokyo)",
                "price": 16.776
              },
              {
                "uuid": "aws::eu-central-1::DC::eu-central-1",
                "displayName": "aws-EU (Frankfurt)",
                "price": 15.292
              },
              {
                "uuid": "aws::ap-south-1::DC::ap-south-1",
                "displayName": "aws-Asia Pacific (Mumbai)",
                "price": 0
              },
              {
                "uuid": "aws::ap-northeast-2::DC::ap-northeast-2",
                "displayName": "aws-Asia Pacific (Seoul)",
                "price": 16.936
              },
              ...
            ]
          }
        ]
      }
    ]
  }
]
```


Editing price adjustments of a specified business unit

To edit a price adjustment for the specified business unit, send a BusinessUnitPriceAdjustmentApiDTO with the corrected information (in this case, changing the percent discount from 25 to 50.

Example: `PUT https://10.10.10.10/api/v3/businessunits/c0acb428d7b447a5d51f9888d2b6660b5fa5225f`

Example Input:


```
{
  "name": "Daltest1",
  "priceAdjustment": {
    "type": "DISCOUNT",
    "value": 50
  },
```

```
  "childrenBusinessUnits": [
    "192821421245"
  ],
  "businessUnitType": "DISCOUNT",
  "uuid": "c0acb428d7b447a5d51f9888d2b6660b5fa5225f"
}
```


Response: The full `BusinessUnitPriceAdjustmentApiDTO` with the corrected values:


```
{
  "uuid": "c0acb428d7b447a5d51f9888d2b6660b5fa5225f",
  "displayName": "Daltest1",
  "className": "BusinessAccount",
  "environmentType": "CLOUD",
  "discount": 50,
  "priceAdjustment": {
    "type": "DISCOUNT",
    "value": 50
  },
  "master": false,
  "hasRelatedTarget": false,
  "budget": {
    "value": 0
  },
  "businessUnitType": "DISCOUNT",
  "cloudType": "AWS",
  "childrenBusinessUnits": [
    "192821421245"
  ],
  "costPrice": 0,
  "severity": "Normal",
  "membersCount": 0,
  "memberType": "Workload",
  "related": {
    "rateCard": null,
    "priceAdjustment": {
      "uuid": "c0acb428d7b447a5d51f9888d2b6660b5fa5225f",
      "displayName": "Daltest1",
      "className": "BusinessAccount"
    }
  },
  "resourceGroupsCount": 0
}
```


Getting templates available to a business unit

Referenced Endpoint: Templates

Gets details about each template available to the specified business unit:

Example: `GET https://10.10.10.10/api/v3/businessunits/192821421245/templates`

Response: A list of `TemplateApiDTO`s representing the available templates for the specified business unit:


```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/templates/B4166E9F-589B-357D-AC74-0A534B53CE85"
      }
    ],
    "uuid": "B4166E9F-589B-357D-AC74-0A534B53CE85",
    "displayName": "db.x1e.32xlarge",
    "className": "DatabaseServerProfile",
    "price": 0,
    "model": "db.x1e.32xlarge",
    "dbEngine": "Oracle",
    "dbEdition": "Enterprise",
    "computeResources": [
      {
        "stats": [
```

```
          {
            "name": "numOfCpu",
            "value": 128
          },
          {
            "name": "cpuSpeed",
            "units": "MHz",
            "value": 384000
          },
          {
            "name": "ioThroughputSize",
            "units": "MB/s",
            "value": 546.875
          },
          {
            "name": "memorySize",
            "units": "MB",
            "value": 3997696
          }
        ]
      }
    ],
    "storageResources": [
      {
        "stats": [
          {
            "name": "diskSize",
            "units": "GB",
            "value": 3840
          },
          {
            "name": "diskIops",
            "value": 35000
          }
        ]
      }
    ],
    "deploymentProfile": {
      "uuid": "5C57DCA5-72AB-34BF-A22D-06719D13182E",
      "displayName": "DEP-RDB-'ap-northeast-1'",
      "className": "ServiceCatalogItem",
      "deployParameters": [
        {
          "targetType": "AWS",
          "providers": [
            {
              "provider": {
                "discoveredBy": {
                  "uuid": "_RRn00PMSEeisB7K_chUWVw",
                  "displayName": "vmturbodev.aws.amazon.com",
                  "type": "AWS"
                }
              }
            }
          ]
        }
      ]
    },
    "discovered": true
  },
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/templates/aws::VMPROFILE::g3.16xlarge"
      }
    ],
    "uuid": "aws::VMPROFILE::g3.16xlarge",
    "displayName": "g3.16xlarge",
    "className": "VirtualMachineProfile",
    "description": "g3.16xlarge",
    "computeResources": [
      {
        "stats": [
          {
            "name": "ioThroughputSize",
            "units": "MB/s",
            "value": 1750
          },
          {
            "name": "networkThroughputSize",
            "units": "MB/s",
```

```
            "value": 2500
          },
          {
            "name": "numOfCpu",
            "value": 64
          },
          {
            "name": "cpuSpeed",
            "units": "MHz",
            "value": 6664.6587
          },
          {
            "name": "cpuConsumedFactor",
            "units": "%",
            "value": 50
          },
          {
            "name": "memorySize",
            "units": "MB",
            "value": 499712
          },
          {
            "name": "memoryConsumedFactor",
            "units": "%",
            "value": 75
          },
          {
            "name": "ioThroughput",
            "units": "MB/s",
            "value": 0
          },
          {
            "name": "networkThroughput",
            "units": "MB/s",
            "value": 0
          }
        ]
      }
    ],
    "storageResources": [
      {
        "stats": [
          {
            "name": "diskSize",
            "units": "GB",
            "value": 976562.5
          },
          {
            "name": "diskIops",
            "value": 20000
          },
          {
            "name": "diskConsumedFactor",
            "units": "%",
            "value": 100
          }
        ]
      }
    ],
    "deploymentProfile": {
      "uuid": "aws::us-east-1::DP::DEP-fake-us-east-1",
      "displayName": "DEP-fake-us-east-1",
      "className": "ServiceCatalogItem",
      "deployParameters": [
        {
          "targetType": "AWS",
          "providers": [
            {
              "provider": {
                "discoveredBy": {
                  "uuid": "_RRn00PMSEeisB7K_chUWVw",
                  "displayName": "vmturbodev.aws.amazon.com",
                  "type": "AWS"
                }
              }
            }
          ]
        }
      ]
    },
    "discovered": true,
    "family": "g3"
```

```
  },
  ...
]
```

Clients endpoint

A secure client, also known as the secure connector for hybrid support, is an agent that you install in your on-prem environment so that a SaaS server can manage on-prem targets, such as VMware vCenter and Red Hat OpenShift.

Tokens are used to establish the secure connection between the client and server.

Using the clients endpoint, you can:

- Get a list of secure client endpoints that need to be accessible to the SaaS server
- Get a list of secure clients
- Remove access for all secure clients
- Get a list of secure client tokens
- Create a token to add a secure client
- Delete a secure client token

Clients requests

Getting the secure client endpoints

Gets a list of secure client endpoints that need to be accessible to the SaaS server.

Example: `GET https://10.10.10.10/api/v3/clients/networks/configuration`

Response:

```
{
  "clientFirewallAllowlist": [
    {
      "name": "claims:0:0",
      "type": "Route",
      "host": "claims-turbonomic.example-host",
      "ip": "9.9.9.9",
      "port": 443
    },
    {
      "name": "skupper:0:0",
      "type": "Route",
      "host": "skupper-turbonomic.example-host",
      "ip": "9.9.9.9",
      "port": 443
    },
    {
      "name": "skupper-edge:0:0",
      "type": "Route",
      "host": "skupper-edge-turbonomic.example-host",
      "ip": "9.9.9.9",
      "port": 443
},
    {
      "name": "skupper-inter-router:0:0",
      "type": "Route",
      "host": "skupper-inter-router-turbonomic.example-host",
      "ip": "9.9.9.9",
      "port": 443
    }
  ]
}
```

Getting a list of secure clients

Gets a list of secure clients.

Example: `GET https://10.10.10.10/api/v3/clients/networks/sites`

Response:

```
[
  {
    "id": "<ID_1>",
    "name": "turbonomic-client"
  },
  {
    "id": "<ID_2>",
    "name": "turbonomic-client"
  }
]
```

Removing access for all secure clients

Removes access for all secure clients.

Example: `DELETE https://10.10.10.10/api/v3/clients/networks/sites`

Response: `Status 200`

Get a list of secure client tokens

Gets a list of secure client tokens.

Example: `GET https://10.10.10.10/api/v3/clients/networks/tokens`

Response:

```
[
  {
    "id": "<ID_1>",
    "created": "1690810016",
    "claimsMade": 0,
    "claimsRemaining": 1,
    "claimExpiration": "1690810916"
  },
  {
    "id": "<ID_2>",
    "created": "1690819931",
    "claimsMade": 0,
    "claimsRemaining": 1,
    "claimExpiration": "1690820831"
  },
  {
    "id": "<ID_3>",
    "created": "1690821508",
    "claimsMade": 1,
    "claimsRemaining": 0,
    "claimExpiration": "1690822408"
  },
 {
    "id": "<ID_4>",
    "created": "1690879957",
    "claimsMade": 1,
    "claimsRemaining": 0,
    "claimExpiration": "1690880857"
  }
]
```

Creating a token to add a secure client

In order to add a new secure client, a token must be generated and then applied on the secure client. A token can be returned in either base-64 encoded or in raw format depending on the request format.

With a payload in the following format:

```
{ "claimLimit": 1, "expiration": 15 }
```

`claimLimit` refers to the number of times a token can be applied.

`expiration` refers to the time in minutes for the token to be claimed, a token that has expired cannot be used to connect client and server.

Example: `POST https://10.10.10.10/api/v3/clients/networks/tokens?output=encoded`

Response (encoded in base-64):

```
{
  "id": "<ID>",
  "tokenData": "<token>"
}
```

Example: `POST https://10.10.10.10/api/v3/clients/networks/tokens?output=raw`

Response (in raw format):

```
{
  "apiVersion": "v1",
  "kind": "Secret",
  "metadata": {
    "name": "2f2cb866-304c-11ee-a817-d283076531fd",
    "annotations": {
      "skupper.io/generated-by": "5e922f3a-9e66-4bd3-85a0-2a93c9c5134e",
      "skupper.io/site-version": "1.2.7",
      "skupper.io/url": "https://10.10.10.10:30373/2f2cb866-304c-11ee-a817-d283076531fd"
    },
    "labels": {
      "skupper.io/type": "token-claim"
    }
  },
  "data": {
    "ca.crt": "<crt>",
    "password": "<password>"
  }
}
```

Removing a secure client token

Secure client tokens can be removed based on their unique token ID.

Example: `DELETE https://10.10.10.10/api/v3/clients/networks/tokens?tokenId=<TOKEN_ID>`

Response: `Status 200`

Entities endpoint

Entities are the objects in your topology that Turbonomic manages. For example, VMs, hosts, datastores, and applications are all entities. Turbonomic discovers entities via targets, it monitors and analyzes their utilization of resources, and it recommends actions to change the placement, configuration, or state of entities.

Each market manages a set of entities. Using a market's UUID, you can get a list of entities managed by that market. For each entity you can get a full range of data, including the resources it buys and sells, the providers it buys resources from, actions for the entity, and other associated information.

Remember that more than one market can be resident in memory at a given time — Turbonomic maintains a real-time market, and there can also be plan markets in memory. As you access market entities, you should keep the following in mind:

- Entities in the real-time market reflect the current state in your physical environment
- You should never add or remove entities in the real-time market
- Entities in a plan market reflect a snapshot of the environment from when the market was created
- Plan markets can be scoped to a subset of your physical environment
- You *can* add or remove entities in a plan market

An interesting point about entities and markets — A plan market can contain entities that are also managed by the real-time market. These entities have the same stats in both markets. Turbonomic independently performs analysis on these entities in both markets.

Using the entities endpoint of the Turbonomic API, you can get a list of entities (from the market or group that contains the entities). For an individual entity you can get:

- Actions and related data for an entity
- Constraints assigned to the entity
- Through the entity constraints, get the available providers and consumers for the entity
- Groups the entity belongs to
- Notifications and notification data
- Workload placement policies that affect this entity
- Automation policies that affect this entity
- Stats for the entity
- The supply chain of entities that stitch directly with this entity
- Any tags that are defined for the entity

This endpoint contains references to start and end times for certain operations. For more information, including valid time formats, see Time in the Turbonomic API

Entities utility requests

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

Getting a list of entry points for entities

The entities endpoint contains a utility request that shows the entry points you can use to get lists of entities. These can be used to find the UUID and details relating to specific entities.

Example: `GET https://10.10.10.10/api/v3/entities`

Response: An array of Link objects, where each object contains a URL to get all entities for a specified market or group.

```
{
  "links":[
    {
      "rel":"Market actions",
      "href":"https://10.10.10.10/api/v3/markets/{uuid}/entities"
    },
    {
      "rel":"Group actions",
      "href":"https://10.10.10.10/api/v3/groups/{uuid}/entities"
    }
  ]
}
```

Getting a list of entry points for a single entity

If you pass an entity UUID to the entities endpoint, then you get a description of the entity, plus a list of entry points for entity data such as stats, constraints, actions, etc.

Example: `GET https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073`

Response: An array of Link objects, where each object contains a URL to get information about the entity.


```
{
  "links": [
    {
     "rel": "self",
     "href": "https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073?
include_aspects=false"
    },
    {
     "rel": "To Stats",
     "href": "https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/stats"
    },
    {
     "rel": "Entity notifications",
     "href": "https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/
notifications"
    },
    {
      "rel": "Entity actions",
      "href": "https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/actions?
ascending=false"
    },
    {
      "rel": "To Constraints",
      "href": "https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/
policies"
    },
    {
      "rel": "To Settings",
      "href": "https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/
settings?include_settingspolicies=false"
    },
    {
     "rel": "To Entity List",
     "href": "https://10.10.10.10/api/v3/entities"
    }
  ],
  ...
}
```


Entities requests

Getting entities

Referenced Endpoint: Groups

To get a list of entities, start with the object that contains those entities. This is usually a market or a group. You work with individual entities via their UUIDs — A list of entities gives you a list of UUIDs that you can work with.

Note:

When working with groups, you should remember that a list of entities in the group is not the same as a list of group members. Assume a group of host clusters. In that case, the list of entities would be all the hosts in those clusters, while the group members would be a list of clusters, plus all the hosts in each cluster. For more information, see Groups Endpoint.

Example: `GET https://10.10.10.10/api/v3/groups/_BZbFsWXUEemoItN1K8jw3Q/entities`

Response: A list of `EntityApiDTO`s.


```
[
  {
    "links": [
      ...
    ],
    "uuid": "azure::VM::2ffc038a-7af3-4d23-86a9-211da2e4d433",
    "displayName": "e47f9c97-b44d-4461-be02-0304b97e201e",
    "className": "VirtualMachine",
    "priceIndex": 1.0067123,
```

```
    "state": "ACTIVE",
    "severity": "Critical",
    "costPrice": 0.64534247,
    "discoveredBy": {
     "uuid": "_-S8GgFYoEemHGIeet0BOnQ",
     "displayName": "758ad253-cbf5-4b18-8863-3eed0825bf07",
     "type": "Azure"
    },
   "remoteId": "azure::VM::2ffc038a-7af3-4d23-86a9-211da2e4d433",
   "providers": [
      {
       "uuid": "azure::ST::eastus2-managed_premium",
       "displayName": "azure-eastus2-Managed Premium",
       "className": "Storage"
      },
      {
       "uuid": "azure::eastus2::PM::eastus2",
       "displayName": "azure-eastus2",
       "className": "PhysicalMachine"
      }
  ],
  "consumers": [
    {
     "uuid": "azure::APP::2ffc038a-7af3-4d23-86a9-211da2e4d433",
     "displayName": "GuestLoad [e47f9c97-b44d-4461-be02-0304b97e201e]",
     "className": "Application"
    }
  ],
  "template": {
    "uuid": "azure::VMPROFILE::Standard_DS13_v2",
    "displayName": "Standard_DS13_v2",
    "discovered": false,
    "enableMatch": false
  },
  "environmentType": "CLOUD",
  "tags": {
    "instance_group": [
     "diego_cell"
    ]
  }
 },
 {
   ...
  }
]
```


Getting actions related to an entity

Referenced Endpoint: Actions

If you provide the entity UUID but do not provide an action UUID, this returns a list of ActionApiDTOs for all the actions that are current for the entity. If you provide both the entity UUID and the action UUID, it returns the ActionApiDTO for the specified action.

This request takes the following parameters:

- entity_Uuid

  The UUID of the entity whose actions will be returned.

- action_Uuid

  The UUID of the action you want to get – Required to get a single action.

- cursor

  When making subsequent calls to retrieve additional results, this is the cursor returned by the last call to this method.

- limit

  The maximum number of items to return. Must be a positive integer. If not set, the server will provide a default (20).

- order_by

The field to order the results by. `[ creation_date, name, risk_category, savings, severity ]`.

- ascending

  Default: true. When false, results will be in descending order.

Examples:

- Get a list of actions for the passed entity: `GET https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/actions?order_by=severity&ascending=true`
- Get a single action for the entity: `GET https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/actions/9223304219064809841`

Response: A list of `ActionApiDTO`s representing the actions that are current for the given entity, or a single ActionApiDTO for the specified action. The following listing shows the list of ActionApiDTOs wrapped in an array.


```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/actions/9223304219106592017"
      },
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/actions/2560158594896"
      }
    ],
    "uuid": "9223304219106592017",
    "createTime": "2019-04-23T10:41:37-04:00",
    "actionType": "RIGHT_SIZE",
    "actionState": "PENDING_ACCEPT",
    "actionMode": "MANUAL",
    "details": "Scale down VMem for VirtualMachine vEOS-4 from 2 GB to 623 MB",
    "importance": 0,
    "target": {
      "uuid": "420b1eeb-6190-a9d2-6436-4327ad3ac073",
      "displayName": "vEOS-4",
      "className": "VirtualMachine",
      "discoveredBy": {
        "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
        "displayName": "vsphere-dc11.dev.mycorp.com",
        "type": "vCenter"
      },
      "aspects": {
        "virtualMachineAspect": {
          "os": "Other 2.6.x Linux (32-bit)",
          "connectedNetworks": [
            {
              "uuid": "d2da5316adfff0bf779652bbe7eb76c3307c2ab5",
              "displayName": "Infrastructure"
            },
            {
              "uuid": "bbef68e6ee0e76ba2c82960012b4ba563b78fdbb",
              "displayName": "vmnet4"
            },
            {
              "uuid": "d59c5e7e738ad9d2e2cb45aac5fe0073d36b49fb",
              "displayName": "vmnet6"
            },
            {
              "uuid": "96a42a4d438d1336e8d2c6962c976d889792af8a",
              "displayName": "vmnet8"
            },
            {
              "uuid": "067bc9d882e9681c55d964295e43134711520145",
              "displayName": "vmnet57"
            },
            {
              "uuid": "05b0ce5abc5cecc4104b26514b11618d77818c6d",
              "displayName": "vmnet59"
            }
          ],
          "numVCPUs": 1,
```

```
          "type": "VMEntityAspectApiDTO"
        }
      },
      "environmentType": "ONPREM"
    },
    "currentEntity": {
      "uuid": "_3vIfwmXVEemoItN1K8jw3Q",
      "className": "VMem"
    },
    "newEntity": {
      "uuid": "_5fnl82XVEemoItN1K8jw3Q",
      "className": "VMem",
      "aspects": {
        "virtualMachineAspect": {
          "os": "Linux (Free)",
          "type": "VMEntityAspectApiDTO"
        }
      }
    },
    "currentValue": "2097152.0",
    "newValue": "637952.0",
    "resizeToValue": "637952.0",
    "risk": {
      "uuid": "_5fr3YWXVEemoItN1K8jw3Q",
      "subCategory": "Efficiency Improvement",
      "description": "Underutilized VMem in Virtual Machine 'vEOS-4'",
      "severity": "MINOR",
      "importance": 0,
      "reasonCommodity": "VMem"
    },
    "stats": [
      {
        "name": "costPrice",
        "filters": [
          {
            "type": "savingsType",
            "value": "savings"
          }
        ],
        "units": "$",
        "value": 69.58008
      }
    ],
    "currentLocation": {
      "uuid": "98e12595449d993db8861c1f94c8f86ba277f17b",
      "displayName": "Cloud",
      "className": "DataCenter"
    },
    "newLocation": {
      "uuid": "98e12595449d993db8861c1f94c8f86ba277f17b",
      "displayName": "Cloud",
      "className": "DataCenter"
    },
    "actionID": 2560158594896
  },
  ...
]
```


Getting a filtered list of actions related to an entity

Referenced Endpoint: Actions

Gets a list of actions, given a market UUID and filtered by an ActionApiInputDTO using one or more of the following criteria:

- actionModeList
  - `DISABLED`:

    Do not recommend or perform the action. When you disable an action, the user interface will no longer display actions of that type.
  - `RECOMMENDED`:

    Recommend the action so a user can execute it via the given hypervisor or by other external means
  - `MANUAL`:

Recommend the action, and provide the option to execute that action through the Turbonomic user interface or an /actions API request.

  - `AUTOMATED`:

    You have directed Turbonomic to execute the action automatically.

  - `COLLECTION` **(UNUSED)**:

    Legacy action mode, no longer used.

- actionStateList

  - `DISABLED`:

    An action whose policy has been disabled.

  - `RECOMMEND`:

    An action that cannot be automated due to policy settings or must be executed outside the system, such as the purchase of hardware.

  - `PENDING_ACCEPT`:

    When the action mode is set to manual, a recommended action that has not been accepted or cleared.

  - `ACCEPTED`:

    An action that has been accepted, but is not yet in progress.

  - `IN_PROGRESS`:

    An action that is being executed.

  - `SUCCEEDED`:

    An action that was accepted and completed successfully.

  - `FAILED`:

    An action that was accepted and did not complete successfully.

  - `REJECTED`:

    When the action mode is set to manual, an action that has been rejected by the user.

  - `CLEARED`:

    When the action mode is set to manual, an action that is no longer recommended by the market.

  - `ACCOUNTING`:

    For cloud entities, an action to resize in order to use a different instance template.

  - `QUEUED`:

    When more than 10 actions are to be executed for a single target, Turbonomic will place the 11th and all subsequent actions into a QUEUED state, and they will execute as the previous actions complete.

- actionTypeList

  - `START`:

    Start an entity, and add it to the Turbonomic market.

  - `MOVE`:

    Move an entity from one provider to another. For example, moving a VM between hosts, or a datastore between disk arrays.

  - `SUSPEND`:

    Suspend the entity to exclude its capacity from your environment. Suspend does not remove the entity from your environment, but the market does not include the entity in its analysis. Based on market conditions, analysis can recommend restarting suspended entities, which includes them in market analysis once again.

  - `ADD PROVIDER`:

    This is equivalent to the START action, but for storage entities.
  - `CHANGE`:

    Move a virtual machine's storage between datastores.
  - `PROVISION`:

    Add capacity to your environment.
  - `RECONFIGURE`:

    Analysis recommends this when a VM is not able to correct a misconfiguration via another type of action. For example, analysis can recommend a RECONFIGURE action when a placement policy requires a VM to move to a host on a particular cluster, but cross-cluster moves are not permitted.
  - `DELETE`:

    This is equivalent to the SUSPEND action, but for storage entities.
  - `RESERVE_ON_PM`:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular host.
  - `RESERVE_ON_DS`:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular datastore.
  - `RESIZE_FOR_EFFICIENCY`:

    A resize that reduces the capacity of a commodity, such as vCPU, or vMEM.
  - `RESIZE_FOR_PERFORMANCE`:

    A resize that increases the capacity of a commodity, such as vCPU, or vMEM.
- cleared

  Default: `false`.

  When `true`, actions that were removed from the action list before being executed will be returned.
- costType
  - `Saving`

    An action that will decrease cloud spending.
  - `Investment`

    An action that will increase cloud spending.
- cumulative

  Default: `false`.

  When `true`, shows the savings for each action in a scope as a cumulative total, rather than showing the savings for individual actions.
- environmentType
  - `ONPREM`:

    Returns only actions related to entities that are part of your on-prem environment.
  - `CLOUD`:

    Returns only actions related to entities that are part of your cloud environment.
- groupBy

  You may group the actions returned by the following criteria: `[actionModes, actionStates, actionTypes, risk, riskSeverity, riskSubCategory]`

- hasReservedInstance

  Default: `false`.

  When `true`, only actions that are related to reserved instances will be returned.
- relatedEntityTypes

  Returns only actions related to entities of these types within the scope. For example, if your scope is a virtual datacenter and you specify VirtualMachine as a related entity type, the request will include only those actions related to VMs in the specified virtual datacenter.

  Choose from the following entity types: `[ Datacenter, PhysicalMachine, VirtualMachine, Storage, Application, Chassis, DiskArray, IOModule, StorageControl, Switch, VirtualDataCenter, VPod, DPod, Container, Database, DatabaseServerContainer, LogicalPool ]`
- riskSeverityList

  Returns only actions of the given severities: `[ Unknown, Normal, Minor, Major, Critical ]`
- riskSubCategoryList
- returns only actions of the given subcategories: `[ Performance Assurance, Efficiency Improvement, Prevention, Compliance ]`

This request takes the following parameters:

- entity_Uuid

  The UUID of the market whose actions will be returned.
- cursor

  When making subsequent calls to retrieve additional results, this is the cursor returned by the last call to this method.
- limit

  The maximum number of items to return. Must be a positive integer. If not set, the server will provide a default (20).
- order_by

  The field to order the results by: `[ creation_date, name, risk_category, savings, severity ]`.
- ascending

  Default: `true`. When `false`, results will be in descending order.

Example: `POST https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/actions?order_by=severity&ascending=true`

Example input: In this example, a list of actions that meet the following criteria will be returned and grouped by the `actionMode` of the action, in order of severity:

- Is in state `PENDING_ACCEPT`
- Is part of an on-prem environment
- Is an action for a virtual machine

```
{
  "actionInput":{
    "actionStateList":[
      "PENDING_ACCEPT"
    ],
    "environmentType":"ONPREM",
    "groupBy":[
      "actionModes"
    ]
  },
  "relatedType":"VirtualMachine",
```

```
  "scopes":[
    "Market"
  ]
}
```

Response: A list of `ActionApiDTO`s that meet the specified criteria.

Getting statistics for a filtered list of actions of an entity

Referenced Endpoint: Actions

Referenced Endpoint: Stats

Gets a list of statistics for a filtered list of actions that are current for the passed entity. Statistics returned include the number of actions, total savings/investment, and the number of entities affected by the actions. In addition to the Entity UUID, provide an ActionApiInputDTO using one or more of the following criteria:

- actionModeList
  - `DISABLED`:

    Do not recommend or perform the action. When you disable an action, the user interface will no longer display actions of that type.
  - `RECOMMENDED`:

    Recommend the action so a user can execute it via the given hypervisor or by other external means
  - `MANUAL`:

    Recommend the action, and provide the option to execute that action through the Turbonomic user interface or an /actions API request.
  - `AUTOMATED`:

    You have directed Turbonomic to execute the action automatically.
  - `COLLECTION` **(UNUSED)**:

    Legacy action mode, no longer used.
- actionStateList
  - `DISABLED`:

    An action whose policy has been disabled.
  - `RECOMMEND`:

    An action that cannot be automated due to policy settings or must be executed outside the system, such as the purchase of hardware.
  - `PENDING_ACCEPT`:

    When the action mode is set to manual, a recommended action that has not been accepted or cleared.
  - `ACCEPTED`:

    An action that has been accepted, but is not yet in progress.
  - `IN_PROGRESS`:

    An action that is being executed.
  - `SUCCEEDED`:

    An action that was accepted and completed successfully.
  - `FAILED`:

    An action that was accepted and did not complete successfully.
  - `REJECTED`:

When the action mode is set to manual, an action that has been rejected by the user.

  - `CLEARED`:

    When the action mode is set to manual, an action that is no longer recommended by the market.

  - `ACCOUNTING`:

    For cloud entities, an action to resize in order to use a different instance template.

  - `QUEUED`:

    When more than 10 actions are to be executed for a single target, Turbonomic will place the 11th and all subsequent actions into a QUEUED state, and they will execute as the previous actions complete.

- actionTypeList

  - `START`:

    Start an entity, and add it to the Turbonomic market.

  - `MOVE`:

    Move an entity from one provider to another. For example, moving a VM between hosts, or a datastore between disk arrays.

  - `SUSPEND`:

    Suspend the entity to exclude its capacity from your environment. Suspend does not remove the entity from your environment, but the market does not include the entity in its analysis. Based on market conditions, analysis can recommend restarting suspended entities, which includes them in market analysis once again.

  - `ADD PROVIDER`:

    This is equivalent to the START action, but for storage entities.

  - `CHANGE`:

    Move a virtual machine's storage between datastores.

  - `PROVISION`:

    Add capacity to your environment.

  - `RECONFIGURE`:

    Analysis recommends this when a VM is not able to correct a misconfiguration via another type of action. For example, analysis can recommend a RECONFIGURE action when a placement policy requires a VM to move to a host on a particular cluster, but cross-cluster moves are not permitted.

  - `DELETE`:

    This is equivalent to the SUSPEND action, but for storage entities.

  - `RESERVE_ON_PM`:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular host.

  - `RESERVE_ON_DS`:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular datastore.

  - `RESIZE_FOR_EFFICIENCY`:

    A resize that reduces the capacity of a commodity, such as vCPU, or vMEM.

  - `RESIZE_FOR_PERFORMANCE`:

    A resize that increases the capacity of a commodity, such as vCPU, or vMEM.

- cleared

  Default: `false`.

When `true`, actions that were removed from the action list before being executed will be returned.

- costType
  - `Saving`

    An action that will decrease cloud spending.
  - `Investment`

    An action that will increase cloud spending.
- cumulative

  Default: `false`.

  When `true`, shows the savings for each action in a scope as a cumulative total, rather than showing the savings for individual actions.
- environmentType
  - `ONPREM`:

    Returns only actions related to entities that are part of your on-prem environment.
  - `CLOUD`:

    Returns only actions related to entities that are part of your cloud environment.
- groupBy

  You may group the actions returned by the following criteria: `[actionModes, actionStates, actionTypes, risk, riskSeverity, riskSubCategory]`
- hasReservedInstance

  Default: `false`.

  When `true`, only actions that are related to reserved instances will be returned.
- relatedEntityTypes

  Returns only actions related to entities of these types within the scope. For example, if your scope is a virtual datacenter and you specify VirtualMachine as a related entity type, the request will include only those actions related to VMs in the specified virtual datacenter.

  Choose from the following entity types: `[ Datacenter, PhysicalMachine, VirtualMachine, Storage, Application, Chassis, DiskArray, IOModule, StorageControl, Switch, VirtualDataCenter, VPod, DPod, Container, Database, DatabaseServerContainer, LogicalPool ]`
- riskSeverityList

  Returns only actions of the given severities: `[ Unknown, Normal, Minor, Major, Critical ]`
- riskSubCategoryList
- returns only actions of the given subcategories: `[ Performance Assurance, Efficiency Improvement, Prevention, Compliance ]`

Example: `POST https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/actions/stats`

Example input: Get a list of stats for actions that meet the following criteria, grouped by the `actionMode` of the action, in order of severity:

- Is in state `PENDING_ACCEPT`
- Is part of an on-prem environment
- Is an action for a virtual machine
- Is an action generated by the Turbonomic real-time market

```
{
  "actionInput":{
```

```
    "actionStateList":[
      "PENDING_ACCEPT"
    ],
    "environmentType":"ONPREM",
    "groupBy":[
      "actionModes"
    ]
  },
  "relatedType":"VirtualMachine"
}
```

Response: A list of `StatSnapshotApiDTO`s that aggregate statistics about the filtered actions.

```
[
  {
    "date": "2019-04-24T03:56:34-04:00",
    "statistics": [
      {
        "name": "numActions",
        "values": {
          "max": 1,
          "min": 1,
          "avg": 1,
          "total": 1
        },
        "value": 1
      },
      {
        "name": "costPrice",
        "filters": [
          {
            "type": "property",
            "value": "savings"
          }
        ],
        "units": "$",
        "values": {
          "max": 69.77539,
          "min": 69.77539,
          "avg": 69.77539,
          "total": 69.77539
        },
        "value": 69.77539
      },
      {
        "name": "numEntities",
        "values": {
          "max": 1,
          "min": 1,
          "avg": 1,
          "total": 1
        },
        "value": 1
      }
    ]
  }
]
```

Getting the Constraints for placement of an entity

When it calculates where to place an entity, Turbonomic respects whatever placement constraints there might be in your environment. For example, unless you have a Merge Cluster policy in effect, VM placement is constrained to hosts in the same cluster.

This call gets a ConstraintApiDTO object, which describes these constraints for the specified entity. The data in this object is equivalent to the Placement Constraints view that you can see in the user interface. To see this view, you drill down to the policy view for a single entity. In this view you can see the entity's related providers and consumers. You can also open a view for each related entity to show the constraints that limit potential placement for the entity. Also in this view, you can test out different placement options. (For information about testing different placement options, see Inspecting placement options within constraints for an element.)

In the same way, the ConstraintApiDTO contains data objects to describe the current placement relationships, as well as potential changes you could make. The DTO delivers an array of objects for each consumer or provider. Within each object:

- `relation` indicates whether the entity is a provider (`bought`) or consumer (`sold`)
- `relatedEntities` describes the current consumer or provider
- `placementOptions` show other placement you could exploit for this relationship, if you were to remove the current constraints (via merge policies or other means)

Example: `GET https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/constraints`

Response: This listing shows a subset of the related entities and placement options:

```
[
  {
    "relation": "bought",
    "entityType": "PhysicalMachine",
    "relatedEntities": [
      {
        "links": [
          {
            "rel": "self",
            "href": "https://10.10.10.10/api/v3/entities/30333436-3638-5355-4532-313159335631?
include_aspects=false"
          }
        ],
        "uuid": "30333436-3638-5355-4532-313159335631",
        "displayName": "hp-dl571.dev.mycorp.com",
        "className": "PhysicalMachine",
        "priceIndex": 6.577457,
        "state": "ACTIVE",
        "severity": "Normal",
        "discoveredBy": {
          "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
          "displayName": "vsphere-dc11.dev.mycorp.com",
          "type": "vCenter"
        },
        "remoteId": "host-30",
        "environmentType": "ONPREM"
      }
    ],
    "numPotentialEntities": 1,
    "placementOptions": [
      {
        "constraintType": "ClusterCommodity",
        "scope": {
          "uuid": "17fe23bc346750d62dce8ba4274fa99b4739975f",
          "displayName": "Cloud\\Physical"
        },
        "target": {
          "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
          "displayName": "vCenter:vsphere-dc11.dev.mycorp.com",
          "type": "vCenter"
        },
        "numPotentialEntities": 3,
        "key": "Cluster::17fe23bc346750d62dce8ba4274fa99b4739975f"
      },
      {
        "constraintType": "NetworkCommodity",
        "scope": {
          "displayName": "NetworkCommodity/vEOS-4"
        },
        "numPotentialEntities": 2,
        "key": "DistributedVirtualPortgroup::vmnet4"
      },
      {
        "constraintType": "DatastoreCommodity",
        "scope": {
          "uuid": "5a822e24-584bce98-ecb2-441ea15e2534",
          "displayName": "SCIO:ESXDC11DS2"
        },
        "target": {
          "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
          "displayName": "vCenter:vsphere-dc11.dev.mycorp.com",
          "type": "vCenter"
        },
```

```
        "numPotentialEntities": 2,
        "key": "Storage::5a822e24-584bce98-ecb2-441ea15e2534"
      },
      {
        "constraintType": "DataCenterCommodity",
        "scope": {
          "uuid": "98e12595449d993db8861c1f94c8f86ba277f17b",
          "displayName": "Cloud"
        },
        "target": {
          "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
          "displayName": "vCenter:vsphere-dc11.dev.mycorp.com",
          "type": "vCenter"
        },
        "numPotentialEntities": 7,
        "key": "DataCenter::98e12595449d993db8861c1f94c8f86ba277f17b"
      }
    ]
  },
  {
    "relation": "sold",
    "entityType": "Application",
      ...
  }
]
```

Inspecting Placement Options Within Constraints for an Element

Within the given constraints that limit placement of an entity, there are usually a number of options for placement. The current placement is just one of many.

For information about getting entity constraints, see Getting the Constraints for Placement of an Entity.

When you get the constraints for an entity, the ConstraintApiDTO includes other placement options for that entity on its providers, and options to place other consumers on the entity. The call in this section passes these placement options and returns the respective providers or consumers that satisfy the resulting constraint changes.

Making this call is equivalent to clicking **Find More Placement Options** in the user interface. To see this in the user interface, set the scope to a single entity, display the POLICIES tab, and scroll down to the Placement Constraints section. Then click **Constraints** next to the provider you want to explore, and click **Find More Placement Options**.

Example: `POST https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/constraints/entities`

Example input: In this example, two placement options that you copied from a ConstraintApiDTO for the entity you are inspecting was passed. Note that each placement option in the ConstraintApiDTO includes a `constraintType` and a `key`. To pass a placement option, these are the only fields you need. For this example pass a `ClusterCommodity` constraint and a `DatastoreCommodity` constraint.

```
{
  "relation": "bought",
  "placementOptions": [
    {
      "constraintType": "ClusterCommodity",
      "key": "Cluster::17fe23bc346750d62dce8ba4274fa99b4739975f"
    },
    {
      "constraintType": "DatastoreCommodity",
      "key": "Storage::5a822e24-584bce98-ecb2-441ea15e2534"
    }
  ]
}
```

Response: In the data posted in the previous example, the cluster commodity has three potential entities, and the datastore commodity has two potential entities. The response brings back the intersection of these three potential provider entities – Two hosts.


```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/entities/30343835-3638-584d-5131-323530394e39?
include_aspects=false"
      }
    ],
    "uuid": "30343835-3638-584d-5131-323530394e39",
    "displayName": "hp-dl569.dev.mycorp.com",
    "className": "PhysicalMachine",
    "priceIndex": 1.031998,
    "state": "ACTIVE",
    "severity": "Normal",
    "discoveredBy": {
      "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
      "displayName": "vsphere-dc11.dev.mycorp.com",
      "type": "vCenter"
    },
    "remoteId": "host-3323",
    "environmentType": "ONPREM"
  },
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/entities/30333436-3638-5355-4532-313159335631?
include_aspects=false"
      }
    ],
    "uuid": "30333436-3638-5355-4532-313159335631",
    "displayName": "hp-dl571.dev.mycorp.com",
    "className": "PhysicalMachine",
    "priceIndex": 6.916899,
    "state": "ACTIVE",
    "severity": "Normal",
    "discoveredBy": {
      "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
      "displayName": "vsphere-dc11.dev.mycorp.com",
      "type": "vCenter"
    },
    "remoteId": "host-30",
    "environmentType": "ONPREM"
  }
]
```


Getting group membership of an entity

Referenced Endpoint: Groups

The Groups endpoint tells you which entities are in a given group. The `/entities/{entity_Uuid}/groups` call tells you which groups a given entity belongs to. You can make this call in two modes:

- Include relationship paths

  In this mode the results include only those groups that are in a buy or sell relationship with the entity. For example, a VM entity might have relationships with a datacenter and a host cluster. The call would return entries for the datacenter, the host cluster, and the VM itself.

- Do not include relationship paths

  In this mode the call returns every group that includes the passed entity as a member entity. This can be a large number of groups, including all the discovered groups as well as any custom groups you created. The results do not show buy/sell relationships.

This request takes the following parameters:

- entity_Uuid

The UUID of the entity whose actions will be returned.

- path

  Default: false. When true, results include only related groups, and they show the providers and consumers for each group when applicable.

Examples:

- All groups (`path = false`): `GET https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/groups?path=false`
- Buy/Sell related groups, only (`path = true`): `GET https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/groups?path=true`

Response: A list of related groups. For the default (`path = false`), this returns an array of group objects.

```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/groups/_lSlr0GddEemDFcIzuV5wcQ?
include_aspects=false"
      }
    ],
    "uuid": "_lSlr0GddEemDFcIzuV5wcQ",
    "displayName": "CUD_Vm_Group_vEOS",
    "className": "Group",
    "environmentType": "ONPREM",
    "entitiesCount": 5,
    "membersCount": 5,
    "groupType": "VirtualMachine",
    "severity": "Normal",
    "isStatic": true,
    "logicalOperator": "OR",
    "activeEntitiesCount": 3
  },
  ...
 ]
```

Getting settings policies affecting an entity

Referenced Endpoint: Settings Policies

This call returns an array of SettingsPolicyApiDTOs for all the Automation Policies that are effective for the given entity. Note that one entity can be in multiple groups, and so it can have policies that set different values for the same setting. In that case, the most conservative setting takes effect. The data returned by this call does not reflect which setting takes precedence in this case. The call merely reflects what you would see in the user interface.

Example: `GET https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/settingspolicies`

Response: An array of `SettingsPolicyApiDTO`s that are effective for the entity.

```
[
  {
    "uuid": "_i-QuUGXUEemoItN1K8jw3Q",
    "displayName": "CUD_VM_AUTOMATION_POLICY",
    "entityType": "VirtualMachine",
    "scopes": [
      {
        "uuid": "_BZbFsWXUEemoItN1K8jw3Q",
        "displayName": "CUD_VM_Group",
        "groupType": "VirtualMachine",
        "isStatic": true,
        "logicalOperator": "AND"
      }
    ],
```

```
    "settingsManagers": [
      {
        "uuid": "automationmanager",
        "displayName": "Action Mode Settings",
        "category": "Automation",
        "settings": [
          {
            "uuid": "startVM",
            "displayName": "Start",
            "value": "MANUAL",
            "defaultValue": "RECOMMEND",
            "valueType": "STRING",
            "options": [
              {
                "label": "Disabled",
                "value": "DISABLED"
              },
              {
                "label": "Recommend",
                "value": "RECOMMEND"
              },
              {
                "label": "Manual",
                "value": "MANUAL"
              },
              {
                "label": "Automated",
                "value": "AUTOMATIC"
              }
            ],
            "entityType": "VirtualMachine",
            "sourceGroupName": "Global"
          }
        ]
      }
    ],
    "disabled": false,
    "readOnly": false,
    "default": false
  }
]
```


Getting statistics of an entity

Referenced Endpoint: Stats

Gets all statistics for the specified entity. This request takes the following parameters:

- entity_uuid

  The entity whose statistics will be returned.

- q

  Used internally by Turbonomic.

Example: `GET https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/stats`

Response: A list of `StatSnapshotApiDTO`s representing the current values of the statistics.


```
[
  {
    "displayName": "AdityaReplica",
    "date": "2019-04-24T20:09:46-04:00",
    "statistics": [
      {
        "name": "priceIndex",
        "relatedEntityType": "VirtualMachine",
        "values": {
          "max": 1.01,
          "min": 1.01,
          "avg": 1.01,
          "total": 1.01
        },
```

```
        "value": 1.01
      },
      {
        "displayName": "CPUProvisioned/AdityaReplica",
        "name": "CPUProvisioned",
        "capacity": {
          "max": 3120000,
          "min": 3120000,
          "avg": 3120000,
          "total": 3120000
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        "units": "MHz",
        "values": {
          "max": 5200,
          "min": 5200,
          "avg": 5200,
          "total": 5200
        },
        "value": 5200
      },
      {
        "displayName": "Swapping/AdityaReplica",
        "name": "Swapping",
        "capacity": {
          "max": 40000000,
          "min": 40000000,
          "avg": 40000000,
          "total": 40000000
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        "units": "bit/sec",
        "values": {
          "max": 0,
          "min": 0,
          "avg": 0,
          "total": 0
        },
        "value": 0
      },
      {
        "displayName": "NumDisk/AdityaReplica",
        "name": "NumDisk",
        "capacity": {
          "max": 2147483650,
          "min": 2147483650,
          "avg": 2147483650,
          "total": 2147483650
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        "values": {
          "max": 1,
          "min": 1,
          "avg": 1,
          "total": 1
        },
        "value": 1
      },
    ...
    ]
  }
]
```

Getting a filtered list of statistics for an entity

Referenced Endpoint: Stats

Gets a list of statistics filtered by a user-created StatPeriodApiInputDTO. For more information about the criteria that make up the input for thi request, see the Swagger documentation for this request, located at `https://<Your_Turbonomic_URL/apidpoc`.

Example: `POST https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/stats`

Example Input: This input will return the values for the VMem, VCPU, and VStorage statistics of the specified entity, ordered first by key, then relatedEntity, then virtualDisk.

```
{
  "statistics": [
    {
      "name": "VMem",
      "groupBy": [
        "key",
        "relatedEntity",
        "virtualDisk"
      ]
    },
    {
      "name": "VCPU",
      "groupBy": [
        "key",
        "relatedEntity",
        "virtualDisk"
      ]
    },
    {
      "name": "VStorage",
      "groupBy": [
        "key",
        "relatedEntity",
        "virtualDisk"
      ]
    }
  ]
}
```

Response: An array of `StatSnapshotApiDTO`s representing the filtered statistics for the entity:

```
[
  {
    "displayName": "vEOS-4",
    "date": "2019-04-25T17:57:02-04:00",
    "statistics": [
      {
        "displayName": "vEOS-4",
        "name": "VCPU",
        "capacity": {
          "max": 2663,
          "min": 2663,
          "avg": 2663,
          "total": 2663
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "virtualDisk",
            "value": ""
          },
          {
            "type": "key",
            "value": null
          },
          {
            "type": "relation",
            "value": "sold"
          }
```

```
        ],
        "relatedEntity": {
          "uuid": "c0849b50a2cce61f57ea0e49e2199503e82f3f54",
          "displayName": "GuestLoad[vEOS-4]",
          "className": "Application"
        },
        "units": "MHz",
        "values": {
          "max": 143,
          "min": 91.47,
          "avg": 91.47,
          "total": 91.47
        },
        "value": 91.47
      },
      {
        "displayName": "vEOS-4",
        "name": "VMem",
        "capacity": {
          "max": 2097152,
          "min": 2097152,
          "avg": 2097152,
          "total": 2097152
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "virtualDisk",
            "value": ""
          },
          {
            "type": "key",
            "value": null
          },
          {
            "type": "relation",
            "value": "sold"
          }
        ],
        "relatedEntity": {
          "uuid": "c0849b50a2cce61f57ea0e49e2199503e82f3f54",
          "displayName": "GuestLoad[vEOS-4]",
          "className": "Application"
        },
        "units": "KB",
        "values": {
          "max": 188740,
          "min": 130022,
          "avg": 130022,
          "total": 130022
        },
        "value": 130022
      }
    ]
  }
]
```


Getting the supply chain of an entity

Referenced Endpoint: Supply Chains

Turbonomic models your environment as a market of buyers and sellers. It maps the entities it discovers to a supply chain that represents the buy/sell relationships between these entities.

In the user interface, the default view on the Home Page shows the global supply chain. You can change scope to drill down to a supply chain for the relationships to a single entity. This is the same as the result when you get a supply chain for an entity via the API. The returned data is a SupplyChainApiDTO that expresses the scope for the passed entity.

The supply chain shows tiers of entity types, where depth indicates where in the supply chain levels a specific tier is. When you get a supply chain for an entity, the entity is at `depth = 0`. The depths for tiers above the entity are negative. For example, if a VM is at `depth = 0`, then the application that consumes from the VM would be at `depth = -1`. Likewise, providers to the VM would have positive depths. For

example, storage that provides resources to the VM would have `depth = 1`, and a disk array that hosts the storage would have `depth = 2`.

The supply chain request takes the following parameters:

- entity_uuid

  The entity that sets the scope of the supply chain.

- types

  An array of strings for the entity types you want to include in the supply chain. If you leave this blank, the call returns the complete supply chain for this entity. If you specify any types, the returned supply chain contains only those types. Note that this rule also applies to the entity that sets the scope (`entity_uuid`) – If you do not include that entity's type, then the supply chain will not include that entity, even though it sets the scope of the request.

- entity_states

  An array of strings. Limit the supply chain to include only entities in the given state. Entity states are:

  – ACTIVE: The entity is actively consuming resources
  – EVACUATED: All consumers have been moved off of the entity, and it can be suspended or terminated
  – FAILOVER: The entity is reserved for failover and will not participate in the market
  – IDLE: The entity is powered off and not actively consuming market resources
  – LAUNCH: The entity is starting up
  – MAINTENANCE: The entity is in Maintenance mode
  – NOT_MONITORED: Turbonomic does not currently monitor the entity
  – QUEUED: More than ten actions are to be executed for a single target, and this entity has pending actions in the queue
  – RESOURCE_ALLOCATION: For internal use
  – RESOURCE_RELEASE: For internal use
  – SUSPEND: The entity has been placed in Suspend mode
  – SUSPEND_PENDING: The entity is making preparations to suspend, or is the next entity of its type to suspend
  – TERMINATE_PENDING: The entity is making preparations to terminate, or is the next entity of its type to terminate
  – UNKNOWN: Turbonomic is unable to obtain the current state of the entity

- environment_type

  Limit the supply chain to a specific environment type. If you do not specify environment type, the supply chain includes all environments, which is the same as HYBRID. The environment types are:

  – HYBRID
  – CLOUD
  – ONPREM

- detail_type

  The level of detail you want in your supply chain. Can be one of:

  – compact: Minimal detail about the supply chain items
  – entity: The supply chain items include full entity descriptions
  – aspects: The supply chain items also include aspects (mata data stored with the entity)

- health

  Whether to include a health summary with the entities in the supply chain. `false` by default.

Example: `GET https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073/supplychains?types=VirtualMachine&types=Storage&types=DiskArray&detail_type=compact&health=true`

This example gets a supply chain for a VM entity, with the following settings: including the VM, storage, and disk arrays:

- Only include the VM, its storage, and the disk arrays
- Give compact details
- Show the entity health

Response: The `SupplyChainApiDTO` for the filtered request:


```
{
  "seMap": {
    "VirtualMachine": {
      "depth": 0,
      "entitiesCount": 1,
      "healthSummary": {
        "Normal": 1
      },
      "stateSummary": {
        "ACTIVE": 1
      },
      "connectedProviderTypes": [
        "PhysicalMachine",
        "Storage",
        "VirtualDataCenter"
      ],
      "connectedConsumerTypes": [
        "Application"
      ],
      "instances": {
        "420b1eeb-6190-a9d2-6436-4327ad3ac073": {
          "links": [
            {
              "rel": "self",
              "href": "https://10.10.10.10/api/v3/entities/420b1eeb-6190-a9d2-6436-4327ad3ac073?
include_aspects=false"
            }
          ],
          "uuid": "420b1eeb-6190-a9d2-6436-4327ad3ac073",
          "displayName": "vEOS-4",
          "className": "VirtualMachine",
          "priceIndex": 1.1463195,
          "state": "ACTIVE",
          "severity": "Normal",
          "discoveredBy": {
            "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
            "displayName": "vsphere-dc11.dev.mycorp.com",
            "type": "vCenter"
          },
          "remoteId": "vm-100",
          "environmentType": "ONPREM"
        }
      }
    },
    "Storage": {
      "depth": 1,
      "entitiesCount": 2,
      "healthSummary": {
        "Normal": 2
      },
      "stateSummary": {
        "ACTIVE": 2
      },
      "connectedProviderTypes": [
        "DiskArray"
      ],
      "connectedConsumerTypes": [
        "PhysicalMachine",
        "VirtualMachine"
      ],
      "instances": {
        "5a822e24-584bce98-ecb2-441ea15e2534": {
          "links": [
```

```
            {
              "rel": "self",
              "href": "https://10.10.10.10/api/v3/entities/5a822e24-584bce98-ecb2-441ea15e2534?
include_aspects=false"
            }
          ],
          "uuid": "5a822e24-584bce98-ecb2-441ea15e2534",
          "displayName": "SCIO:ESXDC11DS2",
          "className": "Storage",
          "priceIndex": 1.2797012,
          "state": "ACTIVE",
          "severity": "Normal",
          "discoveredBy": {
            "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
            "displayName": "vsphere-dc11.dev.mycorp.com",
            "type": "vCenter"
          },
          "remoteId": "datastore-2357",
          "environmentType": "ONPREM",
          "tags": {
            "Datastore_DC11": [
              "True"
            ]
          }
        },
        "9bd4ee88-99c64661": {
          "links": [
            {
              "rel": "self",
              "href": "https://10.10.10.10/api/v3/entities/9bd4ee88-99c64661?
include_aspects=false"
            }
          ],
          "uuid": "9bd4ee88-99c64661",
          "displayName": "QS1:NFSShare",
          "className": "Storage",
          "priceIndex": 20000,
          "state": "ACTIVE",
          "severity": "Normal",
          "discoveredBy": {
            "uuid": "_gOQEYEaREemB8r4uTzazvA",
            "displayName": "vsphere-dc5.dev.mycorp.com",
            "type": "vCenter"
          },
          "remoteId": "datastore-1411",
          "environmentType": "ONPREM",
          "tags": {
            "DimStor": [
              "DimStorTag"
            ],
            "Datastore_DC11": [
              "True"
            ],
            "Test_Tag": [
              "MyStorageTag"
            ]
          }
        }
      }
    },
    "DiskArray": {
      "depth": 2,
      "entitiesCount": 2,
      "healthSummary": {
        "Normal": 2
      },
      "stateSummary": {
        "ACTIVE": 2
      },
      "connectedConsumerTypes": [
        "Storage"
      ],
      "instances": {
        "DiskArray-9bd4ee88-99c64661": {
          "links": [
            {
              "rel": "self",
              "href": "https://10.10.10.10/api/v3/entities/DiskArray-9bd4ee88-99c64661?
include_aspects=false"
            }
          ],
          "uuid": "DiskArray-9bd4ee88-99c64661",
```

```
          "displayName": "DiskArray-QS1:NFSShare",
          "className": "DiskArray",
          "priceIndex": 0,
          "state": "ACTIVE",
          "severity": "Normal",
          "discoveredBy": {
            "uuid": "_oQ2aIEG1EemgjaEqcE-BJw",
            "displayName": "vsphere-dc13.dev.mycorp.com",
            "type": "vCenter"
          },
          "remoteId": "datastore-982",
          "environmentType": "ONPREM"
        },
        "DiskArray-5a822e24-584bce98-ecb2-441ea15e2534": {
          "links": [
            {
              "rel": "self",
              "href": "https://10.10.10.10/api/v3/entities/DiskArray-5a822e24-584bce98-
ecb2-441ea15e2534?include_aspects=false"
            }
          ],
          "uuid": "DiskArray-5a822e24-584bce98-ecb2-441ea15e2534",
          "displayName": "DiskArray-SCIO:ESXDC11DS2",
          "className": "DiskArray",
          "priceIndex": 0,
          "state": "ACTIVE",
          "severity": "Normal",
          "discoveredBy": {
            "uuid": "_FJwg0EzHEemaXZFqO1sC0g",
            "displayName": "vsphere-dc11.dev.mycorp.com",
            "type": "vCenter"
          },
          "remoteId": "datastore-2357",
          "environmentType": "ONPREM"
        }
      }
    }
  }
}
```


Getting the tags on an entity

Referenced Endpoint: Tags

Turbonomic discovers tags and tag values that are assigned to different entities in your environment. This call returns the tag and value sets for the entity you pass.

Example: `GET https://10.10.10.10/api/v3/entities/9bd4ee88-99c64661/tags`

Response: A `TagApiDTO` for the passed entity. This gives an array of key/value pairs, where the value is an array of strings.


```
[
  {
    "key": "Datastore_DC11",
    "values": [
      "True"
    ]
  },
  {
    "key": "Test_Tag",
    "values": [
      "MyStorageTag"
    ]
  },
  {
    "key": "DimStor",
    "values": [
      "DimStorTag"
    ]
  }
]
```

Creating a list of user-defined tags of an entity

Referenced Endpoint: Tags

Turbonomic allows a user to assign tags to a specific entity, in addition to whatever tags are discovered for that entity.

This request takes the following parameters:

- entity_Uuid

  The UUID of the entity.

- body

```
[
  {
    "key": "ExampleTagKey",
    "values": [
      "ExampleTagValue"
    ]
  }
]
```

Example: `POST https://10.10.10.10/api/v3/entities/75694629540194/tags`

Response: Returns the list of tags for an entity that are created.

```
[
  {
    "key": "ExampleTagKey",
    "values": [
      "ExampleTagValue"
    ]
  }
]
```

Deleting a list of user-defined tags of an entity

Referenced Endpoint: Tags

Turbonomic allows a user to delete tags of a specific entity. If a tag key list is given, it deletes all the tags defined on that list. If no list is given, it deletes all the tags for that entity.

This request takes the following parameters:

- (Optional) key **-** array[string]

  List of keys to delete.

Example: `DELETE https://10.10.10.10/api/v3/entities/75694629540194/tags?key=ExampleTagKey`

Response: Empty response upon successful execution.

Groups endpoint

A group is a collection of entities (such as physical machines, containers, clusters, or even other groups) that Turbonomic can work with as a unit. The most common use of groups is to set scope for display in charts or for processing in plans, and to retrieve statistics for a subset of your environment.

This endpoint contains references to start and end times for certain operations. For more information, including valid time formats, see Time in the Turbonomic API

Using the groups endpoint of the Turbonomic API, you can:

- Get a list of all groups on your Turbonomic appliance
- Create a group

- Edit a group
- Delete a group
- Get a list of actions for a specified groups
- Get a list of entities in a specified group
- Get parent groups of a specified group
- Get a list of members of a specified group
- Get a list of notifications for a specified group
- Get a list of settings for a specified group
- Get a list of policies for a specified group
- Get statistics for a specified group
- Get the supply chain related to a specified group
- Get the tags belonging to a specified group

Group types and class names

Turbonomic subdivides the entities on your appliance by the following group class name, to aid in both searching and understanding the purpose of a group:

- Folder

  These appear in the user interface as folders, and are for visual organization. Discovered folders represent the folder structure in the target — for example, the vCenter folder structure. In addition, Turbonomic discovers vCenter Server resource pools, and groups them into folders.

- Group

  Turbonomic places discovered entities into standard groups, and users can create groups of their own with static or dynamic membership.

- RefGroup

  A RefGroup can be a group of groups- for example, a group of PM cluster groups that make up a hypervisor target- or a group defined by a relationship. For example, the built-in group `PMsByDatacenter` is a RefGroup reflecting the relationship between the datacenters and hosts.

- Cluster / StorageCluster

  Groups that correspond to discovered clusters.

- DiscoveredGroup

  Groups that are defined by a target service. For example, Turbonomic can discover DRS domains that were defined in vCenter Server.

- MarketGroup

  A group that is based on the infrastructure cost of that entity type. These groups are based on the Infrastructure Cost settings in the Turbonomic policies.

- StaticMetaGroup / RefMetaGroup / MetaGroup

  Groups used internally by Turbonomic. These should not be modified in any way.

In addition, each group has a `groupType` that identifies the type of members that group contains. Remember that when you create a group in the user interface or via the API, you cannot add more than one type of member. The different group types include:

- Cluster

  A group of clusters. For example, you could create a group of all PM clusters that include the name "Development".

- ServiceEntity

An internal group type reserved for groups that Turbonomic discovers or creates. Note that the group will still contain members of only one type.

- Entity Type
  - APPLICATION
  - APPLICATION_SERVER
  - BUSINESS_ACCOUNT
  - BUSINESS_UNIT
  - CHASSIS
  - CLOUD_SERVICE
  - CONTAINER
  - CONTAINER_POD
  - DATABASE
  - DATABASE_SERVER
  - DATACENTER
  - DISK_ARRAY
  - IO_MODULE
  - LOAD_BALANCER
  - LOGICAL_POOL
  - NETWORK
  - PHYSICAL_MACHINE
  - STORAGE
  - STORAGE_CONTROLLER
  - SWITCH
  - VIRTUAL_APPLICATION
  - VIRTUAL_DATACENTER
  - VIRTUAL_MACHINE

Understanding the different group types helps you find the group you want. If you know the name of a specific group, then you can filter the returned list for a group of that name. Or if you want to step through all the clusters Turbonomic has discovered, you can filter all the entries with a `className` of `Cluster` or `StorageCluster`.

Group filter types

Turbonomic utilizes regex and a `filterType` to build a group. The `filterType` is the criterion to which the regex will match. For example, a `filterType` of `pmsByName` will apply the regex to the display names of physical machines.

The available options for this parameter depend on the entity making up your group:

- ApplicationComponentSpec
  - appComponentSpecsByName
  - appComponentSpecsByBusinessAccountUuid
  - appComponentSpecsByHybridConnectionCount
  - appComponentSpecsByDeploymentSlotCount
- ApplicationComponent
  - appCompsByName
  - applicationComponentByBusinessApplication

  - applicationComponentByBusinessTransaction
  - applicationComponentByService
- AvailabilityZone
  - zonsByName
  - zoneByCloudProvider
- BillingFamily
  - billingFamilyByName
- BusinessAccountFolder
  - businessAccountFolderByName
- BusinessAccount
  - businessAccountByName
  - businessAccountByUuid
  - businessAccountByVendorId
  - businessAccountCloudProvider
  - businessAccountProbeType
  - businessAccountTargetName
  - businessAccountValidationStatus
  - businessAccountByResourceGroup
  - businessAccountByTag
  - subBusinessAccountOfUuid
- BusinessApplication
  - busAppsByName
  - businessApplicationByUserDefinedEntity
- BusinessTransaction
  - busTransByName
  - businessTransactionByBusinessApplication
  - businessTransactionByUserDefinedEntity
- BusinessUser
  - businessUserByName
  - businessUserByDesktopPool
  - businessUserByViewPod
- Chassis
  - chasByName
- CloudService
- Cluster
  - clustersByName
  - clustersByTag
- ComputeTier
  - computeTiersByName
  - computeTiersByMem
  - computeTiersByNumCPUs
  - computeTiersByCloudProvider

- ContainerPlatformCluster
  - containerPlatformClustersByName
- Container
  - containersByName
  - containersByPodName
  - containersByNamespace
  - containersByApplicationComponent
  - containersByBusinessApplication
  - containersByBusinessTransaction
  - containersByService
  - containersByControllerKind
  - containersByTag
  - containersByContainerPlatformCluster
- ContainerPod
  - containerPodsByName
  - containerPodsByVMName
  - containerPodsByNamespace
  - containerPodsByTag
  - containerPodsByControllerKind
  - containerPodsByWorkloadControllerName
  - containerPodsByContainerPlatformCluster
  - containerPodsByState
- ContainerSpec
  - containerSpecsByName
  - containerSpecsByNamespace
  - containerSpecsByControllerKind
  - containerSpecsByContainerPlatformCluster
  - containerSpecsByTag
- DataCenter
  - datacentersByName
  - datacentersByTag
- Database
  - databaseByName
  - databaseByTag
  - databaseByBusinessAccountUuid
  - databaseByResourceGroupName
  - databaseByCloudProvider
  - databaseByApplicationComponent
  - databaseByBusinessApplication
  - databaseByBusinessTransaction
  - databaseByService
  - dbByAvailabilityZoneName
  - dbByRegionName

  - dbByDatabaseServerName
  - databaseByReplicationRole
  - databaseByPricingModel
  - databaseByServiceTier
  - databaseByHoursIdle
  - databaseByAge
- DatabaseServer
  - databaseServerByName
  - databaseServerByTag
  - databaseServerByBusinessAccountUuid
  - databaseServerByResourceGroupName
  - databaseServerByCloudProvider
  - databaseServerByApplicationComponent
  - databaseServerByBusinessApplication
  - databaseServerByBusinessTransaction
  - databaseServerByService
  - databaseServerByEngine
  - databaseServerByEdition
  - databaseServerByVersion
  - databaseServerByState
  - databaseServerByServerConfiguration
  - databaseServerByStorageAutoscaling
  - databaseServerByPerformanceInsights
  - databaseServerByStorageEncryption
  - databaseServerByClusterRole
  - databaseServerByStorageTier
  - databaseServerByAvailabilityZoneName
  - databaseServerByRegionName
  - databaseServerByAge
- DatabaseServerTier
  - databaseServerTiersByName
  - databaseServerTiersByVMem
  - databaseServerTiersByCloudProvider
- DatabaseTier
  - databaseTiersByName
  - databaseTiersByCloudProvider
- DesktopPool
  - desktopPoolByName
  - desktopPoolByViewPod
- DiskArray
  - diskarrayByName
- DocumentCollection
  - dcByName

  - dcByTag
  - dcByBusinessAccount
  - dcByResourceGroup
  - dcByCloudProvider
  - dcByRegion
  - dcByDatabaseServer
- Group
  - groupsByName
- HCIPhysicalMachine
- IOModule
  - ioModuleByName
- Internet
  - internetByName
- LoadBalancer
  - lbsByName
- LogicalPool
  - logicalPoolByName
- Namespace
  - namespacesByName
  - namespacesByContainerPlatformCluster
  - namespacesByTag
- Network
  - netsByName
- NodePool
- PhysicalMachine
  - pmsByName
  - pmsByTargetType
  - pmsByClusterName
  - pmsByDC
  - pmsByState
  - pmsByTag
  - pmsByStorage
  - pmsBySwitch
  - pmsByMem
  - pmsByNumVms
  - pmsByNumCPUs
  - pmsByVendorName
  - pmsByCPUModel
  - pmsByModel
  - pmsByTimezone
  - pmsBySupportedProcessorCompatibilityModes
  - pmsByClusterTag

- ProcessorPool
- Region
  - regsByName
  - regsByCloudProvider
- ReservedInstance
- ResourceGroup
  - resourceGroupByName
  - resourceGroupByTag
  - resourceGroupByBusinessAccountUuid
- Service
  - servicesByName
  - serviceByNamespace
  - serviceByContainerPlatformCluster
  - serviceByKubernetesServiceType
  - serviceByBusinessApplication
  - serviceByTag
  - serviceByBusinessTransaction
  - serviceByUserDefinedEntity
- Storage
  - storageByName
  - storageByVMs
  - storageByPMCluster
  - storageByStorageCluster
  - storageByState
  - storageByTag
  - storageByDC
  - storageWithLocalSupport
- StorageCluster
  - storageClustersByName
- StorageController
  - storagecontrollerByName
- StorageTier
  - storageTierByName
  - storageTiersByCloudProvider
- Switch
  - switchByName
- ViewPod
  - viewPodByName
- VirtualDataCenter
  - vdcsByName
  - vdcsByTag
- VirtualMachine

- vmsByName
- vmsByPMName
- vmsByVDC
- vmsByDCnested
- vmsByDiskArrayName
- vmsByStorage
- vmsByApplicationComponent
- vmsByBusinessApplication
- vmsByBusinessTransaction
- vmsByService
- vmsByClusterName
- vmsByStorageClusterName
- vmsByDC
- vmsByState
- vmsByTag
- vmsConnectedToEphemeralStorage
- vmsHotAddMemory
- vmsHotAddCPU
- vmsHotRemoveCPU
- vmsConnectedToEncryptedVolume
- vmsByGuestName
- vmsBySockets
- vmsByCoresPerSocket
- vmsByNumCPUs
- vmsByAge
- vendorToolsInstalled
- vendorToolsVersion
- vmsByProcessorCompatibilityMode
- vmsBySharingMode
- vmsByHasDedicatedProcessors
- vmByTargetCategory
- vmByTargetName
- vmByTargetType
- vmsByBusinessAccountUuid
- vmsByMem
- vmsByStorageAmount
- vmsByLogicalPoolName
- vmsByNetwork
- vmsByResourceGroupName
- vmsByAvailabilityZoneName
- vmsByRegionName
- vmsByCloudProvider
- vmsByContainerPlatformCluster

  - vmsByActiveSessions
  - vmsByVendorId
  - vmByComputeTier
- VirtualMachineCluster
  - virtualMachineClustersByName
- VirtualMachineSpec
  - virtualMachineSpecsByName
  - virtualMachineSpecsByTier
  - virtualMachineSpecsByAppCount
  - virtualMachineSpecsByBusinessAccountUuid
  - virtualMachineSpecsByRegionName
  - virtualMachineSpecsByResourceGroup
  - virtualMachineSpecsByComputeTier
  - virtualMachineSpecsByTag
  - virtualMachineSpecsByNumDaysEmpty
- VirtualVolume
  - virtualVolumeByName
  - virtualVolumeWithLocalSupport
  - volumeByTag
  - volumeByAccount
  - volumeByConnectedVm
  - volumeByAttachmentState
  - volumeByNumDaysUnattached
  - volumeByTier
  - volumeByRegion
  - volumeByCloudProvider
  - volumeById
  - volumeByResourceGroupName
  - volumeByDeletable
  - volumeByStorage
  - volumeByStorageGroup
  - volumeByAge
- Workload
  - workloadByName
- WorkloadController
  - workloadControllersByName
  - workloadControllersByNamespace
  - workloadControllersByKind
  - workloadControllersByContainerPlatformCluster
  - workloadControllersByTag

Note that in the case of a discovered resource group, group members can be discovered by multiple targets. In that case, the Turbonomic API returns a single target for the group.

Groups requests

Pagination for group requests

When you request group data, the API uses pagination to return subsets of the data. You should always use pagination when available to make sure you get your data in manageable chunks. To paginate returned data, the `limit`, `orderBy`, and `cursor` parameters control how the APIs paginate their data. If you do not specify a `limit` for the following requests, the API will paginate the data to a default limit of records:

- `GET /groups`
- `GET /groups/<groupId>/members`
- `GET /groups/<groupId>/entities`

If your scripts make these requests without setting a limit, you should be sure they check for a pagination cursor in the returned data.

For general information about pagination, see Pagination in the Turbonomic API.

Getting a list of groups

Gets a list of all groups on your Turbonomic appliance. This will include Turbonomic internal and default groups, as well as any groups discovered from your targets, in addition to your custom groups.

Example: `GET https://10.10.10.10/api/v3/groups`

Response: A list of `GroupApiDTO`s representing the groups on your appliance.


```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/groups/_S4k98SFQEd-S4dn3K4_SSA?
include_aspects=false"
      }
    ],
    "uuid": "_S4k98SFQEd-S4dn3K4_SSA",
    "displayName": "Datacenters",
    "className": "Group",
    "environmentType": "ONPREM",
    "entitiesCount": 12,
    "membersCount": 12,
    "groupType": "DataCenter",
    "severity": "Critical",
    "isStatic": false,
    "logicalOperator": "AND",
    "activeEntitiesCount": 44
  },
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/groups/_S4k94SFQEd-S4dn3K4_SSA?
include_aspects=false"
      }
    ],
    "uuid": "_S4k94SFQEd-S4dn3K4_SSA",
    "displayName": "Chassis",
    "className": "Group",
    "environmentType": "ONPREM",
    "entitiesCount": 2,
    "membersCount": 1,
    "groupType": "Chassis",
    "severity": "Normal",
    "isStatic": false,
    "logicalOperator": "AND",
    "activeEntitiesCount": 2
  },
  ...
```

```
]
```

Creating a group

In order to create a group in Turbonomic, provide a GroupApiDTO with a `groupType` and `isStatic`.

For a dynamic group (`isStatis = false`), you also include the `criteriaList`information:

- caseSensitive

  If `true`, the regex will be evaluated as case-sensitive.
- expType
- For strings, the allowable values are EQ and NEQ (Equals, Does Not Equal). For integers, you may additionally use `GT, LT, GTE, LTE` (Greater than, less than, greater than or equal to, and less than or equal to, respectively).
- expVal

  The value of the regular expression.
- filterType

  See **Group Filter Types** for a full list of available options

For a static group:

- Provide UUIDs for each entity that is a member of the group in the `memberUuidList`
- DO NOT provide any `criteriaList` parameters

Example: `POST https://10.10.10.10/api/v3/groups/`

Example input, showing the creation of a dynamic group of VMs whose names start with "AJ":

```
{
  "isStatic": false,
  "displayName": "Dallas-Dynamic",
  "memberUuidList": [],
  "criteriaList": [
    {
      "expType": "EQ",
      "expVal": "AJ.*",
      "filterType": "vmsByName",
      "caseSensitive": false
    }
  ],
  "groupType": "VirtualMachine"
}
```

Response:

```
{
  "uuid": "_4L0PcE9SEemD69d9a0uLJQ",
  "displayName": "Dallas-Dynamic",
  "className": "Group",
  "environmentType": "ONPREM",
  "entitiesCount": 1,
  "membersCount": 1,
  "groupType": "VirtualMachine",
  "severity": "Major",
  "isStatic": false,
  "logicalOperator": "AND",
  "criteriaList": [
    {
      "expVal": "AJ.*",
      "expType": "EQ",
      "filterType": "vmsByName",
      "caseSensitive": false
    }
  ],
  "activeEntitiesCount": 1
```

```
}
```

Example input, showing the creation of a static group of Hosts:

```
{
  "isStatic": true,
  "displayName": "Dallas-Static",
  "memberUuidList": [
    "4C4C4544-0042-5A10-804A-B1C04F384331",
    "4C4C4544-004A-3610-8057-C8C04F464331",
    "Virtual_ESX_42381e13-4256-64bb-f554-d17798ddb2ba"
  ],
  "criteriaList": [],
  "groupType": "PhysicalMachine"
}
```

Response:

```
{
  "links": [
    {
      "rel": "self",
      "href": "https://10.10.10.10/api/v3/groups/_vn_u0FJ0EemD69d9a0uLJQ?include_aspects=false"
    }
  ],
  "uuid": "_vn_u0FJ0EemD69d9a0uLJQ",
  "displayName": "Dallas-Static",
  "className": "Group",
  "environmentType": "ONPREM",
  "entitiesCount": 3,
  "membersCount": 3,
  "groupType": "PhysicalMachine",
  "severity": "Critical",
  "isStatic": true,
  "logicalOperator": "OR",
  "activeEntitiesCount": 3
}
```

Editing a group

In order to edit a user-created group, send the complete `GroupApiDTO` with your changes as the input to this request.

Example: `PUT https://10.10.10.10/api/v3/groups/_Y6BL4FDQEemD69d9a0uLJQ`

Example Input:A modified `GroupApiDTO`. In this case, the `expVal` parameter has been modified.

```
{
  "isStatic": false,
  "displayName": "Dallas-Host",
  "memberUuidList": [],
  "criteriaList": [
    {
      "expType": "EQ",
      "expVal": "hp*change",
      "filterType": "pmsByName",
      "caseSensitive": false
    }
  ],
  "groupType": "PhysicalMachine",
  "logicalOperator": "AND"
}
```

Response: The full `GroupApiDTO` reflecting your changes:

```
{
  "links": [
    {
      "rel": "self",
      "href": "https://10.10.10.10/api/v3/groups/_Y6BL4FDQEemD69d9a0uLJQ?include_aspects=false"
```

```
    }
  ],
  "uuid": "_Y6BL4FDQEemD69d9a0uLJQ",
  "displayName": "Dallas-Host",
  "className": "Group",
  "environmentType": "ONPREM",
  "entitiesCount": 0,
  "membersCount": 0,
  "groupType": "PhysicalMachine",
  "severity": "Normal",
  "isStatic": false,
  "logicalOperator": "AND",
  "criteriaList": [
    {
      "expVal": "hp*change",
      "expType": "EQ",
      "filterType": "pmsByName",
      "caseSensitive": false
    }
  ],
  "activeEntitiesCount": 0
}
```


Deleting a group

To delete a user-created group, pass the UUID of the group in the URL of the request.

Example: `DELETE https://10.10.10.10/api/v3/groups/_vn_u0FJ0EemD69d9a0uLJQ`

Response: A response of 200 indicates successful deletion.

Getting actions related to a group

Referenced Endpoint: Actions

Gets a list of actions relating to the entities in the specified group. To get a single action, include the UUID of the action in the request. This request takes the following parameters:

- group_Uuid

  The UUID of the group whose actions will be returned.

- cursor

  When making subsequent calls to retrieve additional results, this is the cursor returned by the last call to this method.

- limit

  The maximum number of items to return. Must be a positive integer. If not set, the server will provide a default (20).

- order_by

  The field to order the results by. `[ creation_date, name, risk_category, savings, severity ]`.

- ascending

  Default: true. When false, results will be in descending order.

Examples:

All actions: `GET https://10.10.10.10/api/v3/groups/421dd4ce-a4c1-231a-71e7-64fbf598bc65/actions?limit=50&order_by=creation_date&ascending=true`

Single action: `GET https://10.10.10.10/api/v3/groups/421dd4ce-a4c1-231a-71e7-64fbf598bc65/actions/_pJg98EQaEemD69d9a0uLJQ`

Response: A list of ActionApiDTOs related to the specified group:


```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/actions/_pJg98EQaEemD69d9a0uLJQ"
      },
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/actions/2500817580000"
      }
    ],
    "uuid": "_pJg98EQaEemD69d9a0uLJQ",
    "createTime": "2019-03-11T12:28:03-04:00",
    "actionType": "RIGHT_SIZE",
    "actionState": "PENDING_ACCEPT",
    "actionMode": "MANUAL",
    "details": "Scale down Memory reservation for Virtual Machine 'AJ_Test' from 2 GB to 1 GB",
    "importance": 0,
    "target": {
      "uuid": "421dd4ce-a4c1-231a-71e7-64fbf598bc65",
      "displayName": "AJ_Test",
      "className": "VirtualMachine",
      "aspects": {
        "virtualMachineAspect": {
          "os": "CentOS 4/5/6/7 (64-bit)",
          "ip": [
            "10.10.10.107"
          ],
          "connectedNetworks": [
            {
              "uuid": "34f9f5364cd843aa1fe99900621ee3b51a7dc6e5",
              "displayName": "VM Network"
            }
          ],
          "numVCPUs": 1,
          "ebsOptimized": false
        }
      },
      "environmentType": "ONPREM"
    },
    "currentEntity": {
      "uuid": "_pJlPYUQaEemD69d9a0uLJQ",
      "className": "Mem"
    },
    "newEntity": {
      "uuid": "_pLCA40QaEemD69d9a0uLJQ",
      "className": "Mem",
      "aspects": {
        "virtualMachineAspect": {
          "os": "Linux (Free)",
          "ebsOptimized": false
        }
      }
    },
    "currentValue": "2097152.0",
    "newValue": "1048576.0",
    "resizeToValue": "1048576.0",
    "risk": {
      "uuid": "_pL_qMEQaEemD69d9a0uLJQ",
      "subCategory": "Efficiency Improvement",
      "description": "Underutilized Memory reservation",
      "severity": "MINOR",
      "importance": 0
    },
    "currentLocation": {
      "uuid": "cdcd967d1e08d20936c3aa9a0879c635214c79ee",
      "displayName": "ACM",
      "className": "DataCenter"
    },
    "newLocation": {
      "uuid": "cdcd967d1e08d20936c3aa9a0879c635214c79ee",
      "displayName": "ACM",
      "className": "DataCenter"
    },
    "actionID": 2500817580000
  },
  {
    "links": [
```

```
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/actions/_XdP4EEQ8EemD69d9a0uLJQ"
      },
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/actions/2501049335840"
      }
    ],
    "uuid": "_XdP4EEQ8EemD69d9a0uLJQ",
    "createTime": "2019-03-11T16:29:28-04:00",
    "actionType": "RIGHT_SIZE",
    "actionState": "PENDING_ACCEPT",
    "actionMode": "MANUAL",
    "details": "Remove VMem limit for VirtualMachine AJ_Test to maximize efficiency",
    "importance": 0,
    "target": {
      "uuid": "421dd4ce-a4c1-231a-71e7-64fbf598bc65",
      "displayName": "AJ_Test",
      "className": "VirtualMachine",
      "aspects": {
        "virtualMachineAspect": {
          "os": "CentOS 4/5/6/7 (64-bit)",
          "ip": [
            "10.10.10.107"
          ],
          "connectedNetworks": [
            {
              "uuid": "34f9f5364cd843aa1fe99900621ee3b51a7dc6e5",
              "displayName": "VM Network"
            }
          ],
          "numVCPUs": 1,
          "ebsOptimized": false
        }
      },
      "environmentType": "ONPREM"
    },
    "currentEntity": {
      "uuid": "_XdP4EkQ8EemD69d9a0uLJQ",
      "className": "VMem"
    },
    "newEntity": {
      "uuid": "_Xnq_YUQ8EemD69d9a0uLJQ",
      "className": "VMem",
      "aspects": {
        "virtualMachineAspect": {
          "os": "Linux (Free)",
          "ebsOptimized": false
        }
      }
    },
    "currentValue": "2097152.0",
    "newValue": "0.0",
    "resizeToValue": "0.0",
    "risk": {
      "uuid": "_XnuCsUQ8EemD69d9a0uLJQ",
      "subCategory": "Efficiency Improvement",
      "description": "VMem congestion in Virtual Machine 'AJ_Test'",
      "severity": "MAJOR",
      "reasonCommodity": "VMem",
      "importance": 0
    },
    "currentLocation": {
      "uuid": "cdcd967d1e08d20936c3aa9a0879c635214c79ee",
      "displayName": "ACM",
      "className": "DataCenter"
    },
    "newLocation": {
      "uuid": "cdcd967d1e08d20936c3aa9a0879c635214c79ee",
      "displayName": "ACM",
      "className": "DataCenter"
    },
    "actionID": 2501049335840
  }
]
```

Getting the entities in a group

Referenced Endpoint: Entities

Gets all entities contained within a group.

Note: This request will not return the contents of nested groups. In order to see groups of groups, use the `groups/members` request.

Example: `GET https://10.10.10.10/api/v3/groups/_4L0PcE9SEemD69d9a0uLJQ/entities`

Response: A list of `EntityApiDTO`s representing the entities in the specified group.


```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/actions/_pJg98EQaEemD69d9a0uLJQ"
      },
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/actions/2500817580000"
      }
    ],
    "uuid": "_pJg98EQaEemD69d9a0uLJQ",
    "createTime": "2019-03-11T12:28:03-04:00",
    "actionType": "RIGHT_SIZE",
    "actionState": "PENDING_ACCEPT",
    "actionMode": "MANUAL",
    "details": "Scale down Memory reservation for Virtual Machine 'AJ_Test' from 2 GB to 1 GB",
    "importance": 0,
    "target": {
      "uuid": "421dd4ce-a4c1-231a-71e7-64fbf598bc65",
      "displayName": "AJ_Test",
      "className": "VirtualMachine",
      "aspects": {
        "virtualMachineAspect": {
          "os": "CentOS 4/5/6/7 (64-bit)",
          "ip": [
            "10.10.10.107"
          ],
          "connectedNetworks": [
            {
              "uuid": "34f9f5364cd843aa1fe99900621ee3b51a7dc6e5",
              "displayName": "VM Network"
            }
          ],
          "numVCPUs": 1,
          "ebsOptimized": false
        }
      },
      "environmentType": "ONPREM"
    },
    "currentEntity": {
      "uuid": "_pJlPYUQaEemD69d9a0uLJQ",
      "className": "Mem"
    },
    "newEntity": {
      "uuid": "_pLCA40QaEemD69d9a0uLJQ",
      "className": "Mem",
      "aspects": {
        "virtualMachineAspect": {
          "os": "Linux (Free)",
          "ebsOptimized": false
        }
      }
    },
    "currentValue": "2097152.0",
    "newValue": "1048576.0",
    "resizeToValue": "1048576.0",
    "risk": {
      "uuid": "_pL_qMEQaEemD69d9a0uLJQ",
      "subCategory": "Efficiency Improvement",
      "description": "Underutilized Memory reservation",
      "severity": "MINOR",
      "importance": 0
    },
    "currentLocation": {
```

```
      "uuid": "cdcd967d1e08d20936c3aa9a0879c635214c79ee",
      "displayName": "ACM",
      "className": "DataCenter"
    },
    "newLocation": {
      "uuid": "cdcd967d1e08d20936c3aa9a0879c635214c79ee",
      "displayName": "ACM",
      "className": "DataCenter"
    },
    "actionID": 2500817580000
  }
]
```

Getting the parent groups of a group

For a specified group, gets any group that the specified group belongs to.

Note: Turbonomic contains a default parent group for user-created groups called "My Groups". This group should not be modified or deleted.

This request takes the following parameters:

- group_Uuid

  The UUID of the group whose actions will be returned.

- path

  Default: `false`. When `true`, this request returns the parent group, along with any groups traversed to obtain the parent group. For groups that are not nested, this returns the group whose ID you entered and its parent group.

Example: `GET https://10.10.10.10/api/v3/groups/_4L0PcE9SEemD69d9a0uLJQ/groups?path=true`

Response: A list of `GroupApiDTO`s representing the group and parent group.

```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/groups/_S4k68yFvEd-S4dn3T4_SSA?
include_aspects=false"
      }
    ],
    "uuid": "_S4k68yFvEd-S4dn3T4_SSA",
    "displayName": "My Groups",
    "className": "Group",
    "environmentType": "ONPREM",
    "entitiesCount": 1,
    "membersCount": 3,
    "groupType": "ServiceEntity",
    "severity": "Major",
    "isStatic": false,
    "logicalOperator": "AND",
    "activeEntitiesCount": 1
  },
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/groups/_4L0PcE9SEemD69d9a0uLJQ?
include_aspects=false"
      }
    ],
    "uuid": "_4L0PcE9SEemD69d9a0uLJQ",
    "displayName": "Dallas-Dynamic",
    "className": "Group",
    "environmentType": "ONPREM",
    "entitiesCount": 1,
    "membersCount": 1,
    "groupType": "VirtualMachine",
    "severity": "Major",
    "isStatic": false,
```

```
    "logicalOperator": "AND",
    "criteriaList": [
      {
        "expVal": "AJ.*",
        "expType": "EQ",
        "filterType": "vmsByName",
        "caseSensitive": false
      }
    ],
    "activeEntitiesCount": 1
  }
]
```

Getting the members of a group

Referenced Endpoint: Entities

Gets all members of a group.

Note: This request returns nested groups and the member entities of each. To get only the *entities* in the specified group, use the `groups/entities` request.

Example: `GET https://10.10.10.10/api/v3/groups/_4L0PcE9SEemD69d9a0uLJQ/members`

Response: A list of `serviceEntityApiDTO`s representing the members of the group:

```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/entities/183b37c68211668aecd36eadcdce9110ebc58f85?
include_aspects=false"
      }
    ],
    "uuid": "183b37c68211668aecd36eadcdce9110ebc58f85",
    "displayName": "Olga",
    "className": "VirtualDataCenter",
    "priceIndex": 1,
    "state": "ACTIVE",
    "severity": "Normal",
    "discoveredBy": {
      "uuid": "_h3Km4Bs3EemMFrqlXNi1yQ",
      "displayName": "vsphere-dc23.corp.vmturbo.com",
      "type": "vCenter"
    },
    "remoteId": "resgroup-96",
    "environmentType": "ONPREM"
  },
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/entities/2332ba0f9dc14c07f11adb38e12df43c4bc8ca56?
include_aspects=false"
      }
    ],
    "uuid": "2332ba0f9dc14c07f11adb38e12df43c4bc8ca56",
    "displayName": "olga_do_not_touch_1",
    "className": "VirtualDataCenter",
    "priceIndex": 1,
    "state": "ACTIVE",
    "severity": "Normal",
    "discoveredBy": {
      "uuid": "_5kMqwNLnEeiJs9KWCzOdUQ",
      "displayName": "vsphere-dc1.dev.mycorp.com",
      "type": "vCenter"
    },
    "remoteId": "resgroup-1088",
    "environmentType": "ONPREM"
  },
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/entities/6668eb0940bf970d24eb13279b31a52f3c44a91f?
include_aspects=false"
```

```
      }
    ],
    "uuid": "6668eb0940bf970d24eb13279b31a52f3c44a91f",
    "displayName": "Gilad",
    "className": "VirtualDataCenter",
    "priceIndex": 1,
    "state": "ACTIVE",
    "severity": "Normal",
    "discoveredBy": {
      "uuid": "_lj_tkH39EeiKy53b3GBglA",
      "displayName": "vsphere-dc5.dev.mycorp.com",
      "type": "vCenter"
    },
    "remoteId": "resgroup-1953",
    "environmentType": "ONPREM"
  },
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/entities/20b9e0acf4c6f7d5b00f744d1dc954616095f1f2?
include_aspects=false"
      }
    ],
    "uuid": "20b9e0acf4c6f7d5b00f744d1dc954616095f1f2",
    "displayName": "ACM",
    "className": "VirtualDataCenter",
    "priceIndex": 1,
    "state": "ACTIVE",
    "severity": "Normal",
    "discoveredBy": {
      "uuid": "_lj_tkH39EeiKy53b3GBglA",
      "displayName": "vsphere-dc5.dev.mycorp.com",
      "type": "vCenter"
    },
    "remoteId": "resgroup-v1946",
    "environmentType": "ONPREM"
  },
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/entities/56e6cfd2bc88ef5675df9dc3e06796ceb8b6e301?
include_aspects=false"
      }
    ],
    "uuid": "56e6cfd2bc88ef5675df9dc3e06796ceb8b6e301",
    "displayName": "olga_do_not_touch_7",
    "className": "VirtualDataCenter",
    "priceIndex": 1,
    "state": "ACTIVE",
    "severity": "Normal",
    "discoveredBy": {
      "uuid": "_GRNtoNu0Eeilt9UDtI29QA",
      "displayName": "vsphere-dc7.dev.mycorp.com",
      "type": "vCenter"
    },
    "remoteId": "resgroup-1066",
    "environmentType": "ONPREM"
  }
]
```


Getting settings of a group

Referenced Endpoint: Settings

Gets both global and user-modified settings for the specified group. To get a single setting, pass the setting UUID as part of the request URL. This request takes the following parameters:

- group_Uuid

  The Uuid of the group whose settings will be returned

- include_settingspolices

  Default: `false`. When `true`, returns the settings policies affecting the specified group, in addition to the settings.

Example: `GET https://10.10.10.10/api/v3/groups/_WvaGkOBjEeierZ9b1bxmgg/settings?include_settingspolicies=false`

Response: A list of `SettingsManagerApiDTO`s representing the settings, or settings and settings policies for the specified group:

```
[
  {
    "uuid": "appsettingsmanager",
    "displayName": "Application Discovery",
    "category": "AppDiscovery",
    "settings": [
      {
        "uuid": "Username",
        "displayName": "Username",
        "value": "",
        "defaultValue": "",
        "valueType": "STRING",
        "entityType": "VirtualMachine",
        "sourceGroupName": "Global"
      },
      {
        "uuid": "Password",
        "displayName": "Password",
        "value": "",
        "defaultValue": "",
        "valueType": "STRING",
        "entityType": "VirtualMachine",
        "sourceGroupName": "Global"
      },
      ...
    },
    {
    "uuid": "controlmanager",
    "displayName": "Action Workflow",
    "category": "Automation",
    "settings": [
      {
        "uuid": "moveVMWorkflow",
        "displayName": "Move Workflow",
        "value": "",
        "defaultValue": "",
        "valueType": "STRING",
        "entityType": "VirtualMachine",
        "sourceGroupName": "Global"
      }
    ]
  }
  ...
}
```

Editing a specified setting of a group

Referenced Endpoint: Settings

Edits a specified setting for the specified group. This request takes the following parameters:

- group_Uuid

  The Uuid of the group whose setting will be edited
- settingsManager_Uuid

  The Uuid of the settings manager whose setting will be edited
- setting_Uuid

  The Uuid of the setting to change

Example: `PUT https://10.10.10.10/api/v3/groups/_WvaGkOBjEeierZ9b1bxmgg/settings/appsettingsmanager/Retry_Interval`

Example Input:

The new value for the setting:

```
{
  "value": "61"
}
```

Response: A `ServiceEntityApiDTO` representing the specified setting, with your changed value(s):

```
{
  "uuid": "Retry_Interval",
  "displayName": "Retry Interval [mins]",
  "value": "61",
  "defaultValue": "60.0",
  "valueType": "NUMERIC",
  "min": 0,
  "max": 50000,
  "entityType": "VirtualMachine",
  "sourceGroupName": "A_Few_ACM_VMs"
}
```

Getting settings policies of a group

Referenced Endpoint: Settings Policies

Gets the settings policies of the specified group.

Example:

Response:

Getting statistics of a group

Referenced Endpoint: Stats

Gets all statistics for the specified group:

Note:

When you retrieve statistics for a cluster, using the `/groups` endpoint will return the aggregated statistics for all applicable cluster entities. For individual stats per entity, use the `/stats` endpoint.

Some statistics are only valid for certain scopes. When you request given statistics for a scope that doesn't include them, the API does not return any data. For example, the `numVMs` statistic is valid in only the Market and Cluster scopes – the API omits `numVMs` data from the response when you query other scopes.

Example: `https://10.10.10.10/api/v3/groups/_WvaGkOBjEeierZ9b1bxmgg/stats`

Response: A list of `StatSnapshotApiDTO`s representing the statistics for the specified group:

```
[
  {
    "displayName": "A_Few_ACM_VMs",
    "date": "2019-03-31T17:11:35-04:00",
    "statistics": [
      {
        "displayName": "(multiple)",
        "name": "priceIndex",
        "values": {
          "max": 1.44,
          "min": 1.03,
          "avg": 1.19,
          "total": 4.77
        },
        "value": 1.19
      },
      {
        "displayName": "(multiple)",
        "name": "Q1VCPU",
        "capacity": {
```

```
          "max": 20000,
          "min": 20000,
          "avg": 20000,
          "total": 60000
        },
        "reserved": {
          "max": 10000,
          "min": 10000,
          "avg": 10000,
          "total": 30000
        },
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        ...
      }
      ...
      ]
    }
```

Getting a filtered list of statistics of a group

Referenced Endpoint: Stats

Gets a filtered list of statistics for the specified group. For more information about creating a filtered list of statistics, including example POSTs and responses, see *Filtered Statistics* in the Statistics Endpoint

Example: `POST https://10.10.10.10/api/v3/groups/_WvaGkOBjEeierZ9b1bxmgg/stats`

Example input:

```
{
  "endDate": "-1d",
  "startDate": "-1w",
  "statistics": [
    {
      "filters": [
        {
          "type": "relation",
          "value": "bought"
        }
      ]
    }
  ]
}
```

Response: A list of `StatSnapshotApiDTO`s representing the statistics for the specified group:

```
[
  {
    "displayName": "A_Few_ACM_VMs",
    "date": "2019-03-24T00:00:00-04:00",
    "statistics": [
      {
        "name": "Mem",
        "capacity": {
          "max": 134214448,
          "min": 134214448,
          "avg": 134214448,
          "total": 1073715580
        },
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        "units": "KB",
        "values": {
          "max": 2073084,
          "min": 679782.1,
          "avg": 1124840.4,
```

```
          "total": 4499361.5
        },
        "value": 1124840.4
      },
      {
        "name": "MemAllocation",
        "capacity": {
          "max": 504346624,
          "min": 504346624,
          "avg": 504346624,
          "total": 4034772990
        },
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        "units": "KB",
        "values": {
          "max": 2073084,
          "min": 679782.1,
          "avg": 1124840.4,
          "total": 4499361.5
        },
        "value": 1124840.4
      },
      {
        "name": "Q1VCPU",
        "capacity": {
          "max": 20000,
          "min": 20000,
          "avg": 20000,
          "total": 120000
        },
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        "values": {
          "max": 90,
          "min": 14,
          "avg": 41.08,
          "total": 123.23
        },
        "value": 41.08
      }
      ...
    ]
  }
]
```


Getting the supply chain of a group

Referenced Endpoint: Supply Chains

Using one or more of the following parameters, you can filter the supply chain that is returned by this request:

- market_Uuid

  The UUID of the market whose supply chain will be returned

- types

  The entity types whose supply chain will be returned. Choose from:

  - APPLICATION
  - APPLICATION_SERVER
  - BUSINESS_ACCOUNT
  - BUSINESS_UNIT
  - CHASSIS

  - CLOUD_SERVICE
  - CONTAINER
  - CONTAINER_POD
  - DATABASE
  - DATABASE_SERVER
  - DATACENTER
  - DISK_ARRAY
  - IO_MODULE
  - LOAD_BALANCER
  - LOGICAL_POOL
  - NETWORK
  - PHYSICAL_MACHINE
  - STORAGE
  - STORAGE_CONTROLLER
  - SWITCH
  - VIRTUAL_APPLICATION
  - VIRTUAL_DATACENTER
  - VIRTUAL_MACHINE
- entity_states

  Only return entities in the specified states. Choose from:

  - ACTIVE:

    Entity is actively consuming resources.
  - EVACUATED:

    Entity is ready to suspend and will not participate in the market, but will not generate a SUSPEND action.

    Note: This state replaces the SUSPEND state, and requires additional configuration to enable. Turbonomic does not recommend enabling this state.
  - FAILOVER:

    This entity is reserved for failover and will not participate in the market.
  - IDLE:

    Entity is powered off and not actively consuming market resources.
  - LAUNCH:

    Entity is starting.
  - MAINTENANCE:

    Entity is in maintenance mode.
  - NOT_MONITORED:

    Entity is not currently monitored by Turbonomic.
  - QUEUED:

    Entity is in the process of changing from one state to another.

    Note: This state is internal to Turbonomic and will not appear in the UI, but may appear in an API response if the call is executed at the precise moment the entity is changing state.
  - RESOURCE_ALLOCATION:

Cloud entity is being created.

- RESOURCE_RELEASE:

  Cloud Entity is being deleted.

- SUSPEND:

  Entity is suspended or the guest OS is in sleep, standby, or suspended state. The entity is not removed from the environment, but will not be used in market decisions. Based on market conditions, suspended entities can be started, re-adding them to the market.

- SUSPEND_PENDING:

  Entity is making preparations to suspend, or is the next entity of its type to suspend.

- TERMINATE_PENDING:

  Entity is in the process of being deleted.

- UNKNOWN:

  Turbonomic is unable to obtain the current state of the entity.

  This occurs when the entity is reported to the hypervisor as having a state of DISCONNECTED, ORPHANED, or INVALID.

- environment_type
- Only return actions that affect entities in the specified environment type. [CLOUD, HYBRID, ONPREM].
- detail_type

  The level of detail returned for each item in the returned supply chain. [compact, entity, aspects].

- health

  Return a small summary of the health of the returned supply chain.

Example: GET https://10.10.10.10/api/v3/groups/_WvaGkOBjEeierZ9b1bxmgg/supplychains?types=PhysicalMachine&entity_states=ACTIVE&entity_states=SUSPEND&entity_states=MAINTENANCE&environment_type=ONPREM&detail_type=entity&health=true

Response: A SupplychainApiDTO representing the supply chain of the specified group:


```
  "seMap": {
    "PhysicalMachine": {
      "depth": 1,
      "entitiesCount": 3,
      "healthSummary": {
        "Minor": 1,
        "Normal": 2
      },
      "stateSummary": {
        "ACTIVE": 3
      },
      "connectedProviderTypes": [
        "DataCenter",
        "Storage"
      ],
      "instances": {
        "34313836-3333-5553-4537-33364e385146": {
          "links": [
            {
              "rel": "self",
              "href": "https://10.10.10.10/api/v3/entities/34313836-3333-5553-4537-33364e385146?
include_aspects=false"
            }
          ],
          "uuid": "34313836-3333-5553-4537-33364e385146",
          "displayName": "dc17-host-01.dev.mycorp.com",
          "className": "PhysicalMachine",
          "priceIndex": 3.2925103,
          "state": "ACTIVE",
```

```
          "severity": "Normal",
          "discoveredBy": {
            "uuid": "_yXCEIF6hEeiNCKdg4ZVvVQ",
            "displayName": "vsphere-dc17.dev.mycorp.com",
            "type": "vCenter"
          },
          "remoteId": "host-621",
          "environmentType": "ONPREM"
        },
        ...

      }
    }
  }
}
```

Creating a list of user-defined tags for a group

Referenced Endpoint: Tags

Turbonomic allows a user to assign tags to a specific group, in addition to whatever tags are discovered for that group.

This request takes the following parameters:

- entity_Uuid

  The UUID of the group.
- body

```
[
  {
    "key": "ExampleGroupTagKey",
    "values": [
      "ExampleGroupTagValue"
    ]
  }
]
```

Example: `POST https://10.10.10.10/api/v3/groups/286920513901488/tags`

Response: Returns the list of tags for a group that are created.

```
[
  {
    "key": "ExampleGroupTagKey",
    "values": [
      "ExampleGroupTagValue"
    ]
  }
]
```

Deleting a list of user-defined tags of a group

Referenced Endpoint: Tags

Turbonomic allows a user to delete tags of a specific group. If a tag key list is given, it deletes all the tags defined on that list. If no list is given, it deletes all the tags of that group.

This request takes the following parameters:

- (Optional) key **-** array[string]

  List of keys to delete.

Example: `DELETE https://10.10.10.10/api/v3/groups/286920513901488/tags?key=ExampleTagKey`

Response: Empty response upon successful execution.

Groups endpoint tips

GroupBy field criteria

When using the POST requests to query actions and action statistics, you can use the `groupBy` parameter to organize the results. The valid values for this parameter differ by entity type:

Note:

When sending an inputDTO using this field, only the value is needed.

Example:`"groupBy": "vmsByPMName"`

```
{
  "VirtualMachine":{
    "criteria":[
      {
        "filterType":"vmsByName"
      },
      {
        "filterType":"vmsByPMName"
      },
      {
        "filterType":"vmsByStorage"
      },
      {
        "filterType":"vmsByNetwork"
      },
      {
        "filterType":"vmsByApplication"
      },
      {
        "filterType":"vmsByDatabaseServer"
      },
      {
        "filterType":"vmsByDatabaseServerVersion"
      },
      {
        "filterType":"vmsByDC"
      },
      {
        "filterType":"vmsByVDC"
      },
      {
        "filterType":"vmsByDCnested"
      },
      {
        "filterType":"vmsByNumCPUs"
      },
      {
        "filterType":"vmsByMem"
      },
      {
        "filterType":"vmsByGuestName"
      },
      {
        "filterType":"vmsByAltName"
      },
      {
        "filterType":"vmsByClusterName"
      },
      {
        "filterType":"vmsByDiskArrayName"
      },
      {
        "filterType":"vmsByLogicalPoolName"
      },
      {
        "filterType":"vmsByProfileName"
      },
      {
        "filterType":"vmsByTag"
      },
      {
        "filterType":"vmsByState"
      },
      {
```

```
        "filterType":"vmsByBusinessAccountUuid"
      },
      {
        "filterType":"vmsByResourceGroupUuid"
      }
    ]
  },
  "VirtualDataCenter":{
    "criteria":[
      {
        "filterType":"vdcsByName"
      },
      {
        "filterType":"vdcsByVDCName"
      },
      {
        "filterType":"vdcsByTag"
      },
      {
        "filterType":"vdcsByState"
      }
    ]
  },
  "PhysicalMachine":{
    "criteria":[
      {
        "filterType":"pmsByName"
      },
      {
        "filterType":"pmsByStorage"
      },
      {
        "filterType":"pmsByNetwork"
      },
      {
        "filterType":"pmsBySwitch"
      },
      {
        "filterType":"pmsByNumVms"
      },
      {
        "filterType":"pmsByDC"
      },
      {
        "filterType":"pmsByMem"
      },
      {
        "filterType":"pmsByNumCPUs"
      },
      {
        "filterType":"pmsByVendorName"
      },
      {
        "filterType":"pmsByCPUModel"
      },
      {
        "filterType":"pmsByModel"
      },
      {
        "filterType":"pmsByTimezone"
      },
      {
        "filterType":"pmsByClusterName"
      },
      {
        "filterType":"pmsByTag"
      },
      {
        "filterType":"pmsByState"
      }
    ]
  },
  "Storage":{
    "criteria":[
      {
        "filterType":"storageByName"
      },
      {
        "filterType":"storageByTag"
      },
      {
        "filterType":"storageByVMs"
```

```
      },
      {
        "filterType":"storageByDC"
      },
      {
        "filterType":"storageByPMCluster"
      },
      {
        "filterType":"storageByState"
      }
    ]
  },
  "Application":{
    "criteria":[
      {
        "filterType":"appsByName"
      },
      {
        "filterType":"appsByTag"
      }
    ]
  },
  "ApplicationComponent":{
    "criteria":[
      {
        "filterType":"appSrvsByName"
      }
    ]
  },
  "WebServer":{
    "criteria":[
      {
        "filterType":"wbSrvsByName"
      }
    ]
  },
  "BusinessApplication":{
    "criteria":[
      {
        "filterType":"busAppsByName"
      }
    ]
  },
  "Database":{
    "criteria":[
      {
        "filterType":"databaseByName"
      },
      {
        "filterType":"databaseByTag"
      },
      {
        "filterType":"databaseByBusinessAccountUuid"
      },
      {
        "filterType":"databaseByResourceGroupUuid"
      }
    ]
  },
  "DatabaseServer":{
    "criteria":[
      {
        "filterType":"databaseServerByName"
      },
      {
        "filterType":"databaseServerByTag"
      },
      {
        "filterType":"databaseServerByBusinessAccountUuid"
      },
      {
        "filterType":"databaseServerByEngine"
      },
      {
        "filterType":"databaseServerByEdition"
      },
      {
        "filterType":"databaseServerByVersion"
      }
    ]
  },
  "VirtualApplication":{
```

```
    "criteria":[
      {
        "filterType":"vappsByName"
      }
    ]
  },
  "Cluster":{
    "criteria":[
      {
        "filterType":"clustersByName"
      },
      {
        "filterType":"clustersByTag"
      }
    ]
  },
  "DataCenter":{
    "criteria":[
      {
        "filterType":"datacentersByName"
      },
      {
        "filterType":"datacentersByTag"
      }
    ]
  },
  "Group":{
    "criteria":[
      {
        "filterType":"groupsByName"
      },
      {
        "filterType":"groupsByTargetName"
      }
    ]
  },
  "StorageCluster":{
    "criteria":[
      {
        "filterType":"storageClustersByName"
      }
    ]
  },
  "DiskArray":{
    "criteria":[
      {
        "filterType":"diskarrayByName"
      }
    ]
  },
  "Zone":{
    "criteria":[
      {
        "filterType":"zonsByName"
      }
    ]
  },
  "Region":{
    "criteria":[
      {
        "filterType":"regsByName"
      }
    ]
  },
  "Network":{
    "criteria":[
      {
        "filterType":"netsByName"
      }
    ]
  },
  "LoadBalancer":{
    "criteria":[
      {
        "filterType":"lbsByName"
      }
    ]
  },
  "Chassis":{
    "criteria":[
      {
        "filterType":"chasByName"
```

```
      }
    ]
  },
  "IOModule":{
    "criteria":[
      {
        "filterType":"ioModuleByName"
      }
    ]
  },
  "StorageController":{
    "criteria":[
      {
        "filterType":"storagecontrollerByName"
      }
    ]
  },
  "DPod":{
    "criteria":[
      {
        "filterType":"dpodByName"
      }
    ]
  },
  "VPod":{
    "criteria":[
      {
        "filterType":"vpodByName"
      }
    ]
  },
  "LogicalPool":{
    "criteria":[
      {
        "filterType":"logicalPoolByName"
      }
    ]
  },
  "Switch":{
    "criteria":[
      {
        "filterType":"switchByName"
      }
    ]
  },
  "Container":{
    "criteria":[
      {
        "filterType":"containersByName"
      },
      {
        "filterType":"containersByVMName"
      }
    ]
  },
  "ContainerPod":{
    "criteria":[
      {
        "filterType":"containerpodsByName"
      },
      {
        "filterType":"containerpodsByVMName"
      }
    ]
  },
  "ServiceEntity":{
    "criteria":[
      {
        "filterType":"workloadByBusinessAccountUuid"
      },
      {
        "filterType":"workloadByResourceGroupUuid"
      }
    ]
  },
  "Workload":{
    "criteria":[
      {
        "filterType":"workloadByBusinessAccountUuid"
      },
      {
        "filterType":"workloadByResourceGroupUuid"
```

```
      }
    ]
  },
  "ResourceGroup":{
    "criteria":[
      {
        "filterType":"resourceGroupByName"
      },
      {
        "filterType":"resourceGroupByUuid"
      },
      {
        "filterType":"resourceGroupByBusinessAccountUuid"
      },
      {
        "filterType":"resourceGroupByTag"
      }
    ]
  },
  "BusinessAccount":{
    "criteria":[
      {
        "filterType":"businessAccountByName"
      },
      {
        "filterType":"businessAccountByUuid"
      },
      {
        "filterType":"subBusinessAccountOfUuid"
      },
      {
        "filterType":"businessAccountValidationStatus"
      },
      {
        "filterType":"businessAccountTargetName"
      },
      {
        "filterType":"businessAccountCloudProvider"
      }
    ]
  },
  "BillingFamily":{
    "criteria":[
      {
        "filterType":"billingFamilyByName"
      }
    ]
  },
  "ViewPod":{
    "criteria":[
      {
        "filterType":"viewPodByName"
      }
    ]
  },
  "DesktopPool":{
    "criteria":[
      {
        "filterType":"desktopPoolByName"
      },
      {
        "filterType":"desktopPoolByViewPod"
      }
    ]
  },
  "BusinessUser":{
    "criteria":[
      {
        "filterType":"businessUserByName"
      },
      {
        "filterType":"businessUserByDesktopPool"
      }
    ]
  }
}
```

Licenses endpoint

A product license determines the product features you can use, and the number of workloads that you can manage with Turbonomic. To increase the number of workloads you can manage, you can add additional licenses to your Turbonomic installation. Note that if you use multiple licenses, they must all support the same feature set.

In all circumstances, you should contact your sales representative to make sure that you get the correct license, and that you know how to install it properly.

Using the licenses endpoint, you can:

- Get a license summary
- Get a list of licenses currently added to your Turbonomic appliance
- Add a license
- Delete a license

Licenses requests

Getting licenses

Gets a list of licenses. For instances with multiple licenses, the `licenses/summary` request will return a summed total of licensed workload. For example, if you have two licenses for 10,000 workloads each, the `/licenses` request will return both LicenseApiDTOs in a list. The `/licenses/summary` call will return a single summed LicenseApiDTO, which will state that you have capacity for 20,0000 workloads.

Examples:

- **List of Licenses:** `GET https://10.10.10.10/api/v3/licenses`
- **License Summary:** `GET https://10.10.10.10/api/v3/licenses/summary`

Adding a license

Adds a license to your Turbonomic installation. You may optionally specify the `dryRun` parameter which, when set to true, will validate the license file but not apply it to the appliance.

Note:

This request requires a multipart file as input. Due to this requirement, this request cannot be executed directly from the Swagger.

Example: `POST https://10.10.10.10/api/v3/licenses?file=%5Bobject%20File%5D&dryRun=true`

Deleting a license

Removes the specified license from your Turbonomic installation. Deleting a license may reduce your licensed workload capacity below your current usage. In this case, you will be unable to add more targets until more workload capacity is available.

Example: `DELETE https://10.10.10.10/api/v3/licenses/_I_WVcJtWEeiGEs_doVlDOw`

Logs requests

The logs endpoint is not implemented at this time.

Markets endpoint

Turbonomic uses market-based analysis to perform workload management. To do this, it constructs a model of your environment, representing each entity as a buyer and seller in a market. You can access this model via a named market resource. There are two types of markets:

- **Real-Time Market**: The real-time market performs analysis and workload management on your environment. You can use the real-time market to access entities and get current or historical data about them.

  The `displayName` of the real-time market is `Market`. This market should always be in the `RUNNING` state. You should not attempt to use the API to stop this market.

  Turbonomic performs discovery to populate the topology it manages (the collection of entities in the real-time market). For this reason, you should never use the API add or delete entities in the real-time market.
- **Plan Market**: A plan market begins as a snapshot of the real-time market. You POST a scenario to the real-time market, and Turbonomic makes a copy of the real-time market to create a plan market. This plan market is just like the real-time market, except for any changes that were declared in the scenario. For example, if the scenario declares a scope for the plan, then the plan market only contains entities that are within that scope.

  Turbonomic runs economic cycles (buy/sell cycles) against this plan market until there are no more meaningful improvements to be made. At this point the plan run is completed.

  Note that once Turbonomic creates the plan market, that market stays in memory until you delete it. Also note that you can later apply a scenario to the plan market. This is how you run a plan on a plan. The logic flow is the same, and Turbonomic creates yet another plan market to run the analysis.

This endpoint contains references to start and end times for certain operations. For more information, including valid time formats, see Time in the Turbonomic API

Using the markets endpoint of the Turbonomic API, you can:

- Get a list of markets
- Save or Stop a plan market
- Delete a plan market
- See unplaced entities in a plan market
- Get filtered and unfiltered lists of actions, entities, notifications, policies, reservations, and statistics for a particular market
- Get, edit, and delete placement policies in the real-time market

For more information about managing placement policies with the API, see “Working with policies in the API” on page 1299.

Determining the status of the main market

After restarting the Turbonomic instance, the UI and API may be available before the real-time market has finished initializing and loading the topology of your environment. To check market status, execute the `GET /markets/market_uuid` request with a market UUID of `Market`.

If information about the market is returned, it has initialized. If the real-time market is not initialized, the following response will be returned:

```
{
  "type":400,
  "exception":"com.vmturbo.api.exceptions.OperationFailedException: Infrastructure is not
controlled yet. Could not get market.",
  "message":"Infrastructure is not controlled yet. Could not get market."
}
```

Markets requests

Getting a list of markets

Returns all the Markets currently stored on your Turbonomic appliance. To get the details of a single market, include the UUID of the market in the request.

Note:

One market will have the `displayName` of *Market*. This market is the real-time market. You should not add nor delete entities directly to this market using the API, nor should you attempt to execute a STOP action on this market.

Examples:

- All Markets: `GET https://10.10.10.10/api/v3/markets`
- Single Market: `GET https://10.10.10.10/api/v3/markets/_0x3OYUglEd-gHc4L513yOA`

Response: A list of `MarketApiDTO`s representing the markets on your appliance.

```
[
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/markets/_0x3OYUglEd-gHc4L513yOA"
      }
    ],
    "uuid":"_0x3OYUglEd-gHc4L513yOA",
    "displayName":"Market",
    "className":"Market",
    "state":"RUNNING",
    "unplacedEntities":false,
    "environmentType":"HYBRID"
  },
  ...
]
```

Stopping or saving plan markets

For troubleshooting, you may need to stop or save a plan market that is taking a long time to run, or does not seem to complete. These operations should never be used on the real-time market. Note that a manually-stopped plan market may have incomplete data.

This request takes the following parameters:

- market_Uuid

  The UUID of the market that will be stopped or saved.
- operation

  The operation to perform on the market. `[ stop, save ]`

Example: `POST https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw?operation=stop`

Response: The `MarketApiDTO` of the specified market, showing its state as `STOPPING`.

```
{
  "links": [
    {
      "rel": "self",
      "href": "https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw"
    },
    {
      "rel": "Market notifications",
      "href": "https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw/notifications"
    },
    {
```

```
      "rel": "Market actions",
      "href": "https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw/actions?
ascending=true"
    },
    {
      "rel": "To Entities",
      "href": "https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw/entities"
    },
    {
      "rel": "To Constraints",
      "href": "https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw/policies"
    },
    {
      "rel": "To Reservations",
      "href": "https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw/reservations"
    },
    {
      "rel": "To Statistics",
      "href": "https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw/stats"
    },
    {
      "rel": "To Unplaced Entities",
      "href": "https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw/unplacedentities"
    },
    {
      "rel": "To Market List",
      "href": "https://10.10.10.10/api/v3/markets"
    }
  ],
  "uuid": "_NO8_sCi_EemDHtuQzfdzxw",
  "displayName": "ADD_WORKLOAD_administrator_1549313730114",
  "className": "Market",
  "state": "STOPPING",
  "scenario": {
    "uuid": "2452690315520",
    "displayName": "Add Workload 1",
    "owners": [],
    "type": "ADD_WORKLOAD"
  },
  "unplacedEntities": false,
  "runDate": "2019-02-04T15:55:30-05:00",
  "environmentType": "ONPREM"
}
```


Deleting a plan market

Deleting a plan market removes the market entity along with all the topological links, such as actions, notifications, entities, policies and reservations. This operation is permanent. To delete a plan market, pass the UUID of the market in the request.

This request takes the following parameters:

market_Uuid

The UUID of the market that will be deleted.

Example: `DELETE https://10.10.10.10/api/v3/markets/_NO8_sCi_EemDHtuQzfdzxw`

Response: The `MarketApiDTO` of the specified market, showing its state as `DELETING`.


```
{
  "links":[
    {
      "rel":"self",
      "href":"https://10.10.10.10/api/v3/markets/2452692215536"
    }
  ],
  "uuid":"2452692215536",
  "displayName":"ADD_WORKLOAD_administrator_1549313730114",
  "className":"Market",
  "state":"DELETING",
  "scenario":{
    "uuid":"2452692204928",
    "displayName":"Add Workload 1",
    "owners":[
      {
```

```
        "links":[
          {
            "rel":"self",
            "href":"https://10.10.10.10/api/v3/users/_4T_7kwY-Ed-WUKbEYSVIDw"
          }
        ],
        "uuid":"_4T_7kwY-Ed-WUKbEYSVIDw",
        "displayName":"Administrator User",
        "username":"administrator",
        "roleUuid":"_4UAioQY-Ed-WUKbEYSVIDw",
        "roleName":"administrator",
        "loginProvider":"Local",
        "type":"DedicatedCustomer",
        "showSharedUserSC":false
      }
    ],
    "type":"ADD_WORKLOAD"
  },
  "saved":true,
  "unplacedEntities":false,
  "runDate":"2019-02-04T15:55:30-05:00",
  "runCompleteDate":"2019-02-04T15:57:26-05:00",
  "environmentType":"ONPREM"
}
```


Getting a filtered list of actions for a market

Referenced Endpoint: Actions

Gets a list of actions, given a market UUID and filtered by an ActionApiInputDTO using one or more of the following criteria:

- actionModeList
  - `DISABLED`:

    Do not recommend or perform the action. When you disable an action, the user interface will no longer display actions of that type.
  - `RECOMMENDED`:

    Recommend the action so a user can execute it via the given hypervisor or by other external means
  - `MANUAL`:

    Recommend the action, and provide the option to execute that action through the Turbonomic user interface or an /actions API request.
  - `AUTOMATED`:

    You have directed Turbonomic to execute the action automatically.
  - `COLLECTION` **(UNUSED)**:

    Legacy action mode, no longer used.
- actionStateList
  - `DISABLED`:

    An action whose policy has been disabled.
  - `RECOMMEND`:

    An action that cannot be automated due to policy settings or must be executed outside the system, such as the purchase of hardware.
  - `PENDING_ACCEPT`:

    When the action mode is set to manual, a recommended action that has not been accepted or cleared.
  - `ACCEPTED`:

    An action that has been accepted, but is not yet in progress.

  - IN_PROGRESS:

    An action that is being executed.
  - SUCCEEDED:

    An action that was accepted and completed successfully.
  - FAILED:

    An action that was accepted and did not complete successfully.
  - REJECTED:

    When the action mode is set to manual, an action that has been rejected by the user.
  - CLEARED:

    When the action mode is set to manual, an action that is no longer recommended by the market.
  - ACCOUNTING:

    For cloud entities, an action to resize in order to use a different instance template.
  - QUEUED:

    When more than 10 actions are to be executed for a single target, Turbonomic will place the 11th and all subsequent actions into a QUEUED state, and they will execute as the previous actions complete.
- actionTypeList
  - START:

    Start an entity, and add it to the Turbonomic market.
  - MOVE:

    Move an entity from one provider to another. For example, moving a VM between hosts, or a datastore between disk arrays.
  - SUSPEND:

    Suspend the entity to exclude its capacity from your environment. Suspend does not remove the entity from your environment, but the market does not include the entity in its analysis. Based on market conditions, analysis can recommend restarting suspended entities, which includes them in market analysis once again.
  - ADD PROVIDER:

    This is equivalent to the START action, but for storage entities.
  - CHANGE:

    Move a virtual machine's storage between datastores.
  - PROVISION:

    Add capacity to your environment.
  - RECONFIGURE:

    Analysis recommends this when a VM is not able to correct a misconfiguration via another type of action. For example, analysis can recommend a RECONFIGURE action when a placement policy requires a VM to move to a host on a particular cluster, but cross-cluster moves are not permitted.
  - DELETE:

    This is equivalent to the SUSPEND action, but for storage entities.
  - RESERVE_ON_PM:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular host.
  - RESERVE_ON_DS:

When using the reservation functionality, this is the action to reserve the resources of a VM on a particular datastore.

  - `RESIZE_FOR_EFFICIENCY`:

    A resize that reduces the capacity of a commodity, such as vCPU, or vMEM.

  - `RESIZE_FOR_PERFORMANCE`:

    A resize that increases the capacity of a commodity, such as vCPU, or vMEM.

- cleared

  Default: `false`.

  When `true`, actions that were removed from the action list before being executed will be returned.

- costType

  - `Saving`

    An action that will decrease cloud spending.

  - `Investment`

    An action that will increase cloud spending.

- cumulative

  Default: `false`.

  When `true`, shows the savings for each action in a scope as a cumulative total, rather than showing the savings for individual actions.

- environmentType

  - `ONPREM`:

    Returns only actions related to entities that are part of your on-prem environment.

  - `CLOUD`:

    Returns only actions related to entities that are part of your cloud environment.

- groupBy

  You may group the actions returned by the following criteria: `[actionModes, actionStates, actionTypes, risk, riskSeverity, riskSubCategory]`

- hasReservedInstance

  Default: `false`.

  When `true`, only actions that are related to reserved instances will be returned.

- relatedEntityTypes

  Returns only actions related to entities of these types within the scope. For example, if your scope is a virtual datacenter and you specify VirtualMachine as a related entity type, the request will include only those actions related to VMs in the specified virtual datacenter.

  Choose from the following entity types: `[ Datacenter, PhysicalMachine, VirtualMachine, Storage, Application, Chassis, DiskArray, IOModule, StorageControl, Switch, VirtualDataCenter, VPod, DPod, Container, Database, DatabaseServerContainer, LogicalPool ]`

- riskSeverityList

  Returns only actions of the given severities: `[ Unknown, Normal, Minor, Major, Critical ]`

- riskSubCategoryList
- returns only actions of the given subcategories: `[ Performance Assurance, Efficiency Improvement, Prevention, Compliance ]`

This request takes the following parameters:

- market_Uuid

  The UUID of the market whose actions will be returned.

- cursor

  When making subsequent calls to retrieve additional results, this is the cursor returned by the last call to this method.

- limit

  The maximum number of items to return. Must be a positive integer. If not set, the server will provide a default (20).

- order_by

  The field to order the results by. `[ creation_date, disruptiveness, file_size, idleness, name, reversibility, risk_category, savings, severity ]`.

- ascending

  Default: true. When false, results will be in descending order.

Example: `POST https://10.10.10.10/api/v3/markets/Market/actions?order_by=severity&ascending=true`

Example input: In this example, a list of actions that meet the following criteria will be returned and grouped by the `actionMode` of the action, in order of severity:

- Is in state `PENDING_ACCEPT`
- Is part of an on-prem environment
- Is an action for a virtual machine
- Is an action generated by the Turbonomic real-time market

```
{
  "actionInput":{
    "actionStateList":[
      "PENDING_ACCEPT"
    ],
    "environmentType":"ONPREM",
    "groupBy":[
      "actionModes"
    ]
  },
  "relatedType":"VirtualMachine",
  "scopes":[
    "Market"
  ]
}
```

Response: `ActionApiDTO`s meeting the specified criteria.

```
{
..
  "uuid": "_qFuM8CiOEemDHtuQzfdzxw",
  "createTime": "2019-02-04T10:07:58-05:00",
  "actionType": "RIGHT_SIZE",
  "actionState": "PENDING_ACCEPT",
  "actionMode": "MANUAL",
  "details": "Scale VirtualMachine 8f0e3136-b259-43e5-41f8-0d2c64c62acc from Standard_B2ms to
Standard_E2s_v3",
  "importance": 0,
  ...
}
```

For information about filtering actions within a specified time period, see *Filtering Actions by Time* in "Markets endpoint tips" on page 1415.

Getting statistics for a filtered list of actions for a market

Referenced Endpoint: Actions

Referenced Endpoint: Stats

Gets a list of statistics for a filtered list of actions. Statistics returned include the number of actions, total savings/investment, and the number of entities affected by the actions. In addition to the Market UUID, provide an ActionApiInputDTO using one or more of the following criteria:

- actionModeList
  - `DISABLED`:

    Do not recommend or perform the action. When you disable an action, the user interface will no longer display actions of that type.
  - `RECOMMENDED`:

    Recommend the action so a user can execute it via the given hypervisor or by other external means
  - `MANUAL`:

    Recommend the action, and provide the option to execute that action through the Turbonomic user interface or an /actions API request.
  - `AUTOMATED`:

    You have directed Turbonomic to execute the action automatically.
  - `COLLECTION` **(UNUSED)**:

    Legacy action mode, no longer used.
- actionStateList
  - `DISABLED`:

    An action whose policy has been disabled.
  - `RECOMMEND`:

    An action that cannot be automated due to policy settings or must be executed outside the system, such as the purchase of hardware.
  - `PENDING_ACCEPT`:

    When the action mode is set to manual, a recommended action that has not been accepted or cleared.
  - `ACCEPTED`:

    An action that has been accepted, but is not yet in progress.
  - `IN_PROGRESS`:

    An action that is being executed.
  - `SUCCEEDED`:

    An action that was accepted and completed successfully.
  - `FAILED`:

    An action that was accepted and did not complete successfully.
  - `REJECTED`:

    When the action mode is set to manual, an action that has been rejected by the user.
  - `CLEARED`:

    When the action mode is set to manual, an action that is no longer recommended by the market.
  - `ACCOUNTING`:

    For cloud entities, an action to resize in order to use a different instance template.

  - QUEUED:

    When more than 10 actions are to be executed for a single target, Turbonomic will place the 11th and all subsequent actions into a QUEUED state, and they will execute as the previous actions complete.

- actionTypeList
  - START:

    Start an entity, and add it to the Turbonomic market.
  - MOVE:

    Move an entity from one provider to another. For example, moving a VM between hosts, or a datastore between disk arrays.
  - SUSPEND:

    Suspend the entity to exclude its capacity from your environment. Suspend does not remove the entity from your environment, but the market does not include the entity in its analysis. Based on market conditions, analysis can recommend restarting suspended entities, which includes them in market analysis once again.
  - ADD PROVIDER:

    This is equivalent to the START action, but for storage entities.
  - CHANGE:

    Move a virtual machine's storage between datastores.
  - PROVISION:

    Add capacity to your environment.
  - RECONFIGURE:

    Analysis recommends this when a VM is not able to correct a misconfiguration via another type of action. For example, analysis can recommend a RECONFIGURE action when a placement policy requires a VM to move to a host on a particular cluster, but cross-cluster moves are not permitted.
  - DELETE:

    This is equivalent to the SUSPEND action, but for storage entities.
  - RESERVE_ON_PM:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular host.
  - RESERVE_ON_DS:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular datastore.
  - RESIZE_FOR_EFFICIENCY:

    A resize that reduces the capacity of a commodity, such as vCPU, or vMEM.
  - RESIZE_FOR_PERFORMANCE:

    A resize that increases the capacity of a commodity, such as vCPU, or vMEM.
- cleared

  Default: `false`.

  When `true`, actions that were removed from the action list before being executed will be returned.
- costType
  - Saving

    An action that will decrease cloud spending.
  - Investment

An action that will increase cloud spending.

- cumulative

  Default: `false`.

  When `true`, shows the savings for each action in a scope as a cumulative total, rather than showing the savings for individual actions.

- environmentType
  - `ONPREM`:

    Returns only actions related to entities that are part of your on-prem environment.
  - `CLOUD`:

    Returns only actions related to entities that are part of your cloud environment.

- groupBy

  You may group the actions returned by the following criteria: `[actionModes, actionStates, actionTypes, risk, riskSeverity, riskSubCategory]`

- hasReservedInstance

  Default: `false`.

  When `true`, only actions that are related to reserved instances will be returned.

- relatedEntityTypes

  Returns only actions related to entities of these types within the scope. For example, if your scope is a virtual datacenter and you specify VirtualMachine as a related entity type, the request will include only those actions related to VMs in the specified virtual datacenter.

  Choose from the following entity types: `[ Datacenter, PhysicalMachine, VirtualMachine, Storage, Application, Chassis, DiskArray, IOModule, StorageControl, Switch, VirtualDataCenter, VPod, DPod, Container, Database, DatabaseServerContainer, LogicalPool ]`

- riskSeverityList

  Returns only actions of the given severities: `[ Unknown, Normal, Minor, Major, Critical ]`

- riskSubCategoryList
- returns only actions of the given subcategories: `[ Performance Assurance, Efficiency Improvement, Prevention, Compliance ]`

Example: `POST https://10.10.10.10/api/v3/markets/Market/actions/stats`

Example input: In this example, a list of actions that meet the following criteria will be returned and grouped by the `actionMode` of the action, in order of severity:

- Is in state `PENDING_ACCEPT`
- Is part of an on-prem environment
- Is an action for a virtual machine
- Is an action generated by the Turbonomic real-time market

```
{
  "actionInput":{
    "actionStateList":[
      "PENDING_ACCEPT"
    ],
    "environmentType":"ONPREM",
    "groupBy":[
      "actionModes"
    ]
  },
  "relatedType":"VirtualMachine",
  "scopes":[
    "Market"
  ]
```

```
}
```

Response: A list of `StatSnapshotApiDTO` containing statistics about the filtered actions.

```
[
  {
    "date":"2019-02-05T14:40:08-05:00",
    "statistics":[
      {
        "name":"numActions",
        "values":{
          "max":268,
          "min":268,
          "avg":268,
          "total":268
        },
        "value":268
      },
      {
        "name":"costPrice",
        "filters":[
          {
            "type":"property",
            "value":"savings"
          }
        ],
        "units":"$",
        "values":{
          "max":24100,
          "min":24100,
          "avg":24100,
          "total":24100
        },
        "value":24100
      },
      ...
    ]
  }
]
```

Getting all entities within a market

Referenced Endpoint: Entities

Gets a list of all entities in the specified market. This request may be used to retrieve a list of entities used in a plan market.

Example: `GET https://10.10.10.10/api/v3/markets/2452692215536/entities`

Response: A list of `ServiceEntityApiDTOs`, representing the entities in the market. In this case, note that there is an attribute entitled `realtimeMarketRefrence` because these are entities from a plan market.

```
[
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/entities/_QQVtEyi_EemDHtuQzfdzxw?
include_aspects=false"
      }
    ],
    "uuid":"_QQVtEyi_EemDHtuQzfdzxw",
    "displayName":"osp-master-2",
    "className":"VirtualMachine",
    "priceIndex":3.6063216,
    "state":"ACTIVE",
    "severity":"Normal",
    "realtimeMarketReference":{
      "uuid":"42000c5d-9ae5-44f0-252c-daabbb423a86"
    },
    "environmentType":"ONPREM"
  },
  ...
```

```
]
```

Getting statistics of entities within a market

Referenced Endpoint: Stats

Referenced Endpoint: Entities

Gets statistics related to entities in a specified market. For example, this request may be used to see the final calculated statistics for entities in a plan market. For more information about creating a filtered list of statistics, including example POSTs and responses, see *Filtered Statistics* in the Statistics Endpoint.

Getting statistics of a specified group of entities within a market

Referenced Endpoint: Stats

Referenced Endpoint: Groups

Referenced Endpoint: Entities

Gets statistics related to a specific group of entities in a specified market. For example, this request may be used to see the final calculated statistics for a group of hosts in a plan market. For more information about creating a filtered list of statistics, including example POSTs and responses, see *Filtered Statistics* in the Statistics Endpoint.

Getting policies for a market

Referenced Endpoint: Policies

To get a list of all current policies affecting a specified market, make a GET request to the `/markets/{market_uuid}/policies` endpoint. For a plan market, this is a list of policies that you created, enabled, disabled, or changed in the plan configuration. For the real-time market, this is a list of policies currently available on your appliance.

To get a single policy, include the UUID of the policy in the request.

Examples:

- All policies: `GET https://10.10.10.10/api/v3/markets/2481646239616/policies`
- One policy:

  ```
  GET https://10.10.10.10/api/v3/markets/2481646239616/policies/
  _z3LKSTk0EemeycL7hWgJNw
  ```

Response: A list of `PolicyApiDTO`s representing the policies requested.

```
[
   {
      "links":[
         {
            "rel":"self",
            "href":"https://10.10.10.10/api/v3/policies/_z3LxVzk0EemeycL7hWgJNw"
         }
      ],
      "uuid":"_z3LxVzk0EemeycL7hWgJNw",
      "displayName":"DalTestPolMerge",
      "type":"MERGE",
      "name":"DalTestPolMerge",
      "enabled":true,
      "commodityType":"ClusterCommodity",
      "mergeType":"Cluster",
      "mergeGroups":[
         {
            "uuid":"91ff08749fd278925d73b61b0f0f69d79c99fc3a",
            "displayName":"PMs_Beta DC6\\Cluster-1",
            "className":"Group",
            "groupType":"PhysicalMachine",
            "isStatic":true,
            "logicalOperator":"AND"
```

```
         },
         {
            "uuid":"8c17d8e79c3db85ed5d3868de622cb89187849ce",
            "displayName":"PMs_Beta DC6\\Cluster-2",
            "className":"Group",
            "groupType":"PhysicalMachine",
            "isStatic":true,
            "logicalOperator":"AND"
         }
      ]
   },
   {
      "links":[
         {
            "rel":"self",
            "href":"https://10.10.10.10/api/v3/policies/_z3LKSTk0EemeycL7hWgJNw"
         }
      ],
      "uuid":"_z3LKSTk0EemeycL7hWgJNw",
      "displayName":"CommSegmentation/SegmentManager__4T_7kwY-Ed-WUKbEYSVIDw_1551123454966",
      "type":"MUST_RUN_TOGETHER",
      "name":"CommSegmentation/SegmentManager__4T_7kwY-Ed-WUKbEYSVIDw_1551123454966",
      "enabled":false,
      "capacity":1000000000,
      "commodityType":"DrsSegmentationCommodity",
      "consumerGroup":{
         "uuid":"_z2M57Dk0EemeycL7hWgJNw",
         "displayName":"Buyers-Group-GROUP-DRS-KTG-rule/DC13-Cluster/vsphere-
dc13.dev.mycorp.com",
         "className":"DiscoveredGroup",
         "groupType":"VirtualMachine",
         "isStatic":true,
         "logicalOperator":"AND"
      },
      "providerGroup":{
         "uuid":"_z2M57zk0EemeycL7hWgJNw",
         "displayName":"Sellers-Group-GROUP-DRS-KTG-rule/DC13-Cluster/vsphere-
dc13.dev.mycorp.com",
         "className":"DiscoveredGroup",
         "groupType":"PhysicalMachine",
         "isStatic":true,
         "logicalOperator":"AND"
      }
   },
   ...
]
```

Deleting a placement policy for a market

Referenced Endpoint: Policies

To delete a policy from a market, include the UUID of the policy and the market. Note that if you delete an imported policy, it reappears after the next discovery cycle of the target it was imported from. To delete imported policies, delete them on the target directly.

Example: `DELETE https://10.10.10.10/api/v3/markets/2481646239616/policies/_z3LxVzk0EemeycL7hWgJNw`

Response: A successful deletion request returns a Response Code of 200.

Getting reservations for a market

Referenced Endpoint: Reservations

Returns a list of all current reservations for a specified market. For the real-time market, this is a list of reservations currently active on your appliance.

Example: `GET https://10.10.10.10/api/v3/markets/Market/reservations`

Response: A list of DemandReservationApiDTOs representing any current reservations.

```
[
   {
      "uuid":"_I6z38Tk8EemeycL7hWgJNw",
```

```
      "displayName":"DalRes2",
      "count":1,
      "status":"RESERVED",
      "reserveDateTime":"Mon Feb 25 15:30:07 EST 2019",
      "expireDateTime":"Mon Mar 25 08:29:54 EDT 2019",
      "deployDateTime":"Tue Mar 26 08:29:54 EDT 2019",
      "reserveCount":1,
      "deployCount":0,
      "demandEntities":[
         {
            "uuid":"_JEZRwTk8EemeycL7hWgJNw",
            "displayName":"DalRes2_C0",
            "className":"VirtualMachine",
            "template":{
               "uuid":"T-ef47f3dbdde39f888052332cbe7d0cf2",
               "displayName":"AVG:PMs_ACM\\ACM Cluster for last 10 days",
               "className":"VirtualMachineProfile"
            },
            "placements":{
               "computeResources":[
                  {
                     "stats":[
                        {
                           "name":"numOfCpu",
                           "value":2
                        },
                        {
                           "name":"cpuSpeed",
                           "units":"MHz",
                           "value":5320
                        },
                        {
                           "name":"cpuConsumedFactor",
                           "units":"%",
                           "value":4.2516437
                        },
                        {
                           "name":"memorySize",
                           "units":"MB",
                           "value":6631
                        },
                        {
                           "name":"memoryConsumedFactor",
                           "units":"%",
                           "value":46.328953
                        },
                        {
                           "name":"ioThroughput",
                           "units":"MB/s",
                           "value":0.05301181
                        },
                        {
                           "name":"networkThroughput",
                           "units":"MB/s",
                           "value":0.103116125
                        }
                     ],
                     "provider":{
                        "uuid":"30333436-3638-5355-4532-313159335631",
                        "displayName":"hp-dl571.dev.mycorp.com",
                        "className":"PhysicalMachine"
                     }
                  }
               ],
               "storageResources":[
                  {
                     "stats":[
                        {
                           "name":"diskSize",
                           "units":"GB",
                           "value":92.468285
                        },
                        {
                           "name":"diskIops",
                           "value":1.6827557
                        }
                     ],
                     "provider":{
                        "uuid":"5a822e24-584bce98-ecb2-441ea15e2534",
                        "displayName":"SCIO:ESXDC11DS2",
                        "className":"Storage"
                     }
```

```
                  }
               ]
            }
         }
      ]
   }
]
```

Running a scenario on a market

Referenced Endpoint: Scenarios

When you run a scenario on the real-time market, it creates a plan market showing the results of the scenario. If you run a second scenario a plan market, that is called running a *plan over plan*. This runs a plan new configuration on a market that is already the result of a plan run.

Example: `POST https://10.10.10.10/api/v3/markets/Market/scenarios/2452690315520?ignore_constraints=false`

Response: A `MarketApiDTO` showing the new state of the market- if successful, `COPYING`.

```
{
   "links":[
      {
         "rel":"self",
         "href":"https://10.10.10.10/api/v3/markets/_ps_iUDlBEemeycL7hWgJNw"
      },
      {
         "rel":"Market notifications",
         "href":"https://10.10.10.10/api/v3/markets/_ps_iUDlBEemeycL7hWgJNw/notifications"
      },
      {
         "rel":"Market actions",
         "href":"https://10.10.10.10/api/v3/markets/_ps_iUDlBEemeycL7hWgJNw/actions?
ascending=true"
      },
      {
         "rel":"To Entities",
         "href":"https://10.10.10.10/api/v3/markets/_ps_iUDlBEemeycL7hWgJNw/entities"
      },
      {
         "rel":"To Constraints",
         "href":"https://10.10.10.10/api/v3/markets/_ps_iUDlBEemeycL7hWgJNw/policies"
      },
      {
         "rel":"To Reservations",
         "href":"https://10.10.10.10/api/v3/markets/_ps_iUDlBEemeycL7hWgJNw/reservations"
      },
      {
         "rel":"To Statistics",
         "href":"https://10.10.10.10/api/v3/markets/_ps_iUDlBEemeycL7hWgJNw/stats"
      },
      {
         "rel":"To Unplaced Entities",
         "href":"https://10.10.10.10/api/v3/markets/_ps_iUDlBEemeycL7hWgJNw/unplacedentities"
      },
      {
         "rel":"To Market List",
         "href":"https://10.10.10.10/api/v3/markets"
      }
   ],
   "uuid":"_ps_iUDlBEemeycL7hWgJNw",
   "displayName":"API_Plan_Market",
   "className":"Market",
   "state":"COPYING",
   "scenario":{
      "uuid":"2452690315520",
      "displayName":"Add Workload 1",
      "owners":[

      ],
      "type":"ADD_WORKLOAD"
   },
   "unplacedEntities":false,
   "runDate":"2019-02-25T16:09:34-05:00",
   "environmentType":"ONPREM"
```

```
}
```

Getting statistics for a market

Referenced Endpoint: Stats

Gets statistics about the specified market. This call can be used to see statistics from the real-time market, or to see the resultant statistics in a plan market after the scenario has been applied.

Example: `GET https://10.10.10.10/api/v3/markets/Market/stats`

Response: A list of `StatSnapshotApiDTO`s representing the statistics of the real-time market.

```
[
   {
      "displayName":"PMs",
      "date":"2019-02-23T00:00:00-05:00",
      "statistics":[
         {
            "name":"CPUHeadroom",
            "capacity":{
               "max":8610,
               "min":0,
               "avg":1602.22,
               "total":14420
            },
            "units":"VM",
            "values":{
               "max":8595,
               "min":0,
               "avg":1571.44,
               "total":14143
            },
            "value":1571.44
         },
         {
            "name":"CPUExhaustion",
            "units":"Day",
            "values":{
               "max":2147483650,
               "min":0,
               "avg":2147483650,
               "total":2147483650
            },
            "value":2147483650
         },
         ...
    ...
]
```

Getting a filtered list of statistics for a market

Referenced Endpoint: Stats

Gets a list of statistics about the specified market, filtered by the criteria in the StatPeriodApiInputDTO you must pass as an inputDTO to this request. For more information about creating a filtered list of statistics, including example POSTs and responses, see *Filtered Statistics* in the Statistics Endpoint.

Example:

Example Input:

Response: An array of `StatSnapshotApiDTO`s representing the filtered statistics for the market.

```
```

Getting the supply chain for a market

Referenced Endpoint: Supply Chains

Using one or more of the following parameters, you can filter the supply chain that is returned by this request:

- market_Uuid

  The UUID of the market whose supply chain will be returned

- types

  The entity types whose supply chain will be returned. Choose from:

  - `APPLICATION`
  - `APPLICATION_SERVER`
  - `BUSINESS_ACCOUNT`
  - `BUSINESS_UNIT`
  - `CHASSIS`
  - `CLOUD_SERVICE`
  - `CONTAINER`
  - `CONTAINER_POD`
  - `DATABASE`
  - `DATABASE_SERVER`
  - `DATACENTER`
  - `DISK_ARRAY`
  - `IO_MODULE`
  - `LOAD_BALANCER`
  - `LOGICAL_POOL`
  - `NETWORK`
  - `PHYSICAL_MACHINE`
  - `STORAGE`
  - `STORAGE_CONTROLLER`
  - `SWITCH`
  - `VIRTUAL_APPLICATION`
  - `VIRTUAL_DATACENTER`
  - `VIRTUAL_MACHINE`

- entity_states

  Only return entities in the specified states. Choose from:

  - `ACTIVE`:

    Entity is actively consuming resources.

  - `EVACUATED`:

    Entity is ready to suspend and will not participate in the market, but will not generate a `SUSPEND` action.

    Note: This state replaces the `SUSPEND` state, and requires additional configuration to enable. Turbonomic does not recommend enabling this state.

  - `FAILOVER`:

    This entity is reserved for failover and will not participate in the market.

  - `IDLE`:

    Entity is powered off and not actively consuming market resources.
  - `LAUNCH`:

    Entity is starting.
  - `MAINTENANCE`:

    Entity is in maintenance mode.
  - `NOT_MONITORED`:

    Entity is not currently monitored by Turbonomic.
  - `QUEUED`:

    Entity is in the process of changing from one state to another.

    Note: This state is internal to Turbonomic and will not appear in the UI, but may appear in an API response if the call is executed at the precise moment the entity is changing state.
  - `RESOURCE_ALLOCATION`:

    Cloud entity is being created.
  - `RESOURCE_RELEASE`:

    Cloud Entity is being deleted.
  - `SUSPEND`:

    Entity is suspended or the guest OS is in sleep, standby, or suspended state. The entity is not removed from the environment, but will not be used in market decisions. Based on market conditions, suspended entities can be started, re-adding them to the market.
  - `SUSPEND_PENDING`:

    Entity is making preparations to suspend, or is the next entity of its type to suspend.
  - `TERMINATE_PENDING`:

    Entity is in the process of being deleted.
  - `UNKNOWN`:

    Turbonomic is unable to obtain the current state of the entity.

    This occurs when the entity is reported to the hypervisor as having a state of `DISCONNECTED`, `ORPHANED`, or `INVALID`.
- environment_type
- Only return actions that affect entities in the specified environment type. `[ CLOUD, HYBRID, ONPREM ]`.
- detail_type

  The level of detail returned for each item in the returned supply chain. `[ compact, entity, aspects ]`.
- health

  Return a small summary of the health of the returned supply chain.

Example: `GET https://10.10.10.10/api/v3/markets/Market/supplychains?types=VirtualMachine&entity_states=RESOURCE_ALLOCATION&entity_states=ACTIVE&environment_type=ONPREM&detail_type=aspects&health=true`

Response: A SupplychainApiDTO representing the entities meeting your criteria. In the previous example, a supply chain including Virtual Machines in the ACTIVE or RESOURCE_ALLOCATION states, in an ONPREM environment, and a health check will be included at the end of the request.


```
{
   "seMap":{
      "VirtualMachine":{
         "depth":-1,
         "entitiesCount":335,
         "healthSummary":{
            "Major":6,
            "Minor":148,
            "Normal":181
         },
         "stateSummary":{
            "ACTIVE":335
         },
         "connectedProviderTypes":[
            "PhysicalMachine",
            "Storage",
            "VirtualDataCenter"
         ],
         "connectedConsumerTypes":[
            "Application"
         ],
         "instances":{
            "421edc7a-761d-9a79-9378-7140caecd50f":{
               "links":[
                  {
                     "rel":"self",
                     "href":"https://10.10.10.10/api/v3/entities/
421edc7a-761d-9a79-9378-7140caecd50f?include_aspects=false"
                  }
               ],
               "uuid":"421edc7a-761d-9a79-9378-7140caecd50f",
               "displayName":"vm-max-test-netapp",
               "className":"VirtualMachine",
               "priceIndex":1.0070138,
               "state":"ACTIVE",
               "severity":"Normal",
               "discoveredBy":{
                  "uuid":"_8c98wBwnEem2J-ksdl1CgA",
                  "displayName":"vsphere-dc7.dev.mycorp.com",
                  "type":"vCenter"
               },
               "remoteId":"vm-281",
               "aspects":{
                  "virtualMachineAspect":{
                     "os":"SUSE Linux Enterprise 11 (64-bit)",
                     "connectedNetworks":[
                        {
                           "uuid":"7d6d67a50cfe1979797de159220841a37a55a50c",
                           "displayName":"VM network"
                        }
                     ],
                     "numVCPUs":1,
                     "ebsOptimized":false
                  }
               },
               "environmentType":"ONPREM"
            }            ...
         ]
        ...
     ...
   ...
}
```


Markets endpoint tips

Unplaced entities

When running a plan in Turbonomic, you can add workload or remove infrastructure – if the remaining resources are insufficient for the workload, entities may be "unplaced". Turbonomic provides an array of the unplaced workloads.

Note: An example will be forthcoming.

Finding a previously run plan market

While the Turbonomic UI makes a plan look like one item, it is not. A plan comprises two parts:

- A scenario, which represents the configuration options for the plan For more information about plan configuration, see Scenarios.
- The plan market that is created when a scenario is run – this contains the actions, notifications, etc. that make up the state of the market after the plan.

The simplest way to discover a plan market is to execute the `GET https://10.10.10.10/api/v3/markets` and disregard the main market. The main market DTO entry will be similar to the following:

```
{
  "uuid":"777777",
  "displayName":"Market",
  "className":"Market",
  "state":"RUNNING",
  "stateProgress":100,
  "unplacedEntities":false,
  "environmentType":"HYBRID"
}
```

The other markets that the request returns are plan markets. Plan market DTOs are similar to:

```
{
  "uuid":"214172431889696",
  "displayName":"Add Workload 2",
  "className":"Market",
  "state":"SUCCEEDED",
  "stateProgress":100,
  "scenario":{
    "uuid":"214172431885856",
    "displayName":"Add Workload 2",
    "owners":[
      {
        "uuid":"2993567556912",
        "showSharedUserSC":false
      }
    ]
    ...
}
```

The outer-most `uuid` and `displayName` parameters are the UUID and display name of plan market. The `state` field shows the results of the plan run, where `SUCCEEDED` means the plan finished successfully, and `FAILED` means the plan did not finish running- usually because it was canceled by the user.

Within the plan itself, the `scenario` parameter includes information about the scenario used on the market. Note the inner `uuid` and `displayName` are the UUID and display name of the scenario, which you can use to see all markets created using this scenario.

Finally, iterating over the return of the previous `/markets` call, and looking at each market and comparing the UUID of the scenario (in this case, 214172431889696), you can see how many times the scenario was run, and also each market containing the results of that run.

Filtering actions by time

When you make a GET request to the `/actions` endpoint, you can filter the list of returned actions based on a specified start time, end time, or both.

- The `startTime` parameter excludes any actions that were executed (succeeded or failed) before a specified time.
- The `endTime` parameter excludes any actions with a `createTime` value that is after a specified time.

When you define both these parameters, the response includes all current and historical actions that were pending or executed during the specified time period.

For example, the following GET request to the `markets/Market/actions` endpoint returns all actions that were created, pending, or executed between 6:00 and 11:00 on May 8, 2024.

Example: `GET https://10.10.10.10/api/v3/markets/Market/actions`

Example input:

```
{ "startTime": "2024-05-08T06:00:19Z",
  "endTime": "2024-05-08T011:00:19Z",
  ...
}
```

Although the response does not include any actions that were created after the `endTime`, it might include actions that were created before the `startTime`, if they are still pending as of the `startTime`. Therefore, this filtering includes all actions that are either pending or executed during the specified time period, regardless of when those actions were created.

Notifications endpoint

Notifications are system and market messages from your Turbonomic instance.

This endpoint contains references to start and end times for certain operations. For more information, including valid time formats, see Time in the Turbonomic API

Using the notifications endpoint, you can:

- Get all notifications
- Get a filtered list of notifications
- Get related actions for a specified notification

Notification categories

Each notification from your Turbonomic instance belongs to a particular category of the following:

- Discovery
- Monitoring
- Control
- Mediation
- Healthcheck
- InterAppliance
- MarketAction
- MarketProblem
- Presentation
- Deploy
- Security

Notifications requests

Getting notifications

Gets a list of system and market level notifications. For a list of notifications, the API returns an array of LogEntryApiDTOs. Each LogEntryApiDTO will contain details about the notification, including the time, state, severity, and importance of the notification. To get a single notification, include the UUID of the notification in the request.

This request takes the following parameters:

- start_time

  The start time for a historic range of notifications.

- end_time

  The end time for a historic range of notifications.

- category

  The category of notifications to return. `[Discovery, Monitoring, Control, Mediation, Healthcheck, InterAppliance, MarketAction, MarketProblem, Presentation, Deploy, Security]`

- cursor

  When making subsequent calls to retrieve additional results, this is the cursor returned by the last call to this method.

- limit

  The maximum number of items to return. Must be a positive integer. If not set, the server will provide a default (20).

- order_by

  The field to order the results by. This value can be any field found in the response.

- ascending

  Default: true. When false, results will be in descending order.

Examples:

- **All Notifications (no parameters):** `GET https://10.10.10.10/api/v3/notifications`
- **All Notifications (parameters):** `GET https://10.10.10.10/api/v3/notifications?start_time=1237056032&end_time=1238056032&limit=100&order_by=name&ascending=true`
- **Single Notification:** `GET https://10.10.10.10/api/v3/notifications/_5wcf4J5WEeinZ-6Uu4soUw`

Response:


```
[
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/notifications/_5wcf4J5WEeinZ-6Uu4soUw"
      }
    ],
    "uuid":"_5wcf4J5WEeinZ-6Uu4soUw",
    "logActionTime":1534095671518,
    "targetSE":"Hyper-V:hp-cluster.corp.mycorp.com",
    "category":"Discovery",
    "subCategory":"Validation",
    "description":"Validation of target hp-cluster.corp.mycorp.com failed",
    "shortDescription":"Validation Failed",
    "state":"NOTIFY",
    "severity":"MAJOR",
    "importance":0
  },
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/notifications/_sQ7xUJ5XEeinZ-6Uu4soUw"
      }
    ],
    "uuid":"_sQ7xUJ5XEeinZ-6Uu4soUw",
    "logActionTime":1534096010470,
    "targetSE":"Hyper-V:hp-cluster.corp.mycorp.com",
    "category":"Discovery",
    "subCategory":"Discovery",
    "description":"Could not get domain for the 'hp-cluster.corp.mycorp.com' target",
    "shortDescription":"Discovery failed",
```

```
    "state":"NOTIFY",
    "severity":"MAJOR",
    "importance":0
  },
  ...
]
```

Parking endpoint

Turbonomic supports 'parking' actions for cloud resources. These user-initiated actions *stop* your cloud resources for a given period of time to help you reduce your cloud expenses, and then *start* these resources later when you need them. You can enforce parking actions on demand, or automatically through parking schedules and policies.

Using the parking endpoint of the Turbonomic API, you can:

- Get a list of parkable entities
- Enforce parking actions on demand or automatically using schedules

  To enforce parking actions according to a schedule, use `/schedules/timespans/` requests in the `Schedules` endpoint. With these requests, you can configure parking schedules and then attach the schedules to parkable entities. For details, see Parking schedules requests.

  When parking schedules have been attached to entities, you can use the parking endpoint to pause and resume schedules as needed.
- Create parking policies

  Note:

  The parking endpoint does not handle automation policies or parking policies. For more information about managing policies with the API, see Working with policies in the API.

Parking requests

This topic describes the requests that you can use to view parkable workloads, execute parking actions, and pause/resume parking schedules. For more information about managing parking policies, see Parking policies requests.

Pagination for parking requests

When you request parking entities, the API uses pagination to return subsets of the data. You should always use pagination when available to make sure you get your data in manageable chunks. To paginate returned data, the `limit` and `cursor` parameters control how the APIs paginate their data.

Example: `POST https://10.10.10.10/api/v3/parking/entities?cursor=10&limit=5&order_by=DISPLAY_NAME`

This example returns the 11th through 16th entities ordered by display name.

For general information about pagination, see Pagination in the Turbonomic API.

Getting parkable entities

Provides a list of entities that can be stopped or started.

Example: `POST https://10.10.10.10/api/v3/parking/entities`

Results may be sorted by one of the following: `DISPLAY_NAME, ENTITY_TYPE, STATE, ACCOUNT_NAME, PROVIDER, REGION_NAME, INSTANCE_TYPE, COST, SCHEDULE_NAME, POLICY_NAME, GROUP_NAME, CLOUD_SERVICE_NAME, RESTRICTIONS_UNPARKABLE`. The default sort order is `ascending` but it may be reversed by including `ascending=false` in the query string.

Example: `POST https://10.10.10.10/api/v3/parking/entities?order_by=DISPLAY_NAME&ascending=true`

Getting a filtered list of parkable entities

The simplest form of filtering is to provide one of the following main-level keys:

Field	Possible Values	Example
providers	AWS, AZURE, GCP	{"providers": ["AWS", "GCP"]}
status	STOPPED, RUNNING, STARTING, STOPPING	{"status": ["RUNNING", "STARTING"]}
entityTypes	VirtualMachine, DatabaseServer **Note:** DatabaseServer only applies to AWS RDS instances.	{"entityTypes": ["VirtualMachine"]}

Example: POST https://10.10.10.10/api/v3/parking/entities

Example input that fetches all running VMs in all of your AWS targets:

```
{"entityTypes": ["VirtualMachine"], "status":["RUNNING"], "providers": ["AWS"]}
```

Example: POST https://10.10.10.10/api/v3/parking/entities

Example input, showing the same query with the addition of the field logicalOperator. The logicalOperator value can either be AND or OR, and will control whether the supplied filters are all required or are optional. For example, this request now fetches all entities that are a virtual machine, or are running, or are provided by an AWS target:

```
{"entityTypes": ["VirtualMachine"], "status": "RUNNING"], "providers": ["AWS"],
"logicalOperator": "OR"}
```

Advanced Filtering: More powerful filtering is available by using criteriaList. This object supports additional fields and provides more control over your filter expressions, including regular expression filtering. Each element in a criteriaList contains the following keys: expVal, expType, and filterType.

filterType	expType	Constraints
displayName	EQ, NEQ, RXEQ, RXNEQ	
accountName	EQ, NEQ, RXEQ, RXNEQ	
regionName	EQ, NEQ, RXEQ, RXNEQ	
instanceType	EQ, NEQ, RXEQ, RXNEQ	
scheduleName	EQ, NEQ, RXEQ, RXNEQ	
tags	EQ, NEQ, RXEQ, RXNEQ	Discovered tags containing pipes | may produce unexpected results when using an expType of RXEQ or RXNEQ
cost	EQ, NEQ, GT, LT, GTE, LTE	

Example: POST https://10.10.10.10/api/v3/parking/entities

Example input that fetches all entities within the "Engineering" account:

```
{
  "criteriaList": [
    {
      "expVal": "Engineering",
      "expType": "EQ",
      "filterType": "accountName"
    }
  ]
}
```

Advanced filtering may also take advantage of `logicalOperator`, if desired.

Example: POST `https://10.10.10.10/api/v3/parking/entities`

Example input that returns all entities in either the "Engineering" or "Test" accounts:

```
{
  "criteriaList": [
    {
      "expVal": "Engineering",
      "expType": "EQ",
      "filterType": "accountName"
    },
    {
      "expVal": "Test",
      "expType": "EQ",
      "filterType": "accountName"
    }
  ]
  "logicalOperator": "OR"
}
```

Executing parking actions

You may start or stop parkable entities on demand by submitting a manual action request. Manual actions are executed immediately.

Note:

Manual actions will not be accepted if the entities are controlled by a schedule. For these entities, you must first pause schedule enforcement.

Example: POST `https://10.10.10.10/api/v3/parking/bulkActionRequest`

Example input that generates actions to start the provided entities:

```
{"action":"START","entityUuids":["74913973295604"]}
```

Constraints: `action` can either be START or STOP.

Response: This API responds with one element for each `entityUuid` provided and indicates the resulting `state`.

```
[{"state":"STARTING","entityUUID":"74913973295604"}]
```

If an action could not be generated for a particular `entityUuid`, the response will contain an error message:

```
[{"error":"unable to toggle the entity","entityUUID":"74913973295605"}]
```

If an action is requested for multiple entities, a response may contain a partial success. Check the response body to understand whether the action succeeded for a particular entity.

```
[{"state":"STARTING","entityUUID":"74913973295604"}, {"error":"unable to toggle the
entity","entityUUID":"74913973295605"}]
```

Pausing and resuming parking schedules

Parking schedules must first be configured and attached to entities using `/schedules/timespans` requests. For details, see Parking Schedules Requests.

After attaching schedules to entities, you can pause or resume schedules using the Parking endpoint.

Pausing Parking Schedules: There are times when pausing schedule enforcement is useful. The most common use is when you need to pause an environment during non-typical work hours or when an unanticipated maintenance window requires automation outside Turbonomic to start and stop workloads. Like attaching and detaching schedules, you may pause or resume 1-N entities from schedule enforcement.

Example: `POST https://10.10.10.10/api/v3/parking/enforcementPauses`

Example input that pauses schedule enforcement for 1-N entities using a duration in minutes (only supported unit):

```
{"entityUuids":["74913973295604"],"durationMinutes":15}
```

Example input that pauses schedule enforcement for 1-N entities until 12:26 PM Central (UTC-5) in ISO 8601 date and time format:

```
{"entityUuids":["74913973295604"],"until":"2023-05-16T14:26:00-05:00"}
```

Resuming Parking Schedules Example: `DELETE https://10.10.10.10/api/v3/parking/enforcementPauses`

Example input that resumes schedule enforcement for 1-N entities:

```
{"entityUuids":["74913973295604"]}
```

Parking policies requests

This topic describes the requests that you can use to manage parking policies. For more information about viewing parkable workloads, executing parking actions, and pausing/resuming parking schedules, see Parking requests.

A parking policy is a set of rules that Turbonomic must satisfy when enforcing parking actions (stop and start) on public cloud workloads.

The parking policies endpoint (`/parking/policies`) supports the following API requests:

- Create a parking policy (`POST /parking/policies`).
- Get the details of a parking policy (`GET /parking/policies/{policy_Uuid}`).
- Update a parking policy (`PUT /parking/policies/{policy_Uuid}`).
- Delete a parking policy (`DELETE /parking/policies/{policy_Uuid}`).
- Get parking policies (`GET /parking/policies`).

To use the parking policies endpoint, you must have administrator or site administrator access to Turbonomic.

Note:

For more information about managing parking policies through the user interface, see Parking policies.

Creating a parking policy

Example: `POST https://10.10.10.10/api/v3/parking/policies`

Configure parameters in the request, using the following example as reference.

```
{
  "displayName": "My Policy",
```

```
  "impacts": {
    "attachSchedule": {
      "timespanScheduleUuid": "111111111111111"
    },
    "restrictUnparkable": false
  },
  "level": "ACCOUNT",
  "scopes": [
      {
        "uuid": "222222222222222"
      },
      {
        "uuid": "333333333333333"
      }
    ],
   "criteriaList": [
      {
        "expVal": "a1.2xlarge|a2-ultragpu-8g",
        "expType": "EQ",
        "filterType": "instanceType",
        "caseSensitive": false,
        "entityType": null,
        "singleLine": false
      }
    ],
  "priority": 1
}
```

- Display name

  ```
    "displayName": "{policy_name}",
  ```

  Specify a meaningful name for the policy.

- (Optional) Schedule

  ```
    "impacts": {
      "attachSchedule": {
        "timespanScheduleUuid": "{parking_schedule_UUID}"
      },
  ```

  If you set `restrictUnparkable: false` (see the next item), attach an existing parking schedule to the policy by specifying the UUID of the schedule. For more information about creating a schedule, see Parking schedules requests.

- Enforcement of parking actions

  ```
      "restrictUnparkable": false
  ```

  The default value of `false` means that parking actions are enforced on parkable workloads. Removing this parameter has the same effect.

  Set to `true` to disable the enforcement of parking actions. For example, you may want to choose this option for virtual machines in your production environment that must be running all the time.

- Level

  ```
      "level": "{Level}",
      "scopes": [
        {
          "uuid": "{UUID}"
        }
      ],
  ```

  Level represents the first of two tiers of prioritization for policy matching. It mitigates the complexity of prioritizing a large set of policies in a single list.

  Choose from the following levels:

  – Global

    ```
        "level": "GLOBAL",
    ```

Set a policy at the global level if you have multiple cloud providers and all parkable workloads discovered from these providers must apply the same policy. This level has the lowest priority.

No additional parameter is required for this level.

– Cloud provider

```
"level": "CLOUD_PROVIDER",
"scopes": [
  {
    "uuid": "{cloud_provider_UUID}"
  }
],
```

Set a policy at the cloud provider level if all parkable workloads discovered from a specific provider must apply the same policy. This level has a higher priority than Global, but a lower priority than Account or Resource group.

For the `uuid` parameter, choose from the following options:

- AWS
- AZURE
- GCP

Note:

GCP is Google Cloud.

– Account

```
"level": "ACCOUNT",
"scopes": [
  {
    "uuid": "{Account1_UUID}"
  },
  {
    "uuid": "{Account2_UUID}"
  }
],
```

Set a policy at the account level if all parkable workloads discovered from specific accounts must apply the same policy. For AWS and Google Cloud, this level has the highest priority. For Azure, this level has a lower priority than Resource group.

The following cloud provider accounts are supported:

- AWS accounts
- Azure subscriptions
- Google Cloud projects

Specify the UUID of the account. You can add multiple accounts, as shown in the preceding example.

– Resource Group

```
"level": "RESOURCE_GROUP",
"scopes": [
  {
    "uuid": "{ResourceGroup1_UUID}"
  },
  {
    "uuid": "{ResourceGroup2_UUID}"
  }
],
```

Set a policy at the Resource Group level if all parkable workloads discovered from specific Resource Groups must apply the same policy. This level has the highest priority for Azure. If you do not have Azure targets, this level does not apply and you can start at the Account level.

Specify the UUID of the Resource Group. You can add multiple Resource Groups, as shown in the preceding example.

- Filters

```
"criteriaList": [
  {
    "expVal": "string",
    "expType": "EQ",
    "filterType": "string",
    "caseSensitive": true,
    "entityType": "string",
    "singleLine": true
  }
],
```

Each policy can have filters applied on top of the chosen level for more fine-grained control. Filters do not the change the priority of a policy.

Each element in a `criteriaList` contains the `expVal`, `expType`, and `filterType` keys.

filterType	**expType**	**Constraints**
`entityType`	`EQ, NEQ`	None
`instanceType`	`EQ, NEQ, RXEQ, RXNEQ`	None
`name`	`EQ, NEQ, RXEQ, RXNEQ`	None
`regionName`	`EQ, NEQ, RXEQ, RXNEQ`	None
`serviceName`	`EQ, NEQ`	None
`tags`	`EQ, NEQ, RXEQ, RXNEQ`	Discovered tags containing pipes `|` may produce unexpected results when using an `expType` of RXEQ or RXNEQ.

- Priority

```
"priority": {number}
```

Specify a priority for the given level. The lower the value, the higher the priority. For example, if you set `level: ACCOUNT` and `priority: 1`, the policy has the highest priority among parking policies at the account level.

Response: The API responds with information about the new parking policy.

Getting the details of a parking policy

Example: `GET https://10.10.10.10/api/v3//parking/policies/{policy_Uuid}`

Response: The API responds with details for the parking policy, including its display name, enforcement status, schedule, level, filters, and priority.

Updating a parking policy

Example: `PUT https://10.10.10.10/api/v3/parking/policies/{policy_Uuid}`

Response: The API responds with details about the updated parking policy.

Deleting a parking policy

Example: `DELETE https://10.10.10.10/api/v3/parking/policies/{policy_Uuid}`

Response: The API responds with the HTTP status code.

Getting parking policies

Example: `GET https://10.10.10.10/api/v3/parking/policies`

Response: The API responds with details for the parking policies, including their display name, enforcement status, schedule, level, filters, and priority.

- Results may be filtered by one of the following levels: `GLOBAL, CLOUD_PROVIDER, ACCOUNT, RESOURCE_GROUP`.

  Example: `GET https://10.10.10.10/api/v3/parking/policies?level=ACCOUNT`
- Results may be sorted by one of the following parameters: `DISPLAY_NAME, PRIORITY, CREATED_DATE, SCHEDULE_NAME`. The default sort order is `ascending` but it may be reversed by including `ascending=false` in the query string.

  Example: `GET https://10.10.10.10/api/v3/parking/policies?order_by=DISPLAY_NAME&ascending=true`

Pagination for parking policies requests

When you request parking policies, the API uses pagination to return subsets of the data. Always use pagination when available to make sure you get your data in manageable chunks. The `limit` and `cursor` parameters control how the APIs paginate data.

Example: `GET https://10.10.10.10/api/v3/parking/policies?level=ACCOUNT&cursor=5&limit=4&order_by=DISPLAY_NAME&ascending=true`

This example returns the 6th through 10th policies ordered by display name.

For general information about pagination, see Pagination in the API.

Policies endpoint

Use the policies endpoint to manage placement policies.

Note:

The policies endpoint does not handle automation policies or parking policies. For more information about managing policies with the API, see Working with policies in the API.

A placement policy is a set of rules that Turbonomic must satisfy when generating placement recommendations for on-prem and containerized workloads.

Using the policies endpoint, you can:

- Get API entry points for placement policies
- See details about a specific placement policy

The hypervisors that you set as targets can include placement policies of their own. Turbonomic imports these placement policies, and you can choose to enable or disable them as you wish. By default, Turbonomic enables imported placement policies.

Policies utility requests

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

The policies endpoint contains a utility request that shows the entry points for policy-related requests. These can be used to find the UUID and details relating to a specific policy.

Example: `GET https://10.10.10.10/api/v3/policies`

Response: An array of Link objects, where each object contains a URL to get all policies for a specified market, entity, or group.

```
{
  "links":[
    {
      "rel":"Market policies",
      "href":"https://10.10.10.10/api/v3/markets/{uuid}/policies"
    },
    {
      "rel":"Entity policies",
      "href":"https://10.10.10.10/api/v3/entities/{uuid}/policies"
    },
    {
      "rel":"Group policies",
      "href":"https://10.10.10.10/api/v3/groups/{uuid}/policies"
    }
  ],
  "enabled":false
}
```

Placement policies requests

A placement policy is a set of rules that Turbonomic must satisfy when generating placement recommendations for on-prem and containerized workloads.

With these policies, Turbonomic can recommend placement actions that comply with your business rules. For example, a placement policy can constrain VMs to specific hosts. If a VM needs to move to a different host due to resource congestion, Turbonomic will generate an action to move the VM to one of the hosts defined in the policy.

Turbonomic discovers placement policies from your targets, and allows you to create your own policies.

A placement policy exists in the context of a market, either the real-time market or a plan market. Each market has its own set of placement policies. Even if a plan market is an exact copy of the real-time market, the plan market has its own placement policies, each with its own UUID.

Before you work with placement policies via the API, understand how they work in the user interface. You must understand the types of placement policies you can create, and the effect of each on market analysis. For more information, see “Placement policies” on page 1196.

The API supports the following types of placement policies:

- `AT_MOST_N`: Only the given number of consumers can run on a single member of the providers group. This is set in the `capacity` property of the policy object.
- `BIND_TO_GROUP`: The consumers can only run on members of the provider group.
- `BIND_TO_COMPLEMENTARY_GROUP`: The consumers cannot run on any members of the provider group.
- `MUST_RUN_TOGETHER`: These consumers must run on the same provider entity.
- `AT_MOST_N_BOUND`: Only the given number of consumers can run on a single member of the providers group, AND The consumers can only run on members of the provider group.
- `MERGE`: Remove cluster boundaries for the specified clusters.
- `BIND_TO_GROUP_AND_LICENSE`: Create a license group.

You can use the following requests to manage your placement policies with the Turbonomic API.

Getting a list of URLs for placement policy-related requests

The policies endpoint contains a utility request that shows the entry points for policy-related requests. These can be used to find the UUID and details relating to a specific policy.

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

Example: `GET https://10.10.10.10/api/v3/policies`

Response: An array of Link objects, where each object contains a URL to get all policies for a specified market, entity, or group.


```
{
  "links":[
    {
      "rel":"Market policies",
      "href":"https://10.10.10.10/api/v3/markets/{uuid}/policies"
    },
    {
      "rel":"Entity policies",
      "href":"https://10.10.10.10/api/v3/entities/{uuid}/policies"
    },
    {
      "rel":"Group policies",
      "href":"https://10.10.10.10/api/v3/groups/{uuid}/policies"
    }
  ],
  "enabled":false
}
```


Getting the details of a specific placement policy

The single policy request gets the details of a specified policy, given that policy's UUID. After you get the UUIDs for policies that affect a certain scope, you can use this request to view details of those policies.

Example: `GET https://10.10.10.10/api/v3/policies/104:domain-c7:10.10.111.111`

Response: A `PolicyApiDTO` describing details about the policy, including the group information for the provider of the policy.


```
{
  "links":[
    {
      "rel":"self",
      "href":"https://10.10.10.10/api/v3/policies/104:domain-c7:10.10.111.111"
    }
  ],
  "uuid":"104:domain-c7:10.10.111.111",
  "displayName":"GROUP-DRS-LN-rule/Cluster1/10.10.111.111",
  "type":"BIND_TO_GROUP",
  "name":"GROUP-DRS-LN-rule/Cluster1/10.10.111.111",
  "enabled":true,
  "capacity":1000000000,
  "commodityType":"DrsSegmentationCommodity",
  "consumerGroup":{
    "uuid":"7843e134bab4fb841b9cb7dd98ef04acf1f36f7b",
    "displayName":"Buyers-Group-GROUP-DRS-vmGroup1/Cluster1/10.10.111.111",
    "className":"DiscoveredGroup",
    "groupType":"VirtualMachine",
    "isStatic":true,
    "logicalOperator":"AND"
  },
  "providerGroup":{
    "uuid":"597a35089bfe405291911f3191cf24234d4713d1",
    "displayName":"Sellers-Group-GROUP-DRS-hostGrp1/Cluster1/10.10.111.111",
    "className":"DiscoveredGroup",
    "groupType":"PhysicalMachine",
    "isStatic":true,
    "logicalOperator":"AND"
  }
}
```


Creating a placement policy

Policies are created for a specific market- either a plan market, or the real-time market. For more information about policy creation, see the Markets endpoint.

When you get a policy, the returned object describes the policy type, as well as the consumer and provider groups. When you create a policy, you do not have to provide the full data. You provide:

- `buyerUuid`: The group of consumers for this policy.
- `sellerUuid`: The provider group for this policy.
- `type`: The policy type.
- `policyName`: A display name for this policy.
- `enabled`: Whether to enable the policy in the market. Can be `true` or `false`.
- `capacity`: For an `AT_MOST_N` or `AT_MOST_N_BOUND` policy, the number of consumers to allow on a provider entity.
- `mergeType`: For a `MERGE` policy, the type of clusters to merge. Can be one of `Cluster`, `StorageCluster`, DesktopPools, or `DataCenter`.
- `mergeUuids`: The groups that you want to merge. The group type must match the ``mergeType`.

Because placement policies exist in the context of a market, make a POST request to the `/markets` endpoint and specify the UUID of the market where you want to create a policy.

Example: `POST https://10.10.10.10/api/v3/markets/{market_UUID}/policies`

For example, to create a DON'T PLACE policy, include the following DTO in the request body:

```
{
  "buyerUuid": "f82dbbc2b3366052f3bc1ac8a68c9c06b0eb182a",
  "enabled": false,
  "policyName": "PolicyFromApi",
  "sellerUuid": "4a2f5f132ae690af147ccfd6ea9839e79da3db79",
  "type": "BIND_TO_COMPLEMENTARY_GROUP"
}
```

Getting policies for a market

Referenced Endpoint: Policies

To get a list of all current policies affecting a specified market, make a GET request to the `/markets/{market_uuid}/policies` endpoint. For a plan market, this is a list of policies that you created, enabled, disabled, or changed in the plan configuration. For the real-time market, this is a list of policies currently available on your appliance.

To get a single policy, include the UUID of the policy in the request.

Examples:

- All policies: `GET https://10.10.10.10/api/v3/markets/2481646239616/policies`
- One policy:

  `GET https://10.10.10.10/api/v3/markets/2481646239616/policies/_z3LKSTk0EemeycL7hWgJNw`

Response: A list of `PolicyApiDTO`s representing the policies requested.

```
[
   {
      "links":[
         {
            "rel":"self",
            "href":"https://10.10.10.10/api/v3/policies/_z3LxVzk0EemeycL7hWgJNw"
         }
      ],
      "uuid":"_z3LxVzk0EemeycL7hWgJNw",
      "displayName":"DalTestPolMerge",
      "type":"MERGE",
      "name":"DalTestPolMerge",
      "enabled":true,
      "commodityType":"ClusterCommodity",
      "mergeType":"Cluster",
      "mergeGroups":[
         {
            "uuid":"91ff08749fd278925d73b61b0f0f69d79c99fc3a",
            "displayName":"PMs_Beta DC6\\Cluster-1",
            "className":"Group",
```

```
            "groupType":"PhysicalMachine",
            "isStatic":true,
            "logicalOperator":"AND"
         },
         {
            "uuid":"8c17d8e79c3db85ed5d3868de622cb89187849ce",
            "displayName":"PMs_Beta DC6\\Cluster-2",
            "className":"Group",
            "groupType":"PhysicalMachine",
            "isStatic":true,
            "logicalOperator":"AND"
         }
      ]
   },
   {
      "links":[
         {
            "rel":"self",
            "href":"https://10.10.10.10/api/v3/policies/_z3LKSTk0EemeycL7hWgJNw"
         }
      ],
      "uuid":"_z3LKSTk0EemeycL7hWgJNw",
      "displayName":"CommSegmentation/SegmentManager__4T_7kwY-Ed-WUKbEYSVIDw_1551123454966",
      "type":"MUST_RUN_TOGETHER",
      "name":"CommSegmentation/SegmentManager__4T_7kwY-Ed-WUKbEYSVIDw_1551123454966",
      "enabled":false,
      "capacity":1000000000,
      "commodityType":"DrsSegmentationCommodity",
      "consumerGroup":{
         "uuid":"_z2M57Dk0EemeycL7hWgJNw",
         "displayName":"Buyers-Group-GROUP-DRS-KTG-rule/DC13-Cluster/vsphere-
dc13.dev.mycorp.com",
         "className":"DiscoveredGroup",
         "groupType":"VirtualMachine",
         "isStatic":true,
         "logicalOperator":"AND"
      },
      "providerGroup":{
         "uuid":"_z2M57zk0EemeycL7hWgJNw",
         "displayName":"Sellers-Group-GROUP-DRS-KTG-rule/DC13-Cluster/vsphere-
dc13.dev.mycorp.com",
         "className":"DiscoveredGroup",
         "groupType":"PhysicalMachine",
         "isStatic":true,
         "logicalOperator":"AND"
      }
   },
   ...
]
```


Deleting a placement policy for a market

Referenced Endpoint: Policies

To delete a policy from a market, include the UUID of the policy and the market. Note that if you delete an imported policy, it reappears after the next discovery cycle of the target it was imported from. To delete imported policies, delete them on the target directly.

Example: `DELETE https://10.10.10.10/api/v3/markets/2481646239616/policies/_z3LxVzk0EemeycL7hWgJNw`

Response: A successful deletion request returns a Response Code of `200`.

Price lists utility requests

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

The price list endpoint contains a sample rate card file and RateCardApiDTO. To access this sample:

1. Execute the `GET https://10.10.10.10/api/v3/pricelists` request, and locate the price list whose `displayName` begins with `Sample Azure Rate Card`.
2. Record the UUID of this rate card.

3. Execute the `https://10.10.10.10/api/v3/pricelists/{uuid_from_step_2}/export` request, and download the file that makes up the response body.

Price lists requests

The price lists endpoint is not implemented at this time.

Reporting endpoint

You can use the reporting endpoint to enable and manage the Turbonomic SaaS reporting feature in an on-prem environment. SaaS reporting helps you understand trends in application resource management and to share insights with stakeholders through reports and Liveboards.

SaaS reporting is available in both SaaS and on-prem environments with a minimum Turbonomic version of 8.15.4. For more information, see Chapter 14, “Integration - SaaS reporting,” on page 1913.

The SaaS reporting endpoint is used for on-prem instances of Turbonomic. However, to begin the enablement process, you must first open a case with IBM support and work with your Turbonomic representative. After your representative provides an encoded configuration string, you can use the reporting endpoint to complete the following tasks in an on-prem environment:

- Enable the SaaS reporting feature
- Verify your SaaS reporting configuration
- Delete the SaaS reporting feature.

Reporting requests

To enable the SaaS reporting feature in an on-prem environment, you must first open a case with IBM support and work with your Turbonomic representative. After your representative processes your request, you receive an encoded configuration string.

You can also complete this task by using the swagger UI for your instance. For more information, see “Enabling SaaS reporting for on-prem Turbonomic instances” on page 1913.

Enable the SaaS reporting feature (on-prem)

You can use the `reporting/configuration` endpoint to enable on-prem SaaS reporting by providing the encoded configuration string from your Turbonomic representative.

Example: `POST https://10.10.10.10/api/v3/reporting/configuration`

Request body:

```
{
    "configuration":
"<your-configuration-string>",
    "enabled": true
}
```

Verify your on-prem SaaS reporting configuration

To verify your encoded configuration string and confirm whether SaaS reporting is enabled, make a `GET` request to the `reporting/configuration` endpoint.

Example: `GET https://10.10.10.10/api/v3/reporting/configuration`

Response:

```
{
    "configuration":
"<your-configuration-string>",
    "enabled": true
}
```

Delete your on-prem SaaS reporting configuration

To delete your configuration and remove the SaaS reporting feature from your instance, make a `DELETE` request to the `reporting/configuration` endpoint.

Example: `DELETE https://10.10.10.10/api/v3/reporting/configuration`

Response:

```
true
```

Reservations endpoint

Turbonomic uses reservations to save the resources you will need if you want to deploy workloads at a future date. Turbonomic calculates optimal placement for these workloads, and then it reserves whatever resources the different hosts and storage entities will need to support those workloads. For example, you can instruct Turbonomic on 01 January to reserve the resources required for 100 workloads on 07 January.

Using the reservations endpoint of the Turbonomic API, you can:

- Get a list of reservations
- Create a reservation
- Get information about a specific reservation
- Delete a reservation

Reservation states

Reservations will be in one of the following states:

- `FUTURE`

  The reservation is set to occur after the current date.
- `INITIAL`

  The reservation is being created; occurs at the start time of the reservation.
- `INPROGRESS`

  The reservation request has been sent to the market provider(s), and is awaiting results.
- `INVALID`

  This error state occurs when a reservation is placed on an unready market (for example, when the market component is restarting) or one of the constraints or templates used in the reservation has been deleted between the creation of the reservation, and the start date.
- `PLACEMENT_FAILED`

  The reservation has completed, but there were insufficient resources to place all entites.
- `RESERVED`

  The reservation has completed and was able to obtain all needed resources.
- `UNFULFILLED`

  The reservation has been added to the database, but has not yet been sent to the market.

Reservations in the product

From the Workload Placement Page, you can set up reservations to save the resources you will need to deploy workloads at a future date. Turbonomic uses its intelligent workload management to calculate optimal placement for these workloads, and then it reserves whatever resources the different hosts and storage entities will need to support those workloads.

When creating a reservation in the product, you will select the VM template, any constraints (scope, policy, network), then give your reservation a name, virtual machine count, and start and end dates.

Reservations requests

Getting reservations

For a list of reservations, the API returns a list of DemandReservationApiDTOs. Each DemandReservationApiDTO will contain the entity information for the reserved instances, the count to be deployed, and current status. To get the DTO of a particular reservation, include the UUID in the request.

This request takes the following parameters:

Status

Returns only reservations in the specified status. `[PLACEMENT_FAILED, UNFULFILLED, RESERVED, INPROGRESS, INITIAL, FUTURE, INVALID]`

Examples:

- All reservations with a status of RESERVED:

  ```
  GET https://10.10.10.10/api/v3/reservations?status=RESERVED
  ```

- A single reservation:

  ```
  GET https://10.10.10.10/api/v3/reservations/_TszrkcwHEei7A9DOcc4f1g
  ```

Response:


```
[
  {
    "uuid": "215446856141616",
    "displayName": "MyReservation",
    "count": 1,
    "status": "RESERVED",
    "reserveDateTime": "2022-11-01T13:28:00Z",
    "expireDateTime": "2022-12-01T14:28:00Z",
    "constraintInfos": [
      {
        "uuid": "74635127126226",
        "constraintType": "DATA_CENTER"
      }
    ],
    "reservationDeployed": false,
    "mode": "NO_GROUPING",
    "grouping": "NONE",
    "demandEntities": [
      {
        "template": {
          "uuid": "215364307544240",
          "displayName": "headroomVM",
          "className": "VirtualMachineProfile"
        },
        "placements": {
          "computeResources": [
            {
              "stats": [ ...
                  Stats for CPU, CPUProvisioned, Mem, MemProvisioned, IOThroughput,
NetThroughput
              ... ],
              "provider": {
                "uuid": "74635127126242",
                "displayName": "hp-dl3111.eng.vmturbo.com",
                "className": "PhysicalMachine"
              },
              "linkedResources": [
                {
                  "uuid": "285741371272387",
                  "displayName": "DC13/Physical Hosts_Cluster2 - Intel"
                }
              ]
            }
```

```
          ],
          "storageResources": [
            {
              "stats": [ ...
                  Stats for StorageProvisioned, StorageAmount, StorageAccess
              ... ],
              "provider": {
                "uuid": "74635124839222",
                "displayName": "QSGRID01:DC1314Shared",
                "className": "Storage"
              }
            }
          ]
        }
      }
    ]
  }
]
```

Creating a placement reservation

To create a placement reservation, POST a completed DemandReservationApiInputDTO as input to this request. The DTO specifies reservation type, the template to use, the UUIDs of constraints such as placement or network policies, and other attributes of the reservation.

Example: `POST https://10.10.10.10/api/v3/reservations`

Example inputDTO, to execute a reservation for two instances of a template:

```
{
  "demandName": "MyReservation1",
  "action": "RESERVATION",
  "mode": "NO_GROUPING",
  "grouping": "NONE",
  "reserveDateTime": "2022-11-01T21:03:22Z",
  "expireDateTime": "2022-12-01T21:03:22Z",
  "parameters": [
    {
      "placementParameters": {
        "count": 2,
        "templateID": "215364307544240",
        "entityNames": [
          "MyRes1_VM1",
          "MyRes1_VM2"
        ],
        "constraintIDs": [
          "285815619396128"
        ]
      }
    }
  ]
}
```

Response: A `DemandReservationApiDTO` that will include the status of the placement attempt. Note that the status of this reservation is FUTURE. When Turbonomic executes the reservation, if it succeeds in placing the reserved VMs then the `placements` field will contain descriptions of the host and storage providers for the two VMs.

```
{
  "uuid": "215447101807136",
  "displayName": "MyReservation1",
  "count": 2,
  "status": "FUTURE",
  "reserveDateTime": "2022-11-01T21:03:22Z",
  "expireDateTime": "2022-12-01T21:03:22Z",
  "constraintInfos": [
    {
      "uuid": "285815619396128",
      "constraintType": "POLICY"
    }
  ],
  "reservationDeployed": false,
  "mode": "NO_GROUPING",
```

```
  "grouping": "NONE",
  "demandEntities": [
    {
      "template": {
        "uuid": "215364307544240",
        "displayName": "headroomVM",
        "className": "VirtualMachineProfile"
      },
      "placements": {}
    },
    {
      "template": {
        "uuid": "215364307544240",
        "displayName": "headroomVM",
        "className": "VirtualMachineProfile"
      },
      "placements": {}
    }
  ]
}
```

Deleting a reservation

Deletes the specified reservation and removes all reserved resources associated with that reservation. This will also remove any pending deployment related to the deleted reservation.

Example: `DELETE https://10.10.10.10/api/v3/reservations/_i-FLwdLiEeiuY7KDiLoLWg`

Response: A successful deletion request returns a Response Code of `200`.

Reservations endpoint tips

Customizing a reservation

When creating a reservation, you should be familiar with placement policies, templates, scope constraints, and network constraints.

Note:

Use network constraint to limit workload placement to the networks that Turbonomic discovered in your environment.

This request takes the following parameters of the `DemandReservationApiInputDTO`:

demandName
The display name of the reservation — If you do not specify names for the reserved workloads, they will all get this as the root name, with a numeric suffix. For example, MyReservation1, MyReservation2, etc.

action
The action type: This is always `RESERVATION`.

reserveDateTime
The time to calculate the workload placement and create the reservation — This time cannot be earlier than the time that you POST the reservation to the API.

expireDateTime
When to cancel the reservation. At this time, Turbonomic will delete the reservation. To take advantage of the reserved resources, you should deploy the actual VMs to their reserved locations before this time.

parameters
A `DemandReservationParametersDTO`, which is itself composed of a `PlacementParametersDTO` and `DeploymentParametersDTO`.

In order to pass a completed `DemandReservationParametersDTO`, you will create a `PlacementParametersDTO` with the following parameters:

count
The number of workloads in this reservation

templateID
The UUID of the template used for creating the workload required for the placement

entityNames
Name of the placed workload, which will be affixed with a numeric suffix for reservations of more than one workload

constraintIDs
The UUIDs of the scope, group and any policies or network constraints that the reservation will adhere to

For example, the following POST creates a reservation of 10 VMs, scoped to a specific virtual data center with a UUID of 73365042986774 and network with a UUID of 73365048169751. Because it does not specify `entityNames` (an array of names for the VMs), Turbonomic will generate the VM names.

```
{
  "action":"RESERVATION",
  "demandName":"DalReservation",
  "reserveDateTime":"2020-03-11T15:44:41Z",
  "expireDateTime":"2020-04-11T15:44:41Z",
  "parameters":[
    {
      "placementParameters":{
        "count":"10",
        "templateID":"214103358655713",
        "constraintIDs":[
          "73365042986774",
          "73365048169751"
        ]
      }
    }
  ]
}
```

Affinity rules

When creating reservations, you can specify affinity rules for the reserved VMs, to ensure all the VMs in a reservation deploy to the same cluster. If you do not specify affinity, the VMs deploy to any clusters in the reservation's scope. These settings are optional, but if you choose to specify affinity, you must specify compatible values for both settings.

These setting respect other constraints you might have set for the reservation. For example, if you scope the reservation to a group of clusters, with a cluster affinity rule Turbonomic will choose one cluster from that scope and try to place all the VMs on it.

To specify affinity for a reservation, you declare:

- `mode` (optional):

  Can be one of:

  – `AFFINITY` - Group reserved VMs in the same cluster. The `grouping` parameter must be set to `CLUSTER`. If there is no cluster that can host the requested VMs, the reservation will be `UNFULLFILLED`.

  – `NO_GROUPING` - Do not group reserved VMs in the same cluster. The `grouping` parameter must be set to `NONE`.

- `grouping` (optional):

  Can be one of:

  – `CLUSTER` - Group reserved VMs. The `mode` parameter must be set to `CLUSTER`.

– NONE - Do not group reserved VMs. The mode parameter must be set to NO_GROUPING.

```
{
  "demandName": "MyReservation1",
  "action": "RESERVATION",
  "mode": "AFFINITY",
  "grouping": "CLUSTER",
  ...
}
```

Getting reservation information

To get a list of current reservations, execute the GET https://10.10.10.10/api/v3/reservations request. This returns a list of all active reservations. If you know the UUID of the reservation you want to retrieve, you can pass it to get data for just that reservation.

Each reservation object describes:

- Display name
- Status — can be:
  - IN_PROGRESS — Placement calculation is in progress
  - PLACEMENT_SUCCEEDED — For a new reservation, the environment has sufficient resources to place the workload; if you specified a reserve date, this will be an active reservation
  - PLACEMENT_FAILED — For a newly created reservation, the environment doesn't have resources to place the workload; if you specified a reserve date, this will be an unfulfilled reservation
  - RETRYING — Turbonomic is trying to place the workload of an unfulfilled reservation
- Time the reservation was created, and time it will expire
- A description of the reserved workload
- Statistics for the compute and storage resources the reservation sets aside

For example, this listing shows a successful reservation for one VM:

```
{
  "uuid":"_kWZHIDDmEeePgeXuo0RRbw",
  "displayName":"MyReservation",
  "count":1,
  "status":"PLACEMENT_SUCCEEDED",
  "reserveDateTime":"Thu May 04 16:27:29 UTC 2017",
  "expireDateTime":"Thu Aug 31 16:27:29 UTC 2017",
  "deployDateTime":"Thu Aug 31 16:27:29 UTC 2017",
  "reserveCount":1,
  "demandEntities":[
    {
      "uuid":"_kWgb7TDmEeePgeXuo0RRbw",
      "displayName":"MyReservation_C0",
      "className":"VirtualMachine",
      "template":{
        "uuid":"T423f548d-cadc-e525-6df4-1f90724cf696",
        "displayName":"vsphere-dc3.dev.mycorp.com::TMP-SUSE64",
        "className":"VirtualMachineProfile"
      },
      "placements":{
        "computeResources":[
          {
            "stats":[
              {
                "name":"numOfCpu",
                "value":1
              },
              {
                "name":"cpuSpeed",
                "value":2603
              },
              {
                "name":"cpuConsumedFactor",
                "value":0.5
              },
              {
```

```
                "name":"memorySize",
                "value":2097152
              },
              {
                "name":"memoryConsumedFactor",
                "value":0.75
              },
              {
                "name":"ioThroughput",
                "value":0
              },
              {
                "name":"networkThroughput",
                "value":0
              }
            ],
            "provider":{
              "uuid":"Virtual_ESX_42381da5-12fa-1e82-2f1c-887419380d43",
              "displayName":"hp-esx21.corp.vmturbo.com",
              "className":"PhysicalMachine"
            }
          }
        ],
        "storageResources":[
          {
            "stats":[
              {
                "name":"diskSize",
                "value":18432.363
              },
              {
                "name":"diskIops",
                "value":0
              }
            ],
            "provider":{
              "uuid":"10545c15-7687ef26",
              "displayName":"QS2:ESXDC3DS1",
              "className":"Storage"
            }
          }
        ]
      }
    }
  ]
}
```


Creating a reservation

To create a reservation, you POST an input DTO that defines the reservation's:

- `action`: The action type: `RESERVATION`
- `demandName`: The display name of the reservation — If you do not specify names for the added workloads, this will be the root name for new VMs
- `placementParameters`:
  - `constraintIDs`: An array of UUIDs for placement policies that will affect the calculated placement
  - `count`: The number of workloads to place
  - `entityNames`: An array of names for the placed VMs — The array length should equal `count`
  - `geographicRedundancy`: If `true` place the workloads on unique hosts, otherwise Turbonomic can place multiple workloads on the same host
  - `templateID`: The UUID of the template that you will use to place this workload — Note that the template must include a reference to the deployment profile that you specify in `deploymantParameters`
- `expireDateTime`: When to cancel a reservation if Turbonomic cannot place all the workloads by that date — for a `RESERVATION` action, Turbonomic automatically sets the deploy time to equal this time
- `reserveDateTime`: The time to calculate the workload placement and create the reservation — This time cannot be earlier than the time that you POST the reservation to the API

The following listing shows an input DTO that creates a reservation. It will place four workloads, and it gives specific names ot each one.

```
{
  "action": "RESERVATION",
  "demandName": "MyReservation",
  "expireDateTime": "2017-10-10T12:38:17+00:00",
  "parameters": [
    {
      "deploymentParameters": {
        "deploymentProfileID": "_c9CJMDDAEeePgeXuo0RRbw"
      },
      "placementParameters": {
        "geographicRedundancy": false,
        "count": 4,
        "entityNames": [
          "foo","bar","baz","bonk"
        ],
        "templateID": "_UKsnkJkSEeCHcOXEhzJExA"
      }
    }
  ],
  "reserveDateTime": "2017-05-04T18:22:12+00:00"
}
```

Reserved instances endpoint

Turbonomic analysis takes advantage of AWS Reserved Instances and Azure Reserved VM Instances to calculate optimal workload placement and to arrive at the best possible costs for your deployments on the cloud.

Using the reserved instances endpoint of the Turbonomic API, you can:

- Get a single reserved instance or list of reserved instances present in your environment
- Get statistics related to the reserved instances

Reserved instances requests

Getting reserved instances

For a list of reserved instances, the API returns an array of ReservedInstanceApiDTOs. Each ReservedInstanceApiDTO will contain the validation status, target type and category, and input field names and values. To get a single reserved instance, include the UUID of the reserved instance in the request.

This request takes the following parameters:

scope

Returns only reserved instances in the specified scope. This can be the scope of a group of entities or target. For example, to see reserved instances for a single AWS target, use that target's UUID here.

Examples:

- All Reserved instances: `GET https://10.10.10.10/api/v3/reservedinstances`
- Reserved Instances for a Specified Scope: `GET https://10.10.10.10//api/v3/reservedinstances?scope=_zeCtIMAVEei2-fP3xzx1Fw`

Response:

```
[
  {
    "uuid": "aws::eu-west-1::RI::a0643140-85d7-4c5e-a7ba-4ec2fc47bb76",
    "displayName": "i3.large",
    "className": "ReservedInstance",
    "template": {
      "uuid": "aws::VMPROFILE::i3.large",
      "displayName": "i3.large",
      "discovered": false,
```

```
      "family": "i3"
    },
    "platform": "LINUX",
    "type": "STANDARD",
    "location": {
      "uuid": "aws::eu-west-1::DC::eu-west-1",
      "displayName": "aws-EU (Ireland)"
    },
    "accountId": "19265476245",
    "targetId": "_zeCtIMAVEei2-fP3xzx1Fw",
    "masterAccountId": "194356375",
    "term": {
      "units": "Year",
      "values": {
        "max": 1,
        "min": 1,
        "avg": 1,
        "total": 1
      },
      "value": 1
    },
    "payment": "PARTIAL_UPFRONT",
    "onDemandPrice": {
      "units": "$/h",
      "values": {
        "max": 0.172,
        "min": 0.172,
        "avg": 0.172,
        "total": 0.172
      },
      "value": 0.172
    },
    "costPrice": {
      "units": "$/h",
      "values": {
        "max": 0.11193608,
        "min": 0.11193608,
        "avg": 0.11193608,
        "total": 0.11193608
      },
      "value": 0.11193608
    },
    "coupons": {
      "capacity": {
        "max": 16,
        "min": 16,
        "avg": 16
      },
      "units": "RICoupon",
      "values": {
        "max": 0,
        "min": 0,
        "avg": 0,
        "total": 0
      },
      "value": 0
    },
    "totalCoupons": 0,
    "expDate": "Sat Apr 06 19:11:19 EDT 2019",
    "instanceCount": 1,
    "tenancy": "DEFAULT",
    "upFrontCost": 490,
    "actualHourlyCost": 0.0560000017285347,
    "effectiveHourlyCost": 0.11193607478789543
  },
  ...
]
```


Getting entities covered by reserved instances

Given the UUID of a reserved instance, the API returns a list of entities covered by the reserved instance.

This request takes the following parameters:

reservedInstance_Uuid

The UUID of the reserved instance.

Example: `GET https://10.10.10.10/api/v3/reservedinstances/1234/coveredEntities`

Response: An array of `BaseApiDTO`s describing the entities covered by the specified RI.

```
[
   {
      "uuid":"73943736161360",
      "displayName":"pyaus",
      "className":"VirtualMachine"
   },
   {
      "uuid":"73812178348464",
      "displayName":"ptaub807disabled",
      "className":"VirtualMachine"
   },
   {
      "uuid":"73746051159552",
      "displayName":"pymm",
      "className":"VirtualMachine"
   }
]
```

Roles requests

The roles endpoint is not implemented at this time.

Scenarios endpoint

To run a plan scenario, Turbonomic creates a snapshot copy of your real-time market and modifies that snapshot according to the scenario. It then uses the Economic Scheduling Engine to perform analysis on that plan market. A scenario can modify the snapshot market by changing the workload, adding or removing hardware resources, or changing or eliminating constraints such as cluster boundaries or placement policies.

As it runs a plan, Turbonomic continuously analyzes the plan market until it arrives at the optimal conditions that market can achieve. When it reaches that point, the Economic Scheduling Engine cannot find better prices for any of the resources demanded by the workload — the plan stops running, and it displays the results as the plan's desired state.

The display includes the resulting workload distribution across hosts and datastores, as well as a list of actions the plan executed to achieve the desired result.

For example, assume a scenario that adds virtual machines to a cluster. To run the plan, Turbonomic takes a snapshot of the current market, and adds the VMs to the specified cluster. Turbonomic then runs analysis on the plan market, where each entity in the supply chain shops for the resources it needs, always looking for a better price — looking for those resources from less-utilized suppliers. This analysis continues until all the resources are provided at the best possible price.

The results might show that you can add more workload to your environment, even if you reduce compute resources by suspending physical machines. The recommended actions would then indicate which hosts you can take offline, and how to distribute your virtual machines among the remaining hosts.

The settings you make in a scenario correspond to the plan settings you can make in the user interface. These include:

- Plan scope
- Changes to workload (adding, removing, or replacing VMs or containers)
- Changes to supply (adding, removing, replacing PMs or storage)
- Enable/disable placement policies and other constraints
- Changes to action modes
- Enable/disable provisioning of supply
- Enable/disable resizing of workloads

Using the scenarios endpoint of the Turbonomic API, you can:

- Get a list of scenarios on your instance
- Create a scenario
- Get a list of markets that are connected to a particular scenario

Scenarios requests

Getting scenarios

The API returns a ScenarioApiDTO including the following fields when you execute a `GET /scenarios` request:

- uuid

  The unique identifier for this scenario.
- displayName

  By default, Turbonomic creates a display name that matches the scenario type. You may rename the scenario after creation using a PUT request.
- owners

  An array of user accounts that can use the scenario. Currently, this is always the user that created the scenario.
- type

  The type of scenario. When a user creates a scenario in the GUI, this matches the given type that user chooses from the Plan Wizard.
- scope

  An array of scope objects to specify the scope of the plan.
- projectionDays

  An array of days to specify how far into the future to project the plan. For example, a value of `[30, 60, 120]` would return projections after one, two, and four months, respectively.
- topologyChanges

  An array of additions or deletions to the set of entities in the plan scope.
- loadChanges

  An array of changes to the utilization percentage for different groups of entities within the plan scope.
- configChanges

  An array of configuration changes for different groups of entities within the plan scope.
- `changes - DEPRECATED`

  An array of scenario settings that make changes to the market before running the plan.

Example: `GET https://10.10.10.10/api/v3/scenarios?show_all=false`

Response:


```
{
    "uuid": "214704587455872",
    "displayName": "Migrate to Public Cloud 1_CLOUD_MIGRATION__CONSUMPTION",
    "type": "CLOUD_MIGRATION",
    "scope": [
      {
        "uuid": "73935388327649",
        "displayName": "aws-EU (Frankfurt)",
        "className": "Region"
      }
    ],
    "projectionDays": [
```

```
    0
  ],
  "topologyChanges": {
    "migrateList": [
      {
        "projectionDay": 0,
        "source": {
          "uuid": "285041726554576",
          "displayName": "Unknown"
        },
        "destination": {
          "uuid": "73935388327649",
          "displayName": "aws-EU (Frankfurt)",
          "className": "Region",
          "state": "ACTIVE",
          "severity": "Minor",
          "discoveredBy": {
            "uuid": "73935444966640",
            "displayName": "testmachine",
            "category": "Public Cloud",
            "type": "AWS",
            "readonly": false
          },
          "severityBreakdown": {
            "MINOR": 1
          },
          "numRelatedVMs": 2,
          "environmentType": "CLOUD",
          "vendorIds": {
            "vmturboamp": "aws::eu-central-1::DC::eu-central-1"
          }
        },
        "sources": [
          {
            "uuid": "285041726554576",
            "displayName": "Unknown"
          }
        ],
        "destinations": [
          {
            "uuid": "73935388327649",
            "displayName": "aws-EU (Frankfurt)",
            "className": "Region",
            "state": "ACTIVE",
            "severity": "Minor",
            "discoveredBy": {
              "uuid": "73935444966640",
              "displayName": "testmachine",
              "category": "Public Cloud",
              "type": "AWS",
              "readonly": false
            },
            "severityBreakdown": {
              "MINOR": 1
            },
            "numRelatedVMs": 2,
            "environmentType": "CLOUD",
            "vendorIds": {
              "vmturboamp": "aws::eu-central-1::DC::eu-central-1"
            }
          }
        ],
        "destinationEntityType": "VirtualMachine",
        "removeNonMigratingWorkloads": true
      }
    ]
  },
  "loadChanges": {
    "utilizationList": [],
    "maxUtilizationList": []
  },
  "configChanges": {
    "addPolicyList": [],
    "removePolicyList": [],
    "automationSettingList": [],
    "removeConstraintList": [],
    "riSettingList": [
      {
        "uuid": "ri.aws.preferredOfferingClass",
        "displayName": "Type",
        "value": "STANDARD",
        "valueObjectType": "String"
```

```
        },
        {
          "uuid": "ri.aws.preferredPaymentOption",
          "displayName": "Payment",
          "value": "ALL_UPFRONT",
          "valueObjectType": "String"
        },
        {
          "uuid": "ri.aws.preferredTerm",
          "displayName": "Term",
          "value": "YEARS_1",
          "valueObjectType": "String"
        }
      ],
      "osMigrationSettingList": [
        {
          "uuid": "linuxTargetOs",
          "displayName": "linuxTargetOs",
          "value": "LINUX",
          "valueObjectType": "String"
        },
        {
          "uuid": "linuxByol",
          "displayName": "linuxByol",
          "value": "false",
          "valueObjectType": "String"
        },
        {
          "uuid": "rhelTargetOs",
          "displayName": "rhelTargetOs",
          "value": "RHEL",
          "valueObjectType": "String"
        },
        {
          "uuid": "rhelByol",
          "displayName": "rhelByol",
          "value": "false",
          "valueObjectType": "String"
        },
        {
          "uuid": "slesTargetOs",
          "displayName": "slesTargetOs",
          "value": "SLES",
          "valueObjectType": "String"
        },
        {
          "uuid": "slesByol",
          "displayName": "slesByol",
          "value": "false",
          "valueObjectType": "String"
        },
        {
          "uuid": "windowsTargetOs",
          "displayName": "windowsTargetOs",
          "value": "WINDOWS",
          "valueObjectType": "String"
        },
        {
          "uuid": "windowsByol",
          "displayName": "windowsByol",
          "value": "false",
          "valueObjectType": "String"
        },
        {
          "uuid": "selectedMigrationProfileOption",
          "displayName": "selectedMigrationProfileOption",
          "value": "MATCH_SOURCE_TO_TARGET_OS",
          "valueObjectType": "String"
        },
        {
          "uuid": "matchToSource",
          "displayName": "matchToSource",
          "value": "true",
          "valueObjectType": "String"
        }
      ],
      "includedCoupons": {
        "includedCouponOidsList": [],
        "iswhiteList": true
      },
      "subscription": {
        "uuid": "73935382668544",
```

```
        "displayName": "EngineeringSample",
        "className": "BusinessAccount",
        "accountId": "323871187550",
        "master": false,
        "cloudType": "AWS",
        "riSupported": false,
        "hasRelatedTarget": false
      },
      "respectActionModeSetting": false
    }
  }
```

Creating a scenario

In order to create a scenario, see the “Scenario DTO breakdown” on page 1445. For examples of complete ScenarioApiDTOs, see “Scenario configuration examples” on page 1455.

Deleting a scenario

Deletes the specified scenario. Note that this does not delete plan markets you create using that scenario.

Example: `DELETE https://10.10.10.10/api/v3/scenarios/__xwWYAE3EeiNKPbzHdETIA`

Response: A successful deletion request returns a Response Code of `200`.

Scenario DTO breakdown

Scenario DTO overview

A scenario DTO assembles configuration settings that you use to set up and run a plan. To run a plan, you will POST the scenario to a market. For most plans, this will be the real-time market. In the case of a plan-over-plan, this will be the market created from the first scenario.

Turbonomic will then run a plan based on the combination of the scenario and the market state.

These configuration settings fall into one of several groups:

- Configuration Changes (ConfigChangesApiDTO)

  Adding and removing placement policies, removing constraints, and providing automation settings, OS migration settings, reserved instance settings, and associated business units.
- Load Changes (LoadChangesApiDTO)

  Modifications to the current and max utilization values, and peak baseline values.
- Time-Based Topology Changes (TimeBasedTopologyChangesApiDTO)

  Settings to include reserved VMs and periodically add historical workload to the environment.
- Topology Changes (TopologyChangesApiDTO)

  Settings to add, remove, replace, and migrate workload, or to specify the clusters for an Alleviate Pressure plan.

DTO configuration changes

The scenario configuration changes, located in the `configChanges` parameter of the ScenarioApiDTO, enable you to perform the following tasks:

- addPolicyList

  Add a placement policy to the market generated for this plan.
- automationSettingList

  Change one of the available automation plan settings: VM Resize, Host Provision/Suspend, and Storage Provision/Suspend.

- osMigrationSettingList

  For a Migrate to Cloud plan, this parameter will contain the source and destination OS platforms, and their BYOL (Bring Your Own License) status.
- removeConstraintList

  Remove a constraint from the market generated for this plan.
- removePolicyList

  Remove a placement policy from the market generated for this plan.
- riSettingList

  Add information about your reserved cloud instances.
- subscription

  Associate a business unit to the plan.
- respectActionModeSetting

  Indicates whether to import and respect the action mode settings from real-time market automation policies.

Adding and Removing Placement Policies

Referenced Endpoint: Policies

When running plans, you might want to see what your environment would look like with or without certain policies enabled, or with a new policy present. You can see the effects of these changes by passing the UUIDs of current policies, or a PolicyApiDTO for a new policy.

Example input (Current Policies):

```
"addPolicyList": [
      {
        "uuid": "_VnNS0CsqEemvEL3szcXU1Q",
        "uuid": "522a3317-39f5-1a97-3b53-d725927a6ae8",
        "uuid": "52672058-a93b-d0a9-bbd9-5acd33f27882"
      }
    ]
```

Example input (New Policy):

```
"addPolicyList": [
      {
        "capacity": null,
        "consumerGroup": {
          "uuid": "8c17d8e79c3db85ed5d3868de622cb89187849ce",
          "displayName": "PMs_Beta DC6\\Cluster-2",
          "className": "Group",
          "groupType": "PhysicalMachine",
          "isStatic": true,
          "logicalOperator": "AND"
        },
        "enabled": true,
        "mergeGroups": null,
        "mergeType": null,
        "name": "DalTestPolMerge",
        "displayName": "DalTestPolMerge",
        "providerGroup": {
          "uuid": "91ff08749fd278925d73b61b0f0f69d79c99fc3a",
          "displayName": "PMs_Beta DC6\\Cluster-1",
          "className": "Group",
          "groupType": "PhysicalMachine",
          "isStatic": true,
          "logicalOperator": "AND"
        },
        "type": "MERGE",
        "uuid": ""
      }
    ]
```

Setting Plan Automation Settings

Use the plan automation settings to see the effect of your scenario if VM resizes, Host Provisioning / Suspension, and Storage Provisioning / Suspension are enabled or disabled.

Example Input:

```
"automationSettingList":[
  {
    "uuid":"resize",
    "displayName":"Resize for VMS disabled",
    "value":"false",
    "entityType":"VirtualMachine"
  },
  {
    "uuid":"provisionPM",
    "displayName":"Provision for PHYSICALMACHINES disabled",
    "value":"false",
    "entityType":"PhysicalMachine"
  },
  {
    "uuid":"suspendPM",
    "displayName":"Suspend for PHYSICALMACHINES disabled",
    "value":"false",
    "entityType":"PhysicalMachine"
  },
  {
    "uuid":"provisionDS",
    "displayName":"Provision for STS disabled",
    "value":"false",
    "entityType":"Storage"
  },
  {
    "uuid":"suspendDS",
    "displayName":"Suspend for STS enabled",
    "value":"true",
    "entityType":"Storage"
  }
]
```

Additionally, you can use these settings to adjust operational constraints, such as provisioned values for CPU and MEM:

Example Input:

```
"automationSettingList":[
  {
    "uuid":"capacity_MemProvisioned",
    "displayName":"Mem Overprovisioned Percentage",
    "value":"9876.0",
    "defaultValue":"1000.0",
    "categories":[
      "utilizationThresholds"
    ],
    "valueType":"NUMERIC",
    "min":1,
    "max":1000000,
    "entityType":"PhysicalMachine",
    "sourceGroupName":"Global"
  }
```

Providing Cloud Reserved Instance Information

Turbonomic enables you to upload RI pricing information specific to the scenario.

Example Input:

```
"riSettingList": [
      {
        "uuid": "preferredTerm",
        "displayName": "Term",
        "value": "YEARS_1",
        "entityType": "YEARS_1"
      },
      {
```

```
        "uuid": "preferredPaymentOption",
        "displayName": "Payment",
        "value": "PARTIAL_UPFRONT",
        "entityType": "PARTIAL_UPFRONT"
      },
      {
        "uuid": "preferredOfferingClass",
        "displayName": "Type",
        "value": "STANDARD",
        "entityType": "STANDARD"
      },
      {
        "uuid": "preferredCoverage",
        "displayName": "Coverage",
        "value": "80",
        "entityType": "80"
      },
      {
        "uuid": "riCoverageOverride",
        "displayName": "RI Coverage Override",
        "value": "false",
        "entityType": "false"
      }
    ]
```

Removing Constraints

Referenced Endpoint: Entities

When running plans, you might want to see what your environment would look like with or without certain constraints. For example, if a group of virtual machines was allowed to move to any cluster of your environment.

In order to remove a constraint, you must know its UUID. To get a list of constraints for this entity, execute the `/entities/ENTITY_UUID/constraints` request for the entity whose constraint you wish to remove.

```
"removeConstraintList":[
  {
    "constraintType":"ClusterCommodity",
    "projectionDay":0,
    "target":{
      "uuid":"a6d365f4fbf03fb9a2e8cdb29e38045e31cee41b",
      "displayName":"Beta DC6_vm",
      "className":"VirtualMachine"
    }
  }
]
```

Setting OS Migration Settings for a Migrate to Cloud Plan

When performing a Migrate to Cloud plan, you might want to use custom OS migration settings. You can specify both the OS mapping, and the BYOL (bring your own license) status of each OS in the mapping.

Example Input:

```
"osMigrationSettingList":[
  {
    "uuid":"linuxTargetOs",
    "displayName":"linuxTargetOs",
    "value":"LINUX"
  },
  {
    "uuid":"linuxByol",
    "displayName":"linuxByol",
    "value":"true"
  },
  {
    "uuid":"rhelTargetOs",
    "displayName":"rhelTargetOs",
    "value":"RHEL"
  },
  {
    "uuid":"rhelByol",
```

```
    "displayName":"rhelByol",
    "value":"false"
  },
  {
    "uuid":"slesTargetOs",
    "displayName":"slesTargetOs",
    "value":"WINDOWS"
  },
  {
    "uuid":"slesByol",
    "displayName":"slesByol",
    "value":"true"
  },
  {
    "uuid":"windowsTargetOs",
    "displayName":"windowsTargetOs",
    "value":"WINDOWS"
  },
  {
    "uuid":"windowsByol",
    "displayName":"windowsByol",
    "value":"true"
  },
  {
    "uuid":"showMatchSourceToTargetOsOption",
    "displayName":"showMatchSourceToTargetOsOption",
    "value":"true"
  },
  {
    "uuid":"showByolOption",
    "displayName":"showByolOption",
    "value":"false"
  },
  {
    "uuid":"showCustomOsOption",
    "displayName":"showCustomOsOption",
    "value":"true"
  },
  {
    "uuid":"matchToSource",
    "displayName":"matchToSource",
    "value":"false"
  },
  {
    "uuid":"selectedMigrationProfileOption",
    "displayName":"selectedMigrationProfileOption",
    "value":"CUSTOM_OS"
  }
]
```

Scenario DTO load changes

The scenario load changes, located in the `loadChanges` parameter of the ScenarioApiDTO, make workload utilization changes that enable you to change the following values:

- baselineDate

  If your plan should be run using historical data, the date of the data to be used.
- maxUtilizationList

  Changes the max utilization values for a group of entities in your plan.
- peakBaselineList

  Changes the peak baseline values for a group of entities in your plan.
- utilizationList

  Changes the utilization values for a group of entities in your plan.

Changing the Baseline Date

Example input:

```
"baselineDate": 1552622400000
```

Changing the Utilization For a Group of Entities

Example input:

```
    "utilizationList": [
      {
        "percentage": -12,
        "projectionDay": 0,
        "target": {
          "displayName": "Global Environment",
          "uuid": "Market"
        }
      }
    ]
```

Changing the Maximum Utilization For a Group of Entities

Example input:

```
    "maxUtilizationList": [
      {
        "maxPercentage": 82,
        "projectionDay": 0,
        "target": {
          "displayName": "All On-prem Hosts",
          "className": "Group",
          "uuid": "_PE0v-YEUEee_hYfzgV9uYg"
        }
      },
      {
        "maxPercentage": 85,
        "projectionDay": 0,
        "target": {
          "displayName": "ACM_datastore",
          "className": "Group",
          "uuid": "c31a6f99f4409700115fc61fca71e5449ad68700"
        }
      },
      {
        "maxPercentage": 77,
        "projectionDay": 0,
        "target": {
          "displayName": "Beta DC6_datastore",
          "className": "Group",
          "uuid": "5544c35ed3d7fc0d33879d3457c219d7e40fc176"
        }
      }
    ]
  }
```

Scenario DTO time-based topology changes

The scenario time-based topology changes, located in the `timebasedTopologyChanges` parameter of the ScenarioApiDTO, make time-dependent topology changes to your scenario, such as:

- addHistoryVMs

  If `true`, Turbonomic will add additional workload to your scenario each month equal to the workload added to the real-time market in the previous month.

- includeReservation

  If `true`, Turbonomic will take current reservations and reflect their resource utilization when the scenario runs.

Adding Historical VM Usage and/or Including VM Reservations

Example input:

```
  "timebasedTopologyChanges": [
    {
      "addHistoryVMs": "true",
      "includeReservation": "true"
    }
  ]
```

Scenario DTO topology changes

The scenario topology changes, located in the `topologyChanges` parameter of the ScenarioApiDTO, make topology changes to your scenario, such as:

- addList

  A list of entities to add to your environment as part of plan configuration.
- migrateList

  A list of entities to move from one set of providers to another as part of plan configuration.
- relievePressureList

  In a Relieve Pressure plan, the source and target clusters.
- removeList

  A list of entities to be removed from your environment as part of plan configuration.
- replaceList

  A list of entities and the templates to replace them in your environment as part of plan configuration.

Adding Workload

Adding workload to a scenario simulates the effect of increased VM or container density in your environment. You can add multiple copies of VM groups or individual VMs.

Example input:

```
    "addList": [
      {
        "count": 3,
        "projectionDays": [
          0
        ],
        "target": {
          "uuid": "421dd4ce-a4c1-231a-71e7-64fbf598bc65",
          "displayName": "AJ_Test",
          "className": "VirtualMachine"
        }
      },
      {
        "count": 10,
        "projectionDays": [
          0
        ],
        "target": {
          "uuid": "421e503d-d0c0-75f9-8cce-79421924a90d",
          "displayName": "olg_test",
          "className": "VirtualMachine"
        }
      },
      {
        "count": 5,
        "projectionDays": [
          0
        ],
        "target": {
          "uuid": "420f1840-297f-5a39-9543-b72046b9126f",
          "displayName": "opensuse_with_stress_template",
          "className": "VirtualMachine"
        }
      },
      {
```

```
      "count": 1,
      "projectionDays": [
        0
      ],
      "target": {
        "uuid": "53523c9a4c5d67962a96f90f59456b0c3e72aca0",
        "displayName": "Development67_vm",
        "className": "VirtualMachine"
      }
    }
  ]
```

Migrating Workload to the Cloud

Turbonomic includes a special plan to simulate migration of on-prem workload to the cloud, or migration of workloads from one cloud to another. This plan focuses on optimizing your costs on the cloud by choosing the best templates (most adequate compute resources) and regions to host your workloads. If analysis discovers on-prem workloads that are good candidates for Reserved Instances (RIs), then it recommends migrating to those templates, and can even recommend purchasing more RI capacity. For each group of virtual machines to be migrated (source), you must provide a destination.

Example input:

```
  "migrateList": [
    {
      "projectionDay": 0,
      "source": {
        "uuid": "4656c0ba6b976d05d1a9470ee673672022e8e926",
        "displayName": "ACM_vm",
        "className": "VirtualMachine",
        "severity": "Major"
      },
      "destination": {
        "uuid": "GROUP-PMsByTargetType_Azure",
        "displayName": "PMs_Azure",
        "className": "PhysicalMachine"
      }
    },
    {
      "projectionDay": 0,
      "source": {
        "uuid": "fe23c8642efc5ad2ed4fa1cc502dfb179b8dc9d2",
        "displayName": "Development DC1_vm",
        "className": "VirtualMachine",
        "severity": "Major"
      },
      "destination": {
        "uuid": "GROUP-PMsByTargetType_Azure",
        "displayName": "PMs_Azure",
        "className": "PhysicalMachine"
      }
    },
    {
      "projectionDay": 0,
      "source": {
        "uuid": "f0e26cb417d8143173270b7f01c7213e027cd9e5",
        "displayName": "Development DC7_vm",
        "className": "VirtualMachine",
        "severity": "Major"
      },
      "destination": {
        "uuid": "GROUP-PMsByTargetType_Azure",
        "displayName": "PMs_Azure",
        "className": "PhysicalMachine"
      }
    }
  ]
```

Relieve Pressure

Including this parameter creates an Alleviate Pressure plan to find out how to migrate workloads from a stressed or hot cluster over to a cluster with more headroom.

Example input:

```
    "relievePressureList": [
      {
        "projectionDay": 0,
        "sources": [
          {
            "uuid": "11c63ebd77647e7b9c9fb85dd2d4c77464cbeac2",
            "displayName": "DC13\\DC13-Cluster",
            "className": "Cluster"
          }
        ],
        "destinations": [
          {
            "uuid": "57a4026a69ba4b46ebc430cb62d5c55eeea00695",
            "displayName": "DC13\\Physical Hosts_Cluster2 - Intel",
            "className": "Cluster"
          }
        ]
      }
    ]
```

Removing Workload

Removing workload in a scenario configuration simulates the effect of decreased VM or container density in your environment. You can remove VM groups or individual VMs.

Example input:

```
    "removeList": [
      {
        "projectionDay": 0,
        "target": {
          "uuid": "5E4F6056-8783-45BC-9070-35CF09457E98",
          "displayName": "AH-TestHyperV-VM001",
          "className": "VirtualMachine"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "421d62c2-964a-0f4f-32cb-ec4cec9e26db",
          "displayName": "Anshuman_MidServer",
          "className": "VirtualMachine"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "421e048b-f66b-539c-f743-7dcc7b40943b",
          "displayName": "Gilad_XL_Storage_Testing - Pure-1",
          "className": "VirtualMachine"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "421e9199-7159-04f2-29fc-6a501f44d1be",
          "displayName": "Gilad_XL_Storage_Testing - Pure-2",
          "className": "VirtualMachine"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "421d9ec6-ab97-84d0-4b12-2db88ae43e2a",
          "displayName": "MS-SQLServer-3",
          "className": "VirtualMachine"
        }
      }
    ]
```

Replacing Workload

When you replace workload, you select one or more VMs or containers that you want to change (target), and then you select a template to use in their place.

Example input:


```
    "replaceList": [
      {
        "projectionDay": 0,
        "target": {
          "uuid": "421ef003-e152-5a25-6e03-19698340cd01",
          "displayName": "centos_imp_7",
          "className": "VirtualMachine",
          "severity": "Minor"
        },
        "template": {
          "uuid": "_wIBz8Jj4EeC6nYMiQT1jqA",
          "displayName": "Microsoft_SQL2008-small",
          "className": "VirtualMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "42001807-bc8f-b144-8ff6-eb0d0d32cc7a",
          "displayName": "cheuk-OpsMgr-172.113",
          "className": "VirtualMachine",
          "severity": "Minor"
        },
        "template": {
          "uuid": "_wIBz8Jj4EeC6nYMiQT1jqA",
          "displayName": "Microsoft_SQL2008-small",
          "className": "VirtualMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4211806a-65f2-8be1-1015-a4abe1a2ab6f",
          "displayName": "dim-dc13amd-6.2.7_01",
          "className": "VirtualMachine",
          "severity": "Minor"
        },
        "template": {
          "uuid": "_wIBz8Jj4EeC6nYMiQT1jqA",
          "displayName": "Microsoft_SQL2008-small",
          "className": "VirtualMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "42008beb-264e-b0f6-2295-f3443e4ce7d6",
          "displayName": "enOpenShiftInfra1",
          "className": "VirtualMachine",
          "severity": "Minor"
        },
        "template": {
          "uuid": "_wIBz8Jj4EeC6nYMiQT1jqA",
          "displayName": "Microsoft_SQL2008-small",
          "className": "VirtualMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "420036f6-27c3-0252-439d-e23bcff63d02",
          "displayName": "enOpenShiftInfra2",
          "className": "VirtualMachine",
          "severity": "Minor"
        },
        "template": {
          "uuid": "_wIBz8Jj4EeC6nYMiQT1jqA",
          "displayName": "Microsoft_SQL2008-small",
          "className": "VirtualMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4200ba09-71eb-3084-026e-1770c23abaa4",
```

```
          "displayName": "enOpenshiftDNS",
          "className": "VirtualMachine",
          "severity": "Minor"
        },
        "template": {
          "uuid": "_wIBz8Jj4EeC6nYMiQT1jqA",
          "displayName": "Microsoft_SQL2008-small",
          "className": "VirtualMachineProfile"
        }
      }
    ]
```

Scenario configuration examples

Basic scenario POSTs and responses

Most scenarios consist of multiple configuration changes. However, the UI provides several starting points for custom configuration based on plan type.

Turbonomic provides the same starting point in the API via the `/scenarios` endpoint and the following list of POST calls and server responses:

The following examples are simplified in order to accomplish a single goal. There are many other options you might choose in conjunction with those in the following examples.

Add workload

POST:

```
{
  "configChanges": {
    "addPolicyList": [],
    "automationSettingList": [],
    "removeConstraintList": [],
    "removePolicyList": [],
    "riSettingList": [],
    "osMigrationSettingList": [],
    "subscription": {}
  },
  "displayName": "Add Workload- Basic",
  "loadChanges": {
    "utilizationList": [],
    "maxUtilizationList": []
  },
  "projectionDays": [
    0
  ],
  "scope": [
    {
      "uuid": "Market",
      "displayName": "Global Environment",
      "className": "Market"
    }
  ],
  "topologyChanges": {
    "addList": [
      {
        "count": 50,
        "projectionDays": [
          0
        ],
        "target": {
          "uuid": "421dd4ce-a4c1-231a-71e7-64fbf598bc65",
          "displayName": "AJ_Test",
          "className": "VirtualMachine"
        }
      }
    ],
    "migrateList": [],
    "removeList": [],
    "replaceList": [],
    "relievePressureList": []
  },
  "type": "ADD_WORKLOAD"
```

```
}
```

Response:


```
{
  "uuid": "2502057162928",
  "displayName": "Add Workload- Basic",
  "owners": [
    {
      "uuid": "_4T_7kwY-Ed-WUKbEYSVIDw",
      "username": "administrator",
      "showSharedUserSC": false
    }
  ],
  "type": "ADD_WORKLOAD",
  "scope": [
    {
      "uuid": "_0x3OYUglEd-gHc4L513yOA",
      "displayName": "Market",
      "className": "Market"
    }
  ],
  "projectionDays": [
    0
  ],
  "topologyChanges": {
    "addList": [
      {
        "projectionDays": [
          0
        ],
        "target": {
          "uuid": "421dd4ce-a4c1-231a-71e7-64fbf598bc65",
          "displayName": "AJ_Test",
          "className": "VirtualMachine"
        },
        "count": 50
      }
    ]
  },
  "loadChanges": {},
  "configChanges": {
    "respectActionModeSetting": false"
  },
  "changes": [
    {
      "index": 1,
      "type": "SCOPE",
      "scope": [
        {
          "uuid": "_0x3OYUglEd-gHc4L513yOA",
          "displayName": "Market",
          "className": "Market"
        }
      ]
    },
    {
      "index": 2,
      "type": "PROJECTION_PERIODS",
      "projectionDays": [
        0
      ]
    },
    {
      "index": 3,
      "type": "ADDED",
      "description": "50 AJ_Test",
      "targets": [
        {
          "uuid": "421dd4ce-a4c1-231a-71e7-64fbf598bc65",
          "displayName": "AJ_Test",
          "className": "VirtualMachine",
          "severity": "Major"
        }
      ],
      "value": "50",
      "projectionDays": [
        0
      ]
```

```
    }
  ],
  "timebasedTopologyChanges": {}
}
```

Decommission hosts

POST:

```
{
  "configChanges": {
    "addPolicyList": [],
    "automationSettingList": [
      {
        "uuid": "provisionPM",
        "displayName": "PROVISION for PHYSICALMACHINES disabled",
        "value": "false",
        "entityType": "PhysicalMachine"
      }
    ],
    "removeConstraintList": [],
    "removePolicyList": [],
    "riSettingList": [],
    "osMigrationSettingList": [],
    "subscription": {}
  },
  "displayName": "Decommission Hosts- Basic",
  "loadChanges": {
    "utilizationList": [],
    "maxUtilizationList": []
  },
  "projectionDays": [
    0
  ],
  "scope": [
    {
      "uuid": "Market",
      "displayName": "Global Environment",
      "className": "Market"
    }
  ],
  "topologyChanges": {
    "addList": [],
    "migrateList": [],
    "removeList": [
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-0042-5A10-804A-B1C04F384331",
          "displayName": "dell-pe1944.corp.vmturbo.com",
          "className": "PhysicalMachine"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-004A-3610-8057-C8C04F464331",
          "displayName": "dell-pe1953.corp.vmturbo.com",
          "className": "PhysicalMachine"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-0059-4D10-8030-B3C04F464331",
          "displayName": "dell-pe1954.corp.vmturbo.com",
          "className": "PhysicalMachine"
        }
      }
    ],
    "replaceList": [],
    "relievePressureList": []
  },
  "type": "DECOMMISSION_HOST"
}
```

Response:


```
{
  "uuid": "2502061312640",
  "displayName": "Decommission Hosts- Basic",
  "owners": [
    {
      "uuid": "_4T_7kwY-Ed-WUKbEYSVIDw",
      "username": "administrator",
      "showSharedUserSC": false
    }
  ],
  "type": "DECOMMISSION_HOST",
  "scope": [
    {
      "uuid": "_0x3OYUglEd-gHc4L513yOA",
      "displayName": "Market",
      "className": "Market"
    }
  ],
  "projectionDays": [
    0
  ],
  "topologyChanges": {
    "removeList": [
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-0042-5A10-804A-B1C04F384331",
          "displayName": "dell-pe1944.corp.vmturbo.com",
          "className": "PhysicalMachine"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-004A-3610-8057-C8C04F464331",
          "displayName": "dell-pe1953.corp.vmturbo.com",
          "className": "PhysicalMachine"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-0059-4D10-8030-B3C04F464331",
          "displayName": "dell-pe1954.corp.vmturbo.com",
          "className": "PhysicalMachine"
        }
      }
    ]
  },
  "loadChanges": {},
  "configChanges": {
    "automationSettingList": [
      {
        "uuid": "provisionPM",
        "displayName": "Provision",
        "value": "false",
        "defaultValue": "RECOMMEND",
        "valueType": "STRING",
        "options": [
          {
            "label": "Disabled",
            "value": "DISABLED"
          },
          {
            "label": "Recommend",
            "value": "RECOMMEND"
          },
          {
            "label": "Manual",
            "value": "MANUAL"
          },
          {
            "label": "Automated",
            "value": "AUTOMATIC"
          }
        ],
        "entityType": "PhysicalMachine",
        "sourceGroupName": "Global"
      }
```

```
    ],
    "respectActionModeSetting": false
  },
  "changes": [
    {
      "index": 1,
      "type": "SCOPE",
      "scope": [
        {
          "uuid": "_0x3OYUglEd-gHc4L513yOA",
          "displayName": "Market",
          "className": "Market"
        }
      ]
    },
    {
      "index": 2,
      "type": "PROJECTION_PERIODS",
      "projectionDays": [
        0
      ]
    },
    {
      "index": 3,
      "type": "REMOVED",
      "description": "dell-pe1944.corp.vmturbo.com",
      "targets": [
        {
          "uuid": "4C4C4544-0042-5A10-804A-B1C04F384331",
          "displayName": "dell-pe1944.corp.vmturbo.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        }
      ],
      "value": "1",
      "projectionDays": [
        0
      ]
    },
    {
      "index": 4,
      "type": "REMOVED",
      "description": "dell-pe1953.corp.vmturbo.com",
      "targets": [
        {
          "uuid": "4C4C4544-004A-3610-8057-C8C04F464331",
          "displayName": "dell-pe1953.corp.vmturbo.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        }
      ],
      "value": "1",
      "projectionDays": [
        0
      ]
    },
    {
      "index": 5,
      "type": "REMOVED",
      "description": "dell-pe1954.corp.vmturbo.com",
      "targets": [
        {
          "uuid": "4C4C4544-0059-4D10-8030-B3C04F464331",
          "displayName": "dell-pe1954.corp.vmturbo.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        }
      ],
      "value": "1",
      "projectionDays": [
        0
      ]
    },
    {
      "index": 6,
      "type": "SET_ACTION_SETTING",
      "description": "Provision for PhysicalMachine disabled",
      "targets": [
        {
          "uuid": "PhysicalMachine"
        }
      ],
```

```
      "value": "PhysicalMachine",
      "enable": false,
      "name": "provision",
      "projectionDays": [
        0
      ]
    }
  ],
  "timebasedTopologyChanges": {}
}
```

Migrate to cloud

POST:

```
{
  "configChanges": {
    "addPolicyList": [],
    "automationSettingList": [],
    "removeConstraintList": [],
    "removePolicyList": [],
    "riSettingList": [],
    "osMigrationSettingList": [],
    "subscription": {}
  },
  "displayName": "Migrate to Public Cloud- Basic",
  "loadChanges": {
    "utilizationList": [],
    "maxUtilizationList": []
  },
  "projectionDays": [
    0
  ],
  "scope": [
    {
      "uuid": "4656c0ba6b976d05d1a9470ee673672022e8e926",
      "displayName": "ACM_vm",
      "groupType": "VirtualMachine",
      "className": "Group"
    },
    {
      "uuid": "_lRrTYB--EeewItqBJctLGw",
      "displayName": "All Cloud Zones",
      "groupType": "PhysicalMachine",
      "className": "Group"
    }
  ],
  "topologyChanges": {
    "addList": [],
    "migrateList": [
      {
        "projectionDay": 0,
        "source": {
          "uuid": "4656c0ba6b976d05d1a9470ee673672022e8e926",
          "displayName": "ACM_vm",
          "className": "VirtualMachine"
        },
        "destination": {
          "uuid": "_lRrTYB--EeewItqBJctLGw",
          "displayName": "All Cloud Zones",
          "className": "PhysicalMachine"
        }
      }
    ],
    "removeList": [
      {
        "projectionDay": 0,
        "target": {
          "uuid": "_nuoxwETSEemi-4wSMHOOWA",
          "displayName": "All VMs In All Cloud Zones",
          "className": "VirtualMachine"
        }
      }
    ],
    "replaceList": [],
    "relievePressureList": []
  },
  "type": "CLOUD_MIGRATION"
```

```
}
```

Response:


```
{
  "uuid": "2502081867536",
  "displayName": "Migrate to Public Cloud- Basic",
  "owners": [
    {
      "uuid": "_4T_7kwY-Ed-WUKbEYSVIDw",
      "username": "administrator",
      "showSharedUserSC": false
    }
  ],
  "type": "CLOUD_MIGRATION",
  "scope": [
    {
      "uuid": "4656c0ba6b976d05d1a9470ee673672022e8e926",
      "displayName": "ACM_vm",
      "className": "Group",
      "groupType": "VirtualMachine",
      "isStatic": true,
      "logicalOperator": "AND"
    },
    {
      "uuid": "_lRrTYB--EeewItqBJctLGw",
      "displayName": "All Cloud Zones",
      "className": "Group",
      "groupType": "PhysicalMachine",
      "isStatic": true,
      "logicalOperator": "AND"
    }
  ],
  "projectionDays": [
    0
  ],
  "topologyChanges": {
    "removeList": [
      {
        "projectionDay": 0,
        "target": {
          "uuid": "_nuoxwETSEemi-4wSMHOOWA",
          "displayName": "All VMs In All Cloud Zones",
          "className": "Group"
        }
      }
    ],
    "migrateList": [
      {
        "projectionDay": 0,
        "source": {
          "uuid": "4656c0ba6b976d05d1a9470ee673672022e8e926",
          "displayName": "ACM_vm",
          "className": "Group"
        },
        "destination": {
          "uuid": "_lRrTYB--EeewItqBJctLGw",
          "displayName": "All Cloud Zones",
          "className": "Group"
        }
      }
    ]
  },
  "loadChanges": {},
  "configChanges": {
    "osMigrationSettingList": [
      {
        "uuid": "matchToSource",
        "value": "true"
      }
    ],
    "respectActionModeSetting": false
  },
  "changes": [
    {
      "index": 1,
      "type": "SCOPE",
      "scope": [
        {
```

```
        "uuid": "4656c0ba6b976d05d1a9470ee673672022e8e926",
        "displayName": "ACM_vm",
        "className": "Group",
        "entitiesCount": 83,
        "groupType": "VirtualMachine",
        "severity": "Major",
        "logicalOperator": "AND"
      },
      {
        "uuid": "_lRrTYB--EeewItqBJctLGw",
        "displayName": "All Cloud Zones",
        "className": "Group",
        "entitiesCount": 78,
        "groupType": "PhysicalMachine",
        "severity": "Normal",
        "logicalOperator": "AND"
      }
    ]
  },
  {
    "index": 2,
    "type": "PROJECTION_PERIODS",
    "projectionDays": [
      0
    ]
  },
  {
    "index": 3,
    "type": "REMOVED",
    "description": "All VMs In All Cloud Zones",
    "targets": [
      {
        "uuid": "_nuoxwETSEemi-4wSMHOOWA",
        "displayName": "All VMs In All Cloud Zones",
        "className": "Group",
        "entitiesCount": 154,
        "groupType": "VirtualMachine",
        "severity": "Major",
        "logicalOperator": "AND"
      }
    ],
    "value": "1",
    "projectionDays": [
      0
    ]
  },
  {
    "index": 4,
    "type": "MIGRATION",
    "description": "Migrate [ACM_vm] to All Cloud Zones",
    "targets": [
      {
        "uuid": "4656c0ba6b976d05d1a9470ee673672022e8e926",
        "displayName": "ACM_vm",
        "className": "Group",
        "entitiesCount": 83,
        "groupType": "VirtualMachine",
        "severity": "Major",
        "logicalOperator": "AND"
      },
      {
        "uuid": "_lRrTYB--EeewItqBJctLGw",
        "displayName": "All Cloud Zones",
        "className": "Group",
        "entitiesCount": 78,
        "groupType": "PhysicalMachine",
        "severity": "Normal",
        "logicalOperator": "AND"
      }
    ],
    "projectionDays": [
      0
    ]
  },
  {
    "index": 5,
    "type": "SET_TARGET_OS",
    "targets": [
      {
        "uuid": "4656c0ba6b976d05d1a9470ee673672022e8e926",
        "displayName": "ACM_vm",
        "className": "Group",
```

```
          "entitiesCount": 83,
          "groupType": "VirtualMachine",
          "severity": "Major",
          "logicalOperator": "AND"
        }
      ],
      "projectionDays": [
        0
      ]
    }
  ],
  "timebasedTopologyChanges": {}
}
```

Optimize cloud

POST:

```
{
  "configChanges": {
    "addPolicyList": [],
    "automationSettingList": [
      {
        "uuid": "resize",
        "displayName": "resize for VMS enabled",
        "value": "true",
        "entityType": "VirtualMachine"
      }
    ],
    "removeConstraintList": [],
    "removePolicyList": [],
    "riSettingList": [
      {
        "uuid": "preferredOfferingClass",
        "displayName": "Type",
        "value": "STANDARD",
        "entityType": "STANDARD"
      },
      {
        "uuid": "preferredTerm",
        "displayName": "Term",
        "value": "YEARS_3",
        "entityType": "YEARS_3"
      },
      {
        "uuid": "preferredPaymentOption",
        "displayName": "Payment",
        "value": "ALL_UPFRONT",
        "entityType": "ALL_UPFRONT"
      },
      {
        "uuid": "preferredCoverage",
        "displayName": "Coverage",
        "value": "80",
        "entityType": "80"
      },
      {
        "uuid": "riCoverageOverride",
        "displayName": "RI Coverage Override",
        "value": "false",
        "entityType": "false"
      }
    ],
    "osMigrationSettingList": [],
    "subscription": {}
  },
  "displayName": "Optimize Cloud- Basic",
  "loadChanges": {
    "utilizationList": [],
    "maxUtilizationList": []
  },
  "projectionDays": [
    0
  ],
  "scope": [
    {
      "uuid": "GROUP-PMsByTargetType_AWS",
      "displayName": "PMs_AWS",
```

```
      "groupType": "PhysicalMachine",
      "className": "Group"
    }
  ],
  "topologyChanges": {
    "addList": [],
    "migrateList": [],
    "removeList": [],
    "replaceList": [],
    "relievePressureList": []
  },
  "type": "OPTIMIZE_CLOUD"
}
```

Response:

```
{
  "uuid": "2502117573632",
  "displayName": "Optimize Cloud- Basic",
  "owners": [
    {
      "uuid": "_4T_7kwY-Ed-WUKbEYSVIDw",
      "username": "administrator",
      "showSharedUserSC": false
    }
  ],
  "type": "OPTIMIZE_CLOUD",
  "scope": [
    {
      "uuid": "be95c99fc2a2195ac896204766a52d0e141818a6",
      "displayName": "PMs_AWS",
      "className": "Group",
      "groupType": "PhysicalMachine",
      "isStatic": true,
      "logicalOperator": "AND"
    }
  ],
  "projectionDays": [
    0
  ],
  "topologyChanges": {},
  "loadChanges": {},
  "configChanges": {
    "automationSettingList": [
      {
        "uuid": "resize",
        "displayName": "resize",
        "value": "true"
      }
    ],
    "riSettingList": [
      {
        "uuid": "preferredPaymentOption",
        "displayName": "Payment",
        "value": "ALL_UPFRONT",
        "defaultValue": "ALL_UPFRONT",
        "valueType": "STRING",
        "options": [
          {
            "label": "ALL_UPFRONT",
            "value": "ALL_UPFRONT"
          },
          {
            "label": "PARTIAL_UPFRONT",
            "value": "PARTIAL_UPFRONT"
          },
          {
            "label": "NO_UPFRONT",
            "value": "NO_UPFRONT"
          }
        ],
        "sourceGroupName": "Global"
      },
      {
        "uuid": "preferredTerm",
        "displayName": "Term",
        "value": "YEARS_3",
        "defaultValue": "YEARS_1",
        "valueType": "STRING",
```

```
        "options": [
          {
            "label": "YEARS_1",
            "value": "YEARS_1"
          },
          {
            "label": "YEARS_3",
            "value": "YEARS_3"
          }
        ],
        "sourceGroupName": "Global"
      },
      {
        "uuid": "preferredCoverage",
        "displayName": "Coverage",
        "value": "80",
        "defaultValue": "80",
        "valueType": "NUMERIC",
        "min": 0,
        "max": 100,
        "sourceGroupName": "Global"
      },
      {
        "uuid": "riCoverageOverride",
        "displayName": "RI Coverage Override",
        "value": "false",
        "defaultValue": "false",
        "valueType": "BOOLEAN",
        "sourceGroupName": "Global"
      },
      {
        "uuid": "preferredOfferingClass",
        "displayName": "Type",
        "value": "STANDARD",
        "defaultValue": "STANDARD",
        "valueType": "STRING",
        "options": [
          {
            "label": "STANDARD",
            "value": "STANDARD"
          },
          {
            "label": "CONVERTIBLE",
            "value": "CONVERTIBLE"
          }
        ],
        "sourceGroupName": "Global"
      }
    ],
    respectActionModeSetting": false
  },
  "changes": [
    {
      "index": 1,
      "type": "SCOPE",
      "scope": [
        {
          "uuid": "be95c99fc2a2195ac896204766a52d0e141818a6",
          "displayName": "PMs_AWS",
          "className": "Group",
          "entitiesCount": 50,
          "groupType": "PhysicalMachine",
          "severity": "Normal",
          "logicalOperator": "AND"
        }
      ]
    },
    {
      "index": 2,
      "type": "PROJECTION_PERIODS",
      "projectionDays": [
        0
      ]
    },
    {
      "index": 3,
      "type": "ENABLED",
      "description": "Resize enabled",
      "targets": [],
      "enable": true,
      "projectionDays": [
        0
```

```
      ]
    },
    {
      "index": 4,
      "type": "RI_SETTINGS",
      "targets": [
        {
          "uuid": "preferredPaymentOption|ALL_UPFRONT"
        },
        {
          "uuid": "preferredTerm|YEARS_3"
        },
        {
          "uuid": "preferredCoverage|80"
        },
        {
          "uuid": "riCoverageOverride|false"
        },
        {
          "uuid": "preferredOfferingClass|STANDARD"
        }
      ],
      "projectionDays": [
        0
      ]
    }
  ],
  "timebasedTopologyChanges": {}
}
```

On-prem workload migration (cluster to cluster)

POST:

```
{
  "configChanges": {
    "addPolicyList": [],
    "automationSettingList": [],
    "removeConstraintList": [],
    "removePolicyList": [],
    "riSettingList": [],
    "osMigrationSettingList": [],
    "subscription": {}
  },
  "displayName": "On-Prem Workload Migration 1",
  "loadChanges": {
    "utilizationList": [],
    "maxUtilizationList": []
  },
  "projectionDays": [
    0
  ],
  "scope": [
    {
      "uuid": "3e47ff3351f3247367c00c08fdc5bb974695c443",
      "displayName": "DC23-Datacenter\\DC23-cluster1",
      "groupType": "PhysicalMachine",
      "className": "Cluster"
    }
  ],
  "topologyChanges": {
    "addList": [
      {
        "count": 1,
        "projectionDays": [
          0
        ],
        "target": {
          "uuid": "67d24a6f265151876e87f39475fb4ee4b4170098",
          "displayName": "AvailabilitySet::ARSEN-AVS-NCUS",
          "className": "VirtualMachine"
        }
      }
    ],
    "migrateList": [],
    "removeList": [],
    "replaceList": [],
    "relievePressureList": []
```

```
  },
  "type": "WORKLOAD_MIGRATION"
}
```

Response:

```
{
  "uuid": "2502063893936",
  "displayName": "On-Prem Workload Migration- Basic",
  "owners": [
    {
      "uuid": "_4T_7kwY-Ed-WUKbEYSVIDw",
      "username": "administrator",
      "showSharedUserSC": false
    }
  ],
  "type": "WORKLOAD_MIGRATION",
  "scope": [
    {
      "uuid": "_0x3OYUglEd-gHc4L513yOA",
      "displayName": "Market",
      "className": "Market"
    }
  ],
  "projectionDays": [
    0
  ],
  "topologyChanges": {
    "addList": [
      {
        "projectionDays": [
          0
        ],
        "target": {
          "uuid": "a6d365f4fbf03fb9a2e8cdb29e38045e31cee41b",
          "displayName": "Beta DC6_vm",
          "className": "Group"
        },
        "count": 1
      }
    ]
  },
  "loadChanges": {},
  "configChanges": {
    "respectActionModeSetting": false
  },
  "changes": [
    {
      "index": 1,
      "type": "SCOPE",
      "scope": [
        {
          "uuid": "_0x3OYUglEd-gHc4L513yOA",
          "displayName": "Market",
          "className": "Market"
        }
      ]
    },
    {
      "index": 2,
      "type": "PROJECTION_PERIODS",
      "projectionDays": [
        0
      ]
    },
    {
      "index": 3,
      "type": "ADDED",
      "description": "1 Beta DC6_vm",
      "targets": [
        {
          "uuid": "a6d365f4fbf03fb9a2e8cdb29e38045e31cee41b",
          "displayName": "Beta DC6_vm",
          "className": "Group",
          "entitiesCount": 56,
          "groupType": "VirtualMachine",
          "severity": "Major",
          "logicalOperator": "AND"
        }
```

```
      ],
      "value": "1",
      "projectionDays": [
        0
      ]
    }
  ],
  "timebasedTopologyChanges": {}
}
```

Hardware refresh

POST:

```
{
  "configChanges": {
    "addPolicyList": [],
    "automationSettingList": [],
    "removeConstraintList": [],
    "removePolicyList": [],
    "riSettingList": [],
    "osMigrationSettingList": [],
    "subscription": {}
  },
  "displayName": "Hardware Refresh- Basic",
  "loadChanges": {
    "utilizationList": [],
    "maxUtilizationList": []
  },
  "projectionDays": [
    0
  ],
  "scope": [
    {
      "uuid": "Market",
      "displayName": "Global Environment",
      "className": "Market"
    }
  ],
  "topologyChanges": {
    "addList": [],
    "migrateList": [],
    "removeList": [],
    "replaceList": [
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-0042-5A10-804A-B1C04F384331",
          "displayName": "dell-pe1944.corp.vmturbo.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        },
        "template": {
          "uuid": "_juCtwJkfEeCX4rLBy_KD2g",
          "displayName": "HP_DL580 G7",
          "className": "PhysicalMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-004A-3610-8057-C8C04F464331",
          "displayName": "dell-pe1953.corp.vmturbo.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        },
        "template": {
          "uuid": "_juCtwJkfEeCX4rLBy_KD2g",
          "displayName": "HP_DL580 G7",
          "className": "PhysicalMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-0059-4D10-8030-B3C04F464331",
          "displayName": "dell-pe1954.corp.vmturbo.com",
          "className": "PhysicalMachine",
```

```
          "severity": "Critical"
        },
        "template": {
          "uuid": "_juCtwJkfEeCX4rLBy_KD2g",
          "displayName": "HP_DL580 G7",
          "className": "PhysicalMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "Virtual_ESX_4238e933-3923-8334-37fe-3aec8fbd046f",
          "displayName": "hp-esx22.dev.mycorp.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        },
        "template": {
          "uuid": "_juCtwJkfEeCX4rLBy_KD2g",
          "displayName": "HP_DL580 G7",
          "className": "PhysicalMachineProfile"
        }
      }
    ],
    "relievePressureList": []
  },
  "type": "RECONFIGURE_HARDWARE"
}
```

Response:

```
{
  "uuid": "2502070918448",
  "displayName": "Hardware Refresh- Basic",
  "owners": [
    {
      "uuid": "_4T_7kwY-Ed-WUKbEYSVIDw",
      "username": "administrator",
      "showSharedUserSC": false
    }
  ],
  "type": "RECONFIGURE_HARDWARE",
  "scope": [
    {
      "uuid": "_0x3OYUglEd-gHc4L513yOA",
      "displayName": "Market",
      "className": "Market"
    }
  ],
  "projectionDays": [
    0
  ],
  "topologyChanges": {
    "replaceList": [
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-0042-5A10-804A-B1C04F384331",
          "displayName": "dell-pe1944.corp.vmturbo.com",
          "className": "PhysicalMachine"
        },
        "template": {
          "uuid": "_juCtwJkfEeCX4rLBy_KD2g",
          "displayName": "HP_DL580 G7",
          "className": "PhysicalMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-004A-3610-8057-C8C04F464331",
          "displayName": "dell-pe1953.corp.vmturbo.com",
          "className": "PhysicalMachine"
        },
        "template": {
          "uuid": "_juCtwJkfEeCX4rLBy_KD2g",
          "displayName": "HP_DL580 G7",
          "className": "PhysicalMachineProfile"
        }
      },
```

```
      {
        "projectionDay": 0,
        "target": {
          "uuid": "4C4C4544-0059-4D10-8030-B3C04F464331",
          "displayName": "dell-pe1954.corp.vmturbo.com",
          "className": "PhysicalMachine"
        },
        "template": {
          "uuid": "_juCtwJkfEeCX4rLBy_KD2g",
          "displayName": "HP_DL580 G7",
          "className": "PhysicalMachineProfile"
        }
      },
      {
        "projectionDay": 0,
        "target": {
          "uuid": "Virtual_ESX_4238e933-3923-8334-37fe-3aec8fbd046f",
          "displayName": "hp-esx22.dev.mycorp.com",
          "className": "PhysicalMachine"
        },
        "template": {
          "uuid": "_juCtwJkfEeCX4rLBy_KD2g",
          "displayName": "HP_DL580 G7",
          "className": "PhysicalMachineProfile"
        }
      }
    ]
  },
  "loadChanges": {},
  "configChanges": {
    "respectActionModeSetting": false
  },
  "changes": [
    {
      "index": 1,
      "type": "SCOPE",
      "scope": [
        {
          "uuid": "_0x3OYUglEd-gHc4L513yOA",
          "displayName": "Market",
          "className": "Market"
        }
      ]
    },
    {
      "index": 2,
      "type": "PROJECTION_PERIODS",
      "projectionDays": [
        0
      ]
    },
    {
      "index": 3,
      "type": "REPLACED",
      "description": "Replace [dell-pe1944.corp.vmturbo.com, dell-pe1953.corp.vmturbo.com, dell-
pe1954.corp.vmturbo.com, hp-esx... with HP_DL580 G7",
      "targets": [
        {
          "uuid": "4C4C4544-0042-5A10-804A-B1C04F384331",
          "displayName": "dell-pe1944.corp.vmturbo.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        },
        {
          "uuid": "4C4C4544-004A-3610-8057-C8C04F464331",
          "displayName": "dell-pe1953.corp.vmturbo.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        },
        {
          "uuid": "4C4C4544-0059-4D10-8030-B3C04F464331",
          "displayName": "dell-pe1954.corp.vmturbo.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        },
        {
          "uuid": "Virtual_ESX_4238e933-3923-8334-37fe-3aec8fbd046f",
          "displayName": "hp-esx22.dev.mycorp.com",
          "className": "PhysicalMachine",
          "severity": "Critical"
        },
        {
```

```
          "uuid": "_juCtwJkfEeCX4rLBy_KD2g",
          "displayName": "HP_DL580 G7",
          "className": "PhysicalMachineProfile"
        }
      ],
      "value": "0",
      "projectionDays": [
        0
      ]
    }
  ],
  "timebasedTopologyChanges": {}
}
```

Alleviate pressure

POST:

```
{
  "configChanges": {
    "addPolicyList": [],
    "automationSettingList": [],
    "removeConstraintList": [],
    "removePolicyList": [],
    "riSettingList": [],
    "osMigrationSettingList": [],
    "subscription": {}
  },
  "displayName": "Alleviate Pressure- Basic",
  "loadChanges": {
    "utilizationList": [],
    "maxUtilizationList": []
  },
  "projectionDays": [
    0
  ],
  "scope": [
    {
      "uuid": "11c63ebd77647e7b9c9fb85dd2d4c77464cbeac2",
      "displayName": "DC13\\DC13-Cluster",
      "className": "Cluster"
    },
    {
      "uuid": "57a4026a69ba4b46ebc430cb62d5c55eeea00695",
      "displayName": "DC13\\Physical Hosts_Cluster2 - Intel",
      "className": "Cluster"
    }
  ],
  "topologyChanges": {
    "addList": [],
    "migrateList": [],
    "removeList": [],
    "replaceList": [],
    "relievePressureList": [
      {
        "projectionDay": 0,
        "sources": [
          {
            "uuid": "11c63ebd77647e7b9c9fb85dd2d4c77464cbeac2",
            "displayName": "DC13\\DC13-Cluster",
            "className": "Cluster"
          }
        ],
        "destinations": [
          {
            "uuid": "57a4026a69ba4b46ebc430cb62d5c55eeea00695",
            "displayName": "DC13\\Physical Hosts_Cluster2 - Intel",
            "className": "Cluster"
          }
        ]
      }
    ]
  },
  "type": "ALLEVIATE_PRESSURE"
}
```

Response:


```
{
  "uuid": "2531404249984",
  "displayName": "On-Prem Workload Migration 1",
  "owners": [
    {
      "uuid": "_4T_7kwY-Ed-WUKbEYSVIDw",
      "username": "administrator",
      "showSharedUserSC": false
    }
  ],
  "type": "WORKLOAD_MIGRATION",
  "scope": [
    {
      "uuid": "3e47ff3351f3247367c00c08fdc5bb974695c443",
      "displayName": "DC23-Datacenter\\DC23-cluster1",
      "className": "Cluster",
      "groupType": "PhysicalMachine",
      "isStatic": true,
      "logicalOperator": "AND"
    }
  ],
  "projectionDays": [
    0
  ],
  "topologyChanges": {
    "addList": [
      {
        "projectionDays": [
          0
        ],
        "target": {
          "uuid": "67d24a6f265151876e87f39475fb4ee4b4170098",
          "displayName": "AvailabilitySet::ARSEN-AVS-NCUS",
          "className": "DiscoveredGroup"
        },
        "count": 1
      }
    ]
  },
  "loadChanges": {},
  "configChanges": {
    "respectActionModeSetting": false"
  },
  "changes": [
    {
      "index": 1,
      "type": "SCOPE",
      "scope": [
        {
          "uuid": "3e47ff3351f3247367c00c08fdc5bb974695c443",
          "displayName": "DC23-Datacenter\\DC23-cluster1",
          "className": "Cluster",
          "entitiesCount": 3,
          "groupType": "PhysicalMachine",
          "severity": "Critical",
          "logicalOperator": "AND"
        }
      ]
    },
    {
      "index": 2,
      "type": "PROJECTION_PERIODS",
      "projectionDays": [
        0
      ]
    },
    {
      "index": 3,
      "type": "ADDED",
      "description": "1 AvailabilitySet::ARSEN-AVS-NCUS",
      "targets": [
        {
          "uuid": "67d24a6f265151876e87f39475fb4ee4b4170098",
          "displayName": "AvailabilitySet::ARSEN-AVS-NCUS",
          "className": "DiscoveredGroup",
          "entitiesCount": 3,
          "groupType": "VirtualMachine",
          "severity": "Critical",
          "logicalOperator": "AND"
```

```
        }
      ],
      "value": "1",
      "projectionDays": [
        0
      ]
    }
  ],
  "timebasedTopologyChanges": {}
}
```

Optimize current environment

The result of a plan run with no custom configuration is an optimization of your current environment, respecting all current policies and constraints.

POST:

```
{
  "configChanges": {
    "addPolicyList": [],
    "automationSettingList": [],
    "removeConstraintList": [],
    "removePolicyList": [],
    "riSettingList": [],
    "osMigrationSettingList": [],
    "subscription": {}
  },
  "displayName": "Custom- No Config",
  "loadChanges": {
    "utilizationList": [],
    "maxUtilizationList": []
  },
  "projectionDays": [],
  "scope": [
    {
      "uuid": "Market",
      "displayName": "Global Environment",
      "className": "Market"
    }
  ],
  "topologyChanges": {
    "addList": [],
    "migrateList": [],
    "removeList": [],
    "replaceList": [],
    "relievePressureList": []
  },
  "type": "CUSTOM"
}
```

Response:

```
{
  "uuid": "2502119526800",
  "displayName": "Custom- No Config",
  "owners": [
    {
      "uuid": "_4T_7kwY-Ed-WUKbEYSVIDw",
      "username": "administrator",
      "showSharedUserSC": false
    }
  ],
  "type": "CUSTOM",
  "scope": [
    {
      "uuid": "_0x3OYUglEd-gHc4L513yOA",
      "displayName": "Market",
      "className": "Market"
    }
  ],
  "changes": [
    {
      "index": 1,
      "type": "SCOPE",
```

```
      "scope": [
        {
          "uuid": "_0x3OYUglEd-gHc4L513yOA",
          "displayName": "Market",
          "className": "Market"
        }
      ]
    }
  ]
}
```

Scenario future load configuration example

Future load plans

Future load plans are a type of plan in which commodities from active VMs are taken and applied to passive or low-workload VMs, simulating the effect on the environment if these passive VMs became active. For example, you may want to investigate the effect on your environment if your disaster recovery VMs were forced to run at historical peak values. In this case, the production VMs whose commodities will be used are the active VMs, and the currently idle disaster recovery VMs are the passive VMs.

Currently, there are two future load plan types:

- **PEAKOFPEAKS**: The peak-of-peaks future load plan replicates in the passive VMs every commodity of the active VMs reaching its historical peak utilization simultaneously, and the plan result will reflect the effect on your environment.
- **CUSTOM**: The custom future load plan replicates in the passive VMs every commodity of the active VMs reaching its current utilization simultaneously, and the plan result will reflect the effect on your environment.

The setup and configuration for both future load plans differ only in the content of the `type` parameter- all other steps will be identical.

The settings you make in a scenario correspond to the plan settings you can make in the user interface. These include:

- Plan scope
- Changes to workload (adding, removing, or replacing VMs or containers)
- Changes to supply (adding, removing, replacing PMs or storage)
- Enable/disable placement policies and other constraints
- Changes to action modes
- Enable/disable provisioning of supply
- Enable/disable resizing of workloads

Future load prerequisites

In order to run a future load plan, you must retrieve statistics for the entities to be included in the plan. Typically, this is done utilizing the `POST /markets/market_uuid/entities/stats` request, with `startDate` and `scopes` parameters.

Example input for statistic retrieval:

```
{
  "period": {
    "startDate": 1237056032
  },
  "scopes": [
    "4223ae0c-c99b-f6ca-c1ef-3f367bdf5528",
    "4223bb37-0090-f332-0140-cc3cce7b3735",
    "4223dc55-78f2-2cd7-9abf-b09b1be835b4",
    "42231932-83af-10d9-f0cd-26c8a08a9e18",
    "422374a3-2c64-b613-a9a7-1f900a2314b4"
  ]
```

```
}
```

Once the call is completed, copy the response in its entirety- or if using Swagger, click the Download button next to the response and copy the contents of the file.

Running the future load plan

After creating the ScenarioApiDTO, the response will include the uuid of the created scenario. Use this uuid along with the uuid of the market to run the plan against (typically the real-time market) in the `POST /markets/market_uuid/scenarios/scenario_uuid` request.

Viewing the peak of peak plan

After running the plan, the response will include the uuid of the created plan market, which you should note. Results can be viewed using the `GET /markets/market_uuid` request.

Schedules endpoint

Schedules specify a period of time during which certain events can occur. Turbonomic uses the following types of schedules:

- Calendar Schedule

  A calendar schedule is a setting that sets up a window of time when an *automation policy* takes effect. This policy can execute *non-parking* actions on public cloud or on-prem entities, or change settings that affect analysis and action generation.

  To configure calendar schedules, you must use `/schedules` requests. With these requests, you can:

  – Get calendar schedules
  – Create calendar schedules
  – Edit or delete calendar schedules
  – View all policies using a specific calendar schedule

  Note: When you configure a schedule window for a resize action, to ensure Turbonomic will execute the action during the scheduled time, you must turn off the Enforce Non Disruptive Mode setting for that scheduled policy. Even if you turn the setting off for the global policy, you still must turn the setting off for your scheduled policy. Otherwise Turbonomic will not execute the resize action.

- Parking Schedule

  A parking schedule is a setting that sets up a window of time when *parking* actions (stop or start) are enforced on entities in your public cloud environment.

  To configure parking schedules, you must use `/schedules/timespans/` requests. With these requests, you can:

  – Get parking schedules
  – Create parking schedules
  – Attach parking schedules to parkable entities
  – Detach parking schedules from parkable entities
  – Edit or delete parking schedules

Calendar schedules requests

The requests listed in this topic only apply to calendar schedules used in automation policies. For details about parking schedules that can be attached to parkable entities, see Parking Schedules Requests.

Getting calendar schedules

Gets a list of schedules. For a list of schedules, the API returns an array of ScheduleApiDTOs. Each ScheduleApiDTO will contain details about the schedule, including the start and end time and date and recurrence. To get a single schedule, include the UUID of the notification in the request.

Examples:

- **All Schedules:** `GET https://10.10.10.10/api/v3/schedules`
- **Single Schedule:** `GET https://10.10.10.10/api/v3/schedules/284466929277824`

Response:

```
[
  {
    "uuid": "284466927892512",
    "displayName": "DalSched1",
    "startTime": "2020-02-29T16:30",
    "endTime": "2020-02-29T19:30",
    "recurrence": {
      "type": "WEEKLY",
      "daysOfWeek": [
        "Mon",
        "Fri"
      ],
      "interval": 1
    },
    "timeZone": "America/Denver",
    "nextOccurrence": "2020-03-02T16:30:00",
    "nextOccurrenceTimestamp": 1583191800000
  },
  {
    "uuid": "284466929277824",
    "displayName": "DalSched2",
    "startTime": "2020-02-29T16:30",
    "endTime": "2020-02-29T17:45",
    "recurrence": {
      "type": "MONTHLY",
      "daysOfMonth": [
        5
      ],
      "interval": 3
    },
    "timeZone": "America/Denver",
    "nextOccurrence": "2020-05-05T16:30:00",
    "nextOccurrenceTimestamp": 1588717800000
  }
]
```

Adding a calendar schedule

In order to create a schedule in Turbonomic, you must provide a valid ScheduleApiDTO.

Example: `POST https://10.10.10.10/api/v3/schedules`

Example input, showing a schedule to be run weekly on Friday between the hours of 0000 and 0400 that starts on 2019-12-20 and has no end date:

```
{
  "displayName":"DallasSched",
  "endTime":"2019-12-20T04:00",
  "startTime":"2019-12-20T00:00",
  "timeZone":"America/Denver",
  "recurrence":{
    "type":"WEEKLY",
    "interval":1,
    "daysOfWeek":[
      "Fri"
    ]
  }
}
```

Getting policies using a calendar schedule

Given a schedule ID, retrieve any policies associated with that schedule.

Example: `GET https://10.10.10.10/api/v3/schedules/284466929277824/settingsPolicy`

Response: An array of `SettingspolicyApiDTOs` that represent the policies using the specified schedule.


```
[
  {
    "uuid": "284466929430656",
    "displayName": "DalStorage",
    "entityType": "Storage",
    "scopes": [
      {
        "uuid": "284460958528132",
        "displayName": "HawthorneHC_datastore",
        "isStatic": true,
        "logicalOperator": "AND"
      }
    ],
    "settingsManagers": [
      {
        "uuid": "automationmanager",
        "displayName": "Action Mode Settings",
        "category": "Automation",
        "settings": [
          {
            "uuid": "suspend",
            "displayName": "Suspend",
            "value": "MANUAL",
            "defaultValue": "MANUAL",
            "valueType": "STRING",
            "valueObjectType": "String",
            "options": [
              {
                "label": "Disabled",
                "value": "DISABLED"
              },
              {
                "label": "Recommend",
                "value": "RECOMMEND"
              },
              {
                "label": "Manual",
                "value": "MANUAL"
              },
              {
                "label": "Automatic",
                "value": "AUTOMATIC"
              }
            ],
            "entityType": "Storage"
          }
        ]
      }
    ],
    "schedule": {
      "uuid": "284466929277824",
      "displayName": "DalSched2",
      "startTime": "2020-02-29T16:30",
      "endTime": "2020-02-29T17:45",
      "recurrence": {
        "type": "MONTHLY",
        "daysOfMonth": [
          5
        ],
        "interval": 3
      },
      "timeZone": "America/Denver",
      "nextOccurrence": "2020-05-05T16:30:00",
      "nextOccurrenceTimestamp": 1588717800000
    },
    "disabled": false,
    "readOnly": false,
    "default": false
  }
```

```
]
```

Calendar schedules tips

When creating schedules, you can set multiple timing criteria. The following examples show schedules with advanced timing. For basic schedules, see the Schedules Endpoint.

Bi-weekly calendar schedule with specific days

This schedule is set to run biweekly, indicated by the `"type": "WEEKLY"` and `"interval": 2` fields. As indicated by the `daysOfWeek` array, this schedule is active on Monday, Wednesday, and Friday.

InputDTO:

```
{
  "displayName":"Schedule1",
  "endDate":"2020-07-10",
  "endTime":"2020-05-06T19:30",
  "startTime":"2020-05-06T15:30",
  "timeZone":"America/Denver",
  "recurrence":{
    "type":"WEEKLY",
    "interval":2,
    "daysOfWeek":[
      "Mon",
      "Wed",
      "Fri"
    ]
  }
}
```

The API response for this schedule:

```
{
  "uuid":"284559431391120",
  "displayName":"DallasTest",
  "endDate":"2020-07-10",
  "startTime":"2020-05-06T15:30",
  "endTime":"2020-05-06T19:30",
  "recurrence":{
    "type":"WEEKLY",
    "daysOfWeek":[
      "Mon",
      "Wed",
      "Fri"
    ],
    "interval":2
  },
  "timeZone":"America/Denver",
  "nextOccurrence":"2020-05-06T15:30:00",
  "nextOccurrenceTimestamp":1588800600000
}
```

Six-month calendar schedule with specific week of the month

This schedule is set to run every six months, indicated by the `"type": "MONTHLY"` and `"interval": 6` fields. As indicated by the `daysOfWeek` array this schedule is active on Wednesday, and the `"weekOfTheMonth": 2` indicates that this schedule will only run on the second week of the month.

This schedule was created on May 5th, 2020. Converting the timestamp in the API Response to a human-readable date, you see that the next run is on Wednesday, May 13, 2020 3:30:00 PM, the second Wednesday in May.

Input DTO:

```
{
  "displayName":"DallasTest2",
  "endDate":"2020-10-19",
  "endTime":"2020-05-06T19:15",
  "startTime":"2020-05-06T15:30",
  "timeZone":"America/Denver",
  "recurrence":{
    "type":"MONTHLY",
    "interval":6,
    "daysOfWeek":[
      "Wed"
    ],
    "weekOfTheMonth":[
      2
    ]
  }
}
```

The API response for this schedule:

```
{
  "uuid":"284559434037456",
  "displayName":"DallasTest2",
  "endDate":"2020-10-19",
  "startTime":"2020-05-06T15:30",
  "endTime":"2020-05-06T19:15",
  "recurrence":{
    "type":"MONTHLY",
    "daysOfWeek":[
      "Wed"
    ],
    "weekOfTheMonth":[
      2
    ],
    "interval":6
  },
  "timeZone":"America/Denver",
  "nextOccurrence":"2020-05-13T15:30:00",
  "nextOccurrenceTimestamp":1589405400000
}
```

Getting policies attached to a specific calendar schedule

In order to see what policies are using a schedule, use the `https://10.10.10.10/api/v3/schedules/{schedule_UUID}/settingsPolicy` request.

Response: An array of `SettingspolicyApiDTO`s that represent the policies using the specified schedule, whose details are also included.

```
[
  {
    "uuid": "284466929430656",
    "displayName": "DalStorage",
    "entityType": "Storage",
    "scopes": [
      {
        "uuid": "284460958528132",
        "displayName": "HawthorneHC_datastore",
        "isStatic": true,
        "logicalOperator": "AND"
      }
    ],
    "settingsManagers": [
      {
        "uuid": "automationmanager",
        "displayName": "Action Mode Settings",
        "category": "Automation",
        "settings": [
          {
```

```
            "uuid": "suspend",
            "displayName": "Suspend",
            "value": "MANUAL",
            "defaultValue": "MANUAL",
            "valueType": "STRING",
            "valueObjectType": "String",
            "options": [
              {
                "label": "Disabled",
                "value": "DISABLED"
              },
              {
                "label": "Recommend",
                "value": "RECOMMEND"
              },
              {
                "label": "Manual",
                "value": "MANUAL"
              },
              {
                "label": "Automatic",
                "value": "AUTOMATIC"
              }
            ],
            "entityType": "Storage"
          }
        ]
      }
    ],
    "schedule": {
      "uuid": "284466929277824",
      "displayName": "DalSched2",
      "startTime": "2020-02-29T16:30",
      "endTime": "2020-02-29T17:45",
      "recurrence": {
        "type": "MONTHLY",
        "daysOfMonth": [
          5
        ],
        "interval": 3
      },
      "timeZone": "America/Denver",
      "nextOccurrence": "2020-05-05T16:30:00",
      "nextOccurrenceTimestamp": 1588717800000
    },
    "disabled": false,
    "readOnly": false,
    "default": false
  }
]
```


Parking schedules requests

The requests listed in this topic currently only apply to parking schedules and parkable entities. For details about calendar schedules used in automation policies, see Calendar Schedules Requests.

Getting parking schedules

Gets a list of parking schedules. For a list of schedules, the API returns an array of parking schedule metadata. To get a single parking schedule, include the parking schedule UUID in the URL.

Examples:

- **All Schedules:** `GET https://10.10.10.10/api/v3/schedules/timespans`
- **Single Schedule:** `GET https://10.10.10.10/api/v3/schedules/timespans/778668020368144`

All Schedules Response:


```
[
  {
    "uuid": "778601653004480",
    "displayName": "testing",
    "timeZone": "America/New_York"
  },
```

```
  {
    "uuid": "778603154693344",
    "displayName": "everyday of the week",
    "timeZone": "America/New_York"
  },
  {
    "uuid": "778668020368144",
    "displayName": "Sample Schedule",
    "description": "This is a sample schedule where entities are active during normal business
hours but may undergo maintenance in the early hours of Sunday.",
    "timeZone": "America/Chicago"
  }
]
```


Single Schedule Response:


```
{
  "uuid": "778668020368144",
  "displayName": "Sample Schedule",
  "description": "This is a sample schedule where entities are active during normal business
hours but may undergo maintenance in the early hours of Sunday.",
  "timeZone": "America/Chicago",
  "timeSpans": {
    "type": "WeekDayTimeSpansApiDTO",
    "sunday": [
      {
        "begins": "00:00",
        "ends": "01:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      },
      {
        "begins": "02:00",
        "ends": "03:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "IGNORE"
        }
      },
      {
        "begins": "04:00",
        "ends": "23:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      }
    ],
    "monday": [
      {
        "begins": "00:00",
        "ends": "06:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      },
      {
        "begins": "07:00",
        "ends": "16:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "ON"
        }
      },
      {
        "begins": "17:00",
        "ends": "23:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      }
    ],
    "tuesday": [
      {
        "begins": "00:00",
```

```
        "ends": "06:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      },
      {
        "begins": "07:00",
        "ends": "16:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "ON"
        }
      },
      {
        "begins": "17:00",
        "ends": "23:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      }
    ],
    "wednesday": [
      {
        "begins": "00:00",
        "ends": "06:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      },
      {
        "begins": "07:00",
        "ends": "16:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "ON"
        }
      },
      {
        "begins": "17:00",
        "ends": "23:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      }
    ],
    "thursday": [
      {
        "begins": "00:00",
        "ends": "06:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      },
      {
        "begins": "07:00",
        "ends": "16:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "ON"
        }
      },
      {
        "begins": "17:00",
        "ends": "23:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      }
    ],
    "friday": [
      {
        "begins": "00:00",
        "ends": "06:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
```

```
        }
      },
      {
        "begins": "07:00",
        "ends": "16:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "ON"
        }
      },
      {
        "begins": "17:00",
        "ends": "23:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      }
    ],
    "saturday": [
      {
        "begins": "00:00",
        "ends": "23:59",
        "policy": {
          "type": "ParkingItemApiDTO",
          "state": "OFF"
        }
      }
    ]
  }
}
```

Adding a parking schedule

To create a parking schedule, you must provide a valid ParkingItemApiDTO.

Example: `POST https://10.10.10.10/api/v3/schedules/timespans`

Example input, showing a schedule that causes entities to stay powered on Monday through Friday from 7:00 AM to 5:00 PM. Entities are off all other times except for a two-hour window on Sunday morning between 2:00 AM and 4:00 AM where the schedule does not enforce a powered state:

```
{
  "displayName": "Sample Schedule",
  "timeZone": "America/Chicago",
  "description": "This is a sample schedule where entities are active during normal business
hours but may undergo maintenance in the early hours of Sunday.",
  "timeSpans": {
    "type": "WeekDayTimeSpansApiDTO",
    "monday": [
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "00:00",
        "ends": "06:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "ON"},
        "begins": "07:00",
        "ends": "16:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "17:00",
        "ends": "23:59"
      }
    ],
    "tuesday": [
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "00:00",
        "ends": "06:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "ON"},
        "begins": "07:00",
        "ends": "16:59"
      },
      {
```

```
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "17:00",
        "ends": "23:59"
      }
    ],
    "wednesday": [
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "00:00",
        "ends": "06:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "ON"},
        "begins": "07:00",
        "ends": "16:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "17:00",
        "ends": "23:59"
      }
    ],
    "thursday": [
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "00:00",
        "ends": "06:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "ON"},
        "begins": "07:00",
        "ends": "16:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "17:00",
        "ends": "23:59"
      }
    ],
    "friday": [
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "00:00",
        "ends": "06:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "ON"},
        "begins": "07:00",
        "ends": "16:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "17:00",
        "ends": "23:59"
      }
    ],
    "saturday": [
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "00:00",
        "ends": "23:59"
      }
    ],
    "sunday": [
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "00:00",
        "ends": "01:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "IGNORE"},
        "begins": "02:00",
        "ends": "03:59"
      },
      {
        "policy": {"type": "ParkingItemApiDTO", "state": "OFF"},
        "begins": "04:00",
        "ends": "23:59"
      }
    ]
```

```
  }
}
```

Constraints:

- `displayName` value must be unique.
- `timeZone` value must be consistent with the IANA Timezone Database (for example, UTC or America/New_York).
- `begins` and `ends` values are HH:MM representations of a 24-hour clock and are considered inclusive.
- If time spans (or days of the week) are omitted, the system behaves the same as if a corresponding time span with a policy state of `IGNORE` exists.
- Time spans cannot overlap.

Attaching parking schedules to entities

Once a parking schedule is created and your parkable entities are discovered, you can attach a schedule to one or more entities.

Example: `POST https://10.10.10.10/api/v3/schedules/timespans/778668074428784/entities`

Constraints:

- Parking schedule only.
- Entity must be a `VirtualMachine` discovered from cloud targets (AWS, Azure, or GCP), or a `DatabaseServer` (AWS RDS instance).

Example input that attaches an existing parking schedule to 1-N entities:

```
{"entityUuids":["74913973295604","74913973295605","74913973295606"]}
```

Detaching parking schedules

You can detach a parking schedule from one or more entities.

Example: `DELETE https://10.10.10.10/api/v3/schedules/timespans/778668074428784/entities`

Example input that detaches an existing parking schedule from a 1-N entity.

```
{"entityUuids":["74913973295604","74913973295606"]}
```

Constraints: Parking schedule only.

Updating a parking schedule

Update an existing parking schedule. The following example shows a request to update a schedule to enforce a powered off state from midnight Monday to midnight Tuesday. The schedule does not enforce a state any other time.

Example: `PUT https://10.10.10.10/api/v3/schedules/timespans/778668020368144`

Response: An array of `SettingspolicyApiDTO`s that represent the policies using the specified schedule.

```
{
  "timeZone": "America/Chicago",
  "displayName": "Sample Schedule - Updated",
  "description": "Do nothing all week, except on Monday where the attached VMs should be
powered off all day",
  "timeSpans": {
    "type": "WeekDayTimeSpansApiDTO",
    "monday": [
      {
```

```
        "policy": {
          "state": "OFF",
          "type": "ParkingItemApiDTO"
        },
        "begins": "00:00",
        "ends": "23:59"
      }
    ]
  }
}
```

Constraints:

- `displayName` value must be unique.
- `timeZone` value must be consistent with the IANA Timezone Database (for example, UTC or America/New_York).
- `begins` and `ends` values are HH:MM representations of a 24-hour clock and are considered inclusive.
- If time spans (or days of the week) are omitted, the system behaves the same as if a corresponding time span with a policy state of `IGNORE` exists.
- Time spans may not overlap.

Deleting a parking schedule

Remove an unattached parking schedule.

Example: `DELETE https://10.10.10.10/api/v3/schedules/timespans/778668020368144`

Constraints: Parking schedules to be deleted must not be attached to any entity.

Search endpoint

A search is a query performed on the entities that participate in the Turbonomic market. The most common use of search is to locate a particular object, service entity, or group of entities that share a characteristic.

`Object` and `Entity` refer to different items in your Turbonomic topology:

Using the search endpoint of the Turbonomic API, you can:

- Obtain criteria for use in detailed searching or building groups
- Get detailed information about any object in your Turbonomic environment

Search utility requests

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

The search endpoint contains a utility request that shows the various criteria that you may use to search, for each entity type in the Turbonomic supply chain, and a second that shows the options for a specified criterion. You can then use this information to build your search criteria for more detailed groups.

The available filterTypes depend on the entity making up your search:

- ApplicationComponentSpec
  - appComponentSpecsByName
  - appComponentSpecsByBusinessAccountUuid
  - appComponentSpecsByHybridConnectionCount
  - appComponentSpecsByDeploymentSlotCount
- ApplicationComponent
  - appCompsByName
  - applicationComponentByBusinessApplication

  - applicationComponentByBusinessTransaction
  - applicationComponentByService
- AvailabilityZone
  - zonsByName
  - zoneByCloudProvider
- BillingFamily
  - billingFamilyByName
- BusinessAccountFolder
  - businessAccountFolderByName
- BusinessAccount
  - businessAccountByName
  - businessAccountByUuid
  - businessAccountByVendorId
  - businessAccountCloudProvider
  - businessAccountProbeType
  - businessAccountTargetName
  - businessAccountValidationStatus
  - businessAccountByResourceGroup
  - businessAccountByTag
  - subBusinessAccountOfUuid
- BusinessApplication
  - busAppsByName
  - businessApplicationByUserDefinedEntity
- BusinessTransaction
  - busTransByName
  - businessTransactionByBusinessApplication
  - businessTransactionByUserDefinedEntity
- BusinessUser
  - businessUserByName
  - businessUserByDesktopPool
  - businessUserByViewPod
- Chassis
  - chasByName
- CloudService
- Cluster
  - clustersByName
  - clustersByTag
- ComputeTier
  - computeTiersByName
  - computeTiersByMem
  - computeTiersByNumCPUs
  - computeTiersByCloudProvider

- ContainerPlatformCluster
  - containerPlatformClustersByName
- Container
  - containersByName
  - containersByPodName
  - containersByNamespace
  - containersByApplicationComponent
  - containersByBusinessApplication
  - containersByBusinessTransaction
  - containersByService
  - containersByControllerKind
  - containersByTag
  - containersByContainerPlatformCluster
- ContainerPod
  - containerPodsByName
  - containerPodsByVMName
  - containerPodsByNamespace
  - containerPodsByTag
  - containerPodsByControllerKind
  - containerPodsByWorkloadControllerName
  - containerPodsByContainerPlatformCluster
  - containerPodsByState
- ContainerSpec
  - containerSpecsByName
  - containerSpecsByNamespace
  - containerSpecsByControllerKind
  - containerSpecsByContainerPlatformCluster
  - containerSpecsByTag
- DataCenter
  - datacentersByName
  - datacentersByTag
- Database
  - databaseByName
  - databaseByTag
  - databaseByBusinessAccountUuid
  - databaseByResourceGroupName
  - databaseByCloudProvider
  - databaseByApplicationComponent
  - databaseByBusinessApplication
  - databaseByBusinessTransaction
  - databaseByService
  - dbByAvailabilityZoneName
  - dbByRegionName

  - dbByDatabaseServerName
  - databaseByReplicationRole
  - databaseByPricingModel
  - databaseByServiceTier
  - databaseByHoursIdle
  - databaseByAge
- DatabaseServer
  - databaseServerByName
  - databaseServerByTag
  - databaseServerByBusinessAccountUuid
  - databaseServerByResourceGroupName
  - databaseServerByCloudProvider
  - databaseServerByApplicationComponent
  - databaseServerByBusinessApplication
  - databaseServerByBusinessTransaction
  - databaseServerByService
  - databaseServerByEngine
  - databaseServerByEdition
  - databaseServerByVersion
  - databaseServerByState
  - databaseServerByServerConfiguration
  - databaseServerByStorageAutoscaling
  - databaseServerByPerformanceInsights
  - databaseServerByStorageEncryption
  - databaseServerByClusterRole
  - databaseServerByStorageTier
  - databaseServerByAvailabilityZoneName
  - databaseServerByRegionName
  - databaseServerByAge
- DatabaseServerTier
  - databaseServerTiersByName
  - databaseServerTiersByVMem
  - databaseServerTiersByCloudProvider
- DatabaseTier
  - databaseTiersByName
  - databaseTiersByCloudProvider
- DesktopPool
  - desktopPoolByName
  - desktopPoolByViewPod
- DiskArray
  - diskarrayByName
- DocumentCollection
  - dcByName

  - dcByTag
  - dcByBusinessAccount
  - dcByResourceGroup
  - dcByCloudProvider
  - dcByRegion
  - dcByDatabaseServer
- Group
  - groupsByName
- HCIPhysicalMachine
- IOModule
  - ioModuleByName
- Internet
  - internetByName
- LoadBalancer
  - lbsByName
- LogicalPool
  - logicalPoolByName
- Namespace
  - namespacesByName
  - namespacesByContainerPlatformCluster
  - namespacesByTag
- Network
  - netsByName
- NodePool
- PhysicalMachine
  - pmsByName
  - pmsByTargetType
  - pmsByClusterName
  - pmsByDC
  - pmsByState
  - pmsByTag
  - pmsByStorage
  - pmsBySwitch
  - pmsByMem
  - pmsByNumVms
  - pmsByNumCPUs
  - pmsByVendorName
  - pmsByCPUModel
  - pmsByModel
  - pmsByTimezone
  - pmsBySupportedProcessorCompatibilityModes
  - pmsByClusterTag

- ProcessorPool
- Region
  - regsByName
  - regsByCloudProvider
- ReservedInstance
- ResourceGroup
  - resourceGroupByName
  - resourceGroupByTag
  - resourceGroupByBusinessAccountUuid
- Service
  - servicesByName
  - serviceByNamespace
  - serviceByContainerPlatformCluster
  - serviceByKubernetesServiceType
  - serviceByBusinessApplication
  - serviceByTag
  - serviceByBusinessTransaction
  - serviceByUserDefinedEntity
- Storage
  - storageByName
  - storageByVMs
  - storageByPMCluster
  - storageByStorageCluster
  - storageByState
  - storageByTag
  - storageByDC
  - storageWithLocalSupport
- StorageCluster
  - storageClustersByName
- StorageController
  - storagecontrollerByName
- StorageTier
  - storageTierByName
  - storageTiersByCloudProvider
- Switch
  - switchByName
- ViewPod
  - viewPodByName
- VirtualDataCenter
  - vdcsByName
  - vdcsByTag
- VirtualMachine

- vmsByName
- vmsByPMName
- vmsByVDC
- vmsByDCnested
- vmsByDiskArrayName
- vmsByStorage
- vmsByApplicationComponent
- vmsByBusinessApplication
- vmsByBusinessTransaction
- vmsByService
- vmsByClusterName
- vmsByStorageClusterName
- vmsByDC
- vmsByState
- vmsByTag
- vmsConnectedToEphemeralStorage
- vmsHotAddMemory
- vmsHotAddCPU
- vmsHotRemoveCPU
- vmsConnectedToEncryptedVolume
- vmsByGuestName
- vmsBySockets
- vmsByCoresPerSocket
- vmsByNumCPUs
- vmsByAge
- vendorToolsInstalled
- vendorToolsVersion
- vmsByProcessorCompatibilityMode
- vmsBySharingMode
- vmsByHasDedicatedProcessors
- vmByTargetCategory
- vmByTargetName
- vmByTargetType
- vmsByBusinessAccountUuid
- vmsByMem
- vmsByStorageAmount
- vmsByLogicalPoolName
- vmsByNetwork
- vmsByResourceGroupName
- vmsByAvailabilityZoneName
- vmsByRegionName
- vmsByCloudProvider
- vmsByContainerPlatformCluster

  - vmsByActiveSessions
  - vmsByVendorId
  - vmByComputeTier
- VirtualMachineCluster
  - virtualMachineClustersByName
- VirtualMachineSpec
  - virtualMachineSpecsByName
  - virtualMachineSpecsByTier
  - virtualMachineSpecsByAppCount
  - virtualMachineSpecsByBusinessAccountUuid
  - virtualMachineSpecsByRegionName
  - virtualMachineSpecsByResourceGroup
  - virtualMachineSpecsByComputeTier
  - virtualMachineSpecsByTag
  - virtualMachineSpecsByNumDaysEmpty
- VirtualVolume
  - virtualVolumeByName
  - virtualVolumeWithLocalSupport
  - volumeByTag
  - volumeByAccount
  - volumeByConnectedVm
  - volumeByAttachmentState
  - volumeByNumDaysUnattached
  - volumeByTier
  - volumeByRegion
  - volumeByCloudProvider
  - volumeById
  - volumeByResourceGroupName
  - volumeByDeletable
  - volumeByStorage
  - volumeByStorageGroup
  - volumeByAge
- Workload
  - workloadByName
- WorkloadController
  - workloadControllersByName
  - workloadControllersByNamespace
  - workloadControllersByKind
  - workloadControllersByContainerPlatformCluster
  - workloadControllersByTag

Example: `GET https://10.10.10.10/api/v3/search/criteria`

Response: An array of filter criteria, separated by entity type. For example, this is the filter criteria for Virtual Machine entities:


```
"VirtualMachine": {
    "criteria": [
      {
        "inputType": "*",
        "elements": "displayName",
        "filterCategory": "property",
        "filterType": "vmsByName"
      },
      {
        "inputType": "*",
        "elements": "HostedBy:displayName",
        "filterCategory": "entity",
        "filterType": "vmsByPMName"
      },
      {
        "inputType": "*",
        "elements": "LayeredOver:Storage:displayName",
        "filterCategory": "entity",
        "filterType": "vmsByStorage"
      },
      {
        "inputType": "*",
        "elements": "LayeredOver:Network:displayName",
        "filterCategory": "entity",
        "filterType": "vmsByNetwork"
      },
      {
        "inputType": "*",
        "elements": "Hosts:Application:displayName",
        "filterCategory": "entity",
        "filterType": "vmsByApplication"
      },
      {
        "inputType": "*",
        "elements": "HostedBy:PhysicalMachine:HostedBy:DataCenter:displayName",
        "filterCategory": "entity",
        "filterType": "vmsByDC"
      },
      {
        "inputType": "*",
        "elements": "LayeredOver:VirtualDataCenter:displayName",
        "filterCategory": "entity",
        "filterType": "vmsByVDC"
      },
      {
        "inputType": "*",
        "elements": "AllMemberOf:VirtualDataCenter:displayName",
        "filterCategory": "entity",
        "filterType": "vmsByDCnested"
      },
      {
        "inputType": "#",
        "elements": "ExtendedBy:VMTopoExt:numCpus",
        "filterCategory": "property",
        "filterType": "vmsByNumCPUs"
      },
      {
        "inputType": "#",
        "elements": "Commodities:VMem:capacity",
        "filterCategory": "property",
        "filterType": "vmsByMem"
      },
      {
        "inputType": "*",
        "elements": "ExtendedBy:VMTopoExt:guestName",
        "filterCategory": "property",
        "filterType": "vmsByGuestName"
      },
      {
        "inputType": "*",
        "elements": "ExtendedBy:VMTopoExt:altName",
        "filterCategory": "property",
        "filterType": "vmsByAltName"
      },
      {
        "inputType": "*",
        "elements": "HostedBy:PhysicalMachine:MemberOf:Cluster:displayName",
```

```
        "filterCategory": "entity",
        "filterType": "vmsByClusterName"
      },
      {
        "inputType": "*",
        "elements": "LayeredOver:Storage:LayeredOver:DiskArray:displayName",
        "filterCategory": "entity",
        "filterType": "vmsByDiskArrayName"
      },
      {
        "inputType": "*",
        "elements": "LayeredOver:Storage:LayeredOver:LogicalPool:displayName",
        "filterCategory": "entity",
        "filterType": "vmsByLogicalPoolName"
      },
      {
        "inputType": "*",
        "elements": "tags",
        "filterCategory": "property",
        "filterType": "vmsByTag"
      },
      {
        "inputType": "*",
        "elements": "state",
        "filterCategory": "property",
        "filterType": "vmsByState"
      },
      {
        "inputType": "*",
        "elements": "belongsTo:uuid",
        "filterCategory": "entity",
        "filterType": "vmsByBusinessAccountUuid"
      },
      {
        "inputType": "*",
        "elements": "MemberOf:ResourceGroup:uuid",
        "filterCategory": "entity",
        "filterType": "vmsByResourceGroupUuid"
      }
    ]
  }
```

Getting the Options for a Criterion

For a specific search criterion and entity type, you can get the possible values. For example, you may want to retrieve all tags for a particular group of hosts, or see the available state options for a particular group of virtual machines. This request takes the following parameters:

- criteria_key

  The criterion to retrieve options for. Currently, Turbonomic accepts the following criteria keys: `[ state, tags ]`. Any other input that returns results should be considered internal or early access, and should not be relied on for scripting purposes.

- scopes

  The UUID of the scope to retrieve the criterion values for. For example, to retrieve all tags for virtual machines in a particular cluster, put the cluster UUID here.

- entity_type

  The entity whose state or tag options will be returned. For example, if `scopes` is passed with the UUID of a cluster, and this parameter is passed with a value of VirtualMachine, the request will return the options for each virtual machine in that cluster.

- order_by

  The field to order the results by. `[ creation_date, name, risk_category, savings, severity ]`.

- ascending

  Default: true. When false, results will be in descending order.

- environment_type

Returns only objects in the specified environment type. `[ Cloud, Hybrid, Onprem ]`

Example: `GET https://10.10.10.10/api/v3/search/criteria/state/options?scopes=7c8d50fae46c09b68b0091641146977030422 6b4&entity_type=VirtualMachine`

Response: A `CriteriaOptionApiDTO` with the available options. For example, the following response indicates that all virtual machines on the cluster whose UUID was passed are either in active or idle states.

```
[
  {
    "value": "ACTIVE"
  },
  {
    "value": "IDLE"
  }
]
```

Search requests

Getting object information by UUID

Gets information about any object in your Turbonomic environment.

`Object` and `Entity` refer to different items in your Turbonomic topology:

Note:

For searches, the API paginates the returned data. If you do not specify a pagination limit, then the API returns pages set to the default limit of 100. You can set a maximum of 500 for the pagination limit. If you make no settings, and your search results include more than 100 entries, the results will be paginated. Your scripts should always check for a pagination cursor when processing search results.

You can change the default and max settings in the Turbonomic cr.yaml file. Find the entries for:

- `apiPaginationDefaultLimit`
- `apiPaginationMaxLimit`

If you set the `objectType` parameter has a value of `Group`, the response will include objects related to an entity that perform grouping. For example, resource groups and clusters both perform grouping, and will be included in the response.

To further refine your search, include a value for the `group_type` parameter. This parameter describes the membership of the group you are searching for. For example, a search with an `objectType` of `Group` and `group_type` of `VirtualMachine` will return all groups of virtual machines in the current scope.

Example: `https://10.10.10.10/api/v3/search/Market`

Response: A DTO with information about the retrieved object. In this example, the retrieved item is the real-time Market:

```
{
  "links": [
    {
      "rel": "self",
      "href": "https://10.10.10.10/api/v3/markets/_0x3OYUglEd-gHc4L513yOA"
    }
  ],
  "uuid": "_0x3OYUglEd-gHc4L513yOA",
  "displayName": "Market",
  "className": "Market",
  "state": "RUNNING",
  "unplacedEntities": false,
  "environmentType": "HYBRID"
}
```

Note:

When making this request, you can use the groupBy parameter to order the results. The /search endpoint does not support the utilization value for the groupBy parameter. When this value is used, the API will return the results ordered by displayName.

Settings endpoint

The settings endpoint can be used to change settings in a variety of settings managers. For example, using the actionscriptmanager, you can change action script settings, and using the automationmanager, you can change automation settings on your instance.

Using the settings endpoint of the Turbonomic API, you can:

- Get a list of available setting managers and settings
- Edit a setting

The following settings managers are available to be modified:

- actionscriptmanager

  Contains settings related to the use of action scripts.
- automationmanager

  Contains settings related to action modes for entities in your environment.
- cloudsavingsmanager

  Contains settings related to execution retention for cloud entities.
- clusterHeadroomConfig

  Contains settings related to the headroom plans generated by the system.
- controlmanager

  Contains settings related to workflows that replace native action handling for generated actions in your environment.
- emailmanager

  Contains settings related to SMTP email notifications.
- hcisettingsmanager

  Contains settings related to managing hyperconverged infrastructure.
- marketsettingsmanager

  Contains settings related to market analysis.
- osmigrationmanager

  Contains settings realted to operating system migration for cloud entities.
- persistencemanager

  Contains settings related to statistic retention data and reporting.
- reservedinstancemanager

  Contains settings related to reserved instances.
- storagesettingsmanager

  Contains settings related to storage entity statistics and analysis.
- telemetrymanager

  Contains settings related to telemetry opt-in and terms of service.
-

For more information about each settingsmanager, execute the GET https://10.10.10.10/api/v3/settings request.

Settings utility requests

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

The settings endpoint contains a utility request that shows each settings manager, and the settings available.

Example: `GET https://10.10.10.10/api/v3/settings/`

Response: An array of `SettingsManagerApiDTO`s, where each object is a settings manager and contains the settings for that manager.

Settings requests

Getting current settings for a specified settings manager

Gets the current settings for the specified settings manager. This request is often used to see the settings of a particular settings manager before updating.

Example: `GET https://10.10.10.10/api/v3/settings/automationmanager`

Response: A list of `SettingApiDTO`s representing the settings for the specified settings manager:


```
[
  {
    "uuid": "startVM",
    "displayName": "Start",
    "value": "RECOMMEND",
    "defaultValue": "RECOMMEND",
    "valueType": "STRING",
    "options": [
      {
        "label": "Disabled",
        "value": "DISABLED"
      },
      {
        "label": "Recommend",
        "value": "RECOMMEND"
      },
      {
        "label": "Manual",
        "value": "MANUAL"
      },
      {
        "label": "Automated",
        "value": "AUTOMATIC"
      }
    ],
    "entityType": "VirtualMachine",
    "sourceGroupName": "Global"
  },
  {
    "uuid": "startPM",
    "displayName": "Start",
    "value": "RECOMMEND",
    "defaultValue": "RECOMMEND",
    "valueType": "STRING",
    "options": [
      {
        "label": "Disabled",
        "value": "DISABLED"
      },
      {
        "label": "Recommend",
        "value": "RECOMMEND"
      },
      {
        "label": "Manual",
        "value": "MANUAL"
      },
      {
        "label": "Automated",
        "value": "AUTOMATIC"
      }
```

```
    ],
    "entityType": "PhysicalMachine",
    "sourceGroupName": "Global"
  },
  {
    "uuid": "startDS",
    "displayName": "Start",
    "value": "RECOMMEND",
    "defaultValue": "RECOMMEND",
    "valueType": "STRING",
    "options": [
      {
        "label": "Disabled",
        "value": "DISABLED"
      },
      {
        "label": "Recommend",
        "value": "RECOMMEND"
      },
      {
        "label": "Manual",
        "value": "MANUAL"
      },
      {
        "label": "Automated",
        "value": "AUTOMATIC"
      }
    ],
    "entityType": "Storage",
    "sourceGroupName": "Global"
  },
  ...
]
```

Editing settings

To edit a Setting, pass the new value in the body of this request. This request takes the following parameters:

- settingsManager_uuid

  The UUID of the settingsmanager whose setting will be changed.
- setting_uuid

  The setting whose value will be changed.

Example: `PUT https://10.10.10.10/api/v3/settings/automationmanager/moveVM`

Example Input: A SettingApiDTO. In this case, the value is being changed from `MANUAL` to `RECOMMEND`:

```
{
  "uuid": "moveVM",
  "displayName": "Move",
  "value": "RECOMMEND",
  "defaultValue": "MANUAL",
  "valueType": "STRING",
  "options": [
    {
      "label": "Disabled",
      "value": "DISABLED"
    },
    {
      "label": "Recommend",
      "value": "RECOMMEND"
    },
    {
      "label": "Manual",
      "value": "MANUAL"
    },
    {
      "label": "Automated",
      "value": "AUTOMATIC"
    }
  ],
  "entityType": "VirtualMachine",
  "sourceGroupName": "Global"
```

```
}
```

Response: The `SettingApiDTO` reflecting your changes:

```
{
  "uuid": "moveVM",
  "displayName": "Move",
  "value": "RECOMMEND",
  "defaultValue": "MANUAL",
  "valueType": "STRING",
  "options": [
    {
      "label": "Disabled",
      "value": "DISABLED"
    },
    {
      "label": "Recommend",
      "value": "RECOMMEND"
    },
    {
      "label": "Manual",
      "value": "MANUAL"
    },
    {
      "label": "Automated",
      "value": "AUTOMATIC"
    }
  ],
  "entityType": "VirtualMachine",
  "sourceGroupName": "Global"
}
```

Settings policies endpoint

Use the settings policies endpoint to manage automation policies.

Note:

The settings policies endpoint does not handle placement policies or parking policies. For more information about managing policies with the API, see Working with policies in the API.

An automation policy is a set of rules that Turbonomic must satisfy when executing non-parking actions on public cloud or on-prem entities, or changing settings that affect analysis and action generation.

A settings policy is the collection of settings that apply to an entity. In Turbonomic, there is a default policy for each type of entity. Any settings policies that you create overrides the default policies.

When you create a settings policy, you assign a scope, which specifies one or more entity groups that apply the policy. For these policies, you specify just the settings that are different from the default policies.

In addition to global default policies and custom policies, Turbonomic also discovers groups that need special settings, and creates policies for those groups. For example, when Turbonomic discovers Azure Availability Sets and AWS Autoscaling Groups, it generates settings policies for those groups that turn on Consistent Resizing for the group.

Using the settings policies endpoint, you can:

- Get a listing of the settings policies in your environment
- Get details for a single settings policy
- Create settings policies
- Edit settings policies
- Delete settings policies

Automation policies requests

An automation policy is a set of rules that Turbonomic must satisfy when executing non-parking actions on public cloud or on-prem entities, or changing settings that affect analysis and action generation.

For more information about automation policies that are managed through the user interface, see “Automation policies” on page 1204.

You can create and manage your automation policies with the Turbonomic API by making requests to the settings policies endpoint (`/settingspolicies`).

Getting automation policies

This call returns an array of `SettingsPolicyApiDTOs` for all the automation policies in your environment. You can pass entity types to limit the listing to policies for those types, and you can limit the list to only default policies.

This request takes the following parameters:

- `only_defaults`

  If this is `true`, the call only gets the default automation policies
- `entity_type`

  One or more entity types to limit the list of automation policies

Examples:

Request	Example
Get all the automation policies.	`GET https://10.10.10.10/api/v3/settingspolicies`
Get all the default automation policies.	`GET https://10.10.10.10/api/v3/settingspolicies?only_defaults=true`
Get automation policies for VMs and hosts	`GET https://10.10.10.10/api/v3/settingspolicies?entity_type=VirtualMachine&entity_type=PhysicalMachine`
Get a single automation policy (pass the policy UUID)	`GET https://10.10.10.10/api/v3/settingspolicies/_aVDEwGegEemDFcIzuV5wcQ`

Response: An array of `SettingsPolicyApiDTOs`. This listing shows the automation policies for container entities.


```
[
  {
    "uuid": "_vyMHEDnSEemHXrxYkw9y2Q",
    "displayName": "Container defaults",
    "entityType": "Container",
    "settingsManagers": [
      {
        "uuid": "marketsettingsmanager",
        "displayName": "Operational Constraints",
        "category": "Analysis",
        "settings": [
          {
            "uuid": "usedIncrement_Container_VMEM",
            "displayName": "Increment constant for Container VMem [MB]",
            "value": "64.0",
            "defaultValue": "64.0",
            "categories": [
              "resizeRecommendationsConstants"
            ],
            "valueType": "NUMERIC",
            "min": 0,
```

```
            "max": 10000000,
            "entityType": "Container",
            "sourceGroupName": "Global"
          },
          {
            "uuid": "usedIncrement_Container_VCPU",
            "displayName": "Increment constant for Container VCPU [MHz]",
            "value": "100.0",
            "defaultValue": "100.0",
            "categories": [
              "resizeRecommendationsConstants"
            ],
            "valueType": "NUMERIC",
            "min": 0,
            "max": 1000000,
            "entityType": "Container",
            "sourceGroupName": "Global"
          },
          {
            "uuid": "enforceConsistentResizingContainer",
            "displayName": "Consistent Resizing",
            "value": "false",
            "defaultValue": "false",
            "categories": [
              "resizeRecommendationsConstants"
            ],
            "valueType": "BOOLEAN",
            "entityType": "Container",
            "sourceGroupName": "Global"
          },
          {
            "uuid": "resizeTargetUtilization_Container",
            "displayName": "Resize Target Utilization",
            "value": "0.0",
            "defaultValue": "0.0",
            "valueType": "NUMERIC",
            "min": 0,
            "max": 100,
            "entityType": "Container",
            "sourceGroupName": "Global"
          }
        ]
      },
      ...
  }
]
```

Creating an automation policy

Referenced Endpoint: Groups

To create an automation policy, you must do the following:

- Specify a policy name,
- Specify whether to enable or disable the policy in your market.
- Set the policy scope – Specify one or more groups of a given entity type. This determines the entity type for the policy.
- For the of settings you want to make, specify the appropriate settings manager, and then specify an array of setting/value pairs.
- Optionally, specify a schedule window and/or action execution schedule for the policy.

To set the policy scope, you provide an array of objects that contain group Uuids. These objects are effectively abbreviated GroupApiDTOs. Note that the groups must all be for the same type of entity. This specifies the entity type for the automation policy. Use the Groups endpoint to get the Uuid values for the groups you want in the scope. For example, to set the scope to a single group of VMs, use the following scopes object:

```
"scopes": [
    {
      "uuid": "_lSlr0GddEemDFcIzuV5wcQ"
    }
```

```
]
```

To specify the actual settings, you provide an array of SettingsManagerApiDTOs. Each one contains an array of settings that you want to make. To see all the settings for a given entity type, refer to the user interface. The setting managers you can choose from are:

- `actionscriptmanager`

  Contains settings related to the use of action scripts.
- `automationmanager`

  Contains settings related to action modes for entities in your environment.
- `cloudsavingsmanager`

  Contains settings related to execution retention for cloud entities.
- `clusterHeadroomConfig`

  Contains settings related to the headroom plans generated by the system.
- `controlmanager`

  Contains settings related to workflows that replace native action handling for generated actions in your environment.
- `emailmanager`

  Contains settings related to SMTP email notifications.
- `hcisettingsmanager`

  Contains settings related to managing hyperconverged infrastructure.
- `marketsettingsmanager`

  Contains settings related to market analysis.
- `osmigrationmanager`

  Contains settings realted to operating system migration for cloud entities.
- `persistencemanager`

  Contains settings related to statistic retention data and reporting.
- `reservedinstancemanager`

  Contains settings related to reserved instances.
- `storagesettingsmanager`

  Contains settings related to storage entity statistics and analysis.
- `telemetrymanager`

  Contains settings related to telemetry opt-in and terms of service.
-

For example, to set the VM start action to AUTOMATED and to set the VCPU maximum to 16, use the following `settingsManagers` array:

```
"settingsManagers": [
    {
      "uuid": "automationmanager",
      "settings": [
        {
          "uuid": "startVM",
          "value": "AUTOMATIC"
        }
      ]
    },
    {
      "uuid": "marketsettingsmanager",
      "settings": [
```

```
        {
          "uuid": "resizeVcpuMaximum_VM",
          "value": 16
        }
      ]
    }
]
```

You can set up a schedule window that determines when the automation policy takes effect. For that you provide a ScheduleApiDTO for the schedule details. It includes: the start and end dates, start and end times,

- Start and end dates.
- Start and end times.
- An optional RecurrenceApiDTO for schedule recurrence. If you don't provide this you will specify a one-time schedule window.

For example, use this to specify a "perpetual" schedule window (does not expire until 2099) that starts at 12:30 am and ends at 7:30 am every Saturday and Sunday.

```
"schedule": {
  "startDate": "2019-04-28T00:30:00-04:00",
  "endDate": "2099-12-31T07:30:00-05:00",
  "startTime": "2019-04-28T00:30:00-04:00",
  "endTime": "2019-04-28T07:30:00-04:00",
  "recurrence": {
    "type": "WEEKLY",
    "daysOfWeek": [
      "Sat",
      "Sun"
    ]
}
```

An example complete data payload for a policy with a policy-level schedule, action execution instruction, and action execution schedule is as follows:

```
{
   "disabled":false,
   "entityType":"VirtualMachine",
   "displayName":"TestPol",
   "scopes":[
      {
         "uuid":"285041619454560"
      }
   ],
   "settingsManagers":[
      {
         "uuid":"automationmanager",
         "displayName":"Action Mode Settings",
         "category":"Automation",
         "settings":[
            {
               "uuid":"storageMove",
               "value":"RECOMMEND",
               "valueType":"STRING",
               "valueObjectType":"String",
               "defaultValue":"RECOMMEND",
               "entityType":"VirtualMachine",
               "displayName":"Storage Move",
               "options":[
                  {
                     "label":"Disabled",
                     "value":"DISABLED"
                  },
                  {
                     "label":"Recommend",
                     "value":"RECOMMEND"
                  },
                  {
                     "label":"External Approval",
                     "value":"EXTERNAL_APPROVAL"
                  },
                  {
```

```
                     "label":"Manual",
                     "value":"MANUAL"
                  },
                  {
                     "label":"Automatic",
                     "value":"AUTOMATIC"
                  }
               ]
            },
            {
               "uuid":"storageMoveExecutionSchedule",
               "value":"285088800975856",
               "valueType":"LIST",
               "valueObjectType":"String",
               "entityType":"VirtualMachine",
               "displayName":"Execution window for Storage Move action"
            }
         ]
      }
   ],
   "schedule":{
      "uuid":"285088797821136",
      "displayName":"TestSched",
      "endTime":"2021-05-24T16:00",
      "startTime":"2021-05-24T14:30",
      "timeZone":"America/Chicago",
      "recurrence":{
         "type":"MONTHLY",
         "interval":1,
         "daysOfMonth":[
            1
         ]
      }
   }
}
```

Example: Using the previous payload, `POST https://10.10.10.10/api/v3/settingspolicies`

Response: The automation policy that you created. Use this response to get the policy UUID.

Editing an automation policy

To edit an automation policy, push a new set of data to the settings policies endpoint. The best way to push data is to copy the `SettingsPolicyApiDTO` that you want to edit, make changes to that DTO, and then push the DTO out to the API. To remove the policy from the scope, pass the `SettingsPolicyApiDTO`, and also pass `reset_defaults=true` as a parameter.

This request takes the following parameters:

- `settingsPolicy_Uuid`

  The UUID of the policy you want to edit
- `reset_defaults`

  A boolean you can use to reset the policy to default. When `reset_defaults=true`, you remove the policy and reset that scope to default. When you use this setting, pass an empty object `{}` as the data payload.

Examples:

- Assuming you copied a `SettingsPolicyApiDTO` and then made some changes to it, and pass that at the payload: `PUSH https://10.10.10.10/api/v3/settingspolicies/_2-CbQGoNEemyOoIvFZyv1A?reset_defaults=false`
- Assuming you want to remove the policy (and so you pass an empty object as the payload): `PUSH https://10.10.10.10/api/v3/settingspolicies/_2-CbQGoNEemyOoIvFZyv1A?reset_defaults=true`

Response: The modified `SettingsPolicyApiDTO`.

Deleting an automation policy

Delete a policy.

Example: `DELETE https://10.10.10.10/api/v3/settingspolicies/_aV5ucGoTEemyOoIvFZyv1A`

Response: 200 for success.

Statistics endpoint

Statistics overview

Statistics are pieces of information that Turbonomic discovers while continuously analyzing your environment. The most common use of statistics is to look at a particular statistic over a specific time frame.

Using the stats endpoint of the Turbonomic API, you can:

- Get a list of stats requests in other endpoints
- Get a filtered list of statistics over a specific time frame
- Get a list of statistics for a specified entity
- Get a filtered list of statistics for a specified entity

Statistic timings

Statistics associated to headroom and capacity planning are written to the Turbonomic database daily, with a timestamp of midnight. Other statistics are collected and stored in the database in ten-minute increments. When retrieving groups of statistics that include both headroom and actively-monitored information, the timestamps returned will reflect the last-written date for each statistic.

Statistics utility requests

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

The stats endpoint contains a utility request that shows common entry points for statistic-related requests. These can be used to obtain statistics for particular entities or groups.

Example: `GET https://10.10.10.10/api/v3/stats`

Response: An array of Link objects, where each object contains a URL to get all statistics for a specified UUID, entity, or group.

Statistics requests

Getting statistics for an entity

Gets all statistics for the specified entity. This request takes the following parameters:

- entity_uuid

  The entity whose statistics will be returned.

- q

  Used internally by Turbonomic.

Note:

When you retrieve statistics for a cluster, using the `/groups` endpoint will return the aggregated statistics for all applicable cluster entities. For individual stats per entity, use the `/stats` endpoint.

Some statistics are only valid for certain scopes. When you request given statistics for a scope that doesn't include them, the API does not return any data. For example, the numVMs statistic is valid in only the Market and Cluster scopes – the API omits numVMs data from the response when you query other scopes.

Example: https://10.10.10.10/api/v3/stats/423f26a7-1132-df1c-64e2-8a040480b046

Response: A list of StatSnapshotApiDTOs representing the current values of the statistics.


```
[
  {
    "displayName": "AdityaReplica",
    "date": "2019-04-24T20:09:46-04:00",
    "statistics": [
      {
        "name": "priceIndex",
        "relatedEntityType": "VirtualMachine",
        "values": {
          "max": 1.01,
          "min": 1.01,
          "avg": 1.01,
          "total": 1.01
        },
        "value": 1.01
      },
      {
        "displayName": "CPUProvisioned/AdityaReplica",
        "name": "CPUProvisioned",
        "capacity": {
          "max": 3120000,
          "min": 3120000,
          "avg": 3120000,
          "total": 3120000
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        "units": "MHz",
        "values": {
          "max": 5200,
          "min": 5200,
          "avg": 5200,
          "total": 5200
        },
        "value": 5200
      },
      {
        "displayName": "Swapping/AdityaReplica",
        "name": "Swapping",
        "capacity": {
          "max": 40000000,
          "min": 40000000,
          "avg": 40000000,
          "total": 40000000
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        "units": "bit/sec",
        "values": {
          "max": 0,
          "min": 0,
          "avg": 0,
          "total": 0
        },
        "value": 0
      },
      {
        "displayName": "NumDisk/AdityaReplica",
        "name": "NumDisk",
        "capacity": {
```

```
          "max": 2147483650,
          "min": 2147483650,
          "avg": 2147483650,
          "total": 2147483650
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "relation",
            "value": "bought"
          }
        ],
        "values": {
          "max": 1,
          "min": 1,
          "avg": 1,
          "total": 1
        },
        "value": 1
      },
    ...
    ]
  }
]
```

Getting a filtered list of statistics

Gets a list of statistics filtered by a `StatPeriodApiInputDTO` that you create. For more information about the criteria that make up the input for thi request, see the Swagger documentation for this request, located at `https://<Your_Turbonomic_URL/apidpoc`.

Example: POST https://10.10.10.10/api/v3/stats/423f26a7-1132-df1c-64e2-8a040480b046

Example Input: This input will return the values for the VMem, VCPU, and VStorage statistics of the specified entity, ordered first by key, then relatedEntity, then virtualDisk.

```
{
  "statistics": [
    {
      "name": "VMem",
      "groupBy": [
        "key",
        "relatedEntity",
        "virtualDisk"
      ]
    },
    {
      "name": "VCPU",
      "groupBy": [
        "key",
        "relatedEntity",
        "virtualDisk"
      ]
    },
    {
      "name": "VStorage",
      "groupBy": [
        "key",
        "relatedEntity",
        "virtualDisk"
      ]
    }
  ]
}
```

Response: A list of `StatSnapshotApiDTO`s representing the filtered statistics for the entity:

```
[
  {
    "displayName": "AdityaReplica",
    "date": "2019-04-24T20:28:06-04:00",
    "statistics": [
      {
        "displayName": "AdityaReplica",
```

```
        "name": "VMem",
        "capacity": {
          "max": 1048576,
          "min": 1048576,
          "avg": 1048576,
          "total": 1048576
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "virtualDisk",
            "value": ""
          },
          {
            "type": "key",
            "value": null
          },
          {
            "type": "relation",
            "value": "sold"
          }
        ],
        "relatedEntity": {
          "uuid": "5d2f7b69eb3bf506089feba37d31b0c376b02e34",
          "displayName": "GuestLoad[AdityaReplica]",
          "className": "Application"
        },
        "units": "KB",
        "values": {
          "max": 0,
          "min": 0,
          "avg": 0,
          "total": 0
        },
        "value": 0
      },
      {
        "displayName": "AdityaReplica",
        "name": "VCPU",
        "capacity": {
          "max": 5200,
          "min": 5200,
          "avg": 5200,
          "total": 5200
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "virtualDisk",
            "value": ""
          },
          {
            "type": "key",
            "value": null
          },
          {
            "type": "relation",
            "value": "sold"
          }
        ],
        "relatedEntity": {
          "uuid": "5d2f7b69eb3bf506089feba37d31b0c376b02e34",
          "displayName": "GuestLoad[AdityaReplica]",
          "className": "Application"
        },
        "units": "MHz",
        "values": {
          "max": 19,
          "min": 14.93,
          "avg": 14.93,
          "total": 14.93
        },
        "value": 14.93
      }
    ]
  }
]
```

Getting a filtered list of statistics for all objects in a specified scope

For the specified scope, gets a filtered list of statistics for each entity. If you do not provide any statistics in the input, the response contains basic information about each entity. For more information about the criteria that make up the input for thi request, see the Swagger documentation for this request, located at `https://<Your_Turbonomic_URL/apidpoc`.

Example: POST https://10.10.10.10/api/v3/stats?ascending=true

Example Input: This input will return the VMem values for each entity in the real-time market from a period of five days prior to the request, to one day prior to the request.

```
{
  "scopes": [
    "Market"
  ],
  "period": {
    "startDate": "-5d",
    "endDate": "-1d",
    "statistics": [
      {
        "name": "VMem"
      }
    ]
  },
  "relatedType": "VirtualMachine"
}
```

Response: A list of `StatSnapshotApiDTO`s representing the filtered statistics for the entities in the specified scope:

```
[
  {
    "uuid": "4200829c-1272-c0a4-2d17-55f2cc0ef1b4",
    "displayName": "vm-c67a1cdd-609e-4282-809c-e0def5cd0d7a",
    "className": "VirtualMachine",
    "environmentType": "ONPREM",
    "stats": [
      {
        "displayName": "vm-c67a1cdd-609e-4282-809c-e0def5cd0d7a",
        "date": "2019-04-19T00:00:00-04:00",
        "statistics": [
          {
            "name": "VMem",
            "capacity": {
              "max": 16777216,
              "min": 16777216,
              "avg": 16777216,
              "total": 33554432
            },
            "relatedEntityType": "VirtualMachine",
            "filters": [
              {
                "type": "relation",
                "value": "sold"
              }
            ],
            "units": "KB",
            "values": {
              "max": 1509949.5,
              "min": 335544.3,
              "avg": 587202.56,
              "total": 1174405.1
            },
            "value": 587202.56
          }
        ]
      },
      {
        "displayName": "vm-c67a1cdd-609e-4282-809c-e0def5cd0d7a",
        "date": "2019-04-20T00:00:00-04:00",
        "statistics": [
          {
            "name": "VMem",
            "capacity": {
              "max": 16777216,
```

```
            "min": 16777216,
            "avg": 16777216,
            "total": 33554432
          },
          "relatedEntityType": "VirtualMachine",
          "filters": [
            {
              "type": "relation",
              "value": "sold"
            }
          ],
          "units": "KB",
          "values": {
            "max": 1509949.5,
            "min": 369098.75,
            "avg": 587202.56,
            "total": 1174405.1
          },
          "value": 587202.56
        }
      ]
    },
    {
      "displayName": "vm-c67a1cdd-609e-4282-809c-e0def5cd0d7a",
      "date": "2019-04-22T00:00:00-04:00",
      "statistics": [
        {
          "name": "VMem",
          "capacity": {
            "max": 16777216,
            "min": 16777216,
            "avg": 16777216,
            "total": 33554432
          },
          "relatedEntityType": "VirtualMachine",
          "filters": [
            {
              "type": "relation",
              "value": "sold"
            }
          ],
          "units": "KB",
          "values": {
            "max": 12582912,
            "min": 0,
            "avg": 452984.84,
            "total": 905969.7
          },
          "value": 452984.84
        }
      ]
    },
    {
      "displayName": "vm-c67a1cdd-609e-4282-809c-e0def5cd0d7a",
      "date": "2019-04-23T00:00:00-04:00",
      "statistics": [
        {
          "name": "VMem",
          "capacity": {
            "max": 16777216,
            "min": 16777216,
            "avg": 16777216,
            "total": 33554432
          },
          "relatedEntityType": "VirtualMachine",
          "filters": [
            {
              "type": "relation",
              "value": "sold"
            }
          ],
          "units": "KB",
          "values": {
            "max": 838860.8,
            "min": 0,
            "avg": 167772.16,
            "total": 335544.3
          },
          "value": 167772.16
        }
      ]
    }
```

```
    ]
  },
  {
    "uuid": "39EBB726-41D9-4BC8-A41D-CA301854309C",
    "displayName": "Gilad_OM-38071_SMB_VM2",
    "className": "VirtualMachine",
    "environmentType": "ONPREM"
  },
  {
    "uuid": "97A3F420-B7AD-414D-A543-FD308AC774A6",
    "displayName": "Gilad_VMM_Test_VM_2",
    "className": "VirtualMachine",
    "environmentType": "ONPREM"
  },
  {
    "uuid": "4211f183-69a8-3676-3e53-0151365500ff",
    "displayName": "james-ubuntu-dc13",
    "className": "VirtualMachine",
    "environmentType": "ONPREM",
    "stats": [
      {
        "displayName": "james-ubuntu-dc13",
        "date": "2019-04-19T00:00:00-04:00",
        "statistics": [
          {
            "name": "VMem",
            "capacity": {
              "max": 1048576,
              "min": 1048576,
              "avg": 1048576,
              "total": 2097152
            },
            "relatedEntityType": "VirtualMachine",
            "filters": [
              {
                "type": "relation",
                "value": "sold"
              }
            ],
            "units": "KB",
            "values": {
              "max": 83886.08,
              "min": 8388.61,
              "avg": 18874.37,
              "total": 37748.73
            },
            "value": 18874.37
          }
        ]
      },
      {
        "displayName": "james-ubuntu-dc13",
        "date": "2019-04-20T00:00:00-04:00",
        "statistics": [
          {
            "name": "VMem",
            "capacity": {
              "max": 1048576,
              "min": 1048576,
              "avg": 1048576,
              "total": 2097152
            },
            "relatedEntityType": "VirtualMachine",
            "filters": [
              {
                "type": "relation",
                "value": "sold"
              }
            ],
            "units": "KB",
            "values": {
              "max": 62914.56,
              "min": 13631.49,
              "avg": 18874.37,
              "total": 37748.73
            },
            "value": 18874.37
          }
        ]
      },
      {
        "displayName": "james-ubuntu-dc13",
```

```
    "date": "2019-04-21T00:00:00-04:00",
    "statistics": [
      {
        "name": "VMem",
        "capacity": {
          "max": 1048576,
          "min": 1048576,
          "avg": 1048576,
          "total": 2097152
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "relation",
            "value": "sold"
          }
        ],
        "units": "KB",
        "values": {
          "max": 73400.32,
          "min": 11534.34,
          "avg": 17825.79,
          "total": 35651.59
        },
        "value": 17825.79
      }
    ]
  },
  {
    "displayName": "james-ubuntu-dc13",
    "date": "2019-04-22T00:00:00-04:00",
    "statistics": [
      {
        "name": "VMem",
        "capacity": {
          "max": 1048576,
          "min": 1048576,
          "avg": 1048576,
          "total": 2097152
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "relation",
            "value": "sold"
          }
        ],
        "units": "KB",
        "values": {
          "max": 83886.08,
          "min": 8388.61,
          "avg": 19922.94,
          "total": 39845.89
        },
        "value": 19922.94
      }
    ]
  },
  {
    "displayName": "james-ubuntu-dc13",
    "date": "2019-04-23T00:00:00-04:00",
    "statistics": [
      {
        "name": "VMem",
        "capacity": {
          "max": 1048576,
          "min": 1048576,
          "avg": 1048576,
          "total": 2097152
        },
        "relatedEntityType": "VirtualMachine",
        "filters": [
          {
            "type": "relation",
            "value": "sold"
          }
        ],
        "units": "KB",
        "values": {
          "max": 73400.32,
          "min": 3145.73,
          "avg": 18874.37,
```

```
            "total": 37748.73
          },
          "value": 18874.37
        }
      ]
    }
  ]
},
{
  "uuid": "421d2165-36cd-7f6a-3a43-08e09f6f4e0f",
  "displayName": "ACM-LexCorp-Control",
  "className": "VirtualMachine",
  "environmentType": "ONPREM",
  "stats": [
    {
      "displayName": "ACM-LexCorp-Control",
      "date": "2019-04-19T00:00:00-04:00",
      "statistics": [
        {
          "name": "VMem",
          "capacity": {
            "max": 2097152,
            "min": 2097152,
            "avg": 2097152,
            "total": 4194304
          },
          "relatedEntityType": "VirtualMachine",
          "filters": [
            {
              "type": "relation",
              "value": "sold"
            }
          ],
          "units": "KB",
          "values": {
            "max": 167772.16,
            "min": 18874.37,
            "avg": 37748.73,
            "total": 75497.47
          },
          "value": 37748.73
        }
      ]
    },
    {
      "displayName": "ACM-LexCorp-Control",
      "date": "2019-04-20T00:00:00-04:00",
      "statistics": [
        {
          "name": "VMem",
          "capacity": {
            "max": 2097152,
            "min": 2097152,
            "avg": 2097152,
            "total": 4194304
          },
          "relatedEntityType": "VirtualMachine",
          "filters": [
            {
              "type": "relation",
              "value": "sold"
            }
          ],
          "units": "KB",
          "values": {
            "max": 125829.12,
            "min": 18874.37,
            "avg": 37748.73,
            "total": 75497.47
          },
          "value": 37748.73
        }
      ]
    },
    {
      "displayName": "ACM-LexCorp-Control",
      "date": "2019-04-21T00:00:00-04:00",
      "statistics": [
        {
          "name": "VMem",
          "capacity": {
            "max": 2097152,
```

```
            "min": 2097152,
            "avg": 2097152,
            "total": 4194304
          },
          "relatedEntityType": "VirtualMachine",
          "filters": [
            {
              "type": "relation",
              "value": "sold"
            }
          ],
          "units": "KB",
          "values": {
            "max": 104857.6,
            "min": 20971.52,
            "avg": 35651.59,
            "total": 71303.17
          },
          "value": 35651.59
        }
      ]
    },
    {
      "displayName": "ACM-LexCorp-Control",
      "date": "2019-04-22T00:00:00-04:00",
      "statistics": [
        {
          "name": "VMem",
          "capacity": {
            "max": 2097152,
            "min": 2097152,
            "avg": 2097152,
            "total": 4194304
          },
          "relatedEntityType": "VirtualMachine",
          "filters": [
            {
              "type": "relation",
              "value": "sold"
            }
          ],
          "units": "KB",
          "values": {
            "max": 272629.75,
            "min": 20971.52,
            "avg": 73400.32,
            "total": 146800.64
          },
          "value": 73400.32
        }
      ]
    },
    {
      "displayName": "ACM-LexCorp-Control",
      "date": "2019-04-23T00:00:00-04:00",
      "statistics": [
        {
          "name": "VMem",
          "capacity": {
            "max": 2097152,
            "min": 2097152,
            "avg": 2097152,
            "total": 4194304
          },
          "relatedEntityType": "VirtualMachine",
          "filters": [
            {
              "type": "relation",
              "value": "sold"
            }
          ],
          "units": "KB",
          "values": {
            "max": 251658.23,
            "min": 23068.67,
            "avg": 79691.77,
            "total": 159383.55
          },
          "value": 79691.77
        }
      ]
    }
```

```
    ]
  },
  {
    "uuid": "420f3462-ca50-fedf-aa8f-223bd39b1852",
    "displayName": "shai-redhat",
    "className": "VirtualMachine",
    "environmentType": "ONPREM",
    "stats": [
      {
        "displayName": "shai-redhat",
        "date": "2019-04-19T00:00:00-04:00",
        "statistics": [
          {
            "name": "VMem",
            "capacity": {
              "max": 1048576,
              "min": 1048576,
              "avg": 1048576,
              "total": 2097152
            },
            "relatedEntityType": "VirtualMachine",
            "filters": [
              {
                "type": "relation",
                "value": "sold"
              }
            ],
            "units": "KB",
            "values": {
              "max": 20971.52,
              "min": 0,
              "avg": 1048.58,
              "total": 2097.15
            },
            "value": 1048.58
          }
        ]
      },
      ...
    ]
  }
]
```

Complex statistic queries

Using the POST /stats request, you can retrieve a variety of statistics. For example, this is a complex query made by the Turbonomic UI:

```
{
  "statistics":[
    {
      "name":"CPU",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Mem",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
```

```
    },
    {
      "name":"NetThroughput",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"IOThroughput",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Q1VCPU",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Q2VCPU",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Q4VCPU",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Q8VCPU",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
```

```
        }
      ]
    },
    {
      "name":"Q16VCPU",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Q32VCPU",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Ballooning",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Swapping",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"CPUProvisioned",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"MemProvisioned",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
```

```
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Flow",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"FlowAllocation",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Cooling",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Power",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    },
    {
      "name":"Space",
      "relatedEntityType":"PhysicalMachine",
      "groupBy":[
        "key",
        "virtualDisk"
      ],
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ]
    }
  ]
}
```

The previous request is made in context of a datacenter scope, so the response includes information for each host within that datacenter. The `virtualDisk` value for the `groupBy` parameter ensures that all virtual disks appear in the response as individual line items, rather than an aggregation.

Additionally, the `"filters":[{"type":"relation","value":"sold"}]` found throughout the call indicates that the response includes those commodities *sold* by these hosts, rather than purchased from a lower level in the supply chain.

Supplychains endpoint

Turbonomic models your environment as a market of buyers and sellers. It discovers different types of entities in your environment via the targets you have configured for your installation. Discovery maps these entities to the supply chain so Turbonomic can monitor them and manage the workloads they support. For example, for a hypervisor target Turbonomic discovers VMs, the hosts and datastores that provide resources to the VMs, and the applications that use VM resources. The entities in your environment form a chain of supply and demand where some entities provide resources while others consume the supplied resources.

Using the supplychains endpoint, you can:

- Get a full or filtered supply chain related to an entity in your environment
- Get the statistics for the supply chain related to an entity in your environment

Supplychains requests

Getting a supply chain for the specified entity

Gets the full supply chain for the specified entity. For example, if the UUID of a virtual machine is used, you will see the applications that are hosted by that virtual machine, the hosts and datastores that it consumes from, the cluster or datacenter that the hosts consume from, the disk arrays and storage controllers that the datastores consume from, and so on.

In a supply chain, depth refers to the distance in the supply chain from the entity called. For example, if the UUID of a host is used, virtual machines consuming from that host will have a depth of `-1`, applications consuming from those virtual machines will have a depth of `-2`, other hosts that are in the supply chain will have a depth of `0`, and the cluster of the host will have a depth of `1`.

Example: `GET https://10.10.10.10/api/v3/supplychains?uuids=Virtual_ESX_4238042f-7465-e92a-5c28-a7a929307247&health=false`

Response: A `SupplychainApiDTO` representing the supply chain of the specified entity- in this case, a host. Note that when there is a single instance of an entity, Turbonomic will return the details of that entity. In the response, you can see this with the `Storage Controller` entity.


```
{
  "seMap":{
    "Application":{
      "depth":-2,
      "entitiesCount":12,
      "stateSummary":{
        "ACTIVE":5,
        "SUSPEND":1,
        "IDLE":6
      }
    },
    ...
    "StorageController":{
      "depth":3,
      "entitiesCount":1,
      "stateSummary":{
        "ACTIVE":1
      },
      "instances":{
        "be89058c-e786-11e4-82ba-dfc30ffeb3ee":{
          "links":[
            {
```

```
              "rel":"self",
              "href":"https://10.10.10.10/api/v3/entities/be89058c-e786-11e4-82ba-dfc30ffeb3ee?
include_aspects=false"
            }
          ],
          "uuid":"be89058c-e786-11e4-82ba-dfc30ffeb3ee",
          "displayName":"dataontap-mycorp-cm1",
          "className":"StorageController",
          "priceIndex":20000,
          "state":"ACTIVE",
          "severity":"Normal",
          "discoveredBy":{
            "uuid":"_adGLMJwFEei_xJ5Ai6-rnQ",
            "displayName":"data-cluster.eng.mycorp.com",
            "type":"NetApp"
          },
          "remoteId":"be89058c-e786-11e4-82ba-dfc30ffeb3ee",
          "environmentType":"ONPREM"
        }
      }
    }
  }
}
```


Getting leaf entities from a supply chain root

Gets an array of entities that are at the top for a given supply chain scope. You provide an array of UUIDs to *seed* the starting scope. The request then traverses up the supply chain from those seed entities, until it finds entities that have no consumers. These are *leaf* entities for that seeded scope.

- uuids

  A list of UUIDs to seed the request scope. For example, you can give uuids for multiple VMs, and the request will find all leaf entities from all VMs in the list. If you give the uuid for a group, the request treats the group members as a list of entities.

- filterEntityTypes

  A list of entity types to filter out of the request. You can provide any valid entity class name. The request stops traversing the supply chain at the filtered entity. For example, assume you have a supply chain that grows from 1 VM > 4 ApplicationComponent > 2 Service > 1 BusinessApplication. A request starting from the VM would return the single Business Application. If you filter out BusinessApplication, it would return the two Service entities. If you filter out Service, it would return 4 ApplicationComponent entities. Note that the Swagger documentation for this call gives a list of valid filterEntityTypes.

- cursor

  When making subsequent calls to retrieve additional results, this is the cursor returned by the last call to this method.

- limit

  The maximum number of items to return. Must be a positive integer. If not set, the server will provide a service-specific maximum limit.

For example, in the Turbonomic user interface, you can search for the BussinessApplication entity named "App [app]". If you set the scope to that business application, then you can click in the Supply Chain to select the Namespace entity. Find the **Entity Information** chart and copy the ID value. That is the uuid for the app Namespace. If you get the leaf entities for that Namespace, you can see all the Service, ApplicationComponent, and BusinessApplication entities that support the Turbonomic application. For example, the following Service entry is for the Hyper-V target component:


```
  {
    "uuid": "74626820648480",
    "displayName": "Service-app/mediation-hyperv",
    "className": "Service"
  }
```

Now assume you want to use a VM group as the seeded scope, and you want to exclude any Service entities or higher in the Supply Chain:

Example: GET https://10.10.168.102/api/v3/supplychains/leafEntities?uuids=74626820647952&filterEntityTypes=Service

```
[
  {
    "uuid": "74699295824896",
    "displayName": "App-app/cost-67b7b9bb4d-bl8kt/cost",
    "className": "Application Component"
  },
  {
    "uuid": "74699295824908",
    "displayName": "App-app/mediation-vmax-6cd7f4ff-nzqp6/mediation-vmax",
    "className": "Application Component"
  },
  {
    "uuid": "74699295824897",
    "displayName": "App-app/mediation-azurebilling-c585c846f-mvnl7/mediation-azurebilling",
    "className": "Application Component"
  },
  ...
]
```

Getting statistics for a supply chain

Gets a list of statistics filtered by a user-created SupplyChainStatsApiInputDTO. For more information about the criteria that make up the input for thi request, see the Swagger documentation for this request, located at `https://<Your_Turbonomic_URL/apidpoc`.

Example: POST https://10.10.168.102/api/v3/supplychains/stats

Example Input: This input will return the number of applications running on the specified UUID (in this case, a Datacenter):

```
{
  "uuids": [
    "e27e5c2f85d91107ac3f26fd63d4478e808f590b"
  ],
  "types": [
    "Application"
  ],
  "environmentType": "ONPREM"
}
```

Response: A list of `StatSnapshotApiDTO`s representing the filtered statistics for the entity:

```
[
  {
    "date": "2019-05-01T17:45:42-04:00",
    "statistics": [
      {
        "name": "entities",
        "filters": [],
        "value": 25
      }
    ]
  }
]
```

Tags endpoint

Tags overview

Tags are metadata attached to resources that you can use to organize your environment. Tags are most often used to delineate resource groupings for technical, automation, business, or security purposes.

Turbonomic discovers tags for AWS, Azure, and vCenter resources.

Using the tags endpoint, you can:

- Get a list of available tags
- Get the members belonging to a specified tag

Tags requests

Pagination for tags requests

When you request tag data, the API uses pagination to return subsets of the data. You should always use pagination when available to make sure you get your data in manageable chunks. To paginate returned data, the `limit`, `orderBy`, and `cursor` parameters control how the APIs paginate their data. If you do not specify a `limit` for the following requests, the API will paginate the data to a default limit of records:

- `GET /tags`
- `GET /tags/entities?key=<tagKey>`

If your scripts make these requests without setting a limit, you should be sure they check for a pagination cursor in the returned data.

For general information about pagination, see Pagination in the Turbonomic API.

Getting tags

For a list of tags, the API returns a list of TagApiDTOs. Each TagApiDTO will contain the key/value pair of the tag. The key can be used to see the entities belonging to the tag.

Example: `https://10.10.10.10/api/v3/tags`

Response:

```
[
  {
    "key":"owner",
    "values":[
      "QA_Azure"
    ]
  },
  {
    "key":"director",
    "values":[
      "p-bosh",
      "bosh-init"
    ]
  },
  ...
]
```

Filtering tags by scope, entity, and environment

Gets a filtered list of tags for the specified scope, entity type, and/or environment. For example, you can use this to find tags only in a particular cluster, tags that affect at least one VM, and/or limit your result to only cloud entities. This request accepts the following parameters:

- scopes

  A list of UUIDs representing the scope of the request. For example, providing `Market` as a scope will search the entire real-time market. If you provide the UUID of a host, only entities on that host would be in the response.

- entity_type
  - `APPLICATION`
  - `APPLICATION_SERVER`

  - BUSINESS_ACCOUNT
  - BUSINESS_UNIT
  - CHASSIS
  - CLOUD_SERVICE
  - CONTAINER
  - CONTAINER_POD
  - DATABASE
  - DATABASE_SERVER
  - DATACENTER
  - DISK_ARRAY
  - IO_MODULE
  - LOAD_BALANCER
  - LOGICAL_POOL
  - NETWORK
  - PHYSICAL_MACHINE
  - STORAGE
  - STORAGE_CONTROLLER
  - SWITCH
  - VIRTUAL_APPLICATION
  - VIRTUAL_DATACENTER
  - VIRTUAL_MACHINE
- environment_type

  Returns only tags from entities in the specified environment type. `[ Cloud, Hybrid, Onprem ]`

Example: `GET https://10.10.10.10/api/v3/tags?scopes=Market&entity_type=VirtualMachine&environment_type=ONPREM`

Response: a list of VMs running on premises in the real-time market with tags.

Getting Entities For A Specified Tag

Referenced Endpoint: Entities

Gets a list of entities that have the specified tag. An entity is any device in your environment that Turbonomic manages, such as applications, physical machines, VMs, storage, and datacenters.. You can use this information to see where entities with the specified tag are located, what other tags are associated to those entities, and detailed information about each entity.

Example: `GET https://10.10.10.10/api/v3/tags/entities?key=owner`

Response: A list of entities that have the tag `owner`:


```
                [
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/entities/
azure::VM::375ddfb4-3127-44eb-9088-3c7645212f81?include_aspects=false"
      }
    ],
    "uuid":"azure::VM::375ddfb4-3127-44eb-9088-3c7645212f81",
    "displayName":"oleg-westus2-unmanaged-std",
    "className":"VirtualMachine",
    "priceIndex":3.2109845,
    "state":"ACTIVE",
    "severity":"Normal",
    "costPrice":0.024,
```

```
    "discoveredBy":{
      "uuid":"_GLpj4JwIEei_xJ5Ai6-rnQ",
      "displayName":"mycorp.management.core.windows.net",
      "type":"Azure"
    },
    "remoteId":"azure::VM::375ddfb4-3127-44eb-9088-3c7645212f81",
    "template":{
      "uuid":"azure::VMPROFILE::Basic_A0",
      "displayName":"Basic_A0",
      "discovered":false
    },
    "environmentType":"CLOUD",
    "tags":{
      "owner":[
        "PT_Azure"
      ],
      "AutoSuspend":[
        "false"
      ]
    }
  },
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/entities/
azure::VM::c94e96bf-975e-47da-8c52-7ac90161cc87?include_aspects=false"
      }
    ],
    "uuid":"azure::VM::c94e96bf-975e-47da-8c52-7ac90161cc87",
    "displayName":"oleg-westus2-managed-avail-set-1",
    "className":"VirtualMachine",
    "priceIndex":2.446491,
    "state":"ACTIVE",
    "severity":"Normal",
    "costPrice":0.018,
    "discoveredBy":{
      "uuid":"_GLpj4JwIEei_xJ5Ai6-rnQ",
      "displayName":"mycorp.management.core.windows.net",
      "type":"Azure"
    },
    "remoteId":"azure::VM::c94e96bf-975e-47da-8c52-7ac90161cc87",
    "template":{
      "uuid":"azure::VMPROFILE::Standard_B1s",
      "displayName":"Standard_B1s",
      "discovered":false
    },
    "environmentType":"CLOUD",
    "tags":{
      "owner":[
        "PT_Azure"
      ],
      "AutoSuspend":[
        "false"
      ]
    }
  }
]
```


Targets endpoint

A target is a service that performs management functions in your virtual environment. Turbonomic uses targets to monitor workload and to execute actions in your environment.

To discover entities in your environment, you add different targets to your Turbonomic installation. Turbonomic then uses these targets to collect data from your environment.

Turbonomic uses probes to connect to targets and discover their entities. The Turbonomic GUI provides forms for users to specify different target instances. Note that each probe may require different settings, and it's possible that different versions of Turbonomic supports different sets of probes.

For complete target information, review Target Configuration. You should not modify targets without a full understanding of how they interact with Turbonomic.

Using the targets endpoint of the Turbonomic API, you can:

- Get a single target or list of targets present in your environment
- Get a list of probes available in your Turbonomic installation
- Get the actions or statistics for a target
- Get the workflows discovered by a target
- Add or edit a target
- Trigger validation or rediscovery on a target
- Delete a target

Targets utility requests

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

The targets endpoint contains a utility request called `specs` that shows the information required for probe registry for each target available to Turbonomic, including required input fields and expected target type and category values.

Note:

For a list of probes available to your specific Turbonomic installation, review your organization's licensing agreement.

Example: `GET https://10.10.10.10/api/v3/targets/specs`

Response: An array of `TargetApiDTO` objects, where each object contains an array of input fields. This example shows one entry for the JVM probe in the Application Server category.

```
[
 {
  [
    ...
    {
      "category":"Application Server",
      "identifyingFields":[
        "nameOrAddress"
      ],
      "inputFields":[
        {
          "displayName":"Scope/Address",
          "name":"nameOrAddress",
          "isMandatory":true,
          "isSecret":false,
          "valueType":"GROUP_SCOPE",
          "description":"The group of VMs or the address of the JVM server"
        },
        {
          "displayName":"Port Number",
          "name":"targetPort",
          "defaultValue":"1099",
          "isMandatory":true,
          "isSecret":false,
          "valueType":"NUMERIC",
          "description":"The JMX port that connects with the JVM server"
        },
        {
          "displayName":"Username",
          "name":"username",
          "isMandatory":true,
          "isSecret":false,
          "valueType":"STRING",
          "description":"The JVM server JMX password, enter a space if the server has no
security"
        },
        {
          "displayName":"Password",
          "name":"password",
          "isMandatory":true,
          "isSecret":true,
          "valueType":"STRING",
          "description":"The JVM server JMX port, enter a space if the server has no security"
        }
```

```
      ],
      "type":"JVM"
    },
    ...
  }
]
```

Targets requests

Target requests handle creation, configuration, validation, rediscovery, and deletion of targets on your Turbonomic appliance.

Pagination for targets requests

When you request target entities, the API uses pagination to return subsets of the data. You should always use pagination when available to make sure you get your data in manageable chunks. To paginate returned data, the `limit`, `orderBy`, and `cursor` parameters control how the APIs paginate their data.

If you do not specify a `limit` for the `GET /targets/<targetId>/entities` request, the API will paginate the data to a default limit of records. If your script makes this request without setting a limit, you should be sure it checks for a pagination cursor in the returned data.

For general information about pagination, see Pagination in the Turbonomic API.

Getting targets

For a list of targets, the API returns an array of TargetApiDTO objects. Each TargetApiDTO object will contain the validation status, target type and category, and input field names and values. To get a single target, include the UUID of the target in the request.

This request takes the following parameters:

environment_type

Returns only targets in the specified environment type. `[ Cloud, Hybrid, Onprem ]`

Examples:

- All Targets: `GET https://10.10.10.10/api/v3/targets`
- Single Target: `GET https://10.10.10.10/api/v3/targets/__xwWYAE33eku7PbzHdETIA`

Response: A single entry in an array of targets. Notice that the password field is secret, so the response does not contain the password value.

```
[
  {
  "uuid":"__xwWYAE33eku7PbzHdETIA",
  "category":"Hypervisor",
  "type":"vCenter",
  "status":"Validated",
  "lastValidated":"2018-06-26T14:25:08-04:00",
  "inputFields":[
    {
      "displayName":"Address",
      "name":"address",
      "value":"vsphere-demo.mycorp.com",
      "isMandatory":true,
      "isSecret":false,
      "valueType":"STRING",
      "description":"IP or address of vCenter Server",
      "verificationRegex":".*"
    },
    {
      "displayName":"Username",
      "name":"username",
      "value":"corp\\my.user",
      "isMandatory":true,
      "isSecret":false,
      "valueType":"STRING",
```

```
      "description":"User name to use to connect to a target",
      "verificationRegex":".*"
    },
    {
      "displayName":"Password",
      "name":"password",
      "isMandatory":true,
      "isSecret":true,
      "valueType":"STRING",
      "description":"Password to use to connect to a target",
      "verificationRegex":".*"
    }
  ]
```

Adding a target

Adds a target to your Turbonomic installation. You must pass a completed TargetApiDTO object as an inputDTO. For more information about the required fields of the TargetApiDTO for the target you want to add, execute the `/targets/specs` utility request.

Example: `POST https://10.10.10.10/api/v3/targets`

Example inputDTO:

```
{
  "category":"Hypervisor",
  "inputFields":[
    {
      "name":"username",
      "value":"corp\\my.name"
    },
    {
      "name":"password",
      "value":"Octopath"
    },
    {
      "name":"nameOrAddress",
      "value":"10.10.10.10"
    }
  ],
  "type":"vCenter"
}
```

Rediscover or validate all targets

Manually triggers a full rediscovery or validation of all targets in your environment. This is often done to immediately see multiple changes that would normally not be reflected until a daily full discovery is performed, such as displaying updated Orchestrator workflows.

This call takes the following parameters:

- validate

  When true, validates all targets in your environment
- rediscover

  When true, performs a full rediscovery on all targets in your environment

Examples:

- Rediscovery:

  `POST https://10.10.10.10/api/v3/targets?validate=true`
- Validation:

  `POST https://10.10.10.10/api/v3/targets?rediscover=true`

Rediscover or Validate A Single Target

Manually triggers a full rediscovery or validation of a single target in your environment. This is often done to immediately see a change that would normally not be reflected until a daily full discovery is performed, such as displaying updated Orchestrator workflows.

This call takes the following parameters:

- validate

  When true, validates the specified target
- rediscover

  When true, performs a full rediscovery on the specified target

Examples:

- Rediscovery: `https://10.10.10.10/api/v3/targets/__xwWYAE33eku7PbzHdETIA?rediscover=true`
- Validation: `https://10.10.10.10/api/v3/targets/__xwWYAE33eku7PbzHdETIA?validate=true`

Response: The `TargetApiDTO` object of the specified target:

```
{
  "uuid":"__xwWYAE3EeiNKPbzHdETIA",
  "category":"Hypervisor",
  "inputFields":[
    ...
  ],
  "lastValidated":"2018-06-27T21:45:56-04:00",
  "status":"Validated",
  "type":"vCenter"
}
```

Editing target input fields

Edits the input fields of an existing target, and accepts the modified TargetApiDTO as input.

Note: You can GET the target first using the `GET /targets/specs` to see the fields required for that target. Note that you must pass all input fields in the modified TargetApiDTO object.

Example: `PUT https://10.10.10.10/api/v3/targets/_HgzQ4CdCEeaUWYEyo3vsyw`

Example input, editing the username for a single target:

```
[
  {
    "uuid":"_HgzQ4CdCEeaUWYEyo3vsyw",
    "category":"Hypervisor",
    "type":"vCenter",
    "status":"Validated",
    "lastValidated":1474575346109,
    "inputFields":[
      {
        "displayName":"Address",
        "name":"address",
        "value":"vsphere.mycorp.com",
        "isMandatory":true,
        "isSecret":false,
        "valueType":"STRING"
      },
      {
        "displayName":"Username",
        "name":"username",
        "value":"corp\\my.name.changed",
        "isMandatory":true,
        "isSecret":false,
        "valueType":"STRING"
      },
      {
        "displayName":"Password",
        "name":"password",
        "value":"Octopath",
```

```
        "isMandatory":true,
        "isSecret":true
      }
    ]
  }
]
```

Deleting a target

Deletes the specified target and removes all entities managed by that target. This will also remove any pending actions related to the deleted target.

Example: `DELETE https://10.10.10.10/api/v3/targets/__xwWYAE3EeiNKPbzHdETIA`

Response: A successful deletion request returns a Response Code of `200`.

Actions for a specified target

Referenced Endpoint: Actions

These requests get full or filtered lists of actions per target for use in reports, or to find a particular action.

Note: Once you have the action UUID, you can use requests from the actions endpoint to post changes to that action.

Getting an Unfiltered Array of Actions

Gets a list of current actions in your environment associated to entities discovered by the specified target.

Example: `GET https://10.10.10.10/api/v3/targets/__xwWYAE3EeiNKPbzHdETIA/actions`

Getting a Filtered Array of Actions

Gets a filtered list of actions discovered by a single target. Pass an inputDTO containing one or more of the following criteria:

- actionModeList
  - `DISABLED`:

    Do not recommend or perform the action. When you disable an action, the user interface will no longer display actions of that type.
  - `RECOMMENDED`:

    Recommend the action so a user can execute it via the given hypervisor or by other external means
  - `MANUAL`:

    Recommend the action, and provide the option to execute that action through the Turbonomic user interface or an /actions API request.
  - `AUTOMATED`:

    You have directed Turbonomic to execute the action automatically.
  - `COLLECTION` **(UNUSED)**:

    Legacy action mode, no longer used.
- actionStateList
  - `DISABLED`:

    An action whose policy has been disabled.
  - `RECOMMEND`:

    An action that cannot be automated due to policy settings or must be executed outside the system, such as the purchase of hardware.
  - `PENDING_ACCEPT`:

When the action mode is set to manual, a recommended action that has not been accepted or cleared.

  - `ACCEPTED`:

    An action that has been accepted, but is not yet in progress.

  - `IN_PROGRESS`:

    An action that is being executed.

  - `SUCCEEDED`:

    An action that was accepted and completed successfully.

  - `FAILED`:

    An action that was accepted and did not complete successfully.

  - `REJECTED`:

    When the action mode is set to manual, an action that has been rejected by the user.

  - `CLEARED`:

    When the action mode is set to manual, an action that is no longer recommended by the market.

  - `ACCOUNTING`:

    For cloud entities, an action to resize in order to use a different instance template.

  - `QUEUED`:

    When more than 10 actions are to be executed for a single target, Turbonomic will place the 11th and all subsequent actions into a QUEUED state, and they will execute as the previous actions complete.

- actionTypeList

  - `START`:

    Start an entity, and add it to the Turbonomic market.

  - `MOVE`:

    Move an entity from one provider to another. For example, moving a VM between hosts, or a datastore between disk arrays.

  - `SUSPEND`:

    Suspend the entity to exclude its capacity from your environment. Suspend does not remove the entity from your environment, but the market does not include the entity in its analysis. Based on market conditions, analysis can recommend restarting suspended entities, which includes them in market analysis once again.

  - `ADD PROVIDER`:

    This is equivalent to the START action, but for storage entities.

  - `CHANGE`:

    Move a virtual machine's storage between datastores.

  - `PROVISION`:

    Add capacity to your environment.

  - `RECONFIGURE`:

    Analysis recommends this when a VM is not able to correct a misconfiguration via another type of action. For example, analysis can recommend a RECONFIGURE action when a placement policy requires a VM to move to a host on a particular cluster, but cross-cluster moves are not permitted.

  - `DELETE`:

    This is equivalent to the SUSPEND action, but for storage entities.

  - `RESERVE_ON_PM`:

When using the reservation functionality, this is the action to reserve the resources of a VM on a particular host.

  - `RESERVE_ON_DS`:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular datastore.

  - `RESIZE_FOR_EFFICIENCY`:

    A resize that reduces the capacity of a commodity, such as vCPU, or vMEM.

  - `RESIZE_FOR_PERFORMANCE`:

    A resize that increases the capacity of a commodity, such as vCPU, or vMEM.

- cleared

  Default: `false`.

  When `true`, actions that were removed from the action list before being executed will be returned.

- costType

  - `Saving`

    An action that will decrease cloud spending.

  - `Investment`

    An action that will increase cloud spending.

- cumulative

  Default: `false`.

  When `true`, shows the savings for each action in a scope as a cumulative total, rather than showing the savings for individual actions.

- environmentType

  - `ONPREM`:

    Returns only actions related to entities that are part of your on-prem environment.

  - `CLOUD`:

    Returns only actions related to entities that are part of your cloud environment.

- groupBy

  You may group the actions returned by the following criteria: `[actionModes, actionStates, actionTypes, risk, riskSeverity, riskSubCategory]`

- hasReservedInstance

  Default: `false`.

  When `true`, only actions that are related to reserved instances will be returned.

- relatedEntityTypes

  Returns only actions related to entities of these types within the scope. For example, if your scope is a virtual datacenter and you specify VirtualMachine as a related entity type, the request will include only those actions related to VMs in the specified virtual datacenter.

  Choose from the following entity types: `[ Datacenter, PhysicalMachine, VirtualMachine, Storage, Application, Chassis, DiskArray, IOModule, StorageControl, Switch, VirtualDataCenter, VPod, DPod, Container, Database, DatabaseServerContainer, LogicalPool ]`

- riskSeverityList

  Returns only actions of the given severities: `[ Unknown, Normal, Minor, Major, Critical ]`

- riskSubCategoryList

- returns only actions of the given subcategories: `[ Performance Assurance, Efficiency Improvement, Prevention, Compliance ]`

Example: `POST https://10.10.10.10/api/v3/targets/__xwWYAE3EeiNKPbzHdETIA/actions`

Example inputDto, filtering for actions that are pending acceptance:

```
{
  "actionStateList":[
    "PENDING_ACCEPT"
  ]
}
```

Response: An array of actions applicable to the specified target that are in the `PENDING_ACCEPT` state.


```
[
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/actions/_uictIJuHEeics6ASgJ-kdg"
      },
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/actions/2204259249360"
      }
    ],
    "uuid":"_uictIJuHEeics6ASgJ-kdg",
    "createTime":"2018-08-08T23:53:08-04:00",
    "actionType":"RIGHT_SIZE",
    "actionState":"PENDING_ACCEPT",
    "actionMode":"MANUAL",
    "details":"Scale down VMem for VirtualMachine btc-1 from 8 GB to 7 GB",
    "importance":0,
    "target":{
      "uuid":"4223dcd6-0b28-6ae6-1066-689912ced3a1",
      "displayName":"btc-1",
      "className":"VirtualMachine",
      "aspects":{
        "virtualMachineAspect":{
          "os":"Other Linux (64-bit)",
          "connectedNetworks":[
            {
              "uuid":"74c91ba4fa390030f47f91e6e1ee77ff7b4db353",
              "displayName":"VM Network"
            }
          ],
          "numVCPUs":1,
          "ebsOptimized":false
        }
      },
      "environmentType":"ONPREM"
    },
    "currentEntity":{
      "uuid":"_ui4yAZuHEeics6ASgJ-kdg",
      "className":"VMem"
    },
    "newEntity":{
      "uuid":"_ui4yA5uHEeics6ASgJ-kdg",
      "className":"VMem"
    },
    "currentValue":"8388608.0",
    "newValue":"7340032.0",
    "resizeToValue":"7340032.0",
    "risk":{
      "uuid":"_ujB78ZuHEeics6ASgJ-kdg",
      "subCategory":"Efficiency Improvement",
      "description":"Underutilized VMem in Virtual Machine 'btc-1'",
      "severity":"MINOR",
      "reasonCommodity":"VMem",
      "importance":0
    },
    "stats":[
      {
        "name":"costPrice",
        "filters":[
          {
            "type":"savingsType",
```

```
            "value":"savings"
          }
        ],
        "units":"$",
        "value":50
      }
    ],
    "actionID":2204259249360
  },
  ...
  }
]
```

Entities for a specified target

Referenced Endpoint: Entities

Gets a list of entities discovered by the specified target. This is used most often for reporting, dashboard, or inventory purposes.

Getting An Array of Entities: Gets a list of all entities discovered by the specified target.

An entity is any device in your environment that Turbonomic manages, such as applications, physical machines, VMs, storage, and datacenters.

Example:`GET https://10.10.10.10/api/v3/targets/__xwWYAE3EeiNKPbzHdETIA/actions`

Response: An array of `EntityApiDTO` objects belonging to a vCenter target:

```
[
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/entities/vsphere-mycorp.com-ResourcePool-resgroup-8?
include_aspects=false"
      }
    ],
    "uuid":"vsphere-mycorp.com-ResourcePool-resgroup-8",
    "displayName":"Resources-Cluster1",
    "className":"VirtualDataCenter",
    "priceIndex":1e-12,
    "state":"ACTIVE",
    "severity":"Normal",
    "discoveredBy":{
      "uuid":"__xwWYAE3EeiNKPbzHdETIA",
      "displayName":"vsphere-mycorp.com",
      "type":"vCenter"
    },
    "remoteId":"resgroup-8",
    "environmentType":"ONPREM"
  },
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/entities/vsphere-mycorp.com-
GuestLoad-421ddc94-8d0c-504a-62f5-493802e96d0a?include_aspects=false"
      }
    ],
    "uuid":"vsphere-mycorp.com-GuestLoad-421ddc94-8d0c-504a-62f5-493802e96d0a",
    "displayName":"GuestLoad[Nexus1000V]",
    "className":"Application",
    "priceIndex":1,
    "state":"ACTIVE",
    "severity":"Normal",
    "discoveredBy":{
      "uuid":"__xwWYAE3EeiNKPbzHdETIA",
      "displayName":"vsphere-mycorp.com",
      "type":"vCenter"
    },
    "remoteId":"vm-51",
    "environmentType":"ONPREM"
  },
  ...
  }
```

```
]
```

Templates endpoint

Turbonomic uses templates to reserve resources and deploy workload in your environment, to calculate supply or demand changes in a plan, and to calculate workloads for cloud environments.

Using the templates endpoint, you can:

- Get a list of templates available to your Turbonomic appliance
- Get information needed to create templates, such as supported OSes and CPUs
- Create a template
- Edit and Delete templates

Templates utility requests

Utility requests provide metadata that you can use to make related requests, assemble inputDTOs, or see the potential values of a particular DTO.

The templates endpoint contains utility requests that show the possible values for several optional fields used to create detailed templates. These can be used to find the supported cloud OSes and CPU models.

Getting a list of CPU models

Gets a list of valid CPU Models. If given a CPU Model, Turbonomic can more accurately reflect your CPU capacity in market decisions.

Example: `GET https://10.10.10.10/api/v3/templates/physicalmachine/cpumodels`

Result: A list of CpuModelApiDTOs, describing the cores, speed, and internal scaling factor for each CPU. This information will be used to provide more granular CPU data, if passed as part of a TemplateApiInputDTO:

```
[
  {
    "numCores": 4,
    "speed": 3500,
    "scalingFactor": 1.116909571123129,
    "modelName": "AMD A10 PRO-7800B"
  },
  {
    "numCores": 4,
    "speed": 3700,
    "scalingFactor": 0.9745581799084503,
    "modelName": "AMD A10 PRO-7850B"
  },
    ...
      {
    "numCores": 8,
    "speed": 1415,
    "scalingFactor": 1.1638948014750774,
    "modelName": "UltraSPARC T2 Plus"
  }
]
```

Templates requests

Template requests handle creation, configuration, editing and deletion of templates on your Turbonomic appliance.

Getting a List of Templates

Gets a list of templates available to your Turbonomic instance. This list will include both templates discovered from underlying targets and those created through the Turbonomic UI. To get a specific template, pass the template UUID as part of the request.

Examples:

- All Templates: `GET https://10.10.10.10/api/v3/templates`
- Single Target: `GET https://10.10.10.10/api/v3/templates/_v0Q70MpiEd-hypXfJzX8Wg`

Response: A list of `TemplateApiDTO`s:


```
[
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/templates/_v0Q70MpiEd-hypXfJzX8Wg"
      }
    ],
    "uuid": "_v0Q70MpiEd-hypXfJzX8Wg",
    "displayName": "Small",
    "className": "PhysicalMachineProfile",
    "price": 5000,
    "computeResources": [
      {
        "stats": [
          {
            "name": "numOfCores",
            "value": 4
          },
          {
            "name": "cpuSpeed",
            "units": "MHz",
            "value": 2992
          },
          {
            "name": "ioThroughputSize",
            "units": "MB/s",
            "value": 890
          },
          {
            "name": "memorySize",
            "units": "GB",
            "value": 32759
          },
          {
            "name": "networkThroughputSize",
            "units": "MB/s",
            "value": 512
          }
        ]
      }
    ],
    "infrastructureResources": [
      {
        "stats": [
          {
            "name": "powerSize",
            "value": 1
          },
          {
            "name": "spaceSize",
            "value": 1
          },
          {
            "name": "coolingSize",
            "value": 1
          }
        ]
      }
    ],
    "discovered": false
  },
  {
    "links": [
      {
        "rel": "self",
        "href": "https://10.10.10.10/api/v3/templates/_v0Q70cpiEd-hypXfJzX8Wg"
      }
    ],
    "uuid": "_v0Q70cpiEd-hypXfJzX8Wg",
    "displayName": "Medium",
    "className": "PhysicalMachineProfile",
    "price": 10000,
```

```
    "computeResources": [
      {
        "stats": [
          {
            "name": "numOfCores",
            "value": 4
          },
          {
            "name": "cpuSpeed",
            "units": "MHz",
            "value": 2992
          },
          {
            "name": "ioThroughputSize",
            "units": "MB/s",
            "value": 1780
          },
          {
            "name": "memorySize",
            "units": "GB",
            "value": 65519
          },
          {
            "name": "networkThroughputSize",
            "units": "MB/s",
            "value": 512
          }
        ]
      }
    ],
    "infrastructureResources": [
      {
        "stats": [
          {
            "name": "powerSize",
            "value": 1
          },
          {
            "name": "spaceSize",
            "value": 1
          },
          {
            "name": "coolingSize",
            "value": 1
          }
        ]
      }
    ],
    "discovered": false
  },
    ...
]
```

Creating a template

Creates a template for use in your environment. You must pass a completed TemplateApiInputDTO as the inputDTO. You can create templates for virtual machines, physical machines, storage, and containers.

Example: `POST https://10.10.10.10/api/v3/templates`

Example inputs:

- Container

```
{
  "computeResources":[
    {
      "stats":[
        {
          "name":"ioThroughput",
          "value":0
        },
        {
          "name":"networkThroughput",
          "value":0
        },
```

```
        {
          "name":"memorySize",
          "value":8192
        },
        {
          "name":"cpuSpeed",
          "value":1536
        }
      ]
    }
  ],
  "storageResources":[
    {
      "stats":[
        {
          "name":"diskSize",
          "value":20
        }
      ]
    }
  ],
  "className":"ContainerProfile",
  "displayName":"ContainerTemplate1"
}
```


- Physical Machine


```
{
  "cpuModel":"AMD A10-5700",
  "computeResources":[
    {
      "stats":[
        {
          "name":"ioThroughputSize",
          "value":1000
        },
        {
          "name":"networkThroughputSize",
          "value":1000
        },
        {
          "name":"memorySize",
          "value":8192
        },
        {
          "name":"numOfCores",
          "value":4
        },
        {
          "name":"cpuSpeed",
          "value":"3400"
        }
      ]
    }
  ],
  "price":0,
  "className":"PhysicalMachineProfile",
  "displayName":"HostTemplate1"
}
```


- Storage


```
{
  "storageResources":[
    {
      "stats":[
        {
          "name":"diskIops",
          "value":250
        },
        {
          "name":"diskSize",
          "value":750
        }
      ]
    }
```

```
  ],
  "price":0,
  "className":"StorageProfile",
  "displayName":"StorageTemplate1"
}
```

- Virtual Machine

```
{
  "computeResources":[
    {
      "stats":[
        {
          "name":"ioThroughput",
          "value":0
        },
        {
          "name":"networkThroughput",
          "value":500
        },
        {
          "name":"memorySize",
          "value":4096
        },
        {
          "name":"memoryConsumedFactor",
          "value":70
        },
        {
          "name":"numOfCpu",
          "value":2
        },
        {
          "name":"cpuSpeed",
          "value":1024
        },
        {
          "name":"cpuConsumedFactor",
          "value":75
        }
      ]
    }
  ],
  "storageResources":[
    {
      "type":"disk",
      "stats":[
        {
          "name":"diskIops",
          "value":50
        },
        {
          "name":"diskSize",
          "value":20
        },
        {
          "name":"diskConsumedFactor",
          "value":70
        }
      ]
    }
  ]
}
```

Response: The `TemplateApiInputDTO` for the template created. In this case, a virtual machine template:

```
{
  "uuid":"_lmaQoLUtEei5nORDMStFGA",
  "displayName":"VirtualMachineTemplate1",
  "className":"VirtualMachineProfile",
  "computeResources":[
    {
      "stats":[
        {
          "name":"numOfCpu",
```

```
          "value":2.0
        },
        {
          "name":"cpuSpeed",
          "units":"MHz",
          "value":1024.0
        },
        {
          "name":"cpuConsumedFactor",
          "units":"%",
          "value":75.0
        },
        {
          "name":"memorySize",
          "units":"MB",
          "value":4096.0
        },
        {
          "name":"memoryConsumedFactor",
          "units":"%",
          "value":70.0
        },
        {
          "name":"ioThroughput",
          "units":"MB/s",
          "value":0.0
        },
        {
          "name":"networkThroughput",
          "units":"MB/s",
          "value":500.0
        }
      ]
    }
  ],
  "storageResources":[
    {
      "stats":[
        {
          "name":"diskIops",
          "value":50.0
        },
        {
          "name":"diskSize",
          "units":"GB",
          "value":20.0
        },
        {
          "name":"diskConsumedFactor",
          "units":"%",
          "value":70.0
        }
      ],
      "type":"disk"
    }
  ],
  "deploymentProfile":{
    "uuid":"_llbZMrUtEei5nORDMStFGA",
    "displayName":"VirtualMachineTemplate1_DeploymentProfile",
    "className":"ServiceCatalogItem",
    "deployParameters":[
      {
        "targetType":"vCenter",
        "providers":[
          {
            "provider":{
              "uuid":"dfe657268310d8b128d441b2a34ec0721e72ee7e",
              "displayName":"Development DC7",
              "className":"DataCenter",
              "discoveredBy":{
                "uuid":"_OjF3UJwFEei_xJ5Ai6-rnQ",
                "displayName":"vsphere-dc7.mycorp.com",
                "type":"vCenter"
              }
            },
            "parameters":[
              {
                "parameterType":"image",
                "properties":[
                  {
                    "name":"id",
                    "value":"path/to/image"
```

```
                    }
                  ]
                }
              ]
            }
          ]
        }
      ]
    },
    "discovered":false
}
```

Editing a template

Edits a user-created template, given the template UUID and the edited DTO. To edit discovered templates, edit them on the target from which they originate. Note that you must pass a complete TemplateApiDTO with the edited fields.

Example: `PUT https://10.10.10.10/api/v3/templates/_lmaQoLUtEei5nORDMStFGA`

Example input: Note that both the `displayName` and `diskConsumedFactor` values were changed:

```
{
  "computeResources":[
    {
      "stats":[
        {
          "name":"ioThroughput",
          "value":0
        },
        {
          "name":"networkThroughput",
          "value":500
        },
        {
          "name":"memorySize",
          "value":4096
        },
        {
          "name":"memoryConsumedFactor",
          "value":70
        },
        {
          "name":"numOfCpu",
          "value":2
        },
        {
          "name":"cpuSpeed",
          "value":1024
        },
        {
          "name":"cpuConsumedFactor",
          "value":75
        }
      ]
    }
  ],
  "storageResources":[
    {
      "type":"disk",
      "stats":[
        {
          "name":"diskIops",
          "value":50
        },
        {
          "name":"diskSize",
          "value":20
        },
        {
          "name":"diskConsumedFactor",
          "value":75
        }
      ]
    }
  ],
  "className":"VirtualMachineProfile",
```

```
  "displayName":"VirtualMachineTemplate1_Modified",
  "deploymentProfileId":"_llbZMrUtEei5nORDMStFGA"
}
```

Response: The modified `TemplateApiDTO` reflecting the submitted changes.

Deleting a template

Deletes the specified template. This will not delete associated deployment profiles, and deployments already scheduled to use this template will not succeed.

Example: `DELETE https://10.10.10.10/api/v3/templates/_lmaQoLUtEei5nORDMStFGA`

Response: A successful deletion will return `true`.

Topology definitions endpoint

Turbonomic gives you the ability to create your own Business Applications, Business Transactions, and Services without the need to ingest additional application data into the Turbonomic platform. This is especially useful in cases where there are gaps in the application stack shown in the Turbonomic supply chain. For example, in the absence of an application monitoring target such as AppDynamics or Dynatrace, you will not see Business Applications in your supply chain. This feature addresses those gaps.

When you create a new application entity, you identify interrelated application entities and nodes in your existing environment for which you want to measure performance, so Turbonomic can link them in a supply chain and represent them as a unified group. You can monitor overall performance for the group in the context of the new application entity, and drill down to the individual entities and nodes for finer details.

Using the topology definitions endpoint of the Turbonomic API, you can:

- Get a single topology definition or list of topology definitions present in your environment
- Create a topology definition
- Edit a topology definition
- Delete a topology definition

Topology definition requests

Topology definition requests handle creation, configuration, editing, and deletion of topology definitions on your Turbonomic appliance.

Getting Topology Definitions

For a list of topology definitions, the API returns an array of TopologyDataDefinitionApiDTO. Each TopologyDataDefinitionApiDTO will contain the UUID, entity type, and connection attributes for the defined topology. To get a single topology definition, include the UUID of the topology definition in the request.

Examples:

- All Topology Definitions: `GET https://10.10.10.10/api/v3/topologydefinitions`
- Single Topology Definition: `GET https://10.10.10.10/api/v3/topologydefinitions/284711321829968`

Response:

```
{
  "uuid":"284711313871856",
  "displayName":"Daltest1__App",
  "entityType":"BusinessApplication",
  "entityDefinitionData":{
    "namePrefix":"Daltest1_",
    "entityType":"VirtualMachine",
```

```
    "generationAndConnectionAttribute":{
      "expVal":"App",
      "expType":"EQ",
      "filterType":"virtualMachinesByTag",
      "caseSensitive":true,
      "entityType":null,
      "singleLine":false
    }
  }
}
```

Creating a manual topology definition

To create a manual topology definition, you must provide the `entityType` of the definition, and manual mapping data to the entities to use. Consider the following input DTO to create a manual topology definition:

```
{
  "entityType":"BusinessApplication",
  "entityDefinitionData":{
    "manualConnectionData":{
      "BusinessTransaction":{
        "staticConnections":[
          "73585155548499",
          "73585155548500"
        ]
      },
      "Service":{
        "staticConnections":[
          "73585155548487",
          "73585155608929"
        ]
      },
      "ApplicationComponent":{
        "staticConnections":[
          "73596151905427",
          "73596151905425",
          "73596151905428",
          "73596151905429"
        ]
      },
      "VirtualMachine":{
        "staticConnections":[
          "73586657216608",
          "73585300874097",
          "73586657216576"
        ]
      },
      "DatabaseServer":{
        "staticConnections":[
          "73585314633044",
          "73585314632869",
          "73585314632946"
        ]
      },
      "Container":{
        "staticConnections":[
          "73585323455544",
          "73582124234229",
          "73585385438946"
        ]
      },
      "ContainerPod":{
        "staticConnections":[
          "73585334243044",
          "73585314324535",
          "73585314533256"
        ]
      }
    }
  },
  "displayName":"DalManualtest_"
}
```

In the manualConnectionData parameter of the entityDefinitionData, the UUIDs of the related entities have been provided for each level of the supply chain for which the topology should be instantiated.

The creation can be as simple or complex as desired. For example, a simple manual topology may look like the following:

```
{
  "entityType":"BusinessTransaction",
  "entityDefinitionData":{
    "manualConnectionData":{
      "Service":{
        "staticConnections":[
          "73585155548487"
        ]
      },
      "Container":{
        "staticConnections":[
          "73596151905298"
        ]
      }
    }
  },
  "uuid":"284711514573216",
  "displayName":"DalManualSimple"
}
```

Creating an automated topology definition

To create an automated topology definition, you must provide the entityType of the definition, and the data that makes up the regular expression used to determine dynamic group membership. Consider the following input DTO to create an automated topology definition:

```
{
   "entityType":"BusinessApplication",
   "entityDefinitionData":{
      "entityType":"VirtualMachine",
      "namePrefix":"Daltest1_",
      "generationAndConnectionAttribute":{
         "expType":"EQ",
         "expVal":"App",
         "filterType":"vmsByTag",
         "caseSensitive":false
      }
   }
}
```

In the generationAndConnectionAttribute parameter of the entityDefinitionData, the information that makes up the regular expression is sent. In this instance, it will match any virtual machine whose tag value equals "App".

Editing a topology definition

Edits the input fields of an existing topology definition, and accepts the modified TopologyDataDefinitionApiDTO as input.

Note: You can GET the topology definition first using the `GET /topologydefinition/UUID` request to see the current configuration. Note that you must pass all fields in the modified TopologyDataDefinitionApiDTO, not only the changed fields.

A successful edit will return the modified TopologyDataDefinitionApiDTO.

Example: `PUT https://10.10.10.10/api/v3/topologydefinitions`

Deleting a topology definition

Deletes the specified topology definition and removes all created entities. This will also remove any pending actions related to the deleted entities.

Example: `DELETE https://10.10.10.10/api/v3/topologydefinitions/284711321829968`

Response: A successful deletion request returns a Response Code of `200`.

Users endpoint

As an administrator, you specify accounts that grant users access to Turbonomic. Credentials for user accounts can either be saved locally on the Turbonomic server or managed externally through Microsoft Active Directory (AD) or another single sign-on (SSO) provider.

The user account configuration determines the following details:

- User authentication

  To configure an account, you set the type of authentication the account uses:

  – Local user: Configure the username and password and save those credentials on the Turbonomic server.
  – External user: Single user accounts can authenticate through SSO or through AD.
  – External group: Groups of user accounts can authenticate through SSO or AD.
- User authorization

  You can also specify properties that determine the range of access and features for a user:

  – Role: The user role determines access to specific Turbonomic features.
  – Scope: The user scope determines how much of the environment this user can manage.

You can access the users endpoint of the Turbonomic API to create and manage accounts for local users, external users, and external groups. You must have administrator or site administrator role privileges to access the users endpoint.

Managing local users

Administrators can access the users endpoint to create and manage local user accounts. Turbonomic stores the credentials for these accounts on the local server.

Adding a local user

To add a local user, make a POST request to the users endpoint.

```
POST https://{your_instance_address}/api/v3/users
```

Configure the request parameters, using the following example as reference.

```
{
  "username": "<user_name>",
  "password": "<password>",
  "roles": [
    {
      "name": "<ROLE_NAME>"
    }
  ],
  "loginProvider": "Local",
  "type": "DedicatedCustomer",
  "scope": [
  {
    "uuid":"<UUID_1>"
  },
  {
    "uuid":"<UUID_2>"
  }
```

```
  ]
}
```

- (Required) `username`: Specify the username for the account.
- (Required) `password`: Specify the password for the account.
- (Required) `roles`: Specify a user role to grant the user access to specific Turbonomic features. Use uppercase letters only.
  - `ADMINISTRATOR` (only if the logged-in user is an administrator)

    Users with this role can use all Turbonomic features and modify settings to configure the Turbonomic installation. For Turbonomic instances hosted in the public cloud, this role is limited to the Turbonomic representative that manages the instances.
  - `SITE_ADMIN`

    Users with this role can use all Turbonomic features and modify site-specific settings to configure the Turbonomic installation. Users can also administer Groups, Policies, Workflows, Templates, Billing/Costs, and Target Configuration, but not Email, Licenses, Updates, and Maintenance. Users can create other user accounts, except accounts with the Administrator role.
  - `AUTOMATOR`

    Users with this role can use all Turbonomic features (including Plan, Park, and Place), but cannot configure the Turbonomic installation or create policies.
  - `DEPLOYER`

    Users with this role can view all Turbonomic charts and data, use Place to reserve workloads, and create placement policies and templates. However, users cannot run plans or execute any recommended actions.
  - `PARKER`

    Users with this role can park workloads manually or automatically (using parking schedules), manage parking schedules, and view all Turbonomic charts and data. However, users cannot create policies and templates, place workloads, run plans, or execute any recommended actions.
  - `PARKER_AUTOMATOR`

    Users with this role can park workloads manually or automatically (using parking schedules) and view all Turbonomic charts and data. However, users cannot manage parking schedules, create policies and templates, place workloads, run plans, or execute any recommended actions.
  - `ADVISOR`

    Users with this role can view all Turbonomic charts and data, and run plans. However, users cannot use Place to reserve workloads, create policies, or execute any recommended actions.
  - `OBSERVER`

    Users with this role can view the environment, including the Home Page and Dashboards. Users can also use Search to set a scope to the session. For scope, only VM groups and Resource Groups are supported.
  - `OPERATIONAL_OBSERVER`

    Users with this role can view the environment, including the Home Page, Dashboards, Groups, and Policies. Users can also use Search to set a scope to the session.
  - `SHARED_ADVISOR`

    Users with this role are scoped users. They can view the Home Page and Dashboards, but only see VMs and Applications. Users cannot execute Turbonomic actions.
  - `SHARED_OBSERVER`

    Users with this role are scoped users. They can view the Home Page, custom dashboards, and application maps, but can see only VMs and Applications. They cannot view Executive Dashboards or execute Turbonomic actions. Shared Observer is the most restricted user role.

  - REPORT_EDITOR

    Users with this role can create, edit, and delete reports. Due to limits to the reporting license, only one user per instance is allowed to have this role (by default, the local **administrator** user). To assign this role to another user, you must first remove it from the current user. Be sure that the new user is *not* a scoped user.
- (Required) `loginProvider`: For a local user, specify `Local`.
- (Required) `type`: Specify `DedicatedCustomer`.
- (Optional) `scope`: Specify one or more scopes by using the numeric UUID value for each scope. For example, to limit the scope to a group of physical machines that the user manages, specify the UUID for that group. This parameter is not valid for users with the `ADMINISTRATOR` role.

  To get a list of the groups that are available to scope to in your Turbonomic instance, along with their display names and UUIDs, make a `GET` request to the search endpoint, as shown in the following example.

```
GET https://{your_instance_address}/api/v3/search?types=Group
```

Response:

```
{
  "uuid": "4776186363888",
  "displayName": "myuser",
  "username": "myuser",
  "roles": [
    {
      "name": "DEPLOYER"
    }
  ],
  "loginProvider": "Local",
  "type": "DedicatedCustomer",
  "scope": [
    {
      "uuid": "286125332284065",
      "isStatic": true,
      "logicalOperator": "AND",
      "groupClassName": "GroupApiDTO"
    }
    {
      "uuid": "266725335284010",
      "isStatic": true,
      "logicalOperator": "AND",
      "groupClassName": "GroupApiDTO"
    }
  ],
  "showSharedUserSC": false
}
```

Maintaining local users

To edit the details of an existing user or to delete a user, you must know the UUID value for the user.

- To get a list of all the currently configured user accounts, make a `GET` request to the users endpoint:

```
GET https://10.10.10.10/api/v3/users
```

The API returns an array of UserApiDTO objects. Each `UserApiDTO` object includes the UUID, display name, username, roles, scope, and other details about a user account.

Response:

```
[
  {
    "uuid": "0000000000111",
    "displayName": "extTestUser",
    "username": "extTestUser",
    "roles": [
      {
        "name": "DEPLOYER"
```

```
      }
    ],
    "loginProvider": "LDAP",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  },
  {
    "uuid": "0000000000112",
    "displayName": "externaluser1",
    "username": "externaluser1",
    "roles": [
      {
        "name": "OBSERVER"
      }
    ],
    "loginProvider": "LDAP",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  },
  {
    "uuid": "0000000000113",
    "displayName": "administrator",
    "username": "administrator",
    "roles": [
      {
        "name": "ADMINISTRATOR"
      }
    ],
    "loginProvider": "Local",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  },
  {
    "uuid": "0000000000114",
    "displayName": "newTestUser",
    "username": "newTestUser",
    "roles": [
      {
        "name": "DEPLOYER"
      }
    ],
    "loginProvider": "Local",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  }
]
```


  - To get details about a specific user, append the UUID of the user to the request:

    ```
    GET https://10.10.10.10/api/v3/users/0000000000111
    ```

  - To get information about the currently logged-in user, append `/me` to the request:

    ```
    GET https://10.10.10.10/api/v3/users/me
    ```

- To edit the details for a user, make a PUT request to the users endpoint and include the user's UUID:

  ```
  PUT https://10.10.10.10/api/v3/users/0000000000114
  ```

  The following parameters are required in the request body, regardless of whether you want to modify them:

  - `username`
  - `roles`
  - `loginProvider`
  - `type`

  Example:

Include the required parameters in the request body and specify the values that you want to modify. In the following example, the role for the `newTestUser` user is changed from DEPLOYER to OBSERVER. The other parameters remain unchanged:

```
{
  "username": "newTestUser",
  "roles": [
    {
      "name": "OBSERVER"
    }
  ],
  "loginProvider": "Local",
  "type": "DedicatedCustomer"
}
```

Response:

```
{
  "uuid": "0000000000114",
  "displayName": "newTestUser",
  "username": "newTestUser",
  "roles": [
    {
      "name": "OBSERVER"
    }
  ],
  "loginProvider": "Local",
  "type": "DedicatedCustomer",
  "showSharedUserSC": false
}
```

- To delete a user, make a DELETE request to the `users` endpoint and include the user's UUID:

```
DELETE https://10.10.10.10/api/v3/users/0000000000111
```

Managing single sign-on users

Administrators can access the users endpoint to create and manage user accounts that authenticate by single sign-on (SSO) through an external identity provider (IdP). Turbonomic recognizes IdP responses that comply with SAML 2.0.

For information about authenticating with Microsoft Active Directory (AD), see “Managing Active Directory users” on page 1556.

Prerequisites

To enable SSO, you must configure access to the IdP. For information about configuring SSO, see Single Sign-On Authentication.

Important:

For self-hosted Turbonomic instances, you can configure Turbonomic to use SSO authentication. When SSO is enabled, Turbonomic permits logins only through the SSO IdP. Whenever you navigate to your Turbonomic installation, it redirects you to the SSO Identity Provider (IdP) for authentication before it displays the Turbonomic user interface.

Before you enable SSO for your Turbonomic installation, *you must configure at least one SSO user with Turbonomic administrator privileges*. If you do not, then after you enable SSO, you cannot configure any SSO users in Turbonomic. To authorize an SSO user as an administrator, use **External authentication** to do one of the following options:

- Configure a single SSO user with administrator authorization.

  Add an external user. The username must match an account that the IdP manages.
- Configure an SSO user group with administrator authorization.

  Add an external group. The group name must match a user group on the IdP, and that group must have at least one member.

For information about configuring SSO user groups in SAML, see Configuring a Group for SSO Authentication. For information about configuring SSO authentication for Turbonomic, see Single Sign-On Authentication.

To get information about the currently configured SSO provider, make a GET request to the `users/saml` endpoint:

```
GET https://{your_instance_address}/api/v3/users/saml
```

Adding SSO users

To add an SSO user, make a POST request to the users endpoint.

```
POST https://{your_instance_address}/api/v3/users
```

Configure the request parameters, using the following example as reference.

```
{
  "username": "<user_name>",
  "roles": [
    {
      "name": "<ROLE_NAME>"
    }
  ],
  "loginProvider": "LDAP",
  "type": "DedicatedCustomer",
  "scope": [
  {
    "uuid":"<UUID_1>"
  },
  {
    "uuid":"<UUID_2>"
  }
 ]
}
}
```

- (Required) `username`: Specify a username for the account. The username must match a username that is specified on the IdP.
- (Required) `roles`: Specify a user role to grant the user access to specific Turbonomic features. Use uppercase letters only.
  - `ADMINISTRATOR` (only if the logged-in user is an administrator)

    Users with this role can use all Turbonomic features and modify settings to configure the Turbonomic installation. For Turbonomic instances hosted in the public cloud, this role is limited to the Turbonomic representative that manages the instances.
  - `SITE_ADMIN`

    Users with this role can use all Turbonomic features and modify site-specific settings to configure the Turbonomic installation. Users can also administer Groups, Policies, Workflows, Templates, Billing/Costs, and Target Configuration, but not Email, Licenses, Updates, and Maintenance. Users can create other user accounts, except accounts with the Administrator role.
  - `AUTOMATOR`

    Users with this role can use all Turbonomic features (including Plan, Park, and Place), but cannot configure the Turbonomic installation or create policies.
  - `DEPLOYER`

    Users with this role can view all Turbonomic charts and data, use Place to reserve workloads, and create placement policies and templates. However, users cannot run plans or execute any recommended actions.
  - `PARKER`

Users with this role can park workloads manually or automatically (using parking schedules), manage parking schedules, and view all Turbonomic charts and data. However, users cannot create policies and templates, place workloads, run plans, or execute any recommended actions.

  - PARKER_AUTOMATOR

    Users with this role can park workloads manually or automatically (using parking schedules) and view all Turbonomic charts and data. However, users cannot manage parking schedules, create policies and templates, place workloads, run plans, or execute any recommended actions.

  - ADVISOR

    Users with this role can view all Turbonomic charts and data, and run plans. However, users cannot use Place to reserve workloads, create policies, or execute any recommended actions.

  - OBSERVER

    Users with this role can view the environment, including the Home Page and Dashboards. Users can also use Search to set a scope to the session. For scope, only VM groups and Resource Groups are supported.

  - OPERATIONAL_OBSERVER

    Users with this role can view the environment, including the Home Page, Dashboards, Groups, and Policies. Users can also use Search to set a scope to the session.

  - SHARED_ADVISOR

    Users with this role are scoped users. They can view the Home Page and Dashboards, but only see VMs and Applications. Users cannot execute Turbonomic actions.

  - SHARED_OBSERVER

    Users with this role are scoped users. They can view the Home Page, custom dashboards, and application maps, but can see only VMs and Applications. They cannot view Executive Dashboards or execute Turbonomic actions. Shared Observer is the most restricted user role.

  - REPORT_EDITOR

    Users with this role can create, edit, and delete reports. Due to limits to the reporting license, only one user per instance is allowed to have this role (by default, the local **administrator** user). To assign this role to another user, you must first remove it from the current user. Be sure that the new user is *not* a scoped user.

- (Required) `loginProvider`: For an SSO user, specify `LDAP`.
- (Required) `type`: Specify `DedicatedCustomer`.
- (Optional) `scope`: Specify one or more scopes by using the numeric UUID value for each scope. For example, to limit the scope to a group of physical machines that the user manages, specify the UUID for that group. This parameter is not valid for users with the `ADMINISTRATOR` role.

  To get a list of the groups that are available to scope to in your Turbonomic instance, along with their display names and UUIDs, make a `GET` request to the search endpoint, as shown in the following example.

```
GET https://{your_instance_address}/api/v3/search?types=Group
```

Response:

```
curl -X 'POST' \
  'https://{your_instance_address}/api/v3/users' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "username": "mySsoUser",
  "roles": [
    {
      "name": "DEPLOYER"
    }
  ],
  "loginProvider": "LDAP",
```

```
  "type": "DedicatedCustomer",
  "scope": [
    {
      "uuid": "286125332284065",
      "isStatic": true,
      "logicalOperator": "AND",
      "groupClassName": "GroupApiDTO"
    },
    {
      "uuid": "266725335284010",
      "isStatic": true,
      "logicalOperator": "AND",
      "groupClassName": "GroupApiDTO"
    }
  ],
  "showSharedUserSC": false
}'
```

Maintaining SSO users

To edit the details of an existing user or to delete a user, you must know the UUID value for the user.

- To get a list of all the currently configured user accounts, make a GET request to the `users` endpoint:

```
GET https://{your_instance_address}/api/v3/users
```

The API returns an array of UserApiDTO objects. Each `UserApiDTO` object contains the universal unique identifier (UUID), display name, username, encoded password, roles, scope, and other details about a user account.

Response:

```
[
  {
    "uuid": "0000000000111",
    "displayName": "extTestUser",
    "username": "extTestUser",
    "roles": [
      {
        "name": "DEPLOYER"
      }
    ],
    "loginProvider": "LDAP",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  },
  {
    "uuid": "0000000000112",
    "displayName": "externaluser1",
    "username": "externaluser1",
    "roles": [
      {
        "name": "OBSERVER"
      }
    ],
    "loginProvider": "LDAP",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  },
  {
    "uuid": "0000000000113",
    "displayName": "administrator",
    "username": "administrator",
    "roles": [
      {
        "name": "ADMINISTRATOR"
      }
    ],
    "loginProvider": "Local",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  },
  {
    "uuid": "0000000000114",
    "displayName": "newTestUser",
    "username": "newTestUser",
    "roles": [
```

```
      {
        "name": "DEPLOYER"
      }
    ],
    "loginProvider": "Local",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  }
]
```

– To get details about a specific user, append the UUID of the user to the request:

```
GET https://{your_instance_address}/api/v3/users/0000000000111
```

– To get information about the currently logged-in user, append `/me` to the request:

```
GET https://{your_instance_address}/api/v3/users/me
```

- To edit the details for a user, make a PUT request to the `users` endpoint and include the user's UUID:

```
PUT https://{your_instance_address}/api/v3/users/0000000000111
```

The following parameters are required, regardless of whether you want to modify them:

– `username`
– `roles`
– `loginProvider`
– `type`

Include the required parameters in the request body and specify the values that you want to modify. In the following example, the role for the `extTestUser` user is changed from DEPLOYER to OBSERVER. The other parameters remain unchanged:

Example:

```
{
  "username": "extTestUser",
  "roles": [
    {
      "name": "OBSERVER"
    }
  ],
  "loginProvider": "LDAP",
  "type": "DedicatedCustomer"
}
```

Response:

```
{
  "uuid": "0000000000111",
  "displayName": "extTestUser",
  "username": "extTestUser",
  "roles": [
    {
      "name": "OBSERVER"
    }
  ],
  "loginProvider": "LDAP",
  "type": "DedicatedCustomer",
  "showSharedUserSC": false
}
```

- To delete a user, make a DELETE request to the `users` endpoint and include the user's UUID:

```
DELETE https://{your_instance_address}/api/v3/users/0000000000111
```

Adding SSO groups

To add an SSO group, make a POST request to the `/users/ldap/groups` endpoint.

```
POST https://{your_instance_address}/api/v3/users/ldap/groups
```

Configure the request parameters, using the following example as reference.

```
{
  "uuid": "mySsoGroup",
  "displayName: "<display_name>",
  "rolename": "<ROLE_NAME>",
  "type": "DedicatedCustomer"
}
```

- (Required) `displayName`: Specify the name of the SSO group that you are authorizing. The group name must be valid for the authentication system that you are relying on.
- (Required) `rolename`: Specify a role to grant SSO group members access to specific Turbonomic features. Use uppercase letters only.
  - `ADMINISTRATOR` (only if the logged-in user is an administrator)

    Users with this role can use all Turbonomic features and modify settings to configure the Turbonomic installation. For Turbonomic instances hosted in the public cloud, this role is limited to the Turbonomic representative that manages the instances.
  - `SITE_ADMIN`

    Users with this role can use all Turbonomic features and modify site-specific settings to configure the Turbonomic installation. Users can also administer Groups, Policies, Workflows, Templates, Billing/Costs, and Target Configuration, but not Email, Licenses, Updates, and Maintenance. Users can create other user accounts, except accounts with the Administrator role.
  - `AUTOMATOR`

    Users with this role can use all Turbonomic features (including Plan, Park, and Place), but cannot configure the Turbonomic installation or create policies.
  - `DEPLOYER`

    Users with this role can view all Turbonomic charts and data, use Place to reserve workloads, and create placement policies and templates. However, users cannot run plans or execute any recommended actions.
  - `PARKER`

    Users with this role can park workloads manually or automatically (using parking schedules), manage parking schedules, and view all Turbonomic charts and data. However, users cannot create policies and templates, place workloads, run plans, or execute any recommended actions.
  - `PARKER_AUTOMATOR`

    Users with this role can park workloads manually or automatically (using parking schedules) and view all Turbonomic charts and data. However, users cannot manage parking schedules, create policies and templates, place workloads, run plans, or execute any recommended actions.
  - `ADVISOR`

    Users with this role can view all Turbonomic charts and data, and run plans. However, users cannot use Place to reserve workloads, create policies, or execute any recommended actions.
  - `OBSERVER`

    Users with this role can view the environment, including the Home Page and Dashboards. Users can also use Search to set a scope to the session. For scope, only VM groups and Resource Groups are supported.
  - `OPERATIONAL_OBSERVER`

Users with this role can view the environment, including the Home Page, Dashboards, Groups, and Policies. Users can also use Search to set a scope to the session.

  - SHARED_ADVISOR

    Users with this role are scoped users. They can view the Home Page and Dashboards, but only see VMs and Applications. Users cannot execute Turbonomic actions.

  - SHARED_OBSERVER

    Users with this role are scoped users. They can view the Home Page, custom dashboards, and application maps, but can see only VMs and Applications. They cannot view Executive Dashboards or execute Turbonomic actions. Shared Observer is the most restricted user role.

  - REPORT_EDITOR

    Users with this role can create, edit, and delete reports. Due to limits to the reporting license, only one user per instance is allowed to have this role (by default, the local **administrator** user). To assign this role to another user, you must first remove it from the current user. Be sure that the new user is *not* a scoped user.

- (Required) `type`: Specify `DedicatedCustomer`.
- (Optional) `scope`: Specify one or more scopes by using the numeric UUID value for each scope. For example, to limit the scope to a group of physical machines that the SSO group manages, specify the UUID for the physical machine group. This parameter is not valid for SSO groups with the `ADMINISTRATOR` role.

  To get a list of the groups that are available to scope to in your Turbonomic instance, along with their display names and UUIDs, make a `GET` request to the search endpoint, as shown in the following example.

```
GET https://{your_instance_address}/api/v3/search?types=Group
```

Response:

```
curl -X 'POST' \
  'https://{your_instance_address}/api/v3/users' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "uuid": "mySsoGroup",
  "displayName: "mySsoGroup",
  "rolename": "DEPLOYER",
  "type": "DedicatedCustomer"
}'
```

Maintaining SSO groups

To edit the details of an existing SSO group or to delete an SSO group, you must know the UUID value for the group.

- To get a list of all the currently configured SSO groups, make a `GET` request to the `users/ldap/groups` endpoint:

  ```
  GET https://{your_instance_address}/api/v3/users/ldap/groups
  ```

  The API returns an array of ActiveDirectoryApiDTO objects. Each `ActiveDirectoryGroupApiDTO` object contains the UUID, role, and other details about a group.

- To edit the details for a group, make a `PUT` request to the `users/ldap/groups` endpoint.

  ```
  PUT https://{your_instance_address}/api/v3/users/ldap/groups
  ```

  The following parameters are required in the request body, regardless of whether you want to modify them:

  - `displayName`

- roles
- type

Include the required parameters in the request body and specify the values that you want to modify. In the following example, the role for the mySsoGroup group is changed from DEPLOYER to OBSERVER. The other parameters remain unchanged:

```
{
  "displayName": "mySsoGroup",
  "rolename": "OBSERVER",
  "type": "DedicatedCustomer"
}
```

Response:

```
{
"uuid": "mySsoGroup",
"displayName": "mySsoGroup",
"type": "DedicatedCustomer",
"roleName": "DEPLOYER" }
```

- To delete a group, make a DELETE request to the users/ldap/groups endpoint and include the group's UUID:

```
DELETE https://{your_instance_address}/api/v3/users/ldap/groups/mySsoGroup
```

Managing Active Directory users

Administrators can access the users endpoint to create and manage user accounts and groups that authenticate by using Microsoft Active Directory (AD).

Prerequisites

To enable AD you must specify either an AD domain, an AD server, or both. Turbonomic uses this connection for all AD users.

Use a secure connection to communicate with AD servers. The AD domain must be configured to use LDAPS, and you must import a certificate into the Turbonomic server. Turbonomic can support LDAP channel binding and LDAP signing. To support these Active Directory features, you must configure secure access.

For more information, see Enforcing Secure Access.

To connect to an AD server, make a POST request to the users/ldap endpoint.

```
POST https://{your_instance_address}/api/v3/users/ldap
```

Configure your request with the following parameters:

```
{
  "domainName": "<domain_name>",
  "loginProviderURI": "10.10.10.123:444",
  "isSecure": "true"
}
```

- domainName: Specify the domain for AD authentication.

  To authenticate AD groups, specify a domain so that AD can find a user with the User Principal Name (UPN). If you specify a domain, but not a server, authentication uses any AD server from that domain. If you want to support only AD users and not groups, the domain is optional.
- loginProviderURI: Specify the server for AD authentication.

  To disable AD groups, specify a server but do not specify a domain. If you specify a domain and a server, authentication uses that server and also supports groups.

When you configure an AD server, by default Turbonomic assumes the AD server port to be 389 or 636. To specify a custom port for the AD server, add the port number to the AD server IP address. For example, `10.10.10.123:444` sets port 444.

- `isSecure`: Specify `true` for this setting to enable a secure connection to the AD server.

  Use a secure connection to communicate with AD servers. The AD domain must be configured to use LDAPS, and you must import a certificate into the Turbonomic server. Turbonomic can support LDAP channel binding and LDAP signing. To support these Active Directory features, you must configure secure access.

  For more information, see Enforcing Secure Access.

  To get information about the currently configured AD server and AD groups, make a GET request to the `users/ldap` endpoint.

```
GET https://{your_instance_address}/api/v3/users/ldap
```

Response:


```
[
  {
    "loginProviderURI": "ldaps://10.10.10.123:444",
    "isSecure": true,
    "groups": [
      {
        "uuid": "adGroup",
        "displayName": "adGroup",
        "type": "DedicatedCustomer",
        "roleName": "ADVISOR"
      },
      {
        "uuid": "administrator",
        "displayName": "administrator",
        "type": "DedicatedCustomer",
        "roleName": "ADMINISTRATOR"
      },
      {
        "uuid": "advisor",
        "displayName": "advisor",
        "type": "DedicatedCustomer",
        "roleName": "ADVISOR"
      },
      {
        "uuid": "automator",
        "displayName": "automator",
        "type": "DedicatedCustomer",
        "roleName": "AUTOMATOR"
      },
      {
        "uuid": "deployer",
        "displayName": "deployer",
        "type": "DedicatedCustomer",
        "roleName": "DEPLOYER"
      },
      {
        "uuid": "guyexternalgrouptest",
        "displayName": "guyexternalgrouptest",
        "type": "DedicatedCustomer",
        "roleName": "OBSERVER"
      },
      {
        "uuid": "observer",
        "displayName": "observer",
        "type": "DedicatedCustomer",
        "roleName": "OBSERVER"
      },
      {
        "uuid": "operational_observer",
        "displayName": "operational_observer",
        "type": "DedicatedCustomer",
        "roleName": "OPERATIONAL_OBSERVER"
      },
      {
        "uuid": "site_admin",
        "displayName": "site_admin",
        "type": "DedicatedCustomer",
```

```
        "roleName": "SITE_ADMIN"
      },
      {
        "uuid": "testgroop",
        "displayName": "testgroop",
        "type": "DedicatedCustomer",
        "roleName": "OBSERVER"
      }
    ]
  }
]
```

Adding AD users

To add an AD user, make a POST request to the users endpoint.

```
POST https://{your_instance_address}/api/v3/users
```

Configure the request parameters, using the following example as reference.

```
{
  "username": "<user_name>",
  "roles": [
    {
      "name": "<ROLE_NAME>"
    }
  ],
  "loginProvider": "LDAP",
  "type": "DedicatedCustomer"
  "scope": [
  {
    "uuid":"<UUID_1>"
  },
  {
    "uuid":"<UUID_2>"
  }
 ]
}
```

- (Required) `username`: Specify a username for the account. The username must be a valid User Principal Name (UPN), such as `abc@mycompany.com`.
- (Required) `roles`: Specify a user role to grant the user access to specific Turbonomic features. Use uppercase letters only.
  - `ADMINISTRATOR` (only if the logged-in user is an administrator)

    Users with this role can use all Turbonomic features and modify settings to configure the Turbonomic installation. For Turbonomic instances hosted in the public cloud, this role is limited to the Turbonomic representative that manages the instances.
  - `SITE_ADMIN`

    Users with this role can use all Turbonomic features and modify site-specific settings to configure the Turbonomic installation. Users can also administer Groups, Policies, Workflows, Templates, Billing/Costs, and Target Configuration, but not Email, Licenses, Updates, and Maintenance. Users can create other user accounts, except accounts with the Administrator role.
  - `AUTOMATOR`

    Users with this role can use all Turbonomic features (including Plan, Park, and Place), but cannot configure the Turbonomic installation or create policies.
  - `DEPLOYER`

    Users with this role can view all Turbonomic charts and data, use Place to reserve workloads, and create placement policies and templates. However, users cannot run plans or execute any recommended actions.
  - `PARKER`

Users with this role can park workloads manually or automatically (using parking schedules), manage parking schedules, and view all Turbonomic charts and data. However, users cannot create policies and templates, place workloads, run plans, or execute any recommended actions.

  - PARKER_AUTOMATOR

    Users with this role can park workloads manually or automatically (using parking schedules) and view all Turbonomic charts and data. However, users cannot manage parking schedules, create policies and templates, place workloads, run plans, or execute any recommended actions.

  - ADVISOR

    Users with this role can view all Turbonomic charts and data, and run plans. However, users cannot use Place to reserve workloads, create policies, or execute any recommended actions.

  - OBSERVER

    Users with this role can view the environment, including the Home Page and Dashboards. Users can also use Search to set a scope to the session. For scope, only VM groups and Resource Groups are supported.

  - OPERATIONAL_OBSERVER

    Users with this role can view the environment, including the Home Page, Dashboards, Groups, and Policies. Users can also use Search to set a scope to the session.

  - SHARED_ADVISOR

    Users with this role are scoped users. They can view the Home Page and Dashboards, but only see VMs and Applications. Users cannot execute Turbonomic actions.

  - SHARED_OBSERVER

    Users with this role are scoped users. They can view the Home Page, custom dashboards, and application maps, but can see only VMs and Applications. They cannot view Executive Dashboards or execute Turbonomic actions. Shared Observer is the most restricted user role.

  - REPORT_EDITOR

    Users with this role can create, edit, and delete reports. Due to limits to the reporting license, only one user per instance is allowed to have this role (by default, the local **administrator** user). To assign this role to another user, you must first remove it from the current user. Be sure that the new user is *not* a scoped user.

- (Required) `loginProvider`: For an AD user, specify `LDAP`.
- (Required) `type`: Specify `DedicatedCustomer`.
- (Optional) `scope`: Specify one or more scopes by using the numeric UUID value for each scope. For example, to limit the scope to a group of physical machines that the user manages, specify the UUID for that group. This parameter is not valid for users with the `ADMINISTRATOR` role.

  To get a list of the groups that are available to scope to in your Turbonomic instance, along with their display names and UUIDs, make a `GET` request to the search endpoint, as shown in the following example.

```
GET https://{your_instance_address}/api/v3/search?types=Group
```

Response:

```
{
  "uuid": "4776186363888",
  "displayName": "abc",
  "username": "abc@mycompany.com",
  "roles": [
    {
      "name": "DEPLOYER"
    }
  ],
  "loginProvider": "LDAP",
  "type": "DedicatedCustomer",
  "scope": [
```

```
    {
      "uuid": "286125332284065",
      "isStatic": true,
      "logicalOperator": "AND",
      "groupClassName": "GroupApiDTO"
    },
    {
      "uuid": "266725335284010",
      "isStatic": true,
      "logicalOperator": "AND",
      "groupClassName": "GroupApiDTO"
    }
  ],
  "showSharedUserSC": false
}
```

Maintaining AD users

To edit the details of an existing user or to delete a user, you must know the UUID value for the user.

- To get a list of all the currently configured user accounts, make a GET request to the `users` endpoint:

```
GET https://{your_instance_address}/api/v3/users
```

The API returns an array of UserApiDTO objects. Each `UserApiDTO` object contains the universal unique identifier (UUID), display name, username, encoded password, roles, scope, and other details about a user account.

Response:

```
[
  {
    "uuid": "0000000000111",
    "displayName": "extTestUser",
    "username": "extTestUser",
    "roles": [
      {
        "name": "DEPLOYER"
      }
    ],
    "loginProvider": "LDAP",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  },
  {
    "uuid": "0000000000112",
    "displayName": "externaluser1",
    "username": "externaluser1",
    "roles": [
      {
        "name": "OBSERVER"
      }
    ],
    "loginProvider": "LDAP",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  },
  {
    "uuid": "0000000000113",
    "displayName": "administrator",
    "username": "administrator",
    "roles": [
      {
        "name": "ADMINISTRATOR"
      }
    ],
    "loginProvider": "Local",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  },
  {
    "uuid": "0000000000114",
    "displayName": "newTestUser",
    "username": "newTestUser",
    "roles": [
      {
        "name": "DEPLOYER"
```

```
      }
    ],
    "loginProvider": "Local",
    "type": "DedicatedCustomer",
    "showSharedUserSC": false
  }
]
```

– To get details about a specific user, append the UUID of the user to the request:

```
GET https://{your_instance_address}/api/v3/users/0000000000111
```

– To get information about the currently logged-in user, append `/me` to the request:

```
GET https://{your_instance_address}/api/v3/users/me
```

- To edit the details for a user, make a PUT request to the `users` endpoint and include the user's UUID:

```
PUT https://{your_instance_address}/api/v3/users/0000000000111
```

The following parameters are required, regardless of whether you want to modify them:

– `username`
– `roles`
– `loginProvider`
– `type`

Include the required parameters in the request body and specify the values that you want to modify. In the following example, the role for the `extTestUser` user is changed from DEPLOYER to OBSERVER. The other parameters remain unchanged.

Example:

```
{
  "username": "extTestUser",
  "roles": [
    {
      "name": "OBSERVER"
    }
  ],
  "loginProvider": "LDAP",
  "type": "DedicatedCustomer"
}
```

Response:

```
{
  "uuid": "0000000000111",
  "displayName": "extTestUser",
  "username": "extTestUser",
  "roles": [
    {
      "name": "OBSERVER"
    }
  ],
  "loginProvider": "LDAP",
  "type": "DedicatedCustomer",
  "showSharedUserSC": false
}
```

- To delete a user, make a DELETE request to the `users` endpoint and include the user's UUID:

```
DELETE https://{your_instance_address}/api/v3/users/0000000000111
```

Adding AD groups

To add an AD group, make a POST request to the `/users/ldap/groups` endpoint.

```
POST https://{your_instance_address}/api/v3/users/ldap/groups
```

Configure the request parameters, using the following example as reference.

```
{
  "displayName: "<display_name>",
  "rolename": "<ROLE_NAME>",
  "type": "DedicatedCustomer"
}
```

- (Required) `displayName`: Specify the name of the group. The group name must match a group that is accessible from the AD domain and servers that you configured for your Turbonomic instance.
- (Required) `rolename`: Specify a role to grant AD group members access to specific Turbonomic features. Use uppercase letters only.
  - `ADMINISTRATOR` (only if the logged-in user is an administrator)

    Users with this role can use all Turbonomic features and modify settings to configure the Turbonomic installation. For Turbonomic instances hosted in the public cloud, this role is limited to the Turbonomic representative that manages the instances.
  - `SITE_ADMIN`

    Users with this role can use all Turbonomic features and modify site-specific settings to configure the Turbonomic installation. Users can also administer Groups, Policies, Workflows, Templates, Billing/Costs, and Target Configuration, but not Email, Licenses, Updates, and Maintenance. Users can create other user accounts, except accounts with the Administrator role.
  - `AUTOMATOR`

    Users with this role can use all Turbonomic features (including Plan, Park, and Place), but cannot configure the Turbonomic installation or create policies.
  - `DEPLOYER`

    Users with this role can view all Turbonomic charts and data, use Place to reserve workloads, and create placement policies and templates. However, users cannot run plans or execute any recommended actions.
  - `PARKER`

    Users with this role can park workloads manually or automatically (using parking schedules), manage parking schedules, and view all Turbonomic charts and data. However, users cannot create policies and templates, place workloads, run plans, or execute any recommended actions.
  - `PARKER_AUTOMATOR`

    Users with this role can park workloads manually or automatically (using parking schedules) and view all Turbonomic charts and data. However, users cannot manage parking schedules, create policies and templates, place workloads, run plans, or execute any recommended actions.
  - `ADVISOR`

    Users with this role can view all Turbonomic charts and data, and run plans. However, users cannot use Place to reserve workloads, create policies, or execute any recommended actions.
  - `OBSERVER`

    Users with this role can view the environment, including the Home Page and Dashboards. Users can also use Search to set a scope to the session. For scope, only VM groups and Resource Groups are supported.
  - `OPERATIONAL_OBSERVER`

    Users with this role can view the environment, including the Home Page, Dashboards, Groups, and Policies. Users can also use Search to set a scope to the session.

  - SHARED_ADVISOR

    Users with this role are scoped users. They can view the Home Page and Dashboards, but only see VMs and Applications. Users cannot execute Turbonomic actions.
  - SHARED_OBSERVER

    Users with this role are scoped users. They can view the Home Page, custom dashboards, and application maps, but can see only VMs and Applications. They cannot view Executive Dashboards or execute Turbonomic actions. Shared Observer is the most restricted user role.
  - REPORT_EDITOR

    Users with this role can create, edit, and delete reports. Due to limits to the reporting license, only one user per instance is allowed to have this role (by default, the local **administrator** user). To assign this role to another user, you must first remove it from the current user. Be sure that the new user is *not* a scoped user.
- (Required) `type`: Specify `DedicatedCustomer`.
- (Optional) `scope`: Specify one or more scopes by using the numeric UUID value for each scope. For example, to limit the scope to a group of physical machines that the AD group manages, specify the UUID for the physical machine group. This parameter is not valid for AD groups with the `ADMINISTRATOR` role.

  To get a list of the groups that are available to scope to in your Turbonomic instance, along with their display names and UUIDs, make a GET request to the search endpoint, as shown in the following example.

```
GET https://{your_instance_address}/api/v3/search?types=Group
```

Response:

```
curl -X 'POST' \
  'https://{your_instance_address}/api/v3/users' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "uuid": "myAdGroup",
  "displayName: "myAdGroup",
  "rolename": "DEPLOYER",
  "type": "DedicatedCustomer"
}'
```

Note:

If a user is a member of multiple groups, then Turbonomic logs the user on through the first SSO or AD that successfully authenticates the user. Also, Turbonomic does not support nested AD groups. AD logins must be for users in an upper-level group.

Maintaining AD groups

To edit the details of an existing AD group or to delete an AD group, you must know the UUID value for the group.

- To get a list of all the currently configured AD groups, make a GET request to the `users/ldap/groups` endpoint:

```
GET https://{your_instance_address}/api/v3/users/ldap/groups
```

  The API returns an array of ActiveDirectoryApiDTO objects. Each `ActiveDirectoryGroupApiDTO` object contains the UUID, role, and other details about a group.
- To edit the details for a group, make a PUT request to the `users/ldap/groups` endpoint.

```
PUT https://{your_instance_address}/api/v3/users/ldap/groups
```

The following parameters are required in the request body, regardless of whether you want to modify them:

– `displayName`

– `roles`

– `type`

Include the required parameters in the request body and specify the values that you want to modify. In the following example, the role for the `myAdGroup` group is changed from `DEPLOYER` to `OBSERVER`. The other parameters remain unchanged:

Example input:

```
{
  "displayName": "myAdGroupr",
  "rolename": "OBSERVER",
  "type": "DedicatedCustomer"
}
```

- **Response:**

```
{
"uuid": "mySsoGroup",
"displayName": "mySsoGroup",
"type": "DedicatedCustomer",
"roleName": "DEPLOYER" }
```

- To delete a group, make a `DELETE` request to the `users/ldap/groups` endpoint and include the group's UUID:

```
DELETE https://{your_instance_address}/api/v3/users/ldap/groups/myADgroup
```

Workflows endpoint

When you add orchestrator targets to your Turbonomic installation, they assign workflows that run multiple actions to modify your environment. Turbonomic discovers workflows that you define on the orchestrator. You can then set up an automation policy that maps workflows to actions. If the action mode is Manual or Automatic, when Turbonomic recommends the action, it directs the orchestrator to use the mapped workflow to run it.

For more information about specific orchestrator targets, see Orchestrator Targets. For more information about execution of workflows, see Automation Workflow.

You can also use the workflows endpoint to create and manage webhooks.

A webhook is an HTTP custom callback that alters the behavior of a web page or web application. You can use webhooks to integrate Turbonomic with other applications, such as Slack.

You can complete the following tasks by using the workflows endpoint:

- Get a list of all workflows.
- Get detailed information about a specified workflow.
- Get a list of policies that are associated with a specified workflow.
- Create, test, and modify a webhook.

Workflows requests

Getting workflows

For a list of workflows (such as webhooks) that are configured for your Turbonomic instance, the API returns an array of `WorkflowApiDTO` objects. Each `WorkflowApiDTO` object contains detailed information about that workflow, including description and parameters. To get a single workflow, include the UUID of the workflow in the request.

This call takes the following parameter:

Examples:

- All workflows: `GET https://10.10.10.10/api/v3/workflows`
- Single workflow: `GET https://10.10.10.10/api/v3/workflows/637998092280496`

Response:


```
[
  {
    "links":[
      {
        "rel":"self",
        "href":"https://10.10.10.10/api/v3/workflows/637998092280496"
      }
    ],
    "uuid":"637998092280496",
    "displayName":"APIC Container Delete Firewall Rule Action",
    "className":"Workflow",
    "actionType":"NONE",
    "parameters":[
      {
        "name":"APIC_TENANT_L4L7_EPG_ACE_IDENTITY",
        "isMandatory":true,
        "isSecret":false,
        "specificValueType":"ApicTenantL4L7EPGAceIdentity"
      },
      {
        "name":"L4L7Identity",
        "isMandatory":true,
        "isSecret":false,
        "specificValueType":"gen_text_input"
      },
      {
        "name":"ACLName",
        "isMandatory":true,
        "isSecret":false,
        "specificValueType":"gen_text_input"
      },
      {
        "name":"ACEName",
        "isMandatory":true,
        "isSecret":false,
        "specificValueType":"gen_text_input"
      }
    ],
    "discoveredBy":{
      "uuid":"__U1MIJv-Eei_xJ5Ai6-rnQ",
      "displayName":"ucsd.eng.mycorp.com",
      "type":"Cisco UCS Director"
    }
  ],
  ...
    }
]
```


Getting workflows of a specific type

By adding the WorkflowApiDTO `type` parameter to the request, you can filter workflows from a specific source.

For example, the response to the following request is an array of `WorkflowApiDTO` objects that are configured as webhooks:

Example: `GET https://10.10.10.10/api/v3/workflows?type=WEBHOOK`

Getting a list of policies that are associated with a workflow

You can use the `/api/v3/workflows/{workflow_Uuid}/settingspolicies` endpoint to get a list of all the policies that are associated with a workflow that you specify by UUID. The API returns the UUID

and display name for any policies that are associated with the specified workflow. For more information about policies, see “Working with policies” on page 1195.

Example: `GET https://10.10.10.10/api/v3/workflows/637998092280496/settingspolicies`

Response:

```
[
  {
    "uuid": "286242929267568",
    "displayName": "Send message to slack on action generation policy",
    "className": "BaseApiDTO"
  },
  {
    "uuid": "286242929902288",
    "displayName": "Send message to slack before execution policy",
    "className": "BaseApiDTO"
  },
  {
    "uuid": "286242929988464",
    "displayName": "Shutdown VM webhook policy",
    "className": "BaseApiDTO"
  },
  {
    "uuid": "286242930068560",
    "displayName": "Send message to slack after execution policy",
    "className": "BaseApiDTO"
  }
]
```

Creating and testing webhooks

A webhook is an HTTP custom callback that alters the behavior of a web page or web application. You can use webhooks to integrate Turbonomic with other applications, such as Slack.

To create a webhook, make a POST request to the `/workflows` endpoint:

Example: `POST https://10.10.10.10/api/v3/workflows`

The body of the request must include a WorkflowApiDTO that provides details about your webhook.

```
{
    "displayName": "Example webhook workflow DTO",
    "className": "Workflow",
    "description": "Example of workflow request payload",
    "discoveredBy": {
        "readonly": false
    },
    "type": "WEBHOOK",
    "typeSpecificDetails": {
        "url": "<YOUR_WEBHOOK_URL>",
        "method": "POST",
        "template": "$converter.toJson($action)",
        "authenticationMethod": "NONE",
        "trustSelfSignedCertificates": false,
        "type": "WebhookApiDTO"
    }
}
```

For pre-defined request payload examples, see the Turbonomic Orchestration GitHub repository.

In the response body, make note of the UUID value, as you need it to test or modify your webhook.

Response:

```
{
  "uuid": "638205616014777",
  "displayName": "Example webhook workflow DTO",
  "className": "Workflow",
  "description": "Example of workflow request payload",
  "discoveredBy": {
    "readonly": false
  },
  "type": "WEBHOOK",
```

```
  "typeSpecificDetails": {
    "url": "https://webhook.site/182d298f-6089-4b71-a37d-cc756e8ba204",
    "method": "POST",
    "template": "$converter.toJson($action)",
    "authenticationMethod": "NONE",
    "trustSelfSignedCertificates": false,
    "templateValidationStatus": {
      "validationStatus": "VALID",
      "validationErrors": [],
      "lastTemplateValidationTime": "2023-12-01T17:49:46Z"
    },
    "mediaType": "TEXT",
    "type": "WebhookApiDTO"
  },
  "creationTime": "2023-12-01T17:49:46Z",
  "lastModifiedTime": "2023-12-01T17:49:46Z"
}
```

For more information, see “Creating webhook workflows with the API” on page 1229.

To test your webhook, send a POST request to the `/workflows` endpoint and append the UUID value for your webhook:

Example: `POST https://10.10.10.10/api/v3/workflows/{workflow_Uuid}`

In the request body, include the UUID value of a Turbonomic action.

```
{
  "operation": "TEST",
  "actionId": <YOUR_ACTION_ID>
}
```

If the call is successful, you get the following response:

Response:

```
{
    "succeeded": true,
    "details": "The call to the webhook endpoint was successful."
}
```

A successful POST request for a TEST operation also executes the webhook against the configured external endpoint. For example, if the webhook is integrated with Slack, a successful POST request for a TEST operation triggers output in the integrated Slack channel.

Modifying a webhook

To modify an existing webhook, make a PUT request to the `/workflows` endpoint and append the UUID value for your webhook:

Example: `PUT https://10.10.10.10/api/v3/workflows/{workflow_Uuid}`

In the request body, include the full `WorkflowApiDTO` with any changed values:

```
{
    "displayName": "Example webhook workflow DTO",
    "className": "Workflow",
    "description": "updated description of workflow request payload",
    "discoveredBy": {
        "readonly": false
    },
    "type": "WEBHOOK",
    "typeSpecificDetails": {
        "url": "<NEW_WEBHOOK_URL>",
        "method": "POST",
        "template": "$converter.toJson($action)",
        "authenticationMethod": "NONE",
        "trustSelfSignedCertificates": false,
        "type": "WebhookApiDTO"
    }
}
```

If the call is successful, you get the following response:

Response:

```
{
  "succeeded": true,
  "details": "The call to webhook endpoint was successful."
}
```

Turbonomic cookbook

Purpose

The Turbonomic cookbook is intended to assist in performing specific tasks using the API, and answering commonly-asked questions while presenting Turbonomic API best practices.

If there is a situation or task you would like to see included, submit a request through the IBM Ideas portal located at https://ideas.ibm.com/.

Cookbooks

Each cookbook contains one or more recipes. Each recipe will contain all the necessary information to complete the specified task, along with a basic script showing the functionality of the recipe.

Authenticating with the API

Objective

To obtain an authentication token after logging in with the API, and use that token on subsequent API calls.

Authentication in the API

To use the API, you must have a valid user account on the Turbonomic instance. Also note that accounts can have different roles. The API will only execute commands that are valid for your user role. For example, to execute Turbonomic recommended actions, your account must have a role of either `administrator`, `deployer`, or `automator`.

To make API calls, you request an authentication token and pass it with each call to the Turbonomic API. The token request returns a cookie for your authentication. A common way to use this token is to store the cookie locally, and pass it with your API calls.

Example: `curl -s -k -c /tmp/cookies -H 'accept: application/json' 'https://localhost/api/v3/login?hateoas=true' -d 'username=`*`{your_username}`*`&password=`*`{your_password}`*`'`

Then, each request must use the -b `cookie-filename` parameter to use the session cookie delivered by the login request.

Another approach is to get the authentication header and parse out the authentication cookie. Then you can create a header for each API request that includes the cookie. For example, assume you store the value in a variable named `token`. You could use it like this:

```
headers = {'cookie': token}
r = requests.get('https://10.10.123.456/api/v3/targets/specs', headers=headers, verify=False,
stream=True)
```

Prerequisites

In order to obtain an authentication token, you must have the following information:

- The IP address of the Turbonomic instance.

- The username and password of the user who will be logged in.

Procedure

To obtain an authentication token:

1. Construct the API payload.

   The input body must be URL-encoded (`x-www-form-urlencoded`). Use the following format with your credentials in place of `myUsername` and `myPassword`:

   ```
   username=myUsername&password=myPassword
   ```

2. Make the API request.

   Use the `POST https://[INSTANCE_URL]/api/v3/login` request, with the input body constructed in the previous step.

3. Retrieve the authentication token from the response headers.

   Example headers:

   ```
   {
     'Server':'nginx',
     'Date':'Mon, 01 Feb 2021 20:17:02 GMT',
     'Content-Type':'application/json',
     'Transfer-Encoding':'chunked',
     'Connection':'keep-alive',
     'Vary':'Accept-Encoding',
     'Set-Cookie':'JSESSIONID=node0jzerbqte09pe1j29x6ypo92fh17.node0; Path=/; HTTPOnly; Secure',
     'Expires':'Thu, 01 Jan 1970 00:00:00 GMT',
     'X-Content-Type-Options':'nosniff',
     'X-XSS-Protection':'1; mode=block',
     'Strict-Transport-Security':'max-age=63072000; includeSubDomains',
     'X-Frame-Options':'SAMEORIGIN',
     'X-Turbo-Upstream':'API',
     'Content-Encoding':'gzip'
   }
   ```

   The authentication token is the first `;`-separated tocen in the value of the `Set-Cookie` header. In this example, it is `JSESSIONID=node0jzerbqte09pe1j29x6ypo92fh17.node0`.

Using the authentication token in subsequent API requests

Once you have obtained an authentication token, you can use that token to make further API requests. To do so, include a header in your request with the name of `cookie` and a value of the token. For example:

```
{'cookie': 'JSESSIONID=node0jzerbqte09pe1j29x6ypo92fh17.node0' }
```

Script example

These listings show a function that gets the authentication token from the cookie, and a script that invokes the function and prints out the token value.

Function to get the token:

To invoke this function, pass the IP address, the username, and password as arguments. The function builds the authentication payload and requests the token from the given Turbonomic instance. It then parses out the first token in the `Set-Cookie` header and returns that value.

```
import json
import requests
import urllib3
urllib3.disable_warnings()
```

```
def get_cookie(ip, username, password):
    payload = {'username': username, 'password': password}
    r = requests.post(f'https://{ip}/api/v3/login', data=payload, verify=False)
    r.encoding = 'JSON'
    rh = r.headers
    token = rh['Set-Cookie'].split(';')[0]

    return token
```

Script that calls the function:

Assume the script is saved as `use_login.py`. Also assume the function is in the file `./login.py`. To invoke this script, open a shell at the script file location and enter:

`./use-login.py <My_IP_Address> <My_Username> <My_Pwd>`

where the three arguments are the IP address of your Turbonomic, your username, and your password, respectively. The script imports the `./login.py` file and checks for four arguments (the script name and the three arguments you passed). It then executes the `login.get_cookie()` function. The function returns a token, which the script then prints out.

```
#!/usr/bin/env python3

import sys
import login

if len(sys.argv) != 4:
    print("Incorrect arguments list.")
    sys.exit()

token = login.get_cookie(sys.argv[1], sys.argv[2], sys.argv[3])
print(f'TOKEN IS: {token}')
```

Script result

The result of this script displays the authentication token:

```
TOKEN IS: JSESSIONID=node0jzerbqte09pe1j29x6ypo92fh17.node0
```

Authenticating OAuth 2.0 clients with the API

Objective

Create, manage, and authenticate with OAuth 2.0 clients by using authorization requests in the Turbonomic API.

For more information, see Authorization requests.

Plan results cookbook

This cookbook explores plan markets in Turbonomic. Interpreting and parsing this plan market will enable you to better understand how and why Turbonomic has performed a certain action on a particular entity, how the resources- both used and available- have changed, and more as recipes are added to the cookbook.

Note: The Turbonomic UI has certain plan configuration options, available to present a base set of configuration changes to accomplish a particular goal. For example, the Add Workload plan is preconfigured to expect additional workloads to be provisioned. To see how Turbonomic creates these configurations, view the DTOs for the UI plan configurations at Scenario Examples.

Related endpoints

When interpreting plan results, the following endpoints are commonly used to retrieve and clarify data:

- Actions: Retrieving the actions for the plan market will enable you to see what entities were directly affected during the run of the plan.
- Entities: Retrieving the entities for the plan market will enable you to see a full listing of the entities involved in the plan.
- Groups: Retrieving the groups for the plan market will enable you to more easily obtain UUID information for the entities in the specified group.
- Statistics: Retrieving the market statistics for the plan market as a whole, a group, or an entity will enable you to easily see a broad overview of the state of the entity utilization after the plan has finished.

Use case index

The Plan Results Cookbook has the following recipes:

- Seeing Changes to a Subset of Entities

Seeing changes to a subset of entities

Objective

After running a plan, you may want to see the effects of your plan on a particular entity or group of entities. This can help you better understand the effect your configuration changes would have on a subset of plan entities, or to explore why Turbonomic made specific action recommendations.

For example, assume that there is a virtual machine that shares dependencies with a number of other entities. Apart from the overal plan results, you might want to take a closer look at how the plan impacts this VM in particular.

Prerequisites

In order to investigate the changes to the specified entity, you need to know:

- The UUID of a successfully completed plan market

  To obtain this UUID, either get it from the API response sent when you POST a scenario to the market, or execute the `GET /markets` request, and find the market whose name corresponds with the name of the scenario. Alternately, you can find the scenario's UUID in the `scenario` parameter of the plan market. Verify that the `state` parameter has a value of `SUCCEEDED`. For example, the UUID of the scenario that created the following plan market is `214410933912976`, and the UUID of the plan market itself is `214410933923968`:

```
{
    "uuid":"214410933923968",
    "displayName":"Optimize Cloud 2",
    "className":"Market",
    "state":"SUCCEEDED",
    "stateProgress":100,
    "scenario":{
        "uuid":"214410933912976",
        "displayName":"Optimize Cloud 2",
        ...
    }
...
}
```

  For cases where the same scenario has been applied to multiple plans, be sure to check the plan's `runDate` parameter corresponds to the time you expect the plan to have run.

  For more information about creating and running scenarios, see Scenario DTO Breakdown.
- The UUID(s) of the entities you want to inspect

Entity UUIDs in a plan market are identical to their real-time market counterparts. This means you can get the UUIDs either from searching the plan market for a related field (for example, the `displayName` of the entity) or searching in the Turbonomic user interface to find the entity UUID.

You can also use the API's Search endpoint to search for entities by different criteria.

Related endpoints

When interpreting plan results, the following endpoints are commonly used to retrieve and clarify data:

- Actions: Retrieving the actions for the plan market will enable you to see what entities were directly affected during the run of the plan.
- Entities: Retrieving the entities for the plan market will enable you to see a full listing of the entities involved in the plan.
- Groups: Retrieving the groups for the plan market will enable you to more easily obtain UUID information for the entities in the specified group.
- Statistics: Retrieving the market statistics for the plan market as a whole, a group, or an entity will enable you to easily see a broad overview of the state of the entity utilization after the plan has finished.

Procedure

To investigate changes to a particular subset of plan entities:

1. Get the list of UUIDs you want investigate.

   To get all the entities in a market:

   ```
   markets/{market_uuid}/entities
   ```

   If the scope of the plan is small, you can use this to get all the entities in the plan, and then iterate through the results for other attributes such as entity type or display name.

   ```
   markets/{market_uuid}/entities
   ```

   For a large scope, you can use the Search endpoint to find VMs by specific criteria. This works because the plan market and the real-time market use the same UUID values for the same entities. For example, the following call gets all VMs that have the string F00 in their names:

   ```
   https://10.10.123.456/api/v3/search?q=FOO&types=VirtualMachine&query_type=CONTAINS
   ```

2. Get the list of plan actions.

   Run GET `/markets/{market_Uuid}/actions` against the plan market. You can also use POST `/markets/{market_uuid}/actions` to post a filter for advanced action queries, such as only returning actions of a particular type, or particular risk severity.

   Another way to get plan actions is to use the Search endpoint. You can POST search parameters that limit the returned actions by text in the description, cost type, or other data. For example, the following search returns only actions on VMs:

   ```
   searchSpec = {
       "actionRelationTypeFilter": [
           "NONE"
       ],
       "costType": "SAVING",
       "descriptionQuery": {
           "query": "Virtual Machine",
           "type": "CONTAINS",
           "caseSensitive": "true"
       },
       "detailLevel": "STANDARD"
   }
   ```

With the previous payload, the following POST would return VM actions from the plan:

```
requests.post(f'https://{ip}/api/v3/markets/{marketUuid}/actions',
        headers=headers,
        json = searchSpec,
        verify=False,
        stream=True)
```

3. Search the API response and find the actions relevant to your entities.

   Each action will have a `target` parameter, which will in turn have a `uuid`. This UUID identifies the entity the action executed on. This is the best way to locate your entity in a list of actions.

   If no actions have the specified `target > uuid` value, then no actions were performed on the entity in question.

   Note: You can use `GET /markets/{market_Uuid}/entities` to verify that the UUID you are searching for exists in the plan market. If it does not, it means that the entity was not included in the scope of the plan, or the entity is considered an idle workload. An idle workload shows no utilization, so the plan cannot determine optimal placement or what percentage of allocated resources that workload will require when it restarts.

4. Analyze the results.

   Each action contains the information detailed in the ActionApiDTO data, including the details of the action, old/new values for any changed resources, a record of deletion for DELETE actions, etc.

   Search through this DTO and make note of any parameters important to you. (For a DTO reference, see ActionApiDTO.)

Script example

The following script searches through the list of actions for a virtual machine, and returns the requested fields for quick comparison. In this case, the UUID, type, state, reason, risk category, and risk subcategory are returned for the action, and paired with the display name, template UUID, and location of the entity.

This script is easily extensible- for example, if `currentLocation > displayName` is equal to `newLocation > displayName`, you may choose not to show either of those fields, as no change involving the VM location was made. You might also use these results to make further calls- for example, using the template UUID to make a `GET /templates/{template_Uuid}` request to return more information about the template being used for a resize action.

Note that the script uses dot-notation to name the fields you want to inspect. It includes the `getFieldFromDict` function to convert those fields to nested Python dictionary values. Because the API returns JSON data, a function like this could be a generally useful utility.

```
#!/usr/bin/env python3

#
# Commandline args: platformIP, username, pwd, marketID, VmIDs as comma-sep string (no spaces)
# EXAMPLE: ./entity_changes.py 10.10.123.456 MyUser MyPwd 215426452602368
74635124762083,74635124762088,74635124762090
#
import json
import requests
import urllib3
import re

import sys
sys.path.append('../login') #This makes login visible...
import login

#
# This function takes json-like dot-notation fields and returns python dictionary values.
# That makes it easier to specify the DTO fields you want to view.
#
def getFieldFromDict(field, dict):
    fToks = field.split('.')
    l = len(fToks)
    workingDict = dict
```

```
    for item in fToks:
        if item in workingDict:
            ret = workingDict[item]
        else:
            return "ITEM NOT IN OBJECT"
        l -= 1
        if l == 0:
            return ret
        else:
            workingDict = ret
    return ret

#
# Ths list of fields you want to query, in dot-notation
#
RetrieveFields = [
"uuid",
"actionType",
"actionState",
"details",
"currentEntity.displayName",
"newEntity.displayName",
"template.uuid",
"risk.subCategory",
"risk.description",
"currentLocation.displayName",
"newLocation.displayName"
]

#
# A search spec to get actions that are on VMs.
#
searchSpec = {
  "actionRelationTypeFilter": [
    "NONE"
  ],
  "costType": "SAVING",
  "descriptionQuery": {
    "query": "Virtual Machine",
    "type": "CONTAINS",
    "caseSensitive": "true"
  },
  "detailLevel": "STANDARD"
}

#
# Make sure you have the correct count of arguments.
#
if len(sys.argv) != 6:
    print("Incorrect arguments list.")
    sys.exit()

#
# Get the args into well named vars, and also authenticate.
#
ip = sys.argv[1]
marketUuid = sys.argv[4]
ElemsToInspect = sys.argv[5].split(',')

token = login.get_cookie(ip, sys.argv[2], sys.argv[3])
headers = {'cookie': token}

#
# Search plan for actions on VMs
#
actions_r = requests.post(f'https://{ip}/api/v3/markets/{marketUuid}/actions', headers=headers,
json = searchSpec, verify=False, stream=True)

if actions_r.status_code != 200:
    print(f'Market Request Status is {actions_r.status_code}.')
    sys.exit()

#
# For each action that targets one of the entities you pass in the command line,
# parse the object and display the field values.
#
print("Plan Changes for Entity(ies):")
print(sys.argv[5])
print("-----------------------")
r_json = actions_r.json()
for item in r_json:
    target = item['target']
```

```
    if 0 < ElemsToInspect.count(target["uuid"]):
        print(f'VM ACTION: {target["displayName"]}')
        for field in RetrieveFields:
            val = getFieldFromDict(field, item)
            print(f'    {field}: {val}')
```

Script result

The result of this script displays the requested information for each plan entity:

```
Plan Changes for Entity(ies):
74635124762083,74635124762088
----------------------
VM ACTION: test-rosa-28tfg-worker-us-west-2a-5hr8j
    uuid: 145057709553360
    actionType: SCALE
    actionState: SUCCEEDED
    details: Scale Virtual Machine test-rosa-28tfg-worker-us-west-2a-5hr8j from m5.xlarge to
r5a.large in Advanced Engineering
    currentEntity.displayName: m5.xlarge
    newEntity.displayName: r5a.large
    template.uuid: 74635137532084
    risk.subCategory: Efficiency Improvement
    risk.description: Underutilized VCPU, Net Throughput
    currentLocation.displayName: aws-US West (Oregon)
    newLocation.displayName: aws-US West (Oregon)
VM ACTION: test-rosa-28tfg-infra-us-west-2a-mw867
    uuid: 145057709553376
    actionType: SCALE
    actionState: SUCCEEDED
    details: Scale Virtual Machine test-rosa-28tfg-infra-us-west-2a-mw867 from r5.xlarge to
r5a.xlarge in Advanced Engineering
    currentEntity.displayName: r5.xlarge
    newEntity.displayName: r5a.xlarge
    template.uuid: 74635137531741
    risk.subCategory: Efficiency Improvement
    risk.description: Increase RI Utilization
    currentLocation.displayName: aws-US West (Oregon)
    newLocation.displayName: aws-US West (Oregon)
```

Seeing the changes for a group of entities

You can easily modify this script to see the changes for a particular group of entities without needing to know each UUID.

To perform this modification, consume the result of a `GET /groups/{group_uuid}/entities` request. Example result:

```
[
  {
    "uuid":"73422419372894",
    "displayName":"PT_RHEL",
    "className":"VirtualMachine",
    "priceIndex":4.33,
    "state":"ACTIVE",
    "severity":"Normal",
    "costPrice":0.24104111,
    "discoveredBy":{
      "uuid":"73422416392368",
      "displayName":"producttrust",
      "category":"Public Cloud",
      "type":"AWS",
      "readonly":false
    },
    ...
  },
  {
    "uuid":"73422419372988",
    "displayName":"PT_Resize_Up_Target",
```

```
    "className":"VirtualMachine",
    "priceIndex":10074.822,
    "state":"ACTIVE",
    "severity":"Normal",
    "costPrice":0.078387,
    "discoveredBy":{
      "uuid":"73422416392368",
      "displayName":"producttrust",
      "category":"Public Cloud",
      "type":"AWS",
      "readonly":false
    },
    ...
  },
  {
    "uuid":"73422419372921",
    "displayName":"PT_Consistent_Scaling_VM_1",
    "className":"VirtualMachine",
    "priceIndex":100.357,
    "state":"ACTIVE",
    "severity":"Normal",
    "costPrice":0.0159137,
    "discoveredBy":{
      "uuid":"73422416392368",
      "displayName":"producttrust",
      "category":"Public Cloud",
      "type":"AWS",
      "readonly":false
    },
    ...
  }
]
```

Using this result, you can either use each UUID found, or filter based on entities. For example, to iterate through a group and get the UUID of all VM entities:

```
foreach (var entity in groupApiResponse)
{
  if (entity["className"] == VirtualMachine)
  {
    Console.Writeline(entity["uuid"])
  }
}
```

Creating a static group

Objective

To create a static group of entities.

A static group contains a fixed set of entities. This example creates a group of VMs, but you can create a group of any entity type.

The script will get the unique IDs of member entities by their display names, and add them to a list that it then uses to display the group members.

The script also sets up the group type and entity type when it creates the payload data. It then uses the API to create the group, and verify that the group contains the entities you specified.

Prerequisites

In order to create a static group, you must have the following information:

- The display names of the members in the group
- The entity type you will use for this group
- The IP address and login credentials for a Turbonomic instance

Related endpoints

This recipe uses the following endpoints:

- Groups

  It uses the POST /groups request to create the group, and the GET /groups/{group_uuid} request to verify the group membership.
- Search

  It uses the GET /search request to obtain the UUIDs of the group members from the display names.

Script setup

To set up the script, provide values from your environment for the following:

- ip

  The IP address of your Turbonomic instance.
- username

  The username for the account Turbonomic will use to access the API.
- password

  The password for the account Turbonomic will use to access the API.
- names_list

  A list of display names for the entities you want to add to the group. Note that the entities must all be of the same type. If entities of more than one type are provided, group creation will fail.
- group_name

  The display name for the group. This name will appear in the user interface.
- entity_type

  The entity type for the group and its member entities. The script will use this in creating the Group input data, and also when executing a search for the entities' UUID.

  Note:

  Turbonomic does not support creation of groups with more than one entity type at this time.

Authentication

The first step of any script is to authenticate accessc to your Turbonomic instance, retrieve the authentication token and store it for use with subsequent calls. To authenticate, this script uses the values you set for ip, username, and password. For more detail, see the Authentication Recipe.

API Request: https://{ip}/vmturbo/rest/login

UUID retrieval

After it logs in, the script iterates through your list of entity names, passing each in turn to the GET /search request. It uses the returned EntityApiDTO to get the UUIDs for each matching entity.

The search response is an array of entity objects, and each entity object includes the entity's UUID. The script loops through the display names you provide in names_list, and executes a separate search request for each name. Each response should be an array of one item. The script adds the uuid of that item to the group_members_uuids list.

Note:

Another method to obtain the UUIDs is to execute a filtered POST /search request. This script uses repeated GET /search requests for simplicity.

Sample Request: `GET https://10.10.10.10/api/v3/search?q=zrlinuxclone106&types=VirtualMachine&entityTypes=VirtualMachine`

Sample Return:


```
[
  {
    "uuid":"74011863734144",
    "displayName":"zrlinuxclone106",
    "className":"VirtualMachine",
    "state":"ACTIVE",
    "severity":"Critical",
    "discoveredBy":{
      "uuid":"73935378522272",
      "displayName":"vsphere-dc23.eng.vmturbo.com",
      "category":"Hypervisor",
      "type":"vCenter",
      "readonly":false
    },
    "severityBreakdown":{
      "CRITICAL":1
    },
    "providers":[
      {
        "uuid":"73935378875792",
        "displayName":"hp-esx108.eng.vmturbo.com",
        "className":"PhysicalMachine"
      },
      {
        "uuid":"73935378875736",
        "displayName":"QS2_GRID01:DC23DS1",
        "className":"Storage"
      },
      {
        "uuid":"73995333587472",
        "displayName":"Resources-hp-esx108.eng.vmturbo.com",
        "className":"VirtualDataCenter"
      }
    ],
    "environmentType":"ONPREM",
    "vendorIds":{
      "vsphere-dc23.eng.vmturbo.com":"vm-2844"
    }
  }
]
```


Constructing the input DTO and creating the group

Once the script has built the list of UUIDs for the group members, it can build a `GroupApiDTO` to describe the group you want. The DTO needs values for the following properties:

- `isStatic`: true to create a static group
- `displayName`: The group name that you set for the group_name variable
- `memberUuidList`: The array of UUID strings that the script has created
- `criteriaList`: For a static group, pass an empty array
- `groupType`: The entity type for this group -- the value you set for the entity_type variable

Using the obtained UUIDs and the previous information, the script will construct the input GroupApiDTO. The script creates a Python dictionary, which it passes as the input body of the POST `/groups` request.

Example Input DTO:


```
{
  "isStatic":"true",
  "displayName":"TestGrp2",
  "memberUuidList":[
    "74011863734144",
    "73997881750336"
  ],
```

```
  "criteriaList":[

  ],
  "groupType":"VirtualMachine"
}
```

Sample Request: POST https://{ip}/api/v3/groups/

The response to this request will be the GroupApiDTO for the created group.

Sample Response:

```
{
  "uuid":"285123335341792",
  "displayName":"TestGrp2",
  "className":"Group",
  "membersCount":2,
  "entitiesCount":2,
  "groupType":"VirtualMachine",
  "severity":"CRITICAL",
  "environmentType":"ONPREM",
  "isStatic":true,
  "logicalOperator":"AND",
  "memberUuidList":[
    "74011863734144",
    "73997881750336"
  ],
  "temporary":false,
  "activeEntitiesCount":1,
  "cloudType":"UNKNOWN",
  "memberTypes":[
    "VirtualMachine"
  ],
  "entityTypes":[
    "VirtualMachine"
  ],
  "groupClassName":"GroupApiDTO"
}
```

Verification

To verify that the group was created successfully, the script uses the returned UUID to get the group data via the /groups endpoint. If the request gets the group, the returned DTO should be identical to the response from the request to create the group.

If the group does not exist for the given UUID, the API returns the request status 404, with the message, Group not found: <Passed_UUID>. This indicates that the script failed to create the group.

Sample Response: https://{ip}/api/v3/groups/{group_uuid}

You can choose to only use a subset of the provided response. For example, this script prints the displayName and UUID for each group member.

Sample Script Response:

```
Group Members:

Display name: boao.min-spare
OID: 73997881750336
Display name: zrlinuxclone106
OID: 74011863734144
```

Script example

```
import json
import requests
```

```
import urllib3
import re

import sys
sys.path.append('../login') #This makes login visible...
import login


#
# Make sure you have the correct count of arguments.
#
if len(sys.argv) != 7:
    print("Incorrect arguments list.")
    sys.exit()

#
# Get the args into well named vars, and also authenticate.
#
ip = sys.argv[1]
groupName = sys.argv[4]
entityType = sys.argv[5]
entsListByName = sys.argv[6].split(',')

token = login.get_cookie(ip, sys.argv[2], sys.argv[3])
authHeader = {'cookie': token}

#
# Get UUIDs for entities in entsListByName
#
memberIds = []
for name in entsListByName:
    response = requests.get(
        f'https://{ip}/api/v3/search?q={name}&types={entityType}&query_type=EXACT',
        headers=authHeader, verify=False)
    if response.status_code == 200:
        response_json = json.loads(response.text)
        memberIds.append(response_json[0]['uuid'])
    if response.status_code != 200:
        print(f'Search Request Status for {name} is {response.status_code}.')

#
# Assemble the payload object to create the group.
#
group_input_dto = {}
group_input_dto['isStatic'] = 'true'
group_input_dto['displayName'] = groupName
group_input_dto['memberUuidList'] = memberIds
group_input_dto['criteriaList'] = []
group_input_dto['groupType'] = entityType

#
# Create the group
#
response = requests.post(f'https://{ip}/api/v3/groups', json=group_input_dto,
headers=authHeader, verify=False, stream=True)

if response.status_code != 200:
    print(f'Create Group failed. StatusCode: {response.status_code}\n{response.text}')
    sys.exit()

groupResponse_json = json.loads(response.text)
groupId = groupResponse_json['uuid']

#
# Inspect the group
#
groupCheck_response = requests.get(f'https://{ip}/api/v3/groups/{groupId}', headers=authHeader,
verify=False)
groupCheck_json = json.loads(groupCheck_response.text)

if groupCheck_json['uuid']:
    verifiedId = groupCheck_json['uuid']
    groupName = groupCheck_json['displayName']
    print(f'Group \"{groupName}\" successfully created with a UUID of: {verifiedId}')
else:
    print('Error in group creation, no UUID found.')
    sys.exit()

# Get a list of Group members
response = requests.get(
    f'https://{ip}/api/v3/groups/{groupId}/entities', headers=authHeader, verify=False)
if response.status_code != 200:
```

```
    print(f'Could not get group members. Status is {response.status_code}.')
    sys.exit()

groupMembers_json = json.loads(response.text)

# Print a list of group members
print('\nGroup Members:')
for item in groupMembers_json:
    print('Display name: ' + item['displayName'])
    print('OID: ' + item['uuid'])
```

Creating a dynamic group

Objective

To create a dynamic group of entities.

A dynamic group contains a set of entities discovered by filter criteria. In this case, the script will create a group of all virtual machines which reside on a specific host. As virtual machines are added and removed from the host, the membership of this group will reflect those changes.

The script will use the `/search` endpoint to get the unique ID of the host using its display name, and then retrieve a list of expected virtual machines to verify against. The script then creates the criteria list and inserts that value into the rest of the payload data before creating the group and verifying that the membership is as expected.

Prerequisites

In order to create a dynamic group, you must have the following information:

- The display name of the host

  The entity type you will use for this group

  The IP address and login credentials for a Turbonomic instance

Related endpoints

This recipe uses information from the following endpoints:

- Groups

  Note:

  All possible criterion values can be found in the Groups Endpoint Tips.
- Search

Script setup

To set up the script, provide values from your environment for the following:

- `ip`

  The IP of your Turbonomic instance.
- `username`

  The username for the account Turbonomic will use to access the API.
- `password`

  The password for the account Turbonomic will use to access the API.
- `host_name`

  The display name of the host whose virtual machines will be added to the group.

- `group_name`

  The display name for the group. This name will appear in the user interface.

Authentication

The first step of any script is to authenticate to your Turbonomic instance, retrieve the authentication token and store it for use with subsequent calls. To authenticate, this script uses the values you set for `ip`, `username`, and `password`. For more detail, see the Authentication Recipe.

API Request: `https://{ip}/vmturbo/rest/login`

Get expected initial membership

After it logs in, the displays a list of virtual machines that reside on the host. These will be the group members.

The script uses the `/search` endpoint and retrieves the EntityApiDTO for the host. Using the `consumers` parameter, you can see each entity that purchases resources from this host. The script lists the consumers with a `className` that matches the entity type.

Sample Request: `GET https://{ip}/api/v3/search?q={host_name}&types=PhysicalMachine'&detail_type=entity'`

Create the filter criterion

To create a dynamic group, you pass filter criteria to identify matching entities. You pass these criteria as an array. For this script, you pass a single criterion in the array.

Example `criteria_list`:

```
[
  {
    "expType":"RXEQ",
    "expVal":"host_name",
    "filterType":"vmsByPMName",
    "caseSensitive":"false"
  }
]
```

Construct the input DTO and create the group

The script will take the `criterion` parameter and use it to construct the GroupApiDTO, with the following parameters:

- `isStatic`

  When true, the created group is static. The script sets this parameter to `false`.
- `displayName`

  The display name for the group. This name will appear in the user interface. The script sets this parameter
- `memberUuidList`

  For a dynamic group, pass an empty array.
- `criteriaList`

  The filtering criteria used to determine group membership. The script sets this parameter equal to the `criteria_list` variable.
- `groupType`

The entity type of the group members. The script sets this parameter equal to the `group_entity_type` variable.

API Request: `POST https://{ip}/api/v3/groups/`

The response to this request will be the `GroupApiDTO` for the created group.

Sample Response:


```
{
  "uuid":"285126581114848",
  "displayName":"TestGrp2",
  "className":"Group",
  "membersCount":3,
  "entitiesCount":3,
  "groupType":"VirtualMachine",
  "severity":"CRITICAL",
  "environmentType":"ONPREM",
  "isStatic":false,
  "logicalOperator":"AND",
  "criteriaList":[
    {
      "expVal":"hp-esx119.eng.vmturbo.com",
      "expType":"RXEQ",
      "filterType":"vmsByPMName",
      "caseSensitive":false,
      "entityType":null,
      "singleLine":false
    }
  ],
  "memberUuidList":[
    "73943582577376",
    "73943582577568",
    "73943582577744"
  ],
  "temporary":false,
  "activeEntitiesCount":2,
  "cloudType":"UNKNOWN",
  "memberTypes":[
    "VirtualMachine"
  ],
  "entityTypes":[
    "VirtualMachine"
  ],
  "groupClassName":"GroupApiDTO"
}
```


Verify the group

To verify that the group was created successfully, the script uses the returned UUID to get the group data via the `/groups` endpoint. If the request gets the group, the membership of the group should match what was printed to the console in the *Get Expected Initial Membership* portion of the script, assuming that no virtual machines have been added or removed since that time.

If the group does not exist for the given UUID, the API returns the request status 404, with the message, `Group not found: <Passed_UUID>`. This indicates that the script failed to create the group.

API Request: `https://{ip}/api/v3/groups/{group_uuid}`

You might choose to only use a subset of the provided response. For example, this script prints the `displayName` and `UUID` for each group member.

Sample Script Response:


```
Group Members:

Display name: DC26-P02-03
OID: 73943582577376
Display name: DC26-P02-02
OID: 73943582577568
Display name: DC26-P02-01
```

```
OID: 73943582577744
```

Script example

```
#
# Commandline args: platformIP, username, pwd, group_name, entity_type, host_name
# EXAMPLE: ./entity_changes.py 10.10.123.456 MyUser MyPwd MyTestGroup VirtualMachine MyHost
#
import json
import requests
import urllib3
import re

import sys
sys.path.append('../login') #This makes login visible...
import login


#
# Make sure you have the correct count of arguments.
#
if len(sys.argv) != 7:
    print("Incorrect arguments list.")
    sys.exit()

#
# Get the args into well named vars, and also authenticate.
#
ip = sys.argv[1]
group_name = sys.argv[4]
entity_type = sys.argv[5]
host_name = sys.argv[6]

token = login.get_cookie(ip, sys.argv[2], sys.argv[3])
authHeader = {'cookie': token}

#
# Get Expected Initial Membership
#
response = requests.get(
    f'https://{ip}/api/v3/search?q={host_name}&types=PhysicalMachine'
    f'&detail_type=entity',
    headers=authHeader, verify=False)
get_vms_response_json = json.loads(response.text)

print(f'CONSUMERS: {get_vms_response_json[0]["consumers"]}')
print(f'The {entity_type} members of this dynamic group will be:')
for entity in get_vms_response_json[0]['consumers']:
    if entity['className'] == entity_type:
        print(entity)

#
# Create the filter criterion
#
criterion = {}
criterion['expType'] = "RXEQ"
criterion['expVal'] = host_name
criterion['filterType'] = "vmsByPMName"
criterion['caseSensitive'] = "false"

#
# Construct the Input DTO and Create the Group
#
group_input_dto = {}
group_input_dto['isStatic'] = 'false'
group_input_dto['displayName'] = group_name
group_input_dto['memberUuidList'] = []
group_input_dto['criteriaList'] = [criterion]
group_input_dto['groupType'] = entity_type

response = requests.post(
    f'https://{ip}/api/v3/groups', headers=authHeader, json=group_input_dto, verify=False,
stream=True)
group_creation_response_json = json.loads(response.text)
group_uuid = group_creation_response_json['uuid']

#
```

```
# Verify the group
#
response = requests.get(
    f'https://{ip}/api/v3/groups/{group_uuid}/entities', headers=authHeader, verify=False)
group_members_response_json = json.loads(response.text)

# Print a list of group members
print('\nGROUP MEMBERS:')
for item in group_members_response_json:
    print('Display name: ' + item['displayName'])
    print('OID: ' + item['uuid'])
```

Reference: Internal Turbonomic API requests

Internal Requests are requests that are used internally by Turbonomic, but are not intended for customer use. The structure data returned by these requests is subject to change without notice.

The following requests have been marked as Internal Only:

- /entities

  `DELETE /entities/loggingEntities`

  `PUT /entities/{entity_Uuids}/loggingEntities`

  `GET /entities/{entity_Uuid}/aspects/{aspect_name}/actions`

  `POST /entities/{entity_Uuid}/aspects/{aspect_name}/actions`

  `GET /entities/{entity_Uuid}/aspects/{aspect_name}/stats`

  `POST /entities/{entity_Uuid}/aspects/{aspect_name}/stats`

  `POST /entities/{entity_Uuid}/tags`

  `DELETE /entities/{entity_Uuid}/tags`

  `DELETE /entities/{entity_Uuid}/tags/{tag_key}`
- /groups

  `GET /groups/{group_Uuid}/aspects/{aspect_name}/actions`

  `POST /groups/{group_Uuid}/aspects/{aspect_name}/actions`

  `GET /groups/{group_Uuid}/aspects/{aspect_name}/stats`

  `POST /groups/{group_Uuid}/aspects/{aspect_name}/stats`

  `POST /groups/{group_Uuid}/notifications/stats`

  `POST /groups/{group_Uuid}/tags`

  `DELETE /groups/{group_Uuid}/tags`

  `DELETE /groups/{group_Uuid}/tags/{tag_key}`
- /markets

  `PUT /markets/{market_Uuid}/policies/{policy_Uuid}`
- /reports

  `POST /reports/templates`
- /scenarios

  `PUT /scenarios/{scenario_Uuid}`

  `PUT /scenarios/{scenario_Uuid}/entities/{entity_Uuid}`

  `DELETE /scenarios/{scenario_Uuid}/entities/{entity_Uuid}`

  `PUT /scenarios/{scenario_Uuid}/groups/{group_Uuid}`

```
PUT /scenarios/{scenario_Uuid}/policies
DELETE /scenarios/{scenario_Uuid}/policies/{policy_Uuid}
PUT /scenarios/{scenario_Uuid}/templates/{template_name}
```

- /widgetsets

  The entire /widgetsets endpoint is Internal Only.

Reference: Scenario peak-of-peak full scenario input DTO

For reference purposes, see the following example for a complete input DTO for a Peak of Peaks plan.


```
    {
"uuid":"XXXXDemo",
"displayName":"XXXXDemo",
"owners":[
  {
    "uuid":"_4T_7kwY-Ed-WUKbEYSVIDw",
    "username":"administrator",
    "showSharedUserSC":false
  }
],
"type":"PEAKOFPEAKS",
"scope":[
  {
    "uuid":"_PE0v-YEUEee_hYfzgV9uYg",
    "displayName":"All On-prem Hosts",
    "className":"All On-prem Hosts"
  }
],
"topologyChanges":{

},
"loadChanges":{
  "maxUtilizationList":[
    {
      "maxPercentage":70,
      "projectionDay":0,
      "target":{
        "displayName":"All On-prem Hosts",
        "className":"Group",
        "uuid":"_PE0v-YEUEee_hYfzgV9uYg"
      }
    }
  ]
},
"overlayStatsList":[
  {
    "uuid":"4223ae0c-c99b-f6ca-c1ef-3f367bdf5528",
    "displayName":"centos_imp_2",
    "className":"VirtualMachine",
    "environmentType":"ONPREM",
    "stats":[
      {
        "displayName":"centos_imp_2",
        "date":"2019-02-01T00:00:00-05:00",
        "statistics":[
          {
            "displayName":"centos_imp_2",
            "date":"2019-03-14T15:12:36-04:00",
            "statistics":[
              {
                "name":"priceIndex",
                "values":{
                  "max":20000,
                  "min":20000,
                  "avg":20000,
                  "total":20000
                },
                "value":20000
              },
              {
                "name":"CPUProvisioned",
                "capacity":{
                  "max":104000,
                  "min":104000,
                  "avg":104000,
```

```
      "total":104000
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"Virtual_ESX_4238255f-86cc-4cb9-0f47-11ff689f0b22",
      "displayName":"hp-esx4.dev.mycorp.com",
      "className":"PhysicalMachine"
    },
    "units":"MHz",
    "values":{
      "max":5200,
      "min":5200,
      "avg":5200,
      "total":5200
    },
    "value":5200
  },
  {
    "name":"Mem",
    "capacity":{
      "max":16776692,
      "min":16776692,
      "avg":16776692,
      "total":16776692
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"Virtual_ESX_4238255f-86cc-4cb9-0f47-11ff689f0b22",
      "displayName":"hp-esx4.dev.mycorp.com",
      "className":"PhysicalMachine"
    },
    "units":"KB",
    "values":{
      "max":3555836,
      "min":3554799.8,
      "avg":3554799.8,
      "total":3554799.8
    },
    "value":3554799.8
  },
  {
    "name":"MemProvisioned",
    "capacity":{
      "max":167766912,
      "min":167766912,
      "avg":167766912,
      "total":167766912
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"Virtual_ESX_4238255f-86cc-4cb9-0f47-11ff689f0b22",
      "displayName":"hp-esx4.dev.mycorp.com",
      "className":"PhysicalMachine"
    },
    "units":"KB",
    "values":{
      "max":4194304,
      "min":4194304,
      "avg":4194304,
      "total":4194304
    },
    "value":4194304
  },
  {
    "name":"StorageAmount",
    "capacity":{
      "max":3301375,
```

```
      "min":3301375,
      "avg":3301375,
      "total":3301375
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"6f76b47e-e6f0a3fe",
      "displayName":"QSGRID01:ESXDC1DS1",
      "className":"Storage"
    },
    "units":"MB",
    "values":{
      "max":725.32,
      "min":725.32,
      "avg":725.32,
      "total":725.32
    },
    "value":725.32
  },
  {
    "name":"NetThroughput",
    "capacity":{
      "max":3072000,
      "min":3072000,
      "avg":3072000,
      "total":3072000
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"Virtual_ESX_4238255f-86cc-4cb9-0f47-11ff689f0b22",
      "displayName":"hp-esx4.dev.mycorp.com",
      "className":"PhysicalMachine"
    },
    "units":"Kbit/sec",
    "values":{
      "max":0,
      "min":0,
      "avg":0,
      "total":0
    },
    "value":0
  },
  {
    "name":"MemAllocation",
    "capacity":{
      "max":52503552,
      "min":52503552,
      "avg":52503552,
      "total":52503552
    },
    "filters":[
      {
        "type":"key",
        "value":"VirtualDataCenter::2332ba0f9dc14c07f11adb38e12df43c4bc8ca56"
      },
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"2332ba0f9dc14c07f11adb38e12df43c4bc8ca56",
      "displayName":"olga_do_not_touch_1",
      "className":"VirtualDataCenter"
    },
    "units":"KB",
    "values":{
      "max":3555836,
      "min":3554799.8,
      "avg":3554799.8,
      "total":3554799.8
    },
```

```
      "value":3554799.8
    },
    {
      "name":"VCPU",
      "capacity":{
        "max":5200,
        "min":5200,
        "avg":5200,
        "total":5200
      },
      "filters":[
        {
          "type":"relation",
          "value":"sold"
        }
      ],
      "units":"MHz",
      "values":{
        "max":5214,
        "min":5201.9,
        "avg":5201.9,
        "total":5201.9
      },
      "value":5201.9
    },
    {
      "name":"Ballooning",
      "capacity":{
        "max":16776692,
        "min":16776692,
        "avg":16776692,
        "total":16776692
      },
      "filters":[
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"Virtual_ESX_4238255f-86cc-4cb9-0f47-11ff689f0b22",
        "displayName":"hp-esx4.dev.mycorp.com",
        "className":"PhysicalMachine"
      },
      "units":"KB",
      "values":{
        "max":0,
        "min":0,
        "avg":0,
        "total":0
      },
      "value":0
    },
    {
      "name":"StorageAccess",
      "capacity":{
        "max":5000,
        "min":5000,
        "avg":5000,
        "total":5000
      },
      "filters":[
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"6f76b47e-e6f0a3fe",
        "displayName":"QSGRID01:ESXDC1DS1",
        "className":"Storage"
      },
      "units":"IOPS",
      "values":{
        "max":0,
        "min":0,
        "avg":0,
        "total":0
      },
      "value":0
    },
    {
```

```
      "name":"CPUAllocation",
      "capacity":{
        "max":31096,
        "min":31096,
        "avg":31096,
        "total":31096
      },
      "filters":[
        {
          "type":"key",
          "value":"VirtualDataCenter::2332ba0f9dc14c07f11adb38e12df43c4bc8ca56"
        },
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"2332ba0f9dc14c07f11adb38e12df43c4bc8ca56",
        "displayName":"olga_do_not_touch_1",
        "className":"VirtualDataCenter"
      },
      "units":"MHz",
      "values":{
        "max":5214,
        "min":5201.9,
        "avg":5201.9,
        "total":5201.9
      },
      "value":5201.9
    },
    {
      "name":"IOThroughput",
      "capacity":{
        "max":11264000,
        "min":11264000,
        "avg":11264000,
        "total":11264000
      },
      "reserved":{
        "max":9011200,
        "min":9011200,
        "avg":9011200,
        "total":9011200
      },
      "filters":[
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"Virtual_ESX_4238255f-86cc-4cb9-0f47-11ff689f0b22",
        "displayName":"hp-esx4.dev.mycorp.com",
        "className":"PhysicalMachine"
      },
      "units":"Kbit/sec",
      "values":{
        "max":0,
        "min":0,
        "avg":0,
        "total":0
      },
      "value":0
    },
    {
      "name":"StorageLatency",
      "capacity":{
        "max":100,
        "min":100,
        "avg":100,
        "total":100
      },
      "filters":[
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"6f76b47e-e6f0a3fe",
        "displayName":"QSGRID01:ESXDC1DS1",
```

```
      "className":"Storage"
    },
    "units":"msec",
    "values":{
      "max":0.5,
      "min":0.5,
      "avg":0.5,
      "total":0.5
    },
    "value":0.5
  },
  {
    "name":"VMem",
    "capacity":{
      "max":4194304,
      "min":4194304,
      "avg":4194304,
      "total":4194304
    },
    "filters":[
      {
        "type":"relation",
        "value":"sold"
      }
    ],
    "units":"KB",
    "values":{
      "max":3397384,
      "min":3233807,
      "avg":3233807,
      "total":3233807
    },
    "value":3233807
  },
  {
    "name":"CPU",
    "capacity":{
      "max":10400,
      "min":10400,
      "avg":10400,
      "total":10400
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"Virtual_ESX_4238255f-86cc-4cb9-0f47-11ff689f0b22",
      "displayName":"hp-esx4.dev.mycorp.com",
      "className":"PhysicalMachine"
    },
    "units":"MHz",
    "values":{
      "max":5214,
      "min":5201.9,
      "avg":5201.9,
      "total":5201.9
    },
    "value":5201.9
  },
  {
    "name":"VStorage",
    "capacity":{
      "max":2646,
      "min":1014,
      "avg":2238,
      "total":8952
    },
    "filters":[
      {
        "type":"key",
        "value":"VirtualMachine::08bb59f519b55c22415be8d623b5ccdf578b85cc"
      },
      {
        "type":"relation",
        "value":"sold"
      }
    ],
    "units":"MB",
    "values":{
```

```
                "max":1016.14,
                "min":142.4,
                "avg":797.71,
                "total":3190.82
              },
              "value":797.71
            }
          ]
        }
      ]
    },
    {
      "uuid":"4223bb37-0090-f332-0140-cc3cce7b3735",
      "displayName":"centos_imp_3",
      "className":"VirtualMachine",
      "environmentType":"ONPREM",
      "stats":[
        {
          "displayName":"centos_imp_3",
          "date":"2019-02-01T00:00:00-05:00",
          "statistics":[
            {
              "displayName":"centos_imp_3",
              "date":"2019-03-14T15:12:36-04:00",
              "statistics":[
                {
                  "name":"priceIndex",
                  "values":{
                    "max":20000,
                    "min":20000,
                    "avg":20000,
                    "total":20000
                  },
                  "value":20000
                },
                {
                  "name":"CPUProvisioned",
                  "capacity":{
                    "max":104000,
                    "min":104000,
                    "avg":104000,
                    "total":104000
                  },
                  "filters":[
                    {
                      "type":"relation",
                      "value":"bought"
                    }
                  ],
                  "relatedEntity":{
                    "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
                    "displayName":"hp-esx1.dev.mycorp.com",
                    "className":"PhysicalMachine"
                  },
                  "units":"MHz",
                  "values":{
                    "max":5200,
                    "min":5200,
                    "avg":5200,
                    "total":5200
                  },
                  "value":5200
                },
                {
                  "name":"Swapping",
                  "capacity":{
                    "max":40000000,
                    "min":40000000,
                    "avg":40000000,
                    "total":40000000
                  },
                  "filters":[
                    {
                      "type":"relation",
                      "value":"bought"
                    }
                  ],
                  "relatedEntity":{
                    "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
                    "displayName":"hp-esx1.dev.mycorp.com",
                    "className":"PhysicalMachine"
                  },
```

```
      "units":"bit/sec",
      "values":{
        "max":0,
        "min":0,
        "avg":0,
        "total":0
      },
      "value":0
    },
    {
      "name":"NumDisk",
      "capacity":{
        "max":2147483650,
        "min":2147483650,
        "avg":2147483650,
        "total":2147483650
      },
      "filters":[
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
        "displayName":"hp-esx1.dev.mycorp.com",
        "className":"PhysicalMachine"
      },
      "values":{
        "max":1,
        "min":1,
        "avg":1,
        "total":1
      },
      "value":1
    },
    {
      "name":"Mem",
      "capacity":{
        "max":16776692,
        "min":16776692,
        "avg":16776692,
        "total":16776692
      },
      "filters":[
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
        "displayName":"hp-esx1.dev.mycorp.com",
        "className":"PhysicalMachine"
      },
      "units":"KB",
      "values":{
        "max":3069332,
        "min":3062497,
        "avg":3062497,
        "total":3062497
      },
      "value":3062497
    },
    {
      "name":"MemProvisioned",
      "capacity":{
        "max":167766912,
        "min":167766912,
        "avg":167766912,
        "total":167766912
      },
      "filters":[
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
        "displayName":"hp-esx1.dev.mycorp.com",
        "className":"PhysicalMachine"
```

```
      },
      "units":"KB",
      "values":{
        "max":4194304,
        "min":4194304,
        "avg":4194304,
        "total":4194304
      },
      "value":4194304
    },
    {
      "name":"StorageAmount",
      "capacity":{
        "max":3301375,
        "min":3301375,
        "avg":3301375,
        "total":3301375
      },
      "filters":[
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"6f76b47e-e6f0a3fe",
        "displayName":"QSGRID01:ESXDC1DS1",
        "className":"Storage"
      },
      "units":"MB",
      "values":{
        "max":719,
        "min":719,
        "avg":719,
        "total":719
      },
      "value":719
    },
    {
      "name":"NetThroughput",
      "capacity":{
        "max":3072000,
        "min":3072000,
        "avg":3072000,
        "total":3072000
      },
      "filters":[
        {
          "type":"relation",
          "value":"bought"
        }
      ],
      "relatedEntity":{
        "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
        "displayName":"hp-esx1.dev.mycorp.com",
        "className":"PhysicalMachine"
      },
      "units":"Kbit/sec",
      "values":{
        "max":0,
        "min":0,
        "avg":0,
        "total":0
      },
      "value":0
    },
    {
      "name":"MemAllocation",
      "capacity":{
        "max":52503552,
        "min":52503552,
        "avg":52503552,
        "total":52503552
      },
      "filters":[
        {
          "type":"key",
          "value":"VirtualDataCenter::2332ba0f9dc14c07f11adb38e12df43c4bc8ca56"
        },
        {
          "type":"relation",
          "value":"bought"
```

```
      }
    ],
    "relatedEntity":{
      "uuid":"2332ba0f9dc14c07f11adb38e12df43c4bc8ca56",
      "displayName":"olga_do_not_touch_1",
      "className":"VirtualDataCenter"
    },
    "units":"KB",
    "values":{
      "max":3069332,
      "min":3062497,
      "avg":3062497,
      "total":3062497
    },
    "value":3062497
  },
  {
    "name":"VCPU",
    "capacity":{
      "max":5200,
      "min":5200,
      "avg":5200,
      "total":5200
    },
    "filters":[
      {
        "type":"relation",
        "value":"sold"
      }
    ],
    "units":"MHz",
    "values":{
      "max":5203,
      "min":5194.67,
      "avg":5194.67,
      "total":5194.67
    },
    "value":5194.67
  },
  {
    "name":"Ballooning",
    "capacity":{
      "max":16776692,
      "min":16776692,
      "avg":16776692,
      "total":16776692
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
      "displayName":"hp-esx1.dev.mycorp.com",
      "className":"PhysicalMachine"
    },
    "units":"KB",
    "values":{
      "max":0,
      "min":0,
      "avg":0,
      "total":0
    },
    "value":0
  },
  {
    "name":"StorageAccess",
    "capacity":{
      "max":5000,
      "min":5000,
      "avg":5000,
      "total":5000
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
```

```
      "uuid":"6f76b47e-e6f0a3fe",
      "displayName":"QSGRID01:ESXDC1DS1",
      "className":"Storage"
    },
    "units":"IOPS",
    "values":{
      "max":0,
      "min":0,
      "avg":0,
      "total":0
    },
    "value":0
  },
  {
    "name":"CPUAllocation",
    "capacity":{
      "max":31096,
      "min":31096,
      "avg":31096,
      "total":31096
    },
    "filters":[
      {
        "type":"key",
        "value":"VirtualDataCenter::2332ba0f9dc14c07f11adb38e12df43c4bc8ca56"
      },
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"2332ba0f9dc14c07f11adb38e12df43c4bc8ca56",
      "displayName":"olga_do_not_touch_1",
      "className":"VirtualDataCenter"
    },
    "units":"MHz",
    "values":{
      "max":5203,
      "min":5194.67,
      "avg":5194.67,
      "total":5194.67
    },
    "value":5194.67
  },
  {
    "name":"IOThroughput",
    "capacity":{
      "max":11264000,
      "min":11264000,
      "avg":11264000,
      "total":11264000
    },
    "reserved":{
      "max":9011200,
      "min":9011200,
      "avg":9011200,
      "total":9011200
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
      "displayName":"hp-esx1.dev.mycorp.com",
      "className":"PhysicalMachine"
    },
    "units":"Kbit/sec",
    "values":{
      "max":0,
      "min":0,
      "avg":0,
      "total":0
    },
    "value":0
  },
  {
    "name":"StorageLatency",
    "capacity":{
```

```
      "max":100,
      "min":100,
      "avg":100,
      "total":100
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"6f76b47e-e6f0a3fe",
      "displayName":"QSGRID01:ESXDC1DS1",
      "className":"Storage"
    },
    "units":"msec",
    "values":{
      "max":0.33,
      "min":0.33,
      "avg":0.33,
      "total":0.33
    },
    "value":0.33
  },
  {
    "name":"Q2VCPU",
    "capacity":{
      "max":20000,
      "min":20000,
      "avg":20000,
      "total":20000
    },
    "reserved":{
      "max":10000,
      "min":10000,
      "avg":10000,
      "total":10000
    },
    "filters":[
      {
        "type":"relation",
        "value":"bought"
      }
    ],
    "relatedEntity":{
      "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
      "displayName":"hp-esx1.dev.mycorp.com",
      "className":"PhysicalMachine"
    },
    "units":"msec",
    "values":{
      "max":11,
      "min":11,
      "avg":11,
      "total":11
    },
    "value":11
  },
  {
    "name":"VMem",
    "capacity":{
      "max":4194304,
      "min":4194304,
      "avg":4194304,
      "total":4194304
    },
    "filters":[
      {
        "type":"relation",
        "value":"sold"
      }
    ],
    "units":"KB",
    "values":{
      "max":2894068,
      "min":2747267,
      "avg":2747267,
      "total":2747267
    },
    "value":2747267
  },
```

```
                    {
                      "name":"CPU",
                      "capacity":{
                        "max":10400,
                        "min":10400,
                        "avg":10400,
                        "total":10400
                      },
                      "filters":[
                        {
                          "type":"relation",
                          "value":"bought"
                        }
                      ],
                      "relatedEntity":{
                        "uuid":"Virtual_ESX_4238a511-8593-4469-2751-46611e09b6a8",
                        "displayName":"hp-esx1.dev.mycorp.com",
                        "className":"PhysicalMachine"
                      },
                      "units":"MHz",
                      "values":{
                        "max":5203,
                        "min":5194.67,
                        "avg":5194.67,
                        "total":5194.67
                      },
                      "value":5194.67
                    },
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                      "name":"VStorage",
                      "capacity":{
                        "max":2646,
                        "min":1014,
                        "avg":2238,
                        "total":8952
                      },
                      "filters":[
                        {
                          "type":"key",
                          "value":"VirtualMachine::199bb3ec4f8da89f01c126a6b3df4c1321c4b07d"
                        },
                        {
                          "type":"relation",
                          "value":"sold"
                        }
                      ],
                      "units":"MB",
                      "values":{
                        "max":1016.01,
                        "min":142.4,
                        "avg":797.61,
                        "total":3190.42
                      },
                      "value":797.61
                    }
                  ]
                }
              ]
            }
          ]
        },
        "timebasedTopologyChanges":{

        }
      }
```


Reference: Tables

Reference tables overview

This section contains reference tables for common API parameters, separated by endpoint. Note that filter tables may contain other reference tables for completeness.

Action Reference Tables

Action Mode:

- DISABLED:

  Do not recommend or perform the action. When you disable an action, the user interface will no longer display actions of that type.
- RECOMMENDED:

  Recommend the action so a user can execute it via the given hypervisor or by other external means
- MANUAL:

  Recommend the action, and provide the option to execute that action through the Turbonomic user interface or an /actions API request.
- AUTOMATED:

  You have directed Turbonomic to execute the action automatically.
- COLLECTION **(UNUSED)**:

  Legacy action mode, no longer used.

Action State:

- DISABLED:

  An action whose policy has been disabled.
- RECOMMEND:

  An action that cannot be automated due to policy settings or must be executed outside the system, such as the purchase of hardware.
- PENDING_ACCEPT:

  When the action mode is set to manual, a recommended action that has not been accepted or cleared.
- ACCEPTED:

  An action that has been accepted, but is not yet in progress.
- IN_PROGRESS:

  An action that is being executed.
- SUCCEEDED:

  An action that was accepted and completed successfully.
- FAILED:

  An action that was accepted and did not complete successfully.
- REJECTED:

  When the action mode is set to manual, an action that has been rejected by the user.
- CLEARED:

  When the action mode is set to manual, an action that is no longer recommended by the market.
- ACCOUNTING:

  For cloud entities, an action to resize in order to use a different instance template.
- QUEUED:

  When more than 10 actions are to be executed for a single target, Turbonomic will place the 11th and all subsequent actions into a QUEUED state, and they will execute as the previous actions complete.

Action Type:

- START:

  Start an entity, and add it to the Turbonomic market.
- MOVE:

Move an entity from one provider to another. For example, moving a VM between hosts, or a datastore between disk arrays.

- SUSPEND:

  Suspend the entity to exclude its capacity from your environment. Suspend does not remove the entity from your environment, but the market does not include the entity in its analysis. Based on market conditions, analysis can recommend restarting suspended entities, which includes them in market analysis once again.

- ADD PROVIDER:

  This is equivalent to the START action, but for storage entities.

- CHANGE:

  Move a virtual machine's storage between datastores.

- PROVISION:

  Add capacity to your environment.

- RECONFIGURE:

  Analysis recommends this when a VM is not able to correct a misconfiguration via another type of action. For example, analysis can recommend a RECONFIGURE action when a placement policy requires a VM to move to a host on a particular cluster, but cross-cluster moves are not permitted.

- DELETE:

  This is equivalent to the SUSPEND action, but for storage entities.

- RESERVE_ON_PM:

  When using the reservation functionality, this is the action to reserve the resources of a VM on a particular host.

- RESERVE_ON_DS:

  When using the reservation functionality, this is the action to reserve the resources of a VM on a particular datastore.

- RESIZE_FOR_EFFICIENCY:

  A resize that reduces the capacity of a commodity, such as vCPU, or vMEM.

- RESIZE_FOR_PERFORMANCE:

  A resize that increases the capacity of a commodity, such as vCPU, or vMEM.

Action Filter Criteria:

- actionModeList
  - DISABLED:

    Do not recommend or perform the action. When you disable an action, the user interface will no longer display actions of that type.

  - RECOMMENDED:

    Recommend the action so a user can execute it via the given hypervisor or by other external means

  - MANUAL:

    Recommend the action, and provide the option to execute that action through the Turbonomic user interface or an /actions API request.

  - AUTOMATED:

    You have directed Turbonomic to execute the action automatically.

  - COLLECTION **(UNUSED)**:

    Legacy action mode, no longer used.

- actionStateList

  - `DISABLED`:

    An action whose policy has been disabled.
  - `RECOMMEND`:

    An action that cannot be automated due to policy settings or must be executed outside the system, such as the purchase of hardware.
  - `PENDING_ACCEPT`:

    When the action mode is set to manual, a recommended action that has not been accepted or cleared.
  - `ACCEPTED`:

    An action that has been accepted, but is not yet in progress.
  - `IN_PROGRESS`:

    An action that is being executed.
  - `SUCCEEDED`:

    An action that was accepted and completed successfully.
  - `FAILED`:

    An action that was accepted and did not complete successfully.
  - `REJECTED`:

    When the action mode is set to manual, an action that has been rejected by the user.
  - `CLEARED`:

    When the action mode is set to manual, an action that is no longer recommended by the market.
  - `ACCOUNTING`:

    For cloud entities, an action to resize in order to use a different instance template.
  - `QUEUED`:

    When more than 10 actions are to be executed for a single target, Turbonomic will place the 11th and all subsequent actions into a QUEUED state, and they will execute as the previous actions complete.
- actionTypeList
  - `START`:

    Start an entity, and add it to the Turbonomic market.
  - `MOVE`:

    Move an entity from one provider to another. For example, moving a VM between hosts, or a datastore between disk arrays.
  - `SUSPEND`:

    Suspend the entity to exclude its capacity from your environment. Suspend does not remove the entity from your environment, but the market does not include the entity in its analysis. Based on market conditions, analysis can recommend restarting suspended entities, which includes them in market analysis once again.
  - `ADD PROVIDER`:

    This is equivalent to the START action, but for storage entities.
  - `CHANGE`:

    Move a virtual machine's storage between datastores.
  - `PROVISION`:

    Add capacity to your environment.
  - `RECONFIGURE`:

Analysis recommends this when a VM is not able to correct a misconfiguration via another type of action. For example, analysis can recommend a RECONFIGURE action when a placement policy requires a VM to move to a host on a particular cluster, but cross-cluster moves are not permitted.

  - `DELETE`:

    This is equivalent to the SUSPEND action, but for storage entities.

  - `RESERVE_ON_PM`:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular host.

  - `RESERVE_ON_DS`:

    When using the reservation functionality, this is the action to reserve the resources of a VM on a particular datastore.

  - `RESIZE_FOR_EFFICIENCY`:

    A resize that reduces the capacity of a commodity, such as vCPU, or vMEM.

  - `RESIZE_FOR_PERFORMANCE`:

    A resize that increases the capacity of a commodity, such as vCPU, or vMEM.

- cleared

  Default: `false`.

  When `true`, actions that were removed from the action list before being executed will be returned.

- costType

  - `Saving`

    An action that will decrease cloud spending.

  - `Investment`

    An action that will increase cloud spending.

- cumulative

  Default: `false`.

  When `true`, shows the savings for each action in a scope as a cumulative total, rather than showing the savings for individual actions.

- environmentType

  - `ONPREM`:

    Returns only actions related to entities that are part of your on-prem environment.

  - `CLOUD`:

    Returns only actions related to entities that are part of your cloud environment.

- groupBy

  You may group the actions returned by the following criteria: `[actionModes, actionStates, actionTypes, risk, riskSeverity, riskSubCategory]`

- hasReservedInstance

  Default: `false`.

  When `true`, only actions that are related to reserved instances will be returned.

- relatedEntityTypes

  Returns only actions related to entities of these types within the scope. For example, if your scope is a virtual datacenter and you specify VirtualMachine as a related entity type, the request will include only those actions related to VMs in the specified virtual datacenter.

Choose from the following entity types: `[ Datacenter, PhysicalMachine, VirtualMachine, Storage, Application, Chassis, DiskArray, IOModule, StorageControl, Switch, VirtualDataCenter, VPod, DPod, Container, Database, DatabaseServerContainer, LogicalPool ]`

- riskSeverityList

  Returns only actions of the given severities: `[ Unknown, Normal, Minor, Major, Critical ]`

- riskSubCategoryList
- returns only actions of the given subcategories: `[ Performance Assurance, Efficiency Improvement, Prevention, Compliance ]`

Entity Reference Tables

Entity State:

- `ACTIVE`:

  Entity is actively consuming resources.

- `EVACUATED`:

  Entity is ready to suspend and will not participate in the market, but will not generate a `SUSPEND` action.

  Note: This state replaces the `SUSPEND` state, and requires additional configuration to enable. Turbonomic does not recommend enabling this state.

- `FAILOVER`:

  This entity is reserved for failover and will not participate in the market.

- `IDLE`:

  Entity is powered off and not actively consuming market resources.

- `LAUNCH`:

  Entity is starting.

- `MAINTENANCE`:

  Entity is in maintenance mode.

- `NOT_MONITORED`:

  Entity is not currently monitored by Turbonomic.

- `QUEUED`:

  Entity is in the process of changing from one state to another.

  Note: This state is internal to Turbonomic and will not appear in the UI, but may appear in an API response if the call is executed at the precise moment the entity is changing state.

- `RESOURCE_ALLOCATION`:

  Cloud entity is being created.

- `RESOURCE_RELEASE`:

  Cloud Entity is being deleted.

- `SUSPEND`:

  Entity is suspended or the guest OS is in sleep, standby, or suspended state. The entity is not removed from the environment, but will not be used in market decisions. Based on market conditions, suspended entities can be started, re-adding them to the market.

- `SUSPEND_PENDING`:

  Entity is making preparations to suspend, or is the next entity of its type to suspend.

- `TERMINATE_PENDING`:

  Entity is in the process of being deleted.

- `UNKNOWN`:

  Turbonomic is unable to obtain the current state of the entity.

  This occurs when the entity is reported to the hypervisor as having a state of `DISCONNECTED`, `ORPHANED`, or `INVALID`.

Entity Type:

- `APPLICATION`
- `APPLICATION_SERVER`
- `BUSINESS_ACCOUNT`
- `BUSINESS_UNIT`
- `CHASSIS`
- `CLOUD_SERVICE`
- `CONTAINER`
- `CONTAINER_POD`
- `DATABASE`
- `DATABASE_SERVER`
- `DATACENTER`
- `DISK_ARRAY`
- `IO_MODULE`
- `LOAD_BALANCER`
- `LOGICAL_POOL`
- `NETWORK`
- `PHYSICAL_MACHINE`
- `STORAGE`
- `STORAGE_CONTROLLER`
- `SWITCH`
- `VIRTUAL_APPLICATION`
- `VIRTUAL_DATACENTER`
- `VIRTUAL_MACHINE`

Reference: DTO index

Data transfer objects (DTOs) are used to encapsulate and send data to the Turbonomic API. The following DTOs are used by Turbonomic:

Note:

The DTO Index is a work in progress, and included in a draft stage for reference. The information contained in the Index is accurate, but may not be complete.

AcceleratorCardApiDTO

Description

Model to describe info about an Accelerator/GPU Card

Optional Parameters

acceleratorFamily

- **type:** string
- **description:** Accelerator/GPU Micro architecture family E.G NVIDIA_TURING, NVIDIA_AMPERE, AMD_RADEON_PRO

acceleratorModel

- **type:** string
- **description:** Accelerator/GPU Micro model. E.G NVIDIA_T4, NVIDIA_T4G, NVIDIA_A10G, AMD_V520

numCores

- **type:** integer
- **format:** int32
- **description:** Number of Accelerator/GPU cards

memoryPerCore

- **type:** number
- **format:** double
- **description:** Memory in GB per Accelerator/GPU card

partitions

- **type:** integer
- **format:** int32
- **description:** Accelerator partitions if partial GPUs.

ActionApiDTO

Description

Model to describe an Action

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

actionImpactID

- **type:** integer
- **format:** int64
- **description:** The ID for the action, which will persist across restarts

marketID

- **type:** integer

- **format:** int64
- **description:** The ID of the market for which the action was generated

createTime

- **type:** string
- **description:** Creation time

updateTime

- **type:** string
- **description:** Update time

clearTime

- **type:** string
- **description:** Execution time

actionType

- **type:** string
- **description:** Type
- **enum:** ['START', 'MOVE', 'SCALE', 'ALLOCATE', 'SUSPEND', 'PROVISION', 'RECONFIGURE', 'RESIZE', 'DELETE', 'RIGHT_SIZE', 'BUY_RI']

actionState

- **type:** string
- **description:** Action State.
- **enum:** ['ACCEPTED', 'REJECTED', 'PRE_IN_PROGRESS', 'POST_IN_PROGRESS', 'IN_PROGRESS', 'SUCCEEDED', 'FAILED', 'DISABLED', 'QUEUED', 'CLEARED', 'ACCOUNTING', 'READY', 'FAILING', 'BEFORE_EXEC', 'IN_PROGRESS_NON_CRITICAL']

actionMode

- **type:** string
- **description:** Action Mode
- **enum:** ['DISABLED', 'RECOMMEND', 'EXTERNAL_APPROVAL', 'MANUAL', 'AUTOMATIC', 'COLLECTION']

userName

- **type:** string
- **description:** Username of the user that accepted the action [DEPRECATED - Use: userInfo.userName in executionStatus.executorInfo]

details

- **type:** string
- **description:** A user-readable string describing the action

importance

- **type:** number
- **format:** float
- **description:** Numeric value that describes the priority of the action

target

- **description:** Target entity for an action. For example, the VM in a Resize Action, or the host for a VM move
- **reference:** ServiceEntityApiDTO

currentEntity

- **description:** Current entity, such as the current host that a VM resides on for a VM move action
- **reference:** ServiceEntityApiDTO

newEntity

- **description:** Destination entity, such as the host that a VM will move to for a VM move action
- **reference:** ServiceEntityApiDTO

currentValue

- **type:** string
- **description:** Current value of a property, for example vMEM for a VM resize action

newValue

- **type:** string
- **description:** Calculated value to resize to, such as vMEM for a VM resize action

valueUnits

- **type:** string
- **description:** Units of the currentValue and newValue, such as KB for a VM vMEM resize action

resizeAttribute

- **type:** string
- **description:** The commodity attribute to be resized
- **enum:** ['CAPACITY', 'LIMIT', 'RESERVATION']

template

- **description:** Template used for the action, such as in a Cloud entity provision or move action
- **reference:** TemplateApiDTO

risk

- **description:** Cause of the action
- **reference:** LogEntryApiDTO

prerequisites

- **type:** array
- **description:** Description(s) of any prerequisite(s) that must occur before execution is possible. If this field is populated, the action would otherwise be executable but has been blocked by this prerequisite, and is displayed accordingly. If this field is not populated, no prerequisite has caused the action mode to be modified, and the action is displayed as normal.
- – **type:** string

prioritizedAutomationOpportunities

- **type:** array
- **description:** The Action is generated using the 'Prioritze Automation' setting, indicating that it is possible to have more savings if these opportunities are satisfied
- – **type:** string

stats

- **type:** array
- **description:** Statistics, such as Mem, vCPU, costPrice
- – **reference:** StatApiDTO

progress

- **type:** integer
- **format:** int32
- **description:** Moved the 'progress' percentage value inside 'executionStatus', available when action detail level is 'EXECUTION' - [DEPRECATED - Use: executionStatus.progress]

currentLocation

- **description:** The region (DataCenter) where the current service entity is located, for cloud migration actions
- **reference:** DiscoveredEntityApiDTO

newLocation

- **description:** The region, represented as a DataCenter entity, where the target service entity will be located for cloud migration actions
- **reference:** DiscoveredEntityApiDTO

virtualDisks

- **type:** array
- **description:** Virtual Disk used for the action, for example in a storage move
- – **reference:** VirtualDiskApiDTO

policy

- **description:** The policy associated with the action
- **reference:** PolicyApiDTO

relatedSettingsPolicies

- **type:** array
- **description:** The related settings policies associated with the action
- – **reference:** BaseApiDTO

actionSchedule

- **description:** The schedule associated with the action
- **reference:** ActionScheduleApiDTO

compoundActions

- **type:** array
- **description:** Property for compound move actions
- – **reference:** ActionApiDTO

reservedInstance

- **description:** The Reserved Instance associated with the action
- **reference:** ReservedInstanceApiDTO

reservedInstancePlatform

- **type:** string
- **description:** The platform of the recommended RI if the action has a supersaving

externalActionName

- **type:** string
- **description:** The name under which this action appears in third-party software or external approval backend

externalActionUrl

- **type:** string
- **description:** URL to third-party software showing this action. Suitable for external action approval

executionStatus

- **description:** Status details of the execution of the Action. Include percentage progress, error messages and execution times.
- **reference:** ActionExecutionAuditApiDTO

executionCharacteristics

- **description:** Characteristics of the execution of the Action. Include disruptiveness and reversibility.
- **reference:** ActionExecutionCharacteristicApiDTO

entityCharacteristics

- **description:** Characteristics of the entity the action is for.
- **reference:** DBEntityAspectApiDTO

source

- **type:** string
- **description:** Defines what generated this action.
- **enum:** ['MARKET', 'PARKING', 'BUY_RI', 'CLOUD_OPTIMIZER']

relatedActionsCountByType

- **type:** object
- **description:** [BETA]: Map of related actions count by relationship type that have some relation to this Action
- **additionalProperties:** {'type': 'integer', 'format': 'int32'}

actionID

- **type:** integer
- **format:** int64
- **description:** Id of the action

ActionApiInputDTO

Description

Model to describe a Filter for Actions

Required Parameters

startTime

- **type:** string
- **description:** Start time of the period to filter, based on Actions Creation Time

Optional Parameters

actionStateList

- **type:** array
- **description:** List of States to filter.
- – **type:** string

  - **enum:** ['ACCEPTED', 'REJECTED', 'PRE_IN_PROGRESS', 'POST_IN_PROGRESS', 'IN_PROGRESS', 'SUCCEEDED', 'FAILED', 'DISABLED', 'QUEUED', 'CLEARED', 'ACCOUNTING', 'READY', 'FAILING', 'BEFORE_EXEC', 'IN_PROGRESS_NON_CRITICAL']

actionModeList

- **type:** array
- **description:** List of Modes to filter
- – **type:** string
  - **enum:** ['DISABLED', 'RECOMMEND', 'EXTERNAL_APPROVAL', 'MANUAL', 'AUTOMATIC', 'COLLECTION']

riskSeverityList

- **type:** array
- **description:** List of Risk Severities to filter
- – **type:** string

riskSubCategoryList

- **type:** array
- **description:** List of Risk Severities to filter
- – **type:** string

actionRelationTypeFilter

- **type:** array
- **description:** Filters the main list of actions based on the types of relationships they have to other actions.
- – **type:** string
  - **enum:** ['NONE', 'BLOCKED_BY', 'BLOCKING', 'CAUSED_BY', 'CAUSING']

environmentType

- **type:** string
- **description:** Filter the actions by Environment Type
- **enum:** ['ONPREM', 'CLOUD']

costType

- **type:** string
- **description:** Filter the actions by cost Type
- **enum:** ['SAVING', 'INVESTMENT']

descriptionQuery

- **description:** This query will match the details field of the ActionApiDTO to return actions that match the specified expression.
- **reference:** QueryInputApiDTO

riskQuery

- **description:** This query will match the risk description field of the ActionApiDTO to return actions that match the specified expression.
- **reference:** QueryInputApiDTO

executionCharacteristics

- **description:** Characteristics of the execution of the action, including disruptiveness and reversibility.
- **reference:** ActionExecutionCharacteristicInputApiDTO

savingsAmountRange

- **description:** This field will match the costPrice statistic field of the ActionApiDTO to return actions that have a costPrice amount between minValue and maxValue, inclusive
- **reference:** RangeInputApiDTO

effectiveCostSavingsAmountRange

- **description:** This field will match the effectiveCost statistic field of the ActionApiDTO to return actions that have an effectiveCost amount between minValue and maxValue, inclusive
- **reference:** RangeInputApiDTO

unattachedDays

- **description:** [BETA]: Filter the actions based on an unattached days range and is applicable only to delete wasted file actions.
- **reference:** RangeInputApiDTO

hasSchedule

- **type:** boolean
- **description:** If true, only actions that have an associated execution schedule will be returned.

hasPrerequisites

- **type:** boolean
- **description:** If true, only actions that have a prerequisite preventing execution of the action will be returned.

endTime

- **type:** string
- **description:** End time of the period to filter, based on Actions Creation Time

cleared

- **type:** boolean
- **description:** Filter by Actions Executed

cumulative

- **type:** boolean
- **description:** Show the action savings as cumulative values

hasReservedInstance

- **type:** boolean
- **description:** Filter the actions by whether having related reserved instance

excludeNotifications

- **type:** boolean
- **description:** Exclude the notifications from the result

limitEntities

- **type:** integer
- **format:** int32
- **description:** Limit number of entities in the result

detailLevel

- **type:** string
- **description:** Detail level of the Action, 'standard' is the default value

- **enum:** ['STANDARD', 'EXECUTION']

vendorId

- **type:** string
- **description:** Vendor Id indicating the related vendor to the entity in the actions

actionTypeList

- **type:** array
- **description:** List of Types to filter
- – **type:** string
  – **enum:** ['NONE', 'START', 'MOVE', 'SCALE', 'ALLOCATE', 'SUSPEND', 'TERMINATE', 'SPAWN', 'ADD_PROVIDER', 'CHANGE', 'REMOVE_PROVIDER', 'PROVISION', 'RECONFIGURE', 'RESIZE', 'RESIZE_CAPACITY', 'WARN', 'RECONFIGURE_THRESHOLD', 'DELETE', 'RIGHT_SIZE', 'RESERVE_ON_PM', 'RESERVE_ON_DS', 'RESIZE_FOR_EFFICIENCY', 'RESIZE_FOR_PERFORMANCE', 'CROSS_TARGET_MOVE', 'MOVE_TOGETHER', 'ACTIVATE', 'DEACTIVATE', 'BUY_RI', 'GRANT']

relatedEntityTypes

- **type:** array
- **example:** Selecting a VCD, and specifying '"related_entities": ["VirtualMachine"] will include the actions for any VM in the given VDC
- **description:** Related Entity Types to include from the supply chain for this entity
- – **type:** string

targetEntityTypes

- **type:** array
- **example:** Specifying '"target_entities": ["VirtualMachine"] will include the actions where the target entity is a VM.
- **description:** Filter the actions where the target entity of the action matches the provided list of entity types
- – **type:** string

relatedCloudServiceProviderIds

- **type:** array
- **description:** Filter the actions by the related Cloud Service Providers.
- – **type:** string

groupBy

- **type:** array
- **description:** Fields used to Group the Stats
- – **type:** string

instanceTypeFilter

- **description:** [BETA]: Filter the actions based on a list of Instance Type Uuids.
- **reference:** InstanceTypeFilterApiDTO

onDemandCostFilter

- **description:** [BETA]: Filter the actions based on a on-demand cost range.
- **reference:** OnDemandCostFilterApiDTO

effectiveCostFilter

- **description:** Filter the actions based on a effective cost range.

- **reference:** EffectiveCostFilterApiDTO

discountCoverageFilter

- **description:** [BETA]: Filter the actions based on a discount coverage.
- **reference:** DiscountCoverageFilterApiDTO

accountFilter

- **description:** [BETA]: Filter the actions based on a list of account Uuid.
- **reference:** StringListFilterApiDTO

resourceGroupFilter

- **description:** [BETA]: Filter the actions based on a list of resource group Uuid.
- **reference:** StringListFilterApiDTO

tagValueFilter

- **description:** Filter actions based on a list of tag values.
- **reference:** StringListFilterApiDTO

virtualDiskDisplayName

- **description:** This query will match the virtual disk displayName field of the ActionApiDTO to return actions that match the specified expression.
- **reference:** QueryInputApiDTO

ActionDetailsApiDTO

Description

Parent class for action details. Implementing classes include: CloudResizeActionDetailsApiDTO, NoDetailsApiDTO, RIBuyActionDetailsApiDTO, CloudProvisionActionDetailsApiDTO, CloudSuspendActionDetailsApiDTO, OnPremResizeActionDetailsApiDTO and ReconfigureActionDetailsApiDTO.

Required Parameters

actionDetailType

- **type:** string

actionType

- **type:** string

ActionExecutionApiDTO

Description

Model to describe an Action Execution.

Required Parameters

acceptTime

- **type:** string
- **description:** Time when actions were accepted (in ISO8601 format - YYYY-MM-DDTHH:MM:SS)

actionIds

- **type:** array

- **description:** List of action IDs
- – **type:** string

id

- **type:** string
- **description:** Action Execution ID

skippedActions

- **type:** array
- **description:** List of actions that were skipped in execution
- – **reference:** SkippedActionApiDTO

ActionExecutionAuditApiDTO

Description

Model to describe an Action Execution status.

Required Parameters

executionTime

- **type:** string
- **description:** Start of the execution time

executorInfo

- **description:** Information about the executor
- **reference:** ExecutorInfoApiDTO

state

- **type:** string
- **description:** State of the execution
- **enum:** ['ACCEPTED', 'REJECTED', 'PRE_IN_PROGRESS', 'POST_IN_PROGRESS', 'IN_PROGRESS', 'SUCCEEDED', 'FAILED', 'DISABLED', 'QUEUED', 'CLEARED', 'ACCOUNTING', 'READY', 'FAILING', 'BEFORE_EXEC', 'IN_PROGRESS_NON_CRITICAL']

Optional Parameters

completionTime

- **type:** string
- **description:** End of the execution time

message

- **type:** string
- **description:** Status message, if the Action failed, it will contains the Error details

progress

- **type:** integer
- **format:** int32
- **description:** Action execution progress value, from 0 to 100

ActionExecutionCharacteristicApiDTO

Description

Model to describe action execution characteristics

Optional Parameters

disruptiveness

- **type:** string
- **description:** whether action is disruptive.
- **enum:** ['DISRUPTIVE', 'NON_DISRUPTIVE']

reversibility

- **type:** string
- **description:** whether action is reversible.
- **enum:** ['REVERSIBLE', 'IRREVERSIBLE']

ActionExecutionCharacteristicInputApiDTO

Description

Model to describe action execution characteristics

Optional Parameters

disruptiveness

- **type:** string
- **description:** whether action is disruptive.
- **enum:** ['DISRUPTIVE', 'NON_DISRUPTIVE']

reversibility

- **type:** string
- **description:** whether action is reversible.
- **enum:** ['REVERSIBLE', 'IRREVERSIBLE']

ActionExecutionInputApiDto

Description

Model to describe an Action Execution input.

Required Parameters

actionIds

- **type:** array
- **description:** List of action IDs
- – **type:** string

ActionResourceImpactStatApiInputDTO

Description

Model to describe an input for Action Resource Impact Stats

Required Parameters

actionResourceImpactStatList

- **type:** array
- **description:** List of Action Resource Impact Stat Inputs
- – **reference:** ActionResourceImpactStatInput

Optional Parameters

actionStateList

- **type:** array
- **description:** List of States to filter.
- – **type:** string
  – **enum:** ['ACCEPTED', 'REJECTED', 'PRE_IN_PROGRESS', 'POST_IN_PROGRESS', 'IN_PROGRESS', 'SUCCEEDED', 'FAILED', 'DISABLED', 'QUEUED', 'CLEARED', 'ACCOUNTING', 'READY', 'FAILING', 'BEFORE_EXEC', 'IN_PROGRESS_NON_CRITICAL']

actionModeList

- **type:** array
- **description:** List of Modes to filter
- – **type:** string
  – **enum:** ['DISABLED', 'RECOMMEND', 'EXTERNAL_APPROVAL', 'MANUAL', 'AUTOMATIC', 'COLLECTION']

riskSeverityList

- **type:** array
- **description:** List of Risk Severities to filter
- – **type:** string

riskSubCategoryList

- **type:** array
- **description:** List of Risk Severities to filter
- – **type:** string

actionRelationTypeFilter

- **type:** array
- **description:** Filters the main list of actions based on the types of relationships they have to other actions.
- – **type:** string
  – **enum:** ['NONE', 'BLOCKED_BY', 'BLOCKING', 'CAUSED_BY', 'CAUSING']

environmentType

- **type:** string
- **description:** Filter the actions by Environment Type
- **enum:** ['ONPREM', 'CLOUD']

costType

- **type:** string
- **description:** Filter the actions by cost Type
- **enum:** ['SAVING', 'INVESTMENT']

descriptionQuery

- **description:** This query will match the details field of the ActionApiDTO to return actions that match the specified expression.
- **reference:** QueryInputApiDTO

riskQuery

- **description:** This query will match the risk description field of the ActionApiDTO to return actions that match the specified expression.
- **reference:** QueryInputApiDTO

executionCharacteristics

- **description:** Characteristics of the execution of the action, including disruptiveness and reversibility.
- **reference:** ActionExecutionCharacteristicInputApiDTO

savingsAmountRange

- **description:** This field will match the costPrice statistic field of the ActionApiDTO to return actions that have a costPrice amount between minValue and maxValue, inclusive
- **reference:** RangeInputApiDTO

effectiveCostSavingsAmountRange

- **description:** This field will match the effectiveCost statistic field of the ActionApiDTO to return actions that have an effectiveCost amount between minValue and maxValue, inclusive
- **reference:** RangeInputApiDTO

unattachedDays

- **description:** This field will match the unattachedDays resource impacts field of the ActionResourceImpactStatApiInputDTO to return actions that have a unattachedDays count between minValue and maxValue, inclusive
- **reference:** RangeInputApiDTO

hasSchedule

- **type:** boolean
- **description:** If true, only actions that have an associated execution schedule will be returned.

hasPrerequisites

- **type:** boolean
- **description:** If true, only actions that have a prerequisite preventing execution of the action will be returned.

ActionResourceImpactStatInput

Description

Model to describe an Action Resource Impact Stat Input

Required Parameters

actionType

- **type:** string
- **description:** Action Type
- **enum:** ['NONE', 'START', 'MOVE', 'SCALE', 'ALLOCATE', 'SUSPEND', 'TERMINATE', 'SPAWN', 'ADD_PROVIDER', 'CHANGE', 'REMOVE_PROVIDER', 'PROVISION', 'RECONFIGURE', 'RESIZE', 'RESIZE_CAPACITY', 'WARN', 'RECONFIGURE_THRESHOLD', 'DELETE', 'RIGHT_SIZE', 'RESERVE_ON_PM', 'RESERVE_ON_DS', 'RESIZE_FOR_EFFICIENCY', 'RESIZE_FOR_PERFORMANCE', 'CROSS_TARGET_MOVE', 'MOVE_TOGETHER', 'ACTIVATE', 'DEACTIVATE', 'BUY_RI', 'GRANT']

commodityType

- **type:** string
- **example:** VCPU
- **description:** Commodity Type that could be impacted by the action. For example, VCPU or VMEM for a Resize VM action.
- **enum:** ['ACTION_PERMIT', 'ACTIVE_SESSIONS', 'APPLICATION', 'BALLOONING', 'BUFFER_COMMODITY', 'CLUSTER', 'COLLECTION_TIME', 'COOLING', 'CONNECTION', 'COUPON', 'CPU', 'CPU_ALLOCATION', 'CPU_PROVISIONED', 'CPU_REQUEST_ALLOCATION', 'CROSS_CLOUD_MOVE_SVC', 'CROSS_CLUSTER_MOVE_SVC', 'DATACENTER', 'DATASTORE', 'DB_CACHE_HIT_RATE', 'DB_MEM', 'DISK_ARRAY_ACCESS', 'DRS_SEGMENTATION', 'DSPM_ACCESS', 'EXTENT', 'FLOW', 'FLOW_ALLOCATION', 'HEAP', 'HOST_LUN_ACCESS', 'IMAGE_CPU', 'IMAGE_MEM', 'IMAGE_STORAGE', 'INSTANCE_DISK_SIZE', 'INSTANCE_DISK_TYPE', 'IO_THROUGHPUT', 'LICENSE_ACCESS', 'MEM', 'MEM_ALLOCATION', 'MEM_REQUEST_ALLOCATION', 'MEM_PROVISIONED', 'NET_THROUGHPUT', 'NETWORK', 'NUM_DISK', 'NUMBER_CONSUMERS', 'NUM_VCORE', 'POOL_CPU', 'POOL_MEM', 'POOL_STORAGE', 'PORT_CHANNEL', 'POWER', 'Q16_VCPU', 'Q1_VCPU', 'Q2_VCPU', 'Q32_VCPU', 'Q4_VCPU', 'Q64_VCPU', 'Q8_VCPU', 'RESPONSE_TIME', 'SAME_CLUSTER_MOVE_SVC', 'SEGMENTATION', 'SLA_COMMODITY', 'SOFTWARE_LICENSE_COMMODITY', 'SPACE', 'STORAGE_ACCESS', 'STORAGE_ALLOCATION', 'STORAGE_AMOUNT', 'STORAGE_CLUSTER', 'STORAGE_LATENCY', 'STORAGE_PROVISIONED', 'SWAPPING', 'THREADS', 'TRANSACTION', 'TRANSACTION_LOG', 'VCPU', 'VCPU_LIMIT_QUOTA', 'VCPU_REQUEST', 'VCPU_REQUEST_QUOTA', 'VDC', 'VMEM', 'VMEM_LIMIT_QUOTA', 'VMEM_REQUEST', 'VMEM_REQUEST_QUOTA', 'VMPM_ACCESS', 'VSTORAGE', 'ZONE', 'LICENSE_COMMODITY', 'Q3_VCPU', 'NUMBER_CONSUMERS_PM', 'Q6_VCPU', 'Q7_VCPU', 'QN_VCPU', 'RIGHT_SIZE_SVC', 'RIGHT_SIZE_DOWN', 'MOVE', 'Q5_VCPU', 'STORAGE', 'NUMBER_CONSUMERS_STORAGE', 'ACCESS', 'RIGHT_SIZE_UP', 'VAPP_ACCESS', 'HOT_STORAGE', 'HA_COMMODITY', 'NETWORK_POLICY', 'SERVICE_LEVEL_CLUSTER', 'PROCESSING_UNITS', 'TENANCY_ACCESS', 'TEMPLATE_ACCESS', 'BURST_BALANCE', 'DESIRED_COUPON', 'NETWORK_INTERFACE_COUNT', 'BICLIQUE', 'TAINT', 'LABEL', 'PEER_TO_PEER_AFFINITY', 'PEER_TO_PEER_ANTI_AFFINITY', 'ENERGY', 'CARBON', 'VIRTUAL_PROCESSORS', 'CPU_READY', 'PROCESSING_UNITS_PROVISIONED', 'HARDWARE_VERSION', 'GPU_MEM_ALLOCATION', 'GPU_MEM', 'GPU_COUNT', 'GPU_TENSOR_ENGINE', 'GPU_FP64_ENGINE', 'GPU_FP32_ENGINE', 'GPU_FP16_ENGINE', 'GPU_MEMORY_BANDWIDTH', 'STORAGE_ACCESS_OVERALL_VM_MAX_READ', 'STORAGE_ACCESS_OVERALL_VM_MAX_WRITE', 'STORAGE_ACCESS_OVERALL_VM_MAX_READWRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READ', 'IO_THROUGHPUT_OVERALL_VM_MAX_WRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READWRITE', 'STORAGE_ACCESS_HD_BALANCED_READWRITE', 'IO_THROUGHPUT_HD_BALANCED_READWRITE', 'CORE_SPEED', 'VGPU_MEM', 'CONCURRENT_QUERIES', 'LLM_CACHE', 'VGPU_COMPUTE', 'GPU', 'GPU_REQUEST', 'GPU_REQUEST_QUOTA', 'QUEUING_TIME', 'SERVICE_TIME', 'RAID_HOSTS', 'NUMBER_REPLICAS', 'UNKNOWN', 'LOGICAL_BLOCK_SIZE_ACCESS', 'POWER_CAPABILITY_ACCESS']

targetEntityType

- **type:** string

- **description:** Target entity type for an action. For example, VirtualMachine for a resize virtual machine action
- **enum:** ['ApplicationComponent', 'VirtualMachine', 'Storage', 'DiskArray', 'Database', 'DatabaseServer', 'Container', 'LogicalPool']

ActionScheduleApiDTO

Description

Model to describe an Action Schedule.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

nextOccurrence

- **type:** string
- **description:** The time of the next occurrence represented in ISO8601 local time (YYYY-MM-DDTHH:MM:SS). For example, 2019-04-04T19:15.

nextOccurrenceTimestamp

- **type:** integer
- **format:** int64
- **description:** The time of the next occurence of this schedule represented in Timestamp in milliseconds. For example, 1562100619000

timeZoneId

- **type:** string
- **description:** The timezone of the schedule

mode

- **type:** string
- **description:** The action mode affected by the schedule
- **enum:** ['DISABLED', 'RECOMMEND', 'EXTERNAL_APPROVAL', 'MANUAL', 'AUTOMATIC', 'COLLECTION']

acceptedByUserForMaintenanceWindow

- **type:** boolean
- **description:** Boolean signifying acceptance or rejection of the action.

userName

- **type:** string
- **description:** The user who accepted the action for the maintenance window.

remaingTimeActiveInMs

- **type:** integer
- **format:** int64
- **description:** The amount of time that the schedule is going to be active at the time of call in milliseconds. This field will not be populated if the schedule is not currently active.

ActionScopesApiInputDTO

Description

Model to describe the request for Actions in multiple Scopes

Required Parameters

actionInput

- **description:** List of requested actions
- **reference:** ActionApiInputDTO

Optional Parameters

scopes

- **type:** array
- **description:** List of uuid to use as scope
- – **type:** string

relatedType

- **type:** string
- **description:** Get the actions for the entity type related to the scopes

ActionScopesResourceImpactStatApiInputDTO

Description

Model to describe the request for Actions Resource Impacts in multiple Scopes

Required Parameters

actionResourceImpactStatApiInput

- **description:** Requested action resource impact stats
- **reference:** ActionResourceImpactStatApiInputDTO

scopes

- **type:** array
- **description:** List of uuid to use as scope
- – **type:** string

ActionStateChangeApiDTO

Description

Model to describe an Action State Change.

Required Parameters

state

- **type:** string
- **description:** Action state
- **enum:** ['ACCEPTED', 'REJECTED', 'PRE_IN_PROGRESS', 'POST_IN_PROGRESS', 'IN_PROGRESS', 'SUCCEEDED', 'FAILED', 'DISABLED', 'QUEUED', 'CLEARED', 'ACCOUNTING', 'READY', 'FAILING', 'BEFORE_EXEC', 'IN_PROGRESS_NON_CRITICAL']

stateChangeTime

- **type:** string
- **description:** Action state change time

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

ActionWorkflowsApiDTO

Description

Model that describes details associated with a action and workflow pair.

Required Parameters

actionId

- **type:** string
- **description:** ActionId associated with the workflow

actionState

- **type:** string
- **description:** Action state
- **enum:** ['ACCEPTED', 'REJECTED', 'PRE_IN_PROGRESS', 'POST_IN_PROGRESS', 'IN_PROGRESS', 'SUCCEEDED', 'FAILED', 'DISABLED', 'QUEUED', 'CLEARED', 'ACCOUNTING', 'READY', 'FAILING', 'BEFORE_EXEC', 'IN_PROGRESS_NON_CRITICAL']

actionTransitionTime

- **type:** string
- **description:** Action state change time

workflowId

- **type:** string
- **description:** WorkflowId associated with the action

workflowState

- **type:** string
- **description:** Workflow state
- **enum:** ['WORKFLOW_RUNNING', 'WORKFLOW_FAILED', 'WORKFLOW_SUCCEEDED']

workflowTransitionTime

- **type:** string
- **description:** Workflow state change time

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

ActiveDirectoryApiDTO

Description

Model to describe an Active Directory. It contains domain name and server name which are global to Turbonomic server.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

domainName

- **type:** string
- **description:** The domain name of current active directory

loginProviderURI

- **type:** string
- **description:** The server name of current active directory

isSecure

- **type:** boolean
- **description:** The server name of current active directory

groups

- **type:** array
- **description:** Active directory groups defined in Ops Mgr for current active directory
- – **reference:** ActiveDirectoryGroupApiDTO

ActiveDirectoryGroupApiDTO

Description

Model to describe an Active Dierctory Group. Active Directory groups are used to manage user authentication. You can specify role and scope for the group, and any member of the AD group can log into Turbonomic with those privileges.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

type

- **type:** string
- **description:** The user type for current active directory group

roleName

- **type:** string
- **description:** The user role name of current active directory group

scope

- **type:** array
- **description:** The scope of current active directory group
- – **reference:** GroupApiDTO

AddObjectApiDTO

Description

Model to describe an add change in the scenario

Optional Parameters

projectionDays

- **type:** array
- **description:** List of projection days, indicate when to apply the changes
- – **type:** integer
  – **format:** int32

target

- **description:** Target of the change, entity or group
- **reference:** BaseApiDTO

count

- **type:** integer
- **format:** int32
- **description:** How many times to apply the add change

targetEntityType

- **type:** string
- **description:** Type of entity to add from group target

AggregateCommodityFieldApiDTO

Description

Inherits From

This DTO inherits from FieldApiDTO

Required Parameters

aggregation

- **type:** string
- **description:** The type of aggregation.
- **readOnly:** True
- **enum:** ['TOTAL', 'AVERAGE']

commodityAttribute

- **type:** string
- **description:** The commodity attribute.
- **readOnly:** True
- **enum:** ['USED', 'CAPACITY', 'PEAK', 'CURRENT_UTILIZATION', 'WEIGHTED_HISTORICAL_UTILIZATION', 'PERCENTILE_HISTORICAL_UTILIZATION']

commodityType

- **type:** string
- **description:** The commodity name.
- **readOnly:** True
- **enum:** ['ACTION_PERMIT', 'ACTIVE_SESSIONS', 'APPLICATION', 'BALLOONING', 'BUFFER_COMMODITY', 'CLUSTER', 'COLLECTION_TIME', 'COOLING', 'CONNECTION', 'COUPON', 'CPU', 'CPU_ALLOCATION', 'CPU_PROVISIONED', 'CPU_REQUEST_ALLOCATION', 'CROSS_CLOUD_MOVE_SVC', 'CROSS_CLUSTER_MOVE_SVC', 'DATACENTER', 'DATASTORE', 'DB_CACHE_HIT_RATE', 'DB_MEM', 'DISK_ARRAY_ACCESS', 'DRS_SEGMENTATION', 'DSPM_ACCESS', 'EXTENT', 'FLOW', 'FLOW_ALLOCATION', 'HEAP', 'HOST_LUN_ACCESS', 'IMAGE_CPU', 'IMAGE_MEM', 'IMAGE_STORAGE', 'INSTANCE_DISK_SIZE', 'INSTANCE_DISK_TYPE', 'IO_THROUGHPUT', 'LICENSE_ACCESS', 'MEM', 'MEM_ALLOCATION', 'MEM_REQUEST_ALLOCATION', 'MEM_PROVISIONED', 'NET_THROUGHPUT', 'NETWORK', 'NUM_DISK', 'NUMBER_CONSUMERS', 'NUM_VCORE', 'POOL_CPU', 'POOL_MEM', 'POOL_STORAGE', 'PORT_CHANNEL', 'POWER', 'Q16_VCPU', 'Q1_VCPU', 'Q2_VCPU', 'Q32_VCPU', 'Q4_VCPU',

'Q64_VCPU', 'Q8_VCPU', 'RESPONSE_TIME', 'SAME_CLUSTER_MOVE_SVC', 'SEGMENTATION', 'SLA_COMMODITY', 'SOFTWARE_LICENSE_COMMODITY', 'SPACE', 'STORAGE_ACCESS', 'STORAGE_ALLOCATION', 'STORAGE_AMOUNT', 'STORAGE_CLUSTER', 'STORAGE_LATENCY', 'STORAGE_PROVISIONED', 'SWAPPING', 'THREADS', 'TRANSACTION', 'TRANSACTION_LOG', 'VCPU', 'VCPU_LIMIT_QUOTA', 'VCPU_REQUEST', 'VCPU_REQUEST_QUOTA', 'VDC', 'VMEM', 'VMEM_LIMIT_QUOTA', 'VMEM_REQUEST', 'VMEM_REQUEST_QUOTA', 'VMPM_ACCESS', 'VSTORAGE', 'ZONE', 'LICENSE_COMMODITY', 'Q3_VCPU', 'NUMBER_CONSUMERS_PM', 'Q6_VCPU', 'Q7_VCPU', 'QN_VCPU', 'RIGHT_SIZE_SVC', 'RIGHT_SIZE_DOWN', 'MOVE', 'Q5_VCPU', 'STORAGE', 'NUMBER_CONSUMERS_STORAGE', 'ACCESS', 'RIGHT_SIZE_UP', 'VAPP_ACCESS', 'HOT_STORAGE', 'HA_COMMODITY', 'NETWORK_POLICY', 'SERVICE_LEVEL_CLUSTER', 'PROCESSING_UNITS', 'TENANCY_ACCESS', 'TEMPLATE_ACCESS', 'BURST_BALANCE', 'DESIRED_COUPON', 'NETWORK_INTERFACE_COUNT', 'BICLIQUE', 'TAINT', 'LABEL', 'PEER_TO_PEER_AFFINITY', 'PEER_TO_PEER_ANTI_AFFINITY', 'ENERGY', 'CARBON', 'VIRTUAL_PROCESSORS', 'CPU_READY', 'PROCESSING_UNITS_PROVISIONED', 'HARDWARE_VERSION', 'GPU_MEM_ALLOCATION', 'GPU_MEM', 'GPU_COUNT', 'GPU_TENSOR_ENGINE', 'GPU_FP64_ENGINE', 'GPU_FP32_ENGINE', 'GPU_FP16_ENGINE', 'GPU_MEMORY_BANDWIDTH', 'STORAGE_ACCESS_OVERALL_VM_MAX_READ', 'STORAGE_ACCESS_OVERALL_VM_MAX_WRITE', 'STORAGE_ACCESS_OVERALL_VM_MAX_READWRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READ', 'IO_THROUGHPUT_OVERALL_VM_MAX_WRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READWRITE', 'STORAGE_ACCESS_HD_BALANCED_READWRITE', 'IO_THROUGHPUT_HD_BALANCED_READWRITE', 'CORE_SPEED', 'VGPU_MEM', 'CONCURRENT_QUERIES', 'LLM_CACHE', 'VGPU_COMPUTE', 'GPU', 'GPU_REQUEST', 'GPU_REQUEST_QUOTA', 'QUEUING_TIME', 'SERVICE_TIME', 'RAID_HOSTS', 'NUMBER_REPLICAS', 'UNKNOWN', 'LOGICAL_BLOCK_SIZE_ACCESS', 'POWER_CAPABILITY_ACCESS']

AggregatedHealthResponseDTO

Description

A DTO for a single aggregated health response item.

Required Parameters

healthState

- **type:** string
- **description:** The (aggregated) health state for the subcategory.
- **enum:** ['CRITICAL', 'MAJOR', 'MINOR', 'NORMAL']

numberOfItems

- **type:** integer
- **format:** int32
- **description:** Number of entities whose state matches healthState.

subcategory

- **type:** string
- **description:** Health subcategory.

Optional Parameters

recommendations

- **type:** array
- **description:** A list of recommendations specific for each problem (error) type.
- – **reference:** Recommendation

AppliedScopeApiDTOObject

Description

Model to describe the scnope of resources to which a cloud commitment can be applied

Optional Parameters

appliedScopeType

- **type:** object
- **description:** The group or entity type to which the cloud commitment is applicable

appliedScopeIds

- **type:** array
- **description:** The IDs of the group or entity to which the cloud commitment is applicable
- – **type:** string

AppliedScopeApiDTO

Description

Model to describe the scnope of resources to which a cloud commitment can be applied

Optional Parameters

appliedScopeType

- **type:** object
- **description:** The group or entity type to which the cloud commitment is applicable

appliedScopeIds

- **type:** array
- **description:** The IDs of the group or entity to which the cloud commitment is applicable
- – **type:** string

AutomatedEntityDefinitionData

Description

Inherits From

This DTO inherits from IEntityDefinitionData

Required Parameters

entityType

- **type:** string
- **description:** Connected entity type
- **enum:** ['ApplicationComponent', 'BusinessTransaction', 'Container', 'ContainerPod', 'DatabaseServer', 'Service', 'VirtualMachine']

membersCount

- **type:** integer

- **format:** int32
- **description:** Dynamic tag data Members count

namePrefix

- **type:** string
- **description:** Naming prefix for automated entity definition

Optional Parameters

generationAndConnectionAttribute

- **description:** Defined criteria of connected entities
- **reference:** FilterApiDTO

AzureVirtualMachineScaleSetDTO

Description

Model to describe the specific properties of Azure Virtual Machine Scale Set, using which a group of load balanced VMs can be created and managed

Required Parameters

imageReference

- **description:** Image reference properties of Azure Virtual Machine Scale Set
- **reference:** AzureVirtualMachineScaleSetImageReferenceDTO

provisioningState

- **type:** string
- **description:** The provisioning state is the status of a user-initiated, control-plane operation on Azure Virtual Machine Scale Set
- **enum:** ['CREATING', 'UPDATING', 'SUCCEEDED', 'FAILED', 'DELETING', 'MIGRATING']

scalingMode

- **type:** string
- **description:** Scaling mode enables auto scaling to add and remove instances either manually or by custom autoscaling
- **enum:** ['MANUAL', 'AUTOSCALE_METRIC_BASED', 'AUTOSCALE_SPECIFIC_INSTANCE_COUNT']

upgradePolicy

- **description:** Upgrade policy of Azure Virtual Machine Scale Set
- **reference:** AzureVirtualMachineScaleSetUpgradePolicyDTO

Optional Parameters

autoScaleProfileCount

- **type:** integer
- **format:** int32
- **example:** [1]
- **description:** Number of autoscaled profiles in Azure Virtual Machine Scale Set

scaleInPolicy

- **type:** string

- **description:** The order in which the Azure VMs are selected for deletion during Scale-in operation
- **enum:** ['DEFAULT', 'NEWEST_VM', 'OLDEST_VM']

overProvisioningEnabled

- **type:** boolean

singlePlacementGroupEnabled

- **type:** boolean

AzureVirtualMachineScaleSetImageReferenceDTO

Description

Model to represent the property for source OS image of Azure Virtual Machine Scale Set

Required Parameters

publisher

- **type:** string
- **example:** [canonical]
- **description:** Publisher of Azure Virtual Machine Scale Set

sku

- **type:** string
- **example:** [20_04-lts-gen2]
- **description:** SKU of Azure Virtual Machine Scale Set

version

- **type:** string
- **example:** [latest]
- **description:** Version of Azure Virtual Machine Scale Set

Optional Parameters

offer

- **type:** string
- **example:** [0001-com-ubuntu-server-local]
- **description:** Offer of Azure Virtual Machine Scale Set

AzureVirtualMachineScaleSetUpgradePolicyDTO

Description

Model to represent the upgrade policy of Azure Virtual Machine Scale Set determines how virtual machines can be brought up-to-date with the latest scale set model

Required Parameters

upgradePolicyMode

- **type:** string
- **description:** Upgrade policy mode of Azure Virtual Machine Scale Set
- **enum:** ['AUTOMATIC', 'MANUAL', 'ROLLING']

Optional Parameters

autoOSUpgradeEnabled

- **type:** boolean

BaseApiDTO

Description

Basic information of an entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

BaseCommodityApiDTO

Description

Describes a commodity

Required Parameters

type

- **type:** string
- **description:** The type of the commodity
- **enum:** ['ACTION_PERMIT', 'ACTIVE_SESSIONS', 'APPLICATION', 'BALLOONING', 'BUFFER_COMMODITY', 'CLUSTER', 'COLLECTION_TIME', 'COOLING', 'CONNECTION', 'COUPON', 'CPU', 'CPU_ALLOCATION', 'CPU_PROVISIONED', 'CPU_REQUEST_ALLOCATION', 'CROSS_CLOUD_MOVE_SVC', 'CROSS_CLUSTER_MOVE_SVC', 'DATACENTER', 'DATASTORE', 'DB_CACHE_HIT_RATE', 'DB_MEM', 'DISK_ARRAY_ACCESS', 'DRS_SEGMENTATION', 'DSPM_ACCESS', 'EXTENT', 'FLOW', 'FLOW_ALLOCATION', 'HEAP', 'HOST_LUN_ACCESS', 'IMAGE_CPU', 'IMAGE_MEM', 'IMAGE_STORAGE', 'INSTANCE_DISK_SIZE', 'INSTANCE_DISK_TYPE', 'IO_THROUGHPUT', 'LICENSE_ACCESS', 'MEM', 'MEM_ALLOCATION', 'MEM_REQUEST_ALLOCATION', 'MEM_PROVISIONED', 'NET_THROUGHPUT', 'NETWORK', 'NUM_DISK', 'NUMBER_CONSUMERS', 'NUM_VCORE', 'POOL_CPU', 'POOL_MEM', 'POOL_STORAGE', 'PORT_CHANNEL', 'POWER', 'Q16_VCPU', 'Q1_VCPU', 'Q2_VCPU', 'Q32_VCPU', 'Q4_VCPU', 'Q64_VCPU', 'Q8_VCPU', 'RESPONSE_TIME', 'SAME_CLUSTER_MOVE_SVC', 'SEGMENTATION', 'SLA_COMMODITY', 'SOFTWARE_LICENSE_COMMODITY', 'SPACE', 'STORAGE_ACCESS', 'STORAGE_ALLOCATION', 'STORAGE_AMOUNT', 'STORAGE_CLUSTER', 'STORAGE_LATENCY', 'STORAGE_PROVISIONED', 'SWAPPING', 'THREADS', 'TRANSACTION', 'TRANSACTION_LOG', 'VCPU', 'VCPU_LIMIT_QUOTA', 'VCPU_REQUEST', 'VCPU_REQUEST_QUOTA', 'VDC', 'VMEM', 'VMEM_LIMIT_QUOTA', 'VMEM_REQUEST', 'VMEM_REQUEST_QUOTA', 'VMPM_ACCESS', 'VSTORAGE', 'ZONE', 'LICENSE_COMMODITY', 'Q3_VCPU', 'NUMBER_CONSUMERS_PM', 'Q6_VCPU', 'Q7_VCPU', 'QN_VCPU', 'RIGHT_SIZE_SVC', 'RIGHT_SIZE_DOWN', 'MOVE', 'Q5_VCPU', 'STORAGE', 'NUMBER_CONSUMERS_STORAGE', 'ACCESS', 'RIGHT_SIZE_UP', 'VAPP_ACCESS', 'HOT_STORAGE',

'HA_COMMODITY', 'NETWORK_POLICY', 'SERVICE_LEVEL_CLUSTER', 'PROCESSING_UNITS', 'TENANCY_ACCESS', 'TEMPLATE_ACCESS', 'BURST_BALANCE', 'DESIRED_COUPON', 'NETWORK_INTERFACE_COUNT', 'BICLIQUE', 'TAINT', 'LABEL', 'PEER_TO_PEER_AFFINITY', 'PEER_TO_PEER_ANTI_AFFINITY', 'ENERGY', 'CARBON', 'VIRTUAL_PROCESSORS', 'CPU_READY', 'PROCESSING_UNITS_PROVISIONED', 'HARDWARE_VERSION', 'GPU_MEM_ALLOCATION', 'GPU_MEM', 'GPU_COUNT', 'GPU_TENSOR_ENGINE', 'GPU_FP64_ENGINE', 'GPU_FP32_ENGINE', 'GPU_FP16_ENGINE', 'GPU_MEMORY_BANDWIDTH', 'STORAGE_ACCESS_OVERALL_VM_MAX_READ', 'STORAGE_ACCESS_OVERALL_VM_MAX_WRITE', 'STORAGE_ACCESS_OVERALL_VM_MAX_READWRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READ', 'IO_THROUGHPUT_OVERALL_VM_MAX_WRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READWRITE', 'STORAGE_ACCESS_HD_BALANCED_READWRITE', 'IO_THROUGHPUT_HD_BALANCED_READWRITE', 'CORE_SPEED', 'VGPU_MEM', 'CONCURRENT_QUERIES', 'LLM_CACHE', 'VGPU_COMPUTE', 'GPU', 'GPU_REQUEST', 'GPU_REQUEST_QUOTA', 'QUEUING_TIME', 'SERVICE_TIME', 'RAID_HOSTS', 'NUMBER_REPLICAS', 'UNKNOWN', 'LOGICAL_BLOCK_SIZE_ACCESS', 'POWER_CAPABILITY_ACCESS']

Optional Parameters

key

- **type:** string
- **description:** The specific commodity

units

- **type:** string
- **description:** The units in which the commodity is measured, if any

BaseGroupApiDTO

Description

Base model to describe a Group.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

membersCount

- **type:** integer
- **format:** int32
- **description:** Number of members of the Group

BasicActionApiDTO

Description

Model to tersely describe an action, providing only a set of basic fields

Optional Parameters

actionType

- **type:** string
- **description:** Type
- **enum:** ['START', 'MOVE', 'SCALE', 'ALLOCATE', 'SUSPEND', 'PROVISION', 'RECONFIGURE', 'RESIZE', 'DELETE', 'RIGHT_SIZE', 'BUY_RI']

details

- **type:** string
- **description:** A user-readable string describing the action

target

- **description:** Target entity for an action. For example, the VM in a Resize Action, or the host for a VM move
- **reference:** DiscoveredEntityApiDTO

currentEntity

- **description:** Current entity, such as the current host that a VM resides on for a VM move action
- **reference:** DiscoveredEntityApiDTO

newEntity

- **description:** Destination entity, such as the host that a VM will move to for a VM move action
- **reference:** DiscoveredEntityApiDTO

actionID

- **type:** integer
- **format:** int64
- **description:** Id of the action

BillingFamilyApiDTO

Description

Inherits From

This DTO inherits from GroupApiDTO

Optional Parameters

masterAccountUuid

- **type:** string
- **description:** The UUID of the master account.

businessUnitApiDTOList

- **type:** array

- **description:** List of business units, for all accounts in this billing family.
- – **reference:** BusinessUnitApiDTO

billingCostDataExists

- **type:** boolean
- **description:** Indicates if the billing family has associated cost data.

BooleanConditionApiDTO

Description

Inherits From

This DTO inherits from ConditionApiDTO

Required Parameters

value

- **type:** boolean
- **description:** The value of the field must be equal to this boolean value
- **readOnly:** True

BooleanFieldValueApiDTO

Description

Inherits From

This DTO inherits from FieldValueApiDTO

Required Parameters

value

- **type:** boolean
- **description:** The value.
- **readOnly:** True

BulkActionRequestApiDTO

Description

BulkActionRequest

Optional Parameters

state

- **type:** string
- **description:** state
- **enum:** ['RUNNING', 'STOPPED', 'SUSPENDED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'STARTING', 'STOPPING']

error

- **type:** string
- **description:** error

entityUUID

- **type:** string

groupUUID

- **type:** string

BulkActionRequestInputDTO

Description

BulkActionInput

Optional Parameters

action

- **type:** string
- **description:** action
- **enum:** ['START', 'STOP']

entityUuids

- **type:** array
- **description:** entityUuids
- – **type:** string

groupUuids

- **type:** array
- **description:** groupUuid
- – **type:** string

BusinessUnitApiDTO

Description

Model to describe the property of a business unit.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['CLOUD', 'ONPREM', 'HYBRID']

discount

- **type:** number
- **format:** float
- **description:** Entity discount percentage

priceAdjustment

- **description:** Entity price adjustment percentage
- **reference:** PriceAdjustmentDTO

accountId

- **type:** string
- **description:** Account ID

master

- **type:** boolean
- **description:** Flag which shows whether this account is master or not

associatedTargetId

- **type:** integer
- **format:** int64
- **description:** Target associated with this account, if workloads related to this account are being discovered. If no workloads are discovered for this account, the associatedTargetId will be empty.

budget

- **description:** Budget of the business unit, user only need to set value and units(monthly)
- **reference:** StatApiDTO

businessUnitType

- **type:** string
- **description:** Type of the business unit instance
- **enum:** ['BUDGET', 'DISCOUNT', 'PRICE_ADJUSTMENT', 'DISCOVERED']

cloudType

- **type:** string
- **description:** Type of the cloud target which is related to current business account.
- **enum:** ['AWS', 'AZURE', 'GCP', 'UNKNOWN']

accountType

- **type:** string
- **description:** The category of the current cloud account.
- **enum:** ['STANDARD', 'GOVERNMENT_US']

riSupported

- **type:** boolean
- **description:** Flag which shows whether this account is RI supported

childrenBusinessUnits

- **type:** array
- **description:** UUIDs of the discovered children business units that will share the budget or discounts configuration of this business unit.
- – **type:** string

targets

- **type:** array
- **description:** Targets associated with the business unit
- – **reference:** TargetApiDTO

costPrice

- **type:** number
- **format:** float
- **description:** Cost price per Hour

severity

- **type:** string
- **description:** Calculated using the highest severity of the member entities
- **enum:** ['UNKNOWN', 'NORMAL', 'MINOR', 'MAJOR', 'CRITICAL']

membersCount

- **type:** integer
- **format:** int32
- **description:** Number of members of the business unit

memberType

- **type:** string
- **description:** The type of service entities comprising the business unit

resourceGroupsCount

- **type:** integer
- **format:** int32
- **description:** The number of Resource Groups belonging to the business unit

displayable

- **type:** boolean
- **description:** Whether business unit is displayable or not

pricingIdentifiers

- **type:** object
- **description:** Pricing idenifiers for the business unit
- **additionalProperties:** {'type': 'string'}

BusinessUnitApiInputDTO

Description

Model to describe the parameters for creating a business unit.

Required Parameters

name

- **type:** string
- **description:** Name of the business unit

Optional Parameters

budget

- **description:** Budget of the business unit, user only need to set value
- **reference:** StatApiDTO

childrenBusinessUnits

- **type:** array
- **description:** UUIDs of the discovered children business units, that will share the budget or discounts configuration of this business unit.
- – **type:** string

discount

- **type:** number
- **format:** float
- **description:** Business unit enterprise-level discount percentage. It will come to effect if cloud service or template-level discounts are not configured.

priceAdjustment

- **description:** Business unit enterprise-level price adjustment percentage. It will come to effect if cloud service or template-level discounts are not configured.
- **reference:** PriceAdjustmentDTO

targets

- **type:** array
- – **type:** string

nameWithBusinessAccountType

- **type:** string

BusinessUnitPriceAdjustmentApiDTO

Description

Price adjustment values for cloud services and templates related to a business unit. The price adjustment configured on an user-defined business unit apply tocloud services and templates offered by the children business units.

Optional Parameters

servicePriceAdjustments

- **type:** array
- **description:** Price adjustment information for the related cloud services and templates.
- – **reference:** CloudServicePriceAdjustmentApiDTO

serviceDiscounts

- **type:** array

- **description:** Discount information for the related cloud services and templates
- – **reference:** CloudServicePriceAdjustmentApiDTO

BusinessUserEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

sessions

- **type:** array
- **description:** Information about sessions to entities (VM or other) established by this BusinessUser.
- – **reference:** BusinessUserSessionApiDTO

BusinessUserSessionApiDTO

Description

Session information for a connection established by BusinessUser to an entity (VM or other)

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

businessUserUuid

- **type:** string
- **description:** UUID of the connected BusinessUser

connectedEntityUuid

- **type:** string
- **description:** UUID of the entity that user has a connection to

duration

- **type:** integer
- **format:** int64
- **description:** Time elapsed since the session start in seconds- sessions are always timed

ChangePasswordApiDTO

Description

Optional Parameters

oldPassword

- **type:** string

newPassword

- **type:** string

ClientConfigurationApiDTO

Description

Client configuration information

Optional Parameters

clientFirewallAllowlist

- **type:** array
- **description:** Endpoints that client sites need to be able to access
- – **reference:** ClientConfigurationEndpoint

ClientConfigurationEndpoint

Description

Optional Parameters

name

- **type:** string
- **description:** The endpoint name

type

- **type:** string
- **description:** The endpoint type (LoadBalancer or Route)

host

- **type:** string
- **description:** The host string of the endpoint. May be null

ip

- **type:** string
- **description:** The IP address of the endpoint

port

- **type:** integer
- **format:** int32
- **description:** The port of the endpoint

ClientInputDTO

Description

Required Parameters

name

- **type:** string
- **description:** Name of client

supportedServices

- **type:** array
- **description:** List of supported services
- – **type:** string

ClientNetworkRawTokenApiDTO

Description

Inherits From

This DTO inherits from ClientNetworkTokenApiOutputDTO

Optional Parameters

apiVersion

- **type:** string

kind

- **type:** string

metadata

- **reference:** ClientNetworkRawTokenMetadata

data

- **type:** object
- **additionalProperties:** {'type': 'string'}

ClientNetworkRawTokenMetadata

Description

Optional Parameters

name

- **type:** string

annotations

- **type:** object
- **additionalProperties:** {'type': 'string'}

labels

- **type:** object
- **additionalProperties:** {'type': 'string'}

ClientNetworkSitesMetadataApiDTO

Description

Optional Parameters

id

- **type:** string
- **description:** id

name

- **type:** string
- **description:** name

ClientNetworkTokenApiDTO

Description

Inherits From

This DTO inherits from ClientNetworkTokenApiOutputDTO

Optional Parameters

id

- **type:** string
- **description:** id

tokenData

- **type:** string
- **description:** Token data

ClientNetworkTokenApiOutputDTO

Description

One of {ClientNetworkTokenApiDTO, ClientNetworkRawTokenApiDTO}

This is an unspecified object.

ClientNetworkTokenInputDTO

Description

Optional Parameters

claimLimit

- **type:** integer
- **format:** int32
- **example:** 1
- **description:** Claim Limit: Number of times the token can be reused

expiration

- **type:** integer
- **format:** int32
- **example:** 15
- **description:** Expiration time (in minutes) to claim the token

ClientNetworkTokensMetadataApiDTO

Description

Optional Parameters

id

- **type:** string
- **description:** id

created

- **type:** string
- **description:** Created

claimsMade

- **type:** integer
- **format:** int32
- **description:** Claims Made

claimsRemaining

- **type:** integer
- **format:** int32
- **description:** Claims Remaining

claimExpiration

- **type:** string
- **description:** Claim Expiration

ClientServiceApiDTO

Description

Optional Parameters

id

- **type:** string

- **description:** id

supportedServices

- **type:** array
- **description:** Supported services
- – **type:** string

secret

- **type:** string
- **description:** Secret

name

- **type:** string
- **description:** Name of client

CloudAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

zone

- **description:** The zone of the cloud entity
- **reference:** BaseApiDTO

region

- **description:** The region of the cloud entity
- **reference:** BaseApiDTO

businessAccount

- **description:** The BusinessAccount of the cloud entity
- **reference:** BaseApiDTO

resourceGroup

- **description:** The ResourceGroup of the cloud entity
- **reference:** BaseApiDTO

template

- **description:** The template used to generate the entity
- **reference:** BaseApiDTO

billingType

- **type:** string
- **description:** The Cloud Billing type

tier

- **type:** string
- **description:** The Tier of the storage

riCoveragePercentage

- **type:** number
- **format:** float
- **description:** The RI coverage percentage

riCoverage

- **description:** Used value stores VM coupons in use, capacity stores coupon capacity
- **reference:** StatApiDTO

onDemandCost

- **type:** number
- **format:** float
- **description:** On-demand cost paid for workload per Hour (includes Compute and License on-demand costs)

enhancedNetworking

- **type:** string
- **description:** The type of Enhanced Networking supported. If the entity is an AWS VM, this field will be populated by a value describing what kind of enhanced networking, if any, the VM is currently using. If the entity is not an AWS VM, this field is not relevant and will not be populated or displayed.
- **enum:** ['NONE', 'ENA']

enaActive

- **type:** string
- **description:** Whether ENA is active. If the entity is an AWS VM, this field will be populated by a value describing whether ENA is currently in use or not. If the entity is not an AWS VM, this field is not relevant and will not be populated or displayed.
- **enum:** ['Active', 'Inactive']

virtualizationType

- **type:** string
- **description:** The virtualization type. If the entity is an AWS VM, this field will be populated by a value describing what kind of virtualization type, if any, the VM is currently using. If the entity is not an AWS VM, this field is not relevant and will not be populated or displayed.
- **enum:** ['PVM', 'HVM']

architecture

- **type:** string
- **description:** If the entity is an AWS VM, this field will be populated by a value describing what is the architecture of the VM.If the entity is not an AWS VM, this field is not relevant and will not be populated or displayed.
- **enum:** ['32-bit', '64-bit']

nvme

- **type:** string
- **description:** NVMe driver support. If the entity is an AWS VM, this field will be true if NVMe driver installed. If the entity is not an AWS VM, this field is not relevant and will not be populated or displayed.
- **enum:** ['Installed', 'Not installed']

entityUptime

- **description:** The entity uptime

- **reference:** EntityUptimeApiDTO

tenancy

- **type:** string
- **description:** The tenancy type for VMs
- **enum:** ['DEFAULT', 'DEDICATED', 'HOST']

pathSegments

- **type:** array
- **description:** Optional path segments that can represent hierarchical context for this entity such as the names of parents containing this entity.
- – **type:** string

cloudServiceName

- **type:** string
- **description:** Optional parameter that represents the name of the cloud service
- **enum:** ['AMAZON_RDS', 'AMAZON_EC2', 'AMAZON_EC2_AUTO_SCALING', 'AMAZON_REDSHIFT', 'AZURE_APP_SERVICE', 'AZURE_COSMOS_DB', 'AZURE_DISK_STORAGE', 'AZURE_SQL_DATABASE', 'AZURE_SQL_MANAGED_INSTANCE', 'AZURE_SYNAPSE_SQL_POOL', 'AZURE_VIRTUAL_MACHINES', 'AZURE_VIRTUAL_MACHINE_SCALE_SETS', 'GCP_DISK_STORAGE', 'GCP_VIRTUAL_MACHINES', 'GCP_MANAGED_INSTANCE_GROUP']

resourceId

- **type:** string
- **description:** Optional Cloud entity resource id.

CloudCommitmentAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

startTimeInMilliseconds

- **type:** integer
- **format:** int64
- **description:** The start time of the Cloud Commitment in UTC.

expirationTimeInMilliseconds

- **type:** integer
- **format:** int64
- **description:** The expiration time of the Cloud Commitment in UTC.

termInMilliseconds

- **type:** integer
- **format:** int64
- **description:** The term of the Cloud Commitment.

cloudCommitmentScopeType

- **type:** string
- **description:** The scope of the Cloud Commitment.
- **enum:** ['FamilyScoped', 'FamilyAndOSScoped', 'CloudServiceScoped']

cloudCommitmentScopeDTO

- **description:** The info about the scope of the cloud commitment.
- **reference:** CloudCommitmentScopeDTO

cloudCommitmentScopeInfo

- **description:** The commitment scope details.
- **reference:** CloudCommitmentScopeInfoApiDTO

cloudCommitmentPricing

- **description:** The commitment pricing information
- **reference:** CloudCommitmentPricingApiDTO

cloudCommitmentCapacityApiDTO

- **description:** The capacity specific info of the cloud commitment.
- **reference:** CloudCommitmentCapacityApiDTO

payment

- **type:** string
- **description:** Payment option of the cloud commitment.
- **enum:** ['ALL_UPFRONT', 'PARTIAL_UPFRONT', 'NO_UPFRONT']

providerSpecificType

- **type:** string
- **description:** The provider specific type.
- **enum:** ['SAVINGS_PLAN', 'COMMITTED_USE', 'RESERVED_INSTANCE']

commitmentStatus

- **type:** string
- **description:** The cloud commitment status.
- **enum:** ['Unknown', 'Pending', 'Active', 'Expired', 'Canceled']

instanceCount

- **type:** integer
- **format:** int32
- **description:** Number of commitment instances with the same type, template, platform, etc.

term

- **description:** Term, unit is Year
- **reference:** StatApiDTO

CloudCommitmentCapacityApiDTO

Description

Model to describe cloud commitment capacity

Optional Parameters

instanceCapacity

- **type:** integer
- **format:** int64
- **description:** The instance capacity of the cloud commitment.

spendCapacity

- **type:** number
- **format:** double
- **description:** The spend capacity of the cloud commitment.

commoditiesBought

- **type:** object
- **description:** The capacity for each type commodity bought by the commitment.
- **additionalProperties:** {'type': 'number', 'format': 'double'}

CloudCommitmentPricingApiDTO

Description

Model to describe cloud commitment pricing

Optional Parameters

reservedInstanceType

- **type:** string
- **description:** The commitment offering type, e.g. standard or convertible
- **enum:** ['STANDARD', 'CONVERTIBLE']

paymentOption

- **type:** string
- **description:** The commitment payment option
- **enum:** ['ALL_UPFRONT', 'PARTIAL_UPFRONT', 'NO_UPFRONT']

fixedCost

- **type:** number
- **format:** double
- **description:** The fixed cost of the commitment

recurringCost

- **type:** number
- **format:** double
- **description:** The recurring cost of the commitment

effectiveCost

- **description:** Effective hourly cost of the commitment
- **reference:** StatApiDTO

CloudCommitmentScopeDTO

Description

Model to describe cloud committment scope.

Optional Parameters

cloudFamilyReferenceApiDTO

- **description:** The family scoped type specific info of the cloud commitment. If null, indicates service scoped cloud commitment
- **reference:** CloudFamilyReferenceApiDTO

CloudCommitmentScopeInfoApiDTO

Description

A container of scope related objects that may apply to a cloud commitment

Optional Parameters

topologyScopes

- **type:** array
- **description:** A list of topology related scope objects that may apply to a cloud commitment
- – **reference:** TopologyScopeApiDTO

appliedScope

- **description:** A scope representing the eligible resources to which the discount can be applied
- **reference:** AppliedScopeApiDTOObject

CloudFamilyReferenceApiDTO

Description

Model describe a reference to a cloud family

Optional Parameters

family

- **type:** string
- **description:** The family of the Cloud Commitment.

entityType

- **type:** string
- **description:** The entity type the Cloud Commitment applies to.
- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine',

'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

CloudProvisionActionDetailsApiDTO

Description

Inherits From

This DTO inherits from ActionDetailsApiDTO

Optional Parameters

onDemandRate

- **type:** number
- **format:** float

onDemandCost

- **type:** number
- **format:** float

subType

- **type:** string

CloudResizeActionDetailsApiDTO

Description

Inherits From

This DTO inherits from ActionDetailsApiDTO

Optional Parameters

onDemandRateBefore

- **type:** number
- **format:** float

onDemandRateAfter

- **type:** number
- **format:** float

onDemandCostBefore

- **type:** number
- **format:** float

onDemandCostAfter

- **type:** number
- **format:** float

riCoverageBefore

- **description:** RI coupons used and capacity of the target entity before action
- **reference:** StatApiDTO

riCoverageAfter

- **description:** RI coupons used and capacity of the target entity after action
- **reference:** StatApiDTO

Reserved instance rate before

- **type:** number
- **format:** float

Reserved instance rate after

- **type:** number
- **format:** float

commitmentRateBefore

- **type:** number
- **format:** float

commitmentRateAfter

- **type:** number
- **format:** float

effectiveCostBefore

- **type:** number
- **format:** float

effectiveCostAfter

- **type:** number
- **format:** float

commitmentCoverageBefore

- **reference:** StatApiDTO

commitmentCoverageAfter

- **reference:** StatApiDTO

totalDiscountCoverageBefore

- **reference:** StatApiDTO

totalDiscountCoverageAfter

- **reference:** StatApiDTO

iacInfo

- **description:** Iac configuration for cloud VM.
- **reference:** VirtualMachineIacApiDTO

entityUptime

- **description:** The entity uptime which was taken into account when generating the action
- **reference:** EntityUptimeApiDTO

subType

- **type:** string

CloudServicePriceAdjustmentApiDTO

Description

Price adjustments for cloud service and related templates specified for a certain pricing model.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['CLOUD', 'ONPREM', 'HYBRID']

discount

- **type:** number
- **format:** float
- **description:** Entity discount percentage

priceAdjustment

- **description:** Entity price adjustment percentage
- **reference:** PriceAdjustmentDTO

pricingModel

- **type:** string
- **description:** Pricing model, 'On demand' for time-based, 'Reserved' for coupon-based, 'Spot' for customer bidding
- **enum:** ['ON_DEMAND', 'RESERVED', 'SPOT']

templateDiscounts

- **type:** array
- **description:** Discount values for the templates offered by the cloud service under certain pricing model.
- – **reference:** TemplatePriceAdjustmentDTO

templatePriceAdjustments

- **type:** array
- **description:** Price adjustment values for the templates offered by the cloud service under certain pricing model.
- – **reference:** TemplatePriceAdjustmentDTO

CloudServicePricingModelApiDTO

Description

Cloud service + pricing model pair that is available for a business unit.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['CLOUD', 'ONPREM', 'HYBRID']

discount

- **type:** number
- **format:** float
- **description:** Entity discount percentage

priceAdjustment

- **description:** Entity price adjustment percentage
- **reference:** PriceAdjustmentDTO

pricingModel

- **type:** string
- **description:** Pricing model, 'On demand' for time-based, 'Reserved' for coupon-based, 'Spot' for customer bidding
- **enum:** ['ON_DEMAND', 'RESERVED', 'SPOT']

CloudSuspendActionDetailsApiDTO

Description

Inherits From

This DTO inherits from ActionDetailsApiDTO

Optional Parameters

onDemandRate

- **type:** number
- **format:** float

onDemandCost

- **type:** number
- **format:** float

subType

- **type:** string

CommodityDetailsApiDTO

Description

Information on relation type and its related commodities

Required Parameters

commodityType

- **type:** string
- **description:** Relation Type
- **enum:** ['ACTION_PERMIT', 'ACTIVE_SESSIONS', 'APPLICATION', 'BALLOONING', 'BUFFER_COMMODITY', 'CLUSTER', 'COLLECTION_TIME', 'COOLING', 'CONNECTION', 'COUPON', 'CPU', 'CPU_ALLOCATION', 'CPU_PROVISIONED', 'CPU_REQUEST_ALLOCATION', 'CROSS_CLOUD_MOVE_SVC', 'CROSS_CLUSTER_MOVE_SVC', 'DATACENTER', 'DATASTORE', 'DB_CACHE_HIT_RATE', 'DB_MEM', 'DISK_ARRAY_ACCESS', 'DRS_SEGMENTATION', 'DSPM_ACCESS', 'EXTENT', 'FLOW', 'FLOW_ALLOCATION', 'HEAP', 'HOST_LUN_ACCESS', 'IMAGE_CPU', 'IMAGE_MEM', 'IMAGE_STORAGE', 'INSTANCE_DISK_SIZE', 'INSTANCE_DISK_TYPE', 'IO_THROUGHPUT', 'LICENSE_ACCESS', 'MEM', 'MEM_ALLOCATION', 'MEM_REQUEST_ALLOCATION', 'MEM_PROVISIONED', 'NET_THROUGHPUT', 'NETWORK', 'NUM_DISK', 'NUMBER_CONSUMERS', 'NUM_VCORE', 'POOL_CPU', 'POOL_MEM', 'POOL_STORAGE', 'PORT_CHANNEL', 'POWER', 'Q16_VCPU', 'Q1_VCPU', 'Q2_VCPU', 'Q32_VCPU', 'Q4_VCPU', 'Q64_VCPU', 'Q8_VCPU', 'RESPONSE_TIME', 'SAME_CLUSTER_MOVE_SVC', 'SEGMENTATION', 'SLA_COMMODITY', 'SOFTWARE_LICENSE_COMMODITY', 'SPACE', 'STORAGE_ACCESS', 'STORAGE_ALLOCATION', 'STORAGE_AMOUNT', 'STORAGE_CLUSTER', 'STORAGE_LATENCY', 'STORAGE_PROVISIONED', 'SWAPPING', 'THREADS', 'TRANSACTION', 'TRANSACTION_LOG', 'VCPU', 'VCPU_LIMIT_QUOTA', 'VCPU_REQUEST', 'VCPU_REQUEST_QUOTA', 'VDC', 'VMEM', 'VMEM_LIMIT_QUOTA', 'VMEM_REQUEST', 'VMEM_REQUEST_QUOTA', 'VMPM_ACCESS', 'VSTORAGE', 'ZONE', 'LICENSE_COMMODITY', 'Q3_VCPU', 'NUMBER_CONSUMERS_PM', 'Q6_VCPU', 'Q7_VCPU', 'QN_VCPU', 'RIGHT_SIZE_SVC', 'RIGHT_SIZE_DOWN', 'MOVE', 'Q5_VCPU', 'STORAGE', 'NUMBER_CONSUMERS_STORAGE', 'ACCESS', 'RIGHT_SIZE_UP', 'VAPP_ACCESS', 'HOT_STORAGE', 'HA_COMMODITY', 'NETWORK_POLICY', 'SERVICE_LEVEL_CLUSTER', 'PROCESSING_UNITS', 'TENANCY_ACCESS', 'TEMPLATE_ACCESS', 'BURST_BALANCE', 'DESIRED_COUPON', 'NETWORK_INTERFACE_COUNT', 'BICLIQUE', 'TAINT', 'LABEL', 'PEER_TO_PEER_AFFINITY', 'PEER_TO_PEER_ANTI_AFFINITY', 'ENERGY', 'CARBON', 'VIRTUAL_PROCESSORS', 'CPU_READY', 'PROCESSING_UNITS_PROVISIONED', 'HARDWARE_VERSION', 'GPU_MEM_ALLOCATION', 'GPU_MEM', 'GPU_COUNT', 'GPU_TENSOR_ENGINE', 'GPU_FP64_ENGINE', 'GPU_FP32_ENGINE', 'GPU_FP16_ENGINE', 'GPU_MEMORY_BANDWIDTH', 'STORAGE_ACCESS_OVERALL_VM_MAX_READ', 'STORAGE_ACCESS_OVERALL_VM_MAX_WRITE', 'STORAGE_ACCESS_OVERALL_VM_MAX_READWRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READ', 'IO_THROUGHPUT_OVERALL_VM_MAX_WRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READWRITE', 'STORAGE_ACCESS_HD_BALANCED_READWRITE', 'IO_THROUGHPUT_HD_BALANCED_READWRITE', 'CORE_SPEED', 'VGPU_MEM', 'CONCURRENT_QUERIES', 'LLM_CACHE', 'VGPU_COMPUTE', 'GPU', 'GPU_REQUEST', 'GPU_REQUEST_QUOTA', 'QUEUING_TIME', 'SERVICE_TIME', 'RAID_HOSTS', 'NUMBER_REPLICAS', 'UNKNOWN', 'LOGICAL_BLOCK_SIZE_ACCESS', 'POWER_CAPABILITY_ACCESS']

filters

- **type:** array
- **description:** Describes the grouping this commodity is included in, for examplewhether the commodity has real time stats available.
- – **reference:** StatFilterApiDTO

CommodityFieldApiDTO

Description

Inherits From

This DTO inherits from FieldApiDTO

Required Parameters

commodityAttribute

- **type:** string
- **description:** The commodity attribute.
- **readOnly:** True
- **enum:** ['USED', 'CAPACITY', 'PEAK', 'CURRENT_UTILIZATION', 'WEIGHTED_HISTORICAL_UTILIZATION', 'PERCENTILE_HISTORICAL_UTILIZATION']

commodityType

- **type:** string
- **description:** The commodity name.
- **readOnly:** True
- **enum:** ['ACTION_PERMIT', 'ACTIVE_SESSIONS', 'APPLICATION', 'BALLOONING', 'BUFFER_COMMODITY', 'CLUSTER', 'COLLECTION_TIME', 'COOLING', 'CONNECTION', 'COUPON', 'CPU', 'CPU_ALLOCATION', 'CPU_PROVISIONED', 'CPU_REQUEST_ALLOCATION', 'CROSS_CLOUD_MOVE_SVC', 'CROSS_CLUSTER_MOVE_SVC', 'DATACENTER', 'DATASTORE', 'DB_CACHE_HIT_RATE', 'DB_MEM', 'DISK_ARRAY_ACCESS', 'DRS_SEGMENTATION', 'DSPM_ACCESS', 'EXTENT', 'FLOW', 'FLOW_ALLOCATION', 'HEAP', 'HOST_LUN_ACCESS', 'IMAGE_CPU', 'IMAGE_MEM', 'IMAGE_STORAGE', 'INSTANCE_DISK_SIZE', 'INSTANCE_DISK_TYPE', 'IO_THROUGHPUT', 'LICENSE_ACCESS', 'MEM', 'MEM_ALLOCATION', 'MEM_REQUEST_ALLOCATION', 'MEM_PROVISIONED', 'NET_THROUGHPUT', 'NETWORK', 'NUM_DISK', 'NUMBER_CONSUMERS', 'NUM_VCORE', 'POOL_CPU', 'POOL_MEM', 'POOL_STORAGE', 'PORT_CHANNEL', 'POWER', 'Q16_VCPU', 'Q1_VCPU', 'Q2_VCPU', 'Q32_VCPU', 'Q4_VCPU', 'Q64_VCPU', 'Q8_VCPU', 'RESPONSE_TIME', 'SAME_CLUSTER_MOVE_SVC', 'SEGMENTATION', 'SLA_COMMODITY', 'SOFTWARE_LICENSE_COMMODITY', 'SPACE', 'STORAGE_ACCESS', 'STORAGE_ALLOCATION', 'STORAGE_AMOUNT', 'STORAGE_CLUSTER', 'STORAGE_LATENCY', 'STORAGE_PROVISIONED', 'SWAPPING', 'THREADS', 'TRANSACTION', 'TRANSACTION_LOG', 'VCPU', 'VCPU_LIMIT_QUOTA', 'VCPU_REQUEST', 'VCPU_REQUEST_QUOTA', 'VDC', 'VMEM', 'VMEM_LIMIT_QUOTA', 'VMEM_REQUEST', 'VMEM_REQUEST_QUOTA', 'VMPM_ACCESS', 'VSTORAGE', 'ZONE', 'LICENSE_COMMODITY', 'Q3_VCPU', 'NUMBER_CONSUMERS_PM', 'Q6_VCPU', 'Q7_VCPU', 'QN_VCPU', 'RIGHT_SIZE_SVC', 'RIGHT_SIZE_DOWN', 'MOVE', 'Q5_VCPU', 'STORAGE', 'NUMBER_CONSUMERS_STORAGE', 'ACCESS', 'RIGHT_SIZE_UP', 'VAPP_ACCESS', 'HOT_STORAGE', 'HA_COMMODITY', 'NETWORK_POLICY', 'SERVICE_LEVEL_CLUSTER', 'PROCESSING_UNITS', 'TENANCY_ACCESS', 'TEMPLATE_ACCESS', 'BURST_BALANCE', 'DESIRED_COUPON', 'NETWORK_INTERFACE_COUNT', 'BICLIQUE', 'TAINT', 'LABEL', 'PEER_TO_PEER_AFFINITY', 'PEER_TO_PEER_ANTI_AFFINITY', 'ENERGY', 'CARBON', 'VIRTUAL_PROCESSORS', 'CPU_READY', 'PROCESSING_UNITS_PROVISIONED', 'HARDWARE_VERSION', 'GPU_MEM_ALLOCATION', 'GPU_MEM', 'GPU_COUNT', 'GPU_TENSOR_ENGINE',

'GPU_FP64_ENGINE', 'GPU_FP32_ENGINE', 'GPU_FP16_ENGINE', 'GPU_MEMORY_BANDWIDTH', 'STORAGE_ACCESS_OVERALL_VM_MAX_READ', 'STORAGE_ACCESS_OVERALL_VM_MAX_WRITE', 'STORAGE_ACCESS_OVERALL_VM_MAX_READWRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READ', 'IO_THROUGHPUT_OVERALL_VM_MAX_WRITE', 'IO_THROUGHPUT_OVERALL_VM_MAX_READWRITE', 'STORAGE_ACCESS_HD_BALANCED_READWRITE', 'IO_THROUGHPUT_HD_BALANCED_READWRITE', 'CORE_SPEED', 'VGPU_MEM', 'CONCURRENT_QUERIES', 'LLM_CACHE', 'VGPU_COMPUTE', 'GPU', 'GPU_REQUEST', 'GPU_REQUEST_QUOTA', 'QUEUING_TIME', 'SERVICE_TIME', 'RAID_HOSTS', 'NUMBER_REPLICAS', 'UNKNOWN', 'LOGICAL_BLOCK_SIZE_ACCESS', 'POWER_CAPABILITY_ACCESS']

ComputeTierAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Required Parameters

instanceStorageSize

- **type:** number
- **format:** float
- **description:** Size per instance storage in GB. This will be 0 if no instance storage's are present for compute tier.

numInstanceStorages

- **type:** number
- **format:** float
- **description:** Number of instance storage disks. Compute tiers can have 0 or more instance storage disks.

tierFamily

- **type:** string
- **description:** Family of compute tier (for example 'M4' for M4.large)

Optional Parameters

customConfigurable

- **type:** boolean
- **example:** True
- **description:** If the compute tier has commodities that can be configured per VM.

acceleratorCards

- **type:** array
- **description:** The Accelerator/GPU cards
- – **reference:** AcceleratorCardApiDTO

ConditionApiDTO

Description

A condition on a field of an entity or group.

Required Parameters

conditionType

- **type:** string
- **description:** Type of compared values.
- **readOnly:** True
- **enum:** ['NUMBER', 'TEXT', 'INCLUSION', 'INTEGER', 'BOOLEAN']

field

- **description:** Field to compare.
- **readOnly:** True
- **reference:** FieldApiDTO

operator

- **type:** string
- **description:** Operator to use.
- **readOnly:** True
- **enum:** ['EQ', 'NEQ', 'GT', 'LT', 'GE', 'LE', 'LIKE', 'IN']

ConfigChangesApiDTO

Description

Policies and Settings changes in the scenario

Optional Parameters

addPolicyList

- **type:** array
- **description:** List of Placement Policies to add
- – **reference:** PolicyApiDTO

removePolicyList

- **type:** array
- **description:** List of Placement Policies to remove
- – **reference:** PolicyApiDTO

removeAutomationPolicyList

- **type:** array
- **description:** List of Automation Policies to remove
- – **reference:** SettingsPolicyApiDTO

automationSettingList

- **type:** array
- **description:** List of automation settings

- – **reference:** SettingApiDTOString

removeConstraintList

- **type:** array
- **description:** List of constraints to remove
- – **reference:** RemoveConstraintApiDTO

riSettingList

- **type:** array
- **description:** List of Reserved Instance settings
- – **reference:** SettingApiDTO

osMigrationSettingList

- **type:** array
- **description:** List of OS migration settings
- – **reference:** SettingApiDTOString

includedCoupons

- **description:** List of Included Coupons (RIs) and related information
- **reference:** IncludedCouponsApiDTO

subscription

- **description:** Subscription
- **reference:** BusinessUnitApiDTO

respectActionModeSetting

- **type:** boolean
- **description:** Respect action mode settings in the automation policy

ConstraintApiDTO

Description

Model to describe the property of constraints, corresponding to entities constraints view in ui.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

relation

- **type:** string
- **enum:** ['sold', 'bought']

entityType

- **type:** string
- **description:** entity type of providers or consumers

relatedEntities

- **type:** array
- **description:** if relation is sold they are consumers, if bought they are providers
- – **reference:** ServiceEntityApiDTO

numPotentialEntities

- **type:** integer
- **format:** int32
- **description:** how many other providers/consumers can fulfill the constraint

placementOptions

- **type:** array
- **description:** A list of constraints prevent finding better placement for an entity
- – **reference:** PlacementOptionApiDTO

ConstraintApiInputDTO

Description

Model to describe the property of constraints

Optional Parameters

relation

- **type:** string
- **enum:** ['sold', 'bought']

placementOptions

- **type:** array
- **description:** A list of constraints prevent finding better placement for an entity
- – **reference:** PlacementOptionApiInputDTO

entityTypeFilter

- **type:** string
- **description:** potential entity type filter
- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

ContainerPlatformContextAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

namespace

- **type:** string
- **description:** The namespace that a container platform entity belong to - [DEPRECATED - Use: namespaceEntity]

containerPlatformCluster

- **type:** string
- **description:** The container platform cluster name that an entity belongs to - [DEPRECATED - Use: containerClusterEntity]

workloadControllerEntity

- **description:** The workload controller entity that an entity belongs to
- **reference:** BaseApiDTO

namespaceEntity

- **description:** The namespace entity that an entity belongs to
- **reference:** BaseApiDTO

containerClusterEntity

- **description:** The container platform cluster entity that an entity belongs to
- **reference:** BaseApiDTO

serviceEntity

- **description:** The service associated with the entity
- **reference:** BaseApiDTO

CostInputApiDTO

Description

Model to describe the request for Cost Statistics by a Time range

Optional Parameters

tagFilters

- **type:** array
- **description:** List of Tag Filters. NOTE: Required non-empty when used with costGroupBys set to "TAG"
- – **reference:** TagApiDTO

costGroupBys

- **type:** array

- **description:** List of values to group the results by. NOTE: When used non-empty tagFilters required
- – **type:** string
  – **enum:** ['TAG', 'TAG_GROUP', 'TAG_VAL', 'ENTITY', 'ENTITY_TYPE', 'ACCOUNT', 'REGION', 'CLOUD_SERVICE', 'SERVICE_PROVIDER', 'CLOUD_TIER_PROVIDER', 'COST_CATEGORY', 'PRICE_MODEL']

startDate

- **type:** string
- **description:** Start time of the period to filter. If startDate and endDate not provided current stats will be processed

endDate

- **type:** string
- **description:** End time of the period to filter. If startDate and endDate not provided current stats will be processed

CpuModelApiDTO

Description

Model to describe a Template: contains the resources used to Deploy a VM or to Add Workload/Supply in a Plan

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

numCores

- **type:** integer
- **format:** int32
- **description:** Number of Cores

speed

- **type:** integer
- **format:** int32
- **description:** Speed

scalingFactor

- **type:** number
- **format:** double
- **description:** Scaling Factor

modelName

- **type:** string

CriteriaOptionApiDTO

Description

Generic model used to describe a search criteria option

Optional Parameters

value

- **type:** string
- **description:** value

displayName

- **type:** string
- **description:** value to display, more readable

subValues

- **type:** array
- **description:** possible sub-values associated with the main value
- – **type:** string

DBEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

dbEngine

- **type:** string
- **description:** The Database engine

dbEdition

- **type:** string
- **description:** The Database edition

dbVersion

- **type:** string
- **description:** The Database engine version

licenseModel

- **type:** string
- **description:** The license model of the database engine

deploymentType

- **type:** string
- **description:** The deployment type of the database engine

maxConcurrentWorkers

- **type:** integer
- **format:** int32
- **description:** Maximum concurrent workers allowed for the DB

maxConcurrentSessions

- **type:** integer
- **format:** int32
- **description:** Maximum concurrent sessions allowed for the DB

pricingModel

- **type:** string
- **example:** DTU
- **description:** Database Pricing Model.
- **enum:** ['DTU', 'vCore', 'DWU', 'RU']

replicationRole

- **type:** string
- **example:** Primary
- **description:** Replication role if DB is part of geo replicated DB. Currently applies to Azure DB only.
- **enum:** ['Primary', 'Secondary', 'None']

storageTier

- **type:** string
- **example:** GP2
- **description:** Storage Tier

dbServerName

- **type:** string
- **description:** The server name of a database

serviceTier

- **type:** string
- **example:** Hyperscale
- **description:** The service tier of a database, generally defines the storage architecture, space and I/O limits, and business continuity options related to availability and disaster recovery.
- **enum:** ['Free', 'Basic', 'Standard', 'Premium', 'GeneralPurpose', 'BusinessCritical', 'Hyperscale']

computeTier

- **type:** string
- **example:** Provisioned Compute
- **description:** The compute tier of a database. Compute tier options include the provisioned and serverless compute tiers

hardwareGeneration

- **type:** string
- **example:** Gen4
- **description:** Database hardware generation. The generation of hardware that underlies the DB instance

backupPolicy

- **type:** string
- **example:** Enabled
- **description:** The policy to backup and restore a database.

workspace

- **type:** string
- **description:** The resource group where the database is located.

idleTime

- **type:** integer
- **format:** int64
- **description:** Time duration for a database being idle in milliseconds.

poolType

- **type:** string
- **description:** Type of database pool.

resourceId

- **type:** string
- **description:** Resource identifier of the database. For example, the resource id in Azure, the ARN in AWS, or the full resource name in GCP.

serverConfiguration

- **type:** string
- **description:** The Database server configuration

creationTimeStamp

- **type:** integer
- **format:** int64
- **description:** Timestamp in milliseconds for the creation of the database.

DataWarehouseAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

resourceId

- **type:** string
- **description:** Resource ID containing region, account ID etc

nodeCount

- **type:** integer
- **format:** int32
- **description:** Current number of nodes in this DataWarehouse

idleTimeInMilliseconds

- **type:** integer
- **format:** int64
- **description:** Time since last active

trackIdle

- **type:** boolean
- **description:** If the entity is being tracked for being idle

nextMaintenanceWindow

- **type:** string
- **description:** Date and time of next maintenance window dor AWS Redshift entity

address

- **type:** string
- **description:** Address URL for current redshift.

deploymentType

- **type:** string
- **description:** indicates how the data warehouse is deployed
- **enum:** ['SINGLE_AZ', 'MULTI_AZ']

DatabaseServerEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

dbEngine

- **type:** string
- **description:** The Database engine

dbEdition

- **type:** string
- **description:** The Database edition

dbVersion

- **type:** string
- **description:** The Database engine version

licenseModel

- **type:** string
- **description:** The license model of the database engine

deploymentType

- **type:** string
- **description:** The deployment type of the database engine

maxConcurrentWorkers

- **type:** integer
- **format:** int32
- **description:** Maximum concurrent workers allowed for the DB

maxConcurrentSessions

- **type:** integer
- **format:** int32
- **description:** Maximum concurrent sessions allowed for the DB

pricingModel

- **type:** string
- **example:** DTU
- **description:** Pricing Model followed by DB.

storageTier

- **type:** string
- **example:** GP2
- **description:** Storage Tier

clusterRole

- **type:** string
- **description:** Cluster Role
- **enum:** ['Writer', 'Reader']

hourlyBilledOps

- **type:** number
- **format:** double
- **description:** Hourly Billed Ops - Count of operations per hour

featureStateMap

- **type:** object
- **description:** Features
- **additionalProperties:** {'type': 'string', 'enum': ['Enabled', 'Disabled', 'Unknown']}

resourceId

- **type:** string
- **description:** Resource identifier of the database server. For example, the resource id in Azure, the ARN in AWS, or the full resource name in GCP.

creationTimeStamp

- **type:** integer
- **format:** int64
- **description:** Timestamp in milliseconds for the creation of the database server.

serverConfiguration

- **type:** string
- **description:** The Database server configuration

replicas

- **type:** array

- **description:** Replicas associated with this database
- – **reference:** ReplicaApiDTO

databases

- **type:** array
- **description:** Databases that run on this instance
- – **type:** string

fullyQualifiedDomainName

- **type:** string
- **description:** Fully qualified domain name

backupStorageRedundancy

- **type:** string
- **description:** Backup Storage Redundancy - refers to how data is duplicated across storage locations

DatabaseServerTierAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Required Parameters

tierFamily

- **type:** string
- **description:** Family of database server tier (for example 'M5' for db.M5.large)

DatabaseServerTierInfoApiDTO

Description

Model to describe the database server tier to which a cloud commitment applies

Optional Parameters

databaseEngine

- **type:** string
- **description:** The database engine to which a cloud commitment applies
- **enum:** ['UNKNOWN', 'MARIADB', 'MYSQL', 'POSTGRESQL', 'AURORAMYSQL', 'AURORAPOSTGRESQL', 'SQLSERVER', 'ORACLE', 'DB2']

deploymentType

- **type:** string
- **description:** The deployment type that the commitment covers
- **enum:** ['SINGLE_AZ', 'MULTI_AZ']

licenseType

- **type:** string

- **description:** The license type associated with the commitment
- **enum:** ['LICENSE_INCLUDED', 'AHUB', 'BRING_YOUR_OWN_LICENSE', 'NO_LICENSE_REQUIRED']

DatabaseTierAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Required Parameters

tierFamily

- **type:** string
- **description:** Family of database tier (for example 'S' for S4)

DemandEntityInfoDTO

Description

Model to describe a reservation entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

template

- **reference:** BaseApiDTO

deploymentProfile

- **reference:** BaseApiDTO

operationalCost

- **type:** string

placements

- **reference:** PlacementInfoDTO

DemandReservationApiDTO

Description

Basic information of an entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

prefix

- **type:** string

count

- **type:** integer
- **format:** int32

status

- **type:** string

reserveDateTime

- **type:** string

expireDateTime

- **type:** string

deployDateTime

- **type:** string

reserveCount

- **type:** integer
- **format:** int32

deployCount

- **type:** integer
- **format:** int32

placementExpirationDateTime

- **type:** string

placementResultMessage

- **type:** string
- **description:** Details of placement during failure or success

constraintInfos

- **type:** array
- **description:** The constraints user specified when running reservation which limit where templates could be placed.
- – **reference:** ReservationConstraintApiDTO

reservationDeployed

- **type:** boolean

- **description:** When true, the reservation will be included in the system and will keep utilizing the resources from host and storage until the corresponding virtual machine utilization starts to reflect on the historically computed utilization.

mode

- **type:** string
- **description:** Demand related behavior for reservation of NO_GROUPING/AFFINITY. This value interacts with the grouping value which is specifying the type of grouping that the demand should consider the mode for. NO_GROUPING and NONE must be paired together and cannot be paired with other values. Combinations such as: (NO_GROUPING - NONE), (AFFINITY - CLUSTER) are valid. Examples of invalid combinations are (NO_GROUPING - CLUSTER), (AFFINITY - NONE).
- **enum:** ['NO_GROUPING', 'AFFINITY']

grouping

- **type:** string
- **description:** Grouping related behavior for reservation of NONE and CLUSTER. This value interacts with the mode value.
- **enum:** ['NONE', 'CLUSTER']

scope

- **type:** array
- **description:** List of uuids of groups (currently limited to Clusters or Host Groups) whose members are the only possible providers of that type of entity. These are further constrained by the constraintIds passed in placement parameters. If multiple groups of providers are specified then it is possible that entities will be placed within different groups.
- – **type:** string

demandEntities

- **type:** array
- – **reference:** DemandEntityInfoDTO

DemandReservationApiInputDTO

Description

Required Parameters

action

- **type:** string
- **description:** Action requested for the reservation
- **enum:** ['RESERVATION']

demandName

- **type:** string
- **description:** Name of the reservation

Optional Parameters

mode

- **type:** string
- **description:** Demand related behavior for reservation of NO_GROUPING/AFFINITY. This value interacts with the grouping value which is specifying the type of grouping that the demand should

consider the mode for. NO_GROUPING and NONE must be paired together and cannot be paired with other values. Combinations such as: (NO_GROUPING - NONE), (AFFINITY - CLUSTER) are valid. Examples of invalid combinations are (NO_GROUPING - CLUSTER), (AFFINITY - NONE).

- **enum:** ['NO_GROUPING', 'AFFINITY']

grouping

- **type:** string
- **description:** Grouping related behavior for reservation of NONE and CLUSTER. This value interacts with the mode value.
- **enum:** ['NONE', 'CLUSTER']

reserveDateTime

- **type:** string
- **description:** Date to specify when to reserve the Demand

expireDateTime

- **type:** string
- **description:** Date to specify when the Demand request will expire, required if action = RESERVATION

deployDateTime

- **type:** string
- **description:** Date to specify when to deploy the Demand

parameters

- **type:** array
- **description:** Reservation Parameters
- – **reference:** DemandReservationParametersDTO

scope

- **type:** array
- **description:** List of uuids of groups (currently limited to Clusters or Host Groups) whose members are the only possible providers of that type of entity. These are further constrained by the constraintIds passed in placement parameters. If multiple groups of providers are specified then it is possible that entities will be placed within different groups.
- – **type:** string

DemandReservationParametersDTO

Description

Required Parameters

placementParameters

- **description:** Placement parameters
- **reference:** PlacementParametersDTO

Optional Parameters

deploymentParameters

- **description:** Deployment parameters, required if action = DEPLOYMENT, RESERVATION

- **reference:** DeploymentParametersDTO

DeploymentInfoDTO

Description

Contains deployment information about the running Turbonomic instance.

Optional Parameters

clientId

- **type:** string
- **example:** turbonomic1
- **description:** The client Id

clientSvcAccount

- **type:** string
- **example:** turbonomic1-sa
- **description:** The client service account

oidcProviderUrl

- **type:** string
- **example:** https://rh-oidc.s3.us-east-1.amazonaws.com/22ejnvnnturfmt6km08idd0nt4hekbn7
- **description:** The OIDC provider URL

DeploymentParametersDTO

Description

Optional Parameters

deploymentProfileID

- **type:** string
- **description:** Deployment Profile uuid

highAvailability

- **type:** boolean
- **description:** High availability

priority

- **type:** string
- **description:** Workload priority
- **enum:** ['LOW', 'NORMAL', 'MISSION_CRITICAL']

DeploymentProfileApiDTO

Description

Model to describe a Deployment Profile, including target and provider entity parameters

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

account

- **description:** Business account related to the Deployment Profile
- **reference:** BaseApiDTO

deployParameters

- **type:** array
- **description:** Target and Provider parameters of the Deployment Profile
- – **reference:** DeploymentProfileTargetApiDTO

DeploymentProfileParamApiDTO

Description

Provider-specific Deployment Profile parameters

Optional Parameters

parameterType

- **type:** string
- **description:** Parameter type. For vCenter, this can be a folder, osConfiguration, or image. For all other technologies, this must be an image.
- **enum:** ['image', 'folder', 'osConfiguration']

properties

- **type:** array
- **description:** A NameValueInputDTO describing the name and value of the parameter.
- – **reference:** NameValueInputDTO

DeploymentProfileProviderApiDTO

Description

Deployment profile Provider properties

Optional Parameters

provider

- **description:** Provider UUID
- **reference:** BaseApiDTO

parameters

- **type:** array
- **description:** A DeploymentProfileParamApiDTO describing the parameters for the provider entities
- – **reference:** DeploymentProfileParamApiDTO

DeploymentProfileTargetApiDTO

Description

Deployment Profile Target properties

Optional Parameters

targetType

- **type:** string
- **description:** Target type for the deployment profile
- **enum:** ['vCenter', 'AWS', 'Azure', 'Softlayer']

providers

- **type:** array
- **description:** DeploymentProfileProviderApiInputDTO describing the provider entity parameters
- – **reference:** DeploymentProfileProviderApiDTO

DesktopPoolEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

cloneType

- **type:** string
- **description:** Type of a clone provided by the DesktopPool
- **enum:** ['FULL', 'LINKED', 'INSTANT']

provisionType

- **type:** string
- **description:** How clones are provisioned in the DesktopPool
- **enum:** ['UPFRONT', 'ON_DEMAND']

assignmentType

- **type:** string
- **description:** Whether the same clone is assigned each time when user logs back in
- **enum:** ['PERSISTENT', 'DYNAMIC']

masterTemplateUuid

- **type:** string

- **description:** UUID of a master template that is used to create clones. Only one of master VM and master template can be set.

masterVirtualMachineUuid

- **type:** string
- **description:** UUID of a master virtual machine that is used to create clones. Only one of master VM and master template can be set.

masterVirtualMachineSnapshot

- **type:** string
- **description:** Name of the snapshot of a master virtual machine that is used to create clones.

getvCenterClusterName

- **type:** string
- **description:** Name of the vCenter cluster that this DesktopPool draws resources from.

vendorId

- **type:** string
- **description:** Vendor Identifier for the DesktopPool.

DetailDataApiDTO

Description

A single entity detail, represented as key-value pair

Required Parameters

key

- **type:** string
- **description:** Detail key

value

- **type:** string
- **description:** Detail value

Optional Parameters

critical

- **type:** boolean

DevDiagsUploadDTO

Description

Model to determine whether diagnostics needs to be anonymized or not

Optional Parameters

enableAnonymization

- **type:** boolean
- **description:** Boolean signifying diagnostics needs to be anonymized or not

DiagsUploadDTO

Description

Model to describe Diagnostics upload settings, like caseid, transferid, password, email and Upload ClientType

Required Parameters

diagsUploadClientType

- **type:** string
- **description:** diagsUploadClientType
- **enum:** ['ECUREP', 'BLUE_DIAMOND']

password

- **type:** string
- **description:** password

transferId

- **type:** string
- **description:** Transfer Id

Optional Parameters

caseId

- **type:** string
- **description:** Case Id

email

- **type:** string
- **description:** email

DiagsUploadResponseApiDTO

Description

Model to describe Diagnostics upload response settings, like status, message and messagekey

Optional Parameters

status

- **type:** string
- **example:** In Progress
- **description:** success indicating the state of api upload action

message

- **type:** string
- **example:** File Uploading in Progress
- **description:** message indicating the reason for success

messageKey

- **type:** string

- **example:** INVALID_EMAIL_ID
- **description:** message indicating the reason for success
- **enum:** ['INVALID_EMAIL_ID', 'INVALID_CASE_ID', 'INVALID_TRANSFER_ID', 'INVALID_TRANSFER_ID_PASSWORD', 'INVALID_SERVER', 'INVALID_AUTHENTICATION']

DiscountCoverageFilterApiDTO

Description

Model to describe the discount coverage filter information for the action filtering.

Optional Parameters

currentDiscountCoverageFilter

- **description:** Filter the actions based on a current discount coverage.
- **reference:** RangeInputApiDTO

newDiscountCoverageFilter

- **description:** Filter the actions based on a new discount coverage.
- **reference:** RangeInputApiDTO

DiscoveredEntityApiDTO

Description

Model to describe a discovered entity in the system

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['CLOUD', 'ONPREM', 'HYBRID']

discoveredBy

- **description:** Target that discovered the entity
- **reference:** TargetApiDTO

vendorIds

- **type:** object
- **description:** The mapping of target identifier to vendor-provided identity of this entity on the remote target

- **additionalProperties:** {'type': 'string'}

EffectiveCostFilterApiDTO

Description

Model to describe effective cost range for action filtering.

Optional Parameters

currentEffectiveCostRange

- **description:** Filter the actions by the range for current effective cost.
- **reference:** RangeInputApiDTO

newEffectiveCostRange

- **description:** Filter the actions by the range for destination effective cost. Applicable only for scale actions.
- **reference:** RangeInputApiDTO

EntityActionsApiDTO

Description

Basic informations of an entity + actions

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

realtimeMarketReference

- **description:** In case of a Plan Market Entity, this contains the original entity from the Real Time Market
- **reference:** BaseApiDTO

actions

- **type:** array
- **description:** List of actions for this entity
- – **reference:** ActionApiDTO

EntityAspect

Description

Parent class for entity aspects

Required Parameters

type

- **type:** string

Optional Parameters

sli

- **type:** array
- – **type:** string

EntityCostApiDTO

Description

Model to describe the Costs of an entity

Optional Parameters

value

- **type:** number
- **format:** float
- **description:** The cost value

hairCutRatio

- **type:** number
- **format:** float
- **description:** The hair cut ratio, it represents the portion of cost will be taken into consideration

costType

- **type:** string
- **description:** Type of the Cost

costUnit

- **type:** string
- **description:** Unit of the Cost

subCosts

- **type:** array
- **description:** List of sub costs of the current cost
- – **reference:** EntityCostApiDTO

currency

- **type:** integer
- **format:** int32

EntityCountRequestApiDTO

Description

A query used to fetch entity counts.

Optional Parameters

groupBy

- **type:** array
- **description:** Criteria to group the results by.
- **readOnly:** True
- – **type:** string
  – **enum:** ['ENTITY_TYPE', 'ENVIRONMENT_TYPE']

EntityDetailsApiDTO

Description

Model to describe an entity with all its details

Required Parameters

uuid

- **type:** integer
- **format:** int64
- **description:** The internal UUID of the entity

Optional Parameters

details

- **type:** array
- **description:** A list of details, represented as key-value
- – **reference:** DetailDataApiDTO

EntityMetadataRequestApiDTO

Description

A query used to fetch entity metadata.

Required Parameters

entityType

- **type:** string
- **description:** Entity type to fetch metadata for.
- **readOnly:** True
- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

EntityPriceDTO

Description

Context-specific price. For example the base rate of a template in the context of certain business unit.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

price

- **type:** number
- **format:** float
- **description:** Price in the context of a related entity

EntityQueryApiDTO

Description

A query used to search topology entities.

Required Parameters

select

- **description:** Information to be fetched for each entity.
- **readOnly:** True
- **reference:** SelectEntityApiDTO

Optional Parameters

where

- **description:** Conditions that each retrieved entity must satisfy.
- **readOnly:** True
- **reference:** WhereApiDTO

pagination

- **description:** Pagination information.
- **readOnly:** True
- **reference:** PaginationApiDTO

EntityStatsApiDTO

Description

Basic informations of an entity + statistics

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['CLOUD', 'ONPREM', 'HYBRID']

realtimeMarketReference

- **description:** In case of a Plan Market Entity, this contains the original entity from the Real Time Market
- **reference:** BaseApiDTO

stats

- **type:** array
- **description:** Collected monitoring statistics
- – **reference:** StatSnapshotApiDTO

EntityUptimeApiDTO

Description

Model to describe the uptime of an entity

Optional Parameters

creationTimestamp

- **type:** integer
- **format:** int64
- **description:** The creation timestamp of the entity

uptimeDurationInMilliseconds

- **type:** integer
- **format:** int64
- **description:** Uptime duration in milliseconds for the total analyzed duration

totalDurationInMilliseconds

- **type:** integer
- **format:** int64
- **description:** Total analyzed duration in milliseconds for which uptime is calculated

uptimePercentage

- **type:** number
- **format:** double
- **description:** The uptime of the entity as a percentage

EnumFieldValueApiDTO

Description

Inherits From

This DTO inherits from FieldValueApiDTO

Required Parameters

value

- **type:** string
- **description:** The string representation of the value.
- **readOnly:** True

ExecutorInfoApiDTO

Description

Model to describe the entity that triggered action execution. such as a user or a schedule.

Required Parameters

type

- **type:** string
- **description:** The type of the executor. such as a user or a schedule.
- **enum:** ['USER', 'SCHEDULE', 'EXECUTED_EXTERNALLY']

Optional Parameters

scheduleInfo

- **description:** Information about the schedule that triggered this action. Omitted if the action was not triggered by a schedule.
- **reference:** ExecutorScheduleApiDTO

userInfo

- **description:** Information about the user who triggered this action. Omitted if the action was not triggered by a user.
- **reference:** ExecutorUserApiDTO

ExecutorScheduleApiDTO

Description

Model to describe the Schedule that executed the action.

Required Parameters

scheduleName

- **type:** string
- **description:** Name of the schedule

ExecutorUserApiDTO

Description

Model to describe the user who triggered this action.

Required Parameters

userName

- **type:** string
- **description:** Name of the user

FailedResourceApiDTO

Description

A description of the resource that caused an entity to be unplaced

Required Parameters

commodity

- **description:** The commodity that caused the entity to be unplaced
- **reference:** BaseCommodityApiDTO

requestedAmount

- **type:** number
- **format:** double
- **description:** The requested amount of the commodity that caused the entity to be unplaced

Optional Parameters

maxAvailable

- **type:** number
- **format:** double
- **description:** The maximum available amount of the requested commodity. Populated only if there are one or more entities selling the requested commodity.

FamilyScopeApiDTO

Description

Model to describe the instance family of a cloud commitment

Optional Parameters

family

- **type:** string
- **description:** The name of the instance family

FieldApiDTO

Description

A field of an entity or group.

Optional Parameters

fieldType

- **type:** string
- **description:** The kind of the field.
- **enum:** ['PRIMITIVE', 'COMMODITY', 'RELATED_ENTITY', 'RELATED_GROUP', 'RELATED_ACTION', 'AGGREGATE_COMMODITY', 'MEMBER']

FieldValueApiDTO

Description

A field of an entity or group with its value.

Required Parameters

field

- **description:** Field to compare.
- **readOnly:** True
- **reference:** FieldApiDTO

valueType

- **type:** string
- **description:** Type of compared values.
- **readOnly:** True
- **enum:** ['NUMBER', 'TEXT', 'ENUM', 'INTEGER', 'BOOLEAN', 'MULTI_TEXT']

FieldValueTypeApiDTO

Description

A field of an entity or group associated with the type of its values.

Required Parameters

field

- **description:** Field for which information is presented.
- **readOnly:** True
- **reference:** FieldApiDTO

valueType

- **type:** string
- **description:** Type of values of this field.
- **readOnly:** True
- **enum:** ['NUMBER', 'TEXT', 'ENUM', 'INTEGER', 'BOOLEAN', 'MULTI_TEXT']

Optional Parameters

allowableValues

- **type:** array
- **description:** Allowable string values for this field.
- **readOnly:** True
- – **type:** string

FilterApiDTO

Description

Model used to filter Groups or Entities, every filter is built using a filterType + expType + expVal

Required Parameters

caseSensitive

- **type:** boolean
- **description:** Whether the regex is case sensitive

expType

- **type:** string
- **description:** Type of comparator used while building a filter, the allowable values are 'EQ', 'NEQ', 'RXEQ', 'RXNEQ' for strings, 'EQ', 'NEQ', 'GT', 'LT', 'GTE', 'LTE' for numbers'EX', 'NEX' for inclusion in a set of options
- **enum:** ['EQ', 'NEQ', 'GT', 'LT', 'GTE', 'LTE', 'RXEQ', 'RXNEQ', 'EX', 'NEX']

expVal

- **type:** string
- **description:** Regex to be searched for

filterType

- **type:** string
- **description:** Internal name for criterion such as pmsByName, pmsByDC, storageByPMCluster

Optional Parameters

entityType

- **type:** string

- **description:** The entity type which this filter should be applied on

singleLine

- **type:** boolean
- **description:** Whether the regex dot matches all characters including line break characters. Defaults to false.

FrontdoorIdpApiDTO

Description

Model to describe a Frontdoor IDP. It contains IDP URL and is FD enabled or not

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

isFrontdoorEnabled

- **type:** boolean

GcpManagedInstanceGroupDTO

Description

Model to describe specific properties of GCP Managed Instance Groups

Optional Parameters

autoscalingSchedules

- **type:** array
- **description:** Collection of autoscaled schedules in a GCP MIG
- **uniqueItems:** True
- – **type:** string

autoscalingScheduleCount

- **type:** integer
- **format:** int32
- **description:** Number of autoscaled schedules in a GCP MIG

autoScalingMode

- **type:** string
- **description:** Scaling mode enables auto scaling to add and remove instances

deploymentType

- **type:** string
- **description:** Identifies whether MIG is deployed across the multiple/single zone

GroupApiDTO

Description

Model to describe a Group: a collection of Entities. Subclasses include BillingFamilyApiDTO and ResourceGroupApiDTO.

Required Parameters

groupClassName

- **type:** string

isStatic

- **type:** boolean
- **description:** True if group is static

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

membersCount

- **type:** integer
- **format:** int32
- **description:** Number of members of the Group

entitiesCount

- **type:** integer
- **format:** int32
- **description:** Number of entities of the Group

costPrice

- **type:** number
- **format:** float
- **description:** Cost of the Group per Hour: sum of the costs of the member entities

groupType

- **type:** string
- **description:** The type of service entities comprising the group

severity

- **type:** string
- **description:** Calculated using the highest severity of the member entities
- **enum:** ['UNKNOWN', 'NORMAL', 'MINOR', 'MAJOR', 'CRITICAL']

state

- **type:** string
- **description:** Calculated using the state of the member entities
- **enum:** ['UNKNOWN', 'ACTIVE']

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['HYBRID', 'CLOUD', 'ONPREM', 'UNKNOWN']

scope

- **type:** array
- **description:** Scope within which the criteria have to be applied to the list of specific entity type
- – **type:** string

logicalOperator

- **type:** string
- **description:** Logical operator to be applied across all the criteria used to create dynamic group.It can take values 'AND', 'OR', 'XOR'

criteriaList

- **type:** array
- **description:** Criteria list of elements, expression type, expression value, seType - needed to create a dynamic group
- – **reference:** FilterApiDTO

memberUuidList

- **type:** array
- **description:** Uuid list for members of the group - if group is static
- – **type:** string

realtimeMarketReference

- **description:** In case of a Plan Market Group, this contains the original group from the Real Time Market
- **reference:** BaseApiDTO

stats

- **type:** array
- **description:** Collected monitoring statistics
- – **reference:** StatSnapshotApiDTO

temporary

- **type:** boolean
- **description:** The Group is only valid for a limited period of time, it will be removed automatically

activeEntitiesCount

- **type:** integer
- **format:** int32
- **description:** The active entities count of a group

cloudType

- **type:** string
- **description:** Type of the Cloud if environmentType = CLOUD.[DEPRECATED - Use: targetType
- **enum:** ['AWS', 'AZURE', 'GCP', 'HYBRID', 'UNKNOWN']

source

- **description:** The Source of the Group
- **reference:** TargetApiDTO

memberTypes

- **type:** array
- **description:** The types for immediate members of the group.
- – **type:** string

entityTypes

- **type:** array
- **description:** The types of entities contained in the group. This includes types of entities in nested levels of the group if the group is nested.
- – **type:** string

aspects

- **type:** object
- **description:** Additional info about the Group categorized as Aspects
- **additionalProperties:** {'$ref': '#/definitions/EntityAspect'}

vendorIds

- **type:** object
- **description:** The mapping of target identifier to vendor-provided identity of this group, if the group is discovered
- **additionalProperties:** {'type': 'string'}

groupOrigin

- **type:** string
- **description:** Origin of entities and groups
- **enum:** ['DISCOVERED', 'USER', 'SYSTEM']

targetType

- **type:** string
- **description:** Target type of group entities. TargetType UNKNOWN if multiple target types

GroupCountRequestApiDTO

Description

A query used to fetch group counts.

Optional Parameters

groupBy

- **type:** array
- **description:** Criteria to group the results by.
- **readOnly:** True
- **uniqueItems:** True
- – **type:** string
  – **enum:** ['GROUP_TYPE', 'ORIGIN']

GroupMetadataRequestApiDTO

Description

A query used to fetch group metadata.

Required Parameters

groupType

- **type:** string
- **description:** Group type to fetch metadata for.
- **readOnly:** True
- **enum:** ['Group', 'Resource', 'Cluster', 'VMCluster', 'StorageCluster', 'BusinessAccountFolder', 'BillingFamily', 'NodePool']

GroupQueryApiDTO

Description

A query used to search groups.

Required Parameters

select

- **description:** Information to be fetched for each group.
- **readOnly:** True
- **reference:** SelectGroupApiDTO

Optional Parameters

where

- **description:** Conditions that each retrieved group must satisfy.
- **readOnly:** True
- **reference:** WhereApiDTO

pagination

- **description:** Pagination information.
- **readOnly:** True
- **reference:** PaginationApiDTO

HealthCategoryResponseDTO

Description

A DTO to collect the aggregated health data for one health category.

Required Parameters

categoryDisplayName

- **type:** string
- **description:** The reported health category name.

categoryHealthState

- **type:** string
- **description:** The health state for the reported category.
- **enum:** ['CRITICAL', 'MAJOR', 'MINOR', 'NORMAL']

healthCategory

- **type:** string
- **description:** The reported health category.
- **enum:** ['TARGET', 'ACTION', 'COMPONENT']

Optional Parameters

responseItems

- **type:** array
- **description:** A list of individual health response items for this health category.
- – **reference:** AggregatedHealthResponseDTO

HttpProxyDTO

Description

Model to describe http proxy settings, like username, password, port number and host

Required Parameters

isProxyEnabled

- **type:** boolean
- **description:** Whether proxy is enabled or not

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

proxyHost

- **type:** string
- **description:** Proxy host

proxyPortNumber

- **type:** integer
- **format:** int32
- **description:** Proxy port number

userName

- **type:** string
- **description:** Proxy username

password

- **type:** string
- **description:** Proxy password

isProxySecure

- **type:** boolean
- **description:** Whether proxy is secure: SSL is used to connect to the proxy host. Defaults for 'false'.

IEntityDefinitionData

Description

Parent class for topology data definition

Required Parameters

definitionType

- **type:** string
- **enum:** ['MANUAL', 'AUTOMATED']

IManualConnectionsData

Description

Parent class for manual connections.

Required Parameters

connectionType

- **type:** string
- **enum:** ['DYNAMIC', 'STATIC', 'GROUP', 'GROUPS']

membersCount

- **type:** integer
- **format:** int32

IacGrantActionDetailsApiDTO

Description

Inherits From

This DTO inherits from ActionDetailsApiDTO

Optional Parameters

terraformStateFileSpec

- **description:** Terraform state file related properties
- **reference:** TerraformStateFileSpecApiDTO

IncludedCouponsApiDTO

Description

Included RI/Coupon and related information in the scenario

Optional Parameters

includedCouponOidsList

- **type:** array
- **description:** User selected list of RIs to be included in plans
- – **type:** integer
  – **format:** int64

iswhiteList

- **type:** boolean

InclusionConditionApiDTO

Description

Inherits From

This DTO inherits from ConditionApiDTO

Required Parameters

value

- **type:** array
- **description:** A set of strings. The value must belong to one of those.
- **readOnly:** True
- – **type:** string

InputFieldApiDTO

Description

Basic information of an entity

Required Parameters

name

- **type:** string
- **description:** Name of the field, used for field identification.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

value

- **type:** string
- **description:** Field value. Used if field holds primitive value (String, number or boolean

defaultValue

- **type:** string
- **description:** Default value of the field

isMandatory

- **type:** boolean
- **description:** Whether the field is mandatory. Valid targets must have all the mandatory fields set.

isSecret

- **type:** boolean
- **description:** Whether the field is secret. This means, that field value is stored in an encrypted value and not shown in any logs.

isMultiline

- **type:** boolean
- **description:** Whether the field can have multiline values. Can be only true when the valueType field is set to STRING.

isTargetDisplayName

- **type:** boolean
- **description:** Whether the field is used as a target display name.

valueType

- **type:** string

- **description:** Type of the value this field holds
- **enum:** ['STRING', 'BOOLEAN', 'NUMERIC', 'INTEGER', 'LIST', 'OBJECT', 'GROUP_SCOPE', 'SECRET']

specificValueType

- **type:** string
- **description:** A specific Type of the value that can't be described in valueType

description

- **type:** string
- **description:** Additional information about what the input to the field should be

verificationRegex

- **type:** string
- **description:** The regex pattern that needs to be satisfied for the input field text

groupProperties

- **type:** array
- **description:** Group scope structure, filled if this field represents group scope value
- – **type:** array
  - – - **type:** string

allowedValues

- **type:** array
- **description:** Potential values for this field. If nonempty, value must be one of the choices in the list.
- – **type:** string

dependencyKey

- **type:** string
- **description:** A key of a dependency account value field. If this value is not set, field is declared without any dependencies. If an annotated account value field has a dependency it means that it's presence is logically driven by the dependency field. This field is expected to be available ONLY when field specified by {@code dependencyKey} has value {@code dependencyValue}. If this field is not available because of unmet dependency, this field will not be validated and stored in target configuration data. If a dependency for this field is unmet value of this field will be null

dependencyValue

- **type:** string
- **example:** one|two|four
- **description:** Value of a field of account values referenced by {@code dependencyKey} to enable (activate) the annotated field. Value is specified using a regular expression. Specifying just a string will also work correctly. If a referenced account value field is a boolean field, only "true" or "false" are supported as values. If a referenced account value is an enumeration ({@code allowedValues} is specified) this value is only restricted to contain a subset of the allowed values united using a pipe {@code one|two|four}.

InstanceTypeFilterApiDTO

Description

Model to describe instanceType filter information for the action filtering.

Optional Parameters

currentInstanceTypeFilter

- **description:** Filter the actions by matching current instanceType and given the instanceType Uuid list.
- **reference:** InstanceTypeIdsApiDTO

newInstanceTypeFilter

- **description:** Filter the actions by matching new instanceType and given the instanceType Uuid list.
- **reference:** InstanceTypeIdsApiDTO

InstanceTypeIdsApiDTO

Description

Model to describe instanceType Uuid list and operator for the action filtering.

Optional Parameters

instanceTypeIds

- **type:** array
- **description:** InstanceType Uuid list for the action filter.
- – **type:** string

operator

- **type:** string
- **description:** Operator for instanceType Uuid filter.the allowable values are 'EQ' and 'NEQ'
- **enum:** ['EQ', 'NEQ']

IntegerConditionApiDTO

Description

Inherits From

This DTO inherits from ConditionApiDTO

Required Parameters

value

- **type:** integer
- **format:** int64
- **description:** Value to compare against.
- **readOnly:** True

IntegerFieldValueApiDTO

Description

Inherits From

This DTO inherits from FieldValueApiDTO

Required Parameters

value

- **type:** integer
- **format:** int64
- **description:** The value.
- **readOnly:** True

Optional Parameters

units

- **type:** string
- **description:** Units of measurement.
- **readOnly:** True

KubeturboConfigInputDTO

Description

Required Parameters

clusterName

- **type:** string
- **example:** dev-ocp
- **description:** Name of container platform cluster. Will show up as Target Name in the target page.

deployThroughTSC

- **type:** boolean
- **description:** Determines whether to deploy kubeturbo through TSC

targetSubtype

- **type:** string
- **example:** RedHatOpenShift
- **description:** Target subtype

Optional Parameters

namespace

- **type:** string
- **example:** turbo
- **description:** Namespace where kubeturbo will be deployed

permission

- **type:** string
- **description:** Permission for kubeturbo deployed in a cluster
- **enum:** ['OPTIMIZE', 'MONITOR']

privateRegistry

- **type:** string
- **example:** icr.io/cpopen/turbonomic/kubeturbo
- **description:** Private image registry for kubeturbo

privateRegistryUsername

- **type:** string
- **example:** username
- **description:** Private image registry username for kubeturbo

privateRegistryPassword

- **type:** string
- **example:** password
- **description:** Private image registry password for kubeturbo

proxyServerURL

- **type:** string
- **example:** http://proxyserver:proxyport
- **description:** Proxy server address

serverURL

- **type:** string
- **example:** https://Turbo_server_URL
- **description:** Turbo server address

serverVersion

- **type:** string
- **example:** 8.13.5
- **description:** Turbo server version

oauthClientID

- **type:** string
- **description:** OAuth client ID

oauthSecret

- **type:** string
- **description:** OAuth client secret

tscToken

- **type:** string
- **description:** TSC Token

KubeturboDeployOutputDTO

Description

Bash script to deploy kubeturbo

Optional Parameters

script

- **type:** string
- **example:** #!/usr/bin/env bash echo "Sample"
- **description:** Bash script to deploy kubeturbo

LicenseApiDTO

Description

Model to describe properties of a license

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

licenseOwner

- **type:** string
- **description:** The owner of current license

email

- **type:** string
- **description:** The email of current license

customerId

- **type:** string
- **description:** The customer id

expirationDate

- **type:** string
- **description:** The expiration date of current license

features

- **type:** array
- **description:** List of features contained in current license
- **uniqueItems:** True
- – **type:** string

numLicensedEntities

- **type:** integer
- **format:** int32
- **description:** The number of licensed sockets

numInUseEntities

- **type:** integer
- **format:** int32
- **description:** The number of sockets in use

edition

- **type:** string
- **description:** The edition of the current license

filename

- **type:** string
- **description:** The name of the uploaded file

countedEntity

- **type:** string
- **description:** Indicates which entity type is counted with respect to licensing
- **enum:** ['VM', 'SOCKET']

licenseSchemaVersion

- **type:** string
- **description:** License format version

engagementType

- **type:** string
- **description:** Engagement type

companyName

- **type:** string
- **description:** Company name

parentCompanyName

- **type:** string
- **description:** Parent company name

territory

- **type:** string
- **description:** Sales territory

iscId

- **type:** string
- **description:** ISC ID

iscPovId

- **type:** string
- **description:** ISC POV ID

parentCompanyAccountNumber

- **type:** string
- **description:** Global account number that starts with 'GB' or 'GC'

companyAccountNumber

- **type:** string

- **description:** Country account number that starts with 'DB' or 'DC'

maxReportEditorsCount

- **type:** integer
- **format:** int32
- **description:** Maximum number of report editors

expirationDateTime

- **type:** string
- **format:** date-time

errorReasons

- **type:** array
- **description:** Indicate why the license is invalid
- **uniqueItems:** True
- – **type:** string
  – **enum:** ['DUPLICATE_LICENSE', 'EXPIRED', 'INVALID_EMAIL', 'INVALID_CONTENT_TYPE', 'INVALID_LICENSE_KEY', 'INVALID_LICENSE_TYPE_CWOM_ONLY', 'INVALID_LICENSE_TYPE_FREEMIUM_NOT_ALLOWED', 'INVALID_FEATURE_SET', 'INCOMPATIBLE']

isValid

- **type:** boolean
- **description:** Indicate if current license is valid or not
- **readOnly:** True

isExpired

- **type:** boolean
- **description:** Indicate if current license is expired
- **readOnly:** True

LinkNotificationCriteria

Description

Model to describe criteria information for a notification category.

Optional Parameters

severities

- **type:** array
- **description:** Health state criteria to filter when linking notifications.
- – **type:** string
  – **enum:** ['CRITICAL', 'MAJOR', 'MINOR', 'NORMAL']

LinkNotificationRequestApiDTO

Description

Model to describe criteria for notification link request.

Required Parameters

criteria

- **type:** object
- **description:** Notification criteria for a category
- **additionalProperties:** {'$ref': '#/definitions/LinkNotificationCriteria'}

LinkRelation

Description

This is an unspecified object.

Link

Description

Optional Parameters

rel

- **readOnly:** True
- **reference:** LinkRelation

href

- **type:** string
- **readOnly:** True

hreflang

- **type:** string
- **readOnly:** True

media

- **type:** string
- **readOnly:** True

title

- **type:** string
- **readOnly:** True

type

- **type:** string
- **readOnly:** True

deprecation

- **type:** string
- **readOnly:** True

profile

- **type:** string
- **readOnly:** True

name

- **type:** string
- **readOnly:** True

LoadChangesApiDTO

Description

Workload utilization changes in the scenario

Optional Parameters

baselineDate

- **type:** string
- **description:** Baseline date

peakBaselineList

- **type:** array
- **description:** List of peak baseline changes
- – **reference:** PeakBaselineApiDTO

utilizationList

- **type:** array
- **description:** List of utilization changes
- – **reference:** UtilizationApiDTO

maxUtilizationList

- **type:** array
- **description:** List of max utilization changes
- – **reference:** MaxUtilizationApiDTO

overlayStatsList

- **type:** array
- **description:** List of stat changes to override the plan commodities values with the provided input
- – **reference:** EntityStatsApiDTO

LogEntryApiDTO

Description

Basic information of an entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

logActionTime

- **type:** integer
- **format:** int64

targetSE

- **type:** string

targetUuid

- **type:** string

currentSE

- **type:** string

currentSEUuid

- **type:** string

newSE

- **type:** string

newSEUuid

- **type:** string

category

- **type:** string

subCategory

- **type:** string

description

- **type:** string

shortDescription

- **type:** string

state

- **type:** string

severity

- **type:** string

count

- **type:** integer
- **format:** int64

total

- **type:** integer
- **format:** int64

user

- **type:** string

type

- **type:** string

importance

- **type:** number
- **format:** float

reasonCommodities

- **type:** array
- **description:** The distinct set of commodities that were the reason for the action. Not all actions are driven by commodities. So its possible that this can be an empty list.
- **uniqueItems:** True
- – **type:** string

reasonCommodityRelationType

- **type:** string
- **description:** Relation type (bought/sold) of the reason commodity
- **enum:** ['sold', 'bought']

logDBId

- **type:** integer
- **format:** int64

LoggingApiDTO

Description

Model to describe the component logging level

Optional Parameters

componentLoggingLevel

- **type:** object
- **description:** A map from component name to logging level, component could be Discovery, Monitoring,Presentation, Abstraction, Analysis, Extension; logging levels could be INFO, WARN, DEBUG,TRACE
- **additionalProperties:** {'type': 'string', 'enum': ['FATAL', 'ERROR', 'WARN', 'INFO', 'DEBUG', 'TRACE', 'UNKNOWN']}

ManualDynamicConnections

Description

Inherits From

This DTO inherits from IManualConnectionsData

Optional Parameters

dynamicConnectionCriteria

- **type:** array
- **description:** Connection criteria to define dynamically connected entities.
- – **reference:** FilterApiDTO

ManualEntityDefinitionData

Description

Inherits From

This DTO inherits from IEntityDefinitionData

Required Parameters

manualConnectionData

- **type:** object
- **description:** Entity connections set manually
- **additionalProperties:** {'$ref': '#/definitions/IManualConnectionsData'}

ManualGroupConnections

Description

Inherits From

This DTO inherits from IManualConnectionsData

Required Parameters

connectedGroup

- **description:** Group for a connection.
- **reference:** BaseGroupApiDTO

ManualGroupsConnections

Description

Inherits From

This DTO inherits from IManualConnectionsData

Required Parameters

connectedGroups

- **type:** array
- **description:** Groups for a connection.
- – **reference:** BaseGroupApiDTO

ManualStaticConnections

Description

Inherits From

This DTO inherits from IManualConnectionsData

Optional Parameters

staticConnections

- **type:** array
- **description:** Oids to be statically connected to the created entity
- – **type:** string

MarketApiDTO

Description

Model to describe a Market

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

state

- **type:** string
- **description:** State
- **enum:** ['CREATED', 'READY_TO_START', 'RUNNING', 'COPYING', 'SUCCEEDED', 'STOPPING', 'STOPPED', 'DELETING']

stateProgress

- **type:** integer
- **format:** int32
- **description:** State progress indicated by a percentage, used in RUNNING state

scenario

- **description:** Scenario used to create the Plan, only for Plan Market
- **reference:** ScenarioApiDTO

saved

- **type:** boolean
- **description:** Shows if a Plan Market is saved

unplacedEntities

- **type:** boolean
- **description:** Unplaced entities, shows if there are entities that could not be placed in the Plan Market

runDate

- **type:** string
- **description:** Shows when a Plan Market started running

runCompleteDate

- **type:** string
- **description:** Shows when a Plan Market was completed

violations

- **type:** array
- **description:** Violations generated in a Plan Market
- – **reference:** ServiceEntityApiDTO

relatedPlanMarkets

- **type:** array
- **description:** Related Plan Markets generated by particular Scenarios
- – **reference:** MarketApiDTO

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['HYBRID', 'CLOUD', 'ONPREM']

planDestination

- **description:** Plan destination where the plan results would be stored, only for Plan Market.
- **reference:** BaseApiDTO

MasterImageEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

displayName

- **type:** string
- **description:** DisplayName of the masterImage used to create new VMs

numVcpus

- **type:** integer
- **format:** int32
- **description:** The number of vCPUs allocated to each VM generated from the master image

mem

- **type:** number
- **format:** float
- **description:** The amount of Mem allocated to each VM generated from the master image

storage

- **type:** number
- **format:** float
- **description:** The amount of Storage allocated to each VM generated from the master image

MaxUtilizationApiDTO

Description

Model to describe a max utilization change in the scenario

Optional Parameters

projectionDay

- **type:** integer
- **format:** int32
- **description:** Projection day, indicate when to apply the change

target

- **description:** Target of the change, entity or group. If empty, assumes global scope
- **reference:** BaseApiDTO

maxPercentage

- **type:** integer
- **format:** int32
- **description:** Max utilization threshold percentage change

selectedEntityType

- **type:** string
- **description:** Type of entity to select in global scope configuration

globalScope

- **type:** boolean

MemberFieldApiDTO

Description

Inherits From

This DTO inherits from FieldApiDTO

Required Parameters

property

- **type:** string
- **description:** The property of interest.
- **readOnly:** True
- **enum:** ['COUNT']

Optional Parameters

entityType

- **type:** string
- **description:** Restricts the property to entities of this type.
- **readOnly:** True
- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

direct

- **type:** boolean
- **description:** Whether this field refers to direct or indirect members (default = direct).
- **readOnly:** True

MetricApiDTO

Description

Model to describe a Metric

Required Parameters

name

- **type:** string
- **description:** Name of the metric

units

- **type:** string
- **description:** Units, E.G. Mhz, KB

utilization

- **description:** Utilization values, capacity and percentage
- **reference:** MetricUtilizationApiDTO

Optional Parameters

percentile

- **description:** Percentile values, capacity, percentage, rank and observation days
- **reference:** MetricPercentileApiDTO

MetricPercentileApiDTO

Description

Model to describe a single percentile metric, e.g. the VCPU 95th percentile over 30 days of a VM.

Required Parameters

capacity

- **type:** number
- **format:** float
- **description:** Capacity

observationDays

- **type:** integer
- **format:** int32
- **description:** Number of observation days used to calculate the percentile

percentage

- **type:** number
- **format:** float
- **description:** Percentage, usage over capacity

rank

- **type:** integer
- **format:** int32
- **description:** Rank. E.G. 95th percentile

value

- **type:** number
- **format:** float
- **description:** Value of the percentile

MetricUtilizationApiDTO

Description

Model to describe a single utilization metric, e.g. the VCPU of a VM.

Required Parameters

capacity

- **type:** number
- **format:** float
- **description:** Capacity

percentage

- **type:** number
- **format:** float
- **description:** Percentage of utilization over capacity

value

- **type:** number
- **format:** float
- **description:** Utilization value

MetricsSnapshotApiDTO

Description

Model to describe the Metrics of a snapshot, contains a list of Metrics for a moment in time

Required Parameters

time

- **type:** integer
- **format:** int64
- **description:** When the metrics were recorded, in milliseconds

Optional Parameters

metrics

- **type:** array
- **description:** List of metrics, e.g. VCPU, VMem, , ...
- – **reference:** MetricApiDTO

MigrateObjectApiDTO

Description

Model to describe a replace change in the scenario

Optional Parameters

projectionDay

- **type:** integer
- **format:** int32
- **description:** Projection day, indicate when to apply the change

source

- **description:** Source of the migration, entity or group - [DEPRECATED - Use: sources]
- **reference:** BaseApiDTO

destination

- **description:** Destination of the migration - [DEPRECATED - Use: destinations]
- **reference:** BaseApiDTO

sources

- **type:** array
- **description:** Sources of the migration, entities or groups.
- – **reference:** BaseApiDTO

destinations

- **type:** array
- **description:** Destinations of the migration, entities or groups.
- – **reference:** BaseApiDTO

destinationEntityType

- **type:** string
- **description:** Entity type of migration destination, default is VirtualMachine. Case 1, migration source VirtualMachine: destination entity must be VirtualMachine. Case 2, migration source Database Server: destination entity can be VirtualMachine or DatabaseServer.
- **enum:** ['VirtualMachine', 'DatabaseServer']

removeNonMigratingWorkloads

- **type:** boolean
- **description:** If true, remove the existing Virtual Machines, Databases and Database Servers from the scope that are not migrating, so that only the newly migrated entities are included in the plan results.

MultiTextFieldValueApiDTO

Description

Inherits From

This DTO inherits from FieldValueApiDTO

Required Parameters

value

- **type:** array
- **description:** The values associated with the field.
- **readOnly:** True
- – **type:** string

NameValueInputDTO

Description

Deployment Profile provider parameters

Optional Parameters

name

- **type:** string
- **description:** Parameter type. For vCenter, this can be a name or id. For all other target types, this must be an id.
- **enum:** ['name', 'id']

value

- **type:** string

NetworkEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

networkName

- **type:** string
- **description:** Network's name.

clusterName

- **type:** string
- **description:** Network's cluster name. [DEPRECATED - EOL: 8.15.1 - Use: clusterNames]

dataCenterName

- **type:** string
- **description:** Network's data center name.

clusterNames

- **type:** array
- **description:** Clusters that the network is part of.
- – **type:** string

vlanId

- **type:** integer
- **format:** int32

vswitchType

- **type:** string

NoSqlEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

accountThroughputLimit

- **type:** integer
- **format:** int32
- **description:** The throughput capacity limit that is set at account-level.

noSqlApiType

- **type:** string

- **description:** The NoSql engine or api the current entity is using.
- **enum:** ['NOSQL', 'MONGODB', 'CASSANDRA', 'GREMLIN', 'TABLE']

throughputType

- **type:** string
- **description:** Indicates the way of how throughput is allocate to current entity.
- **enum:** ['DEDICATED', 'SHARED', 'NONE']

autoscaling

- **type:** string
- **description:** Indicates whether the entity has autoscale setting enabled.
- **enum:** ['Enabled', 'Disabled', 'Unknown']

maxThroughput

- **type:** integer
- **format:** int32
- **description:** The maximum throughput amount provisioned for current entity.

minThroughput

- **type:** integer
- **format:** int32
- **description:** The minimum throughput amount provisioned for current entity.

effectiveThroughput

- **type:** number
- **format:** double
- **description:** The throughput amount provisioned for current entity, or the amount from host entity or sum of children entities.

databaseName

- **type:** string
- **description:** The associated database name.

NumberConditionApiDTO

Description

Inherits From

This DTO inherits from ConditionApiDTO

Required Parameters

value

- **type:** number
- **format:** double
- **description:** Value to compare against.
- **readOnly:** True

NumberFieldValueApiDTO

Description

Inherits From

This DTO inherits from FieldValueApiDTO

Required Parameters

value

- **type:** number
- **format:** double
- **description:** The value.
- **readOnly:** True

Optional Parameters

units

- **type:** string
- **description:** Units of measurement.
- **readOnly:** True

OAuth2ClientApiDTO

Description

Optional Parameters

clientName

- **type:** string
- **description:** The name of the client

clientId

- **type:** string
- **description:** The ID of the client

grantTypes

- **type:** array
- **description:** The OAuth2 grants that the client can perform in order to acquire a token
- **uniqueItems:** True
- – **type:** string
  – **enum:** ['client_credentials']

clientAuthenticationMethods

- **type:** array
- **description:** The authentication methods that the client supports when acquiring a token. * client_secret_basic: Provide the credentials using the HTTP Basic Authentication Scheme * client_secret_post: Provide the credentials in the HTTP request body as a form parameter

- **uniqueItems:** True
- – **type:** string
  – **enum:** ['client_secret_basic', 'client_secret_post']

scopes

- **type:** array
- **description:** The scopes that the client can request when acquiring a token
- **uniqueItems:** True
- – **type:** string

audience

- **type:** array
- **description:** The audience that the client can request when acquiring a token
- **uniqueItems:** True
- – **type:** string

createdAt

- **type:** integer
- **format:** int64
- **description:** Creation timestamp

tokenSettings

- **reference:** OAuth2ClientOutputTokenSettings

OAuth2ClientInputAccessToken

Description

Optional Parameters

ttlSeconds

- **type:** integer
- **format:** int64
- **example:** 600
- **description:** The lifespan of access tokens acquired by the client in seconds

OAuth2ClientInputTokenSettings

Description

Optional Parameters

accessToken

- **reference:** OAuth2ClientInputAccessToken

OAuth2ClientOutputAccessToken

Description

Optional Parameters

ttlSeconds

- **type:** integer
- **format:** int64
- **description:** The lifespan of access tokens acquired by the client in seconds

OAuth2ClientOutputTokenSettings

Description

Optional Parameters

accessToken

- **reference:** OAuth2ClientOutputAccessToken

OAuth2CreateClientApiDTO

Description

Optional Parameters

clientName

- **type:** string
- **description:** The name of the client

clientId

- **type:** string
- **description:** The ID of the client

grantTypes

- **type:** array
- **description:** The OAuth2 grants that the client can perform in order to acquire a token
- **uniqueItems:** True
- – **type:** string
  – **enum:** ['client_credentials']

clientAuthenticationMethods

- **type:** array
- **description:** The authentication methods that the client supports when acquiring a token. * client_secret_basic: Provide the credentials using the HTTP Basic Authentication Scheme * client_secret_post: Provide the credentials in the HTTP request body as a form parameter
- **uniqueItems:** True
- – **type:** string
  – **enum:** ['client_secret_basic', 'client_secret_post']

scopes

- **type:** array
- **description:** The scopes that the client can request when acquiring a token

- **uniqueItems:** True
- – **type:** string

audience

- **type:** array
- **description:** The audience that the client can request when acquiring a token
- **uniqueItems:** True
- – **type:** string

createdAt

- **type:** integer
- **format:** int64
- **description:** Creation timestamp

tokenSettings

- **reference:** OAuth2ClientOutputTokenSettings

clientSecret

- **type:** string
- **description:** The client secret

OAuth2CreateClientApiInputDTO

Description

Required Parameters

clientName

- **type:** string
- **description:** The name of the client

grantTypes

- **type:** array
- **description:** The OAuth2 grants that the client can perform in order to acquire a token
- **uniqueItems:** True
- – **type:** string
  – **enum:** ['client_credentials']

scopes

- **type:** array
- **description:** The scopes that the client can request when acquiring a token
- **uniqueItems:** True
- – **type:** string

Optional Parameters

clientAuthenticationMethods

- **type:** array

- **description:** The authentication methods that the client supports when acquiring a token. * client_secret_basic: Provide the credentials using the HTTP Basic Authentication Scheme * client_secret_post: Provide the credentials in the HTTP request body as a form parameter
- **uniqueItems:** True
- – **type:** string
  – **enum:** ['client_secret_basic', 'client_secret_post']

tokenSettings

- **reference:** OAuth2ClientInputTokenSettings

OAuthDataApiDTO

Description

Model to define the oAuth data.

Required Parameters

authorizationServerUrl

- **type:** string
- **description:** The URL of the authorization server.

clientId

- **type:** string
- **description:** The client id used for oAuth authorization.

clientSecret

- **type:** string
- **description:** The client secret used for oAuth authorization.

grantType

- **type:** string
- **description:** The oAuth grant type.
- **enum:** ['CLIENT_CREDENTIALS']

Optional Parameters

scope

- **type:** string
- **description:** The oAuth scope.

OnDemandCostFilterApiDTO

Description

Model to describe on-demand cost range for action filtering.

Optional Parameters

currentOnDemandCostRange

- **description:** Filter the actions by the range for current on-demand cost.
- **reference:** RangeInputApiDTO

newOnDemandCostRange

- **description:** Filter the actions by the range for destination on-demand cost. Applicable only for scale actions.
- **reference:** RangeInputApiDTO

OnPremCostsManagerApiDTO

Description

Model to describe a Category Manager for the on prem cost manager

Required Parameters

category

- **type:** string
- **description:** Category of the Manager

settings

- **type:** array
- **description:** List of Settings contained in the Manager
- – **reference:** SettingApiDTOSerializable

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

OnPremResizeActionDetailsApiDTO

Description

Inherits From

This DTO inherits from ActionDetailsApiDTO

Optional Parameters

vcpuBefore

- **type:** integer
- **format:** int32
- **description:** vCPUs before the change

vcpuAfter

- **type:** integer

- **format:** int32
- **description:** vCPUs after the change

socketsBefore

- **type:** integer
- **format:** int32
- **description:** sockets before the change

socketsAfter

- **type:** integer
- **format:** int32
- **description:** sockets after the change

coresPerSocketBefore

- **type:** integer
- **format:** int32
- **description:** cores per socket before the change

coresPerSocketAfter

- **type:** integer
- **format:** int32
- **description:** cores per socket after the change

nonDisruptiveMaxPU

- **type:** number
- **format:** double
- **description:** Non-disruptive Processing Units max resize constraint

nonDisruptiveMinPU

- **type:** number
- **format:** double
- **description:** Non-disruptive Processing Units min resize constraint

subType

- **type:** string

OrderByApiDTO

Description

A field to order results by.

Required Parameters

field

- **description:** The field to order by.
- **readOnly:** True
- **reference:** FieldApiDTO

Optional Parameters

ascending

- **type:** boolean
- **description:** Whether the order is ascending (defaults to false).
- **readOnly:** True

PMDiskAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

diskCapacity

- **type:** integer
- **format:** int64
- **description:** Disk Capacity

diskRole

- **type:** string
- **description:** Disk Role
- **enum:** ['CAPACITY', 'CACHE']

PMDiskGroupAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

disks

- **type:** array
- **description:** Disks
- – **reference:** PMDiskAspectApiDTO

PMEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

diskGroups

- **type:** array
- **description:** Disk Groups
- – **reference:** PMDiskGroupAspectApiDTO

dedicatedFailoverHost

- **type:** boolean
- **description:** If true, this host is a failover host.

cpuModel

- **type:** string
- **example:** Intel(R) Xeon(R) CPU E5-2670 0 @ 2.60GHz
- **description:** The CPU model of the host

connectedNetworks

- **type:** array
- **example:** Broadcom BCM5708C NetXtreme II GigE
- **description:** Connected Networks
- – **type:** string

automationLevel

- **type:** string
- **example:** Fully Automated
- **description:** Vendor provided automation level
- **enum:** ['FULLY_AUTOMATED', 'PARTIALLY_AUTOMATED', 'NOT_AUTOMATED', 'DISABLED']

migrationLevel

- **type:** string
- **example:** Conservative
- **description:** Vendor provided migration level

machineTypeAndModel

- **type:** string
- **example:** 9040-MR9
- **description:** Machine Type and Model

serialNumber

- **type:** string
- **example:** 130330X
- **description:** Serial Number

totalInstalledPhysicalProcessorUnits

- **type:** integer
- **format:** int32
- **example:** 80
- **description:** Total Installed Physical Processor Units

supportedProcessorCompatibilityModes

- **type:** string

- **example:** POWER7, POWER8, POWER9
- **description:** Supported Processor Compatibility Modes

latestSupportedProcessorGeneration

- **type:** string
- **example:** 9
- **description:** Latest Supported Processor Generation

totalInstalledPhysicalMemory

- **type:** integer
- **format:** int32
- **example:** 4096
- **description:** Total Installed Physical Memory

hypervisorUsedMemory

- **type:** integer
- **format:** int32
- **example:** 4096
- **description:** Hypervisor Used Memory

memoryAssignedToVMs

- **type:** integer
- **format:** int32
- **example:** 4096
- **description:** Memory Assigned To VMs

logicalMemoryBlockSize

- **type:** integer
- **format:** int32
- **example:** 256
- **description:** Logical Memory Block Size

supportedVirtualGpuTypes

- **type:** string
- **example:** 1q, 2q, 4q, 8q, 16q
- **description:** Comma separated list of Supported vGPU Types

installedGpuDevices

- **type:** string
- **example:** T4, A16
- **description:** Comma separated list of Installed GPU Devices

hardwareVersion

- **type:** string
- **example:** 19.0
- **description:** Hardware Compatibility Version

vendorLink

- **type:** string
- **description:** Link to navigate to the entity in vendor UI.

hostFailuresClusterTolerates

- **type:** integer
- **format:** int32
- **description:** Host failures that the cluster tolerates

processorPools

- **type:** array
- – **type:** string

PaginationApiDTO

Description

Pagination parameters.

Optional Parameters

orderBy

- **type:** array
- **description:** Order by these fields.
- **readOnly:** True
- – **reference:** OrderByApiDTO

cursor

- **type:** string
- **description:** Cursor for the next batch.
- **readOnly:** True

limit

- **type:** integer
- **format:** int32
- **description:** Maximum number of records to fetch.
- **readOnly:** True

ParkableEntityApiDTO

Description

Parkable Entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

environmentType

- **type:** string
- **example:** ONPREM
- **description:** On-Prem environment type
- **enum:** ['HYBRID', 'CLOUD', 'ONPREM', 'UNKNOWN']

accountOID

- **type:** integer
- **format:** int64
- **example:** 74613608411667
- **description:** Unique ID of the account

accountName

- **type:** string
- **example:** dev-account
- **description:** Display name of the account

provider

- **type:** string
- **example:** AWS
- **description:** Name of the cloud provider
- **enum:** ['AWS', 'AZURE', 'GCP', 'HYBRID', 'UNKNOWN']

state

- **type:** string
- **example:** STOPPED
- **description:** Last known state of the entity
- **enum:** ['RUNNING', 'STOPPED', 'SUSPENDED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'STARTING', 'STOPPING']

parkable

- **type:** boolean
- **example:** True
- **description:** Signifies whether the entity can be parked. [DEPRECATED - EOL: 8.15.1 - Use: restrictions]

entityType

- **type:** string
- **example:** COMPUTE
- **description:** Type of the entity
- **enum:** ['VirtualMachine', 'DatabaseServer', 'Database', 'VirtualMachineSpec']

regionOID

- **type:** integer
- **format:** int64
- **example:** 74613608411668
- **description:** Unique ID of the region, the entity belongs to

regionName

- **type:** string

- **example:** us-east-1
- **description:** Name of the region

instanceType

- **type:** string
- **example:** t3.micro
- **description:** Type of the instance

targetOID

- **type:** integer
- **format:** int64
- **example:** 746136084116679
- **description:** Unique ID of the target

datacenterName

- **type:** string
- **example:** VC1
- **description:** Display name of the Data Center

cost

- **type:** number
- **format:** double
- **example:** 10.24
- **description:** On demand hourly cost of the entity

tags

- **type:** array
- **description:** List of tags attached to the entity
- – **reference:** TagApiDTO

schedules

- **type:** array
- **description:** List of schedules attached to the entity
- – **reference:** ParkableEntityScheduleApiDTO

groups

- **type:** array
- **description:** List of the Groups attached to the entity
- – **reference:** ParkableEntityGroupApiDTO

guestOSName

- **type:** string
- **example:** Linux
- **description:** Guest OS Name of the entity

restrictions

- **type:** array
- **description:** List of the restrictions placed on the entity
- – **reference:** ParkableEntityRestrictionApiDTO

controlledBy

- **type:** integer
- **format:** int64
- **example:** 74613608411668
- **description:** Unique ID of the entity that this entity is controlled by. For example, if this is a Virtual Machine that is part of a scale group, this would be the ID of the scale group

smartParkingRecommendation

- **description:** [BETA]: Smart Parking Schedule Recommendation for the entity
- **reference:** ParkableEntitySmartParkingRecommendationApiDTO

schedulePausedUntil

- **type:** integer
- **format:** int64
- **example:** 1673273639
- **description:** The time until which the schedule is paused only applicable if entity has parking schedule, in epoch time format

cloudServiceName

- **type:** string
- **description:** Cloud service name is the resource bought from the cloud service provider
- **enum:** ['AMAZON_RDS', 'AMAZON_EC2', 'AMAZON_EC2_AUTO_SCALING', 'AMAZON_REDSHIFT', 'AZURE_APP_SERVICE', 'AZURE_COSMOS_DB', 'AZURE_DISK_STORAGE', 'AZURE_SQL_DATABASE', 'AZURE_SQL_MANAGED_INSTANCE', 'AZURE_SYNAPSE_SQL_POOL', 'AZURE_VIRTUAL_MACHINES', 'AZURE_VIRTUAL_MACHINE_SCALE_SETS', 'GCP_DISK_STORAGE', 'GCP_VIRTUAL_MACHINES', 'GCP_MANAGED_INSTANCE_GROUP']

customInstanceType

- **type:** string
- **example:** n2d-custom-2-1024
- **description:** The entity's custom instance type. Similar to instanceType but provides customization details about the type of instance the workload is based on

startDelay

- **type:** integer
- **format:** int32
- **example:** 3
- **description:** [BETA]: The number of minutes a START action is delayed on this entity. The delay is relative to when a START action is executed on the parking group that contains it

stopDelay

- **type:** integer
- **format:** int32
- **example:** 3
- **description:** [BETA]: The number of minutes a STOP action is delayed on this entity. The delay is relative to when a STOP action is executed on the parking group that contains it

ParkableEntityDetailsApiDTO

Description

Details about a Parkable Entity. This includes the summary information from /parking/entities API along with additional details specific to the entity.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

environmentType

- **type:** string
- **example:** ONPREM
- **description:** On-Prem environment type
- **enum:** ['HYBRID', 'CLOUD', 'ONPREM', 'UNKNOWN']

accountOID

- **type:** integer
- **format:** int64
- **example:** 74613608411667
- **description:** Unique ID of the account

accountName

- **type:** string
- **example:** dev-account
- **description:** Display name of the account

provider

- **type:** string
- **example:** AWS
- **description:** Name of the cloud provider
- **enum:** ['AWS', 'AZURE', 'GCP', 'HYBRID', 'UNKNOWN']

state

- **type:** string
- **example:** STOPPED
- **description:** Last known state of the entity
- **enum:** ['RUNNING', 'STOPPED', 'SUSPENDED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'STARTING', 'STOPPING']

parkable

- **type:** boolean

- **example:** True
- **description:** Signifies whether the entity can be parked. [DEPRECATED - EOL: 8.15.1 - Use: restrictions]

entityType

- **type:** string
- **example:** COMPUTE
- **description:** Type of the entity
- **enum:** ['VirtualMachine', 'DatabaseServer', 'Database', 'VirtualMachineSpec']

regionOID

- **type:** integer
- **format:** int64
- **example:** 74613608411668
- **description:** Unique ID of the region, the entity belongs to

regionName

- **type:** string
- **example:** us-east-1
- **description:** Name of the region

instanceType

- **type:** string
- **example:** t3.micro
- **description:** Type of the instance

targetOID

- **type:** integer
- **format:** int64
- **example:** 746136084116679
- **description:** Unique ID of the target

datacenterName

- **type:** string
- **example:** VC1
- **description:** Display name of the Data Center

cost

- **type:** number
- **format:** double
- **example:** 10.24
- **description:** On demand hourly cost of the entity

tags

- **type:** array
- **description:** List of tags attached to the entity
- – **reference:** TagApiDTO

schedules

- **type:** array

- **description:** List of schedules attached to the entity
- – **reference:** ParkableEntityScheduleApiDTO

groups

- **type:** array
- **description:** List of the Groups attached to the entity
- – **reference:** ParkableEntityGroupApiDTO

guestOSName

- **type:** string
- **example:** Linux
- **description:** Guest OS Name of the entity

restrictions

- **type:** array
- **description:** List of the restrictions placed on the entity
- – **reference:** ParkableEntityRestrictionApiDTO

controlledBy

- **type:** integer
- **format:** int64
- **example:** 74613608411668
- **description:** Unique ID of the entity that this entity is controlled by. For example, if this is a Virtual Machine that is part of a scale group, this would be the ID of the scale group

smartParkingRecommendation

- **description:** [BETA]: Smart Parking Schedule Recommendation for the entity
- **reference:** ParkableEntitySmartParkingRecommendationApiDTO

schedulePausedUntil

- **type:** integer
- **format:** int64
- **example:** 1673273639
- **description:** The time until which the schedule is paused only applicable if entity has parking schedule, in epoch time format

cloudServiceName

- **type:** string
- **description:** Cloud service name is the resource bought from the cloud service provider
- **enum:** ['AMAZON_RDS', 'AMAZON_EC2', 'AMAZON_EC2_AUTO_SCALING', 'AMAZON_REDSHIFT', 'AZURE_APP_SERVICE', 'AZURE_COSMOS_DB', 'AZURE_DISK_STORAGE', 'AZURE_SQL_DATABASE', 'AZURE_SQL_MANAGED_INSTANCE', 'AZURE_SYNAPSE_SQL_POOL', 'AZURE_VIRTUAL_MACHINES', 'AZURE_VIRTUAL_MACHINE_SCALE_SETS', 'GCP_DISK_STORAGE', 'GCP_VIRTUAL_MACHINES', 'GCP_MANAGED_INSTANCE_GROUP']

customInstanceType

- **type:** string
- **example:** n2d-custom-2-1024
- **description:** The entity's custom instance type. Similar to instanceType but provides customization details about the type of instance the workload is based on

startDelay

- **type:** integer
- **format:** int32
- **example:** 3
- **description:** [BETA]: The number of minutes a START action is delayed on this entity. The delay is relative to when a START action is executed on the parking group that contains it

stopDelay

- **type:** integer
- **format:** int32
- **example:** 3
- **description:** [BETA]: The number of minutes a STOP action is delayed on this entity. The delay is relative to when a STOP action is executed on the parking group that contains it

virtualMachineSpecDetails

- **description:** Additional details about the entity if the ParkingEntityType is VirtualMachineSpec
- **reference:** ParkableVirtualMachineSpecApiDTO

ParkableEntityEnforcementPauseApiDTO

Description

represents the partial failure per entity for enforcement pause on list of parkable entities

Optional Parameters

error

- **type:** string
- **example:** entityUuid not found
- **description:** error indicating the reason for failure

entityUUID

- **type:** string

ParkableEntityEnforcementPauseInputDTO

Description

represents the enforcement pause on list of parkable entities either until a given time or in duration of minutes

Optional Parameters

until

- **type:** string
- **example:** 2022-11-29T01:02:03+00:00
- **description:** Time when the enforcement pause on parkable entities ends. Should be in ISO 8601 format

durationMinutes

- **type:** integer
- **format:** int32

- **example:** 60
- **description:** duration in minutes to keep enforcement pause on parkable entities

entityUuids

- **type:** array
- **description:** list of parkable entities to pause enforcement on
- – **type:** string

ParkableEntityGroupApiDTO

Description

represents groups which are attached to the parkable entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

groupType

- **type:** string
- **example:** Resource
- **description:** Type of the Group
- **enum:** ['RESOURCE']

ParkableEntityInputDTO

Description

Optional Parameters

entityTypes

- **type:** array
- – **type:** string
  – **enum:** ['VirtualMachine', 'DatabaseServer', 'Database', 'VirtualMachineSpec']

status

- **type:** array
- – **type:** string
  – **enum:** ['RUNNING', 'STOPPED', 'SUSPENDED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'STARTING', 'STOPPING']

providers

- **type:** array
- **description:** Parking Cloud Providers
- – **type:** string
  – **enum:** ['AWS', 'AZURE', 'GCP', 'HYBRID', 'UNKNOWN']

parkableState

- **type:** string
- **description:** Parkable state
- **enum:** ['UNPARKABLE', 'PARKABLE']

criteriaList

- **type:** array
- – **reference:** FilterApiDTO

logicalOperator

- **type:** string
- **description:** Logical operator to be applied across all the criteria. It can take values 'AND', 'OR'. If omitted, defaults to 'AND'
- **enum:** ['AND', 'OR']

ParkableEntityRestrictionApiDTO

Description

Parent class for restrictions details. Implementing classes include: ParkableEntityUnparkableRestrictionApiDTO.

Required Parameters

restrictionType

- **type:** string
- **enum:** ['UNPARKABLE']

ParkableEntityScheduleApiDTO

Description

Represents time span based schedule display name and uuid of the time span based schedule attached to the parkable entity

Optional Parameters

displayName

- **type:** string
- **example:** Up 7-5 on weekdays
- **description:** Display name of the time span based schedule the parkable entities is attached to

uuid

- **type:** string
- **example:** 74613608411667

- **description:** UUID of the time span based schedule the parkable entities are attached to

pauseEnforcementUntil

- **type:** integer
- **format:** int64
- **example:** 1673273639
- **description:** Pause Enforcement time over the entity, in epoch time format

attachedBy

- **description:** Details of the user/policy responsible for attaching the schedule to entity
- **reference:** ParkingEntitySchedulerApiDTO

source

- **type:** string
- **description:** [BETA]: Time span schedule source
- **enum:** ['USER', 'SMART_PARK_RECOMMENDATION']

ParkableEntitySmartParkingRecommendationApiDTO

Description

[BETA]: Smart parking recommendation details

Optional Parameters

uuid

- **type:** string
- **description:** [BETA]: Smart Parking schedule uuid

displayName

- **type:** string
- **description:** [BETA]: Smart Parking schedule name

ParkableEntityStatsInputDTO

Description

Required Parameters

groupBy

- **type:** array
- **description:** List of groupBys to filter
- – **type:** string
  – **enum:** ['ENTITY_TYPE', 'CSP']

ParkableEntityUUIDSetDTO

Description

list of parkable entity uuids

Required Parameters

entityUuids

- **type:** array
- **description:** list of parkable entity uuids
- – **type:** string

ParkableEntityUnparkableRestrictionApiDTO

Description

Inherits From

This DTO inherits from ParkableEntityRestrictionApiDTO

Optional Parameters

attachedBy

- **description:** Details of the user/policy responsible for adding the unparkable restriction
- **reference:** ParkingEntitySchedulerApiDTO

ParkableVirtualMachineSpecApiDTO

Description

Details about a parkable Virtual Machine Spec entity such as a cloud scale group.A cloud scale group is is a collection of virtual machine (VM) instances that are created from a common source. Ex: AWS Auto Scaling Groups, Azure Virtual Machine Scale Sets.

Optional Parameters

instanceTemplateName

- **type:** string
- **description:** Name of template or configuration used to create the scale group. Ex: Name of Launch Template or Launch Configuration in AWS, Auto Scale Profile in Azure or the Instance Template name in Google."

currentConfiguration

- **description:** The current minimum group size, desired group size and maximum group size for a scale group
- **reference:** ParkableVirtualMachineSpecConfigApiDTO

startConfiguration

- **description:** The minimum group size, desired group size and maximum group size that will be set for the scale group when a START action is executed
- **reference:** ParkableVirtualMachineSpecConfigApiDTO

instances

- **type:** array
- **description:** The virtual machine instances in a cloud scale group
- – **reference:** ParkableVirtualMachineSpecInstanceApiDTO

scalingMode

- **type:** string
- **description:** The scaling mode of a cloud scale group
- **enum:** ['AZURE_AUTOSCALE_METRIC_BASED', 'AZURE_AUTOSCALE_SPECIFIC_INSTANCE_COUNT', 'AZURE_MANUAL_SCALE']

ParkableVirtualMachineSpecConfigApiDTO

Description

The minimum group size, desired group size and maximum group size for a scale group

Optional Parameters

minSize

- **type:** integer
- **format:** int32
- **description:** Minimum group size

maxSize

- **type:** integer
- **format:** int32
- **description:** Maximum group size

desiredSize

- **type:** integer
- **format:** int32
- **description:** Default size of group when no provider-initiated scaling actions are applied

ParkableVirtualMachineSpecInstanceApiDTO

Description

Details about a virtual machine instance within a parkable Virtual Machine Spec entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

instanceType

- **type:** string
- **example:** t3.micro
- **description:** Type of the instance

state

- **type:** string
- **example:** STOPPED
- **description:** Last known state of the instance
- **enum:** ['RUNNING', 'STOPPED', 'SUSPENDED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'STARTING', 'STOPPING']

tags

- **type:** array
- **description:** List of tags attached to the instance
- – **reference:** TagApiDTO

availabilityZoneName

- **type:** string
- **example:** aws-us-west-2a
- **description:** Name of the availability zone the instance belongs to

ParkingEntitySchedulerApiDTO

Description

Represents the details of the user/policy responsible for attaching the restriction/schedule to parkable entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

type

- **type:** string
- **example:** POLICY
- **description:** Whether user or policy was responsible for attaching the restriction/schedule
- **enum:** ['USER', 'POLICY']

ParkingGroupApiDTO

Description

Model to describe a Parking Group

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

state

- **type:** string
- **example:** RUNNING, STOPPED, STARTING, STOPPING
- **description:** The power state of the group
- **enum:** ['RUNNING', 'STOPPED', 'SUSPENDED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'STARTING', 'STOPPING']

cost

- **type:** number
- **format:** double
- **example:** 20.25
- **description:** On demand hourly cost of the member entities

memberOids

- **type:** array
- **description:** Oid set for members of the group
- **uniqueItems:** True
- – **type:** integer
  – **format:** int64

schedule

- **description:** Schedule attached to the group
- **reference:** ParkableEntityScheduleApiDTO

providers

- **type:** array
- **example:** AWS, GCP
- **description:** Parking group Cloud Providers
- – **type:** string
  – **enum:** ['AWS', 'AZURE', 'GCP', 'HYBRID', 'UNKNOWN']

createdAt

- **type:** integer
- **format:** int64
- **example:** 1742763366
- **description:** The time at which the parking group was created, in epoch format

updatedAt

- **type:** integer
- **format:** int64
- **example:** 1742763366
- **description:** The time at which the parking group was last updated, in epoch format

ParkingGroupInputDTO

Description

Optional Parameters

criteriaList

- **type:** array
- **description:** Criteria list of elements, expression type, expression value to get the desired group
- – **reference:** FilterApiDTO

logicalOperator

- **type:** string
- **description:** Logical operator to be applied across all the criteria. It can take values 'AND', 'OR'. If omitted, defaults to 'AND'
- **enum:** ['AND', 'OR']

providers

- **type:** array
- **description:** Parking Cloud Providers
- – **type:** string
  – **enum:** ['AWS', 'AZURE', 'GCP', 'HYBRID', 'UNKNOWN']

states

- **type:** array
- **example:** [RUNNING, STOPPED, STARTING, STOPPING]
- **description:** The power states of the group
- – **type:** string
  – **enum:** ['RUNNING', 'STOPPED', 'SUSPENDED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'STARTING', 'STOPPING']

ParkingGroupMemberUpdateInputDTO

Description

Updated member information

Optional Parameters

entityUuid

- **type:** integer
- **format:** int64
- **description:** Parking group member uuid

startDelay

- **type:** integer
- **format:** int32
- **example:** 3
- **description:** The number of minutes a START action is delayed on this entityThe delay is relative to when a START action is executed on the parking group that contains it

stopDelay

- **type:** integer
- **format:** int32
- **example:** 3
- **description:** The number of minutes a STOP action is delayed on this entityThe delay is relative to when a STOP action is executed on the parking group that contains it

ParkingGroupUpdateInputDTO

Description

Updated group information

Optional Parameters

initialState

- **type:** string
- **example:** RUNNING, STOPPED, STARTING, STOPPING
- **description:** The initial power states of the group
- **enum:** ['RUNNING', 'STOPPED', 'SUSPENDED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'STARTING', 'STOPPING']

memberInfo

- **type:** array
- **description:** List of parking group members with start and stop delay information
- **uniqueItems:** True
- – **reference:** ParkingGroupMemberUpdateInputDTO

ParkingItemApiDTO

Description

Inherits From

This DTO inherits from ScheduleItemApiDTO

Required Parameters

state

- **type:** string
- **description:** expected state during the time span
- **enum:** ['IGNORE', 'ON', 'OFF']

ParkingPolicyApiDTO

Description

Model to describe a parking policy, used to assign timespan based schedules to entities based on global, cloud provider, account and resource group level filters

Required Parameters

impacts

- **description:** Impacts to be taken by the policy.
- **reference:** ParkingPolicyImpactsApiDTO

level

- **type:** string
- **description:** Specifies the level at which the policy is applicable.
- **enum:** ['GLOBAL', 'CLOUD_PROVIDER', 'ACCOUNT', 'RESOURCE_GROUP', 'DATACENTER', 'TARGET']

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

scopes

- **type:** array
- **description:** List of scope object to which the parking policy is applicable, like details of account for account level policies.
- – **reference:** BaseApiDTO

criteriaList

- **type:** array
- **description:** List of filters to get the desired the entities to which the policy is applicable.
- – **reference:** FilterApiDTO

priority

- **type:** integer
- **format:** int32
- **description:** Priority is used to identify the applicable priority if there are multiple policies at same level

createdAt

- **type:** integer
- **format:** int64
- **example:** 1673273639

- **description:** The time at which the policy was created, in epoch format

updatedAt

- **type:** integer
- **format:** int64
- **example:** 1673273639
- **description:** The time at which the policy was last updated, in epoch format

ParkingPolicyImpactsApiDTO

Description

Impacts to be taken by the policy.

Optional Parameters

attachSchedule

- **description:** time span based schedule to be attached by policy.
- **reference:** ParkingPolicyTimeSpansApiDTO

restrictUnparkable

- **type:** boolean
- **description:** to mark unparkable by policy.

ParkingPolicyStatsInputDTO

Description

Required Parameters

groupBy

- **type:** array
- **description:** List of groupBys to filter
- – **type:** string
  – **enum:** ['LEVEL']

ParkingPolicyTimeSpansApiDTO

Description

Parking Policy time span schedule

Optional Parameters

timespanScheduleUuid

- **type:** string
- **description:** timespanScheduleUuid used to uniquely recognize the time span Schedule.

displayName

- **type:** string

- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

PatchedTargetDataApiDTO

Description

Model to describe a patched target data.

Required Parameters

patchedFields

- **type:** array
- **description:** Patched target fields.
- – **reference:** PatchedTargetFieldApiDTO

probeType

- **type:** string
- **description:** Probe type of the patched target.

PatchedTargetFieldApiDTO

Description

Model to describe a field in patched target data.

Optional Parameters

fieldName

- **type:** string
- **description:** Patched field name.

fieldValue

- **type:** string
- **description:** Patched field value.

PeakBaselineApiDTO

Description

Model to describe a peak baseline change in the scenario

Optional Parameters

date

- **type:** string
- **description:** Date of the utilization peak

target

- **description:** Target of the change
- **reference:** BaseApiDTO

PlacementInfoDTO

Description

Optional Parameters

computeResources

- **type:** array
- **description:** Compute resources that the reservation is placed in
- – **reference:** ResourceApiDTO

storageResources

- **type:** array
- **description:** Storage resources that the reservation is placed in
- – **reference:** ResourceApiDTO

networkResources

- **type:** array
- **description:** Network resources that the reservation is placed in
- – **reference:** ResourceApiDTO

failureInfos

- **type:** array
- **description:** Details about why the reservation failed. This will only be populated if the reservation failed.
- – **reference:** ReservationFailureInfoDTO

invalidInfo

- **description:** Details about why the reservation is invalid. This will only be populated if the reservation is invalid.
- **reference:** ReservationInvalidInfoApiDTO

PlacementOptionApiDTO

Description

Model to describe the property of provider options.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

constraintType

- **type:** string
- **description:** constraint type, if a commodity is AccessCommodity, then the class name of that commodity is a constraint type
- **enum:** ['ApplicationCommodity', 'ClusterCommodity', 'DataCenterCommodity', 'DatastoreCommodity etc']

scope

- **description:** scope or placement policy
- **reference:** BaseApiDTO

target

- **description:** for placement policy, if it's an user placement policy is null.
- **reference:** TargetApiDTO

numPotentialEntities

- **type:** integer
- **format:** int32
- **description:** example: other VMs, which buys this constraint but not on this host

key

- **type:** string
- **description:** commodity key, this information is needed when trying to retrieve the intersaction of providers or consumers

currentEntities

- **type:** array
- **description:** for consumers only. example: VMs, which are currently on this host per constraint they are buying
- – **reference:** ServiceEntityApiDTO

PlacementOptionApiInputDTO

Description

Model to describe the property of provider options.

Optional Parameters

constraintType

- **type:** string
- **description:** constraint type, if a commodity is AccessCommodity, then the class name of that commodity is a constraint type
- **enum:** ['ApplicationCommodity', 'ClusterCommodity', 'DataCenterCommodity', 'DatastoreCommodity etc']

key

- **type:** string
- **description:** commodity key, this information is needed when trying to retrieve the intersection of providers or consumers

PlacementParametersDTO

Description

Optional Parameters

count

- **type:** integer
- **format:** int32
- **description:** Number of Virtual Machines to Place

templateID

- **type:** string
- **description:** Template Uuid used for the Placement

entityNames

- **type:** array
- **description:** Name of the Placed Virtual Machine
- – **type:** string

constraintIDs

- **type:** array
- **description:** List of uuids that will define the constraints that new VMs should be compliant to, allowed uuids are: Cluster, Datacenter, VirtualDataCenter, Network, Placement Policy. The possible providers are the intersection of all constraints passed.
- **uniqueItems:** True
- – **type:** string

geographicRedundancy

- **type:** boolean
- **description:** Enable Geographic Redundancy

PlanDestinationApiDTO

Description

Model to describe a plan destination. A plan destination is an external destination where plan results can be uploaded.

Required Parameters

businessUnit

- **description:** Business account to plan destination.
- **reference:** BaseApiDTO

exportDescription

- **type:** string
- **description:** Plan export description message on this plan destination.

exportProgressPercentage

- **type:** integer

- **format:** int32
- **description:** Plan export progress percentage on this plan destination.

exportState

- **type:** string
- **description:** Plan export state on this plan destination.
- **enum:** ['NONE', 'REJECTED', 'IN_PROGRESS', 'SUCCEEDED', 'FAILED']

hasExportedData

- **type:** boolean
- **description:** Boolean, true if this plan destination has uploaded data.

market

- **description:** Plan market currently uploaded to plan destination.
- **reference:** BaseApiDTO

numErrors

- **type:** integer
- **format:** int32
- **description:** Number of errors on the plan destination.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

PolicyApiDTO

Description

Model to describe a Workload policy, used to manage the entities in the Environment. e.g. Restrict the number of VMs that can use a given datastore, Limit VM mobility by restricting a collection of VMs to a specified cluster or group of hosts, Expand VM mobility by merging a number of clusters, Keep workload running on hosts that have the required licenses

Required Parameters

type

- **type:** string
- **description:** Placement Policy Type
- **enum:** ['AT_MOST_N', 'BIND_TO_GROUP', 'BIND_TO_COMPLEMENTARY_GROUP', 'MUST_RUN_TOGETHER', 'AT_MOST_N_BOUND', 'MERGE', 'BIND_TO_GROUP_AND_LICENSE', 'MUST_NOT_RUN_TOGETHER', 'EXCLUSIVE_BIND_TO_GROUP']

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

name

- **type:** string
- **description:** Name of the policy

enabled

- **type:** boolean
- **description:** Shows if the policy is enabled

capacity

- **type:** integer
- **format:** int32
- **description:** Indicate the max workload entities at the same time per placement entites, used when type is AT_MOST_N_BOUND

commodityType

- **type:** string
- **description:** Commodity type used for the policy

mergeType

- **type:** string
- **description:** Type of Group that is merged, used only when type is MERGE
- **enum:** ['Cluster', 'StorageCluster', 'DataCenter', 'DesktopPool']

consumerGroup

- **description:** Consumer Group, used when type is any but MERGE
- **reference:** BaseApiDTO

providerGroup

- **description:** Provider Group, used when type is any but MERGE
- **reference:** BaseApiDTO

mergeGroups

- **type:** array
- **description:** List of Groups to merge, used when type is MERGE
- – **reference:** BaseApiDTO

providerEntityType

- **type:** string
- **description:** The entity type for the provider. This property is only applicable when policy type is set to MUST_RUN_TOGETHER or MUST_NOT_RUN_TOGETHER. When policy type is set to

MUST_RUN_TOGETHER or MUST_NOT_RUN_TOGETHER, this field should be set with type of provider where members of consumer group must run or must not run together on it depending on the policy type.

- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

enableCreateResourcePool

- **type:** boolean
- **description:** Enable create resource pool during cross VM moves, used when type is any but MERGE.When enabled resource pool, of which VM is part of at source , is created in destination if it does not exists in destination during a cross cluster or vcenter VM move.By default this is disabled(false)

PolicyApiInputDTO

Description

Model to describe a Placement Policy

Required Parameters

policyName

- **type:** string
- **description:** Name of the policy

type

- **type:** string
- **description:** Placement Policy Type
- **enum:** ['AT_MOST_N', 'BIND_TO_GROUP', 'BIND_TO_COMPLEMENTARY_GROUP', 'MUST_RUN_TOGETHER', 'AT_MOST_N_BOUND', 'MERGE', 'BIND_TO_GROUP_AND_LICENSE', 'MUST_NOT_RUN_TOGETHER', 'EXCLUSIVE_BIND_TO_GROUP']

Optional Parameters

sellerUuid

- **type:** string
- **description:** Uuid of the provider Group in a Placement Policy

buyerUuid

- **type:** string
- **description:** Uuid of the consumer Group in a Placement Policy

mergeUuids

- **type:** array
- **description:** Uuids of the Groups in a Merge Policy

- – **type:** string

mergeType

- **type:** string
- **description:** Group Type in a Merge Policy
- **enum:** ['Cluster', 'StorageCluster', 'DataCenter', 'DesktopPool']

capacity

- **type:** integer
- **format:** int32
- **description:** For an AT_MOST_N or AT_MOST_N_BOUND policy, the number of consumers to allow on a provider entity

enableCreateResourcePool

- **type:** boolean
- **description:** Enable create resource pool during cross VM moves, used when type is any but MERGE.When enabled resource pool, of which VM is part of at source , is created in destination if it does not exists in destination during a cross cluster or vcenter VM move.By default this is disabled(false)

enabled

- **type:** boolean
- **description:** Whether to enable the policy

providerEntityType

- **type:** string
- **description:** The entity type for the provider. This property is only applicable when the policy type is set to MUST_RUN_TOGETHER or MUST_NOT_RUN_TOGETHER. This field is set with the provider type (for example, PhysicalMachine) the policy should respect.
- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

PortChannelApiDTO

Description

Model to describe a Port Channel

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string

- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

capacity

- **description:** Capacity values
- **reference:** StatValueApiDTO

values

- **description:** Contains Average, Min, Max, Total
- **reference:** StatValueApiDTO

units

- **type:** string
- **description:** Units, used for Commodities stats. E.G. $/h

ports

- **type:** array
- **description:** List of ports in the channel
- – **reference:** StatApiDTO

filters

- **type:** array
- **description:** describe the grouping options used to generate the output
- – **reference:** StatFilterApiDTO

PortsAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

portChannels

- **type:** array
- **description:** Port Channels
- – **reference:** PortChannelApiDTO

ports

- **type:** array
- **description:** Independent ports
- – **reference:** StatApiDTO

PowerVmPMEntityAspectDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

activeMemoryExpansionCapable

- **type:** boolean
- **example:** True
- **description:** Declares if PM has active memory expansion

lparSuspendCapable

- **type:** boolean
- **example:** True
- **description:** Declares if PM can suspend VMs

ibmiLparSuspendCapable

- **type:** boolean
- **example:** True
- **description:** Declares if PM can suspend Ibmi VMs

trustedPlatformModuleCapable

- **type:** boolean
- **example:** True
- **description:** Declares if PM has trusted boot

remoteRestartCapable

- **type:** boolean
- **example:** True
- **description:** Declares if PM has remote restart

secureBootCapable

- **type:** boolean
- **example:** True
- **description:** Declares if PM has secure boot

hasMspEnabledVios

- **type:** boolean
- **example:** True
- **description:** Declares if PM has any MSP enabled VIOS

PowerVmVMEntityAspectDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

activeMemoryExpansionEnabled

- **type:** boolean
- **example:** True
- **description:** Declares if VM has active memory expansion enabled

lparSuspendCapable

- **type:** boolean
- **example:** True
- **description:** Declares if VM can suspend

ibmiLparSuspendCapable

- **type:** boolean
- **example:** True
- **description:** Declares if Ibmi VM can suspend

trustedPlatformModuleCapable

- **type:** boolean
- **example:** True
- **description:** Declares if VM has trusted boot

remoteRestartCapable

- **type:** boolean
- **example:** True
- **description:** Declares if VM host has remote restart

secureBootCapable

- **type:** boolean
- **example:** True
- **description:** Declares if VM has secure boot

redundantErrorPathReportingEnabled

- **type:** boolean
- **example:** True
- **description:** Declares if VM has redundant error path reporting enabled

hasLastActivatedProfile

- **type:** boolean
- **example:** True
- **description:** Declares if VM has last activated profile

hasPhysicalIo

- **type:** boolean
- **example:** True
- **description:** Declares if VM has physical adapters

hasHugePageSizesConfigured

- **type:** boolean
- **example:** True
- **description:** Declares if VM has huge page sizes configured

PriceAdjustmentDTO

Description

Optional Parameters

type

- **type:** string
- **enum:** ['DISCOUNT', 'INCREASE']

value

- **type:** number
- **format:** float

PrimitiveFieldApiDTO

Description

Inherits From

This DTO inherits from FieldApiDTO

Required Parameters

fieldName

- **type:** string
- **description:** The name of the primitive field.
- **readOnly:** True

ProbeApiDTO

Description

Model to describe a probe.

Required Parameters

category

- **type:** string
- **description:** Probe category.

id

- **type:** integer
- **format:** int64
- **description:** Probe id.

type

- **type:** string
- **description:** Probe type.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

error

- **type:** string
- **description:** Unexpected error while collecting probe information

ProbeDeploymentApiDTO

Description

Optional Parameters

name

- **type:** string
- **description:** Name of the Probe

version

- **type:** string
- **description:** Version of the Probe

image

- **type:** string
- **description:** Probe image path

imageHash

- **type:** string
- **description:** Probe image hash

ProbePropertyApiDTO

Description

Model to describe a probe property

Required Parameters

name

- **type:** string
- **description:** Probe property name.

value

- **type:** string

- **description:** Probe property value.

Optional Parameters

probeId

- **type:** integer
- **format:** int64
- **description:** Probe id.

targetId

- **type:** integer
- **format:** int64
- **description:** Target id.

ProbePropertyNameValuePairApiDTO

Description

Model to describe a name/value pair for probe properties.

Required Parameters

name

- **type:** string
- **description:** Probe property name.

value

- **type:** string
- **description:** Probe property value.

ProductCapabilityDTO

Description

Model to describe Product Capability information.

Optional Parameters

deploymentMode

- **type:** string
- **description:** Deployment mode, affects hiding / showing multiple UI views, options
- **enum:** ['HYBRID_SAAS', 'SERVER', 'SAAS_TRIAL', 'CLOUDABILITY_PREMIUM']

searchApiEnabled

- **type:** boolean
- **description:** Controls use of new search api

parkingEnabled

- **type:** boolean
- **description:** Controls hiding/showing parking related pages in UI

frontEndTelemetryEnabled

- **type:** boolean
- **description:** Controls enabling/disabling front end telemetry in UI

jdbcHttpSessionEnabled

- **type:** boolean
- **description:** Controls use of JDBC persistent HTTP Session

saasReportingEnabled

- **type:** boolean
- **description:** Controls hiding/showing SaaS reports page in UI

trialExpirationTime

- **type:** string
- **description:** [BETA]: When the user's trial expires in ISO 8601 Format. For SAAS_TRIAL deploymentMode

featureFlags

- **type:** object
- **description:** [BETA]: A map of feature flags. DISCLAIMER: Do not rely on these for scripting, can be removed at any time.
- **additionalProperties:** {'type': 'boolean'}

reportingEnabled

- **type:** boolean

ProductVersionDTO

Description

Contains version information about the running Turbonomic instance.

Optional Parameters

versionInfo

- **type:** string
- **example:** "versionInfo": "Turbonomic Operations Manager 8.0.1-SNAPSHOT (Build \"20200922092548000\") \"2020-09-22 20:10:04\"\n\naction-orchestrator: 8.0.1-SNAPSHOT\napi: 8.0.1-SNAPSHOT\nauth: 8.0.1-SNAPSHOT\ncost: 8.0.1-SNAPSHOT\nextractor: 8.0.1-SNAPSHOT\ngroup: 8.0.1-SNAPSHOT\nhistory: 8.0.1-SNAPSHOT\nmarket: 8.0.1-SNAPSHOT\nmediation-actionscript: 8.0.1-SNAPSHOT\nmediation-appdynamics: 8.0.1-SNAPSHOT\nmediation-appinsights: 8.0.1-SNAPSHOT\nmediation-aws: 8.0.1-SNAPSHOT\nmediation-awsbilling: 8.0.1-SNAPSHOT\nmediation-awscost: 8.0.1-SNAPSHOT\nmediation-azure: 8.0.1-SNAPSHOT\nmediation-azurecost: 8.0.1-SNAPSHOT\nmediation-azureea: 8.0.1-SNAPSHOT\nmediation-azuresp: 8.0.1-SNAPSHOT\nmediation-azurevolumes: 8.0.1-SNAPSHOT\nmediation-dynatrace: 8.0.1-SNAPSHOT\nmediation-hpe3par: 8.0.1-SNAPSHOT\nmediation-hyperv: 8.0.1-SNAPSHOT\nmediation-netapp: 8.0.1-SNAPSHOT\nmediation-nutanix: 8.0.1-SNAPSHOT\nmediation-oneview: 8.0.1-SNAPSHOT\nmediation-pure: 8.0.1-SNAPSHOT\nmediation-ucs: 8.0.1-SNAPSHOT\nmediation-vcenter: 8.0.1-SNAPSHOT\nmediation-vcenterbrowsing: 8.0.1-SNAPSHOT\nmediation-vmax: 8.0.1-SNAPSHOT\nmediation-vmm: 8.0.1-SNAPSHOT\nplan-orchestrator: 8.0.1-SNAPSHOT\nrepository: 8.0.1-SNAPSHOT\ntopology-processor: 8.0.1-SNAPSHOT"

- **description:** A multi-line string describing the version information, including that of the various internal components.

version

- **type:** string
- **example:** 8.0.1-SNAPSHOT
- **description:** The Operations Manager product version, including major and minor versions

build

- **type:** string
- **example:** 20200922211058775
- **description:** The build number associated with this deployment.

commit

- **type:** string
- **example:** 1183dd2
- **description:** The commit associated with this deployment.

branch

- **type:** string
- **example:** 8.0.1
- **description:** The branch associated with this deployment.

gitDescription

- **type:** string
- **example:** 1183dd2bb8ebca3dbcaf708e0c323cabc7b8e352
- **description:** Specific source control information associated with this deployment.

updates

- **type:** string
- **description:** Description of available updates--not yet implemented in XL.

marketVersion

- **type:** integer
- **format:** int32
- **example:** 2
- **description:** The version of the market

apiVersion

- **type:** string
- **example:** 3.1
- **description:** The version of the REST API.

PurchaseProfileDTO

Description

Model to describe a set of preferences for purchasing RIs

Required Parameters

cloudType

- **type:** string
- **description:** Cloud type
- **enum:** ['AWS', 'AZURE', 'GCP', 'HYBRID', 'UNKNOWN']

paymentOption

- **type:** string
- **description:** Preferred payment option
- **enum:** ['ALL_UPFRONT', 'PARTIAL_UPFRONT', 'NO_UPFRONT']

reservedInstanceType

- **type:** string
- **description:** Preferred offering class (Standard or Convertible)
- **enum:** ['STANDARD', 'CONVERTIBLE']

termYears

- **type:** integer
- **format:** int32
- **description:** Number of years in preferred term

QueryInputApiDTO

Description

Model to describe an input query

Required Parameters

query

- **type:** string
- **description:** Query string. The format of this depends on the QueryType chosen. For example, for a REGEX QueryType, this should be a regular expression.

type

- **type:** string
- **description:** Query type
- **enum:** ['EXACT', 'REGEX', 'CONTAINS']

Optional Parameters

caseSensitive

- **type:** boolean
- **description:** If true, the query string will match case-sensitivity. This field is only used when QueryType is not REGEX.

RIBuyActionDetailsApiDTO

Description

Inherits From

This DTO inherits from ActionDetailsApiDTO

Optional Parameters

estimatedOnDemandCost

- **type:** number
- **format:** float

estimatedRICoverage

- **type:** number
- **format:** float

historicalDemandDataset

- **type:** array
- – **reference:** StatSnapshotApiDTO

subType

- **type:** string

RIPurchaseProfilesSettingApiDTO

Description

Inherits From

This DTO inherits from SettingApiDTO

Required Parameters

value

- **description:** Value of the RI Purchase Profiles of type ReservedInstancePurchaseSettingsDTO.
- **reference:** ReservedInstancePurchaseSettingsDTO

Optional Parameters

defaultValue

- **description:** Default value before any change
- **readOnly:** True
- **reference:** ReservedInstancePurchaseSettingsDTO

RangeApiDTO

Description

Model to describe the Range of a Value in the Settings

Optional Parameters

step

- **type:** number

- **format:** double
- **description:** The Step scale factor of the slider

labels

- **type:** array
- **description:** Labels to show in the values of the slider
- – **type:** string

customStepValues

- **type:** array
- **description:** Custom step values when the slider step is not constant - [DEPRECATED - Use: stepValues]
- – **type:** integer
  – **format:** int32

stepValues

- **type:** array
- **description:** Custom step values when the slider step is not constant
- – **type:** number
  – **format:** float

RangeInputApiDTO

Description

Model to describe an input range. Either minValue or maxValue needs to be defined.

Optional Parameters

minValue

- **type:** number
- **format:** float
- **description:** Minimum value of the input range

maxValue

- **type:** number
- **format:** float
- **description:** Maximum value of the input range

Recommendation

Description

A DTO for a single recommendation to an error.

Required Parameters

description

- **type:** string
- **description:** The recommended user action to investigate/mitigate the problem.

errorType

- **type:** string
- **description:** The problem this recommendation is produced for.

ReconfigureActionDetailsApiDTO

Description

Inherits From

This DTO inherits from ActionDetailsApiDTO

Optional Parameters

vcpuBefore

- **type:** integer
- **format:** int32
- **description:** vCPUs before the change

vcpuAfter

- **type:** integer
- **format:** int32
- **description:** vCPUs after the change

socketsBefore

- **type:** integer
- **format:** int32
- **description:** sockets before the change

socketsAfter

- **type:** integer
- **format:** int32
- **description:** sockets after the change

coresPerSocketBefore

- **type:** integer
- **format:** int32
- **description:** cores per socket before the change

coresPerSocketAfter

- **type:** integer
- **format:** int32
- **description:** cores per socket after the change

RecurrenceApiDTO

Description

Model to describe the Schedule for a Settings Policy

Required Parameters

type

- **type:** string
- **description:** Type of the recurrence, Enum: DAILY, WEEKLY, MONTHLY
- **enum:** ['DAILY', 'WEEKLY', 'MONTHLY']

Optional Parameters

daysOfWeek

- **type:** array
- **description:** The day of week that the recurrence will occur. Required if the type of recurrence is WEEKLY, or if the type is MONTHLY and a week for the monthly recurrence has been set.
- – **type:** string
  – **enum:** ['Sun', 'Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat']

daysOfMonth

- **type:** array
- **description:** When recurrence type is MONTHLY, the date to apply the monthly recurrence
- – **type:** integer
  – **format:** int32

weekOfTheMonth

- **type:** array
- **description:** When recurrence type is MONTHLY, which week of the month to apply the recurrence. A value of -1 indicates last week, 0 indicates this week, and so on.
- – **type:** integer
  – **format:** int32

interval

- **type:** integer
- **format:** int32
- **description:** Frequency of the recurrence based on the recurrence type. For example, an interval of '2' with a recurrence type of MONTHLY will occur every two months.

RegionAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

latitude

- **type:** number
- **format:** double
- **description:** The latitude of the region used along with longitude to determine physical location.

longitude

- **type:** number
- **format:** double
- **description:** The longitude of the region used along with latitude to determine physical location.

numWorkloads

- **type:** integer
- **format:** int32
- **description:** Number of workloads (VMs, databases, etc.) in this region.

RelatedActionApiDTO

Description

Model to describe the relationship between two actions

Optional Parameters

action

- **description:** The basic action DTO for the related action
- **reference:** BasicActionApiDTO

actionRelationType

- **type:** string
- **description:** Action relation type
- **enum:** ['BLOCKED_BY', 'BLOCKING', 'CAUSED_BY', 'CAUSING']

RelatedEntityFieldApiDTO

Description

Inherits From

This DTO inherits from FieldApiDTO

Required Parameters

property

- **type:** string
- **description:** The related entity property of interest.
- **readOnly:** True
- **enum:** ['NAMES', 'COUNT']

relatedEntityType

- **type:** string
- **description:** The type of related entities of interest.
- **readOnly:** True
- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database',

'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

RelatedGroupFieldApiDTO

Description

Inherits From

This DTO inherits from FieldApiDTO

Required Parameters

property

- **type:** string
- **description:** The related group property of interest.
- **readOnly:** True
- **enum:** ['NAMES', 'COUNT']

relatedGroupType

- **type:** string
- **description:** The type of related groups of interest.
- **readOnly:** True
- **enum:** ['Group', 'Resource', 'Cluster', 'VMCluster', 'StorageCluster', 'BusinessAccountFolder', 'BillingFamily', 'NodePool']

RelatedStatsApiDTO

Description

Information on relation type and its related commodities

Required Parameters

commodityDetailsList

- **type:** array
- **description:** Commodity detail list
- – **reference:** CommodityDetailsApiDTO

relationType

- **type:** string
- **description:** Relation Type
- **enum:** ['sold', 'bought']

RelievePressureObjectApiDTO

Description

Model to describe a relieve pressure change in the scenario: minimum amount of Actions Moves that will relieve the pressure on the over-utilized clusters

Optional Parameters

projectionDay

- **type:** integer
- **format:** int32
- **description:** Projection day, indicate when to apply the change

sources

- **type:** array
- **description:** Sources of the change, List of Clusters that are overloaded
- – **reference:** BaseApiDTO

destinations

- **type:** array
- **description:** Destinations of the change, List of Clusters where VMs can move to
- – **reference:** BaseApiDTO

RemoveConstraintApiDTO

Description

Model to describe a constraint change in the scenario

Optional Parameters

constraintType

- **type:** string
- **description:** Type of the constraint to remove. To ignore all constraints - use AllCommodities. This will ignore all constraints not just Network/Cluster/StorageCluster/DC. GlobalIgnoreConstraint - deprecated - use AllCommodities instead.
- **enum:** ['DataCenterCommodity', 'ClusterCommodity', 'NetworkCommodity', 'DatastoreCommodity', 'StorageClusterCommodity', 'GlobalIgnoreConstraint', 'AllCommodities']

projectionDay

- **type:** integer
- **format:** int32
- **description:** Projection day, indicate when to apply the change

target

- **description:** Target of the change, when not provided change applied to plan scope
- **reference:** BaseApiDTO

targetEntityType

- **type:** string
- **description:** EntityType to target with constraint configuration

- **enum:** ['VirtualMachine', 'Container', 'ContainerPod']

RemoveObjectApiDTO

Description

Model to describe a remove change in the scenario

Optional Parameters

projectionDay

- **type:** integer
- **format:** int32
- **description:** Projection day, indicate when to apply the change

targetEntityType

- **type:** string
- **description:** Type of entity to remove from group target

target

- **description:** Target of the change, entity or group
- **reference:** BaseApiDTO

ReplaceObjectApiDTO

Description

Model to describe a migration change in the scenario

Optional Parameters

projectionDay

- **type:** integer
- **format:** int32
- **description:** Projection day, indicate when to apply the change

target

- **description:** Target of the change, entity or group
- **reference:** BaseApiDTO

template

- **description:** Template used for the replace
- **reference:** BaseApiDTO

targetEntityType

- **type:** string
- **description:** Type of entity to remove from group target

ReplaceWebhookBodyRequestApiDTO

Description

The details of a request to substitute action fields in a Webhook body template.

Required Parameters

webhookBody

- **type:** string
- **description:** The Webhook body being substituted.

ReplaceWebhookBodyResponseApiDTO

Description

Result of substituting action field values in a Webhook body template.

Required Parameters

succeeded

- **type:** boolean
- **description:** The Status of webhook body replacement.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

error

- **type:** string
- **description:** The error that occurred during replacement, if any.

replacedBody

- **type:** string
- **description:** The replaced body after substitution of action fields.

ReplicaApiDTO

Description

Model to describe Replica attributes

Required Parameters

type

- **type:** string

Optional Parameters

name

- **type:** string

- **description:** Replica name

failoverPriority

- **type:** string
- **description:** Failover priority level of this replica

clusterRole

- **type:** string
- **description:** Role of this replica.
- **enum:** ['Writer', 'Reader']

deploymentType

- **type:** string
- **description:** Deployment type
- **enum:** ['SINGLE_AZ', 'MULTI_AZ']

sli

- **type:** array
- – **type:** string

RepositoryAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

owner

- **type:** string
- **description:** Owner of the repository.

url

- **type:** string
- **description:** URL to access the repository.

numberOfTfResources

- **type:** integer
- **format:** int32
- **description:** Number of TF resources in the repository.

iacType

- **type:** string
- **description:** Iac type of the repository.
- **enum:** ['TERRAFORM', 'UNKNOWN']

RequestHeader

Description

Model to describe a request header.

Required Parameters

name

- **type:** string
- **description:** Request header name

Optional Parameters

value

- **type:** string
- **description:** Request header value. On create, required. On update, not provided means re-use existing.

isSecret

- **type:** boolean
- **description:** Whether the request header value should be encrypted

ReservationConstraintApiDTO

Description

The constraints user specified when running reservation which limit where templates could be placed.

Required Parameters

constraintType

- **type:** string
- **description:** Type of constraint
- **enum:** ['CLUSTER', 'DATA_CENTER', 'VIRTUAL_DATA_CENTER', 'POLICY', 'NETWORK']

uuid

- **type:** string
- **description:** UUID of the constraint

Optional Parameters

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

ReservationFailureInfoDTO

Description

Failure information when reservation fails

Required Parameters

closestSeller

- **description:** provider that came closest to meeting the requirements
- **reference:** BaseApiDTO

closestSellerCluster

- **description:** Closest provider's cluster information
- **reference:** BaseApiDTO

maxQuantityAvailable

- **type:** number
- **format:** double
- **description:** Quantity of resource available on closestSeller

quantityRequested

- **type:** number
- **format:** double
- **description:** Quantity of resource requested

resource

- **type:** string
- **description:** Most constrained resource

timespan

- **type:** string
- **description:** Failure is in historical or current
- **enum:** ['HISTORICAL', 'CURRENT', 'PROJECTED', 'PLAN_SOURCE', 'PLAN_PROJECTED']

Optional Parameters

units

- **type:** string
- **description:** Units for the resource

ReservationInvalidInfoApiDTO

Description

The reason the reservation is invalid

Optional Parameters

invalidReason

- **type:** string
- **description:** The reason why the reservation is invalid

- **enum:** ['MARKET_CONNECTIVITY_ERROR', 'MARKET_NOT_READY', 'INVALID_CONSTRAINTS']

ReservedInstanceApiDTO

Description

Model to describe a Reserved Instance

Required Parameters

cloudType

- **type:** string
- **description:** Cloud Type
- **enum:** ['AWS', 'AZURE', 'GCP', 'HYBRID', 'UNKNOWN']

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

template

- **description:** Related template
- **reference:** BaseApiDTO

platform

- **type:** string
- **description:** Platform name
- **enum:** ['UNKNOWN', 'LINUX', 'RHEL', 'SUSE', 'WINDOWS', 'WINDOWS_WITH_SQL_STANDARD', 'WINDOWS_WITH_SQL_WEB', 'WINDOWS_WITH_SQL_SERVER_ENTERPRISE', 'LINUX_WITH_SQL_STANDARD', 'LINUX_WITH_SQL_WEB', 'LINUX_WITH_SQL_SERVER_ENTERPRISE']

type

- **type:** string
- **description:** Reserved instance type
- **enum:** ['STANDARD', 'CONVERTIBLE']

location

- **description:** Location, either a zone or a data center if zone is not available
- **reference:** BaseApiDTO

accountId

- **type:** string
- **description:** Account ID

accountDisplayName

- **type:** string
- **description:** Account display name

targetId

- **type:** string
- **description:** The associated target ID of the account

masterAccountId

- **type:** string
- **description:** Master account ID

term

- **description:** Term, unit is Year
- **reference:** StatApiDTO

payment

- **type:** string
- **description:** Payment option
- **enum:** ['ALL_UPFRONT', 'PARTIAL_UPFRONT', 'NO_UPFRONT']

onDemandPrice

- **description:** On-demand price, saving is RI price minus on-demand price
- **reference:** StatApiDTO

costPrice

- **description:** Total monthly cost
- **reference:** StatApiDTO

coupons

- **description:** Total coupons and used coupons, utilization = numOfCouponsUsed / numOfCoupons
- **reference:** StatApiDTO

totalCoupons

- **type:** number
- **format:** float
- **description:** Total coupons provided by the cbtp the vm matched with

expDate

- **type:** string
- **description:** Expiration date

instanceCount

- **type:** integer
- **format:** int32
- **description:** Number of reserved instance with the same type, template, platform etc..

tenancy

- **type:** string
- **description:** Tenancy
- **enum:** ['DEFAULT', 'DEDICATED', 'HOST']

upFrontCost

- **type:** number
- **format:** double
- **description:** Up front cost

actualHourlyCost

- **type:** number
- **format:** double
- **description:** Actual hourly cost

effectiveHourlyCost

- **type:** number
- **format:** double
- **description:** Effective hourly cost

isExpired

- **type:** boolean
- **description:** Whether the RI be expired by a date provided

expDateEpochTime

- **type:** integer
- **format:** int64
- **description:** Expiration Date in epoch time

scopeType

- **type:** string
- **description:** Whether the RI scope is Single or Shared (Azure)
- **enum:** ['SINGLE', 'SHARED', 'RESOURCE_GROUP', 'UNKNOWN']

appliedScopes

- **type:** array
- **description:** List of applied scope UUIDs - only provided when scope is Single or Resource_Group (Azure)
- – **type:** string

expandedAppliedScopes

- **type:** array
- **description:** List of applied scopes with additional information about each scope - only provided when scope is Single or Resource_Group (Azure)
- – **reference:** BaseApiDTO

orderID

- **type:** string
- **description:** The Order ID of the RI (Azure)

trueID

- **type:** string
- **description:** The ID of the RI

coveredEntityCount

- **type:** integer

- **format:** int32
- **description:** Count of workload entities covered by the reserved instance

undiscoveredAccountsCoveredCount

- **type:** integer
- **format:** int32
- **description:** Count of undiscovered accounts covered by the reserved instance

toBuy

- **type:** boolean

sizeFlexible

- **type:** boolean

ReservedInstancePurchaseSettingsDTO

Description

Optional Parameters

awsProfile

- **reference:** PurchaseProfileDTO

azureProfile

- **reference:** PurchaseProfileDTO

ResizeActionDetailsApiDTO

Description

Parent class for resize action details. Implementing classes include: CloudResizeActionDetailsApiDTO.

Required Parameters

actionDetailType

- **type:** string

actionType

- **type:** string

Optional Parameters

subType

- **type:** string

ResourceApiDTO

Description

Model to describe the Resources of a Template

Optional Parameters

stats

- **type:** array
- **description:** Statistics, e.g. Number of CPU, Memory size, Disk size, ...
- – **reference:** StatApiDTO

provider

- **description:** Entity provider
- **reference:** BaseApiDTO

type

- **type:** string
- **description:** Type of the resource

template

- **type:** string
- **description:** Template used to fulfill the resources

relatedResources

- **type:** array
- **description:** ID of resources related to this resource
- – **type:** string

linkedResources

- **type:** array
- **description:** resources related to this resource. For now, this will only contain the cluster
- – **reference:** BaseApiDTO

ResourceGroupApiDTO

Description

Inherits From

This DTO inherits from GroupApiDTO

Optional Parameters

parentUuid

- **type:** string
- **description:** Parent Business Account uuid (e.g subscription in Azure) where the Resource Group belongs to.

parentDisplayName

- **type:** string
- **description:** Parent Business Account display name (e.g subscription in Azure) where the Resource Group belongs to.

RoleApiDTO

Description

Model to describe a User Role, it describe the permissions of an user over the Ops. Manager capabilities

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

name

- **type:** string
- **description:** Name of the role

SAMLIdpApiDTO

Description

Model to describe an SAML IDP. It contains IDP URL and is SAML ONLY enabled or not.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

idpURL

- **type:** string
- **description:** The SAML IDP URL

isSAMLOnly

- **type:** boolean

isSingleLogoutEnabled

- **type:** boolean

STEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

displayName

- **type:** string
- **description:** Display Name

name

- **type:** string
- **description:** Name

productName

- **type:** string
- **description:** Product Name

productType

- **type:** string
- **description:** Product Type

deviceId

- **type:** string
- **description:** Device ID

deviceName

- **type:** string
- **description:** Device Name

externalNames

- **type:** array
- **description:** External Names
- – **type:** string

altName

- **type:** string
- **description:** Alternative Name

ssdDiskCount

- **type:** integer
- **format:** int64
- **description:** SSD Disk Count

rpm7200DiskCount

- **type:** integer
- **format:** int64
- **description:** RPM 7200 Disk Count

rpm10KDiskCount

- **type:** integer
- **format:** int64
- **description:** RPM 10K Disk Count

rpm15KDiskCount

- **type:** integer
- **format:** int64
- **description:** RPM 15K Disk Count

getvSeriesDiskCount

- **type:** integer
- **format:** int64
- **description:** VSeries Disk Count

ssdDiskIOPSCapacity

- **type:** number
- **format:** double
- **description:** SSD Disk IOPS Capacity

rpm7200DiskIOPSCapacity

- **type:** number
- **format:** double
- **description:** RPM 7200 Disk IOPS Capacity

rpm10KDiskIOPSCapacity

- **type:** number
- **format:** double
- **description:** RPM 10K Disk IOPS Capacity

rpm15KDiskIOPSCapacity

- **type:** number
- **format:** double
- **description:** RPM 15K Disk IOPS Capacity

getvSeriesDiskIOPSCapacity

- **type:** number
- **format:** double
- **description:** VSeries Disk IOPS Capacity

technologyType

- **type:** string
- **description:** HCI Technology Type

rawCapacity

- **type:** integer
- **format:** int64
- **description:** HCI Raw Capacity

rawFreespace

- **type:** integer

- **format:** int64
- **description:** HCI Raw Freespace

rawUncommitted

- **type:** integer
- **format:** int64
- **description:** HCI Raw Uncommitted

redundancyMethod

- **type:** string
- **description:** HCI Redundancy Method

failuresToTolerate

- **type:** integer
- **format:** int32
- **description:** HCI Failures To Tolerate

spaceReservationPct

- **type:** integer
- **format:** int32
- **description:** HCI Space Reservation Percentage

minimumHosts

- **type:** integer
- **format:** int32
- **description:** HCI Minimum Hosts Required

isHciStorage

- **type:** boolean

vendorLink

- **type:** string
- **description:** Link to navigate to the entity in vendor UI.

SaaSReportingConfigApiDTO

Description

SaaS Reporting Configuration Information

Required Parameters

configuration

- **type:** string
- **description:** SaaS Reporting Encoded Configuration

enabled

- **type:** boolean
- **description:** SaaS Reporting Configuration Information

SavingsInvestmentsApiDTO

Description

Model to describe Savings/Investments

Required Parameters

savingsInvestmentsTotal

- **description:** The total Saving/Investment value
- **reference:** SavingsInvestmentsSnapshotApiDTO

Optional Parameters

csp

- **description:** The csp associated with this savings/investments value
- **reference:** BaseApiDTO

account

- **description:** The account associated with this savings/investments value
- **reference:** BaseApiDTO

resourceGroup

- **description:** The resourceGroup associated with this savings/investments value
- **reference:** BaseApiDTO

region

- **description:** The region associated with this savings/investments value
- **reference:** BaseApiDTO

entity

- **description:** The entity associated with this savings/investments value
- **reference:** BaseApiDTO

actionChainId

- **type:** string
- **description:** The actionChainId associated with this savings/investments value

entityType

- **type:** string
- **description:** The entityType associated with this savings/investments value
- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

actionType

- **type:** string
- **description:** The actionType associated with this savings/investments value
- **enum:** ['NONE', 'START', 'MOVE', 'SCALE', 'ALLOCATE', 'SUSPEND', 'TERMINATE', 'SPAWN', 'ADD_PROVIDER', 'CHANGE', 'REMOVE_PROVIDER', 'PROVISION', 'RECONFIGURE', 'RESIZE', 'RESIZE_CAPACITY', 'WARN', 'RECONFIGURE_THRESHOLD', 'DELETE', 'RIGHT_SIZE', 'RESERVE_ON_PM', 'RESERVE_ON_DS', 'RESIZE_FOR_EFFICIENCY', 'RESIZE_FOR_PERFORMANCE', 'CROSS_TARGET_MOVE', 'MOVE_TOGETHER', 'ACTIVATE', 'DEACTIVATE', 'BUY_RI', 'GRANT']

savingsInvestmentsSnapshot

- **type:** array
- **description:** The list of Savings/Investments snapshots based on the selected time period
- – **reference:** SavingsInvestmentsSnapshotApiDTO

tag

- **description:** The tag key and value associated with this savings/investments value
- **reference:** SavingsInvestmentsTagApiDTO

SavingsInvestmentsApiInputDTO

Description

Model to describe the request of a SavingsInvestment.

Optional Parameters

scopes

- **type:** array
- **description:** List of uuid to use as scope
- – **type:** string

filters

- **type:** array
- **description:** Filters to apply to the requested savings/investment. Supported values: CSP, ACCOUNT, RESOURCE_GROUP, REGION, ENTITY, ACTION_CHAIN, ENTITY_TYPE, ACTION_TYPE
- – **reference:** SavingsInvestmentsFilterApiDTO

groupBy

- **type:** array
- **description:** Execute a groupBy on the values of the savings/investment. Supported values: CSP, ACCOUNT, RESOURCE_GROUP, REGION, ENTITY, ACTION_CHAIN, ENTITY_TYPE, ACTION_TYPE
- – **type:** string
  - **enum:** ['CSP', 'ACCOUNT', 'RESOURCE_GROUP', 'REGION', 'ENTITY', 'ACTION_CHAIN', 'ENTITY_TYPE', 'ACTION_TYPE', 'TAG_KEY', 'TAG_VAL', 'TAG_TYPE']

startDate

- **type:** string
- **description:** Start time of the period to filter, if empty the current savings/investment will be processed

endDate

- **type:** string

- **description:** End time of the period to filter, if empty the current savings/investment will be processed

includeSnapshots

- **type:** boolean
- **description:** Include time period snapshots of savings/investment. Could be daily or monthly

includeOnlySavings

- **type:** boolean
- **description:** Include only the Savings values

includeOnlyInvestments

- **type:** boolean
- **description:** Include only the Investments values

SavingsInvestmentsFilterApiDTO

Description

Model to describe the filter applied to a Savings/Investments response

Required Parameters

type

- **type:** string
- **description:** type of the filter, E.G: ACTION_TYPE, RESOURCE_GROUP ...
- **enum:** ['CSP', 'ACCOUNT', 'RESOURCE_GROUP', 'REGION', 'ENTITY', 'ACTION_CHAIN', 'ENTITY_TYPE', 'ACTION_TYPE', 'TAG_KEY', 'TAG_VAL', 'TAG_TYPE']

value

- **type:** string
- **description:** value of the filter

SavingsInvestmentsSnapshotApiDTO

Description

Model to describe the Saving/Investment of a snapshot, contains a list of Savings/Investments for a moment in time

Required Parameters

currency

- **type:** string
- **description:** Currency

investments

- **type:** number
- **format:** double
- **description:** Investments value

savings

- **type:** number

- **format:** double
- **description:** Savings value

Optional Parameters

date

- **type:** string
- **description:** When the Saving/Investment was recorded

SavingsInvestmentsTagApiDTO

Description

Tag information

Optional Parameters

key

- **type:** string
- **description:** Tag key

value

- **type:** string
- **description:** Tag value

ScenarioApiDTO

Description

Model to describe a What-If Scenario, composed by a list of changes

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

owners

- **type:** array
- **description:** Users that owns the scenario
- – **reference:** UserApiDTO

type

- **type:** string
- **description:** Type of the scenario

scope

- **type:** array
- **description:** List of entities/group that define the Scope
- – **reference:** BaseApiDTO

projectionDays

- **type:** array
- **description:** List of days that define when the scenario should run
- – **type:** integer
  – **format:** int32

topologyChanges

- **description:** Changes to apply to the topology of the plan market
- **reference:** TopologyChangesApiDTO

loadChanges

- **description:** Workload utilization changes in the scenario
- **reference:** LoadChangesApiDTO

configChanges

- **description:** Configuration changes in the scenario
- **reference:** ConfigChangesApiDTO

timebasedTopologyChanges

- **reference:** TimeBasedTopologyChangesApiDTO

ScheduleApiDTO

Description

Model to describe the Schedule for a Settings Policy

Required Parameters

endTime

- **type:** string
- **format:** date-time
- **example:** 2019-04-04T19:15
- **description:** End time in hours and minutes for the scheduled period in ISO8601. For example, 2019-04-04T19:15

startDate

- **type:** string
- **format:** date-time
- **example:** 2019-04-04T19:15
- **description:** Start date of the schedule period, expressed using ISO8601 Format (YYYY-MM-DDTHH:MM:SS). This date is local to the timezone of the instance.

startTime

- **type:** string
- **format:** date-time

- **example:** 2019-04-04T19:15
- **description:** Start time in hours and minutes for the scheduled period in ISO8601. For example, 2019-04-04T17:15

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

endDate

- **type:** string
- **format:** date
- **example:** 2019-04-04
- **description:** End date of the schedule period, expressed using ISO8601 Format (YYYY-MM-DDTHH:MM:SS). This date is local to the timezone of the instance.

recurrence

- **description:** Recurrence setup
- **reference:** RecurrenceApiDTO

timeZone

- **type:** string
- **description:** The time zone name based on tz database

nextOccurrence

- **type:** string
- **description:** The date and time of the next occurence of this schedule represented in ISO8601 local time (YYYY-MM-DDTHH:MM:SS). For example, 2019-04-04T19:15

nextOccurrenceTimestamp

- **type:** integer
- **format:** int64
- **description:** The date and time of the next occurence of this schedule represented in Timestamp in milliseconds. For example, 1562100619000

remaingTimeActiveInMs

- **type:** integer
- **format:** int64
- **description:** The amount of time that the schedule is going to be active at the time of call in milliseconds. This field will not be populated if the schedule is not currently active.

ScheduleEntityResponseApiDTO

Description

represents a response of the attach/detach time span schedule to/from the parkable entity

Optional Parameters

error

- **type:** string
- **example:** entityUuid not found
- **description:** error indicating the reason for failure

entityUUID

- **type:** string

ScheduleItemApiDTO

Description

Base model to describe a ScheduleItem.

Required Parameters

type

- **type:** string
- **example:** ParkingItemApiDTO
- **description:** ScheduleItem type discriminator. Describes which subclass of ScheduleItem is returned.

ScheduleTimeSpansApiDTO

Description

Parking time span schedule

Required Parameters

timeZone

- **type:** string
- **example:** America/New_York
- **description:** Timezone for the time spans in this schedule. Should be chosen from the IANA Timezone Database here: [https://www.iana.org/time-zones]. See the list in wikipedia here: [https://en.wikipedia.org/wiki/List_of_tz_database_time_zones]

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string

- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

description

- **type:** string
- **example:** Up 7-5 on weekdays
- **description:** Description of the schedule

timeSpans

- **description:** Weekly schedule time spans
- **reference:** TimeSpanGroupApiDTO

source

- **type:** string
- **description:** [BETA]: Time span schedule source
- **enum:** ['USER', 'SMART_PARK_RECOMMENDATION']

ScheduleTimeSpansInputApiDTO

Description

Model to describe a filter for schedule timespans.

Optional Parameters

criteriaList

- **type:** array
- **description:** The list of filters to apply in the search request.
- – **reference:** FilterApiDTO

logicalOperator

- **type:** string
- **description:** Logical operator to be applied across all the criteria.
- **enum:** ['AND', 'OR']

ScopeUuidsApiInputDTO

Description

Model to describe the request by UUIDs

Required Parameters

uuids

- **type:** array
- **description:** List of uuids
- – **type:** string

Optional Parameters

marketId

- **type:** string
- **description:** Market ID associated with the requested items

SearchAllQueryApiDTO

Description

A query used to search across the entire topology (entities and groups)

Required Parameters

select

- **description:** Defines what subset of types (entities and groups) should be returned.
- **readOnly:** True
- **reference:** SelectAllApiDTO

where

- **description:** Conditions that each retrieved object must satisfy (currently only name is supported).
- **readOnly:** True
- **reference:** WhereApiDTO

Optional Parameters

pagination

- **description:** Pagination information.
- **readOnly:** True
- **reference:** PaginationApiDTO

SearchCountRecordApiDTO

Description

A count of entities.

Required Parameters

count

- **type:** integer
- **format:** int32
- **description:** Count of entities.
- **readOnly:** True

groupBys

- **type:** array
- **description:** List of fields describing the groupBys that this count represents
- **readOnly:** True
- – **reference:** FieldValueApiDTO

SearchQueryRecordApiDTO

Description

A record with information about an entity or a group.

Required Parameters

oid

- **type:** integer
- **format:** int64
- **description:** Unique identifier for the entity or group.
- **readOnly:** True

values

- **type:** array
- **description:** Information about the entity or group.
- **readOnly:** True
- – **reference:** FieldValueApiDTO

SelectAllApiDTO

Description

Specification of what subset of types (entities and groups) should be returned.

Optional Parameters

entityTypes

- **type:** array
- **description:** Entity types to include in the response.
- **readOnly:** True
- – **type:** string
  – **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

groupTypes

- **type:** array
- **description:** Group types to include in the response.
- **readOnly:** True
- – **type:** string

  - **enum:** ['Group', 'Resource', 'Cluster', 'VMCluster', 'StorageCluster', 'BusinessAccountFolder', 'BillingFamily', 'NodePool']

SelectEntityApiDTO

Description

Specification of what fields should be fetched in an entity search response.

Required Parameters

entityType

- **type:** string
- **description:** Entity Type
- **readOnly:** True
- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

Optional Parameters

fields

- **type:** array
- **description:** Fields to fetch.
- **readOnly:** True
- – **reference:** FieldApiDTO

SelectGroupApiDTO

Description

Specification of what fields should be fetched in a group search response.

Required Parameters

groupType

- **type:** string
- **description:** Group Type.
- **readOnly:** True
- **enum:** ['Group', 'Resource', 'Cluster', 'VMCluster', 'StorageCluster', 'BusinessAccountFolder', 'BillingFamily', 'NodePool']

Optional Parameters

fields

- **type:** array
- **description:** Fields to fetch.
- **readOnly:** True
- – **reference:** FieldApiDTO

ServiceEntityApiDTO

Description

Model to describe an Entity, e.g. Virtual Machine, Physical Machine, Storage

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['CLOUD', 'ONPREM', 'HYBRID']

discoveredBy

- **description:** Target that discovered the entity
- **reference:** TargetApiDTO

vendorIds

- **type:** object
- **description:** The mapping of target identifier to vendor-provided identity of this entity on the remote target
- **additionalProperties:** {'type': 'string'}

priceIndex

- **type:** number
- **format:** float
- **description:** Price index of the Entity

state

- **type:** string
- **description:** State

- **enum:** ['IDLE', 'RESOURCE_ALLOCATION', 'LAUNCH', 'ACTIVE', 'RESOURCE_RELEASE', 'SUSPEND', 'SUSPEND_PENDING', 'TERMINATE_PENDING', 'NOT_MONITORED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'EVACUATED', 'QUEUED', 'NOT_FOUND']

severity

- **type:** string
- **description:** Highest severity of the Actions related to this entity
- **enum:** ['UNKNOWN', 'NORMAL', 'MINOR', 'MAJOR', 'CRITICAL']

unplacementDetails

- **description:** Additional information provided for unplaced entities in a plan
- **reference:** UnplacementDetailsApiDTO

placedOn

- **type:** string
- **description:** The comma-separated classNames of entities successfully placed on, used for Unplaced entities in a Plan Market - [DEPRECATED - EOL: 8.15.1 - Use: unplacementDetails.placedOn]

notPlacedOn

- **type:** string
- **description:** The comma-separated classNames of entities could not be placed on, used for Unplaced entities in a Plan Market - [DEPRECATED - EOL: 8.15.1 - Use: unplacementDetails.reasons]

unplacedExplanation

- **type:** string
- **description:** The reason(s) the entity could not be placed - [DEPRECATED - EOL: 8.15.1 - Use: unplacementDetails.reasons]

costPrice

- **type:** number
- **format:** float
- **description:** Cost price per Hour

severityBreakdown

- **type:** object
- **description:** A Map of severity along with their count. E.g. {CRITICAL: 1, NORMAL: 10}
- **additionalProperties:** {'type': 'integer', 'format': 'int64'}

remoteId

- **type:** string
- **description:** The identifier of this instance on the remote target from which it was discovered. [DEPRECATED - EOL: 8.15.1 - Use: vendorIds]

realtimeMarketReference

- **description:** In case of a Plan Market Entity, this contains the original entity from the Real Time Market
- **reference:** BaseApiDTO

providers

- **type:** array
- **description:** List of Providers

- – **reference:** BaseApiDTO

consumers

- **type:** array
- **description:** List of Consumers
- – **reference:** BaseApiDTO

template

- **description:** Related Template
- **reference:** TemplateApiDTO

numRelatedVMs

- **type:** integer
- **format:** int32
- **description:** Number of related VMs

connectedEntities

- **type:** array
- **description:** Limited list of connected entities relevant to the service entity. This will not contain all connected entities.
- – **reference:** BaseApiDTO

aspects

- **type:** object
- **description:** Additional info about the Entity categorized as Aspects
- **additionalProperties:** {'$ref': '#/definitions/EntityAspect'}

tags

- **type:** object
- **description:** Tags are the metadata defined in name/value pairs. Each name can have multiple values.
- **additionalProperties:** {'type': 'array', 'items': {'type': 'string'}}

onDemandRateBefore

- **type:** number
- **format:** float
- **description:** On-demand rate before actions

onDemandRateAfter

- **type:** number
- **format:** float
- **description:** On-demand rate after actions

staleness

- **type:** string
- **description:** [BETA]: Whether the entity is outdated, i.e. discovered too long ago to be considered for analytical purposes
- **enum:** ['CURRENT', 'STALE']

entityCosts

- **type:** array
- **description:** Cost info for the entity

- – **reference:** EntityCostApiDTO

controllable

- **type:** boolean

SettingActivePolicyApiDTO

Description

Model to describe a Settings Policy that affect the value of a Setting

Required Parameters

numEntities

- **type:** integer
- **format:** int32
- **description:** Number of entities affected by the Settings Policy value

settingsPolicy

- **description:** Settings Policy that affects the final value of a Setting
- **reference:** BaseApiDTO

value

- **type:** string
- **description:** Setting value declared in the Settings Policy

SettingApiDTOSerializable

Description

Generic model to describe a Setting. Sub types include StringSettingApiDTO and RIPurchaseProfilesSettingApiDTO.

Required Parameters

value

- **description:** Value of the setting. Type indicated by the `valueObjectType` field. The default type is `String`
- **$ref:** #/definitions/object

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

valueDisplayName

- **type:** string
- **description:** Display Name of the Value, in case the value is a Template or another entity
- **readOnly:** True

defaultValue

- **description:** Default value before any change
- **readOnly:** True
- **$ref:** #/definitions/object

categories

- **type:** array
- **description:** Category and sub categories of the setting
- **readOnly:** True
- – **type:** string

valueType

- **type:** string
- **description:** Define the type of the value
- **enum:** ['STRING', 'NUMERIC', 'INTEGER', 'BOOLEAN', 'LIST', 'SECRET']

valueObjectType

- **type:** string
- **description:** The type of complex object values. Defaults to 'String'.
- **enum:** ['String', 'ReservedInstancePurchaseSettingsDTO']

options

- **type:** array
- **description:** List of allowable value for this setting
- **readOnly:** True
- – **reference:** SettingOptionApiDTO

min

- **type:** number
- **format:** double
- **description:** If type is NUMERIC, this define the min limit
- **readOnly:** True

max

- **type:** number
- **format:** double
- **description:** If type is NUMERIC, this define the max limit
- **readOnly:** True

entityType

- **type:** string
- **description:** Entity type for this Setting

range

- **description:** Range options for the value of the setting

- **readOnly:** True
- **reference:** RangeApiDTO

scope

- **type:** string
- **description:** Define the visibility of the setting. E.G. a Local Setting is only visible when associated with an Entity/Group
- **readOnly:** True
- **enum:** ['GLOBAL', 'LOCAL']

activeSettingsPolicies

- **type:** array
- **description:** List of Settings Policies that are affecting the final value of this setting
- – **reference:** SettingActivePolicyApiDTO

sourceGroupName

- **type:** string
- **description:** Source group name for setting value

sourceGroupUuid

- **type:** string
- **description:** Source group uuid for setting value

SettingApiDTOString

Description

Generic model to describe a Setting. Sub types include StringSettingApiDTO and RIPurchaseProfilesSettingApiDTO.

Required Parameters

value

- **description:** Value of the setting. Type indicated by the `valueObjectType` field. The default type is `String`
- **$ref:** #/definitions/object

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

valueDisplayName

- **type:** string
- **description:** Display Name of the Value, in case the value is a Template or another entity

- **readOnly:** True

defaultValue

- **description:** Default value before any change
- **readOnly:** True
- **$ref:** #/definitions/object

categories

- **type:** array
- **description:** Category and sub categories of the setting
- **readOnly:** True
- – **type:** string

valueType

- **type:** string
- **description:** Define the type of the value
- **enum:** ['STRING', 'NUMERIC', 'INTEGER', 'BOOLEAN', 'LIST', 'SECRET']

valueObjectType

- **type:** string
- **description:** The type of complex object values. Defaults to 'String'.
- **enum:** ['String', 'ReservedInstancePurchaseSettingsDTO']

options

- **type:** array
- **description:** List of allowable value for this setting
- **readOnly:** True
- – **reference:** SettingOptionApiDTO

min

- **type:** number
- **format:** double
- **description:** If type is NUMERIC, this define the min limit
- **readOnly:** True

max

- **type:** number
- **format:** double
- **description:** If type is NUMERIC, this define the max limit
- **readOnly:** True

entityType

- **type:** string
- **description:** Entity type for this Setting

range

- **description:** Range options for the value of the setting
- **readOnly:** True
- **reference:** RangeApiDTO

scope

- **type:** string
- **description:** Define the visibility of the setting. E.G. a Local Setting is only visible when associated with an Entity/Group
- **readOnly:** True
- **enum:** ['GLOBAL', 'LOCAL']

activeSettingsPolicies

- **type:** array
- **description:** List of Settings Policies that are affecting the final value of this setting
- – **reference:** SettingActivePolicyApiDTO

sourceGroupName

- **type:** string
- **description:** Source group name for setting value

sourceGroupUuid

- **type:** string
- **description:** Source group uuid for setting value

SettingApiDTO

Description

Generic model to describe a Setting. Sub types include StringSettingApiDTO and RIPurchaseProfilesSettingApiDTO.

Required Parameters

value

- **description:** Value of the setting. Type indicated by the `valueObjectType` field. The default type is `String`
- **$ref:** #/definitions/object

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

valueDisplayName

- **type:** string
- **description:** Display Name of the Value, in case the value is a Template or another entity
- **readOnly:** True

defaultValue

- **description:** Default value before any change
- **readOnly:** True
- **$ref:** #/definitions/object

categories

- **type:** array
- **description:** Category and sub categories of the setting
- **readOnly:** True
- – **type:** string

valueType

- **type:** string
- **description:** Define the type of the value
- **enum:** ['STRING', 'NUMERIC', 'INTEGER', 'BOOLEAN', 'LIST', 'SECRET']

valueObjectType

- **type:** string
- **description:** The type of complex object values. Defaults to 'String'.
- **enum:** ['String', 'ReservedInstancePurchaseSettingsDTO']

options

- **type:** array
- **description:** List of allowable value for this setting
- **readOnly:** True
- – **reference:** SettingOptionApiDTO

min

- **type:** number
- **format:** double
- **description:** If type is NUMERIC, this define the min limit
- **readOnly:** True

max

- **type:** number
- **format:** double
- **description:** If type is NUMERIC, this define the max limit
- **readOnly:** True

entityType

- **type:** string
- **description:** Entity type for this Setting

range

- **description:** Range options for the value of the setting
- **readOnly:** True
- **reference:** RangeApiDTO

scope

- **type:** string

- **description:** Define the visibility of the setting. E.G. a Local Setting is only visible when associated with an Entity/Group
- **readOnly:** True
- **enum:** ['GLOBAL', 'LOCAL']

activeSettingsPolicies

- **type:** array
- **description:** List of Settings Policies that are affecting the final value of this setting
- – **reference:** SettingActivePolicyApiDTO

sourceGroupName

- **type:** string
- **description:** Source group name for setting value

sourceGroupUuid

- **type:** string
- **description:** Source group uuid for setting value

SettingOptionApiDTO

Description

Optional Parameters

label

- **type:** string

value

- **type:** string

SettingsManagerApiDTO

Description

Model to describe a Category Manager for the settings

Required Parameters

category

- **type:** string
- **description:** Category of the Manager

settings

- **type:** array
- **description:** List of Settings contained in the Manager
- – **reference:** SettingApiDTOSerializable

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

SettingsPolicyApiDTO

Description

Model to describe a Setting Policy: a group of Settings that are applied based on Scope and/or Schedule

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

entityType

- **type:** string
- **description:** Type of the Settings Policy. E.G. VirtualMachine, PhysicalMachine, ...

scopes

- **type:** array
- **description:** List of scopes where the settings are applied, on create/edit, only Uuid is required. If null it's considered Global scope
- – **reference:** GroupApiDTO

settingsManagers

- **type:** array
- **description:** List of groups of Settings to activate in this Policy
- – **reference:** SettingsManagerApiDTO

onPremCostsManagers

- **type:** array
- **description:** List of groups of Settings to activate for on prem costs policy
- – **reference:** OnPremCostsManagerApiDTO

schedule

- **description:** Schedule to define when the settings are applied
- **reference:** ScheduleApiDTO

disabled

- **type:** boolean

- **description:** Control if the policy is Enabled or Disabled

note

- **type:** string
- **description:** Note for this Policy

readOnly

- **type:** boolean
- **description:** If true, prevents users from deleting or modifying the SettingPolicy.

default

- **type:** boolean

SiteInformationApiDTO

Description

Hybrid SaaS Network Site information

Optional Parameters

tunnelComponents

- **description:** Skupper components image version
- **reference:** TunnelComponents

siteId

- **type:** string
- **description:** Site Id

clientOperatorVersion

- **type:** string
- **description:** Client Operator Image version

autoUpdateEnabled

- **type:** boolean
- **description:** Is Auto Update Enabled?

lastUpdated

- **type:** integer
- **format:** int64
- **description:** Site Information Last Updated in unix epoch seconds format

probes

- **type:** array
- **description:** List of Probes
- – **reference:** ProbeDeploymentApiDTO

SkippedActionApiDTO

Description

Model to describe an action that was skipped in execution.

Required Parameters

actionId

- **type:** string
- **description:** Action ID

Optional Parameters

reason

- **type:** string
- **description:** Reason why action was skipped

StatApiDTO

Description

Model to describe a Statistic

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

name

- **type:** string
- **description:** Name

capacity

- **description:** Capacity values, used for Commodities stats
- **reference:** StatValueApiDTO

reserved

- **description:** Reserved values, used for Commodities stats
- **reference:** StatValueApiDTO

relatedEntityType

- **type:** string
- **description:** E.G.virtual machine

filters

- **type:** array
- **description:** describe the grouping options used to generate the output
- – **reference:** StatFilterApiDTO

relatedEntity

- **description:** Provider or Consumer of the Statistic, 'relatedEntity' and 'numRelatedEntities' are mutually exclusive
- **reference:** BaseApiDTO

numRelatedEntities

- **type:** integer
- **format:** int32
- **description:** Number of providers or consumers for this statistic, 'relatedEntity' and 'numRelatedEntities' are mutually exclusive

units

- **type:** string
- **description:** Units, used for Commodities stats. E.G. $/h

values

- **description:** Contains Average, Min, Max, Total
- **reference:** StatValueApiDTO

value

- **type:** number
- **format:** float
- **description:** Simple value, equal to values.avg.

commoditySource

- **description:** Source for a commodity's capacity and usage, used for Commodities stats
- **reference:** StatCommoditySourceApiDTO

histUtilizations

- **type:** array
- **description:** Commodity historical utilization values
- – **reference:** StatHistUtilizationApiDTO

StatApiInputDTO

Description

Model to describe the request of a Stat. Please note that the costPrice stat is deprecated and will EOL in 8.12.1. Use the dedicated 'cost' endpoints in /entities, /groups and /markets instead.

Optional Parameters

name

- **type:** string
- **description:** Name of the stat

relatedEntityType

- **type:** string
- **description:** Used to get stats from different entity types in the supply chain of {uuid} E.G.costPrice of the VMs where the scope is a PM

filters

- **type:** array
- **description:** Filters to apply to the requested stats. Commodities supported values: key = {commodity_key}, relation = sold|bought, virtualDisk = {uuid}. Cost Price supported values: cumulative = true|false, cloudService = {uuid}, target = {uuid}, CSP = {name}, category = {name}, businessUnit = {uuid}, costComponent = COMPUTE|IP|LICENSE|STORAGE, costComponentArtifact = {key}. Please note that the costPrice stat is deprecated and will EOL in 8.12.1. Use the dedicated 'cost' endpoints in /entities, /groups and /markets instead.Action stats supported values: actionTypes = {actionType}, actionModes: {action_mode}, actionStates: {action_state}, riskSubCategory: Performance Assurance|Efficiency Improvement| Prevention|Compliance, riskSeverity: UNKNOWN|NORMAL|MINOR|MAJOR|CRITICAL
- – **reference:** StatFilterApiDTO

groupBy

- **type:** array
- **description:** Execute a groupBy on the values of the stats. Commodities supported values: key, relatedEntity, virtualDisk. Cost Price supported values: cloudService, target, CSP, category, businessUnit, costComponent, costComponentArtifact. Please note that the costPrice stat is deprecated and will EOL in 8.12.1. Use the dedicated 'cost' endpoints in /entities, /groups and /markets instead.Action stats supported values: actionTypes, actionModes, actionStates, risk, riskSubCategory, riskSeverity
- – **type:** string

historyType

- **type:** string
- **description:** Query the histUtilization table for these historyTypes
- **enum:** ['Percentile', 'Smoothed']

StatCommoditySourceApiDTO

Description

Model to describe commodity source for a single statistic, e.g. commodity source for VCPU (capacity and usage)

Optional Parameters

capacityCommoditySource

- **type:** string
- **description:** Commodity source for capacity

usedCommoditySource

- **type:** string
- **description:** Commodity source for usage

capacitySourceIndex

- **type:** integer
- **format:** int32
- **description:** Commodity source operation index for capacity

usedSourceIndex

- **type:** integer
- **format:** int32

- **description:** Commodity source operation index for usage

StatFilterApiDTO

Description

Model to describe the filter applied to a Statistic

Required Parameters

type

- **type:** string
- **description:** type of the filter, E.G: actionTypes, category, ...

Optional Parameters

value

- **type:** string
- **description:** value of the filter

displayName

- **type:** string
- **description:** display name of the value, E.G: displayName if 'value' is an oid or an enum

StatHistUtilizationApiDTO

Description

Model to describe a historical utilization statistic, e.g. the P95 vCPU or P95 vMEM of a VM over (up to) NN days observation period.

Optional Parameters

type

- **type:** string
- **description:** Historical utilization type

usage

- **type:** number
- **format:** float
- **description:** Historical usage

capacity

- **type:** number
- **format:** float
- **description:** Historical capacity

resizeMaxScalingObservationPeriod

- **type:** number
- **format:** float
- **description:** Resize max scaling observation period. Affects on amount of data points considered for action generation process. Value in days.

resizeScalingAggressiveness

- **type:** number
- **format:** float
- **description:** Resize scaling aggressiveness. Describes how aggressively Turbonomic will resize in response of resource utilization. Value in percents. 100% least aggressive, 95% most aggressive.

StatPeriodApiInputDTO

Description

Model to describe the request for Statistics by a Time range

Required Parameters

statistics

- **type:** array
- **description:** List of requested statistics, if empty all stats will be processed
- – **reference:** StatApiInputDTO

Optional Parameters

startDate

- **type:** string
- **description:** Start time of the period to filter, if empty the current stats will be processed

endDate

- **type:** string
- **description:** End time of the period to filter, if empty the current stats will be processed

StatScopesApiInputDTO

Description

Model to describe the request for Statistics in multiple Scopes

Optional Parameters

scopes

- **type:** array
- **description:** List of uuid to use as scope
- – **type:** string

period

- **description:** List of requested statistics
- **reference:** StatPeriodApiInputDTO

relatedType

- **type:** string
- **description:** Get the stats for the entity type related to the scopes

StatSnapshotApiDTO

Description

Model to describe the Statistics of a snapshot, contains a list of Statistics for a moment in time

Required Parameters

statistics

- **type:** array
- **description:** List of statistics, e.g. Price Index, Commodities values, Cost price, Number of VMs in a Group, ...
- – **reference:** StatApiDTO

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

date

- **type:** string
- **description:** When the statistics were recorded

epoch

- **type:** string
- **description:** An indicator of whether the snapshot represents historical, projected or plan data. Should be used in conjunction with the date field to understand the context of this stat snapshot.
- **enum:** ['HISTORICAL', 'CURRENT', 'PROJECTED', 'PLAN_SOURCE', 'PLAN_PROJECTED']

StatValueApiDTO

Description

Model to describe a single statistic, e.g. the CPU of a PM Group

Optional Parameters

max

- **type:** number
- **format:** float
- **description:** Max value

min

- **type:** number
- **format:** float

- **description:** Min value

avg

- **type:** number
- **format:** float
- **description:** Average value

total

- **type:** number
- **format:** float
- **description:** Total value - sum of average values in a group

totalMax

- **type:** number
- **format:** float
- **description:** Total max value - sum of max values in a group

totalMin

- **type:** number
- **format:** float
- **description:** Total min value - sum of min values in a group

StringListFilterApiDTO

Description

Model to describe a filter by a string list and a operator.

Required Parameters

values

- **type:** array
- **description:** String list for a comparison.
- – **type:** string

Optional Parameters

operator

- **type:** string
- **description:** Operator for a comparison.
- **enum:** ['EQ', 'NEQ']

caseSensitive

- **type:** boolean
- **description:** If true, the query string will match case-sensitivity.

StringSettingApiDTO

Description

Inherits From

This DTO inherits from SettingApiDTO

Required Parameters

value

- **type:** string
- **description:** Value of the setting. Type indicated by the `valueObjectType` field. The default type is `String`

Optional Parameters

defaultValue

- **type:** string
- **description:** Default value before any change
- **readOnly:** True

SupplyChainStatsApiInputDTO

Description

Model to describe a Filter for Entities

Required Parameters

uuids

- **type:** array
- **description:** List of uuids used to build the supply chain
- – **type:** string

Optional Parameters

types

- **type:** array
- **description:** List of Types to filter
- – **type:** string

states

- **type:** array
- **description:** List of States to filter
- – **type:** string
  – **enum:** ['IDLE', 'RESOURCE_ALLOCATION', 'LAUNCH', 'ACTIVE', 'RESOURCE_RELEASE', 'SUSPEND', 'SUSPEND_PENDING', 'TERMINATE_PENDING', 'NOT_MONITORED', 'MAINTENANCE', 'FAILOVER', 'UNKNOWN', 'EVACUATED', 'QUEUED', 'NOT_FOUND']

groupBy

- **type:** array
- **description:** List of fields used to group the Stats
- – **type:** string
  – **enum:** ['entityType', 'state', 'severity', 'riskSubCategory', 'template', 'target', 'businessUnit', 'resourceGroup', 'nodePool', 'attachment']

environmentType

- **type:** string
- **description:** Filter the actions by Environment Type
- **enum:** ['ONPREM', 'CLOUD']

SupplychainApiDTO

Description

Model to describe the Supply Chain: the Environment is described as a chain between Producers and Consumers

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

seMap

- **type:** object
- **description:** Model to describe the type of Entity associated with a SupplyChainEntry
- **additionalProperties:** {'$ref': '#/definitions/SupplychainEntryDTO'}

SupplychainEntryDTO

Description

Model to describe a Supply Chain Entry

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

depth

- **type:** integer
- **format:** int32
- **description:** Number to indicate the level of depth of the Entity type in the Supply Chain

entitiesCount

- **type:** integer
- **format:** int32
- **description:** Number of entity in this level of depth

healthSummary

- **type:** object
- **description:** Entities count by Severity
- **additionalProperties:** {'type': 'integer', 'format': 'int32'}

stateSummary

- **type:** object
- **description:** Entities count by State
- **additionalProperties:** {'type': 'integer', 'format': 'int32'}

connectedProviderTypes

- **type:** array
- **description:** List of provider types connected to this level of depth
- **uniqueItems:** True
- – **type:** string

connectedConsumerTypes

- **type:** array
- **description:** List of consumer types connected to this level of depth
- **uniqueItems:** True
- – **type:** string

instances

- **type:** object
- **description:** List of entities in this level of depth
- **additionalProperties:** {'$ref': '#/definitions/ServiceEntityApiDTO'}

SustainabilityInputApiDTO

Description

Model to describe the request for Sustainability Metrics by a Time range

Optional Parameters

sustainabilityGroupBys

- **type:** array
- **description:** List of values to group the results by.
- – **type:** string
  – **enum:** ['ACCOUNT', 'AVAILABILITY_ZONE', 'ENTITY', 'ENTITY_TYPE', 'REGION', 'SERVICE_PROVIDER']
- **maxItems:** 1
- **minItems:** 0

startDate

- **type:** string
- **description:** Start time of the period to filter. If startDate and endDate not provided current metrics will be processed

endDate

- **type:** string
- **description:** End time of the period to filter. If startDate and endDate not provided current metrics will be processed

TagApiDTO

Description

Tag information

Optional Parameters

key

- **type:** string
- **description:** Tag key

values

- **type:** array
- **description:** Tag value
- – **type:** string

TagKeyApiDTO

Description

Optional Parameters

key

- **type:** string
- **description:** Tag key

TargetApiDTO

Description

Model to describe a Target. Targets are used to discover the components to create a connected virtual environment

Required Parameters

type

- **type:** string
- **description:** Probe type. Turbonomic ships with the following probe types: AppDynamics, Dynatrace, Datadog, Apptio Cloudability, Arista, AWS, Azure, Cisco APIC, Cisco HyperFlex, Cisco UCS Central, Cisco UCS Director, Cisco UCS Manager, CloudFoundry, CloudStack, Dell Compellent, EMC ScaleIO, EMC VMAX, EMC VNX, EMC VPLEX, EMC XtremIO, Hitachi Vantara, HP OneView, HPE 3PAR,

Hyper-V, IBM PowerVM, Instana, JVM, MSExchange, MySQL, NetApp, NetFlow, NetScaler, Nutanix, OpenStack, Oracle, Pivotal Ops Manager, Pure, Red Hat Virtualization Manager, sFlow, SNMP, SoftLayer, SQLServer, Tetration, Tomcat, vCenter, vCloudDirector, VMM, WebLogic, WebSphere, WMI, XenServer,

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

category

- **type:** string
- **example:** Hypervisor
- **description:** Probe category.

isProbeRegistered

- **type:** boolean
- **example:** False
- **description:** Indicator that is used to determine whether the associated probe is running and registered with the system

uiCategory

- **type:** string
- **example:** Application Server, Billing, Cloud Management, Applications and Databases, Fabric and Network, Guest OS Processes, HYPERCONVERGED, HYPERVISOR, Load Balancer, Operations Manager Appliance, ORCHESTRATOR, PaaS, Storage, Storage Browsing, WINDOWS APPLICATION. There are also categories Custom and UNKNOWN to be used for new probe development.
- **description:** Probe UI category.

identifyingFields

- **type:** array
- **description:** Array of field names used to differentiate probes of the same category. Most typically either the name/address field, or a combination of address and port.
- – **type:** string

inputFields

- **type:** array
- **description:** List of all InputFieldApiDTO objects for the target, analogous to the target addition form fields in the GUI.
- – **reference:** InputFieldApiDTO

lastValidated

- **type:** string
- **description:** Date of the last validation - [DEPRECATED - Use: healthSummary.timeOfLastSuccessfulDiscovery]

status

- **type:** string
- **example:** Validated, VALIDATION FAILED, UNKNOWN
- **description:** Description of the status of the most recent Validation Operation - [DEPRECATED - Use: healthSummary.healthState]

derivedTargets

- **type:** array
- **description:** Targets derived (created by discovery of) from this target. Populated depending on the TargetRelationship parameter.
- – **reference:** TargetApiDTO

patchedTargets

- **type:** array
- **description:** Targets patched by this target
- – **reference:** PatchedTargetDataApiDTO

parentTargets

- **type:** array
- **description:** Targets that created this target. Populated depending on the TargetRelationship parameter.
- – **reference:** TargetApiDTO

subType

- **type:** string
- **example:** RedHatOpenShift
- **description:** Target subType.

readonly

- **type:** boolean
- **description:** Whether the target cannot be changed through public APIs.

health

- **description:** [BETA]: A summary of the overall health of the target. Only returned if TargetDetailLevel is HEALTH or HEALTH_DETAILS
- **reference:** TargetHealthApiDTO

lastTargetOperationStages

- **type:** array
- **description:** [BETA]: The status at each stage of the most recent validation or discovery, which ever was more recent. Only returned if TargetDetailLevel is HEALTH_DETAILS
- – **reference:** TargetOperationStageApiDTO

lastEditTime

- **type:** string
- **example:** 2021-07-07T20:33:49Z
- **description:** [BETA]: The last time the target was edited in ISO-8601 Format (YYYY-MM-DDTHH:MM:SSZ). For example 2021-07-07T20:33:49Z

lastEditUser

- **type:** string

- **description:** [BETA]: The username for last user who edited the target.

healthSummary

- **description:** The health summary of the target.
- **reference:** TargetHealthSummaryApiDTO

TargetErrorDetailsApiDTO

Description

Additional details related to a target error.

Required Parameters

targetErrorDetailsClass

- **type:** string

TargetHealthApiDTO

Description

A DTO that describes the health of a single target.

Required Parameters

healthCategory

- **type:** string
- **description:** The type of the health checks that are done.
- **enum:** ['TARGET', 'ACTION', 'COMPONENT']

healthClassDiscriminator

- **type:** string
- **example:** TargetHealthApiDTO
- **description:** Health type discriminator. Describes which subclass of BaseHealthApiDTO is returned.

rollupState

- **type:** string
- **example:** NORMAL
- **description:** The health state of the target including its derived targets.
- **enum:** ['CRITICAL', 'MAJOR', 'MINOR', 'NORMAL']

targetName

- **type:** string
- **description:** Target display name.

uuid

- **type:** string
- **description:** UUID of the target.

Optional Parameters

errorText

- **type:** string
- **description:** Error text (if there's any).

timeOfFirstFailure

- **type:** string
- **description:** Time of first failure of validation/discovery (if the target is in failed state; ISO-8601 formatted String).

timeOfLastSuccessfulDiscovery

- **type:** string
- **example:** 2021-07-07T20:33:49Z
- **description:** [BETA]: The last time a successful discovery finished on the target in ISO-8601 Format (YYYY-MM-DDTHH:MM:SSZ). For example 2021-07-07T20:33:49Z

targetErrorDetails

- **type:** array
- **description:** Error details related to a target. Supports multiple errors details.
- – **reference:** TargetErrorDetailsApiDTO

targetStatusSubcategory

- **type:** string
- **enum:** ['DISCOVERY', 'VALIDATION', 'DUPLICATION', 'DELAYED_DATA']

healthState

- **type:** string
- **enum:** ['CRITICAL', 'MAJOR', 'MINOR', 'NORMAL']

TargetHealthSummaryApiDTO

Description

A DTO that describes target health.

Required Parameters

healthState

- **type:** string
- **example:** NORMAL
- **description:** The target health state.
- **enum:** ['CRITICAL', 'MAJOR', 'MINOR', 'NORMAL']

rollupState

- **type:** string
- **example:** NORMAL
- **description:** The health state of the target including its derived targets.
- **enum:** ['CRITICAL', 'MAJOR', 'MINOR', 'NORMAL']

Optional Parameters

timeOfLastSuccessfulDiscovery

- **type:** string
- **example:** 2021-07-07T20:33:49Z
- **description:** The last time a successful discovery finished on the target in ISO-8601 Format (YYYY-MM-DDTHH:MM:SSZ). For example 2021-07-07T20:33:49Z

TargetOperationStageApiDTO

Description

Holds all information related to a stage of discovery or validation.

Optional Parameters

description

- **type:** string
- **description:** The description of what this stage is doing (ideally in less that 70 characters). Example of description are "validating credentials" or "retrieving business applications".

status

- **description:** Holds all the information of this particular instance of the discovery or validation stage.
- **reference:** TargetOperationStageStatusApiDTO

subStages

- **type:** array
- **description:** List with the subStages of he current stage.
- – **reference:** TargetOperationStageApiDTO

optional

- **type:** boolean
- **description:** Is stage marked as optional.

TargetOperationStageStatusApiDTO

Description

Holds all the information of this particular instance of the discovery or validation stage.

Optional Parameters

state

- **type:** string
- **description:** The state of this stage at the end of the discovery or validation.
- **enum:** ['DID_NOT_RUN', 'SUCCESS', 'FAILURE']

summary

- **type:** string
- **description:** The short message (ideally 70 characters or less) explaining the stage status. Example of such messages are "invalid credentials" or "4 business application retrieved".

fullExplanation

- **type:** string
- **description:** The full message explaining the stage status. This explanation may include the instructions or hints on how to address issue if there is one.

TargetPropertiesApiDTO

Description

Model for registered Target properties (categories, types, subtypes)

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

categories

- **type:** array
- **example:** Container Platform
- **description:** Target categories
- **uniqueItems:** True
- – **type:** string

types

- **type:** array
- **example:** Kubernetes
- **description:** Target types
- **uniqueItems:** True
- – **type:** string

subtypes

- **type:** array
- **example:** RedHatOpenShift
- **description:** Target subtypes
- **uniqueItems:** True
- – **type:** string

TargetThirdPartyErrorDetailsApiDTO

Description

Inherits From

This DTO inherits from TargetErrorDetailsApiDTO

Optional Parameters

targetErrorType

- **type:** string
- **description:** Target validation/discovery error type.
- **enum:** ['INTERNAL_PROBE_ERROR', 'CONNECTIVITY_ERROR', 'UNAUTHENTICATED', 'TOKEN_UNAVAILABLE', 'VERSION_NOT_SUPPORTED', 'DATA_ACCESS_ERROR', 'DUPLICATION', 'DELAYED_DATA', 'THIRD_PARTY_FAILURE']

errorCode

- **type:** string
- **example:** 0x423232, 404, 400
- **description:** Error code related to the failure.

endPoint

- **type:** string
- **description:** End point which caused the third party api failure. This can be a complete URLor a method name while using the third party SDK.

TelemetryDTO

Description

Model to describe the status of telemetry

Optional Parameters

telemetryTermsViewed

- **type:** boolean
- **description:** Controls when an administrator should be shown the telemetry opt in

telemetryEnabled

- **type:** boolean
- **description:** Controls whether analytics and telemetry are enabled

tenantId

- **type:** string
- **description:** [BETA]: tenantId for Pendo agent. Only available in SAAS_TRIAL deploymentMode

ibmUniqueId

- **type:** string
- **description:** [BETA]: ibmUniqueId (also called iui) for Pendo agent. Only available in SAAS_TRIAL deploymentMode

TemplateApiDTO

Description

Model to describe a Template: contains the resources used to Deploy a VM or to Add Workload/Supply in a Plan

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

price

- **type:** number
- **format:** float
- **description:** The estimated cost of the template. Used for Physical Machine and Storage templates where the hardware cost is known.

model

- **type:** string
- **example:** Cisco UCS B200 M5 (Intel Xeon Gold 5115, 2.40 GHz), PowerEdge R940 (Intel Xeon Gold 6136, 3.00 GHz), etc.
- **description:** The model of the entire Physical Machine system.

cpuModel

- **type:** string
- **example:** Intel Xeon Gold 6130, AMD Opteron 6166 HE, etc.
- **description:** The model of the CPU in the Physical Machine.

vendor

- **type:** string
- **example:** Cisco, Microsoft, HP, Novell
- **description:** Vendor of the templated entity

description

- **type:** string
- **description:** Description

image

- **type:** string
- **description:** Container Profile image

imageTag

- **type:** string
- **description:** (UNUSED) Container Profile image tag

cmdWithArgs

- **type:** string
- **description:** (UNUSED) Container command with Arguments

dbEngine

- **type:** string
- **example:** PostgreSql, AuroraMySQL, SqlServer
- **description:** Database engine, used for Database templates

dbEdition

- **type:** string
- **example:** Standard One, Enterprise, Premium_P2
- **description:** Database edition, used for Database templates

computeResources

- **type:** array
- **description:** ResourceApiDTO describing the compute resources. Max 1 per template.
- – **reference:** ResourceApiDTO

storageResources

- **type:** array
- **description:** ResourceApiDTO describing the storage resources
- – **reference:** ResourceApiDTO

networkResources

- **type:** array
- **description:** ResourceApiDTO describing the network resources
- – **reference:** ResourceApiDTO

infrastructureResources

- **type:** array
- **description:** ResourceApiDTO describing the infrastructure resources. Max 1 per template
- – **reference:** ResourceApiDTO

deploymentProfile

- **description:** Deployment profile associated with the template
- **reference:** DeploymentProfileApiDTO

discovered

- **type:** boolean
- **example:** False
- **description:** Indicates if the template is discovered or manually created
- **enum:** [True, False]

family

- **type:** string
- **example:** standardDSv2Family, standardNCFamily, r5d, r3
- **description:** For cloud templates, the instance family.

enableMatch

- **type:** boolean

- **example:** False
- **description:** Add to Infrastructure Cost Policy. Infrastructure Cost policies group hardware devices according to their cost

TemplateApiInputDTO

Description

Model to describe a Template: contains the resources used to Deploy a VM or to Add Workload/Supply in a Plan

Optional Parameters

displayName

- **type:** string
- **description:** Name of the template

className

- **type:** string
- **description:** className
- **enum:** ['VirtualMachine', 'PhysicalMachine', 'Storage', 'Container']

price

- **type:** number
- **format:** float
- **description:** Cost price

model

- **type:** string
- **description:** The Model of the entire Physical Machine system. For example: Cisco UCS B200 M5 (Intel Xeon Gold 5115, 2.40 GHz) vs. PowerEdge R940 (Intel Xeon Gold 6136, 3.00 GHz)

cpuModel

- **type:** string
- **description:** The model of the CPU in the Physical Machine. This is separate from the model of the entire Physical Machine. For example: Intel Xeon Gold 6130 vs. AMD Opteron 6166 HE

vendor

- **type:** string
- **description:** Vendor

description

- **type:** string
- **description:** Description

image

- **type:** string
- **description:** Profile image, used for Container templates

imageTag

- **type:** string
- **description:** Profile image tag, used for Container templates

cmdWithArgs

- **type:** string
- **description:** Command with Arguments, used for Container templates

computeResources

- **type:** array
- **description:** Compute resources: Number of CPU, CPU speed, Memory size, ...
- – **reference:** ResourceApiDTO

storageResources

- **type:** array
- **description:** Storage resources: Disk I/0, Disk Size, Percentage of Disk consumed
- – **reference:** ResourceApiDTO

networkResources

- **type:** array
- **description:** Network resources
- – **reference:** ResourceApiDTO

infrastructureResources

- **type:** array
- **description:** Infrastructure resources: Power, Size, Cooling
- – **reference:** ResourceApiDTO

deploymentProfileId

- **type:** string
- **description:** Ids of the Deployment Profiles associated with this Template

enableMatch

- **type:** boolean
- **description:** Add to Infrastructure Cost Policy. Infrastructure Cost policies group hardware devices according to their cost

TemplatePriceAdjustmentDTO

Description

Price adjustment information for a template within a business unit and specified pricing model.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['CLOUD', 'ONPREM', 'HYBRID']

discount

- **type:** number
- **format:** float
- **description:** Entity discount percentage

priceAdjustment

- **description:** Entity price adjustment percentage
- **reference:** PriceAdjustmentDTO

family

- **type:** string
- **description:** Family of a template, group of similar purpose templates. Available only for virtual machine templates.

pricesPerDatacenter

- **type:** array
- **description:** Template base rates per data center, provided for information purpose.
- – **reference:** EntityPriceDTO

TemplateValidationStatusApiDTO

Description

Model to define the template validation status data.

Required Parameters

validationStatus

- **type:** string
- **description:** The ValidationStatus is the result of template validation.
- **enum:** ['NOTFOUND', 'INVALID', 'VALID']

Optional Parameters

validationErrors

- **type:** array
- **description:** The list of validation errors by action type.
- – **type:** string

lastTemplateValidationTime

- **type:** string
- **description:** The lastTemplateValidationTime is the time in ISO-8601 format when the Webhook template was last validated.

TerraformHcpBackendApiDTO

Description

Model to describe terraform HCP backend config.

Optional Parameters

organization

- **type:** string
- **description:** HCP organization

hostname

- **type:** string
- **description:** HCP hostname

workspace

- **type:** string
- **description:** HCP Workspace

TerraformS3BackendApiDTO

Description

Model to describe terraform s3 backend config.

Optional Parameters

bucket

- **type:** string
- **description:** S3 bucket

region

- **type:** string
- **description:** S3 region

TerraformStateFileLocationApiDTO

Description

Model to describe terraform state file locations.

Optional Parameters

repositoryName

- **type:** string
- **description:** The name of repository

repositoryUrl

- **type:** string
- **description:** The url of repository.

repositoryVendorId

- **type:** string
- **description:** The vendor id of repository

keys

- **type:** array
- **description:** The state file location paths.
- – **type:** string

resourceCount

- **type:** integer
- **format:** int32
- **description:** The count of Terraform resources in the repository

TerraformStateFileSpecApiDTO

Description

Model to describe terraform state file attributes from Repositories

Optional Parameters

s3Config

- **description:** S3 Config
- **reference:** TerraformS3BackendApiDTO

repositoryStateFileLinks

- **type:** array
- **description:** Links of repositories to various state file locations
- – **reference:** TerraformStateFileLocationApiDTO

hcpConfig

- **description:** HCP Config
- **reference:** TerraformHcpBackendApiDTO

TextConditionApiDTO

Description

Inherits From

This DTO inherits from ConditionApiDTO

Required Parameters

value

- **type:** string
- **description:** A regular expression to be matched against the value of the field.
- **readOnly:** True

TextFieldValueApiDTO

Description

Inherits From

This DTO inherits from FieldValueApiDTO

Required Parameters

value

- **type:** string
- **description:** The value.
- **readOnly:** True

TimeBasedTopologyChangesApiDTO

Description

Topology changes based on a timeframe in the scenario

Optional Parameters

includeReservation

- **type:** boolean
- **description:** Add the workload that is currently reserved

reservationCount

- **type:** integer
- **format:** int32
- **description:** # of currently reserved VMs
- **readOnly:** True

addHistoryVMs

- **type:** boolean
- **description:** Should historical vmNetGrowth be considered

vmNetGrowth

- **type:** integer
- **format:** int32
- **description:** Number of workloads added during the last month
- **readOnly:** True

TimeSpanApiDTO

Description

Schedule Time span information

Required Parameters

begins

- **type:** string
- **description:** Start time of the time span

ends

- **type:** string
- **description:** End time of the time span

policy

- **description:** policy to be applied during this time span
- **reference:** ScheduleItemApiDTO

TimeSpanGroupApiDTO

Description

Base model to describe a TimeSpanGroup.

Required Parameters

type

- **type:** string
- **example:** WeekDayTimeSpansApiDTO
- **description:** TimeSpan group type discriminator. Describes which subclass of TimeSpanGroupApiDTO is returned.

TopologyChangesApiDTO

Description

Topology changes in the scenario

Optional Parameters

addList

- **type:** array
- **description:** List of add changes in the Topology
- – **reference:** AddObjectApiDTO

removeList

- **type:** array
- **description:** List of remove changes in the Topology
- – **reference:** RemoveObjectApiDTO

replaceList

- **type:** array
- **description:** List of replace changes in the Topology
- – **reference:** ReplaceObjectApiDTO

migrateList

- **type:** array
- **description:** List of migrate changes in the Topology
- – **reference:** MigrateObjectApiDTO

relievePressureList

- **type:** array
- **description:** List of relieve pressure changes in the Topology
- – **reference:** RelievePressureObjectApiDTO

TopologyDataDefinitionApiDTO

Description

Model to describe a user-defined topology definition.

Required Parameters

entityDefinitionData

- **description:** Entity definition data
- **reference:** IEntityDefinitionData

entityType

- **type:** string
- **description:** The type of service entity being defined
- **enum:** ['BusinessApplication', 'BusinessTransaction', 'Service']

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

customEntityUuidList

- **type:** array
- – **type:** string

discoveryState

- **type:** string
- **description:** Discovery status for application topology.
- **enum:** ['IN_PROGRESS', 'CREATED', 'DETACHED']

contextBased

- **type:** boolean
- **description:** Defines if this Definition is Context-Based.

TopologyScopeApiDTO

Description

Model to describe cloud commitment's topology scoped attributes

Optional Parameters

cloudServiceId

- **type:** string
- **description:** The cloud service ID of the cloud commitment, e.g. aws::CS::AmazonRDS

cloudTierId

- **type:** string
- **description:** The cloud tier ID of the cloud commitment, e.g. db.t3.micro

tenancy

- **type:** string
- **description:** The tenancy of the cloud commitment, e.g. shared or dedicated
- **enum:** ['DEFAULT', 'DEDICATED', 'HOST']

familyScope

- **description:** The family scope of the cloud commitment if applicable
- **reference:** FamilyScopeApiDTO

databaseServerTierInfo

- **description:** The database server tier info of the cloud commitment if applicable
- **reference:** DatabaseServerTierInfoApiDTO

TunnelComponents

Description

Optional Parameters

router

- **type:** string
- **description:** Skupper Router image information

siteController

- **type:** string
- **description:** Skupper Site Controller image information

serviceController

- **type:** string
- **description:** Skupper service controller image information

configSync

- **type:** string
- **description:** Skupper Config Sync image information

UnplacementDetailsApiDTO

Description

Additional information provided for entities in a plan that are unplaced

Required Parameters

reasons

- **type:** array
- **description:** Detailed information about why the entity could not be placed
- – **reference:** UnplacementReasonApiDTO

Optional Parameters

placedOn

- **type:** array
- **description:** For Unplaced entities in a Plan Market, the providers that this entity was able to place on
- – **reference:** BaseApiDTO

UnplacementReasonApiDTO

Description

Details to describe why an entity could not be placed

Optional Parameters

failedResources

- **type:** array
- **description:** A list of resources that caused the entity not to be placed
- – **reference:** FailedResourceApiDTO

resourceOwner

- **description:** The entity owning the resources that caused unplacement, if different than this entity. For example, a volume that caused a VM to be unplaced.
- **reference:** BaseApiDTO

providerType

- **type:** string
- **description:** The entity type that sells the commodities that caused unplacement.
- **enum:** ['Application', 'ApplicationComponent', 'ApplicationServer', 'AvailabilityZone', 'BusinessAccount', 'BusinessApplication', 'BusinessTransaction', 'BusinessUser', 'Chassis', 'CloudService', 'ComputeTier', 'Container', 'ContainerPod', 'ContainerSpec', 'Database', 'DatabaseServer', 'DatabaseServerTier', 'DatabaseTier', 'DataCenter', 'DesktopPool', 'DiskArray', 'DPod', 'HCIPhysicalMachine', 'HypervisorServer', 'Internet', 'IOModule', 'LoadBalancer', 'LogicalPool', 'Namespace', 'Network', 'PhysicalMachine', 'ProcessorPool', 'Region', 'Repository', 'ReservedInstance', 'Service', 'ServiceProvider', 'Storage', 'StorageController', 'StorageTier', 'Switch', 'Unknown', 'ViewPod', 'VirtualApplication', 'VirtualDataCenter', 'VirtualMachine', 'VirtualVolume', 'VPod', 'WorkloadController', 'ContainerPlatformCluster', 'CloudCommitment', 'ApplicationComponentSpec', 'VirtualMachineSpec', 'DocumentCollection', 'NodeGroup', 'PowerVMPhysicalMachine', 'DataWarehouse', 'DataWarehouseTier']

closestSeller

- **description:** The seller that came closest to satisfying the commodities. This information is not always available.
- **reference:** BaseApiDTO

placementProblem

- **type:** string
- **description:** Categorizes the reason which caused unplacement
- **enum:** ['COSTS_NOT_FOUND', 'NOT_CONTROLLABLE', 'UNSATISFIED_COMMODITIES']

UserApiDTO

Description

Model to describe an User

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

username

- **type:** string
- **description:** Username

password

- **type:** string
- **description:** Encoded password

roles

- **type:** array
- **description:** Roles assigned to the user.
- – **reference:** RoleApiDTO

loginProvider

- **type:** string
- **description:** Provider for the login
- **enum:** ['Local', 'LDAP']

type

- **type:** string
- **description:** type
- **enum:** ['DedicatedCustomer', 'SharedCustomer']

scope

- **type:** array
- **description:** List of scopes that the User is allowed to see
- – **reference:** GroupApiDTO

atdScope

- **type:** array
- **description:** List of Application Topology Definitions that the User is allowed to see
- – **reference:** BaseApiDTO

features

- **type:** array
- **description:** List of features that the user is allowed to see
- – **type:** string

authToken

- **type:** string
- **description:** The secured user token. [DEPRECATED - EOL: 8.15.1]

showSharedUserSC

- **type:** boolean

UtilizationApiDTO

Description

Model to describe a utilization change in the scenario

Optional Parameters

projectionDay

- **type:** integer
- **format:** int32
- **description:** Projection day, indicate when to apply the change

target

- **description:** Target of the change, entity or group
- **reference:** BaseApiDTO

percentage

- **type:** integer
- **format:** int32
- **description:** Utilization percentage change

VMEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

os

- **type:** string
- **description:** The operating system type. E.G. 'Linux', 'Windows' and 'Unknown' if not found.

ip

- **type:** array
- **description:** The IP address. Can be IPv4 or IPv6
- – **type:** string

macAddress

- **type:** array
- **description:** The MAC address
- – **type:** string

connectedNetworks

- **type:** array
- **description:** The connected Networks
- – **reference:** BaseApiDTO

numVCPUs

- **type:** integer
- **format:** int32
- **description:** Number of vCPUs

acceleratorCards

- **type:** array
- **description:** The Accelerator/GPU cards
- – **reference:** AcceleratorCardApiDTO

coresPerSocketRatio

- **type:** integer
- **format:** int32
- **description:** Cores per socket ratio

processorPoolName

- **type:** string
- **description:** Processor Pool

hasDedicatedProcessors

- **type:** boolean
- **example:** True
- **description:** Has Dedicated Processors

processorCompatibilityMode

- **type:** string
- **example:** POWER8
- **description:** Processor Compatibility Mode

activeMemoryExpansionEnabled

- **type:** boolean
- **example:** True
- **description:** Active Memory Expansion Enabled

resourceMonitoringControlState

- **type:** string
- **example:** active
- **description:** Resource Monitoring Control (RMC) State

sharingMode

- **type:** string
- **example:** uncapped
- **description:** Sharing Mode

dedicatedSharingMode

- **type:** string
- **example:** share_idle_pros
- **description:** Dedicated Sharing Mode

nonDisruptiveMaxPU

- **type:** number
- **format:** double
- **description:** Non-disruptive Processing Units max resize constraint

nonDisruptiveMinPU

- **type:** number
- **format:** double
- **description:** Non-disruptive Processing Units min resize constraint

ebsOptimized

- **type:** boolean
- **description:** EBS Optimized

sessions

- **type:** array
- **description:** Information about sessions established by a BusinessUser to this VirtualMachine.
- – **reference:** BusinessUserSessionApiDTO

vendorToolsVersion

- **type:** string
- **description:** Vendor Tools Version

resourceId

- **type:** string
- **description:** Resource identifier of the virtual machine. For example, the resource id in Azure, the ARN in AWS, or the full resource name in GCP.

creationTimeStamp

- **type:** integer
- **format:** int64

- **description:** Timestamp in milliseconds for the creation of the virtual machine.

vgpuProfile

- **type:** string
- **description:** vGPU profile used by virtual machine.

hardwareVersion

- **type:** string
- **example:** 19.0
- **description:** Hardware Compatibility Version

passthroughGPUIds

- **type:** array
- **example:** 0000:81:00.0
- **description:** The PCI address ID for passthrough GPUs
- – **type:** string

passthroughGPUModels

- **type:** array
- **example:** NVIDIA A100-PCIE-40GB
- **description:** The Model for passthrough GPUs
- – **type:** string

minimumHosts

- **type:** integer
- **format:** int32
- **description:** Minimum hosts required based on the HCI storage policy.

vendorLink

- **type:** string
- **description:** Link to navigate to the entity in vendor UI.

agentMode

- **type:** string
- **description:** The Mode of the installed APM Host Agent.

iacInfo

- **description:** Iac configuration for the VM.
- **reference:** VirtualMachineIacApiDTO

ahublicense

- **type:** boolean

VirtualDiskApiDTO

Description

Model to describe a Virtual Disk

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

tier

- **type:** string
- **description:** The tier of the Virtual Disk

actions

- **type:** array
- **description:** The related action: Move or Delete
- – **reference:** ActionApiDTO

stats

- **type:** array
- **description:** The related statistics for this virtual disk
- – **reference:** StatApiDTO

attachedVirtualMachine

- **description:** The Virtual Machine if the disk is attached
- **reference:** BaseApiDTO

provider

- **description:** The Storage provider
- **reference:** BaseApiDTO

dataCenter

- **description:** The DataCenter where the VirtualDisk resides
- **reference:** BaseApiDTO

environmentType

- **type:** string
- **description:** Environment type
- **enum:** ['CLOUD', 'ONPREM', 'HYBRID']

lastModified

- **type:** integer
- **format:** int64
- **description:** Last Modified

businessAccount

- **description:** Business Account
- **reference:** BaseApiDTO

resourceGroup

- **description:** Resource Group
- **reference:** BaseApiDTO

creationTime

- **type:** string
- **description:** Creation time

skuName

- **type:** string
- **description:** Sku name

snapshotId

- **type:** string
- **description:** Snapshot id

encryption

- **type:** string
- **description:** Encryption

attachmentState

- **type:** string
- **description:** Attachment state
- **enum:** ['AVAILABLE', 'IN_USE', 'UNKNOWN']

diskRole

- **type:** string
- **description:** Disk Role
- **enum:** ['OS Disk', 'Data Disk']

name

- **type:** string
- **description:** Name

ephemeral

- **type:** string
- **description:** Ephemeral

lastAttachedVm

- **type:** string
- **description:** Name of the VM this volume was last attached to.

numDaysUnattached

- **type:** string
- **description:** Number of days ago this volume was last attached.

hourlyBilledOps

- **type:** number
- **format:** double
- **description:** Average number of billed operations per hour.

creationTimeStamp

- **type:** integer
- **format:** int64
- **description:** Timestamp in milliseconds for the creation of the virtual disk.

attachedVMStorageCompatibility

- **type:** string
- **description:** Storage compatibility for virtualDisks's attached VM(s). The compatibility will affect scale analysis for the volume. If an attached VM is not PREMIUM compatible, the volume will never scale to a premium StorageTier
- **enum:** ['PREMIUM', 'STANDARD']

storageUsageType

- **type:** string
- **description:** Usage type for the volume, i.e., what the volume is used for.
- **enum:** ['SITE_RECOVERY', 'BACKUP', 'STANDARD']

resourceId

- **type:** string
- **description:** Resource identifier of the volume. For example, the resource id in Azure, the ARN in AWS, or the full resource name in GCP.

redundancyType

- **type:** string
- **description:** Specifies the redundancy type for the volume

VirtualDisksAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

virtualDisks

- **type:** array
- **description:** List of Virtual Disks
- – **reference:** VirtualDiskApiDTO

VirtualMachineIacApiDTO

Description

Model to describe Iac configuration for VMs

Optional Parameters

terraformStateFileSpec

- **description:** Terraform state file specifications

- **reference:** TerraformStateFileSpecApiDTO

VirtualMachineSpecAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

cloudTierIds

- **type:** string
- **example:** [t3.micro]
- **description:** Type of the instances

minSize

- **type:** integer
- **format:** int32
- **description:** Minimum group size

maxSize

- **type:** integer
- **format:** int32
- **description:** Maximum group size

desiredSize

- **type:** integer
- **format:** int32
- **description:** Default size of group when no provider-initiated scaling actions are applied

instanceTemplateName

- **type:** string
- **description:** Name of template or configuration used to create the scale group. Ex: Name of Launch Template or Launch Configuration in AWS or the Instance Template name in Google

instanceTemplateInstanceType

- **type:** string
- **description:** Instance type defined in the template or configuration used to create the scale group. Ex: Instance type defined in Launch Template or Launch Configuration in AWS or the Instance type defined in Instance Template in Google

instanceTemplateVersion

- **type:** string
- **description:** Version of template or configuration used to create the scale group. Ex: Version of Launch Template in AWS

availabilityZones

- **type:** string
- **example:** [us-east-1a]
- **description:** Available Zones

terminationPolicies

- **type:** string
- **example:** [Default]
- **description:** Termination policies

resourceId

- **type:** string
- **description:** Resource ID containing region, account ID, AWS Auto Scaling Group ID etc

accountId

- **type:** string
- **description:** Account ID

currentInstanceCount

- **type:** integer
- **format:** int32
- **description:** Current number of instances in this Scale Group

suspendedProcesses

- **type:** string
- **description:** List of suspended processes

creationTimeStamp

- **type:** integer
- **format:** int64
- **description:** Date/time first discovered

azureVirtualMachineScaleSet

- **description:** Azure Virtual Machine Scale Set related properties
- **reference:** AzureVirtualMachineScaleSetDTO

gcpManagedInstanceGroupAspect

- **description:** GCP MIG Scale set related properties
- **reference:** GcpManagedInstanceGroupDTO

VirtualVolumeEntityAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

files

- **type:** array
- **description:** Files comprised in the volume
- – **type:** string

WebhookApiDTO

Description

Inherits From

This DTO inherits from WorkflowAspect

Required Parameters

method

- **type:** string
- **description:** The http method used to make the request.
- **enum:** ['GET', 'POST', 'PUT', 'DELETE', 'PATCH']

url

- **type:** string
- **description:** The URL that HTTP request is made to.

Optional Parameters

template

- **type:** string
- **description:** The template for the body of request.

authenticationMethod

- **type:** string
- **description:** The authentication method to use for the request.
- **enum:** ['NONE', 'BASIC', 'OAUTH']

username

- **type:** string
- **description:** The username for the authenticated request.

password

- **type:** string
- **description:** The password for the authenticated request.

trustSelfSignedCertificates

- **type:** boolean
- **description:** If true, self-signed certificates will be trusted when using HTTPS connections. Defaults to 'false'.

headers

- **type:** array
- **description:** The request headers.
- – **reference:** RequestHeader

templateValidationStatus

- **description:** The status of Webhook Template Validation.
- **reference:** TemplateValidationStatusApiDTO

policyValidationStatus

- **type:** object
- **description:** The status of Webhook Template Validation by action types in related policy(s). The key is the policyId.
- **additionalProperties:** {'$ref': '#/definitions/TemplateValidationStatusApiDTO'}

mediaType

- **type:** string
- **description:** The media type of the workflow template
- **enum:** ['JSON', 'XML', 'TEXT']

proxy

- **description:** The proxy settings.
- **reference:** HttpProxyDTO

oauthData

- **reference:** OAuthDataApiDTO

WeekDayTimeSpansApiDTO

Description

Inherits From

This DTO inherits from TimeSpanGroupApiDTO

Optional Parameters

sunday

- **type:** array
- **description:** Schedule time spans information for sunday
- – **reference:** TimeSpanApiDTO

monday

- **type:** array
- **description:** Schedule time spans information for monday
- – **reference:** TimeSpanApiDTO

tuesday

- **type:** array
- **description:** Schedule time spans information for tuesday
- – **reference:** TimeSpanApiDTO

wednesday

- **type:** array
- **description:** Schedule time spans information for wednesday
- – **reference:** TimeSpanApiDTO

thursday

- **type:** array
- **description:** Schedule time spans information for thursday

- – **reference:** TimeSpanApiDTO

friday

- **type:** array
- **description:** Schedule time spans information for friday
- – **reference:** TimeSpanApiDTO

saturday

- **type:** array
- **description:** Schedule time spans information for saturday
- – **reference:** TimeSpanApiDTO

WhereApiDTO

Description

Filter for entity or group queries.

Optional Parameters

relatedToScopes

- **type:** array
- **description:** Restrict to entities related to these scopes.
- **readOnly:** True
- – **type:** integer
  – **format:** int64

conditions

- **type:** array
- **description:** Restrict to entities satisfying these conditions.
- **readOnly:** True
- – **reference:** ConditionApiDTO

WidgetApiDTO

Description

Basic information of an entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

type

- **type:** string

scope

- **reference:** BaseApiDTO

startPeriod

- **type:** string

endPeriod

- **type:** string

row

- **type:** integer
- **format:** int32

column

- **type:** integer
- **format:** int32

sizeRows

- **type:** integer
- **format:** int32

sizeColumns

- **type:** integer
- **format:** int32

widgetsetId

- **type:** integer
- **format:** int64

widgetElements

- **type:** array
- – **reference:** WidgetElementApiDTO

widgetSource

- **type:** string
- **enum:** ['UI_DEFAULT', 'USER']

WidgetElementApiDTO

Description

Basic information of an entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string

- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

row

- **type:** integer
- **format:** int32

column

- **type:** integer
- **format:** int32

type

- **type:** string

widgetId

- **type:** integer
- **format:** int64

properties

- **type:** object
- **additionalProperties:** {'type': 'string'}

stats

- **type:** array
- – **reference:** StatApiInputDTO

datasets

- **type:** array
- – **type:** string

WidgetsetApiDTO

Description

Basic information of an entity

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

scope

- **type:** string

startPeriod

- **type:** string

endPeriod

- **type:** string

widgets

- **type:** array
- – **reference:** WidgetApiDTO

category

- **type:** string

scopeType

- **type:** string

username

- **type:** string

isSharedWithAllUsers

- **type:** boolean

WorkflowApiDTO

Description

Model to describe a Workflow

Required Parameters

discoveredBy

- **description:** Target that discovered this workflow
- **reference:** TargetApiDTO

type

- **type:** string
- **description:** The type of this workflow
- **enum:** ['UCSD', 'ACTION_SCRIPT', 'SERVICENOW', 'ACTIONSTREAM_KAFKA', 'WEBHOOK']

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

description

- **type:** string

- **description:** Description of the workflow

entityType

- **type:** string
- **description:** Type of the entity associated

actionType

- **type:** string
- **description:** Type of the action associated

parameters

- **type:** array
- **description:** Input parameters for the workflow
- – **reference:** InputFieldApiDTO

scriptPath

- **type:** string
- **description:** Full path to workflow script (e.g. path to an ActionScript executable

actionPhase

- **type:** string
- **description:** Action Phase to which workflow applies

timeLimitSeconds

- **type:** integer
- **format:** int64
- **description:** Time limit for workflow execution, in seconds

typeSpecificDetails

- **description:** The information about this workflow that is specific to its type.
- **reference:** WorkflowAspect

creationTime

- **type:** string
- **description:** The creationTime is the time in ISO-8601 format when the workflow was originally created.

lastModifiedTime

- **type:** string
- **description:** The lastModifiedTime is the time in ISO-8601 format when the workflow was last modified.

WorkflowAspect

Description

Parent model to describe a type-specific workflow details

Required Parameters

type

- **type:** string

WorkflowOperationRequestApiDTO

Description

The details of the requested operation on the workflow.

Required Parameters

actionId

- **type:** integer
- **format:** int64
- **description:** The ID of the action that we perform the workflow operation on

operation

- **type:** string
- **description:** The operation that is being performed on the workflow
- **enum:** ['TEST']

WorkflowOperationResponseApiDTO

Description

Result of a workflow operation.

Required Parameters

succeeded

- **type:** boolean
- **description:** True if the workflow operation has succeeded.

Optional Parameters

uuid

- **type:** string

displayName

- **type:** string
- **description:** A user readable name of the api object. Cannot be blank. The UTF-8 encoding must be at most 255 bytes.

className

- **type:** string

details

- **type:** string
- **description:** Details of workflow operation result.

WorkloadControllerAspectApiDTO

Description

Inherits From

This DTO inherits from EntityAspect

Optional Parameters

controllerType

- **type:** string
- **description:** Information about workload controller type.
- **enum:** ['CRON_JOB_INFO', 'CUSTOM_CONTROLLER_INFO', 'DAEMON_SET_INFO', 'DEPLOYMENT_INFO', 'JOB_INFO', 'REPLICA_SET_INFO', 'REPLICATION_CONTROLLER_INFO', 'STATEFUL_SET_INFO']

customControllerType

- **type:** string
- **description:** Information about custom controller type. Only set when the controllerType is CUSTOM_CONTROLLER_INFO.

controllerReplicaCount

- **type:** integer
- **format:** int32
- **description:** The number of active replicas controlled by the workload controller.

Reference: Entity statistics

Turbonomic monitors the following statistics for entities in your environment.

ApplicationComponent statistics

- `Heap`

  Heap is the portion of a VM or container's memory allocated to individual applications.
- `RemainingGcCapacity`

  Remaining GC capacity is the measurement of Application Component uptime that is *not* spent on garbage collection (GC).
- `ResponseTime`

  Response time is the elapsed time between a request and the response to that request. Response time is typically measured in seconds (s) or milliseconds (ms).
- `Threads`

  Threads is the measurement of thread capacity utilized by applications.
- `Transaction`

  Transaction is a value that represents the per-second utilization of the transactions that are allocated to a given entity.
- `VCPU`

  Virtual CPU is the measurement of CPU that is in use.
- `VMem`

  Virtual memory (vMem) is the measurement of memory that is in use.

ApplicationComponentSpec statistics

- `ResponseTime`

Response time is the elapsed time between a request and the response to that request. Response time is typically measured in seconds (s) or milliseconds (ms).

- VCPU

  Virtual CPU is the measurement of CPU that is in use.

BusinessApplication statistics

- ResponseTime

  Response time is the elapsed time between a request and the response to that request. Response time is typically measured in seconds (s) or milliseconds (ms).

- Transaction

  Transaction is a value that represents the per-second utilization of the transactions that are allocated to a given entity.

BusinessTransaction statistics

- ResponseTime

  Response time is the elapsed time between a request and the response to that request. Response time is typically measured in seconds (s) or milliseconds (ms).

- Transaction

  Transaction is a value that represents the per-second utilization of the transactions that are allocated to a given entity.

Chassis statistics

- Cooling

  Cooling is the percentage of the acceptable temperature range that is utilized by the entity. As the temperature nears the high or low running temperature limits, this percentage increases.

- Power

  Power is the measurement of electricity consumed by a given entity, expressed in watts.

- Space

  Not implemented

Container statistics

- VCPU

  Virtual CPU is the measurement of CPU that is in use.

- VCPULimitQuota

  vCPU limit quota is the total amount of virtual CPU limits (in mCores) for all pods allocated to the namespace against the namespace quota.

- VCPURequest

  If applicable, vCPU request is the virtual CPU (in mCores) utilized by a container against the CPU request.

- VCPURequestQuota

  If applicable, vCPU request quota is the amount of virtual CPU request (in mCores) a pod has allocated against the namespace quota.

- VCPUThrottling

vCPU throttling is the throttling of container virtual CPU that could impact response time, expressed as the percentage of throttling for all containers associated with a Container Spec. In the Capacity and Usage chart for containers, *used* and *utilization* values reflect the actual throttling percentage, while *capacity* value is always 100%.

- `VMem`

  Virtual memory (vMem) is the measurement of memory that is in use.

- `VMemLimitQuota`

  If applicable, vMem limit quota is the amount of virtual memory limit a pod has allocated against the namespace quota.

- `VMemRequest`

  If applicable, vMem request is the virtual memory utilized by a container against the memory request.

- `VMemRequestQuota`

  If applicable, vMem request quota is the amount of virtual memory request a pod has allocated against the namespace quota.

ContainerPlatformCluster statistics

- `NumberConsumers`

  Number of consumers is the total number of pods against the VM (Kubernetes node) capacity.

- `VCPU`

  Virtual CPU is the measurement of CPU that is in use.

- `VCPURequest`

  If applicable, vCPU request is the virtual CPU (in mCores) utilized by a container against the CPU request.

- `VMem`

  Virtual memory (vMem) is the measurement of memory that is in use.

- `VMemRequest`

  If applicable, vMem request is the virtual memory utilized by a container against the memory request.

ContainerPod statistics

- `NumberConsumers`

  Number of consumers is the total number of pods against the VM (Kubernetes node) capacity.

- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.

- `VCPU`

  Virtual CPU is the measurement of CPU that is in use.

- `VCPULimitQuota`

  vCPU limit quota is the total amount of virtual CPU limits (in mCores) for all pods allocated to the namespace against the namespace quota.

- `VCPURequest`

  If applicable, vCPU request is the virtual CPU (in mCores) utilized by a container against the CPU request.

- `VCPURequestQuota`

  If applicable, vCPU request quota is the amount of virtual CPU request (in mCores) a pod has allocated against the namespace quota.

- `VMem`

  Virtual memory (vMem) is the measurement of memory that is in use.
- `VMemLimitQuota`

  If applicable, vMem limit quota is the amount of virtual memory limit a pod has allocated against the namespace quota.
- `VMemRequest`

  If applicable, vMem request is the virtual memory utilized by a container against the memory request.
- `VMemRequestQuota`

  If applicable, vMem request quota is the amount of virtual memory request a pod has allocated against the namespace quota.
- `VStorage`

  Virtual storage is the measurement of virtual storage capacity that is in use.

ContainerSpec statistics

- `VCPU`

  Virtual CPU is the measurement of CPU that is in use.
- `VCPURequest`

  If applicable, vCPU request is the virtual CPU (in mCores) utilized by a container against the CPU request.
- `VCPUThrottling`

  vCPU throttling is the throttling of container virtual CPU that could impact response time, expressed as the percentage of throttling for all containers associated with a Container Spec. In the Capacity and Usage chart for containers, *used* and *utilization* values reflect the actual throttling percentage, while *capacity* value is always 100%.
- `VMem`

  Virtual memory (vMem) is the measurement of memory that is in use.
- `VMemRequest`

  If applicable, vMem request is the virtual memory utilized by a container against the memory request.

Database statistics

- `Connection`

  Connection is the measurement of database connections utilized by applications.
- `costPrice`

  CostPrice is the on-demand monthly cost of an entity.
- `DTU`

  DTU is the measurement of compute capacity for the database. DTU represents CPU, memory, and IOPS/IO Throughput bundled as a single commodity.
- `DWU`

  DWU (Data Warehousing Unit) is the measurement of compute capacity for the dedicated SQL pool. DWU represents CPU, memory, and IO Throughput bundled as a single commodity.
- `IOThroughput`

  I/O throughput is the measurement of an entity's throughput to the underlying storage.
- `LicenseAccess`

Not implemented

- `RU`

  Request unit (RU) is a performance currency that abstracts CPU, IOPS, and memory that are required to perform the database operations supported by Azure Cosmos DB. Azure Cosmos DB normalizes the cost of all database operations using RUs.

- `StorageAccess`

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.

- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.

- `VCPU`

  Virtual CPU is the measurement of CPU that is in use.

- `VMem`

  Virtual memory (vMem) is the measurement of memory that is in use.

DatabaseServer statistics

- `Connection`

  Connection is the measurement of database connections utilized by applications.

- `costPrice`

  CostPrice is the on-demand monthly cost of an entity.

- `DBCacheHitRate`

  DB cache hit rate is the measurement of Database Server accesses that result in cache hits, measured as a percentage of hits versus total attempts. A high cache hit rate indicates efficiency.

- `DBMem`

  Database memory (or DBMem) is the measurement of memory that is utilized by a Database Server.

- `LicenseAccess`

  Not implemented

- `ResponseTime`

  Response time is the elapsed time between a request and the response to that request. Response time is typically measured in seconds (s) or milliseconds (ms).

- `RU`

  Request unit (RU) is a performance currency that abstracts CPU, IOPS, and memory that are required to perform the database operations supported by Azure Cosmos DB. Azure Cosmos DB normalizes the cost of all database operations using RUs.

- `StorageAccess`

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.

- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.

- `Transaction`

  Transaction is a value that represents the per-second utilization of the transactions that are allocated to a given entity.

- `VCPU`

  Virtual CPU is the measurement of CPU that is in use.

- `VMem`

  Virtual memory (vMem) is the measurement of memory that is in use.

DataCenter statistics

- `Ballooning`

  Balloon is the measurement of memory that is shared by VMs running on a host.
- `CarbonFootprint`

  Carbon footprint is the measurement of carbon dioxide equivalent (CO2e) emissions for a given entity. Turbonomic measures carbon footprint in grams.
- `Cooling`

  Cooling is the percentage of the acceptable temperature range that is utilized by the entity. As the temperature nears the high or low running temperature limits, this percentage increases.
- `CoreSpeed`

  CoreSpeed is CPU Clock Speed of the entity.
- `CPU`

  CPU is the measurement of CPU that is reserved or in use.
- `CPUAllocation`

  CPU allocation is the aggregate CPU capacity and utilization on the host
- `CPUProvisioned`

  CPU provisioned is the aggregate overprovisioned CPU capacity and utilization on the host
- `CPUReady`

  CPU ready is the measurement of a host's ready queue capacity that is in use.
- `Energy`

  Energy is the measurement of electricity used by a given entity over a period of time, expressed in watt-hours (Wh).
- `GpuMemAllocation`

  Not implemented
- `HardwareVersion`

  Hardware version specifies hardware versions that are compatible with VMs
- `HostLunAccess`

  Not implemented
- `IOThroughput`

  I/O throughput is the measurement of an entity's throughput to the underlying storage.
- `LogicalBlockSizeAccessCommodity`

  Not implemented
- `Mem`

  Memory is the measurement of memory that is reserved or in use.
- `MemAllocation`

  Memory allocation is the aggregate memory capacity and utilization on the host
- `MemProvisioned`

  Memory provisioned is the aggregate overprovisioned memory capacity and utilization on the host
- `NetThroughput`

Net throughput is the rate of message delivery over a port.

- `numCPUs`

  `numCPUs` is the number of CPUs in the datacenter.

- `numHosts`

  `NumHosts` is the number of hosts in the datacenter.

- `numSockets`

  `numSockets` is the number of sockets in the datacenter.

- `NumVCore`

  `NumVCore` is the number of CPU cores on the entity.

- `PCIPassthroughAccessCommodity`

  `PCIPassthroughAccessCommodity` specifies whether to prevent move actions on entities with Passthrough PCI devices.

- `Power`

  Power is the measurement of electricity consumed by a given entity, expressed in watts.

- `PowerCapabilityAccessCommodity`

  Not implemented

- `ProcessingUnits`

  For PowerVM, processing unit is a unit of measurement for shared processing power across one or more virtual processors. One shared processing unit on one virtual processor accomplishes approximately the same work as one dedicated processor.

- `ProcessingUnitsProvisioned`

  `ProcessingUnitsProvisioned` is the processing units provisioned for the host.

- `RaidHosts`

  Not implemented

- `Space`

  Not implemented

- `StorageAccess`

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.

- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.

- `StorageLatency`

  Storage latency is the capacity, reserved amount, and utilized amount of latency for storage transactions, in msecs.

- `StorageProvisioned`

  Storage provisioned is the utilization of the entity's capacity, including overprovisioning.

- `Swapping`

  Swap is the measurement of a host's swap space that is in use.

- `VirtualProcessors`

  For PowerVM, virtual processor (VP) is the guest OS representation of the capacity of the VM. VPs have a one-to-one correlation with PUs. Therefore ,1 VP that is utilized in a VM equates to 1 PU being used. However, VPs do not have to be equal to PUs on a system. VPs are a virtual representation. Therefore, they can be overprovisioned.

DataWarehouse statistics

- `Connection`

  Connection is the measurement of database connections utilized by applications.
- `costPrice`

  CostPrice is the on-demand monthly cost of an entity.
- `LicenseAccess`

  Not implemented
- `NumberReplicas`

  Number of replicas is the total number of nodes (leader node and compute nodes) in a Redshift provisioned cluster.
- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.

DiskArray statistics

- `CPU`

  CPU is the measurement of CPU that is reserved or in use.
- `DiskArrayAccess`

  Not implemented
- `Extent`

  Not implemented
- `StorageAccess`

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.
- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.
- `StorageLatency`

  Storage latency is the capacity, reserved amount, and utilized amount of latency for storage transactions, in msecs.
- `StorageProvisioned`

  Storage provisioned is the utilization of the entity's capacity, including overprovisioning.

DocumentCollection statistics

- `costPrice`

  CostPrice is the on-demand monthly cost of an entity.
- `LicenseAccess`

  Not implemented
- `RU`

  Request unit (RU) is a performance currency that abstracts CPU, IOPS, and memory that are required to perform the database operations supported by Azure Cosmos DB. Azure Cosmos DB normalizes the cost of all database operations using RUs.

IOModule statistics

- `Cooling`

  Cooling is the percentage of the acceptable temperature range that is utilized by the entity. As the temperature nears the high or low running temperature limits, this percentage increases.

- `NetThroughput`

  Net throughput is the rate of message delivery over a port.

- `PortChannel`

  `PortChannel` is the amalgamation of ports with a shared net throughput and utilization.

- `Power`

  Power is the measurement of electricity consumed by a given entity, expressed in watts.

Namespace statistics

- `VCPU`

  Virtual CPU is the measurement of CPU that is in use.

- `VCPULimitQuota`

  vCPU limit quota is the total amount of virtual CPU limits (in mCores) for all pods allocated to the namespace against the namespace quota.

- `VCPURequest`

  Virtual CPU request is the CPU currently guaranteed by all containers on the node with a CPU request. The capacity for this resource is the node allocatable capacity, which is the amount of resources available for pods and can be less than the physical capacity.

- `VCPURequestQuota`

  If applicable, vCPU request quota is the amount of virtual CPU request (in mCores) a pod has allocated against the namespace quota.

- `VMem`

  Virtual memory (vMem) is the measurement of memory that is in use.

- `VMemLimitQuota`

  If applicable, vMem limit quota is the amount of virtual memory limit a pod has allocated against the namespace quota.

- `VMemRequest`

  Virtual memory request is the memory currently guaranteed by all containers on the node with a memory request. The capacity for this resource is the Node Allocatable capacity, which is the amount of resources available for pods and can be less than the physical capacity.

- `VMemRequestQuota`

  vMem request quota is the total amount of virtual memory request for all pods allocated to the namespace against the namespace quota.

PhysicalMachine statistics

- `Ballooning`

  Balloon is the measurement of memory that is shared by VMs running on a host.

- `CarbonFootprint`

  Carbon footprint is the measurement of carbon dioxide equivalent (CO2e) emissions for a given entity. Turbonomic measures carbon footprint in grams.

- `Cooling`

Cooling is the percentage of the acceptable temperature range that is utilized by the entity. As the temperature nears the high or low running temperature limits, this percentage increases.

- `CoreSpeed`

  CoreSpeed is CPU Clock Speed of the entity.

- `CPU`

  CPU is the measurement of CPU that is reserved or in use.

- `CPUAllocation`

  CPU allocation is the aggregate CPU capacity and utilization on the host

- `CPUProvisioned`

  CPU provisioned is the aggregate overprovisioned CPU capacity and utilization on the host

- `CPUReady`

  CPU ready is the measurement of a host's ready queue capacity that is in use.

- `Energy`

  Energy is the measurement of electricity used by a given entity over a period of time, expressed in watt-hours (Wh).

- `GpuMemAllocation`

  Not implemented

- `HardwareVersion`

  Hardware version specifies hardware versions that are compatible with VMs

- `HostLunAccess`

  Not implemented

- `IOThroughput`

  I/O throughput is the measurement of an entity's throughput to the underlying storage.

- `LogicalBlockSizeAccessCommodity`

  Not implemented

- `Mem`

  Memory is the measurement of memory that is reserved or in use.

- `MemAllocation`

  Memory allocation is the aggregate memory capacity and utilization on the host

- `MemProvisioned`

  Memory provisioned is the aggregate overprovisioned memory capacity and utilization on the host

- `NetThroughput`

  Net throughput is the rate of message delivery over a port.

- `numCPUs`

  `numCPUs` is the number of CPUs on the entity.

- `numSockets`

  `numSockets` is the number of sockets in the datacenter.

- `NumVCore`

  `NumVCore` is the number of CPU cores on the entity.

- `PCIPassthroughAccessCommodity`

  `PCIPassthroughAccessCommodity` specifies whether to prevent move actions on entities with Passthrough PCI devices.

- `Power`

  Power is the measurement of electricity consumed by a given entity, expressed in watts.
- `PowerCapabilityAccessCommodity`

  Not implemented
- `ProcessingUnits`

  For PowerVM, processing unit is a unit of measurement for shared processing power across one or more virtual processors. One shared processing unit on one virtual processor accomplishes approximately the same work as one dedicated processor.
- `ProcessingUnitsProvisioned`

  `ProcessingUnitsProvisioned` is the processing units provisioned for the host.
- `RaidHosts`

  Not implemented
- `Space`

  Not implemented
- `StorageAccess`

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.
- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.
- `StorageLatency`

  Storage latency is the capacity, reserved amount, and utilized amount of latency for storage transactions, in msecs.
- `StorageProvisioned`

  Storage provisioned is the utilization of the entity's capacity, including overprovisioning.
- `Swapping`

  Swap is the measurement of a host's swap space that is in use.
- `VirtualProcessors`

  For PowerVM, virtual processor (VP) is the guest OS representation of the capacity of the VM. VPs have a one-to-one correlation with PUs. Therefore ,1 VP that is utilized in a VM equates to 1 PU being used. However, VPs do not have to be equal to PUs on a system. VPs are a virtual representation. Therefore, they can be overprovisioned.

Service statistics

- `NumberReplicas`

  Number of replicas is the number of Application Component replicas running over a given time period.
- `ResponseTime`

  Response time is the elapsed time between a request and the response to that request. Response time is typically measured in seconds (s) or milliseconds (ms).
- `Transaction`

  Transaction is a value that represents the per-second utilization of the transactions that are allocated to a given entity.

Storage statistics

- `Extent`

Not implemented

- `RaidHosts`

  Not implemented

- `StorageAccess`

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.

- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.

- `StorageLatency`

  Storage latency is the capacity, reserved amount, and utilized amount of latency for storage transactions, in msecs.

- `StorageProvisioned`

  Storage provisioned is the utilization of the entity's capacity, including overprovisioning.

StorageController statistics

- `CPU`

  CPU is the measurement of CPU that is reserved or in use.

- `StorageAccess`

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.

- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.

- `StorageLatency`

  Storage latency is the capacity, reserved amount, and utilized amount of latency for storage transactions, in msecs.

Switch statistics

- `NetThroughput`

  Net throughput is the rate of message delivery over a port.

- `PortChannel`

  `PortChannel` is the amalgamation of ports with a shared net throughput and utilization.

VirtualDataCenter statistics

- `CPUAllocation`

  CPU allocation is the aggregate CPU capacity and utilization on the host

- `MemAllocation`

  Memory allocation is the aggregate memory capacity and utilization on the host

VirtualMachine statistics

- `Ballooning`

  Balloon is the measurement of memory that is shared by VMs running on a host.

- `CarbonFootprint`

Carbon footprint is the measurement of carbon dioxide equivalent (CO2e) emissions for a given entity. Turbonomic measures carbon footprint in grams.

- `CoreSpeed`

  CoreSpeed is CPU Clock Speed of the entity.

- `costPrice`

  CostPrice is the on-demand monthly cost of an entity.

- `CPU`

  CPU is the measurement of CPU that is reserved or in use.

- `CPUAllocation`

  CPU allocation is the aggregate CPU capacity and utilization on the host

- `CPUProvisioned`

  CPU provisioned is the aggregate overprovisioned CPU capacity and utilization on the host

- `CPUReady`

  CPU ready is the measurement of a host's ready queue capacity that is in use.

- `Energy`

  Energy is the measurement of electricity used by a given entity over a period of time, expressed in watt-hours (Wh).

- `GpuMemAllocation`

  Not implemented

- `HardwareVersion`

  Hardware version specifies hardware versions that are compatible with VMs

- `HostLunAccess`

  Not implemented

- `IopsSSDRead`

  IopsSSDRead is the per-second measurement of read operations on a Google Cloud SSD disk.

- `IopsSSDWrite`

  IopsSSDWrite is the per-second measurement of write operations on a Google Cloud SSD disk.

- `IopsStandardRead`

  IopsStandardRead is the per-second measurement of read operations on a Google Cloud standard disk.

- `IopsStandardWrite`

  IopsStandardWrite is the per-second measurement of write operations on a Google Cloud standard disk.

- `IOThroughput`

  I/O throughput is the measurement of an entity's throughput to the underlying storage.

- `IOThroughputRead`

  IOThroughputRead is the measurement of an entity's read operations from the underlying storage.

- `IOThroughputWrite`

  IOThroughputWrite is the measurement of an entity's write operations from the underlying storage.

- `LicenseAccess`

  Not implemented

- `Mem`

  Memory is the measurement of memory that is reserved or in use.

- `MemAllocation`

  Memory allocation is the aggregate memory capacity and utilization on the host
- `MemProvisioned`

  Memory provisioned is the aggregate overprovisioned memory capacity and utilization on the host
- `NetThroughput`

  Net throughput is the rate of message delivery over a port.
- `NetThroughputInbound`

  Net throughput inbound is the rate of inbound message delivery over a port.
- `NetThroughputOutbound`

  Net throughput outbound is the rate of outbound message delivery over a port.
- `NumberConsumers`

  Number of consumers is the total number of pods against the VM (Kubernetes node) capacity.
- `NumVCore`

  `NumVCore` is the number of CPU cores on the entity.
- `numVCPUs`

  `NumVCPUs` is the number of CPUs on the entity.
- `PCIPassthroughAccessCommodity`

  `PCIPassthroughAccessCommodity` specifies whether to prevent move actions on entities with Passthrough PCI devices.
- `StorageAccess`

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.
- `StorageAmount`

  Storage amount is the measurement of storage capacity that is in use.
- `StorageLatency`

  Storage latency is the capacity, reserved amount, and utilized amount of latency for storage transactions, in msecs.
- `StorageProvisioned`

  Storage provisioned is the utilization of the entity's capacity, including overprovisioning.
- `Swapping`

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.

  Storage amount is the measurement of storage capacity that is in use.

  Storage latency is the capacity, reserved amount, and utilized amount of latency for storage transactions, in msecs.

  Storage provisioned is the utilization of the entity's capacity, including overprovisioning.

  Swap is the measurement of a host's swap space that is in use.
- `VCPU`

  Virtual CPU is the measurement of CPU that is in use.
- `VCPURequest`

  If applicable, vCPU request is the virtual CPU (in mCores) utilized by a container against the CPU request.

- `VMem`

  Virtual memory (vMem) is the measurement of memory that is in use.
- `VMemRequest`

  Virtual memory request is the memory currently guaranteed by all containers on the VM (Kubernetes node) with a memory request.
- `VStorage`

  Virtual storage is the measurement of virtual storage capacity that is in use.
- `Zone`

  Zone is the location within a given region that serves as a datacenter for the entity.

VirtualMachineSpec statistics

- `costPrice`

  CostPrice is the on-demand monthly cost of an entity.
- `CPU`

  CPU is the measurement of CPU that is reserved or in use.
- `IopsSSDRead`

  IopsSSDRead is the per-second measurement of read operations on a Google Cloud SSD disk.
- `IopsSSDWrite`

  IopsSSDWrite is the per-second measurement of write operations on a Google Cloud SSD disk.
- `IopsStandardWrite`

  IopsStandardWrite is the per-second measurement of write operations on a Google Cloud standard disk.
- `IOThroughput`

  I/O throughput is the measurement of an entity's throughput to the underlying storage.
- `IOThroughputRead`

  IOThroughputRead is the measurement of an entity's read operations from the underlying storage.
- `IOThroughputWrite`

  IOThroughputWrite is the measurement of an entity's write operations from the underlying storage.
- `LicenseAccess`

  Not implemented
- `Mem`

  Memory is the measurement of memory that is reserved or in use.
- `NetThroughput`

  Net throughput is the rate of message delivery over a port.
- `NetThroughputInbound`

  Net throughput inbound is the rate of inbound message delivery over a port.
- `NetThroughputOutbound`

  Net throughput outbound is the rate of outbound message delivery over a port.
- `NumberReplicas`

  Number of replicas is the total number of VM instances underlying an App Service plan.
- `NumDisk`

NumDisks is the number of disks on the entity.

- StorageAccess

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.

- StorageAmount

  Storage amount is the measurement of storage capacity that is in use.

- VCPU

  Virtual CPU is the measurement of CPU that is in use.

- VMem

  Virtual memory (vMem) is the measurement of memory that is in use.

- Zone

  Zone is the location within a given region that serves as a datacenter for the entity.

VirtualVolume statistics

- StorageAccess

  Storage access, also known as IOPS, is the per-second measurement of read and write access operations on a storage entity.

- StorageAmount

  Storage amount is the measurement of storage capacity that is in use.

- StorageLatency

  Storage latency is the capacity, reserved amount, and utilized amount of latency for storage transactions, in msecs.

- StorageProvisioned

  Storage provisioned is the utilization of the entity's capacity, including overprovisioning.

WorkloadController statistics

- VCPULimitQuota

  vCPU limit quota is the total amount of virtual CPU limits (in mCores) for all pods allocated to the namespace against the namespace quota.

- VCPURequestQuota

  If applicable, vCPU request quota is the amount of virtual CPU request (in mCores) a pod has allocated against the namespace quota.

- VMemLimitQuota

  If applicable, vMem limit quota is the amount of virtual memory limit a pod has allocated against the namespace quota.

- VMemRequestQuota

  If applicable, vMem request quota is the amount of virtual memory request a pod has allocated against the namespace quota.

Chapter 11. Data Ingestion Framework

Data Ingestion Framework (DIF) is a Turbonomic integration that defines custom entities and metrics in your environment that are not discovered through any of the supported targets. DIF loads these custom entities and metrics to Turbonomic for analysis.

DIF uses the patented Turbonomic analysis engine to provide visibility and control across the entire application stack.

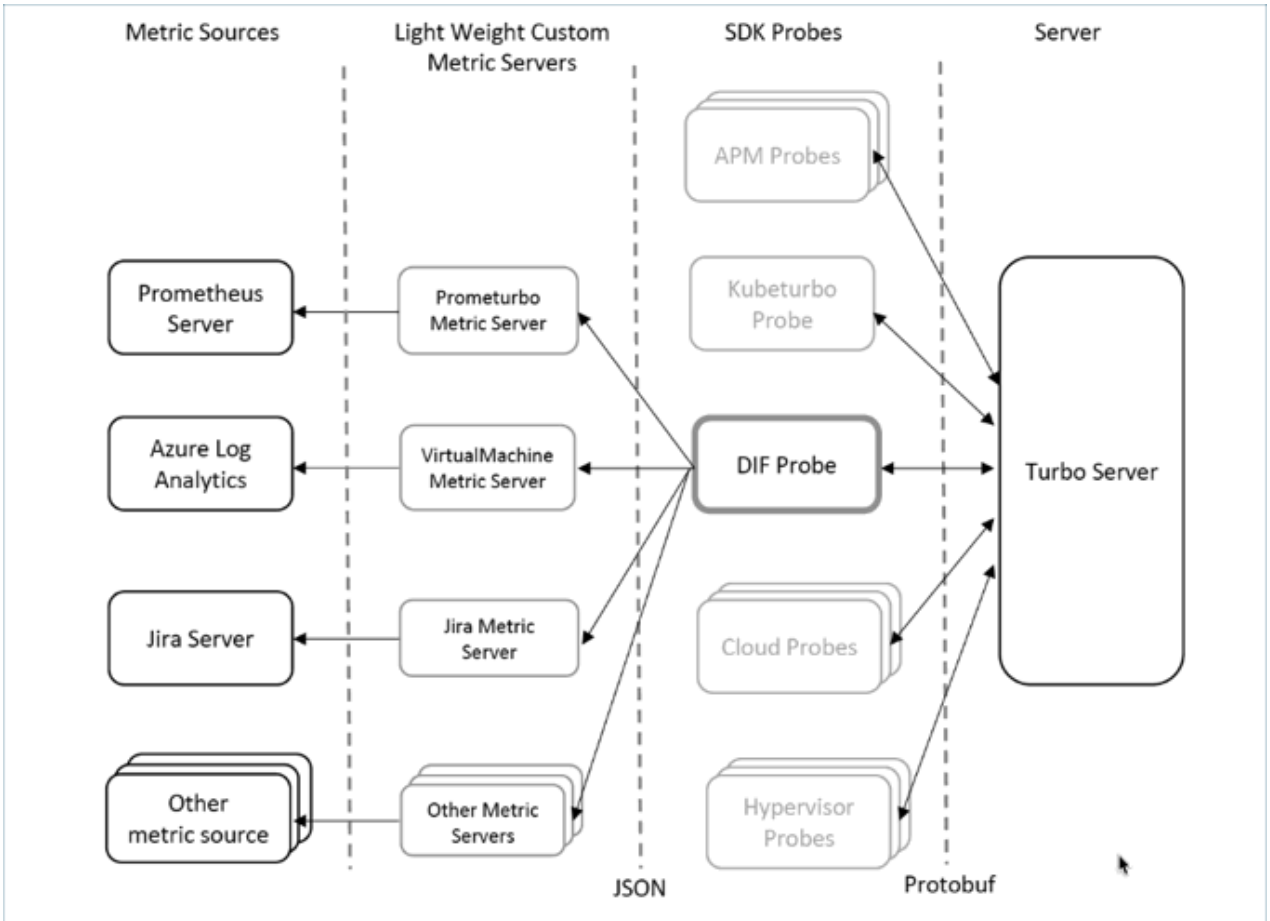
DIF consists of the following components:

- A JSON schema that declares the structure of the data that you loaded to Turbonomic
- A component that runs on the Turbonomic platform and periodically requests updated data to ingest to the platform

How DIF works

Custom entities and metrics are declared in a predefined JSON schema. DIF takes the JSON input, converts it into data structures that are known to Turbonomic, and then pushes it to Turbonomic. DIF performs supply chain and entity validations and participates in the broader mediation framework to resolve conflicts and maintain consistency.

The following diagram shows how Turbonomic uses DIF to query data from metric servers. These servers collect data from specific metric sources.



Chapter 12. Integration - IBM Terraform

Turbonomic Terraform Provider makes Turbonomic the source of truth for **resource optimization decisions** in Terraform-managed environments. It enables infrastructure to self-optimize by pulling recommendations directly into your **plan or apply runs** based on real-time analytics. Reducing manual effort, improving performance and cost efficiency, and consistently enforcing optimization policies as code.

Use the following two methods to optimize Turbonomic by using HashiCorp Terraform:

1. **IBM Turbonomic Provider for Terraform**

   For more information, see IBM Turbonomic Provider.

2. **GitHub and Terraform integration**

   Use the built-in target, such as GitHub within Turbonomic to discover the repositories that contain Terraform code (HCL). For more information, see the following sections:

   - Adding GitHub as a target
   - Adding Terraform connection

Chapter 13. Integration - Data exporter

The Turbonomic platform includes the data exporter, which you can choose to enable when you install the platform. The data exporter streams Turbonomic data about entities and actions as JSON documents that it then publishes to a Kafka topic.

The exporter publishes at different frequencies, depending on the type of data:

- Entity data: Published every 10 minutes.
- Actions data:
  - Pending actions: Published every 6 hours. You can configure this via the `actionExtractionIntervalMins` variable.
  - Executed actions: Published immediately after the action completes, either with success or failure.

You can load the exported data into search and analytics services such as Elasticsearch, and use those services to visualize or analyze the entities and actions in your Turbonomic environment.

With the data exporter you can build custom tracking of changes to your environment, and follow the details as Turbonomic keeps your applications healthy.

Enabling the Data Exporter

To support the Data Exporter, Turbonomic provides an extractor component that can stream data to a standard format. You can load that data into search and analytics services such as Elasticsearch.

To enable the Data Exporter, you must complete the following tasks:

- Enable the extractor component to extract data.

  The extractor is a component that runs as part of your Turbonomic installation. The extractor is enabled by default, but you must configure a property setting so that it can extract data to a connector.
- Deploy a connector that delivers the extractor's stream to your data service.

  The extractor publishes Turbonomic data as Kafka topics. The connector enables your data service to consume the data topic.

Enabling the extractor component

The first step to enable the Data Exporter is to configure a property setting for the extractor component.

1. Open an SSH terminal session to your Turbonomic instance.

   Log in with the System Administrator that you set up when you installed Turbonomic:

   Username: `turbo`

   Password: `[your_private_password]`
2. Open the Turbonomic custom resource.

   In the same SSH session, open the `cr.yaml` file for editing. For example:

   ```
   vi /opt/turbonomic/kubernetes/operator/deploy/crds/charts_v1alpha1_xl_cr.yaml
   ```

   Note:

   If your Turbonomic instance is deployed to Red Hat OpenShift through the Operator Hub, you can access the custom resource by navigating to **Installed Operators** > **Turbonomic Platform Operator** > **Turbonomic Platform Operator tab** > **<xl-release>**.

The default name for the Turbonomiccustom resource in an OpenShift environment is `xl-release`. Select the name of the resource for your instance.

Note:

The extractor component is enabled by default and no explicit definition is required in your `cr.yaml` file. However, if the extractor component was manually disabled in your environment previously, you can re-enable it. Search for the following extractor entry in the `cr.yaml` file:

```
extractor:
  enabled: false
```

Change the entry to `true`.

3. Edit the entry for the extractor properties.

   Search for the extractor entry in the `cr.yaml` file. It appears as the following code:

   ```
   properties:
     extractor:
       enableDataExtraction: false
   ```

   Change the entry to `true`.
4. Save and apply your changes to the platform.

   After you save your changes, use `kubectl` to apply the changes:

   ```
   kubectl apply -f \
   /opt/turbonomic/kubernetes/operator/deploy/crds/charts_v1alpha1_xl_cr.yaml
   ```
5. Verify that the extractor component is running.

   Give the platform enough time to restart the components. Then run the command:

   `kubectl get pods -n turbonomic`

   Look for output similar to the following example:

   ```
   NAME                                          READY   STATUS    RESTARTS
   ...
   extractor-5f41dd61c4-4d6lq                    1/1     Running   0
   ...
   ```

   Look for an entry for the `extractor` component. If the entry is present, ready, and running, then the extractor component is installed and running.

Deploying a connector

The extractor publishes Turbonomic data as Kafka topics. To load this data into a search and analysis service, you must deploy a connector to that service. For example, if you want to load the data into Elasticsearch, then you must deploy an Elasticsearch connector.

You deploy the connector in the same Kubernetes node that runs the Turbonomic platform. To deploy the connector, create a Kubernetes *Deployment* that declares the pods you need for the connector. The following code is a sample deployment of a connector to Elasticsearch.

To deploy the connector, you create a deployment yaml file on the same host that is running the extractor component, and run the command:

```
kubectl create -f <MyConnectorDeployment.yaml>
```

In this example, <`MyConnectorDeployment.yaml`> is the name of the deployment file.

Assume that the name of the deployed pod is `es-kafka-connect`. To verify that the connector is running, run `kubectl get pods -n turbonomic`. Look for output similar to the following example:

```
NAME                                                  READY   STATUS    RESTARTS
...
es-kafka-connect-5f41dd61c4-4d6lq                     1/1     Running   0
...
```

After you deploy the connector, wait for a cycle of Turbonomic analysis (approximately ten minutes). Then, you can see the entities and actions from your Turbonomic environment, loaded as JSON in your data service.

Connector deployment example

If you want to deploy a connector to Elasticsearch so that the service can process the exported data, you might use Kibana with Elasticsearch to display data dashboards.

Before you can deploy the connector, you complete the following prerequisites:

- Deploy Elasticsearch to a VM on the network where you are running Turbonomic. The Elasticsearch host is visible from the Turbonomic Kubernetes node. You will specify this host address in the connector deployment.
- Set up an Elasticsearch index to load the Turbonomic data. You specify this index in the connector deployment.

The following listing is a deployment that uses a Logstash image to collect the extractor data and pipe it to the Elasticsearch host. The deployment also sets up storage volumes, configures the input from the extractor, and configures output to the Elasticsearch instance.

As you go over the listing, pay attention to the following details:

- The location of the Elasticsearch host and the login credentials:

```
...
  env:
    - name: ES_HOSTS
      value: "<urltomyelasticsearchhost>"
    - name: ES_USER
      value: "<myelasticsearchuser>"
    - name: ES_PASSWORD
      valueFrom:
        secretKeyRef:
          name: <MyES_KeyName>
          key: <MyES_Key>
...
```

  Logstash uses the following environment variables:

  – ES_HOSTS: to identify where to pipe the exported data.
  – ES_USER: to identify the user account on Elasticsearch.
  – ES_PASSWORD: for the account login. This connector example assumes that you store the Elasticsearch password as a Kubernetes Secret.

- The name of the Kafka topic:

```
...
  output {
    elasticsearch {
      index => "<MyElasticsearchIndex>"
      hosts => [ "${ES_HOSTS}" ]
    }
  }
...
```

  The Logstash input configuration expects a single topic named `turbonomic.exporter`.

- The Logstash output configuration is to the Elasticsearch server that is identified by the ES_HOSTS environment variable. You specify your own Elasticsearch index in place of <MyElasticsearchIndex>.

  ```
  ... output { elasticsearch { index => "<MyElasticsearchIndex>" hosts => [ "$
  {ES_HOSTS}" ] } } ...
  ```

Sample listing: Elasticsearch connector

This listing is a sample of a deployment file that can work to create an Elasticsearch connector for the Data Exporter. You need to change some settings, such as username and password. You might also need to specify ports and other settings to make the connector comply with your specific environment.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: elasticsearch-kafka-connect
  labels:
    app.kubernetes.io/name: elasticsearch-kafka-connect
spec:
  replicas: 1
  selector:
    matchLabels:
      app.kubernetes.io/name: elasticsearch-kafka-connect
  template:
    metadata:
      labels:
        app.kubernetes.io/name: elasticsearch-kafka-connect
    spec:
      containers:
      - name: logstash
        image: docker.elastic.co/logstash/logstash:7.10.1
        ports:
          - containerPort: 25826
        env:
          - name: ES_HOSTS
            value: "<UrlToMyElasticsearchHost>"
          - name: ES_USER
            value: "<MyElasticsearchUser>"
          - name: ES_PASSWORD
            valueFrom:
              secretKeyRef:
                name: <MyES_KeyName>
                key: <MyES_Key>
        resources:
          limits:
            memory: 4Gi
        volumeMounts:
          - name: config-volume
            mountPath: /usr/share/logstash/config
          - name: logstash-pipeline-volume
            mountPath: /usr/share/logstash/pipeline
      volumes:
      - name: config-volume
        configMap:
          name: logstash-configmap
          items:
            - key: logstash.yml
              path: logstash.yml
      - name: logstash-pipeline-volume
        configMap:
          name: logstash-configmap
          items:
            - key: logstash.conf
              path: logstash.conf
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: logstash-configmap
data:
  logstash.yml: |
    http.host: "0.0.0.0"
    path.config: /usr/share/logstash/pipeline
  logstash.conf: |
    input {
      kafka {
```

```
        topics => ["turbonomic.exporter"]
        bootstrap_servers => "kafka:9092"
        client_id => "logstash"
        group_id => "logstash"
        codec => "json"
        type => "json"
        session_timeout_ms => "60000"   # Rebalancing if consumer is found dead
        request_timeout_ms => "70000"   # Resend request after 70 seconds
      }
    }
    filter {
    }
    output {
      elasticsearch {
        index => "<MyElasticsearchIndex>"
        hosts => [ "${ES_HOSTS}" ]
        user => "${ES_USER}"
        password => "${ES_PASSWORD}"
      }
    }
---
apiVersion: v1
kind: Service
metadata:
  labels:
    app: elasticsearch-kafka-connect
  name: elasticsearch-kafka-connect
spec:
  ports:
    - name: "25826"
      port: 25826
      targetPort: 25826
  selector:
    app: elasticsearch-kafka-connect
```

Increasing the Elasticsearch field limit

For exports to Elasticsearch, you might encounter the following error:

`Elasticsearch: Limit of total fields [1000] has been exceeded`

In the connector's Logstash logs, you might see an error similar to the following example:

```
[2021-05-28T17:16:15,252][INFO ][o.e.a.b.TransportShardBulkAction] [node1] [turbonomic6][0]
mapping update rejected by primary
java.lang.IllegalArgumentException: Limit of total fields [1000] has been exceeded
        at org.elasticsearch.index.mapper.MappingLookup.checkFieldLimit(MappingLookup.java:200)
~[elasticsearch-7.12.0.jar:7.12.0]
        at org.elasticsearch.index.mapper.MappingLookup.checkLimits(MappingLookup.java:192)
~[elasticsearch-7.12.0.jar:7.12.0]
        at org.elasticsearch.index.mapper.DocumentMapper.validate(DocumentMapper.java:152)
~[elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.index.mapper.MapperService.newDocumentMapper(MapperService.java:429)
~[elasticsearch-7.12.0.jar:7.12.0]
        at org.elasticsearch.index.mapper.MapperService.applyMappings(MapperService.java:407)
~[elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.index.mapper.MapperService.mergeAndApplyMappings(MapperService.java:390)
~[elasticsearch-7.12.0.jar:7.12.0]
        at org.elasticsearch.index.mapper.MapperService.merge(MapperService.java:384)
~[elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.action.bulk.TransportShardBulkAction.executeBulkItemRequest(TransportShardBulk
Action.java:281) [elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.action.bulk.TransportShardBulkAction$2.doRun(TransportShardBulkAction.java:164
) [elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.common.util.concurrent.AbstractRunnable.run(AbstractRunnable.java:26)
[elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.action.bulk.TransportShardBulkAction.performOnPrimary(TransportShardBulkAction
.java:209) [elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.action.bulk.TransportShardBulkAction.dispatchedShardOperationOnPrimary(Transpo
rtShardBulkAction.java:115) [elasticsearch-7.12.0.jar:7.12.0]
        at
```

```
org.elasticsearch.action.bulk.TransportShardBulkAction.dispatchedShardOperationOnPrimary(Transpo
rtShardBulkAction.java:74) [elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.action.support.replication.TransportWriteAction$1.doRun(TransportWriteAction.j
ava:168) [elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.common.util.concurrent.ThreadContext$ContextPreservingAbstractRunnable.doRun(T
hreadContext.java:732) [elasticsearch-7.12.0.jar:7.12.0]
        at
org.elasticsearch.common.util.concurrent.AbstractRunnable.run(AbstractRunnable.java:26)
[elasticsearch-7.12.0.jar:7.12.0]
        at java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1128) [?:?]
        at java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:628) [?:?]
        at java.lang.Thread.run(Thread.java:834) [?:?]
```

If you encounter this error, Elastic search might not ingest the full set of data. For example, Elasticsearch might not show some entities that you can see in the Turbonomic user interface.

This error occurs when the number of fields in a single JSON object exceeds the default limit of 1000. For information about this limit, see Mapping limit settings in the Elasticsearch documentation.

To correct this issue, you can increase the field limit:

- In Kibana:

  Go to **Index Management > Settings** and increase the limit. For more information, see Index management in the Elasticsearch documentation.

- Directly in Elasticsearch

  You can run the following command:

  ```
  curl -X PUT -H "Content-Type: application/json" -d "{\"index.mapping.total_fields.limit\":
  2000}" \
  http://{elasticsearch_ip}:9200/{index_name}/_settings
  ```

  Where `{elasticsearch_ip}` is the IP address of your Elasticsearch, and `{index_name}` is the name of the index that exceeds the field limit.

The extractor publishes Turbonomic data as Kafka topics. To load this data into a search and analysis service, you must deploy a connector to that service. For example, if you want to load the data into Elasticsearch, then you must deploy an Elasticsearch connector.

You deploy the connector in the same Kubernetes node that runs the Turbonomic platform. To deploy the connector, create a Kubernetes *Deployment* that declares the pods you need for the connector. For an example of an ElasticSearch connector, see Connector Deployment Example

To deploy the connector, you create a deployment yaml file on the same host that is running the extractor component, and execute the command:

```
kubectl create -f &lt;MyConnectorDeployment.yaml&gt;
```

In this example, <`MyConnectorDeployment.yaml`> is the name of the deployment file.

Assume that the name of the deployed pod is `es-kafka-connect`. To verify that the connector is running, run `kubectl get pods -n turbonomic`. You can see output similar to:

```
NAME                                                READY   STATUS    RESTARTS
...
es-kafka-connect-5f41dd61c4-4d6lq                   1/1     Running   0
...
```

After you deploy the connector, wait for a cycle of Turbonomic analysis (approximately ten minutes). Then, you can see the entities and actions from your Turbonomic environment, loaded as JSON in your data service.

Configuring data export intervals

By default, the Data Exporter sends JSON data for entities and groups at 10-minute intervals. This interval usually gives a good flow of data that your data search and analysis services can ingest.

For large environments, the quantity of data can overwhelm the ability of your service to ingest it. In the worst case, this scenario can result in Out Of Memory errors in your service connector. If these errors occur, lengthening the export interval can reduce the overall volume of data your service must ingest.

You can configure the data export interval in the extractor component for action data, and for entity and group data.

1. Open an SSH terminal session to your Turbonomic instance. Log in with the System Administrator that you set up when you installed Turbonomic:

   Username: `turbo`

   Password: `[your_private_password]`
2. Edit the `cr.yaml` file to add export intervals to the extractor component.

   In the same SSH session, open the `cr.yaml` file for editing. For example:

   ```
   vi /opt/turbonomic/kubernetes/operator/deploy/crds/
   charts_v1alpha1_xl_cr.yaml
   ```
3. Edit the entry for the extractor properties.

   Search for the extractor properties entry in the `cr.yaml` file. It appears as the flowing code:

   ```
   properties:
     extractor:
       enableDataExtraction: true
   ```
4. Add the properties to control the export intervals.

   By default, the extractor exports entity and group data every 10 minutes, and action data every 6 hours. You can add properties to the extractor to change the interval. Each setting is in units of minutes.

   With these settings, you can specify intervals:

   - Globally

     `globalExtractionIntervalMins`

     If you want to set a single interval for all extractor data exports (entities, groups, and actions), use this property.
   - For Entities and Groups

     `entityExtractionIntervalMins`

     To specify an export interval for entity and group data, use this property. If it is present in the extractor properties, this setting takes precedence for entities and groups.
   - For Actions

     `actionExtractionIntervalMins`

     To specify an export interval for actions data, use this property. If it is present in the extractor properties, this setting takes precedence for actions.

   For example, the following specification sets a 30-minute interval on entities and groups, and a one-hour interval on actions:

   ```
   properties:
     extractor:
       enableDataExtraction: true
       entityExtractionIntervalMins: 30
       actionExtractionIntervalMins: 60
   ```

5. Save and apply your changes to the platform.

   After you save your changes, use `kubectl` to apply the changes:

   ```
   kubectl apply -f \
   /opt/turbonomic/kubernetes/operator/deploy/crds/charts_v1alpha1_xl_cr.yaml
   ```

6. Restart the extractor component.

   First, get the name of the extractor pod. Run the command:

   ```
   kubectl get pods -n turbonomic
   ```

   Look for output similar to the following example:

   ```
   NAME                                          READY   STATUS    RESTARTS
   ...
   extractor-5f41dd61c4-4d6lq                    1/1     Running   0
   ...
   ```

   Search the listing for the extractor pod, and use that as the pod name. Then delete the pod to force it to restart. For the previous example, enter the command:

   `kubectl delete pod extractor-5f41dd61c4-4d6lq`

7. Verify that the extractor component is running.

   Give the platform enough time to restart the components. Then run the command:

   ```
   kubectl get pods -n turbonomic
   ```

   You can see the extractor pod in the listing, and its status is `RUNNING`.

Entity object

Every 10 minutes, the extractor sends JSON data to Kafka for individual entities. Each entity object has two main fields:

- `timestamp`: When the entity was broadcast
- `entity`: The object that describes the given entity

Sample Entity object


```
{
  "timestamp": "2021-01-15T19:17:13",
  "entity": {
    "environment": "ON_PREM",
    "name": "test-rdm-shared2",
    "attrs": {
      "guest_os_type": "WINDOWS",
      "connected_networks": [
        "VM Network"
      ],
      "num_cpus": 1
    },
    "state": "POWERED_OFF",
    "oid": 73754533904736,
    "type": "VIRTUAL_MACHINE",
    "related": {
      "PHYSICAL_MACHINE": [
        {
          "name": "hp-esx46.corp.vmturbo.com",
          "oid": 73754533901584
        }
      ],
      "COMPUTE_CLUSTER": [
        {
          "name": "Datacenter DC7/Cluster01",
          "oid": 284907641559697
```

```
        }
      ],
      "DATACENTER": [
        {
          "name": "Datacenter DC7",
          "oid": 73754533900625
        }
      ],
      "VIRTUAL_DATACENTER": [
        {
          "name": "Resources-Cluster01",
          "oid": 73754533901777
        }
      ],
      "VIRTUAL_VOLUME": [
        {
          "name": "Vol-test-rdm-shared2-QSGRID01:DC7DS2",
          "oid": 73754533909440
        },
        {
          "name": "Vol-test-rdm-shared2-test-rdm-shared1 : [QSGRID01:DC7DS2] test-rdm-shared1/
test-rdm-shared1_1.vmdk",
          "oid": 73754533909439
        }
      ],
      "STORAGE": [
        {
          "name": "test-rdm-shared1 : [QSGRID01:DC7DS2] test-rdm-shared1/test-rdm-
shared1_1.vmdk",
          "oid": 73754533901780
        },
        {
          "name": "QSGRID01:DC7DS2",
          "oid": 73754533900756
        }
      ],
      "DISK_ARRAY": [
        {
          "name": "test-rdm-shared1",
          "oid": 73754533901786
        },
        {
          "name": "DiskArray-QSGRID01:DC7DS2",
          "oid": 73754533900667
        }
      ]
    },
    "metric": {
      "STORAGE_AMOUNT": {
        "consumed": 3
      },
      "MEM_ALLOCATION": {
        "consumed": 0
      },
      "CPU_PROVISIONED": {
        "consumed": 2600
      },
      "VCPU": {
        "current": 0,
        "capacity": 2600,
        "utilization": 0
      },
      "BALLOONING": {
        "consumed": 0
      },
      "STORAGE_PROVISIONED": {
        "consumed": 15367.5751953125
      },
      "SWAPPING": {
        "consumed": 0
      },
      "IO_THROUGHPUT": {
        "consumed": 0
      },
      "MEM": {
        "consumed": 0
      },
      "NET_THROUGHPUT": {
        "consumed": 0
      },
      "MEM_PROVISIONED": {
        "consumed": 4194304
```

```
      },
      "VMEM": {
        "current": 0,
        "capacity": 4194304,
        "utilization": 0
      },
      "STORAGE_ACCESS": {
        "consumed": 0
      },
      "CPU_ALLOCATION": {
        "consumed": 0
      },
      "STORAGE_LATENCY": {
        "consumed": 0
      },
      "CPU": {
        "consumed": 0
      }
    }
  }
}
```


Common Entity object fields

Every `entity` object includes the following fields:

JSON Object	Description	Type	Repeated	MapKeyType	Reference
accountExpenses		/exporter/ AccountExpenses	False		AccountExpenses
attrs	Additional descriptions for a given entity. The values of this are different for different entity types.	/shared/ entity.attrs	False		entity.attrs
cost		/exporter/ EntityCost	False		EntityCost
environment	The environment type - Hybrid, Cloud, On-prem, etc.	/enums/ environment_type	False		environment_type
metric	For the given metric type, the utilization and capacity of the commodity.	/exporter/ Commodity	False	/enums/ metric_type	Commodity
name	The entity name.	String	False		
newMetric		JSON	True		Commodity
newRelated		JSON	True		RelatedEntity
oid	The entity ID.	Long	False		
related	For the current entity, another entity that is connected to it in the supply chain.	/exporter/ RelatedEntity	True	/enums/ entity_type	RelatedEntity

JSON Object	Description	Type	Repeated	MapKeyType	Reference
state	The current state of the entity - POWERED_ON, SUSPENDED, MAINTENANCE, etc.	/enums/ entity_state	False		entity_state
type	The entity type.	/enums/ entity_type	False		entity_type

Entity types:

Turbonomic can export data for the following types of entities.

Metric types:

Turbonomic tracks different metrics for different types of entities. For a given entity object, you get the metric data that is appropriate for that entity. This list is all the metric types that the Data Exporter supports.

Related entity types:

Related entities are entities that are connected to the given entity in the supply chain. For example, a VM entity can have containers and applications that are running on it, while it uses disk space from related storage. For a given object, the data includes all the related entities. The supported related types can be any of the supported entities, or any of the group types listed here.

Entity cost data

The JSON for an entity can include cost data. This following information describes the cost data as it relates to an entity.

Sample cost data

This listing shows a sample entity object, with cost data included.

```
{
  "timestamp" : "2021-05-21T19:59:33",
  "entity": {
    "oid": 73978235908269,
    "name": "jjbosvm4-aws",
    "state": "POWERED_ON",
    "environment": "CLOUD",
    "type": "VIRTUAL_MACHINE",
    "cost": {
      "unit": "$/h",
      "total": 0.0070848083,
      "category": {
        "ON_DEMAND_COMPUTE": {
          "total": 0.0,
          "source": {
            "ON_DEMAND_RATE": 0.0126,
            "RI_INVENTORY_DISCOUNT": -0.0126
          }
        },
        "RESERVED_LICENSE": {
          "total": 0.0,
          "source": {
            "RI_INVENTORY_DISCOUNT": 0.0
          }
        },
        "STORAGE": {
          "total": 0.00078480854,
```

```
          "source": {
            "ON_DEMAND_RATE": 0.00078480854
          }
        },
        "RI_COMPUTE": {
          "total": 0.0063,
          "source": {
            "UNCLASSIFIED": 0.0063
          }
        }
      }
    },
    "metric": {
      ...
    },
    "related": {
      ...
    },
    "attrs": {
      ...
    }
  }
}
```

`entity.cost` fields

The `cost` field contains the following data:

- `unit`: The unit of the cost value; for example, `$/h`.
- `total`: The total cost for the given entity.
- `category`: A cost breakdown by category. One entity can include one or more of these categories:
  - `ON_DEMAND_COMPUTE`
  - `ON_DEMAND_LICENSE`
  - `RESERVED_LICENSE`
  - `STORAGE`
  - `RI_COMPUTE`
  - `IP`
  - `SPOT`

Each category can include the following fields to give a total for that category, and to itemize the costs that contribute to that total:

- `total`: The total cost for given category.
- `source`: A breakdown of the total by source. The source can include the following types:
  - `ON_DEMAND_RATE`
  - `RI_INVENTORY_DISCOUNT`
  - `BUY_RI_DISCOUNT`
  - `UNCLASSIFIED`

Each source type gives the cost that is associated with that type. For example, the sample listing includes the following code for on-demand compute costs:

```
"ON_DEMAND_COMPUTE": {
  "total": 0.0,
  "source": {
    "ON_DEMAND_RATE": 0.0126,
    "RI_INVENTORY_DISCOUNT": -0.0126
  }
}
```

This example gives two sources for the category cost. The total is zero because the `RI_INVENTORY_DISCOUNT` cancels the `ON_DEMAND_RATE`.

The cost in the sample listing matches this Workload Cost Breakdown chart:

- The total of $5.17 is the product of `entity.cost.total` (which is $0.0070848083 per hour) times the hours per month (730).
- The cost breakdown in the chart matches the costs for the same categories; `ON_DEMAND_COMPUTE`, `STORAGE`, `RI_COMPUTE`, and `RESERVED_LICENSE`.

The cost total and the breakdown by category give you the data that you need to replicate the charts in Turbonomic. The breakdown by source also exposes how the costs are calculated, which is extra information that does not display in the user interface.

Action object

The extractor sends action data to Kafka the following times:

- Every 6 hours for pending actions.
- Immediately after an action completes, whether completion is success or failure.

Each action object has two main fields:

- `timestamp`: When the action object was broadcast
- `action`: The data that describes the given action

Sample Action object


```
{
  "timestamp": "2020-12-07T18:20:10Z",
  "action": {
    "oid": "144118995513593",
    "creationTime": "2020-12-07T18:13:10Z",
    "type": "MOVE",
    "state": "READY",
    "mode": "MANUAL",
    "category": "PERFORMANCE_ASSURANCE",
    "severity": "CRITICAL",
    "description": "Move Virtual Machine hp-esx59 from hp-bl433.eng.vmturbo.com to hp-
bl435.eng.vmturbo.com",
    "explanation": "Q2 VCPU Congestion",
    "target": {
      "oid": 73741823907376,
      "name": "hp-esx59",
      "type": "VirtualMachine"
    },
    "moveInfo": {
      "from": {
        "oid": "73741823904848",
        "name": "hp-bl433.eng.vmturbo.com",
        "type": "PHYSICAL_MACHINE"
      },
      "to": {
        "oid": "73741823904864",
        "name": "hp-bl435.eng.vmturbo.com",
        "type": "PHYSICAL_MACHINE"
      }
    },
    "related": {
      "PHYSICAL_MACHINE": [
        {
          "name": "hp-bl433.eng.vmturbo.com",
          "oid": 73741823904848
        }
      ],
      "DATACENTER": [
        {
          "name": "GTDC",
          "oid": 73599609177857
        }
      ],
      "VIRTUAL_DATACENTER": [
        {
          "name": "Resources-NACGTDHORDSKBZ",
          "oid": 73599609178290
```

```
        }
      ],
      "STORAGE": [
        {
          "name": "XIO-GTDBZVOLDS01",
          "oid": 73599609177895
        }
      ],
      "DISK_ARRAY": [
        {
          "name": "DiskArray-XIO-GTDBZVOLDS01",
          "oid": 73599609177870
        }
      ],
      "APPLICATION_COMPONENT": [
        {
          "name": "GuestLoad[H04BZFT02UAT005]",
          "oid": 73599609183789
        }
      ],
      "COMPUTE_CLUSTER": [
        {
          "name": "GTDC/NACGTDHORDSKBZ",
          "oid": 284705842023005
        }
      ],
      "STORAGE_CLUSTER": [
        {
          "name": "storageCluster1",
          "oid": 284705842023006
        }
      ],
      "K8S_CLUSTER": [
        {
          "name": "k8sCluster1",
          "oid": 284705842023007
        }
      ]
    }
  }
}
```


Common Action object fields

JSON Object	Description	Type	Repeated	Format	MapKeyType	Reference
buyRiInfo	Data that analysis uses to calculate Buy RI actions.	/shared/BuyRiInfo	False			BuyRiInfo
category	The category of action.	/enums/action_category	False			action_category
creationTime	When analysis created the action.	String	False	yyyy-MM-dd'T'HH:mm:ss'Z'		
deleteInfo	A description of a file is unattached and should be deleted.	/shared/DeleteInfo	False			DeleteInfo

JSON Object	Description	Type	Repeated	Format	MapKeyType	Reference
description	The description that analysis generates for the action.	String	False			
environment		String				environment_type
executionError		String				
executionTime		String				
explanation	The condition that the action is addressing. For example, Q2 VCPU CONGESTION.	String	False			
mode	The mode for the action execution.	/enums/action_mode	False			action_mode
moveInfo	Information to describe a MOVE action.	/shared/MoveChange	False			MoveChange
newRelated		JSON	True			
oid	The action object identifier.	Long	False			
related	An array of entities or groups that are related to the target entity.	/exporter/RelatedEntity	True		/enums/entity_type	RelatedEntity
resizeInfo	Information to describe a RESIZE action.	/shared/CommodityChange	False			CommodityChange
savings	The cost change for the action.	/exporter/CostAmount	False			CostAmount
scaleInfo	Information to describe a SCALE action.	/shared/MoveChange	False			MoveChange

JSON Object	Description	Type	Repeated	Format	MapKeyType	Reference
severity	For a given action, the severity of the issue that it addresses.	/enums/severity	False			severity
stableOid		Long				
state	The different states for the action, such as completed or pending.	/enums/action_state	False			action_state
target	The entity that the action operates on.	/shared/ActionImpactedEntity	False			ActionImpactedEntity
type	The action type.	/enums/action_type	False			action_type
userAccount		String				

Enums

The exported data can include the following enums.

AttachmentState

- ATTACHED:
- UNATTACHED:

OSType

- LINUX:
- LINUX_WITH_SQL_ENTERPRISE:
- LINUX_WITH_SQL_STANDARD:
- LINUX_WITH_SQL_WEB:
- RHEL:
- SUSE:
- UBUNTU_PRO:
- UNKNOWN_OS:
- WINDOWS:
- WINDOWS_BYOL:
- WINDOWS_WITH_SQL_ENTERPRISE:
- WINDOWS_WITH_SQL_STANDARD:
- WINDOWS_WITH_SQL_WEB:

action_category

- COMPLIANCE:

- EFFICIENCY_IMPROVEMENT:
- PERFORMANCE_ASSURANCE:
- PREVENTION:
- UNKNOWN:
- SAVING:

action_mode

- AUTOMATIC:
- DISABLED:
- EXTERNAL_APPROVAL:
- MANUAL:
- RECOMMEND:

action_state

- ACCEPTED:
- CLEARED:
- FAILED:
- FAILING:
- IN_PROGRESS:
- POST_IN_PROGRESS:
- PRE_IN_PROGRESS:
- QUEUED:
- READY:
- REJECTED:
- SUCCEEDED:
- BEFORE_EXEC:
- IN_PROGRESS_NON_CRITICAL:

action_type

- ACTIVATE:
- ALLOCATE:
- BUY_RI:
- DEACTIVATE:
- DELETE:
- MOVE:
- NONE:
- PROVISION:
- RECONFIGURE:
- RESIZE:
- SCALE:
- START:
- SUSPEND:

attr_type

- ENTITY_STATE:
- NUM_VCPU:
- VOLUME_ATTACHED:
- IDLE_POWER:

cost_category

- IP:
- ON_DEMAND_COMPUTE:
- ON_DEMAND_LICENSE:
- RESERVED_LICENSE:
- RI_COMPUTE:
- SPOT:
- STORAGE:
- TOTAL:
- CLOUD_COMMITMENT_USAGE:

cost_source

- BUY_RI_DISCOUNT:
- ENTITY_UPTIME_DISCOUNT:
- ON_DEMAND_RATE:
- RI_INVENTORY_DISCOUNT:
- TOTAL:
- UNCLASSIFIED:
- CLOUD_COMMITMENT_DISCOUNT:

entity_state

- FAILOVER:
- MAINTENANCE:
- POWERED_OFF:
- POWERED_ON:
- SUSPENDED:
- UNKNOWN:

entity_type

- ACTION_MANAGER:
- APPLICATION:
- APPLICATION_COMPONENT:
- APPLICATION_COMPONENT_SPEC:
- APPLICATION_SERVER:
- AVAILABILITY_ZONE:
- BILLING_FAMILY:
- BUSINESS:

- BUSINESS_ACCOUNT:
- BUSINESS_APPLICATION:
- BUSINESS_ENTITY:
- BUSINESS_TRANSACTION:
- BUSINESS_USER:
- CHASSIS:
- CLOUD_COMMITMENT:
- CLOUD_SERVICE:
- COMPUTE_CLUSTER:
- COMPUTE_RESOURCE:
- COMPUTE_TIER:
- CONTAINER:
- CONTAINER_PLATFORM_CLUSTER:
- CONTAINER_POD:
- CONTAINER_SPEC:
- DATABASE:
- DATABASE_SERVER:
- DATABASE_SERVER_TIER:
- DATABASE_TIER:
- DATACENTER:
- DESIRED_RESERVED_INSTANCE:
- DESKTOP_POOL:
- DISK_ARRAY:
- DISTRIBUTED_VIRTUAL_PORTGROUP:
- DOCUMENT_COLLECTION:
- DPOD:
- GROUP:
- HCI_PHYSICAL_MACHINE:
- HYPERVISOR_SERVER:
- INFRASTRUCTURE:
- INTERNET:
- IO_MODULE:
- IP:
- K8S_CLUSTER:
- LICENSING_SERVICE:
- LOAD_BALANCER:
- LOGICAL_POOL:
- MAC:
- MOVER:
- NAMESPACE:
- NETWORK:
- NETWORKING_ENDPOINT:
- NODE_POOL:

- OPERATOR:
- PHYSICAL_MACHINE:
- PORT:
- PROCESSOR_POOL:
- REGION:
- RESERVED_INSTANCE:
- RESERVED_INSTANCE_SPECIFICATION:
- RESOURCE_GROUP:
- RIGHT_SIZER:
- SAVINGS:
- SERVICE:
- SERVICE_ENTITY_TEMPLATE:
- SERVICE_PROVIDER:
- STORAGE:
- STORAGE_CLUSTER:
- STORAGE_CONTROLLER:
- STORAGE_TIER:
- STORAGE_VOLUME:
- SWITCH:
- TARGET:
- THIS_ENTITY:
- THREE_TIER_APPLICATION:
- VIEW_POD:
- VIRTUAL_APPLICATION:
- VIRTUAL_DATACENTER:
- VIRTUAL_MACHINE:
- VIRTUAL_MACHINE_SPEC:
- VIRTUAL_VOLUME:
- VLAN:
- VM_SPEC:
- VPOD:
- WEB_SERVER:
- WORKLOAD_CONTROLLER:
- _NONE_:
- DATA_WAREHOUSE:
- DATA_WAREHOUSE_TIER:
- NODE_GROUP:
- POWERVM_PHYSICAL_MACHINE:

environment_type

- CLOUD:
- HYBRID:
- ON_PREM:

- UNKNOWN_ENV:

file_type

- CONFIGURATION:
- DISK:
- ESXCONSOLE:
- ISO:
- LOG:
- MEMORY:
- SNAPSHOT:
- SWAP:

metric_type

- ACTIVE_SESSIONS:

  Count of the number of actively logged in sessions Count
- BALLOONING:

  Number of kb the host is dedicated for ballooning kilobytes
- BUFFER_COMMODITY:

  Utilization of a port buffer (fabric, UCS specific) megabytes
- CARBON:

  Carbon dioxide equivalent (CO2e) emissions for a given entity grams
- CONNECTION:

  Number of connections onto the application Count
- COOLING:

  acceptable temperature range that is utilized by an entity percentage
- CPU:

  CPU used by host (current column) or host CPU consumed by VMs (consumed column). MHz
- CPU_ALLOCATION:

  For VMs (consumed column) - VM buys from VDC or from desktop pool (VDI). For hosts - PMs sells to VDCs. MHz
- CPU_HEADROOM:

  Number of VMs that can fit in all of the available CPU (on a cluster) Count
- CPU_PROVISIONED:

  Capacity is based on the over-provisioned value in policy MHz
- CPU_READY:

  Ready queue ms
- DB_CACHE_HIT_RATE:

  The percentage of accesses that result in cache hits. Percentage
- DB_MEM:

  DB mem utilization or DM capacity kilobytes
- DTU:

  Capacity defined in the cloud Count

- ENERGY:

  Electricity consumed by a given entity over a period of time watt-hours
- EXTENT:

  Internal NA
- FLOW:

  Amount of network traffic per second between two nodes kilobytes/sec
- FLOW_ALLOCATION:

  bytes
- HEAP:

  Amount of heap configured for an application kilobytes
- IMAGE_CPU:

  For VDI, usage of the business user MHz
- IMAGE_MEM:

  For VDI, usage of the business user kilobytes
- IMAGE_STORAGE:

  For VDI, usage of the business user megabytes
- IO_THROUGHPUT:

  Amount of traffic through the IO adapter kilobytes/sec
- MEM:

  Physical machine - usage of memory. VM - consumed memory. kilobytes
- MEM_ALLOCATION:

  Physical machine - what the host provides to VDCs. VM (consumed column) - what the VM buys from VDC or desktop pool. kilobytes
- MEM_HEADROOM:

  Number of VMs that can fit in the available memory on the cluster Count
- MEM_PROVISIONED:

  Capacity, based on the over-provisioned value in policy kilobytes
- NET_THROUGHPUT:

  Amount of traffic through the Network adapter kilobytes/sec
- POOL_CPU:

  For VDI, usage of the desktop pool MHz
- POOL_MEM:

  For VDI, usage of the desktop pool kilobytes
- POOL_STORAGE:

  For VDI, usage of the desktop pool megabytes
- PORT_CHANEL:

  Traffic on ports for UCS Mgr/HPE1 kilobytes/sec
- POWER:

  Electricity consumed by a given entity watts
- PROCESSING_UNITS:

  unit of measure for shared processing power across one or more virtual processors PU

- Q16_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q1_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q2_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q32_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q3_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q4_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q5_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q64_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q6_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q7_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q8_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- QN_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- REMAINING_GC_CAPACITY:

  Percentage of time not spent on garbage collection Percentage
- RESPONSE_TIME:

  Application response ms
- SLA_COMMODITY:

  User-defined value Number
- STORAGE_ACCESS:

  Number of I/O (read + write) within two discovery cycles count
- STORAGE_ALLOCATION:

  Allocated storage megabytes
- STORAGE_AMOUNT:

  Used storage megabytes
- STORAGE_HEADROOM:

  Count of VM that can fit for the storage in a cluster in terms of storage Count
- STORAGE_LATENCY:

  Acceptable latency. User can set capacity for most storage, is discovered for NetApp. ms
- STORAGE_PROVISIONED:

  Capacity, based on the over-provisioned value in policy megabytes
- SWAPPING:

  Memory swapping bytes/sec
- THREADS:

  Threads in use Count
- TOTAL_HEADROOM:

  Min value of CPU, Mem, IO, of the headrooms Count
- TOTAL_SESSIONS:

  Total number of possible sessions Count
- TRANSACTION:

  Transactions for the application transactions/sec
- TRANSACTION_LOG:

  Size of the transaction log megabytes
- VCPU:

  CPU in use MHz
- VCPU_LIMIT_QUOTA:

  Amount of virtual CPU limit a pod has allocated against the namespace quota millicores
- VCPU_REQUEST:

  Virtual CPU request allocated by a pod against the node allocatable capacity millicores
- VCPU_REQUEST_QUOTA:

  Amount of virtual CPU request a pod has allocated against the namespace quota millicores
- VCPU_THROTTLING:

  Throttling of container virtual CPU that could impact response time percentage
- VIRTUAL_PROCESSORS:

Guest OS representation of the capacity of the VM VP

- VMEM:

  Memory sold by a VM. Memory sold and bought for a container kilobytes
- VMEM_LIMIT_QUOTA:

  Total amount of virtual memory limit for all pods allocated to the namespace against the namespace quota kilobytes
- VMEM_REQUEST:

  Virtual memory request allocated by a pod against the node allocatable capacity kilobytes
- VMEM_REQUEST_QUOTA:

  Amount of virtual memory request a pod has allocated against the namespace quota kilobytes
- VSTORAGE:

  Storage utilized by an entity megabytes
- CONCURRENT_QUERIES:
- GPU:
- GPU_MEM:
- GPU_REQUEST:
- GPU_REQUEST_QUOTA:
- LLM_CACHE:
- QUEUING_TIME:
- SERVICE_TIME:

savings_type

- MISSED_INVESTMENTS:

  Actions you could have taken that would inscrease your investment to provide more resources to your applications.
- MISSED_SAVINGS:

  Actions you could have taken that could have reduced your costs.
- REALIZED_INVESTMENTS:

  Actions you have taken that inscrease your investment to provide more resources to your applications.
- REALIZED_SAVINGS:

  Actions you have taken that have reduced your costs.

severity

- CRITICAL:
- MAJOR:
- MINOR:
- NORMAL:

terminal_state

- FAILED:
- SUCCEEDED:

Data exporter objects

The data exporter generates the following JSON objects.

AccountExpenses

JSON Object	Description	Type	Repeated	MapKeyType	Reference
expenseDate		String	False		
serviceExpenses		/exporter/CostAmount	False	String	CostAmount
serviceList		String	True		
newServiceExpenses		JSON	True		CostAmount

Action

JSON Object	Description	Type	Repeated	Format	MapKeyType	Reference
buyRiInfo	Data that analysis uses to calculate Buy RI actions.	/shared/BuyRiInfo	False			BuyRiInfo
category	The category of action.	/enums/action_category	False			action_category
creationTime	When analysis created the action.	String	False	yyyy-MM-dd'T'HH:mm:ss'Z'		
deleteInfo	A description of a file is unattached and should be deleted.	/shared/DeleteInfo	False			DeleteInfo
description	The description that analysis generates for the action.	String	False			
environment		String				environment_type
executionError		String				
executionTime		String				

JSON Object	Description	Type	Repeated	Format	MapKeyType	Reference
explanation	The condition that the action is addressing. For example, Q2 VCPU CONGESTION.	String	False			
mode	The mode for the action execution.	/enums/action_mode	False			action_mode
moveInfo	Information to describe a MOVE action.	/shared/MoveChange	False			MoveChange
newRelated		JSON	True			
oid	The action object identifier.	Long	False			
related	An array of entities or groups that are related to the target entity.	/exporter/RelatedEntity	True		/enums/entity_type	RelatedEntity
resizeInfo	Information to describe a RESIZE action.	/shared/CommodityChange	False			CommodityChange
savings	The cost change for the action.	/exporter/CostAmount	False			CostAmount
scaleInfo	Information to describe a SCALE action.	/shared/MoveChange	False			MoveChange
severity	For a given action, the severity of the issue that it addresses.	/enums/severity	False			severity
stableOid		Long				
state	The different states for the action, such as completed or pending.	/enums/action_state	False			action_state

JSON Object	Description	Type	Repeated	Format	MapKeyType	Reference
target	The entity that the action operates on.	/shared/ ActionImpactedEntity	False			ActionImpactedEntity
type	The action type.	/enums/ action_type	False			action_type
userAccount		String				

CloudSavings

JSON Object	Description	Type	Reference
accountId		Long	
actionChainId		Long	
actionInitiator		String	
actionType		String	action_type
availabilityZoneId		Long	
billingFamilyId		Long	
cloudServiceId		Long	
currency		Integer	
entityId		Long	
entityType		String	entity_type
realizedInvestments		Double	
realizedSavings		Double	
regionId		Long	
resourceGroupId		Long	
serviceProviderId		Long	
tagGroupId		Long	
timestamp		String	yyyy-MM-dd'T'HH:mm:ss'Z'

Commodity

JSON Object	Description	Type
capacity		Double
consumed		Double
current		Double
utilization		Double
peakConsumed		Double
type		String

CostAmount

JSON Object	Description	Type
amount		Double
unit		String
type		String

CostByCategory

JSON Object	Description	Type	MapKeyType
source		Float	/enums/cost_source
total		Float	

Entity

JSON Object	Description	Type	Repeated	MapKeyType	Reference
accountExpenses		/exporter/ AccountExpenses	False		AccountExpenses
attrs	Additional descriptions for a given entity. The values of this are different for different entity types.	/shared/ entity.attrs	False		entity.attrs
cost		/exporter/ EntityCost	False		EntityCost
environment	The environment type - Hybrid, Cloud, On-prem, etc.	/enums/ environment_type	False		environment_type
metric	For the given metric type, the utilization and capacity of the commodity.	/exporter/ Commodity	False	/enums/ metric_type	Commodity
name	The entity name.	String	False		
newMetric		JSON	True		Commodity
newRelated		JSON	True		RelatedEntity
oid	The entity ID.	Long	False		
related	For the current entity, another entity that is connected to it in the supply chain.	/exporter/ RelatedEntity	True	/enums/ entity_type	RelatedEntity

JSON Object	Description	Type	Repeated	MapKeyType	Reference
state	The current state of the entity - POWERED_ON, SUSPENDED, MAINTENANCE, etc.	/enums/ entity_state	False		entity_state
type	The entity type.	/enums/ entity_type	False		entity_type

EntityCost

JSON Object	Description	Type	MapKeyType	Reference
category		/exporter/CostByCategory	/enums/cost_category	CostByCategory
total		Float		
unit		String		

Group

JSON Object	Description	Type	Reference
attrs		/shared/group.attrs	group.attrs
name		String	
oid		Long	
type		/enums/entity_type	entity_type

RelatedEntity

JSON Object	Description	Type
name		String
oid		Long
type		String
relatedEntityIdentifier		String

Shared data objects

The data exporter generates the following JSON objects. Some of these objects are shared with other data management components in Turbonomic.

ActionCommodity

Data Object	Description	Type
capacity	The available capacity of the given commodity.	float
percentileUtilization	For entities and resources that support it, the calculated percentile of utilization.	Double
used	The amount of the resource currently in use.	float

Data Object	Description	Type
utilization	The percentage of capacity in use.	float

ActionEntity

Data Object	Description	Type	Reference	Repeated
oid	The ID of the entity.	Long		
name	The entity name. This corresponds to the name that displays in the user interface.	String		
type	The entity type.	/enums/entity_type	entity_type	
attrs		JSON	entity.attrs	
newMetric		JSON	Commodity	True
newRelated		JSON	RelatedEntity	True

ActionImpactedCost

Data Object	Description	Type
onDemandCost	For on-demand billing, the entity cost.	Float
onDemandRate	For on-demand billing, the rate of cost.	Float
riCoveragePercentage	For RIs, the percentage of RI coverage.	Float

ActionImpactedCosts

Data Object	Description	Type	Reference
afterActions	A description of what the cost will be after the action executes.	/shared/ ActionImpactedCost	ActionImpactedCost
beforeActions	A description of the current cost, before the action executes.	/shared/ ActionImpactedCost	ActionImpactedCost

ActionImpactedEntity

Data Object	Description	Type	MapKeyType	Reference
affectedCosts	A description of the before and after costs for this entity.	/shared/ ActionImpactedCosts		ActionImpactedCosts
affectedMetrics	A description of the before and after metrics for this entity.	/shared/ ImpactedMetric	/enums/ metric_type	ImpactedMetric
name	The entity name.	String		
oid	The entity ID.	Long		
settings	For entities that support percentile calculations, the	/shared/EntitySettings		EntitySettings

Data Object	Description	Type	MapKeyType	Reference
	aggressiveness and observation period for this entity.			
type	The entity type.	/enums/entity_type		entity_type
attrs		JSON		entity.attrs
newMetric		JSON	Commodity	True
newRelated		JSON	RelatedEntity	True

BuyRiInfo

Data Object	Description	Type	Reference
computeTier	The compute tier, or workload family for the action. The ActionEntity object will be for an entity of type COMPUTE_TIER.	/shared/ActionEntity	ActionEntity
count		Integer	
masterAccount	The account that manages the RI billing.	/shared/ActionEntity	ActionEntity
region	The region that will host the RIs.	/shared/ActionEntity	ActionEntity
target	The cloud account or subscription that will manage the RIs.	/shared/ActionEntity	ActionEntity

Commodity

Data Object	Description	Type
capacity		Double
consumed		Double
current		Double
peakConsumed		Double
type		String
utilization		Double

CommodityChange

Data Object	Description	Type	Reference
attribute		String	
commodityType	The type of commodity the action will change - MEM, IO_THROUGHPUT, etc.	/enums/metric_type	metric_type
from	The commodity value before the action executes.	Float	

Data Object	Description	Type	Reference
percentileChange	For entities and commodities that support percentile calculations, the before and after values for percentile utilization and settings.	/shared/ CommodityPercentileChange	CommodityPercentileChange
target	A description of the entity this action will change.	/shared/ActionEntity	ActionEntity
to	The commodity value after the action executes.	Float	
unit	The disp;layed units of measurement.	String	

CommodityPercentileChange

Data Object	Description	Type
after	The percentile utilization after the action executes	Double
aggressiveness	The aggressiveness setting used to calculate the percentile.	Integer
before	The percentile utilization before the action executes.	Double
observationPeriodDays	The observation period used to calculate the percentile.	Integer

DeleteInfo

Data Object	Description	Type	Format
filePath	The path to the file.	String	
fileSize	The size of the file.	Double	
lastModifiedTimestamp	When the file was last modified.	String	yyyy-MM-dd'T'HH:mm:ss'Z'
unattachedDays	How many days the file has been unattached.	Integer	
unit	The displayed unit of measure for the file size.	String	

entity.attrs

Data Object	Description	Type	SupportedEntityTypes	Repeated	Reference
attachment_state	Whether a given volume is attached or unattached.	/enums/ AttachmentState	VIRTUAL_VOLUME	False	AttachmentState

Data Object	Description	Type	SupportedEntityTypes	Repeated	Reference
connected_networks	A list of network names that a given VM is connected to.	MULTI_TEXT	VIRTUAL_MACHINE	True	
cpu_model	For a physical host, the model of CPU on that host.	TEXT	PHYSICAL_MACHINE	False	
guest_os_name	The name of the OS running on a given VM.	TEXT	VIRTUAL_MACHINE	False	
guest_os_type	The specific type for the OS running on a given VM.	/enums/OSType	VIRTUAL_MACHINE	False	OSType
is_local	Whether a storage is local to the machine, or networked.	BOOLEAN	STORAGE	False	
model	The specific model of the given physical host.	TEXT	PHYSICAL_MACHINE	False	
num_cpus	The number of CPUs on the given physical host.	NUMBER	PHYSICAL_MACHINE	False	
tags	A list of tags that are assigned to the given entity.	TEXT_MULTIMAP (Embedded Reporting) or List (Data Exporter)	ALL	False	
targets	The target that manages the given entity.	/shared/Target	ALL	True	Target
timezone	The timezone that is set for the clock on the physical machine.	TEXT	PHYSICAL_MACHINE	False	
vendor_id		TEXT	VIRTUAL_VOLUME,BUSINESS_ACCOUNT		

EntitySettings

Data Object	Description	Type
percentileAggressiveness	The current aggressiveness setting.	Integer
percentileObservationPeriodDays	The current observation period.	Integer

group.attrs

Data Object	Description	Type	Repeated	Reference	SupportedEntityTypes	SupportedGroupTypes
dynamic	Whether this group is static or dynamic. true for dynamic.	BOOLEAN	False		STORAGE_CLUSTER COMPUTE_CLUSTER GROUP K8S_CLUSTER BILLING_FAMILY RESOURCE_GROUP	RESOURCE_GROUP COMPUTE_CLUSTER BILLING_FAMILY GROUP K8S_CLUSTER STORAGE_CLUSTER
member_types		MULTI_TEXT	True		STORAGE_CLUSTER COMPUTE_CLUSTER GROUP K8S_CLUSTER BILLING_FAMILY RESOURCE_GROUP	RESOURCE_GROUP COMPUTE_CLUSTER BILLING_FAMILY GROUP K8S_CLUSTER STORAGE_CLUSTER
origin		TEXT	False		STORAGE_CLUSTER COMPUTE_CLUSTER GROUP K8S_CLUSTER BILLING_FAMILY RESOURCE_GROUP	RESOURCE_GROUP COMPUTE_CLUSTER BILLING_FAMILY GROUP K8S_CLUSTER STORAGE_CLUSTER

Data Object	Description	Type	Repeated	Reference	SupportedEntityTypes	SupportedGroupTypes
tags		TEXT_MULTIMAP (Embedded Reporting) or List (Data Exporter)	False		GROUP RESOURCE_GROUP COMPUTE_CLUSTER K8S_CLUSTER STORAGE_CLUSTER BILLING_FAMILY	ALL
targets		/shared/Target	False	Target	GROUP RESOURCE_GROUP COMPUTE_CLUSTER K8S_CLUSTER STORAGE_CLUSTER BILLING_FAMILY	ALL

ImpactedMetric

Data Object	Description	Type	Reference
afterActions	A description of what the metric utilization will be after the action executes.	/shared/ActionCommodity	ActionCommodity
beforeActions	A description of the current utilization, before the action executes.	/shared/ActionCommodity	ActionCommodity

MoveChange

Data Object	Description	Type	Repeated	Reference
from	Where the entity will move from.	/shared/ActionImpactedEntity	False	ActionImpactedEntity
resource		/shared/ActionEntity	True	ActionEntity
to	Where the entity will move to.	/shared/ActionImpactedEntity	False	ActionImpactedEntity

RelatedEntity

Data Object	Description	Type
name		String
oid		Long
type		String
relatedEntityIdentifier		String

Target

Data Object	Description	Type
category		String
entityVendorId		String
name		String
oid		Long
type		String

action.attrs

JSON Object	Description	Type	MapKeyType	Reference
buyRiInfo	For an action to buy RIs, additional information about the RIs.	/shared/BuyRiInfo		BuyRiInfo
deleteInfo	For an action to delete a file, a description of that file.	/shared/DeleteInfo		DeleteInfo
moveInfo	For a move action, a description of from and to for the move.	/shared/MoveChange	/enums/ entity_type	MoveChange
resizeInfo	For a resize action, a description of the commodity changes.	/shared/ CommodityChange	/enums/ metric_type	CommodityChange
scaleInfo	For scaling a cloud VM, a description of the from and to for the VM move.	/shared/MoveChange	/enums/ entity_type	MoveChange
target	Extra information about the entity this action impacts.	/shared/ ActionImpactedEntity		ActionImpactedEntity

Reference: Working with YAML files

YAML is the primary file format to create and configure resources on kubernetes, including everything to do with the Turbonomic platform. The Custom Resource YAML provides a convenient, single place to

define the majority of configuration details for Turbonomic. When editing YAML files, you must be careful to respect the file syntax, including indents in the file. General rules for edits include:

- Always uses spaces, not tabs, for all indentation. Since spacing and indentations matter, and can yield an invalid result or a parameter completely skipped over, you should work with YAML files with an editor that supports using vertical lines associated with indentations to visually spot a misalignment. If your editor of choice makes this difficult, you can use the linux `expand` utility when you're done, to convert tabs to equivalent spaces.
- Indention uses two spaces per level.
- Be careful to keep the same indentation for all properties in a given section.
- Never use the same property name twice in the same section. Doing this will render the YAML file invalid, though you will likely not see any notification of a problem. Rather, all but one of the property definitions will be silently ignored.

Spacing matters

For the Turbonomic Custom Resource, indentation defines where parameters are applied (globally or to specific components), so ensure you line up the text appropriately. The following example shows examples of specifications that are applied at a global level (`{“spec”:{“global”:[{“tag”:”8.16.4”}]}}`) which sets the container image tag for all instances. Then indented you see properties that are global for the remote database (`{“spec”:{“properties”:{"global":[{“dbPort”:”6033”}]}}}`) describes a property of `dbPort` that would be set for remote DB connections.

Each line is indented two spaces from the higher level.

```
spec:
  global:
    repository: turbonomic
    tag: 8.16.4
  properties:
    global:
      dbPort: 6033
  kubeturbo:
    enabled: true
  aws:
    enabled: true
```

Combine properties correctly

A YAML file is read beginning to end, and if there are different parameters that apply to the same component, they need to be combined. The following example shows a YAML where properties for the `ui` component of image tag *and* memory limit resources.

```
spec:
  global:
    repository: turbonomic
    tag: 8.16.4
  ui:
    image:
      tag: 8.16.4
  properties:
    global:
      dbPort: 6033
  kubeturbo:
    enabled: true
  aws:
    enabled: true
  ui:
    resources:
      limits:
        memory: 4Gi
```

This YAML will not set both image tag and memory limit resources for the `ui` component because they are set in two different sections. The memory limits would be applied since that is the last section to be

read, and it would overwrite the first set of properties with the image tag. The following YAML will set both image tag and memory limit resources for the `ui` component:

```
spec:
  global:
    repository: turbonomic
    tag: 8.16.4
  ui:
    image:
      tag: 8.16.4
    resources:
      limits:
        memory: 4Gi
  properties:
    global:
      dbPort: 6033
  kubeturbo:
    enabled: true
  aws:
    enabled: true
```

To avoid having the same property name twice in the same section, search for an existing instance of the key you want to add. With the complex embedded structure in many k8s resources, this can be quite challenging; however, you can use an online validator that does not accept duplicate keys to check your entire CR file (such as YAMLint - The YAML Validator).

Finding a property

This documentation refers to specific properties in the CR file using a "path" expression. For example, assume the path `/spec/global/repository` to designate a `repository` property in the file. You can find the property in the file as follows:

1. Find a line that says `spec:` with no indentation at all.
2. Between that line and the next unindented line (not counting comments, which start with #), find a line that says `global:` and is at the next level of indentation.
3. Between that line and the next line with the same indentation, find a line that starts with `repository:`. That line is where the addressed property is defined.

In this example, find the `repository` property specified by `/spec/global/repository`:

```
apiVersion: charts.helm.k8s.io/v1alpha1
kind: Xl
metadata:
  name: xl-release
spec:
  properties:
    global:
      repository:        # This is NOT the correct property
        ...

  # Global settings
  global:
    repository:          # This is the one you want
```

The first `repository` property is not at `/spec/global/repository`, but at `/spec/properties/global/repository`.

Chapter 14. Integration - SaaS reporting

The Turbonomic platform includes an optional SaaS-based reporting component that you can choose to enable. You can use SaaS reporting to understand trends in application resource management and to share insights with stakeholders through reports and Liveboards.

SaaS reporting is provided by the ThoughtSpot analytics software and runs on the public cloud. It stores a history of your managed environment and then presents various views of this history through a set of standard Liveboards. You can create your own Liveboards and reports to focus on other areas of concern.

Enabling SaaS reporting

The steps to enable the optional SaaS reporting component depend on whether your Turbonomic instance is on-prem or hosted on the public cloud. In either case, your Turbonomic instance must meet certain requirements and you must request access from your Turbonomic representative.

Requirements

- Turbonomic version 8.15.4 or newer

  You can check your version of the product by selecting the **Updates** option from the **Settings** page of the Turbonomic user interface (UI) and scrolling to the **About** section.
- Operator version 42.76 or newer

Restriction: Turbonomic hosts all instance data in US or EU region-based data centers. If you are based in a region outside of the mentioned regions and have data residency restrictions, work with your employer to obtain the necessary approvals for SaaS reporting enablement. Alternatively, if you have data residency restrictions, Turbonomic can remove the SaaS reporting option from your instance. Customers from EMEA, APAC, and other regions are welcome to enable SaaS reporting if they have no such restrictions.

Reporting user roles

Contact your Turbonomic representative to request access to SaaS reporting. Specify an email that is registered with IBM Security Verify.

Two different user roles are available for the SaaS reporting environment:

- Report Editors can view, modify, and create charts and Liveboards.
- Report Admins have all the privileges of Report Editors, but they can also create and modify Report Editor users. For more information, see “Managing users” on page 1919.

Turbonomic allows one user per role, per instance. Work with your representative to grant a stakeholder in your organization the Report Admin role.

Enabling SaaS reporting for Turbonomic instances hosted in the public cloud

If your Turbonomic instance is hosted in the public cloud (SaaS) and maintained by your Turbonomic representative, your representative must enable SaaS reporting for you. After SaaS reporting is enabled, you can access the ThoughtSpot UI by selecting the **Reports** option in the Turbonomic UI sidebar menu and entering your IBM Security Verify email and password. For more information, see “Getting started with SaaS reporting” on page 1915.

Enabling SaaS reporting for on-prem Turbonomic instances

If you installed Turbonomic in an on-prem environment, complete the following steps to enable SaaS reporting.

1. To request a new reporting environment, go to the Turbonomic IBM support page and click the **Open a case** button.
2. Submit the case with a severity of 2 and provide the following details in the case description:
   - The full name of your company
   - The name and email address of your company's Turbonomic administrator
   - Your Turbonomic instance ID. To obtain this value, SSH to your instance and run the following command:

     ```
     kubectl exec consul[TAB] -- consul kv get instanceID
     ```

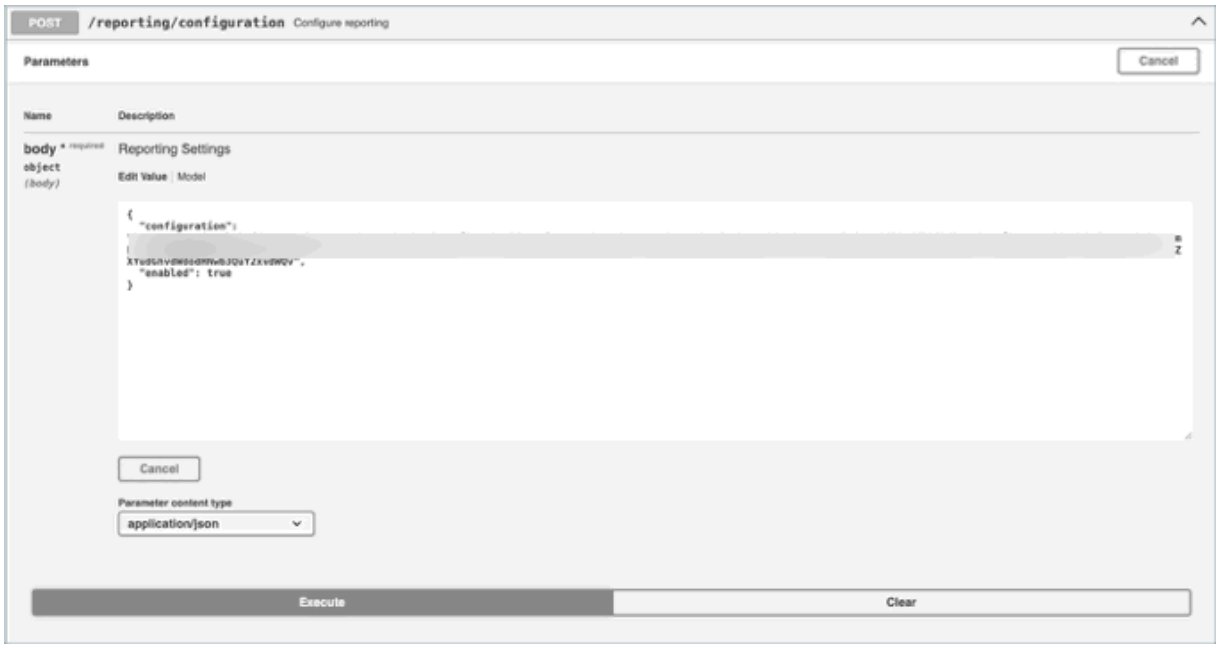
   - Verify that your company has an active IBM Passport Advantage agreement.

   After your request is processed, your Turbonomic representative provides an encoded configuration string that you use in the following steps.
3. Log in to your Turbonomic instance.
4. Open the Turbonomic swagger at the following URL:

   ```
   <Your/Turbo/URL.com>/swagger/#/
   ```

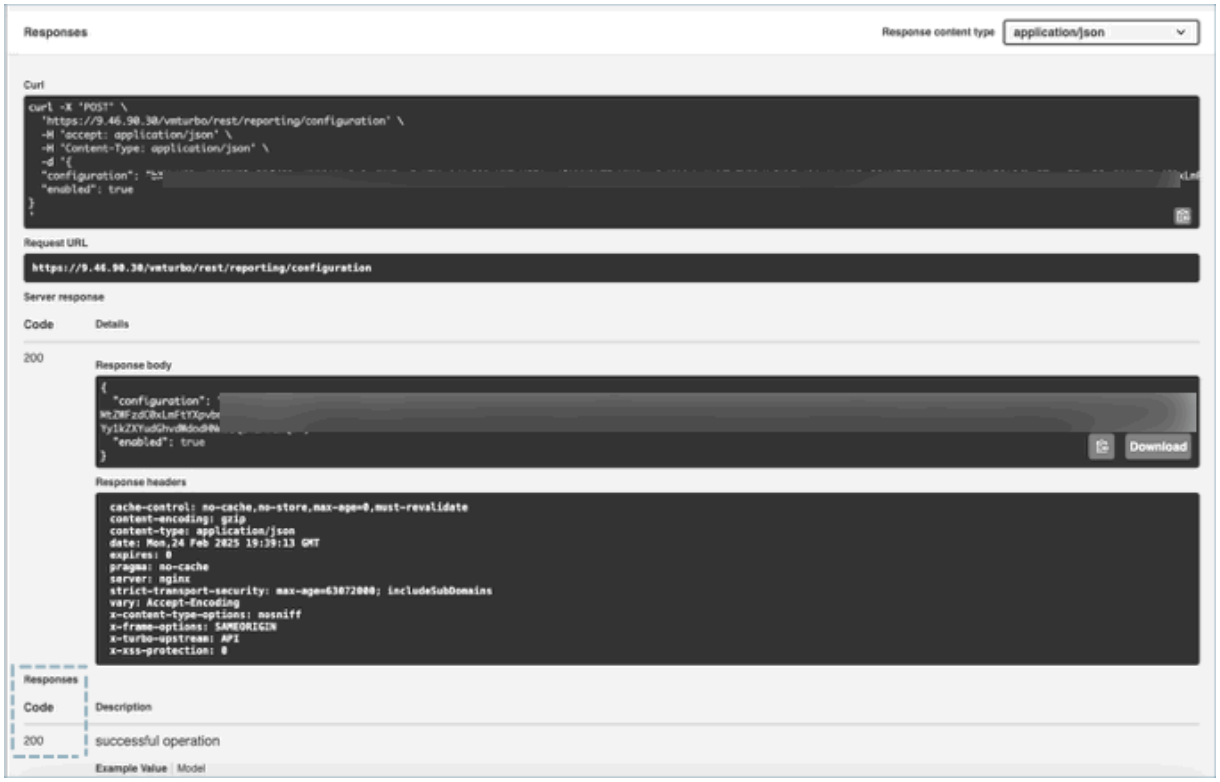
5. In the **Reporting** section, select the **POST /reporting/configuration** endpoint and click **Try it out**.
6. In the **body** field, enter the encoded configuration string that your Turbonomic representative provided in the following format:

   ```
   {
       "configuration":
   "<your-configuration-string>",
       "enabled": true
   }
   ```



7. Click **Execute**.
8. In the response window, look for a 200 response code, which verifies that the encoded configuration string is valid and the POST operation was successful.

.



9. Go back to the home page for your Turbonomic instance and refresh the page. SaaS reporting is now enabled.
10. Select **REPORTS (NEW)** from the sidebar menu.
11. Log in to ThoughtSpot with your IBM Security Verify email and password.

    The initial login must use the same email that you specified for your Turbonomic administrator in your enablement request to IBM support (see step 2).

For more information, see Getting started with SaaS reporting.

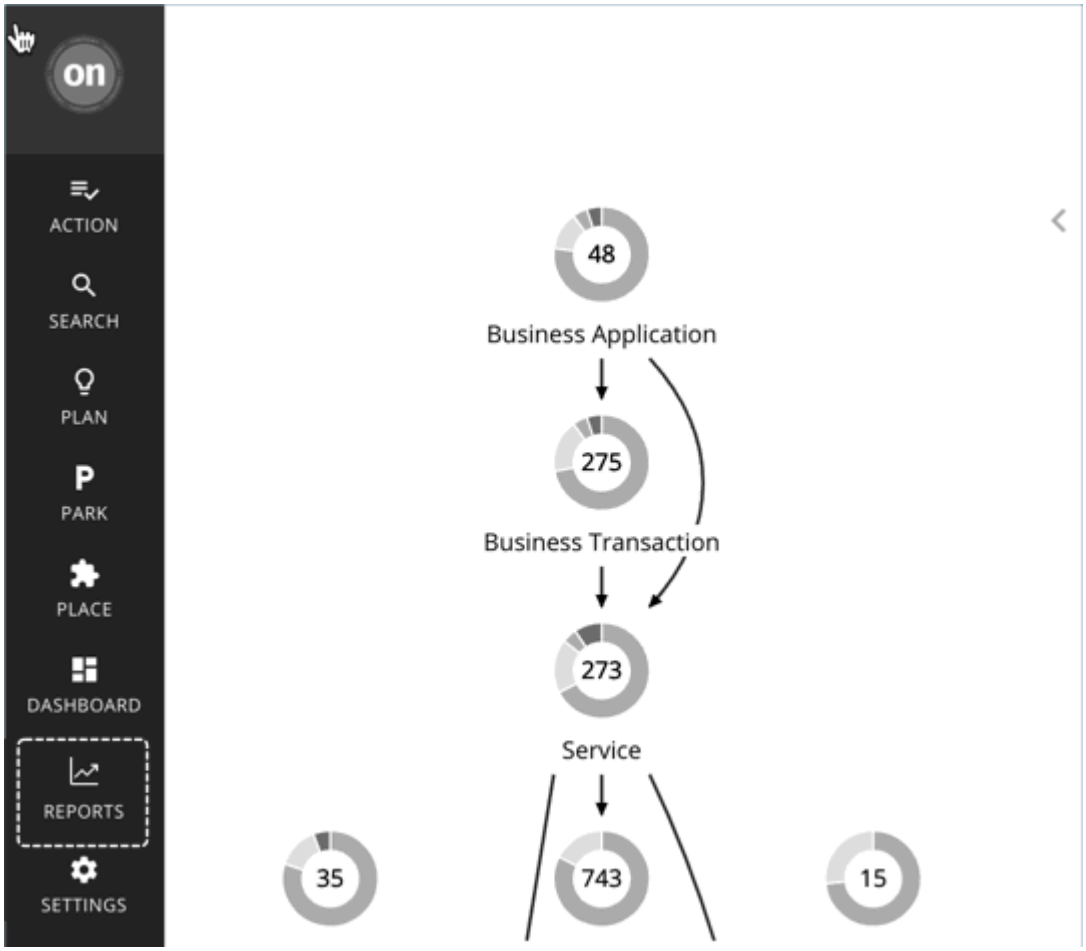
Getting started with SaaS reporting

After you enable SaaS reporting, you can view charts, Liveboards, and other historical data that is provided by ThoughtSpot. You can also schedule reports for any Liveboard to be regularly emailed to stakeholders.

If you are new to ThoughtSpot and need help with getting started, see the ThoughtSpot documentation.

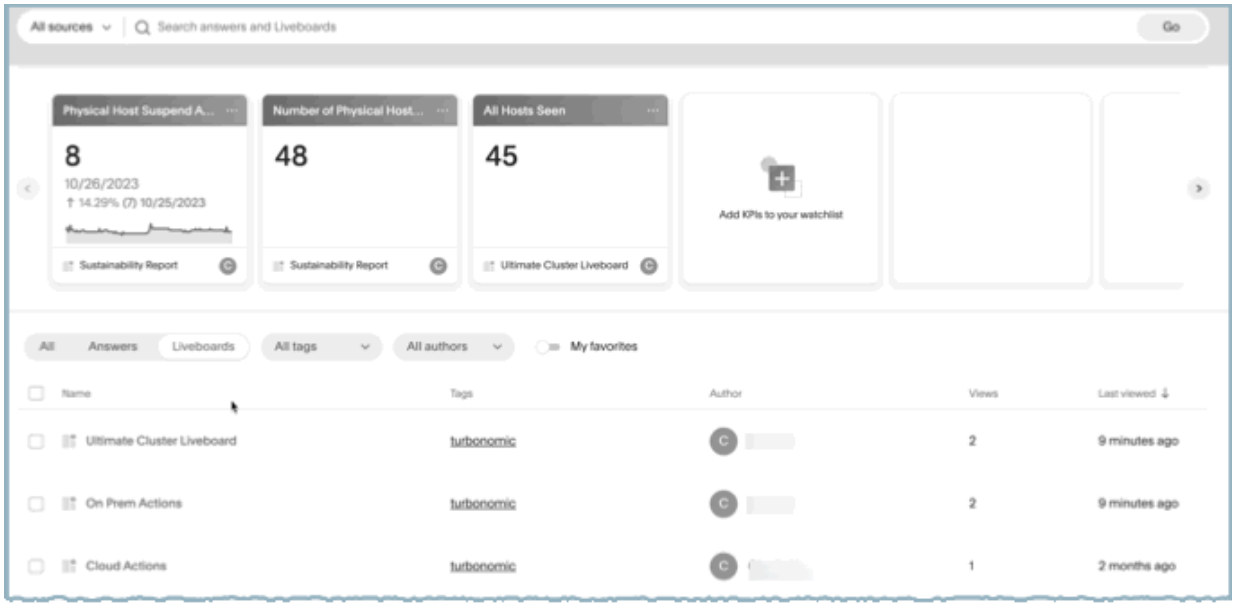
Viewing default Liveboards

1. In the Turbonomic user interface (UI), click **Reports** in the navigation sidebar.



This option opens the login page for the ThoughtSpot UI.

2. Click **Log in using Single Sign On** and specify your IBM Security Verify email and password.
3. After you log in to ThoughtSpot, you can see a list of the default Turbonomic Liveboards. Select a Liveboard from the list to review charts and other historical data about your managed environment.



For more information about the default Liveboards, see “Overview of default Liveboards” on page 1922.

Because Turbonomic regularly updates the default Liveboards, editing these Liveboards is not recommended. Instead, make a copy of the board that you want to customize and edit the copy. For more information, see “Creating custom Liveboards” on page 1941.

ThoughtSpot Liveboard queries that take more than 10 minutes to complete result in a query timeout. If a timeout occurs on a custom chart that you created, review the chart configuration to ensure it's not trying to pull too much data or otherwise improperly configured.

Sharing Liveboards

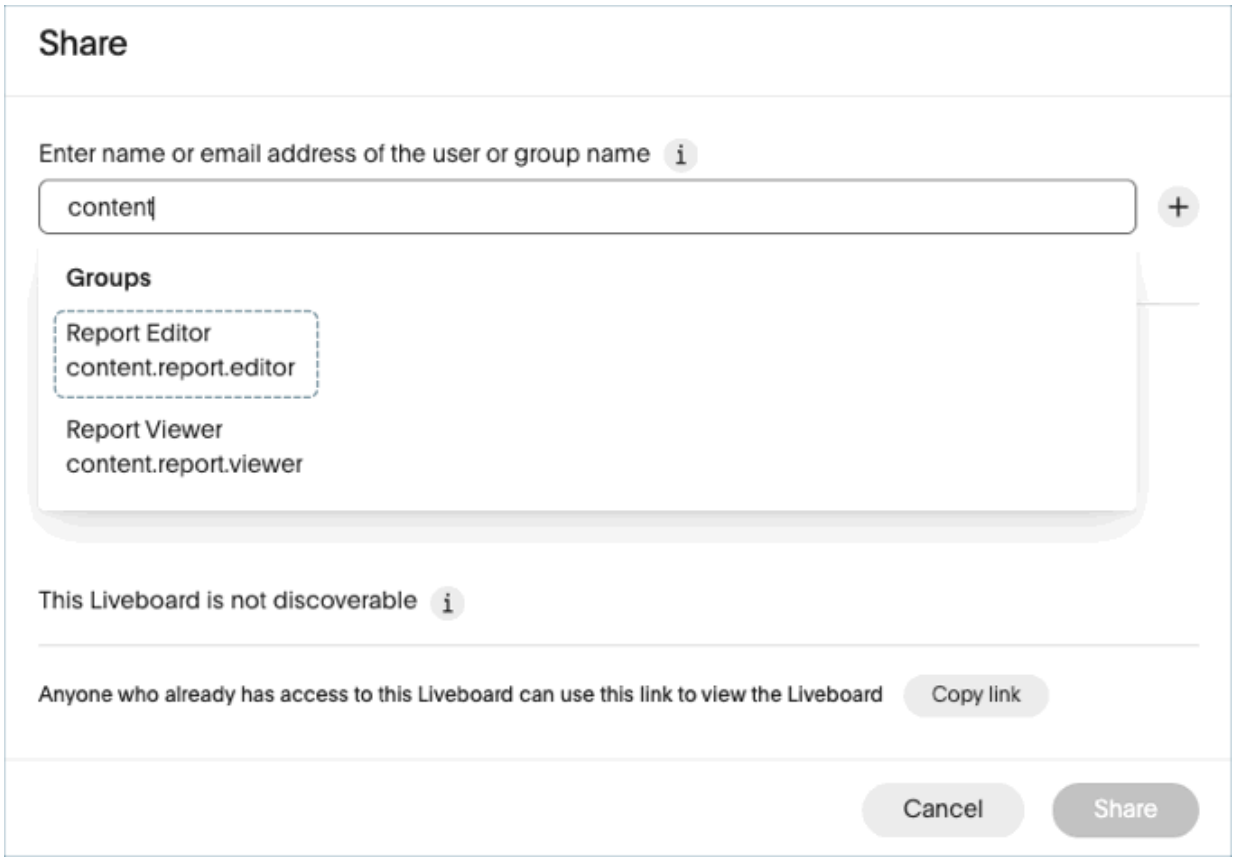
You can share any custom or default Liveboard with other users that have the Report Editor role in your ThoughtSpot instance.

For more information about user roles, see “Managing users” on page 1919.

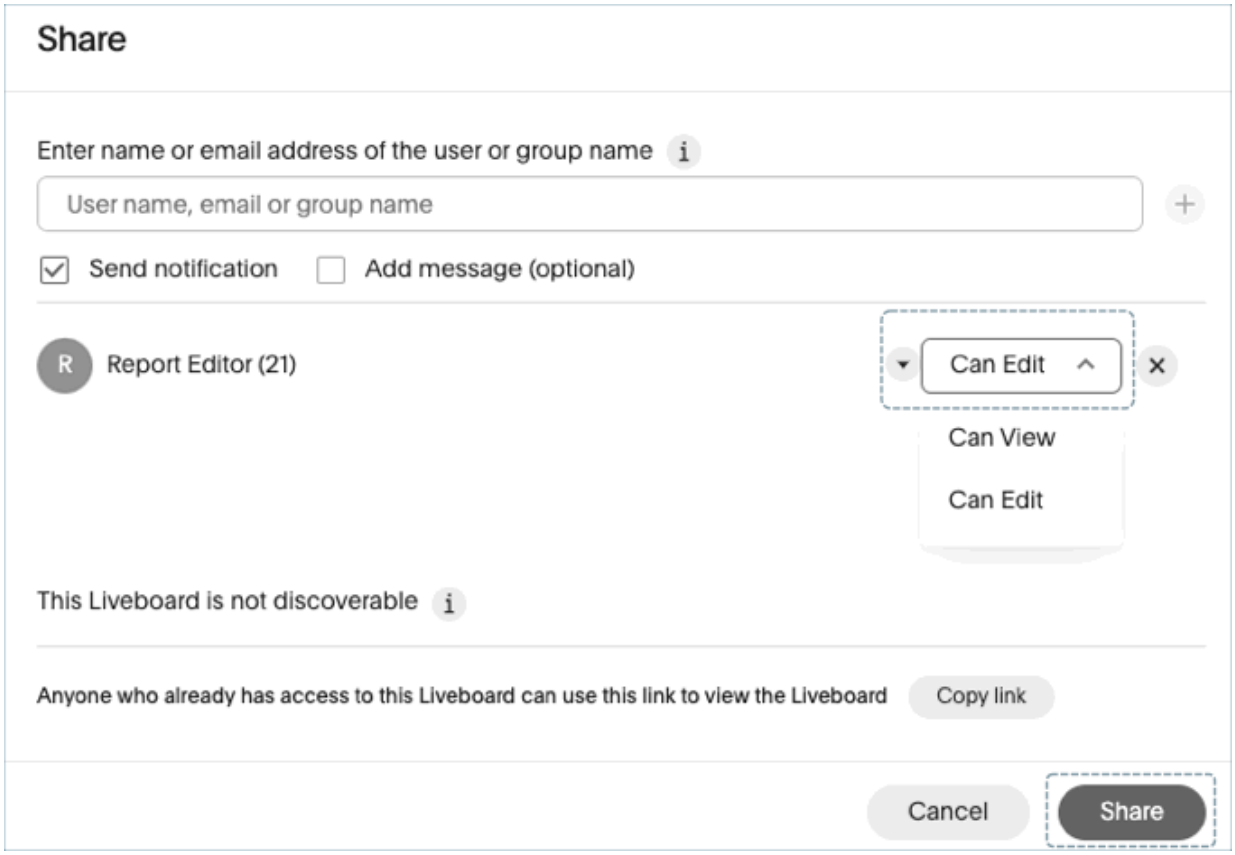
1. Log in to ThoughtSpot and select the Liveboard that you want to share.
2. Click the **Share** icon on the Liveboard overview page.



   The **Share** window opens.
3. In the search dialog box, type **content** and select **Report Editor** from the list.



4. Select **Can Edit** from the drop-down menu.



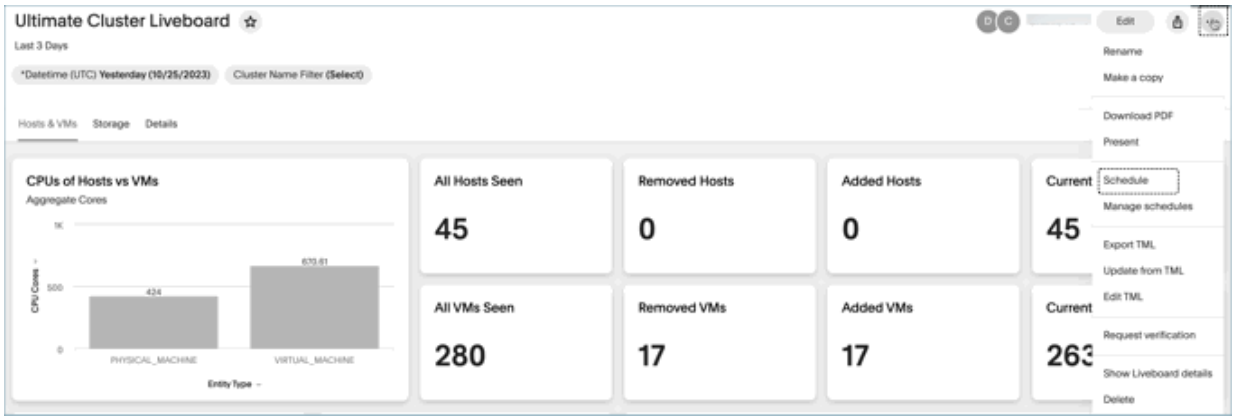
5. Click the **Share** button.

   All Report Editors in your ThoughSpot instance can now view and edit this Liveboard.

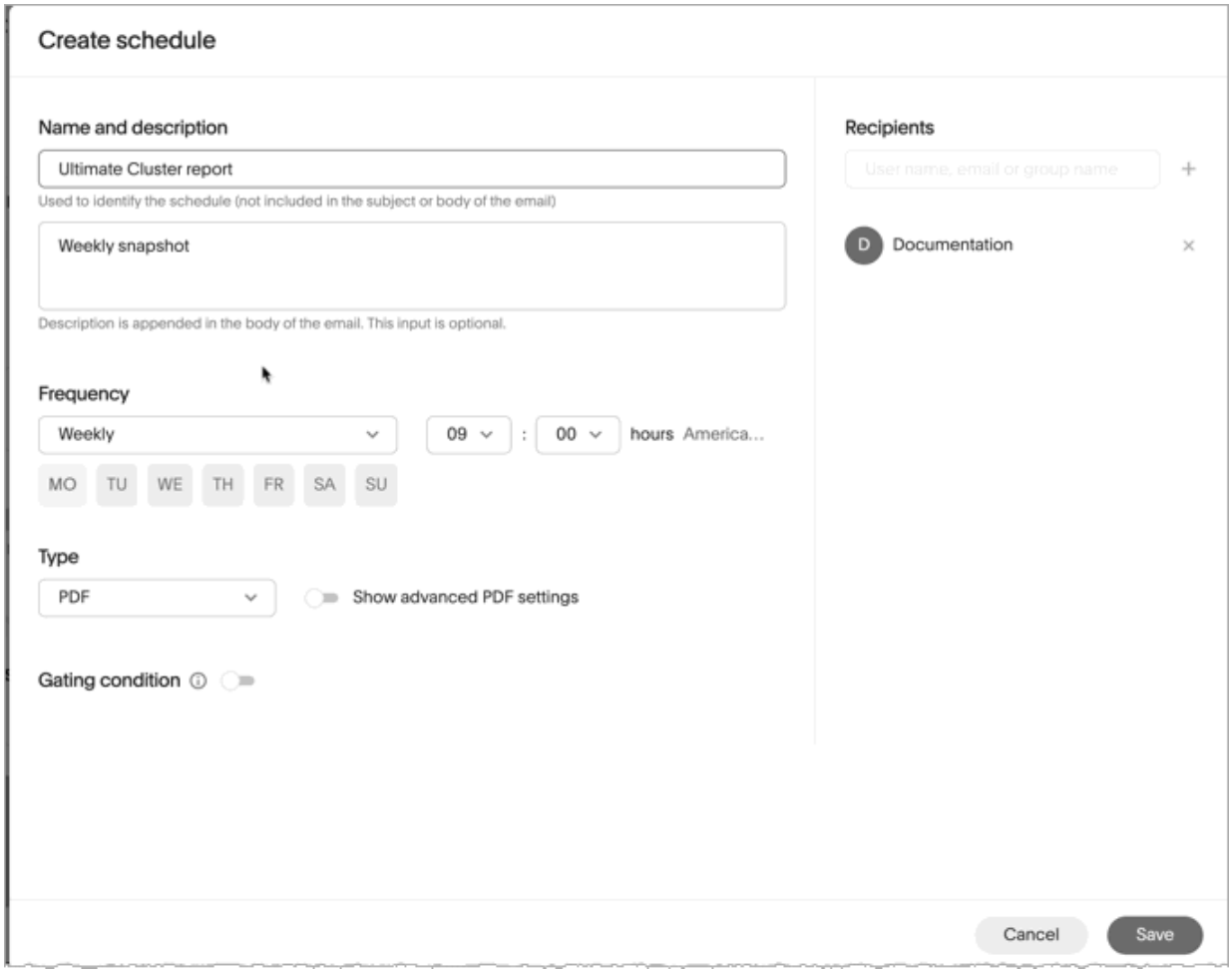
Creating scheduled reports

You can create a recurring PDF or CSV snapshot of any Liveboard that is automatically emailed to your chosen recipients on a specified schedule. To create a schedule, you must have Report Editor or Report Admin permissions.

1. In the ThoughSpot UI, select **Liveboards** from the main menu.
2. From the list of default and custom Liveboards, select the Liveboard that you want schedule a report for.
3. Click the three-dot icon for your chosen Liveboard and select **Schedule** from the drop-down menu.



4. Specify your preferences for frequency, report type, and recipients.



5. Click **Save**.

Your chosen recipients receive regular snapshots of the Liveboard according to the schedule that you specified.

Managing users

If you have Report Admin permissions in ThoughtSpot, you can create and modify users. All users must specify an email address that is registered with IBM Security Verify.

When you create a new user, you grant them the Report Editor User Role by adding them to the access group and a content group that correspond to that role. The access group determines the user permissions. The content group determines what kinds of resources are shared with the user. Turbonomic allows one user per role, per instance. The following user roles are available:

Reporting user Role	Privileges	Content group	Access group
Report editor	Report editors can view default charts, Liveboards, and underlying data tables, and can create new charts and Liveboards. They can also schedule reports for other users.	content.report.editor	access.report.editor

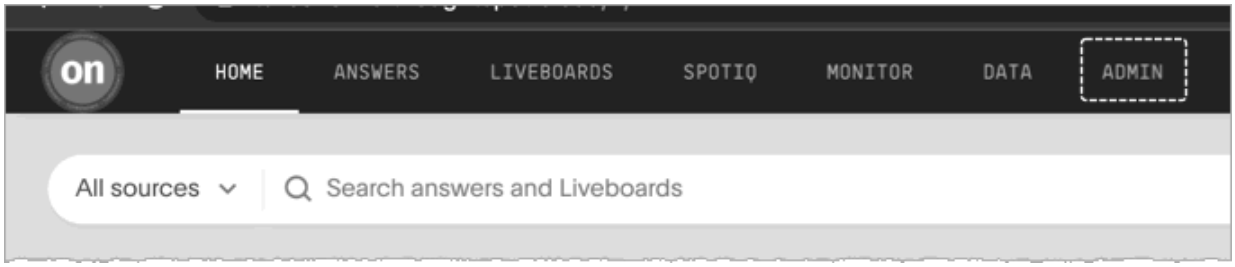
Reporting user Role	Privileges	Content group	Access group
Report admin	Report admins have all the privileges of Report editors, but they can also create and modify Report editor users. Report admins are responsible for adding any Report editor users to their Turbonomic instance.	Administrator	Administrator

Note: Report Admins must be careful not to delete or modify any of the default Liveboards or worksheets that Turbonomic provides. Deleting or modifying these resources might cause issues in the next Turbonomic version upgrade. Report Editors or Report Admins can create custom Liveboards or worksheets by making a copy of a default Liveboard or worksheet and modifying the copy. For more information, see “Creating custom Liveboards” on page 1941.

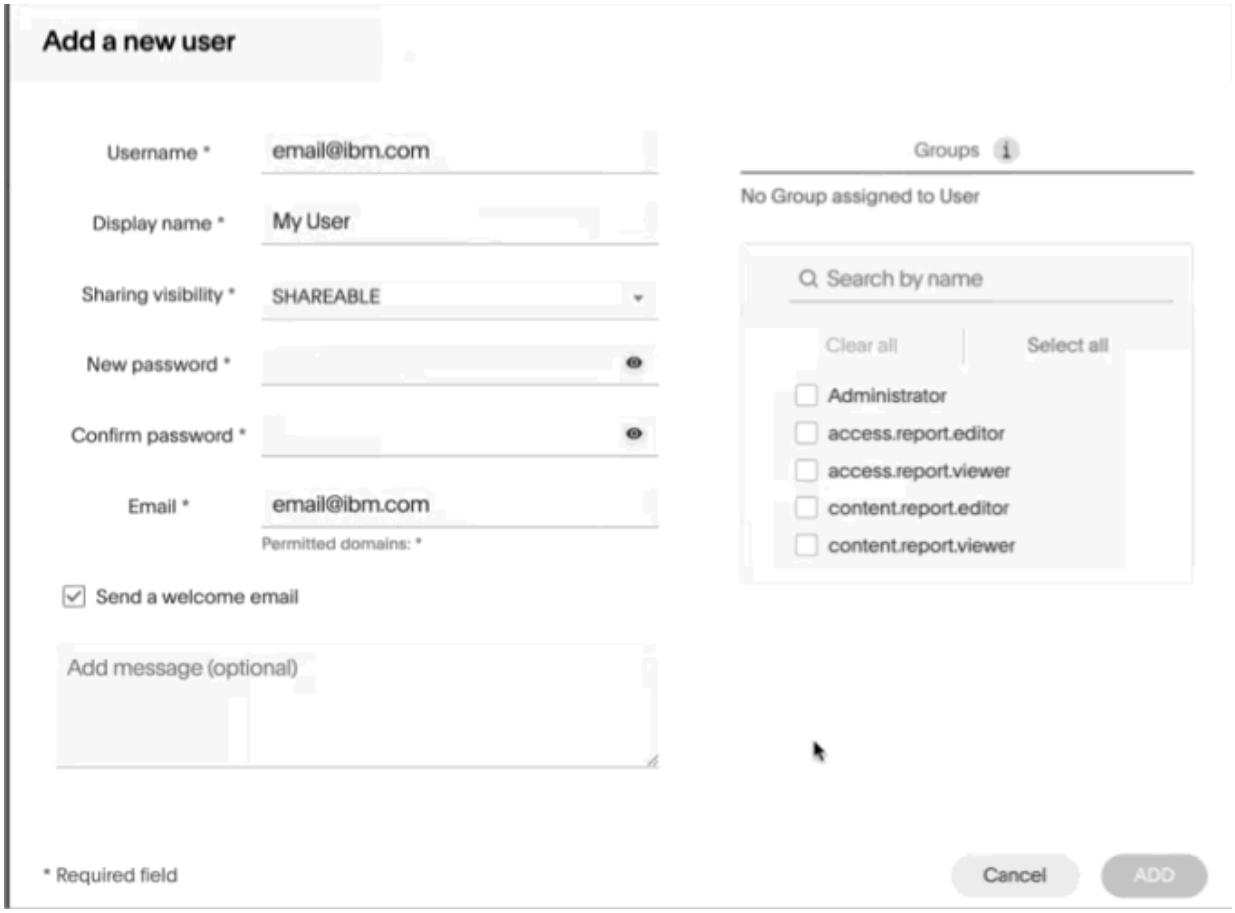
Creating a user

To create a user, complete the following steps.

1. In the ThoughtSpot UI, select **Admin** from the main menu.



2. Click **Users** in the **Admin** sidebar menu.
3. Click **Add user**.
4. In the **Add a new user** window, specify the following details.



- **Username**: The username must be an email address that is registered with IBM Security Verify. It must match the Email that is specified for the user.
- **Display name**: Specify a display name that appears in the UI.
- **Sharing visibility**: Leave this box set to the default **SHAREABLE** setting.
- **New password**: Specify an internal password for the user. The user does not use this password to sign in and it is for ThoughSpot account creation purposes only. All users sign in with IBM Security Verify credentials.
- **Confirm password**: Reenter the password from the previous field.
- **Email**: Specify an email address that is registered with IBM Security Verify. It must match the email address that is specified as the Username.
- **Groups**: Grant the user Report Editor privileges by selecting **access.report.editor** and **content.report.editor**.

5. To create the user, click **Add**.

Modifying a user

To modify an existing user, complete the following steps:

1. In the ThoughtSpot UI, select **Admin** from the main menu.
2. Click **Users** in the **Admin** sidebar menu.
3. Select the user that you want to edit.
4. Make changes as needed to the user details, content group, or access group.

   For more information, see step 4 in the previous section.
5. When you are finished, click **Update**.

Overview of default Liveboards

Turbonomic provides a collection of default charts and Liveboards in ThoughtSpot that are automatically available when you enable the SaaS reporting feature. These default resources cover of many aspects of your Turbonomic environment, including actions, virtual machines, storage, and more.

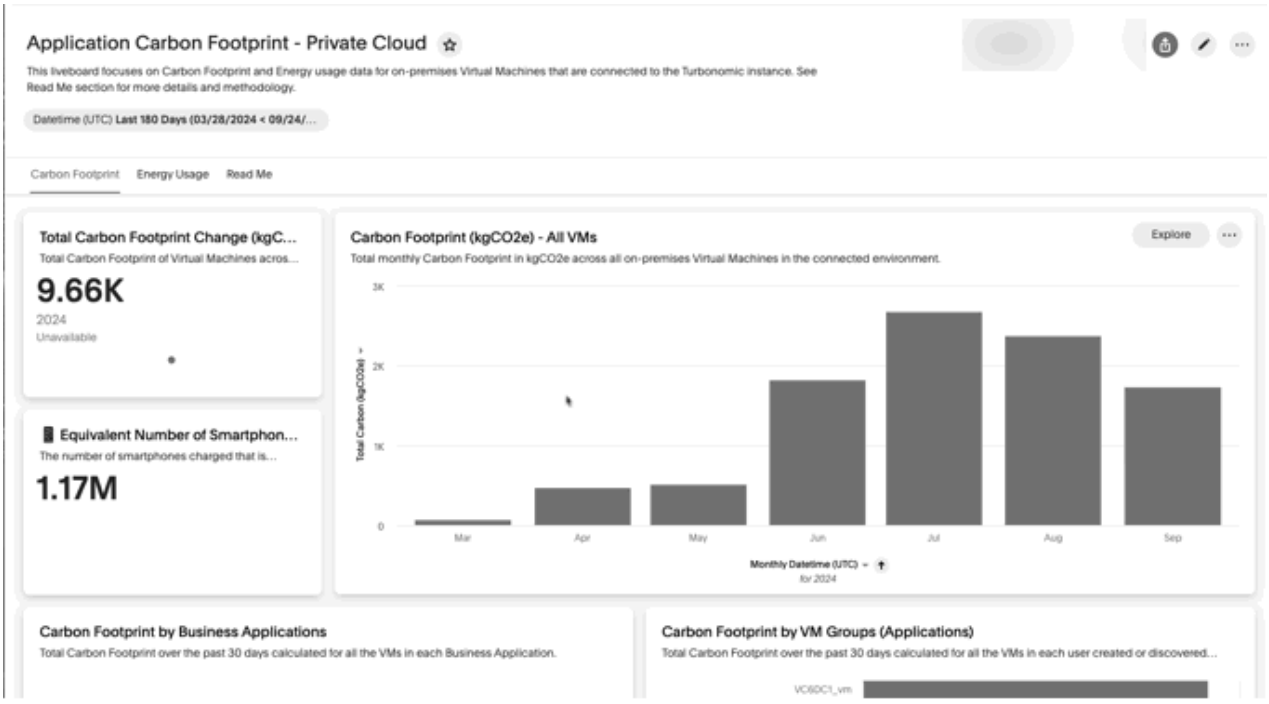
Because Turbonomic regularly updates the default Liveboards, editing these Liveboards is not recommended. Instead, make a copy of the Liveboard that you want to customize and edit the copy. For more information, see “Creating custom Liveboards” on page 1941.

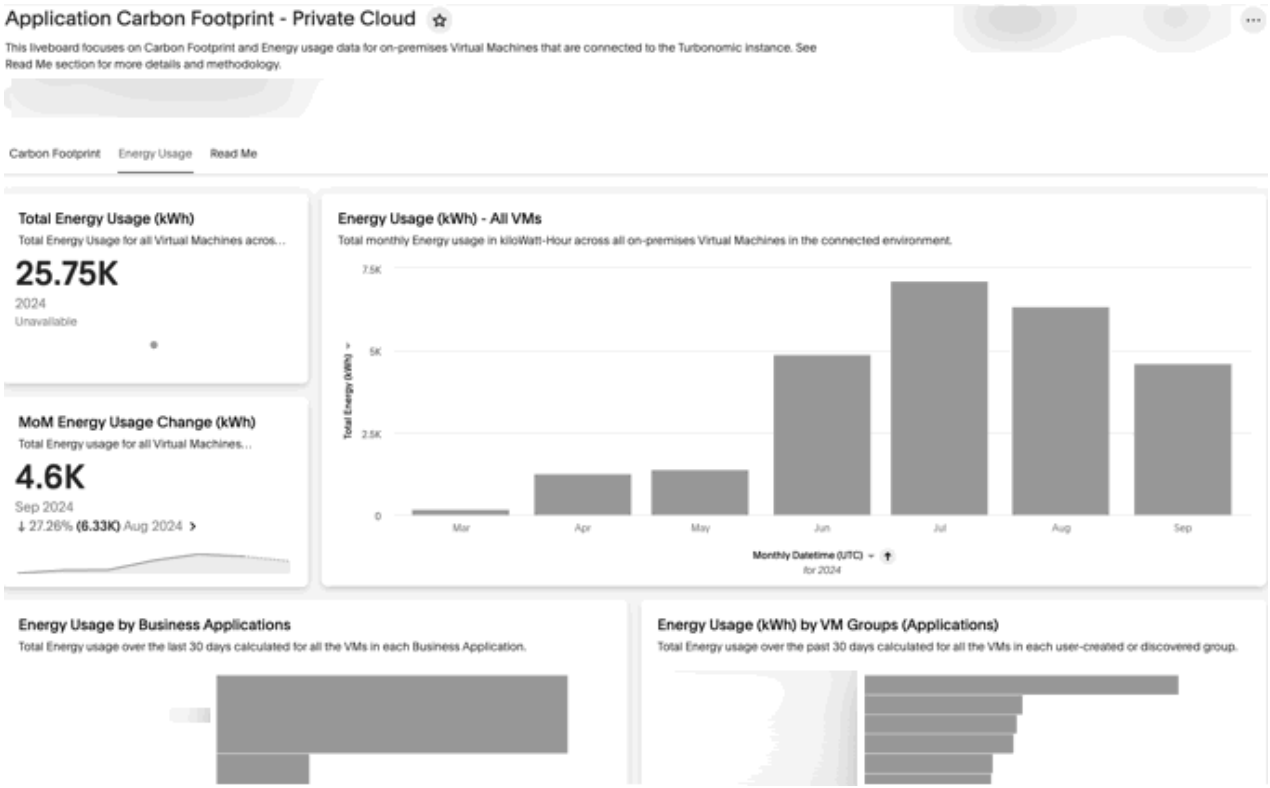
All default Liveboard date and time information is specified in Coordinated Universal Time (UTC).

The following default Turbonomic Liveboards are available:

- Application Carbon Footprint - Private Cloud
- “Application resource management” on page 1924
- Carbon Footprint - Public Cloud
- “Cloud actions and savings” on page 1927
- Container Cluster and Namespace: Capacity and Utilization
- Container Cluster and Namespace: Showback Cost Report
- Container Cluster Workload: Executed Resize Actions
- Container Cluster Workload: Latest Pending Resize Actions
- Container Workload by Label: Showback Cost Report
- “On-prem actions” on page 1935
- SaaS reporting metadata
- Sustainable IT Liveboard - Private Cloud
- “Ultimate cluster Liveboard” on page 1939
- “VM details” on page 1941

Application carbon footprint - private cloud



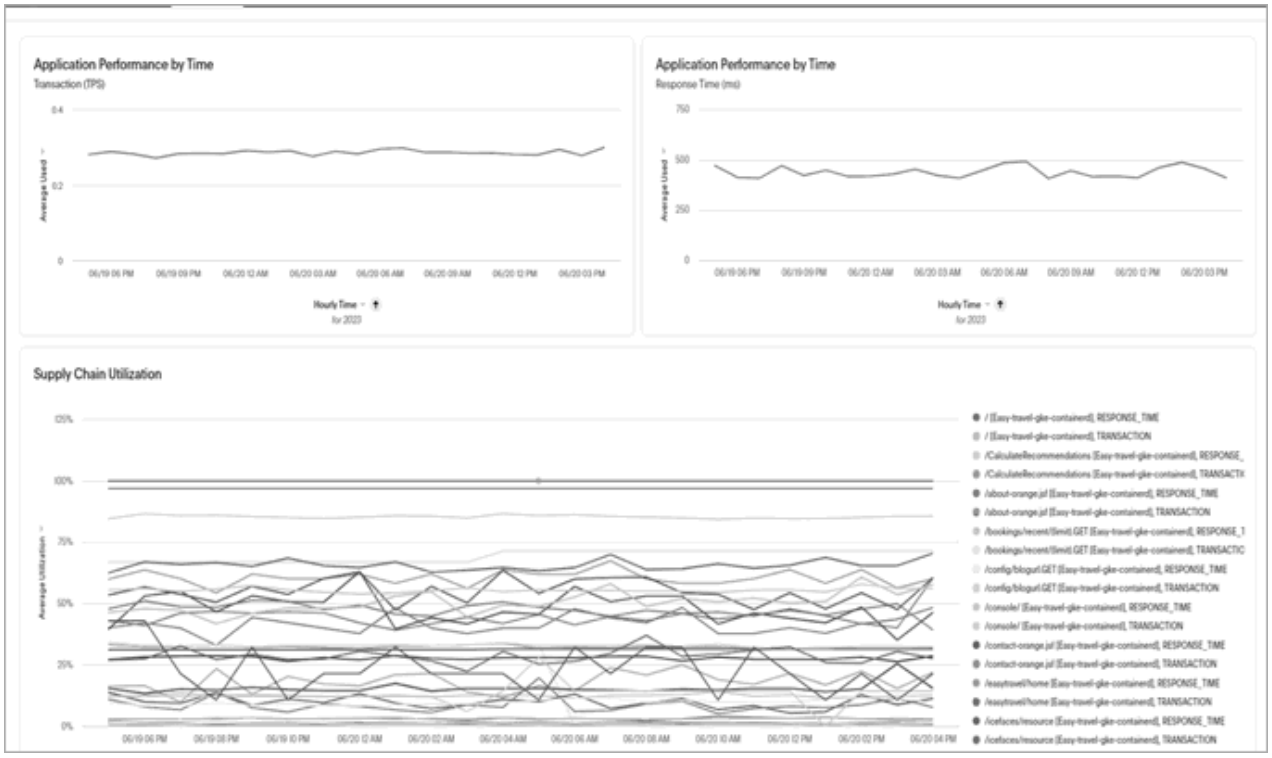


This Liveboard focuses on carbon footprint and energy usage data for on-premises virtual machines and the impact of Turbonomic actions on these metrics. It includes two tabs, **Carbon Footprint** and **Energy Usage**, which include the following charts.

Chart name	**Description**
Total carbon footprint change (kgCO2e)	The total carbon footprint of virtual machines in kgCO2e over the specified duration.
Carbon footprint (kgCO2e) - all VMs	Total monthly carbon footprint in kgCO2e across all on-premises virtual machines in the connected environment.
Equivalent number of smartphone charged	The number of smartphones charged that is equivalent to the total carbon emissions over the specified duration. For more information, see EPA: Greenhouse Gases Equivalencies Calculator - Calculations and References
Carbon footprint by business applications	Total carbon footprint over the past 30 days, calculated for all the VMs in each business application.
Carbon footprint by VM groups (applications)	Total carbon footprint over the past 30 days, calculated for all the VMs in each user-created or discovered group.
Daily count of unique virtual machines	Daily count of unique `Powered_On` and `Powered_Off` on-premises virtual machines in the connected environment.
Pending VM resize actions	Number of current pending VM resize actions for the connected on-premises resources.

Chart name	Description
vCPU net resources change	Net vCPU resources change from pending VM resize actions.
VMEM net resources change	Net vMEM resources change from pending VM resize actions. Each thousand units represent a terabyte (TB) of virtual memory.
Pending VM resizes & resources change	List of VM resize actions and their associated resources change, grouped by clusters.
Total energy usage (kWh)	Total energy usage in kilowatt-hours for all virtual machines over the specified duration.
Energy usage (kWh) - all VMs	Total monthly energy usage in kilowatt-hours across all on-premises virtual machines in the connected environment.
Energy usage by business applications	Total energy usage in kilowatt-hours over the last 30 days calculated for all the VMs in each Business Application.
Energy usage (kWh) by VM groups (applications)	Total energy usage in kilowatt-hours over the past 30 days, calculated for all the VMs in each user-created or discovered group.
MoM energy usage change (kWh)	Comparison of total energy usage for all virtual machines between current month and previous month.

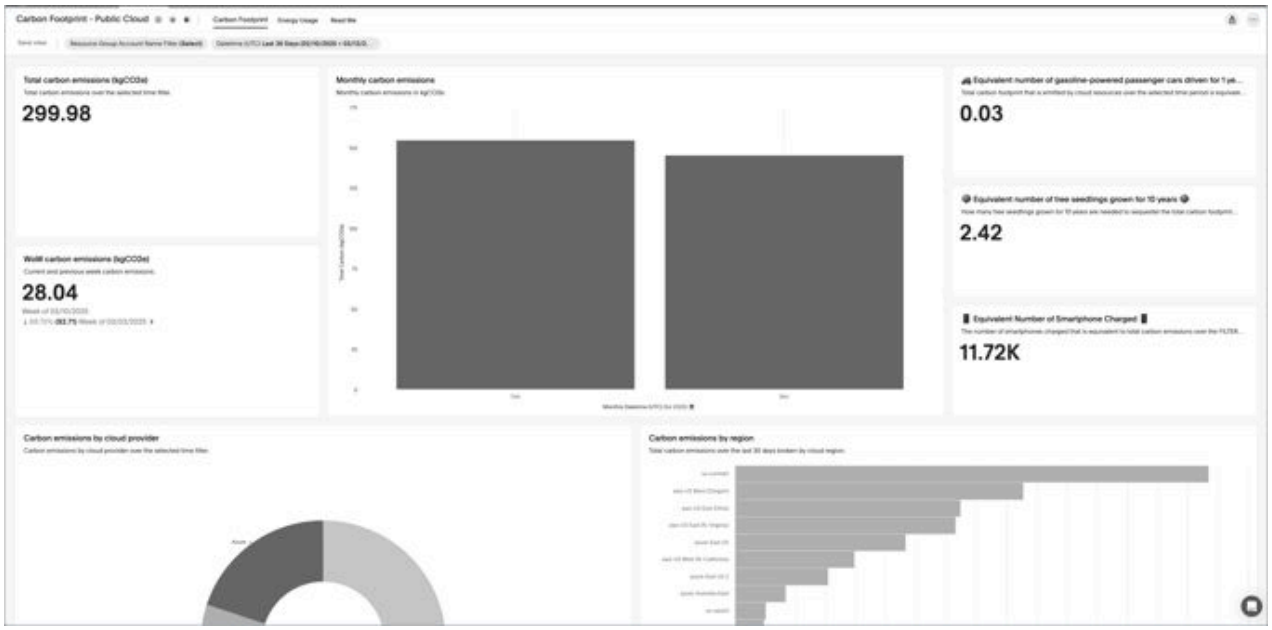
Application resource management

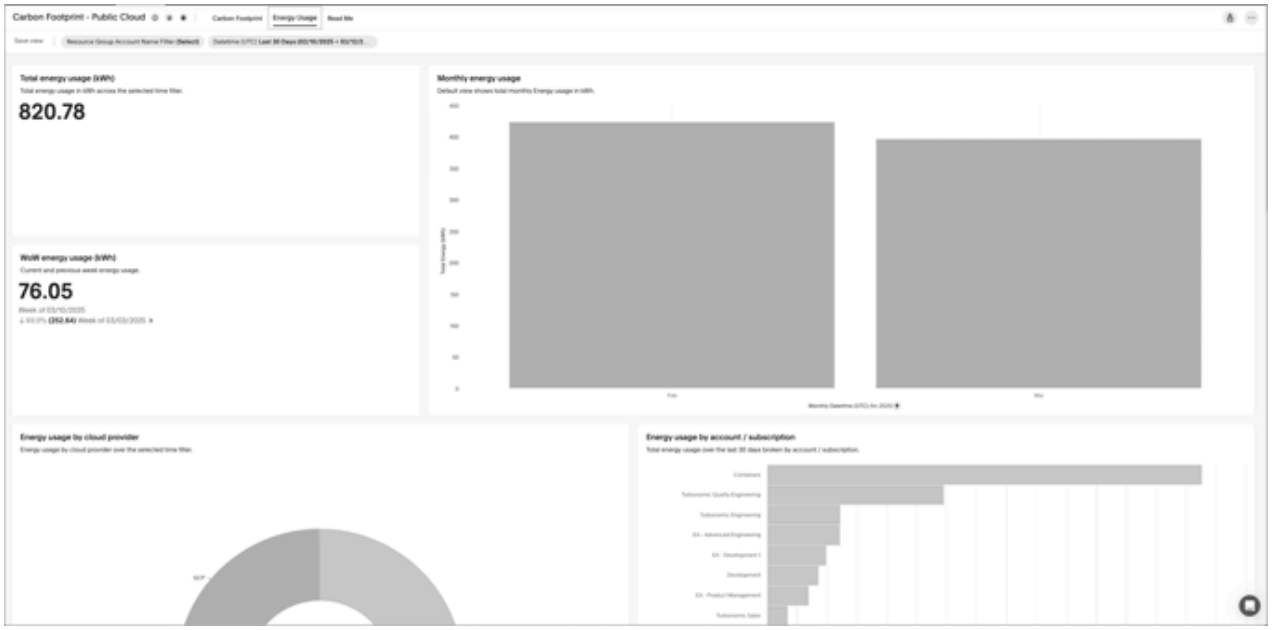


This Liveboard shows data about applications and the entities in the supply chain that make up these applications. It includes the following charts:

Chart name	Description
Application performance by time: transaction (TPS)	Average transactions per second over time
Application performance by time: response time (ms)	Average response time in milliseconds over time
Supply chain utilization	*Supply chain* refers to any entity that is related to the scoped Business Application
Application performance	Response time (ms), transaction (TPS), SLO violations, and average and peak values for response time and transactions by application
Supply chain performance	Average capacity, maximum used, and average used by metric in the context of entities related to the scoped business application

Carbon Footprint - Public Cloud



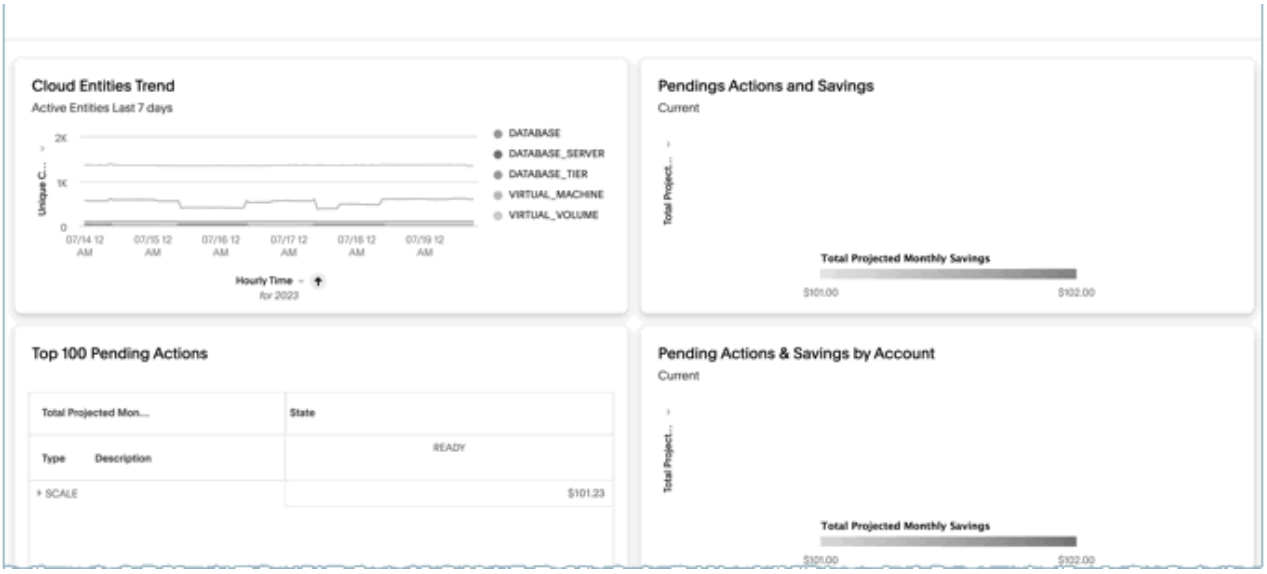


This Liveboard focuses on carbon footprint and energy usage data for public cloud (AWS, Azure, and GCP) virtual machines and the impact of Turbonomic actions on these metrics. It includes two tabs, **Carbon Footprint** and **Energy Usage**, which include the following charts:

Chart name	Description
Total carbon emissions (kgCO2e)	Total carbon emissions over the selected time filter.
Monthly carbon emissions	Monthly carbon emissions in kgCO2e.
Equivalent number of gasoline-powered passenger cars driven for 1 year	Total carbon footprint that is emitted by cloud resources over the selected time period is equivalent to greenhouse gas emissions from U.S. gasoline-powered passenger vehicles driven for 1 year. For more information, see Gasoline-powered passenger vehicles per year.
WoW carbon emissions (kgCO2e)	Current and previous week carbon emissions.
Equivalent number of tree seedlings grown for 10 years	How many tree seedlings grown for 10 years are needed to sequester the total carbon footprint emitted by cloud resources over the selected time period?
Equivalent Number of Smartphone Charged	The number of smartphones charged that is equivalent to total carbon emissions over the FILTERED TIME SCOPE. For more information, see Number of smartphones charged.
Carbon emissions by cloud provider	Carbon emissions by cloud provider over the selected time filter.
Carbon emissions by region	Total carbon emissions over the last 30 days broken by cloud region.
Carbon emissions by account / subscription	Total carbon emissions over the last 30 days broken by account/subscription.
Carbon emissions by azure resource group	Total carbon emissions over the last 30 days broken by azure resource group.

Chart name	Description
Carbon emissions by virtual machines	Total carbon emissions over the last 30 days broken by virtual machines.
List of pending VM scale actions by potential savings	List of pending VM Scale actions sorted by 'Total Projected Monthly Savings'. Optimize your cloud VM resources to reduce cost and Carbon Footprint.
Total energy usage (kWh)	Total energy usage in kWh across the selected time filter.
Monthly energy usage	Default view shows total monthly Energy usage in kWh.
WoW energy usage (kWh)	Current and previous week energy usage.
Energy usage by cloud provider	Energy usage by cloud provider over the selected time filter.
Energy usage by account / subscription	Total energy usage over the last 30 days broken by account / subscription.
Energy usage by region	Total energy usage over the last 30 days broken by region.
Energy usage by azure resource group	Total energy usage over the last 30 days broken by azure resource group.
Energy usage by virtual machines	Total energy usage over the last 30 days broken by virtual machines.
List of pending VM scale actions by potential savings	List of pending VM Scale actions sorted by 'Total Projected Monthly Savings'. Optimize your cloud VM resources to reduce cost and Carbon Footprint.

Cloud actions and savings



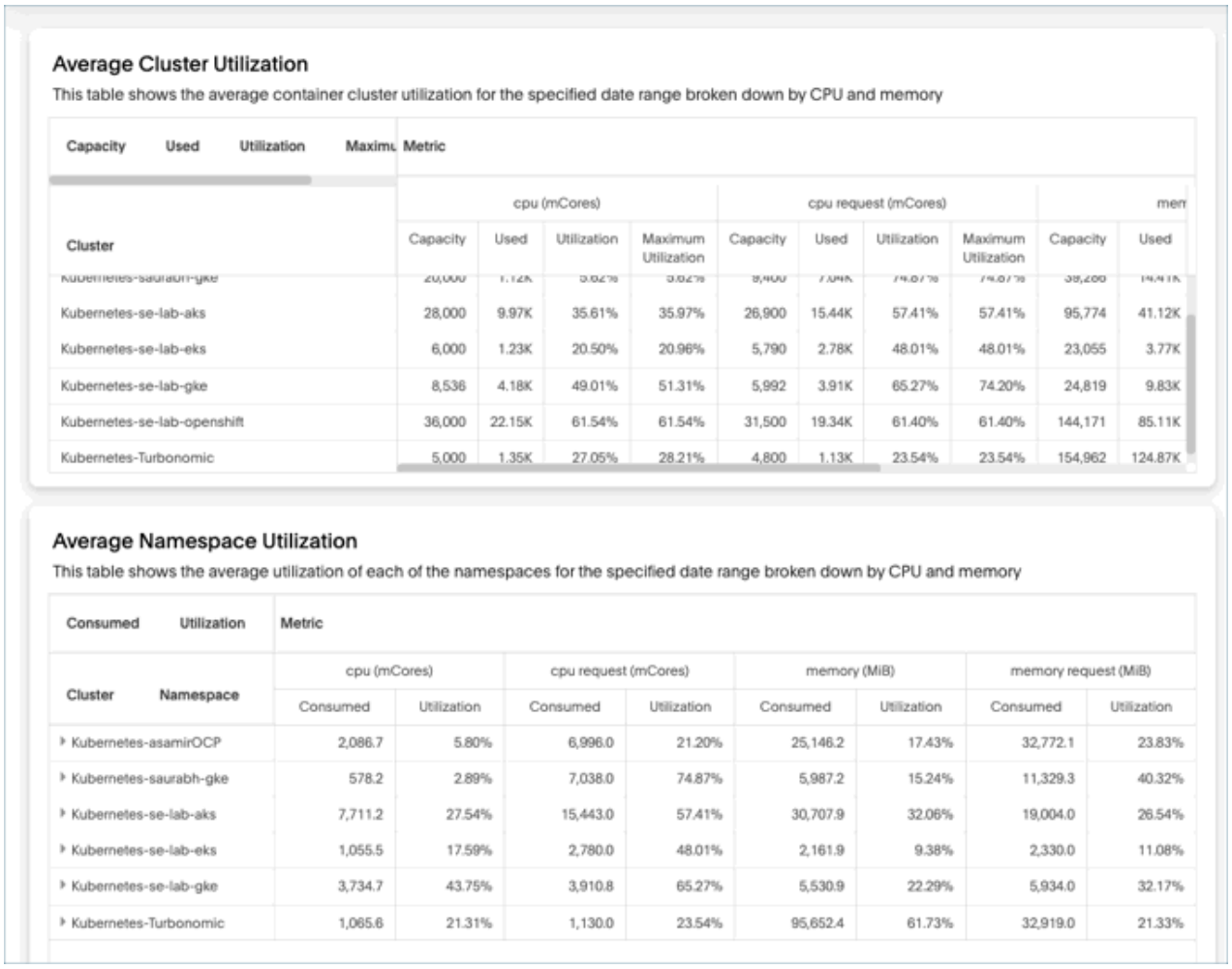


This Liveboard shows information about cloud actions and the potential and realized savings and investments that are associated with these actions. It includes the following charts:

Chart name	Description
Cloud entities trend	Unique counts of active database servers, virtual machines (VM), and virtual volumes over time
Pending actions and savings	Projected monthly savings grouped by action and entity type
Top 100 Pending Actions	Top 100 Pending Actions ranked by projected savings
Pending Actions & savings by account	Projected monthly savings grouped by cloud account. Turbonomic discovers the following cloud accounts: • AWS accounts • Azure subscriptions • GCP projects
Potential savings	Breakdown by action type of the dollar amount that can be saved if the recommended actions are executed

Chart name	Description
Total potential savings	Total dollar amount that can be saved if all recommended actions are executed
Necessary investments	Breakdown by action type of the dollar amount that can be invested to improve performance by executing recommended scale-up actions
Total Necessary investments	Total dollar amount that can be invested to improve performance by executing recommended scale-up actions
Savings by entity type	Daily realized savings by entity type
Savings by action type	Daily realized savings for `scale`, `delete`, and `suspend` action types
Investments by entity type	Daily realized investments by entity type to improve performance through scale-up actions
Savings by account	Daily realized savings by Turbonomic account

Container cluster and namespace: capacity and utilization



Average Cluster Utilization

This table shows the average container cluster utilization for the specified date range broken down by CPU and memory

Capacity Used Utilization Maximu Metric

Cluster	cpu (mCores)				cpu request (mCores)				mem	
	Capacity	Used	Utilization	Maximum Utilization	Capacity	Used	Utilization	Maximum Utilization	Capacity	Used
Kubernetes-saurabh-gke	20,000	1.12K	5.62%	5.62%	9,400	7.04K	74.87%	74.87%	38,288	14.41K
Kubernetes-se-lab-aks	28,000	9.97K	35.61%	35.97%	26,900	15.44K	57.41%	57.41%	95,774	41.12K
Kubernetes-se-lab-eks	6,000	1.23K	20.50%	20.96%	5,790	2.78K	48.01%	48.01%	23,055	3.77K
Kubernetes-se-lab-gke	8,536	4.18K	49.01%	51.31%	5,992	3.91K	65.27%	74.20%	24,819	9.83K
Kubernetes-se-lab-openshift	36,000	22.15K	61.54%	61.54%	31,500	19.34K	61.40%	61.40%	144,171	85.11K
Kubernetes-Turbonomic	5,000	1.35K	27.05%	28.21%	4,800	1.13K	23.54%	23.54%	154,962	124.87K

Average Namespace Utilization

This table shows the average utilization of each of the namespaces for the specified date range broken down by CPU and memory

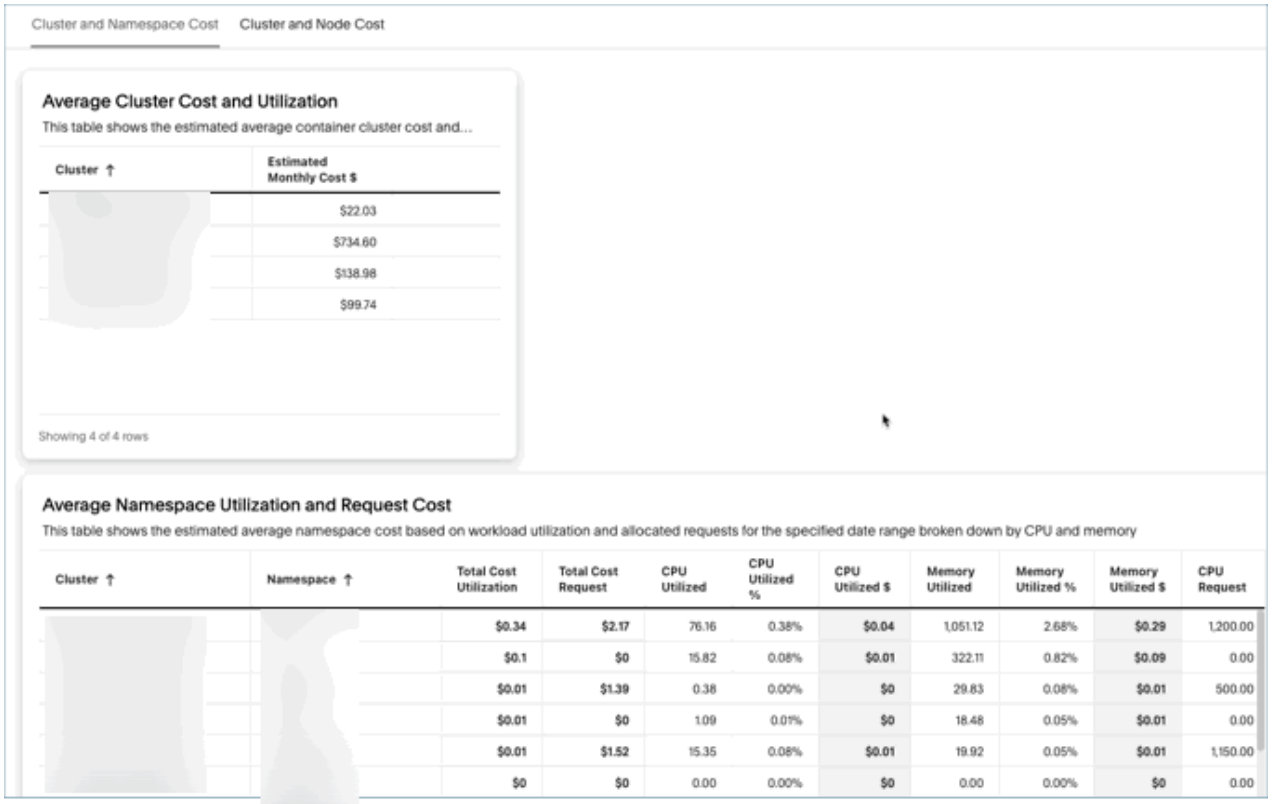
Consumed Utilization Metric

Cluster Namespace	cpu (mCores)		cpu request (mCores)		memory (MiB)		memory request (MiB)	
	Consumed	Utilization	Consumed	Utilization	Consumed	Utilization	Consumed	Utilization
Kubernetes-asamirOCP	2,086.7	5.80%	6,996.0	21.20%	25,146.2	17.43%	32,772.1	23.83%
Kubernetes-saurabh-gke	578.2	2.89%	7,038.0	74.87%	5,987.2	15.24%	11,329.3	40.32%
Kubernetes-se-lab-aks	7,711.2	27.54%	15,443.0	57.41%	30,707.9	32.06%	19,004.0	26.54%
Kubernetes-se-lab-eks	1,055.5	17.59%	2,780.0	48.01%	2,161.9	9.38%	2,330.0	11.08%
Kubernetes-se-lab-gke	3,734.7	43.75%	3,910.8	65.27%	5,530.9	22.29%	5,934.0	32.17%
Kubernetes-Turbonomic	1,065.6	21.31%	1,130.0	23.54%	95,652.4	61.73%	32,919.0	21.33%

This Liveboard shows the average cluster utilization and average namespace utilization for a specified date range, which is broken down by CPU and memory. It includes the following charts.

Chart name	Description
Average cluster utilization	Average container cluster utilization for the specified date range, which is broken down by CPU and memory
Average namespace utilization	Average utilization of each of the namespaces for the specified date range, which is broken down by CPU and memory

Container cluster and namespace: showback cost report

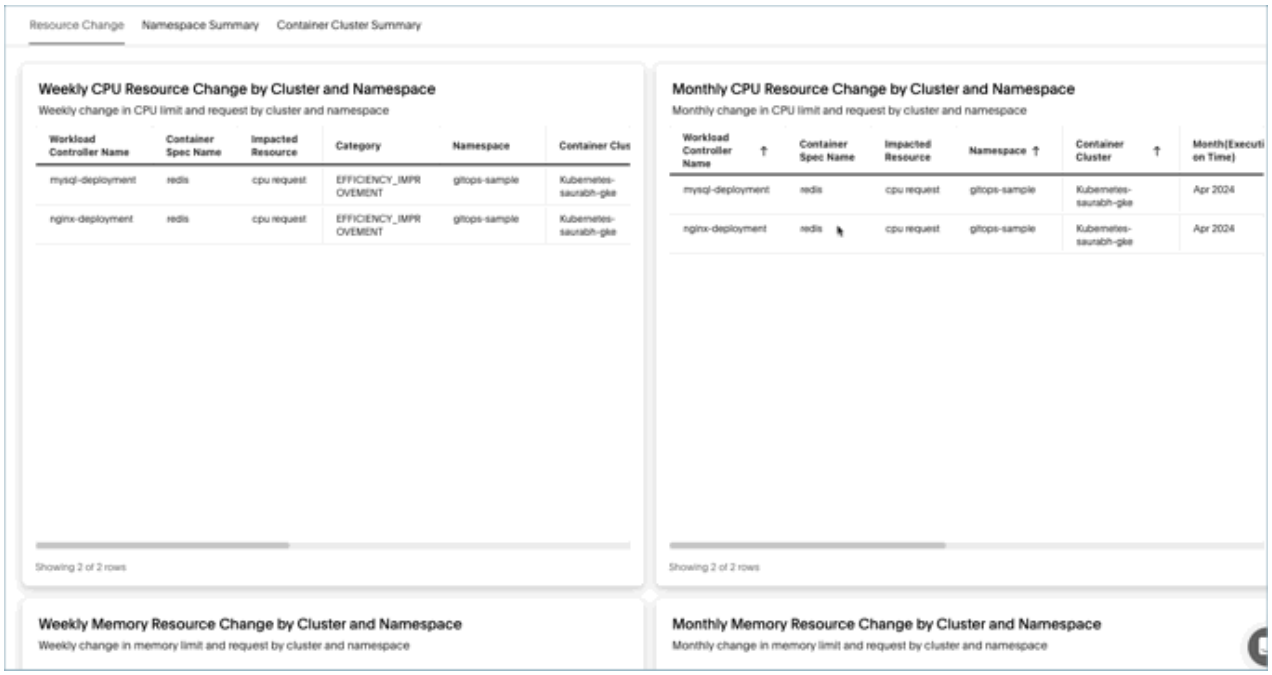


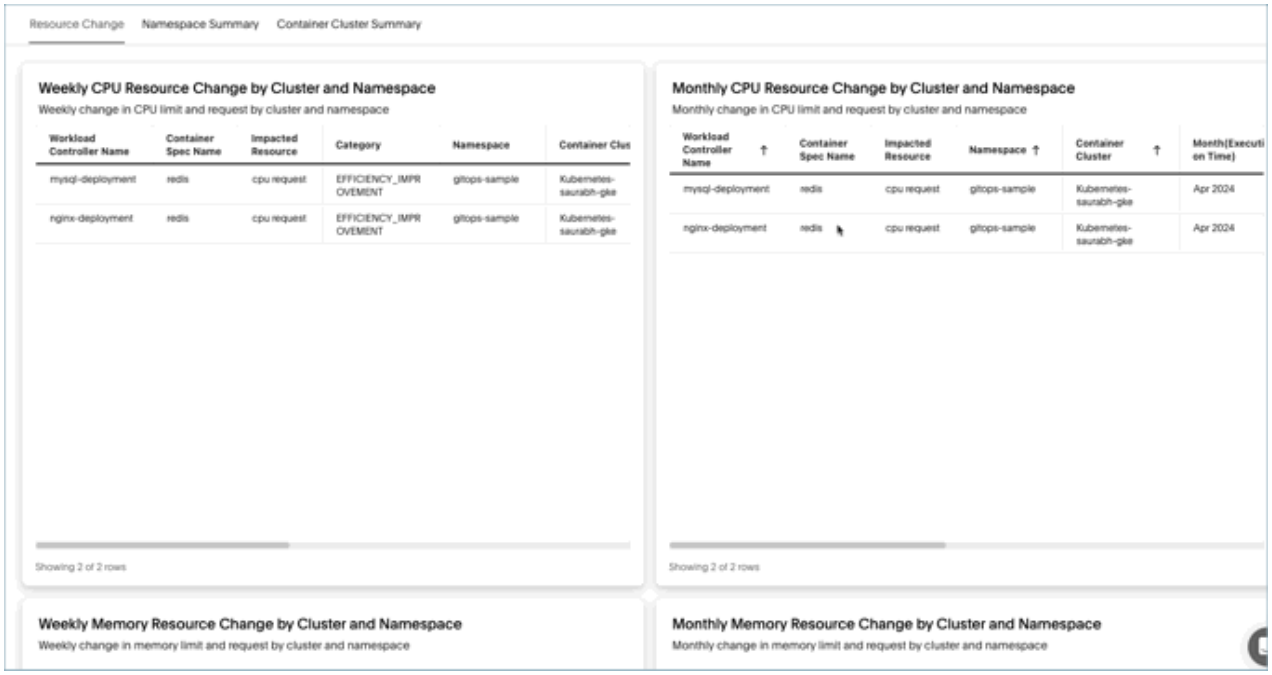


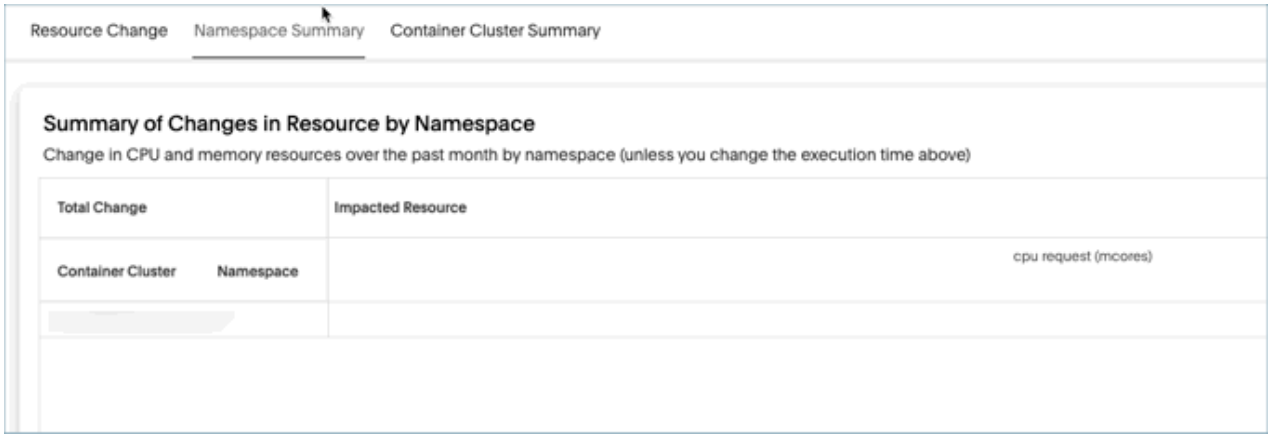
This Liveboard shows bottom-up cost based on the cloud provider on-demand rates. Cost is calculated based on virtual CPU and virtual memory utilization in the namespace by using a selected weighted average according to the total cost of the nodes in the container cluster. The Liveboard includes two tabs, **Cluster and Namespace Cost** and **Cluster and Node Cost**, which include the following charts.

Chart name	Description
Average cluster cost and utilization	Estimated average container cluster cost and utilization for the past 30 days
Average namespace utilization and request cost	Estimated average namespace cost based on workload utilization and allocated requests for the specified date range, which are broken down by CPU and memory
Cluster and node cost	Estimated cluster monthly cost based on an average of the last 30 days of hourly costs of nodes in the cluster

Container cluster workload: executed resize actions







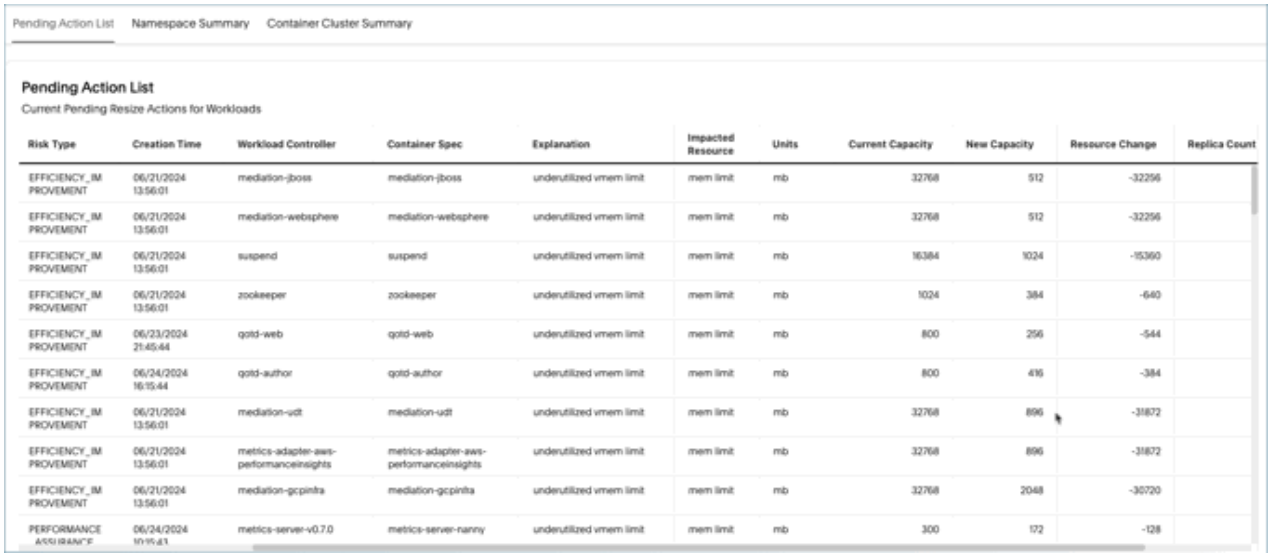


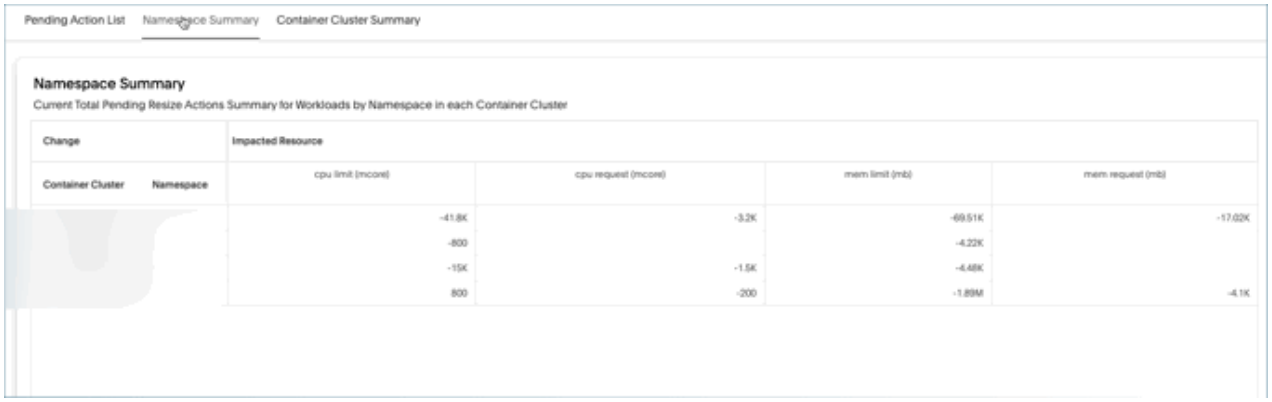
This Liveboard shows all successfully executed container workload resize actions. You can filter these actions by container cluster, namespace, and execution time. The Liveboard includes three tabs, **Resource Change**, **Namespace Summary**, and **Container Cluster Summary**, which include the following charts.

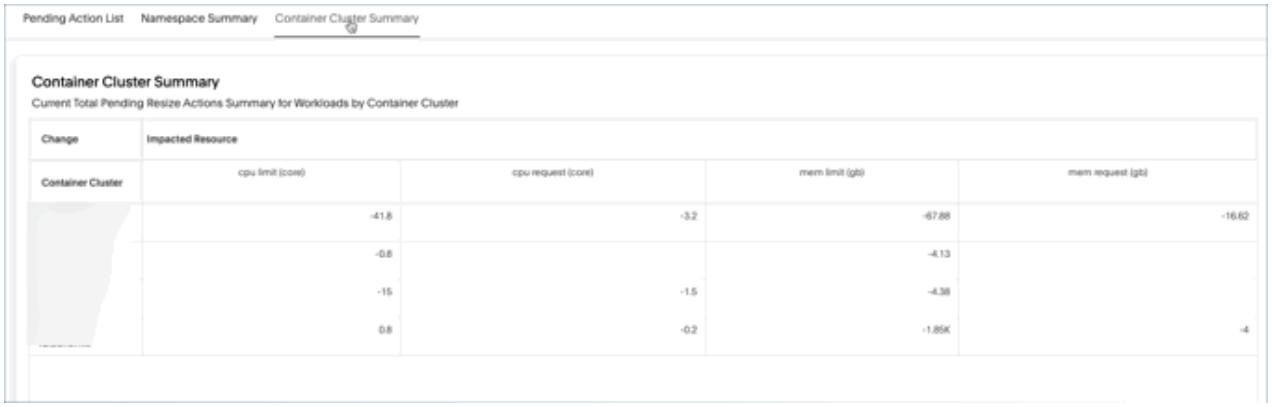
Chart name	Description
Weekly CPU resource change by cluster and namespace	Weekly change in CPU limit and request by cluster and namespace
Monthly CPU resource change by cluster and namespace	Monthly change in CPU limit and request by cluster and namespace

Chart name	Description
Weekly memory resource change by cluster and namespace	Weekly change in memory limit and request by cluster and namespace
Monthly memory resource change by cluster and namespace	Monthly change in memory limit and request by cluster and namespace
Summary of changes in resource by namespace	Change in CPU and memory resources over the past month by namespace (unless you change the default execution time)
Summary of changes in resources by container cluster	Change in CPU and memory resources over the past month by cluster (unless you change the default execution time)

Container cluster workload: latest pending resize actions



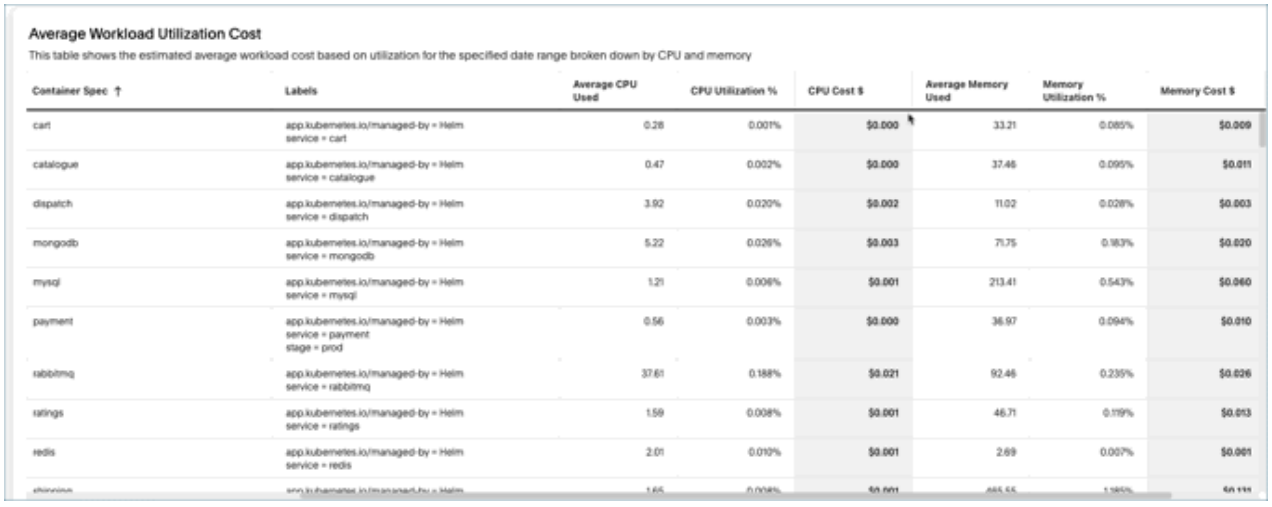




This Liveboard shows all current pending resize actions for container workloads. It includes three tabs: **Pending Action List**, **Namespace Summary**, and **Container Cluster Summary**, which include the following charts.

Chart name	Description
Pending action list	Current pending resize actions for all container workloads
Namespace summary	Current pending resize actions for workloads by namespace in each container cluster
Container cluster summary	Current pending resize actions for workloads by container cluster

Container workload by label: showback cost report



This Liveboard shows bottom-up cost based on the cloud provider on-demand rates. You can filter the data by the labels that are applied to the container workload. The cost is calculated based on the CPU and memory average utilization, compared to the container cluster capacity and its total cost. It includes the following chart.

Chart name	Description
Average workload utilization cost	Estimated average workload cost based on utilization for the specified date range, which is broken down by CPU and memory

On-prem actions



This Liveboard shows on-prem actions, savings, and more. It includes the following charts:

Chart name	Description
Executed actions (bar and pie charts)	Actions for on-prem workloads that were executed manually or automatically
Executed actions by risk	Actions grouped by risk type
Resources allocated and reclaimed	Workload resources that were allocated through resize or provision actions and reclaimed through delete or suspend actions
Executed resizes and showback	Actions to resize quotas for Kubernetes namespaces and the effect of these actions on showback
Delete actions by size	Resources that were reclaimed through delete actions
Storage waste	Storage space that can be reclaimed by deleting files that are not required for operations
Pending Actions	Actions for on-prem workloads that are not yet executed
Pending Actions by type and risk	Pending actions grouped by type and risk

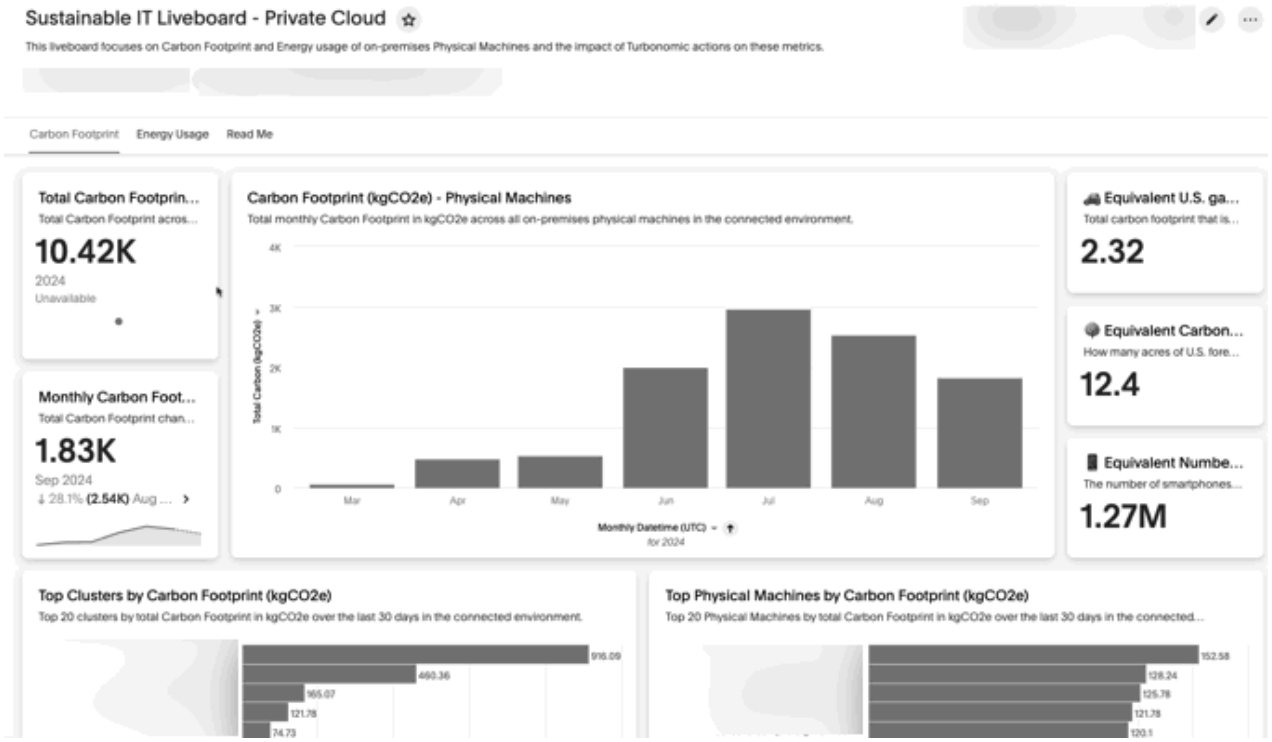
SaaS reporting metadata

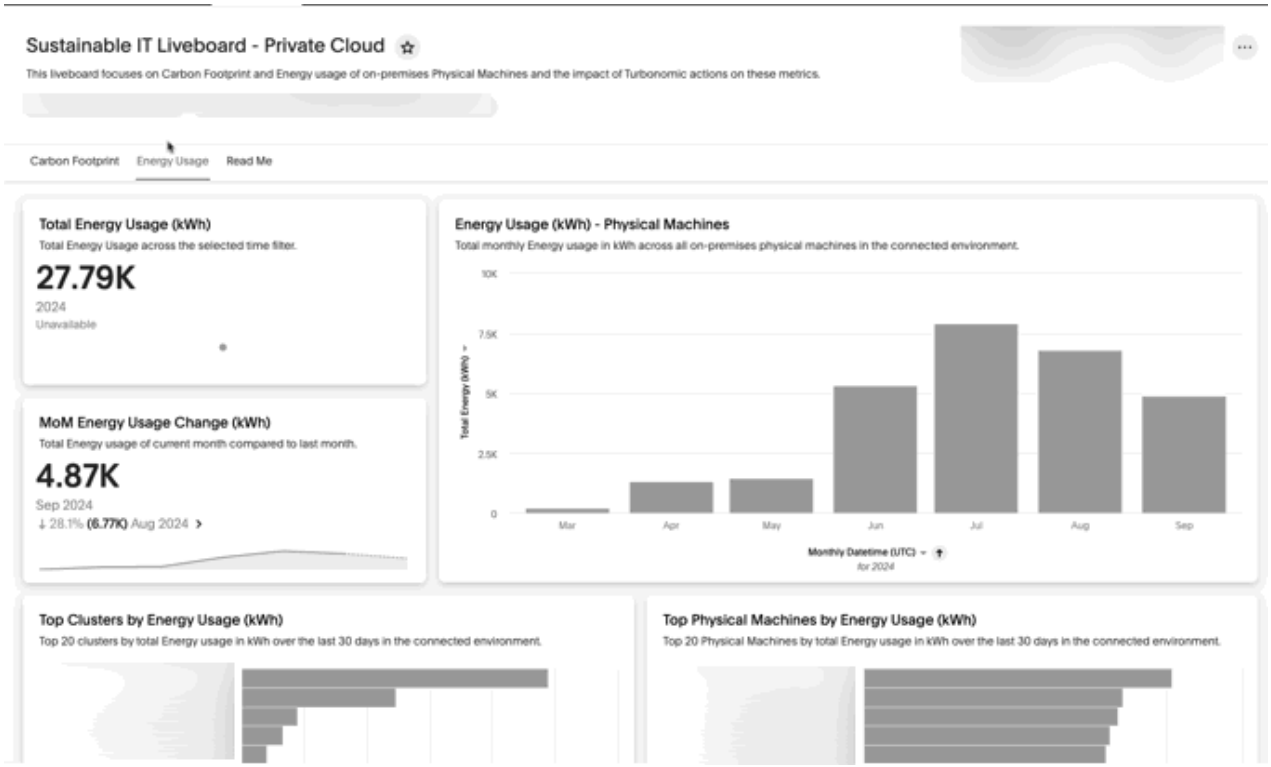


This Liveboard helps determine the current SaaS reporting version for your Turbonomic instance. For more information, see SaaS reporting versions and release notes.

Chart name	Description
What is my SaaS reporting version?	Displays the current SaaS reporting release version for your Turbonomic instance.

Sustainable IT Liveboard - private cloud



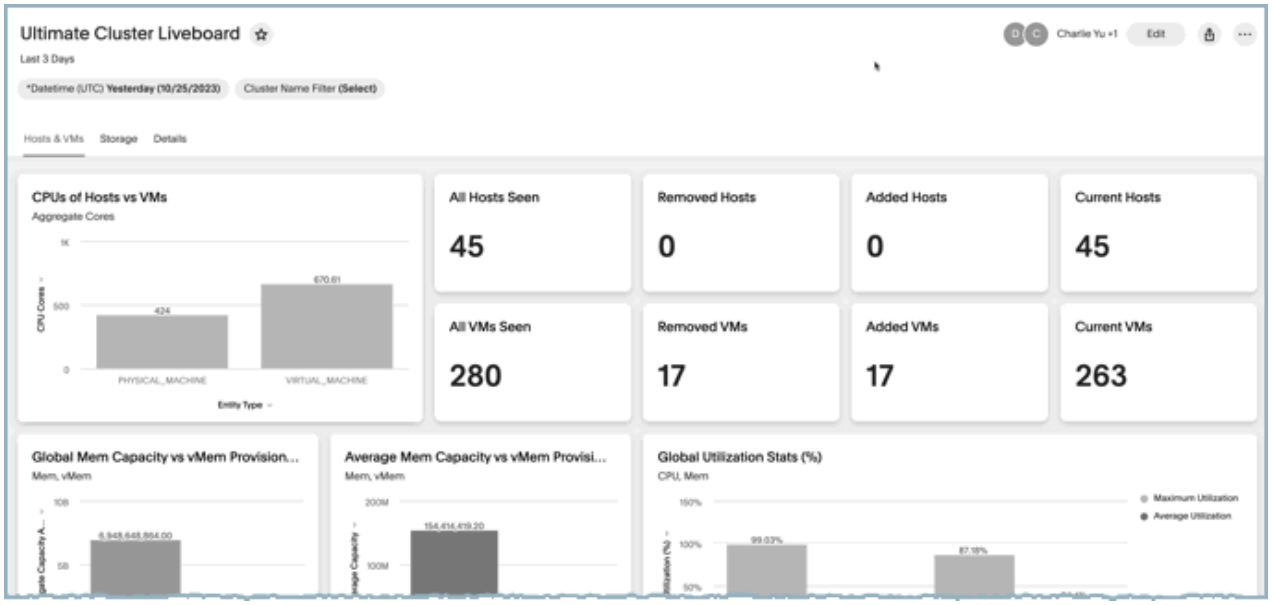


This liveboard focuses on carbon footprint and energy usage of on-premises physical machines and the impact of Turbonomic actions on these metrics. It includes two tabs, **Carbon Footprint** and **Energy Usage**, which include the following charts.

Chart name	Description
Total carbon footprint change (kgCO2e)	The total carbon footprint of physical machines in kgCO2e over the specified duration.
Carbon footprint (kgCO2e) - physical machines	Total monthly carbon footprint in kgCO2e across all on-premises physical machines in the connected environment.
Equivalent number of smartphone charged	The number of smart phones charged that is equivalent to the total carbon emissions over the specified duration. For more information, see EPA: Greenhouse Gases Equivalencies Calculator - Calculations and References.
Equivalent U.S. gasoline-powered passenger vehicles driven for 1 year	Total carbon footprint that is emitted by cloud resources over the specified duration is equivalent to greenhouse gas emissions from this number of U.S. gasoline-powered passenger vehicles driven for 1 year. For more information, see EPA: Greenhouse Gases Equivalencies Calculator - Calculations and References.
Equivalent carbon sequestered by acres of US forest/year	How many acres of U.S. forest per year it takes to sequester the total carbon footprint emitted by cloud resources over the specified duration. For more information, see EPA: Greenhouse Gases Equivalencies Calculator - Calculations and References.

Chart name	Description
Monthly carbon footprint change (kgCO2e)	Comparison of total carbon footprint for all physical machines between current month and previous month.
Top Clusters by Carbon Footprint (kgCO2e)	Top 20 clusters by total carbon footprint in kgCO2e over the last 30 days in the connected environment.
Top physical machines by carbon footprint (kgCO2e)	Top 20 Physical Machines by total carbon footprint in kgCO2e over the last 30 days in the connected environment.
Daily count for unique virtual machines, unique physical machines	Daily count of unique virtual and physical machines in the connected environment.
Pending physical machine suspend (Deactivate) actions	Pending physical machine host suspend actions and the estimated energy and carbon footprint savings that are associated with each action.
Physical machine spec with idle power	List of physical machines and their idle power values (in watts) . Server idle power is the amount of power a host uses when it has no load. Server idle power per CPU is an indication of server energy efficiency, which can help identify energy inefficient servers as candidates for replacement.
Total carbon footprint, VM:host density by monthly datetime (UTC)	Monthly view of total carbon footprint data of physical machines in the connected environment., overlayed with vm:host density ratio. With Turbonomic actions, you can increase vm:host density ratio while reducing carbon footprint through resource optimization and hardware consolidation.
Total energy (kWh), VM:host density by monthly datetime (UTC)	Monthly view of total energy usage across all physical machines in the connected environment, overlayed with monthly vm:host density ratio. With Turbonomic actions, you can increase vm:host density ratio while reducing energy usage through resource optimization and hardware consolidation.
Total energy usage (kWh)	Total energy usage in kilowatt-hours for all virtual machines over sent specified duration.
Energy Usage (kWh) - Physical Machines	Total monthly energy usage in kilowatt-hours across all on-premises physical machines in the connected environment.
Top clusters by energy usage (kWh)	Top 20 clusters by total Energy usage in kilowatt-hours over the last 30 days in the connected environment.
Top Physical machines by energy usage (kWh)	Top 20 physical machines by total energy usage in kilowatt-hours over the last 30 days in the connected environment.
MoM energy usage change (kWh)	Comparison of total energy usage between current month and previous month.

Ultimate cluster Liveboard



This Liveboard shows details of individual clusters in an on-prem environment, including VMs and Storage. It includes three tabs: **Hosts & VMs**, **Storage**, and **Details**. Each tab displays a collection of charts.

The **Hosts & VMs** tab includes the following charts:

Chart name	Description
CPUs of hosts vs VMs	Total number of CPU cores for hosts and VCPU cores for virtual machines
Global Mem capacity vs vMem provisioned	Comparison of the aggregate average capacity for memory and virtual memory for all hosts that are related to the scoped business application
Average Mem capacity vs vMem provisioned	Comparison of the average capacity for memory and virtual memory for all hosts that are related to the scoped business application
Global utilization stats (%)	Average and maximum percentage utilization of CPU and memory
Global CPU stats	Aggregate average capacity, average used, and maximum used CPU
Global Mem stats	Aggregate average capacity, average used, and maximum used Global Memory
Top 20 Most consumed vCPU by Virtual Machine	Based on average vCPU utilization
Top 20 Most Consumed vMem by virtual machine	Based on average vMem utilization

The **Hosts & VMs** tab also includes the counters for the following statistics:

Counter names	Description
All hosts seen and all VMs seen	Hosts or VMs are counted as seen within the time frame of the Liveboard, which is 72 hours by default.

Counter names	Description
Removed hosts and removed VMs	Removed hosts or VMs were seen within the time frame, but not within the last hour of the time frame.
Added hosts and A=added VMs	Added hosts or VMs were seen for the first time within the last hour of the time frame.
Current hosts and current VMs	Current hosts or VMs were seen within the last hour of the time frame.

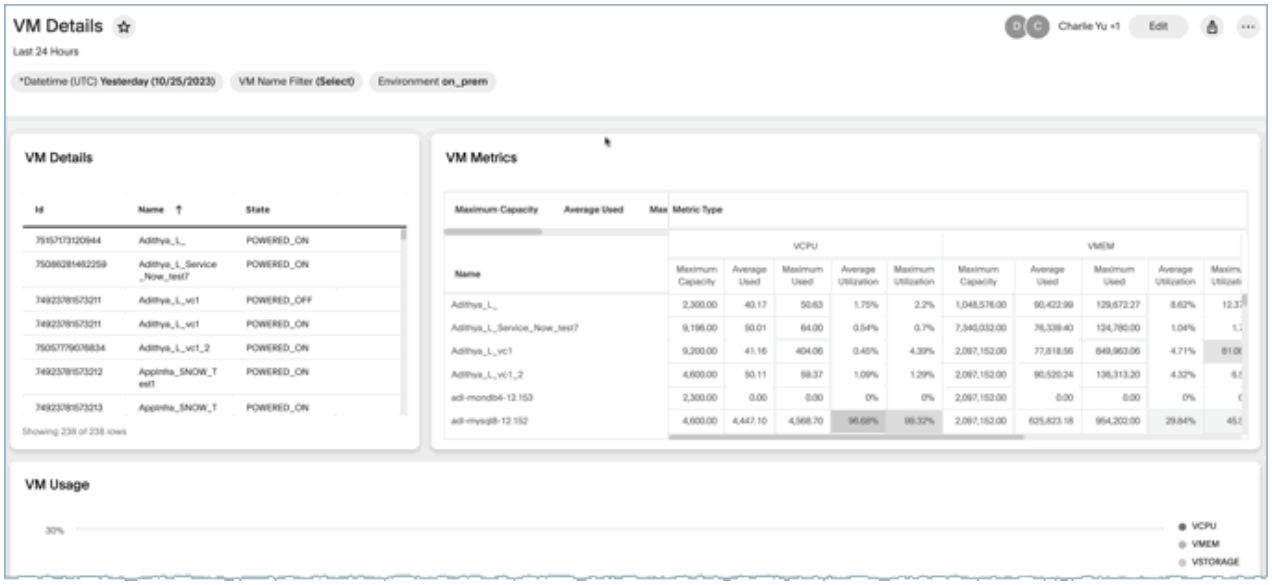
The **Storage** tab includes the following charts:

Chart name	Description
Global storage total vs available	Comparison of the average aggregate capacity for storage and total storage that is available for all data stores that are related to the scoped cluster
Average storage total vs available	Comparison of the average capacity for storage and total storage that is available for all data stores that are related to the scoped cluster
Global storage used vs provisioned	Comparison of the aggregate average capacity for storage and storage that is provisioned for all data stores that are related to the scoped cluster
Average storage used vs provisioned	Comparison of the average capacity for storage and storage that is provisioned for all data stores that are related to the scoped cluster
Storage latency	Comparison of maximum used and average used storage latency for all data stores that are related to the scoped cluster
Storage performance	Comparison of maximum used and average used storage access (IOPS) for all data stores that are related to the scoped cluster
Hourly datastore performance & latency	Hourly comparison of average storage access and average storage latency for all data stores that are related to the scoped cluster

The **Details** tab includes the following charts:

Chart name	Description
CPU cores by cluster	Comparison of the number of CPU cores that are used by each cluster
Host Mem summary	Maximum and average utilization, capacity and used memory for each cluster and its related hosts
Host CPU summary	Maximum and average utilization, capacity, and used CPU for each cluster and its related hosts
Datastore performance summary	Maximum and average used storage and storage latency for each cluster and its related data stores
Datastore storage summary	Maximum and average used storage amount and storage provisioned for each cluster and its related data stores

VM details



This Liveboard shows details of individual on-prem VMs. It includes the following charts:

Chart name	Description
VM details	VM name, ID, and state
VM metrics	Maximum capacity, average used, maximum used, average utilization, and maximum utilization for VCPU, VSTORAGE, and VMEM per VM
VM usage	Average utilization over time for VCPU, VSTORAGE, and VMEM
Related host details	ID, name, and VM name by host
Host metrics	Maximum capacity, average used, maximum used, average utilization, and maximum utilization for VCPU, VSTORAGE, and VMEM per host
Host usage	Average aggregate utilization for BALLOONING, CPU, MEM, and SWAPPING
Related storage details	ID, name, VM name and state per storage volume
Storage metrics	Maximum capacity, average used, maximum used, average utilization, and maximum utilization for storage access and storage latency by data store
Storage usage	Average utilization for storage access and storage latency

Creating custom Liveboards

You can create your own custom charts and Liveboards to focus on areas of specific interest in your managed environment. You must have Report Editor or Report Admin privileges to create or modify charts, Liveboards, or worksheets.

Report Editors and Report Admins have the following options to create custom charts and Liveboards:

- Make a copy of one of the default Liveboards and edit the copy.
- Use one of the default Turbonomic worksheets to create charts and Liveboards.

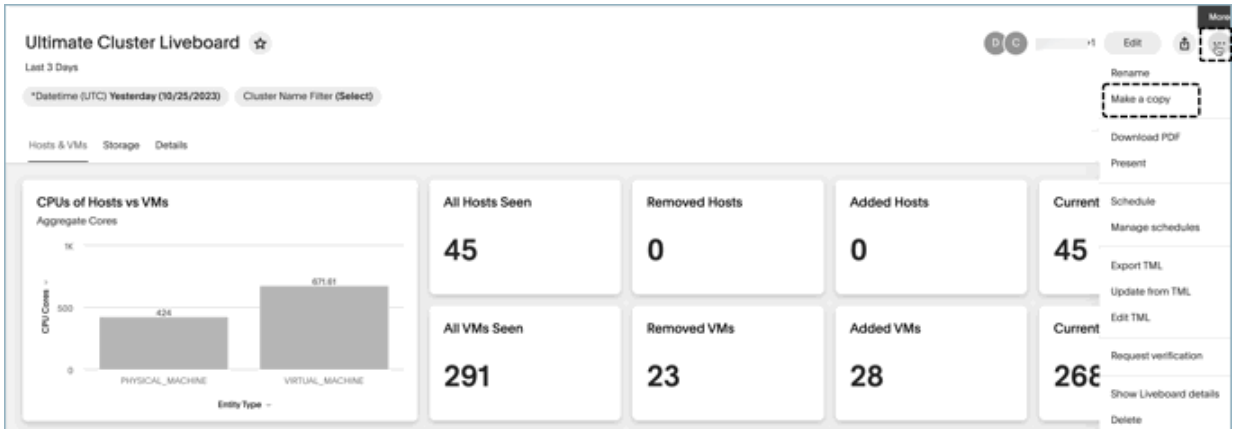
- Create a custom worksheet and use it to create charts and Liveboards.

For information about sharing your custom Liveboards with other Report Editors, see “Sharing Liveboards” on page 1917.

Creating a custom Liveboard from a copy of a default Liveboard

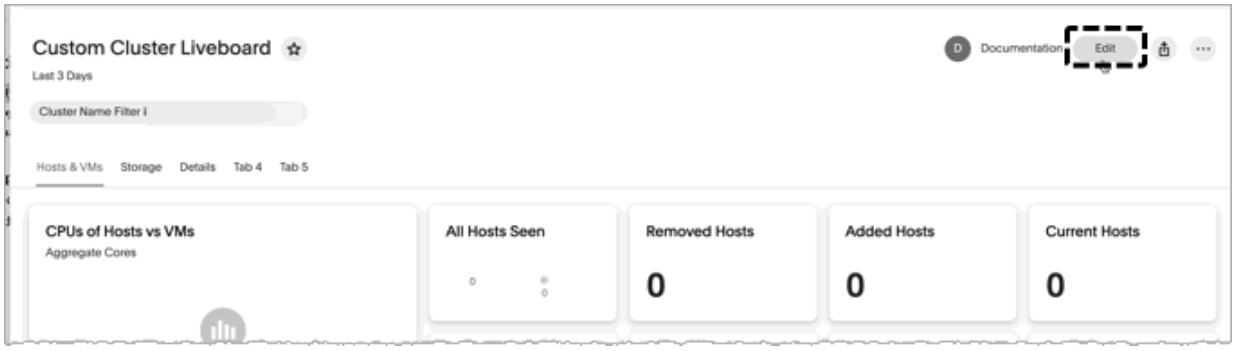
Because Turbonomic regularly updates the default Liveboards, editing these Liveboards is not recommended. Instead, you can make a copy of the Liveboard that you want to customize and edit the copy. For example, you might make a copy of the Ultimate Cluster Liveboard, save it under a new name, change the visualization of existing charts, or add your own charts.

1. In the ThoughSpot UI, select **Liveboards** from the main menu.
2. Select the Liveboard that you want to customize.
3. Click the three-dot icon for your chosen Liveboard and select **Make a copy** from the drop-down menu.



4. Specify a name and description for your new Liveboard.
5. To edit the Liveboard, click **Edit**.

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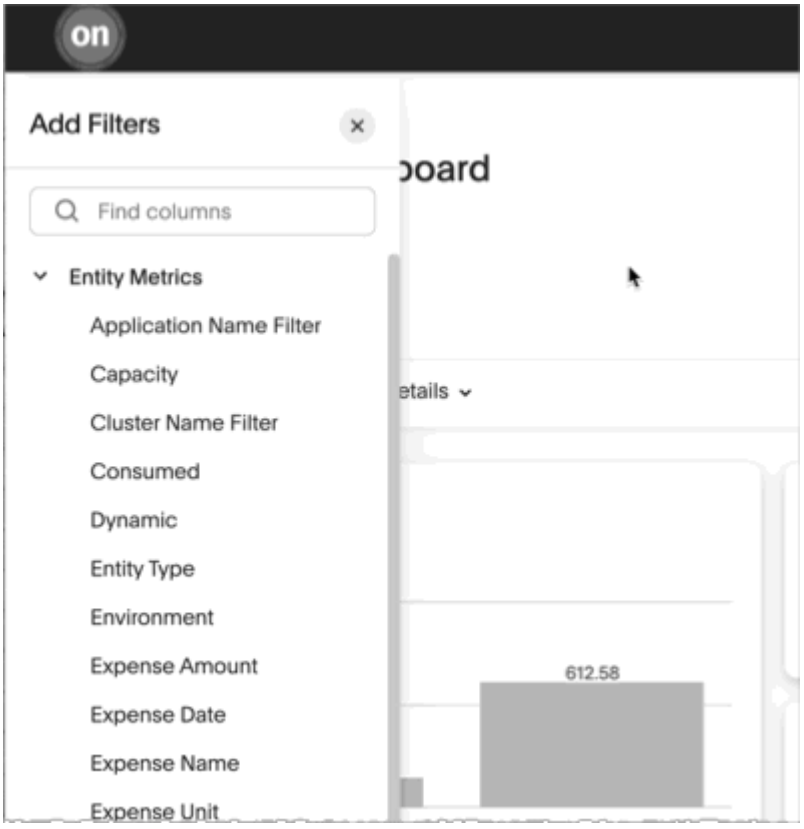


An Edit menu appears in the main navigation bar.

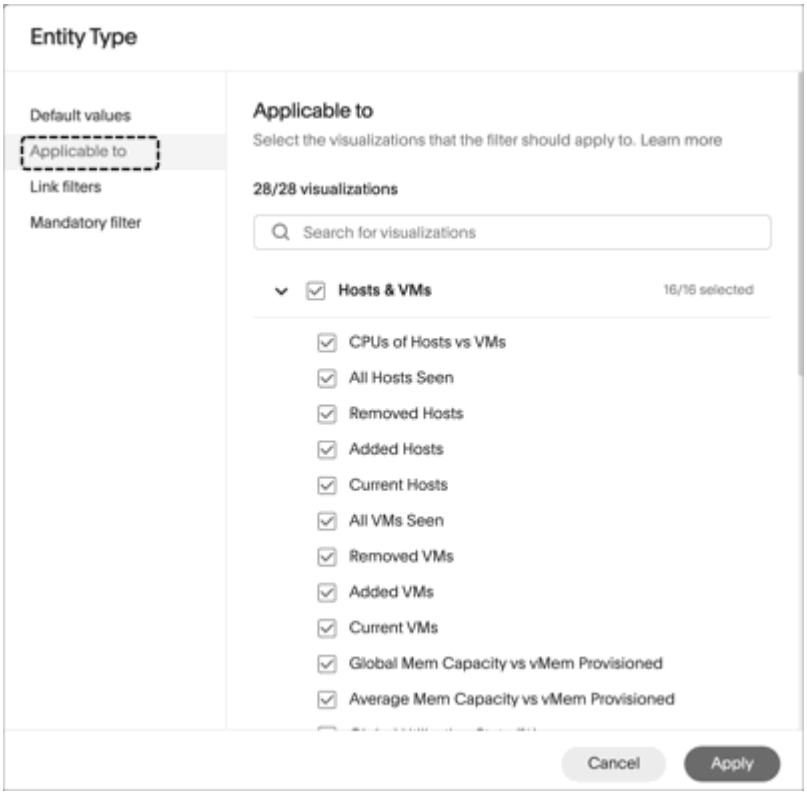
a. To set your own filters for the Liveboard, select **Add Filter** from the Edit menu in the main navigation bar.



b. Select the filter that you want to add from the sidebar **Add Filters** menu.



When you select a Liveboard-level filter, you can optionally specify a subset of charts in the Liveboard to apply it to. Click **Applicable to** and select only the charts you want to filter.



c. When you finish configuring your filters, click **Apply**.

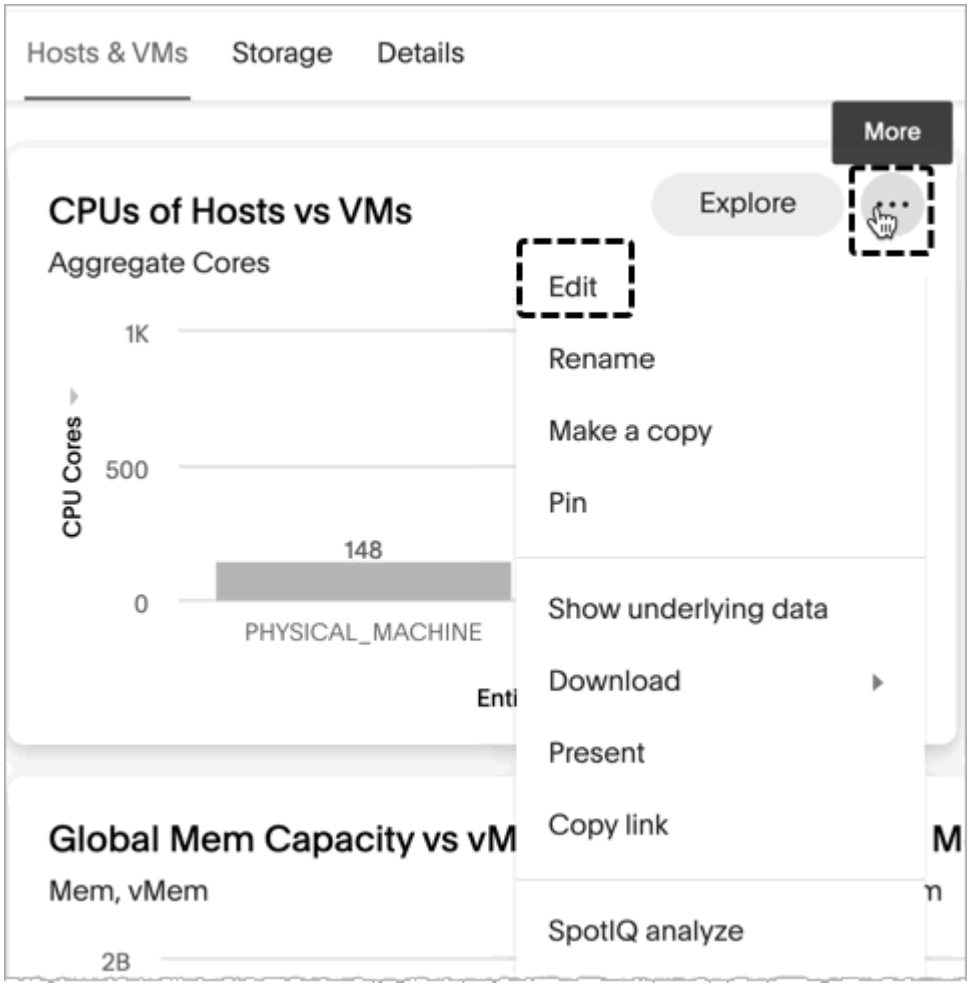
d. When you finish editing the Liveboard, click **Save** in the Edit menu.



Liveboard-level filters override any filters that you set at the chart level.

For more information about editing Liveboards, see the ThoughtSpot documentation.

6. To edit any individual chart in the custom Liveboard, click the three-dot icon for the chart that you want to edit and select **Edit** from the drop-down menu.



a. To set your own filters for the chart, double-click a filter from the list of **Sources** in the **Data** sidebar to add it to the chart.

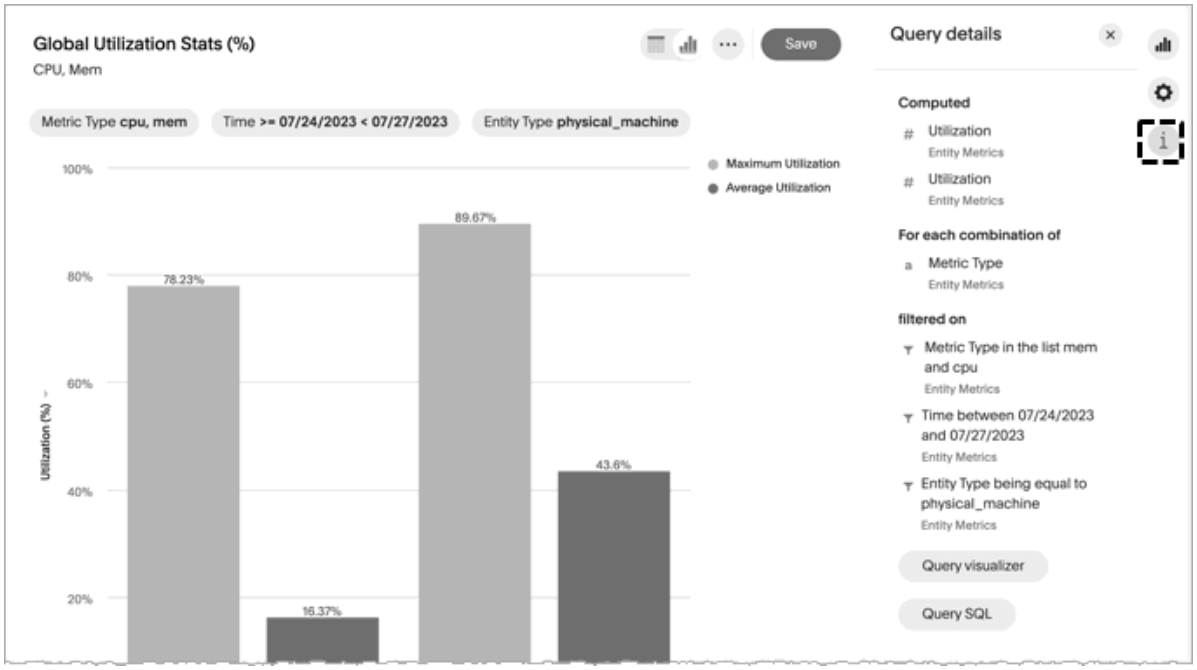


Chart-level filters are overridden by any relevant filters that you set at the Liveboard level.

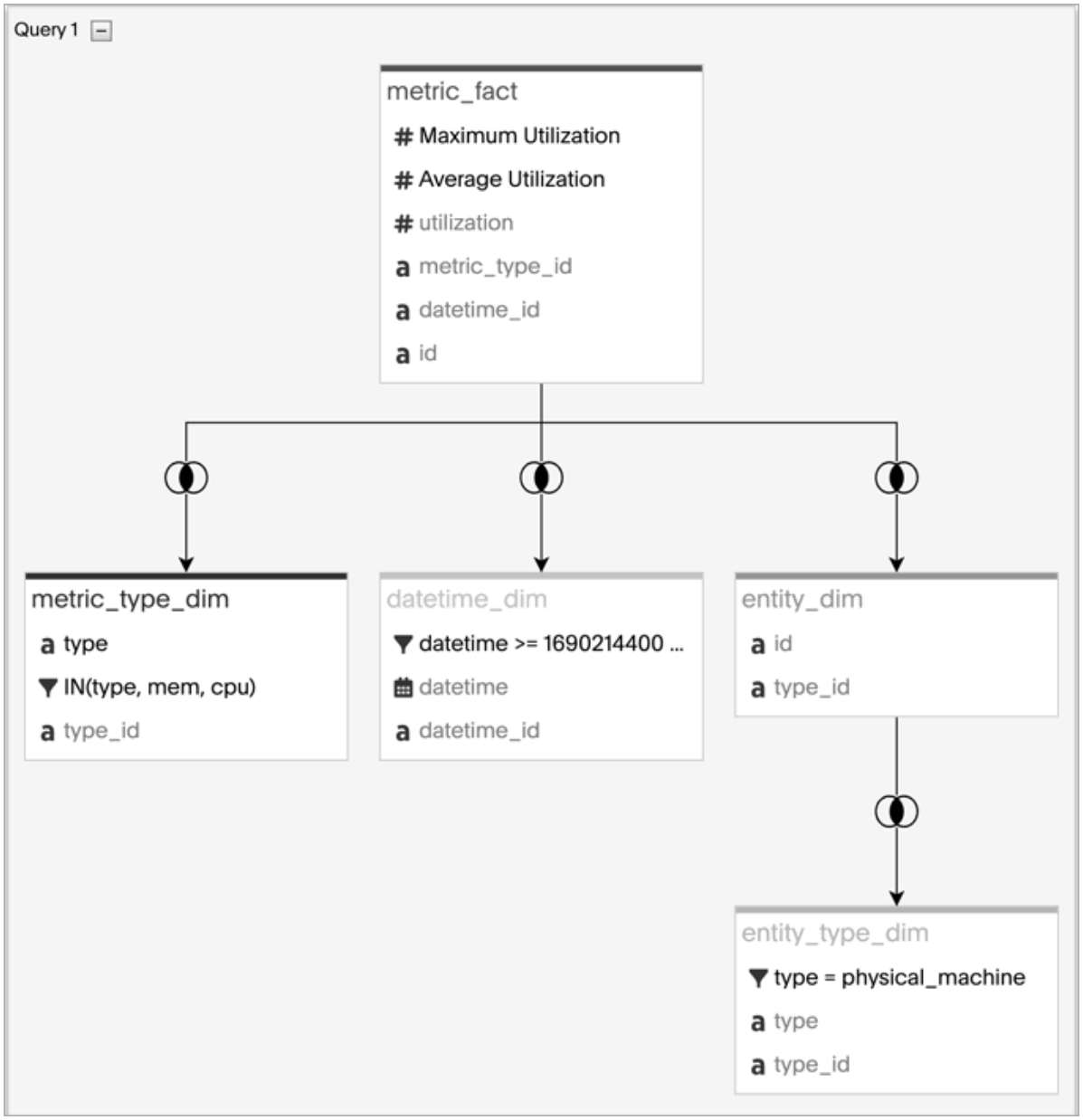
b. When you finish editing the chart, click **Save**.

c. From the Edit window, you can also explore the details of the underlying SQL query for any chart.

After you select **Edit** for a chart, click the  icon in the chart sidebar navigation to view the query details.



d. To view a graphical representation of the query, click **Query visualizer**.



e. To view the SQL code, click **Query SQL**.

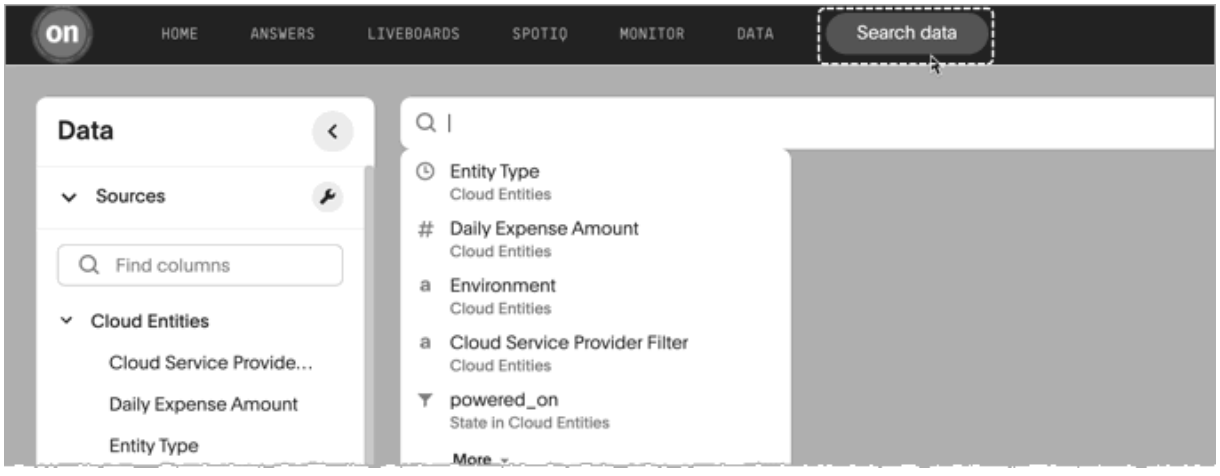


For more information about editing charts, see the ThoughtSpot documentation.

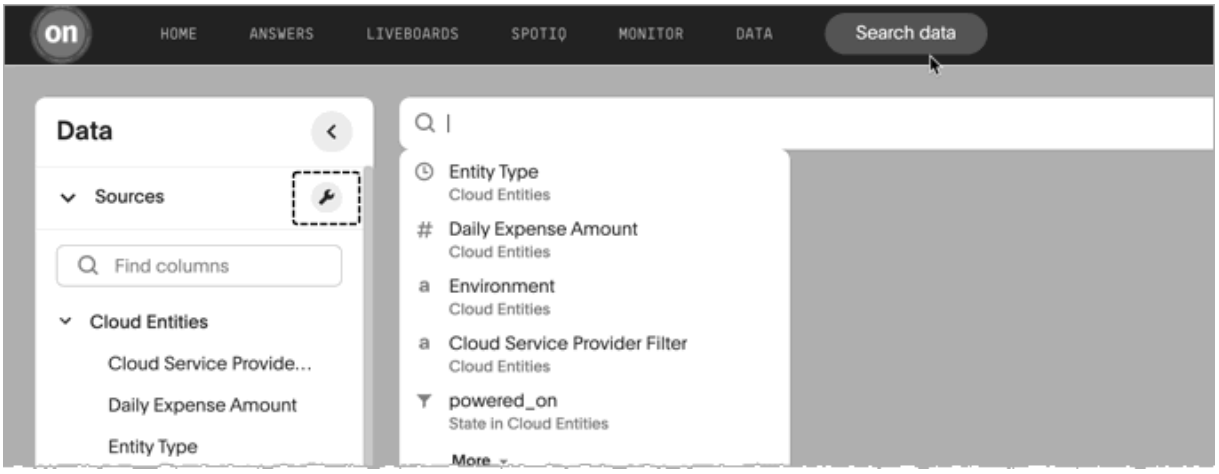
Creating a custom chart or Liveboard by using a default worksheet

ThoughtSpot worksheets create logical views of complex data sets that incorporate multiple entities and relationships. You can use these worksheets to create new charts and Liveboards in your Turbonomic ThoughSpot environment.

1. In the ThoughSpot main menu, click **Search Data**.



2. In the **Sources** sidebar, click the wrench icon.

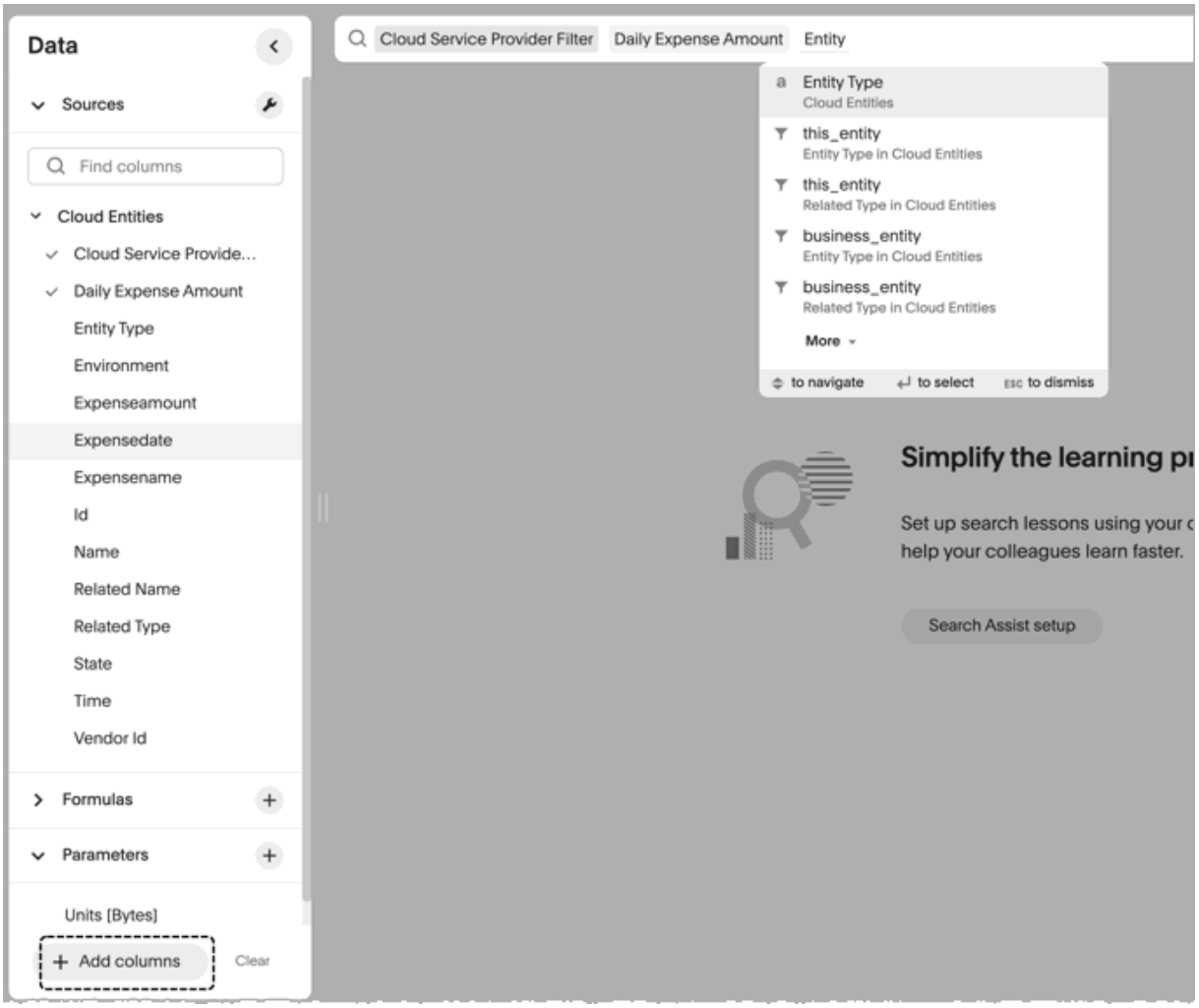


3. In the **Choose Sources** window, the **Worksheets** section lists all the available worksheets. Select the worksheet that you want to use and click **Done**.



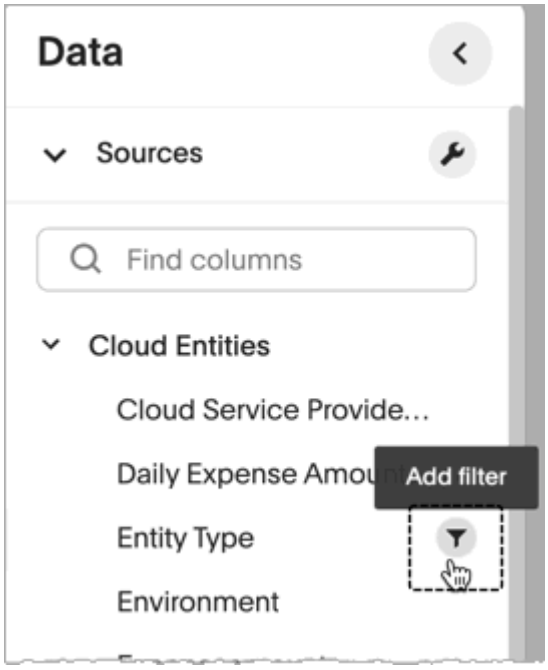
4. To add values to the chart, select from the **Sources** list in the **Data** sidebar and click **Add columns**.

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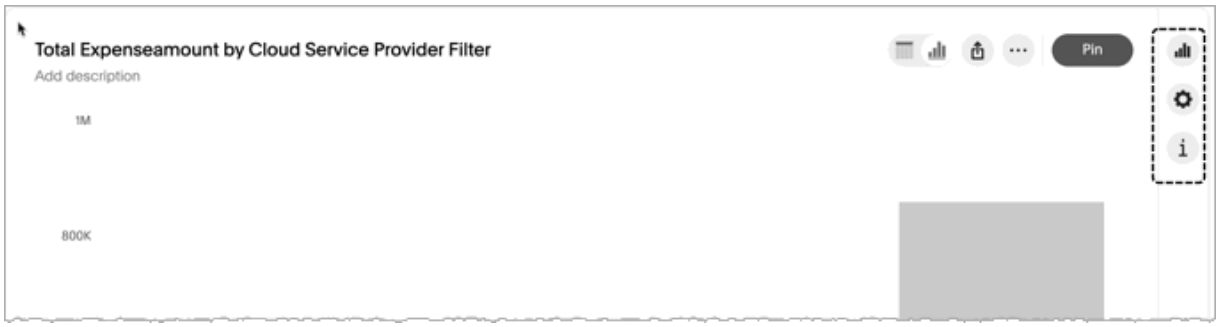
Alternatively, you can specify the name of a value in the main search bar and press **Enter**.

5. To add filters to your chart, hover your cursor over any option in the Sources list and click the filter icon. In the resulting window, select the values that you want to filter on.

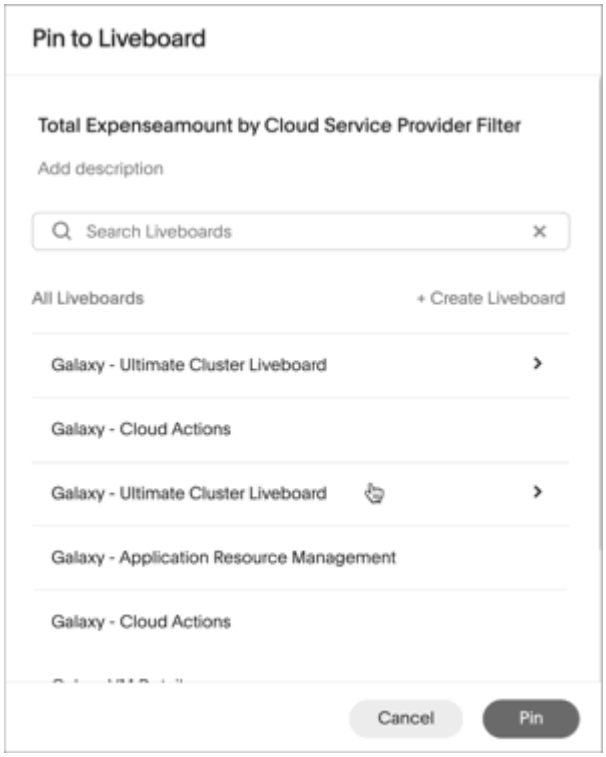


Alternatively, you can specify the name of a column into the main search bar followed by the equals sign (=) and the value that you want to filter on. For example, to filter on powered-on VMs, you might specify `State =powered_on`. To filter on dates, you can specify natural language phrases like `today` or `last 3 weeks`. For more information, see the ThoughtSpot documentation.

6. When you are finished adding values and filters to your chart, click **Go** in the main search bar.
7. ThoughtSpot chooses a default chart type but you can change the visualization, configure the chart, or view the underlying query details by selecting from the icon sidebar menu.



8. To add your chart to a Liveboard, click **Pin**.
9. In the **Pin to Liveboard** window, select from the available Liveboards or click **Create Liveboard** to start a new Liveboard.

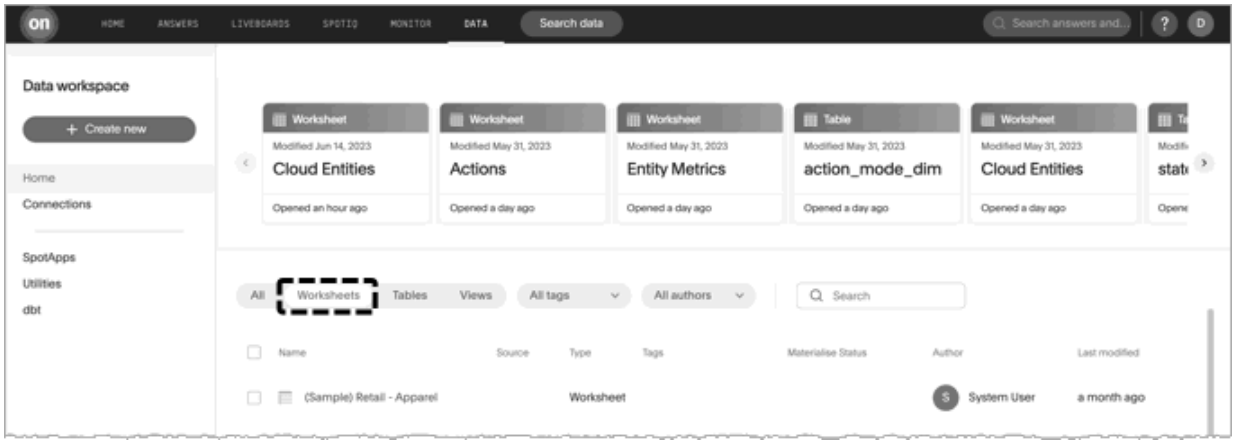


10. After you select a Liveboard, click **Pin.**

Creating a custom worksheet

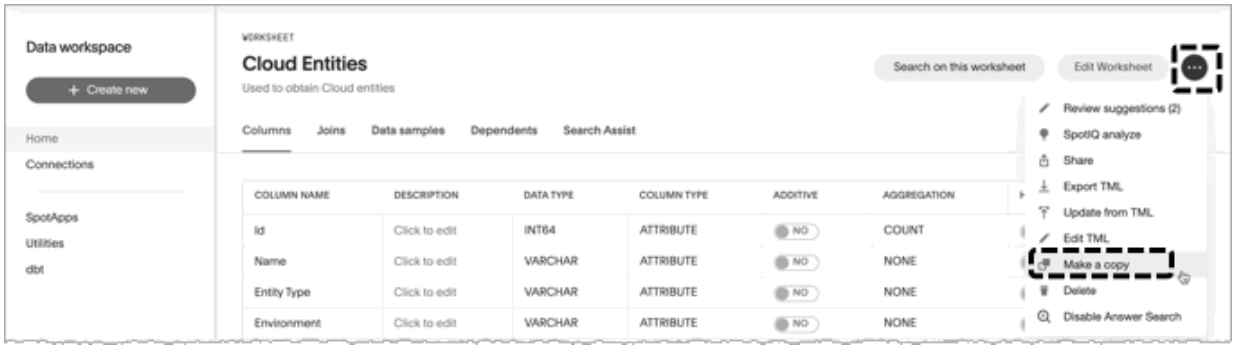
If the default worksheets don't provide the resources that you need, you can create a custom worksheet and use it to create a custom chart or Liveboard. The quickest way to create a custom worksheet is to make a copy of one of the default worksheets and edit the copy.

1. Select **Data** from the ThoughtSpot main menu.
2. Select **Worksheets** from the filter menu.



You can see a list of the default worksheets.

3. Select the worksheet that you want to use.
4. Click the three-dot icon for your chosen worksheet and select **Make a copy** from the drop-down menu.



For more information about editing worksheets, see the ThoughSpot documentation.

5. To use the custom worksheet to create charts and Liveboards, follow the steps in the previous Creating a custom chart or Liveboard by using a default worksheet section.

SaaS reporting versions and release notes

Although versioning conventions are the same between IBM Turbonomic and the SaaS reporting integration, the integration maintains a separate release cycle. Therefore, the current SaaS reporting version might differ from the current IBM Turbonomic version.

Each new SaaS reporting version is deployed to each supported version of Turbonomic as soon as testing and verification for that specific version are complete. For example, if the current Turbonomic version is 8.14.1, the current SaaS reporting version might still be 8.13.6. When SaaS reporting version 8.14.1 is ready to release, it is applied to your instance automatically.

Determining your current SaaS reporting version

To determine the current SaaS reporting version for your Turbonomic instance, review the **SaaS reporting metadata** Liveboard:

1. In the Turbonomic user interface (UI), click **Reports** in the navigation sidebar.
2. Click **Log in using Single Sign On** and specify your IBM Security Verify email and password.
3. Select **SaaS reporting metadata** from the list of default Liveboards.

   The **What is my SaaS reporting version?** window displays your current SaaS reporting version.

SaaS reporting release notes

- Version 8.15.4
  - **New sustainability default Reporting Liveboard "Carbon Footprint - Public Cloud"**: The new default Reporting Liveboard provides visibility into energy consumption and carbon emissions from public cloud VMs (AWS, Azure and Google Cloud). This Liveboard helps users track their environmental impact while uncovering opportunities to reduce energy usage and optimize workloads efficiently. For more information, see Carbon Footprint - Public Cloud.
- Version 8.14.2
  - **Filter actions by tag values**: You can now filter actions in your charts based on the tags that are associated with an action's target entity. You can filter any chart or Liveboard by tag keys or tag values by clicking **Add filter** from the **Edit** menu in the main navigation bar and selecting **Tag key** or **Tag Value** in the **Add Filter** sidebar menu.
  - **Improved performance for showback cost reports**: The performance of these reports is now improved so that they are more responsive and load quickly.
- Version 8.14.1
  - **New SaaS reporting metadata Liveboard**: The 8.14.1 release of SaaS reporting introduces the **SaaS reporting metadata** Liveboard, which helps determine the current SaaS reporting version for your Turbonomic instance.

Schema tables for SaaS reporting

The SaaS reporting data set includes the following tables, views, and worksheets to represent Turbonomic data.

ThoughtSpot schema tables

action_metric_state_dim

Represents metric states linked to pending actions

Column Name	description	type
state_id	The key joining the action metric state enum	INT32
state	The state of a metric to its associated pending action	VARCHAR

pending_action_fact

Tracks pending actions on entities with associated metric usage, capacity, and relationship details

Column Name	description	type
id	The action ID	INT64
entity_id	The entity ID	INT64
datetime_id	The key joining the time that the action data was exported, in UTC time zone	INT32
metric_type_id	The key joining the metric type enum	INT32
action_metric_state_id	The key joining the action metric state enum	INT32
action_metric_relationship_type_id	The key joining the action metric relationship enum	INT32
used	The average usage of the commodity for the entity	DOUBLE

Column Name	description	type
capacity	The capacity of the commodity	DOUBLE
utilization	The utilization of the commodity as a percent	DOUBLE

entity_state_dim

Captures entity state types with identifiers for classification and dimensional analysis

Column Name	description	type
state_id	The key joining the entity state enum	INT32
state	The entity state type	VARCHAR

namespace_cluster_dim

Maps namespaces to their associated clusters with identifying details

Column Name	description	type
cluster_id	The cluster ID	INT64
namespace_id	The namespace ID	INT64
namespace	The namespace name	VARCHAR

cloudsavings_dim

Captures metadata for cloud savings, linking entities, actions, and cloud infrastructure attributes

Column Name	description	type
id	The cloud savings ID	INT64
entity_id	The entity ID	INT64
entity_type_id	The key joining the entity type enum	INT32
action_chain_id	The action chain ID	INT64
action_type_id	The key joining the action type enum	INT32
action_initiator_type_id	The key joining the action initiator enum	INT32
currency	The currency of cloud savings as an integer (ISO Code)	INT32
account_id	The associated account ID	INT64
region_id	The associated region ID	INT64
cloud_service_id	The associated cloud service ID	INT64
availability_zone_id	The associated availability zone ID	INT64
service_provider_id	The associated service provider ID	INT64
resource_group_id	The associated resource group ID	INT64
billing_family_id	The associated billing family ID	INT64

entity_dim

Represents core entity metadata with links to type and environment dimensions for contextual insights

Column Name	description	type
id	The entity ID	INT64
name	The entity name	VARCHAR
type_id	The key joining the entity type enum	INT32
environment_id	The key joining the environment enum	INT32

group_origin_dim

Defines group origin types with identifiers for mapping group creation sources

Column Name	description	type
origin_id	The key joining the group origin enum	INT32
origin	The origin of the group	VARCHAR

latest_action_state_dim

Represents the latest state of actions with corresponding identifiers

Column Name	description	type
state_id	The key joining the action state enum	INT32
state	The state of the action	VARCHAR

workload_controller_attrs_dim

Describes workload controller attributes linked to actions, including namespace and cluster

Column Name	description	type
action_id	The action ID	INT64
namespace	The namespace name	VARCHAR
container_cluster	The associated container cluster of an action	VARCHAR

related_entity_type_dim

Represents types of related entities with identifiers for dimensional mapping and classification

Column Name	description	type
type_id	The key joining the related entity type enum	INT32
type	The type of a related entity	VARCHAR

cluster_cost_fact

Stores monthly cost totals for container clusters based on daily aggregations

Column Name	description	type
container_cluster_id	The cluster ID	INT64
cluster_cost_monthly	The sum of daily cost for the last 30 days by cluster	DOUBLE

entity_tag_fact

Tracks entity-to-tag mappings over time for classification and analysis

Column Name	description	type
entity_id	The entity ID	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
id	The tag ID	INT64

action_category_dim

Defines action categories with identifiers for classification and dimensional linking

Column Name	description	type
category_id	The key joining the category enum	INT32
category	The action category	VARCHAR

entity_environment_dim

Defines environment types with unique identifiers for entity classification and dimensional joins

Column Name	description	type
environment_id	The key joining the environment enum	INT32
environment	The environment type	VARCHAR

related_entity_view

View of related entities with type mapping for contextual reference

Column Name	description	type
type_id	The key joining the related entity type enum	INT32
id	The entity ID	INT64
name	The entity name	VARCHAR

action_tag_fact

[deprecated] table for storing action tag associations over time

Column Name	description	type
action_id	[Deprecated]	INT64
datetime_id	[Deprecated]	INT32
id	[Deprecated]	INT64

idle_power_fact

Stores daily idle power consumption of entities, measured at zero cpu usage

Column Name	description	type
id	The entity ID	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
idle_power	The power in watts consumed by the host at 0 CPU utilization	DOUBLE

cloudsavings_zone_view

Represents area zones referenced in cloud savings analysis

Column Name	description	type
id	The area zone ID, referenced by cloud savings	INT64
name	The area zone name	VARCHAR

group_tag_fact

Links groups to tags with export time context

Column Name	description	type
group_id	The group ID	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
id	The tag ID	INT64

cloudsavings_provider_view

Contains cloud provider details referenced in savings data

Column Name	description	type
id	The provider ID, referenced by cloud savings	INT64
name	The provider name	VARCHAR

action_related_entity_fact

Maps actions to related entities with export time and target flag context

Column Name	description	type
source_id	The action ID	INT64
datetime_id	The key joining the time that the action data was exported, in UTC time zone	INT32
related_id	The ID of the action’s related entity	INT64
is_target	The boolean value representing whether the related entity is the action’s target entity	BOOL

cloudsavings_region_view

Represents regions used in cloud savings calculations

Column Name	description	type
id	The region ID, referenced by cloud savings	INT64

Column Name	description	type
name	The region name	VARCHAR

action_mode_dim

Defines action mode types with identifiers for use in dimensional joins and analysis

Column Name	description	type
mode_id	The key joining the action mode enum	INT32
mode	The action mode	VARCHAR

action_type_dim

Defines action types with identifiers for categorization and dimensional analysis

Column Name	description	type
type_id	The key joining the action type enum	INT32
action_type	The type of action	VARCHAR

group_tag_dim

Defines key-value tag pairs used to categorize and describe group characteristics

Column Name	description	type
id	The tag ID, references group types	INT64
key	The key of the tag	VARCHAR
value	The value of the tag	VARCHAR

_state_fact

Tracks entity state changes over time for historical analysis

Column Name	description	type
id	The entity ID	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
state_id	The key joining the entity state enum	INT32

cloudsavings_fact

Captures daily realized cloud savings and investments by id and time

Column Name	description	type
id	The cloud savings ID	INT64
datetime_id	The key joining the date referencing the cloud savings, in UTC time zone	INT32
realized_savings	The realized savings daily	DOUBLE
realized_investments	The realized investments daily	DOUBLE

action_initiator_dim

Defines types of action initiators used in cloud savings context

Column Name	description	type
type_id	The key joining the action initiator enum	INT32
type	The type of action initiation referenced by cloud savings	VARCHAR

num_cpus_fact

Tracks the number of cpu cores per entity over time for performance and capacity analysis

Column Name	description	type
id	The entity ID	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
num_cpus	The number of cpu cores of an entity	INT32

action_severity_dim

Defines action severity levels with identifiers for impact classification

Column Name	description	type
severity_id	The key joining the action severity enum	INT32
severity	The severity of the issue that the associated action addresses	VARCHAR

tag_dim

Defines tag key-value pairs for entity classification and metadata enrichment

Column Name	description	type
id	The tag ID, references entity types	INT64
key	The key of the tag	VARCHAR
value	The value of the tag	VARCHAR

related_fact

[deprecated] fact table that linked entities, groups, and timestamps for analytical joins across related dimensions

Column Name	description	type
source_id	[Deprecated]	INT64
datetime_id	[Deprecated]	INT32
related_id	[Deprecated]	INT64
related_id_group	[Deprecated]	INT64

action_savings_fact

Stores projected hourly savings per action over time

Column Name	description	type
id	The action ID	INT64
datetime_id	The key joining the time that the action data was exported, in UTC time zone	INT32
savings_hourly	The projected hourly savings of the action	DOUBLE

bottom_up_entity_cost_daily_fact

Stores daily bottom-up cost and uptime data for entities, including cost sources, categories, and associated clusters

Column Name	description	type
value	The cost value	DOUBLE
uptime_hours	The total number of uptime hours for the day	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
entity_id	The entity ID	INT64
source	The source of the cost calculation, such as a discount or rate	VARCHAR
category	The category of the bottom up cost	VARCHAR
container_cluster_id	The container cluster ID	INT64
entity_type_id	The key joining the entity type enum	INT32

bottom_up_entity_cost_fact

Tracks bottom-up cost values per entity over time with source and category details

Column Name	description	type
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
entity_id	The entity ID	INT64
value	The cost value	DOUBLE
source	The source of the cost calculation, such as a discount or rate	VARCHAR
category	The category of the bottom up cost	VARCHAR
unit	The unit of the cost	VARCHAR

metric_daily_fact

Captures daily resource metrics like capacity, usage, and utilization for entities by metric type and entity type

Column Name	description	type
id	The entity ID	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
metric_type_id	The key joining the metric type enum	INT32
capacity	The capacity of the commodity	DOUBLE

Column Name	description	type
used	The average usage of the commodity for the entity	DOUBLE
utilization	The utilization of the commodity as a percent	DOUBLE
consumed	The amount of the commodity used by the consumer entity	DOUBLE
peak_consumed	The max usage of the commodity for the entity	DOUBLE
entity_type_id	The key joining the entity type enum	INT32

cloudsavings_service_view

Lists cloud services referenced in cloud savings data

Column Name	description	type
id	The service ID, referenced by cloud savings	INT64
name	The cloud service name	VARCHAR

group_dim

Defines group entities with attributes for type, origin, and member classifications, supporting dynamic and static classifications

Column Name	description	type
id	The group ID	INT64
group name	The group name	VARCHAR
dynamic	True if the group definition is dynamic; false indicates a static group	BOOL
type_id	The key joining the group type enum	INT32
origin_id	The key joining the group origin enum	INT32
member_type_id	The key joining the group member type enum	INT32

action_metric_relationship_dim

Defines how metrics relate to pending actions using relationship types and enums

Column Name	description	type
type_id	The key joining the action metric relationship enum	INT32
metric_relationship_type	The relationship of a metric to its associated pending action	VARCHAR

cloudsavings_resourcegroup_view

Lists resource groups referenced in cloud savings data

Column Name	description	type
id	The resource group ID, referenced by cloud savings	INT64
name	The resource group name	VARCHAR

action_state_dim

Defines action states with unique identifiers for status tracking and dimensional joins

Column Name	description	type
state_id	The key joining the action state enum	INT32
state	The state of the action	VARCHAR

_action_state_fact

Tracks action states over time for status history and analysis

Column Name	description	type
id	The action ID	INT64
datetime_id	The key joining the time that the action data was exported, in UTC time zone	INT32
state_id	The key joining the action state enum	INT32

entity_type_dim

Defines entity type classifications with ids for joining and analytical grouping

Column Name	description	type
type_id	The key joining the entity type enum	INT32
type	The type of entity	VARCHAR

related_entity_fact

Tracks relationships between entities with export time context

Column Name	description	type
source_id	The entity ID	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
related_id	The ID of the entity’s related entity	INT64

group_member_type_dim

Defines the entity types that can be members of a group

Column Name	description	type
type_id	The key joining the group member type enum	INT32
group_member_type	The entity type of the group’s members	VARCHAR

cloudsavings_account_view

Lists cloud accounts referenced in cloud savings data

Column Name	description	type
id	The account ID, referenced by cloud savings	INT64

Column Name	description	type
name	The account name	VARCHAR

_membership_fact

Tracks entity-group memberships over time with member type classification

Column Name	description	type
source_id	The entity ID	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
group_id	The group ID	INT64
group_member_type_id	The key joining the group member type enum	INT32

group_type_dim

Defines group type categories with identifiers for use in classification and joins

Column Name	description	type
type_id	The key joining the group type enum	INT32
type	The type of group	VARCHAR

related_fact_last_seen

Records the latest timestamp of entity-to-related entity relationships

Column Name	description	type
source_id	The entity ID	INT64
related_id	The ID of the entity’s related entity	INT64
ts	The latest timestamp for the associated relationship	DATE_TIME

action_membership_fact

Captures group associations for actions over time, enabling analysis by group, action, and export timestamp

Column Name	description	type
source_id	The action ID	INT64
datetime_id	The key joining the time that the action data was exported, in UTC time zone	INT32
group_id	The group ID	INT64

action_dim

Stores detailed info about actions, including type, state, target, execution, and context metadata

Column Name	description	type
execution_time	The time at which an action execution was initiated, in UTC time zone	DATE_TIME

Column Name	description	type
target_replica_count	The replica count of the entity that the action operates on	INT32
state_id	The key joining the action state enum	INT32
first_seen	[Internal Use - Not Supported]	DATE_TIME
user_account	The user account that triggered an action execution	VARCHAR
target_name	The name of the entity that the action operates on	VARCHAR
execution_error	The error message associated with a failed action	VARCHAR
resizeinfo_target_type	The type of entity that a resize action operates on	VARCHAR
resizeinfo_unit	The unit measurement of the commodity type associated with a resize action	VARCHAR
id	The action object identifier	INT64
resizeinfo_target_name	The name of entity that a resize action operates on	VARCHAR
creation_time	When the action was recommended, in UTC time zone	DATE_TIME
description	The text description of the action	VARCHAR
explanation	The condition that the action is addressing	VARCHAR
type_id	The key joining the action type enum	INT32
category_id	The key joining the category enum	INT32
mode_id	The key joining the action mode enum	INT32
severity_id	The key joining the action severity enum	INT32
target_type	The name of the entity that the action operates on	VARCHAR
environment_id	The key joining the environment enum	INT32
moveinfo_destination_type	The entity type of the destination associated with a move action	VARCHAR
deleteinfo_filepath	The file path associated with a delete action	VARCHAR
deleteinfo_filesize	The file size associated with a delete action	DOUBLE
resizeinfo_commodity	The commodity type associated with a resize action	VARCHAR
resizeinfo_oldcapacity	The old capacity size associated with a resize action	DOUBLE
resizeinfo_newcapacity	The new capacity size associated with a resize action	DOUBLE

metric_fact

Stores time-based metrics like capacity, usage, and utilization by entity and type

Column Name	description	type
id	The entity ID	INT64
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	INT32
metric_type_id	The key joining the metric type enum	INT32
capacity	The capacity of the commodity	DOUBLE
used	The average usage of the commodity for the entity	DOUBLE

Column Name	description	type
utilization	The utilization of the commodity as a percent	DOUBLE
consumed	The amount of the commodity used by the consumer entity	DOUBLE

datetime_dim

Centralized mapping of utc timestamps with unique identifiers, used in most time-based filters and essential for accurate time-based joins across datasets

Column Name	description	type
datetime_id	The key joining the datetime	INT32
datetime	The timestamp, in UTC	DATE_TIME

metric_type_dim

Identifies commodity types with byte-based unit flag

Column Name	description	type
type_id	The key joining the metric type enum	INT32
is_byte	Whether the commodity type is measured in bytes, units of 1024	BOOL
metric_type	The metric type, otherwise known as the commodity type	VARCHAR

action_related_fact

[deprecated] table for tracking relationships between actions and related entities or groups

Column Name	description	type
source_id	[Deprecated]	INT64
datetime_id	[Deprecated]	INT32
related_id	[Deprecated]	INT64
is_target	[Deprecated]	BOOL

ThoughtSpot schema views

namespace_consumed_daily_fact

Daily namespace resource usage by cluster and metric type

Column Name	description	source	type
cluster_id	The cluster ID	namespace_cluster_dim	INT64
namespace_id	The namespace ID	namespace_cluster_dim	INT64
namespace	The namespace name	namespace_cluster_dim	VARCHAR
metric_type_id	The key joining the metric type enum	metric_daily_fact	INT32
datetime_id	The ID representing the time that the data was exported; convert by joining with datetime_dim	metric_daily_fact	INT32

Column Name	description	source	type
consumed_day	The average daily amount of the commodity used by the consumer entity	metric_daily_fact	DOUBLE

cspec_latest_tags

Latest tags for container spec

Column Name	description	source	type
entity_id	The entity ID	entity_tag_fact	INT64
Label	The latest key-value pairs for the container spec	FORMULA	FORMULA

cluster_cost_dim_old

[deprecated] hourly and monthly cluster cost calculations from bottom-up data

Column Name	description	source	type
cluster_id	[Deprecated]	related_fact_last_seen	INT64
cluster_name	[Deprecated]	related_entity_view	VARCHAR
cluster_cost_hourly	[Deprecated]	FORMULA	FORMULA
cluster_cost_monthly	[Deprecated]	FORMULA	FORMULA

Action affected metrics generic view

Generic view that contains all the information around pending action metrics

Column Name	description	source	type
Action Id		action_dim	INT64
Affected Metric State		action_metric_state_dim	VARCHAR
Action Type		action_type_dim	VARCHAR
Affected Metric Type		metric_type_dim	VARCHAR
Affected Metric Relation		action_metric_relationship_dim	VARCHAR
Affected Entity Id		pending_action_fact	INT64
Used		pending_action_fact	DOUBLE
Utilization		pending_action_fact	DOUBLE
Capacity		pending_action_fact	DOUBLE

Latest entity tags

Latest tags per entity based on most recent assignment

Column Name	description	source	type
entity_id	The entity ID	entity_tag_fact	INT64
id	The tag ID, references entity types	entity_tag_fact	INT64

Resource group name to account view

Mapping resource group names to business accounts using membership and entity relationships

Column Name	description	source	type
Group Name	The group name	group_dim	VARCHAR
Group Id	The group ID	FORMULA	INT64
Resource Group Account Name	The resource group account name	related_entity_view	VARCHAR
Account Id	The account ID	related_entity_fact	INT64
Number of id	Count of entity ids	FORMULA	INT64
Maximum datetime_id	The key joining the latest time that the entity data was exported	datetime_dim	INT32

Latest action state

Latest state of each action based on most recent timestamp

Column Name	description	source	type
state	The state of the action	action_state_dim	VARCHAR
datetime	The timestamp, in UTC	datetime_dim	DATE_TIME
datetime_id	The key joining the time that the action data was exported, in UTC time zone	_action_state_fact	INT32
id	The action ID	_action_state_fact	INT64

Monthly Carbon energy view

Monthly carbon and energy usage for physical machines

Column Name	description	source	type
Month(datetime)	The monthly date	datetime_dim	DATE_TIME
Entity Id	The entity ID	entity_dim	INT64
Carbon sum	The sum of carbon used by month in kg	FORMULA	FORMULA
Energy sum	The sum of energy used by month in kWh over a 10 min interval	FORMULA	FORMULA

cluster_capacity_by_date

Daily cluster capacity and usage metrics by environment and type

Column Name	description	source	type
cluster_id	The cluster ID	entity_dim	INT64
cluster_name	The cluster name	entity_dim	VARCHAR
date_daily	The day of the aggregated values	datetime_dim	DATE_TIME
environment_id	The key joining the environment enum	entity_dim	INT32
metric_type_id	The key joining the metric type enum	metric_daily_fact	INT32

Column Name	description	source	type
datetime_id	The key joining the time that the entity data was exported, in UTC time zone	metric_daily_fact	INT32
capacity_hour	The daily capacity of the commodity	metric_daily_fact	DOUBLE
used_hour	The daily average usage of the commodity for the entity	metric_daily_fact	DOUBLE

cluster_node_cost

Monthly node cost and uptime metrics by cluster using bottom-up cost data

Column Name	description	source	type
node_name	The node name	entity_dim	VARCHAR
node_id	The node ID	bottom_up_entity_cost_daily_fact	INT64
cluster_id	The cluster ID	bottom_up_entity_cost_daily_fact	INT64
hostname	The host name	FORMULA	FORMULA
node_cost	[Static Value] 0.0	FORMULA	FORMULA
cost_time	[Static Value] 1.0	FORMULA	FORMULA
node_cost_monthly	The sum of node cost for the last 30 days	FORMULA	FORMULA
up_time_hours	The total number of uptime hours for the last 30 days	bottom_up_entity_cost_daily_fact	INT64
cluster_cost_monthly	The sum of daily cost for the last 30 days by cluster	FORMULA	FORMULA

cspec_used_daily

Daily container spec usage and capacity metrics by cluster

Column Name	description	source	type
date_daily	The date of the aggregated consumption	datetime_dim	DATE_TIME
metric_type_id	The key joining the metric type enum	metric_daily_fact	INT32
cspec	The container spec name	entity_dim	VARCHAR
cspec_id	The container spec ID	entity_dim	INT64
cluster_id	The cluster ID	related_fact_last_seen	INT64
used_day	The average daily usage of the commodity for the entity	metric_daily_fact	DOUBLE
capacity_day	The average daily capacity of the commodity	metric_daily_fact	DOUBLE

cspec_all_labels

All latest tags for container spec

Column Name	description	source	type
entity_id	The entity ID	entity_tag_fact	INT64

Column Name	description	source	type
all_labels	The aggregation of all latest tag key-value pairs for the container spec	FORMULA	FORMULA

Host : number of related VMs

Latest vm (virtual machine) count per physical host based on entity relationships

Column Name	description	source	type
Host Name	The host name	entity_dim	VARCHAR
Number of Related VMs	The number of related virtual machines according to the latest broadcast	related_entity_view	INT64

Average idle power per host for action

Average idle power per host for actions (yesterday only)

Column Name	description	source	type
id	The entity ID	entity_dim	INT64
Average idle_power	The average idle power (watts) per host	idle_power_fact	DOUBLE

Latest action targeted entity

Earliest and latest timestamps of actions targeting entities

Column Name	description	source	type
id		action_dim	INT64
entity id	The entity ID	entity_dim	INT64
last seen	The latest timestamp for the associated relationship	datetime_dim	DATE_TIME
min datetime	The earliest timestamp for the associated relationship	datetime_dim	DATE_TIME

namespace_consumed_daily

Daily namespace consumption metrics by cluster and metric type

Column Name	description	source	type
cluster_id	The cluster ID	related_entity_view	INT64
date_daily	The date of the aggregate consumption	datetime_dim	DATE_TIME
metric_type_id	The key joining the metric type enum	metric_daily_fact	INT32
namespace	The namespace name	entity_dim	VARCHAR
consumed_day	The amount of the commodity used by the consumer entity	metric_daily_fact	DOUBLE

cspec_namespace

Latest container spec to namespace mapping

Column Name	description	source	type
cspec_id	The container spec ID	entity_dim	INT64
namespace_id	The namespace ID	related_fact_last_seen	INT64
namespace	The namespace name	related_entity_view	VARCHAR

ThoughtSpot schema worksheets

Cluster capacity, cost, and utilization by namespace

Column Name	description	aggregation	synonyms	source	type
Date Range	The date of the aggregated usage	NONE		cluster_capacity_by_date	DATE_TIME
Cluster	The cluster name	NONE		cluster_capacity_by_date	VARCHAR
Metric_ID	The key joining the metric type enum	NONE		cluster_capacity_by_date	INT32
Capacity	The capacity of the commodity	AVERAGE		cluster_capacity_by_date	DOUBLE
Used	The average usage of the commodity for the entity	AVERAGE		cluster_capacity_by_date	DOUBLE
Environment_ID	The key joining the environment enum	COUNT		cluster_capacity_by_date	INT32
cluster_cost_monthly	The sum of daily cost for the last 30 days by cluster	NONE		cluster_cost_fact	DOUBLE
Namespace	The namespace name	NONE		namespace_consumed_daily_fact	VARCHAR
Consumed	The daily amount of the commodity used by the consumer entity	AVERAGE		namespace_consumed_daily_fact	DOUBLE

Column Name	description	aggregation	synonyms	source	type
Utilization	The utilization of the commodity as a percent	SUM		FORMULA	DOUBLE
capacity_by_unit	The capacity of the commodity	SUM		cluster_capacity_by_date	FORMULA
consumed_by_unit	The amount of the commodity used by the consumer entity	SUM		FORMULA	FORMULA
cpu_cost_factor	The cpu cost factor of the cluster based on the CPU weight in cost calculation paramater	NONE		cluster_cost_fact	DOUBLE
memory_cost_factor	The memory cost factor of the cluster based on the CPU weight in cost calculation paramater	NONE		cluster_cost_fact	DOUBLE
metric_friendly	[Formula] The metric type, reformatted	NONE		cluster_capacity_by_date	INT32
used_by_unit	The average usage of the commodity for the entity	SUM		cluster_capacity_by_date	FORMULA

cluster_capacity_namespace_consumed

Column Name	description	aggregation	synonyms	source	type
Cluster	The cluster name	NONE		cluster_capacity_by_date	VARCHAR

Column Name	description	aggregation	synonyms	source	type
Metric_ID	The key joining the metric type enum	COUNT		cluster_capacity_by_date	INT32
Date Range	The date of the aggregated usage	NONE		cluster_capacity_by_date	DATE_TIME
Capacity	The capacity of the commodity	AVERAGE		cluster_capacity_by_date	DOUBLE
Used	The average usage of the commodity for the entity	AVERAGE		cluster_capacity_by_date	DOUBLE
Namespace	The namespace name	NONE		namespace_consumed_daily	VARCHAR
Consumed	The amount of the commodity used by the consumer entity	AVERAGE		namespace_consumed_daily	DOUBLE
Utilization	The utilization of the commodity as a percent	SUM		FORMULA	DOUBLE
metric_friendly	[Formula] The metric type, reformatted	NONE		cluster_capacity_by_date	INT32
capacity_by_unit	The capacity of the commodity	SUM		cluster_capacity_by_date	FORMULA
consumed_by_unit	The amount of the commodity used by the consumer entity	SUM		FORMULA	FORMULA
used_by_unit	The average usage of the commodity for the entity	SUM		cluster_capacity_by_date	FORMULA

Column Name	description	aggregation	synonyms	source	type
cpu_cost_factor	The cpu cost factor of the cluster based on the CPU weight in cost calculation paramater	NONE		cluster_node_cost	FORMULA
memory_cost_factor	The memory cost factor of the cluster based on the CPU weight in cost calculation paramater	NONE		cluster_node_cost	FORMULA
cluster_cost_monthly	The sum of daily cost for the last 30 days by cluster	NONE		cluster_node_cost	FORMULA
node_name	The node name	NONE		cluster_node_cost	VARCHAR
Uptime_hours	The total number of uptime hours for the last 30 days	NONE		cluster_node_cost	INT64
node_cost_monthly	The sum of node cost for the last 30 days	NONE		cluster_node_cost	FORMULA
Hostname	The host name	NONE		cluster_node_cost	FORMULA
Environment_ID	The key joining the environment enum	COUNT		cluster_capacity_by_date	INT32

Actions

Column Name	description	aggregation	synonyms	source	type
Datetime (UTC)	The timestamp	NONE	timestamp	datetime_dim	DATE_TIME
Related Entity Name	The name of the primary entity's related entity	NONE		entity_dim	VARCHAR
Category	The action category	NONE	action category	action_category_dim	VARCHAR
Latest Succeeded	Whether to filter on the latest succeeded actions	NONE		FORMULA	FORMULA
Projected Monthly Savings	The projected savings for the action over a month	SUM		action_savings_fact	DOUBLE
Resizeinfo Commodity	The commodity type associated with a resize action	NONE		action_dim	VARCHAR
State	The latest state of the action	NONE		latest_action_state_dim	VARCHAR
Tag Value	The key of the latest tags of the action's target/ primary entity	NONE		tag_dim	VARCHAR
Desired Size	The after capacity of a resize action	AVERAGE		action_dim	FORMULA
Is Primary Entity (deprecated)	[Deprecated]	NONE		action_related_fact	BOOL
Related Entity Type	The type of the primary entity's related entity	NONE		entity_type_dim	VARCHAR
Type	The type of action	NONE	action type	action_type_dim	VARCHAR
Description	The text description of the action	NONE	action description	action_dim	VARCHAR
Savings Hourly	The projected hourly savings of the action	AVERAGE	projected savings hourly	action_savings_fact	DOUBLE
Tag Key	The key of the latest tags of the action's target/ primary entity	NONE		tag_dim	VARCHAR

Column Name	description	aggregation	synonyms	source	type
Entity Name	The entity name	NONE	primary entity name	action_dim	VARCHAR
Group Name	The group name	NONE		group_dim	VARCHAR
Execution Error	The error message associated with a failed action	NONE		action_dim	VARCHAR
Formatted Action Description	[Formula] Formatted description or explanation of action based on action type, category	NONE		FORMULA	VARCHAR
Severity	The severity of the issue that the associated action addresses	NONE	action severity	action_severity_dim	VARCHAR
Group Id	The group ID	COUNT		action_membership_fact	INT64
Deleted File Size (GB)	The file size associated with a delete action, in GB	SUM		action_dim	FORMULA
Formatted Action Type	[Formula] Formatted action type based on action type, target type	NONE		FORMULA	VARCHAR
Group Type	The type of group	NONE		group_type_dim	VARCHAR
Group Dynamic	True if the group definition is dynamic; false indicates a static group	NONE		group_dim	BOOL
Showback	[Formula] Resource change based on resize commodity type	AVERAGE		action_dim	VARCHAR
Latest Pending	Whether to filter on the latest pending actions	NONE		FORMULA	FORMULA
Deleteinfo Filepath	The file path associated with a delete action	NONE		action_dim	VARCHAR
Projected Monthly Investments	The projected investments for	SUM		action_savings_fact	DOUBLE

Column Name	description	aggregation	synonyms	source	type
	the action over a month				
Formatted Resize Commodity	[Formula] Formatted commodity for a resize action based on description	NONE		action_dim	VARCHAR
Environment	The environment type	NONE		entity_environment_dim	VARCHAR
Current Size	The before capacity of a resize action	AVERAGE		action_dim	FORMULA
Latest Accepted	Whether to filter on the latest accepted actions	NONE		FORMULA	FORMULA
Deleteinfo Filesize	The file size associated with a delete action	AVERAGE		action_dim	DOUBLE
Explanation	The condition that the action is addressing	NONE	action explanation	action_dim	VARCHAR
State Over Time	The state of the action, over time	NONE	action state	action_state_dim	VARCHAR
Mode	The action mode	NONE	action mode	action_mode_dim	VARCHAR
Id	The action ID	COUNT	action id, action oid	action_dim	INT64
Entity Type	The type of the entity that the action operates on	NONE	primary entity type	action_dim	VARCHAR
Resource Change	The difference between the old and new capacity for a resize action	AVERAGE		action_dim	DOUBLE
Labor Saved (Hours)	[Formula] Estimated labor cost based on action information	SUM		FORMULA	VARCHAR
Related Entity Id	The ID of the primary entity's related entity	NONE		action_related_entity_fact	INT64

Column Name	description	aggregation	synonyms	source	type
Related Entity Id (deprecated)	[Deprecated]	NONE		action_related_fact	INT64
Is Primary Entity	Whether the related entity refers to the action's targeted/ primary entity	NONE		action_related_entity_fact	BOOL
Average Idle Power	The average idle power of an entity, over the last 24 hours	AVERAGE		Average idle power per host for action	FORMULA
Unique Projected Monthly Savings	[Formula] Average projected monthly savings by id	SUM		Latest action state	INT64
Affected Metric State		NONE		Action affected metrics generic view	FORMULA
Affected Metric Type		NONE		Action affected metrics generic view	FORMULA
Used		AVERAGE		Action affected metrics generic view	FORMULA
Affected Metric Relation		NONE		Action affected metrics generic view	FORMULA
User Account	The user account that triggered an action execution	NONE		action_dim	VARCHAR

Entity metrics

Used to obtain metrics for an entity.

Column Name	description	aggregation	synonyms	source	type
Id	The entity ID	COUNT	entity id, entity oid	entity_dim	INT64
Name	The entity name	NONE	entity name	entity_dim	VARCHAR
Entity Type	The type of entity	NONE		entity_type_dim	VARCHAR
Environment	The environment type	NONE		entity_environment_dim	VARCHAR
Metric Type	The metric type, otherwise known as the commodity type	NONE		metric_type_dim	VARCHAR
Num Cpus	The number of cpu cores of an entity	AVERAGE		num_cpus_fact	INT32

Column Name	description	aggregation	synonyms	source	type
Related Id (deprecated)	[Deprecated]	COUNT	related entity id, related entity oid	related_fact	INT64
State	The state of the entity	NONE	entity state	entity_state_dim	VARCHAR
Datetime (UTC)	The timestamp	NONE	timestamp	datetime_dim	DATE_TIME
VM Name Filter	Lists names of all virtual machines	NONE		FORMULA	FORMULA
Utilization	The utilization of the commodity as a percent	AVERAGE		metric_fact	DOUBLE
Consumed	The amount of the commodity used by the consumer entity	AVERAGE		metric_fact	DOUBLE
Group Origin	The origin of the group	NONE		group_origin_dim	VARCHAR
Group Type	The type of group	NONE		group_type_dim	VARCHAR
Application Name Filter	Lists names of all applications	NONE		FORMULA	FORMULA
Group Name	The group name	NONE		group_dim	VARCHAR
Dynamic	True if the group definition is dynamic; false indicates a static group	NONE	group dynamic	group_dim	BOOL
Capacity	The capacity of the commodity	AVERAGE		FORMULA	FORMULA
Used	The average usage of the commodity for the entity	AVERAGE		FORMULA	FORMULA
Related Name	The name of the entity's related entity	NONE	related entity name	related_entity_view	VARCHAR
Related Type	The type of the entity's related entity	NONE	related entity type	related_entity_type_dim	VARCHAR
Cluster Name Filter	Lists names of all compute clusters	NONE		FORMULA	FORMULA
Entity Tag Key	The key of the tag referenced by entities	NONE		tag_dim	VARCHAR

Column Name	description	aggregation	synonyms	source	type
Entity Tag Value	The value of the tag referenced by entities	NONE		tag_dim	VARCHAR
Group Id	The group ID	COUNT		_membership_fact	INT64
Related Id	The ID of the entity's related entity	COUNT		related_entity_fact	INT64
Group Tag Key	The key of the tag referenced by groups	NONE		group_tag_dim	VARCHAR
Group Tag Value	The value of the tag referenced by groups	NONE		group_tag_dim	VARCHAR
Idle Power	The power of an idle entity	AVERAGE		idle_power_fact	DOUBLE

Workload controller resize action worksheet

Column Name	description	aggregation	synonyms	source	type
Namespace	The namespace name	NONE		workload_controller_attrs_dim	VARCHAR
Container Cluster	The associated container cluster of an action	NONE		workload_controller_attrs_dim	VARCHAR
Type	The type of the action	NONE		action_type_dim	VARCHAR
Severity	The severity of the issue that the associated action addresses	NONE		action_severity_dim	VARCHAR
Mode	The action mode	NONE		action_mode_dim	VARCHAR
Category	The action category	NONE		action_category_dim	VARCHAR
Execution Time	When an action execution was initiated, in UTC time zone	NONE		action_dim	DATE_TIME
User Account	The user account that triggered an	NONE		action_dim	VARCHAR

Column Name	description	aggregation	synonyms	source	type
	action execution				
Workload Controller Name	The workload controller name	NONE		action_dim	VARCHAR
Resizeinfo Target Type	The type of entity that a resize action operates on	NONE		action_dim	VARCHAR
Units	The unit measurement of commodity for a resize action	NONE		action_dim	VARCHAR
Container Spec Name	The associated container spec of an action	NONE		action_dim	VARCHAR
Creation Time	When the action was recommended, in UTC time zone	NONE	recommendation_time	action_dim	DATE_TIME
Description	The text description of the action	NONE		action_dim	VARCHAR
Explanation	The condition that the action is addressing	NONE		action_dim	VARCHAR
Target Type	The name of the entity that the action operates on	NONE		action_dim	VARCHAR
Resize Commodity	The commodity type associated with a resize action	NONE		action_dim	VARCHAR
Current Capacity	The before capacity of a resize action	AVERAGE		action_dim	DOUBLE
New Capacity	The new capacity size associated with a resize action	AVERAGE		action_dim	DOUBLE

Column Name	description	aggregation	synonyms	source	type
Replica Count	The replica count of the target entity	SUM		action_dim	INT32
Resource Change	The difference between the old and new capacity	SUM		FORMULA	FORMULA
Resource	[Formula] The commodity type of a resize action, reformatted	NONE		action_dim	VARCHAR
Change Memory to MB	[Formula] The capacity change divided by 1024 for mem resize actions	SUM		action_dim	VARCHAR
Total Change	[Formula] The capacity change value multiplied by the target replica count	SUM		action_dim	INT32
Total Change Memory to MB	[Formula] The capacity change value multiplied by the target replica count, divided by 1024 for mem resize actions	SUM		action_dim	VARCHAR
Total Change Mem and CPU Conversion	[Formula] The capacity change value multiplied by the target replica count, divided by 1024 for mem resize actions or divided by 1000 for cpu resize actions	SUM		action_dim	VARCHAR
Units to MB	[Formula] The units of a resize action, reformatted to	NONE		action_dim	VARCHAR

Column Name	description	aggregation	synonyms	source	type
	use MB instead of KB				
State	The latest state of the action	NONE		action_state_dim	VARCHAR
Latest	The latest timestamp for the associated relationship	NONE		_action_state_fact	INT32

Linking: entity metrics

This worksheet is designed to use the liveboard filter linking feature to discover metrics about related entities. when using this worksheet to search, assume that an entity metrics "entity" is equivalent to a "related entity" in this worksheet, and vice versa. to link the filters in a liveboard, add a filter on the primary worksheet using an "entity" type field and link to the equivalent "related entity" type field from this worksheet.

Column Name	description	aggregation	synonyms	source	type
Id	The entity ID	COUNT	entity id, entity oid	entity_dim	INT64
Entity Type	The type of entity	NONE		entity_type_dim	VARCHAR
Environment	The environment type	NONE		entity_environment_dim	VARCHAR
Metric Type	The metric type, otherwise known as the commodity type	NONE		metric_type_dim	VARCHAR
Num Cpus	The number of cpu cores of an entity	AVERAGE		num_cpus_fact	INT32
Related Id (Deprecated)	[Deprecated]	COUNT	related entity id, related entity oid	related_fact	INT64
State	The state of the entity	NONE	entity state	entity_state_dim	VARCHAR
Datetime (UTC)	The timestamp	NONE	timestamp	datetime_dim	DATE_TIME
Utilization	The utilization of the commodity as a percent	AVERAGE		metric_fact	DOUBLE
Consumed	The amount of the commodity used by the consumer entity	AVERAGE		metric_fact	DOUBLE
Name	The entity name	NONE	entity name	entity_dim	VARCHAR

Column Name	description	aggregation	synonyms	source	type
Capacity	The capacity of the commodity	AVERAGE		FORMULA	FORMULA
Used	The average usage of the commodity for the entity	AVERAGE		FORMULA	FORMULA
Related Name	The name of the entity's related entity	NONE	related entity name	related_entity_view	VARCHAR
Related Type	The type of the entity's related entity	NONE	related entity type	related_entity_type_dim	VARCHAR
Related Id	The ID of the entity's related entity	COUNT		related_entity_fact	INT64

Metadata worksheet

Column Name	description	aggregation	synonyms	source	type
Datetime (UTC)	The timestamp	NONE		datetime_dim	DATE_TIME
Entity Id	The entity ID	COUNT		entity_dim	INT64
Action Id	The action ID	COUNT	actionid, actionoid	action_dim	INT64
Release Version	[Internal Use - Do not edit]	NONE		FORMULA	FORMULA

Cloud realized savings

Column Name	description	aggregation	synonyms	source	type
Id	The cloud savings ID	NONE		cloudsavings_dim	INT64
Action Initiator Type	The type of action initiation referenced by cloud savings	NONE		action_initiator_dim	VARCHAR
Action Type	The type of action	NONE		action_type_dim	VARCHAR
Currency	The currency of cloud savings as an integer (ISO Code)	NONE		cloudsavings_dim	INT32
Realized Savings	The realized savings daily	SUM		cloudsavings_fact	DOUBLE

Column Name	description	aggregation	synonyms	source	type
Realized Investments	The realized investments daily	SUM		cloudsavings_fact	DOUBLE
Service Provider Name	The service provider name	NONE		cloudsavings_provider_view	VARCHAR
Account Id	The account ID	NONE		cloudsavings_dim	INT64
Region Id	The region ID	NONE		cloudsavings_dim	INT64
Cloud Service Id	The cloud service ID	NONE		cloudsavings_dim	INT64
Availability Zone Id	The availability zone ID	NONE		cloudsavings_dim	INT64
Service Provider Id	The service provider ID	NONE		cloudsavings_dim	INT64
Resource Group Id	The resource group ID	NONE		cloudsavings_dim	INT64
Billing Family Id	The billing family ID	NONE		cloudsavings_dim	INT64
Region Name	The region name	NONE		cloudsavings_region_view	VARCHAR
Resource Group Name	The resource group name	NONE		cloudsavings_resourcegroup_view	VARCHAR
Cloud Service Name	The cloud service name	NONE		cloudsavings_service_view	VARCHAR
Availability Zone Name	The availability zone name	NONE		cloudsavings_zone_view	VARCHAR
Datetime (UTC)	The timestamp	NONE		datetime_dim	DATE_TIME
Entity Id	The entity ID	NONE		cloudsavings_dim	INT64
Entity Type	The type of entity	NONE		entity_type_dim	VARCHAR
Entity Name	The entity name	NONE		entity_dim	VARCHAR
Action Chain Id	The action chain ID	NONE		cloudsavings_dim	INT64
Entity Tag Key	The key of the tag referenced by entities	NONE		tag_dim	VARCHAR

Column Name	description	aggregation	synonyms	source	type
Entity Tag Value	The value of the tag referenced by entities	NONE		tag_dim	VARCHAR
Related Entity Id	The ID of the entity's related entity	COUNT		entity_dim	INT64
Account Name	The account name	NONE		cloudsavings_account_view	VARCHAR

Cloud entities

Used to obtain cloud entities

Column Name	description	aggregation	synonyms	source	type
Id	The entity ID	COUNT	entity id, entity oid	entity_dim	INT64
Name	The entity name	NONE	entity name	entity_dim	VARCHAR
Entity Type	The type of entity	NONE		entity_type_dim	VARCHAR
Environment	The environment type, ie. CLOUD	NONE		entity_environment_dim	VARCHAR
State	The state of the entity	NONE	entity state	entity_state_dim	VARCHAR
Datetime (UTC)	The timestamp	NONE	timestamp	datetime_dim	DATE_TIME
Related Name	The name of the entity's related entity	NONE	related entity name	related_entity_view	VARCHAR
Related Type	The type of the entity's related entity	NONE	related entity type	related_entity_type_dim	VARCHAR
Tag Key	The key of the tag referenced by entities	NONE		tag_dim	VARCHAR
Tag Value	The value of the tag referenced by entities	NONE		tag_dim	VARCHAR

Chapter 15. Integration - Embedded reporting

The Turbonomic platform includes an embedded reporting component that you can choose to enable when you install the platform. Embedded reporting stores a history of your managed environment and then presents selective snapshots of this history with a set of standard dashboards and reports. You can also create your own dashboards and reports to focus on other areas of interest.

Dashboards and charts in Turbonomic embedded reporting are powered by the Grafana observability platform. With Grafana, it's easy to navigate the existing dashboards, and to make your own charts and dashboards with no coding required. You can also create custom reports via SQL queries against the embedded reporting database. For more information about creating custom reports, see Custom reports.

The embedded reporting feature uses a PostgreSQL database with the TimescaleDB extension installed to manage the reporting data. The OVA-based installation of Turbonomic includes both a PostgreSQL database server and TimescaleDB extension. Alternatively, you can deploy a containerized or an external PostgreSQL database with the TimescaleDB extension.

For instructions to enable embedded reporting, see Enabling embedded reporting.

Enabling embedded reporting

Turbonomic includes an embedded reporting component that you can choose to enable when you install the platform. Use embedded reporting to understand application resource management trends and to share insights with stakeholders through reports and dashboards.

Embedded reporting runs as its own component, as part of the Turbonomic platform. This architecture enhances performance and reduces storage requirements. It stores a history of your managed environment and then presents selective snapshots of this history through a set of standard dashboards and reports. You can create your own dashboards and reports to focus on other areas of concern.

Embedded reports use TimescaleDB, which is an extension that is installed on a PostgreSQL database sever . These instructions are for the default configuration of TimescaleDB. As an alternative, you can configure an external deployment of TimescaleDB to manage your embedded reports data. For more information, see Configuring an External TimescaleDB.

The method that you use to enable embedded reports depends on the version status of your Turbonomic instance. Two options are available:

- Run the enable_reporting.py script

  You can use this option if you installed Turbonomic as a new VM image (OVA or VHD; see Installing on a Virtual Machine Image).
- Edit the `charts_v1alpha1_xl_cr.yaml` file.

  You can use this option if you installed Turbonomic as a Kubernetes cluster, not using the VM image for your Turbonomic installation. For more information, see “Enabling embedded reporting in Kubernetes and Red Hat OpenShift Container Platform” on page 1990

  You can also use this option if you installed Turbonomic as a new VM image. For more information, see Editing the Turbonomic charts_v1alpha1_xl_cr.yaml file

Using the script interface

If you installed Turbonomic as a VM image, the script to enable embedded reports is already installed on your installation at `/opt/local/bin/enable_reporting.py`.

1. Open an SSH terminal session to your Turbonomic instance.

   Log in with the System Administrator that you set up when you installed Turbonomic:

- Username: `turbo`
- Password: `[your_private_password]`

2. Go to the script directory.

```
cd /opt/local/bin
```

3. Run the script.

```
./enable_reporting.py
```

The script prompts you for two passwords:

- Grafana admin password

  This password enables access to Grafana from external URLs and also from the extractor component that feeds data to Grafana.

  Do not use special characters.

  Important:

  This step is the only time that you should change the Grafana Admin password.

- Grafana database password

  This password enables communication between Grafana and the PostgreSQL database that stores the reporting data.

After you supply the passwords, the script displays a confirmation message similar to the following example:

```
Succesfully applied new changes to /opt/turbonomic/kubernetes/operator/deploy/crds/
charts_v1alpha1_xl_cr.yaml.
Backup written to /opt/turbonomic/kubernetes/operator/deploy/crds/
charts_v1alpha1_xl_cr.yaml.bak
```

This indicates that the script successfully updated the Turbonomic configuration. The script then applies the changed configuration to enable the embedded reports feature. It should display messages similar to the following example:

```
Applying CR file /opt/turbonomic/kubernetes/operator/deploy/crds/charts_v1alpha1_xl_cr.yaml
Warning: kubectl apply should be used on resource created by either
kubectl create --save-config or kubectl apply
xl.charts.helm.k8s.io/xl-release configured
Waiting for changes to take effect...
Restarting api pod to apply configuration changes.
pod "api-65cf47986f-jxszd" deleted
Changes have been successfully applied. Embedded reporting is now enabled.
```

4. Verify your installation.

```
./enable_reporting.py --validate
```

If embedded reports are successfully enabled, the script output should be similar to the following example:

```
No obvious embedded reporting installation errors detected.
```

Editing the Turbonomic YAML file

These instructions describe how to locate and edit the `charts_v1alpha1_xl_cr.yaml` for the VM image installation. If you installed on a Kubernetes node cluster, the Turbonomic configuration file is likely in a different location.

To enable embedded reports, complete the following tasks:

- Enable the components that implement the embedded reporting.

- Restart the API pod.
- Double-check the installation.
- Optionally enable email subscriptions.

Enable the Grafana Exporter, TimescaleDB, and data extraction processes by editing the `charts_v1alpha1_xl_cr.yaml` file.

1. Open an SSH terminal session to your Turbonomic instance.

   Log in with the System Administrator that you set up when you installed Turbonomic:

   - Username: `turbo`
   - Password: `[your_private_password]`
2. Open the following file in a text editor:

   ```
   /opt/turbonomic/kubernetes/operator/deploy/crds/charts_v1alpha1_xl_cr.yaml
   ```
3. Specify the IP address of the Turbonomic instance for external access to the embedded reporting database.

   In the `global:` section of the file, add the following line, where *<Platform_IP>* is the IP address of your instance:

   ```
   global:
     externalTimescaleDBIP: <Platform_IP>
   ```

   If you are using a cloud-based TimescaleDB server, you can alternatively specify your instance by its fully qualified domain name (FQDN):

   ```
   global:
     externalTimescaleDBName: <FQDN>
   ```

   You can also customize port for the TimescaleDB instance by specifying a port value for `externalTimescaleDBPort`.

   ```
   global:
     externalTimescaleDBName: <FQDN>
     externalTimescaleDBPort: 1143
   ```

   Note: If you intend to use an instance with a custom DB root user, you must also specify the PostgreSQL default root user credentials, as shown in the following example:

   ```
     properties:
       global:
         dbRootUsername: admin
         dbRootPassword: MyComplexPass
         dbs.postgresDefault:
           rootUserName: postgres
           rootPassword: <The PostgreSQL user's password. The default password in OVA is
   vmturbo.>
   ```
4. Enable the Grafana component.

   Find the `grafana:` section in the `crds/charts_v1alpha1_xl_cr.yaml` file, and uncomment the line, `enabled: true`.
5. Enable `postgres` as the database type.

   Enabling `postgres` sets persistent storage of historical data for embedded reports. In the `grafana:` section, find the subsection for `grafana.ini: database:` and uncomment the line, `type: postgres`.

   The changes that you have made so far should be similar to:

   ```
   global:
       externalTimescaleDBIP: <Platform_IP>
       ...
   ```

```
grafana:
    enabled: true
    adminPassword: admin
    grafana.ini:
      database:
        type: postgres
        ...
```

6. Change the admin and database passwords.

   It is good practice to change any passwords, and not keep their default values.

   Important:

   Use only alpha-numeric characters for these passwords.

   These passwords enable communication between the various embedded reports components. Some of the components accept only alpha-numeric characters. If you use special characters, then the components cannot communicate. Further, the steps to correct these passwords require assistance from your Support representative.

   To set the passwords:

   - Set the Grafana admin password.

     This password enables access to Grafana from external URLs and also from the extractor component that feeds data to Grafana. In the `grafana:` section, change the value of `adminPassword`.

     Do not use special characters.

     Assume that your password is `MyNewGrafanaPassword`. Then, you would set `adminPassword:` **`MyNewGrafanaPassword`**

     Important:

     This step is the only time that you should change the Grafana Admin password.

   - Set the Grafana database password.

     This password enables communication between Grafana and the PostgreSQL database that stores the reporting data. In the `grafana:` section, find the subsection for `grafana.ini: database: password:` and change the password value.

     ```
     grafana:
         enabled: true
         adminPassword: MyNewGrafanaPassword
         grafana.ini:
           database:
             type: postgres
             password: MyNewDatabasePassword
     ```

7. Enable the reporting processes.

   Just after the `properties:` section that you added, and at the same level to it, add the following entries to enable the reporting processes:

   ```
     reporting:
       enabled: true
     timescaledb:
       enabled: true
   ```

   It is important that you align these entries with the indentation for the `grafana:` section and the `properties:` section. The changes that you have made should now be similar to the following example:

   ```
   global:
       externalTimescaleDBIP: <Platform_IP>
       ...
   ```

```
grafana:
    enabled: true
    adminPassword: MyNewGrafanaPassword
    grafana.ini:
      database:
        type: postgres
        password: MyNewDatabasePassword

properties:
    extractor:
        grafanaAdminPassword: MyNewGrafanaPassword

reporting:
    enabled: true
timescaledb:
    enabled: true
```

Note:

The extractor component is enabled by default and no explicit definition is required in your `cr.yaml` file. However, if the extractor component was manually disabled in your environment previously, you can re-enable it. Search for the following extractor entry in the `cr.yaml` file:

```
extractor:
  enabled: false
```

Change the entry to `true`.

8. When you are done editing the `charts_v1alpha1_xl_cr.yaml` file, save and apply your changes.
   - Save your changes and quit the text editor.
   - Run the command to apply the changes.

     ```
     kubectl apply -f \
       /opt/turbonomic/kubernetes/operator/deploy/crds/charts_v1alpha1_xl_cr.yaml
     ```

   - Delete the `api` pod. If extractor was previously running, delete the `extractor` pod.

     Deleting these pods triggers them to restart, which loads the changes you made.

     Run the following command to get the full pod names, run the command,

     ```
     kubectl get pods -n turbonomic
     ```

     Find the entries for the pods that begin with `api` and `extractor`. For example, assume that the entries are:

     ```
     ...
     api-7887c66f4b-shndq                          1/1     Running   0
     ...
     extractor-5b86976bc8-vxwz4                    1/1     Running   0
     ...
     ```

     Run the following commands:

     ```
     kubectl delete pod -n turbonomic api-7887c66f4b-shndq
     ```

     ```
     kubectl delete pod -n turbonomic extractor-5b86976bc8-vxwz4
     ```

9. In an SSH terminal session for your Turbonomic instance, set the password for PostgreSQL and enable access to psql on the appliance's local host. Run the following commands:

   ```
   sudo -iu postgres psql -c "ALTER ROLE postgres WITH PASSWORD 'vmturbo'"
   ```

   ```
   sudo -iu postgres psql -c "CREATE ROLE turbo WITH SUPERUSER CREATEDB CREATEROLE LOGIN
   REPLICATION BYPASSRLS PASSWORD 'vmturbo'"
   ```

10. Verify your installation.

- Verify that the embedded reports pods are running.

```
kubectl get pods -n turbonomic
```

The output is similar to the following example:

```
NAME                                        READY   STATUS    RESTARTS
extractor-7759dbcb47-vs6hr                  1/1     Running   0
grafana-84ccb4bfb-l7sp7                     1/1     Running   0
```

- Verify that PostgreSQL is running.

The PostgreSQL database should be running as a daemon on the Turbonomic server machine. To check the status, run the command:

```
sudo systemctl status postgresql-15.service
```

In versions prior to Turbonomic 8.14.2, the supported PostgreSQL version was 12.x. If you are using an older version of PostgreSQL, then update to PostgreSQL version 15. For more information, see Upgrade PostgreSQL.

Run the `sudo systemctl status postgresql-15.service` command.

The output is similar to the following example:

```
[turbo@upg ~]$ sudo systemctl status postgresql-15.service
● postgresql-15.service - PostgreSQL 15 database server
     Loaded: loaded (/usr/lib/systemd/system/postgresql-15.service; enabled; preset:
disabled)
     Active: active (running) since Tue 2024-11-05 12:12:15 EST; 6 days ago
       Docs: https://www.postgresql.org/docs/15/static/
    Process: 61045 ExecStartPre=/usr/pgsql-15/bin/postgresql-15-check-db-dir ${PGDATA}
(code=exited, status=0/SUCCESS)
   Main PID: 61050 (postmaster)
      Tasks: 12 (limit: 209714)
     Memory: 1.3G
        CPU: 37min 29.975s
     CGroup: /system.slice/postgresql-15.service
             ├─53565 "postgres: grafana_backend grafana 10.11.12.13(52082) idle"
             ├─61050 /usr/pgsql-15/bin/postmaster -D /var/lib/pgsql/15/data/
             ├─61052 "postgres: logger "
             ├─61053 "postgres: checkpointer "
             ├─61054 "postgres: background writer "
             ├─61056 "postgres: walwriter "
             ├─61057 "postgres: autovacuum launcher "
             ├─61058 "postgres: TimescaleDB Background Worker Launcher "
             ├─61059 "postgres: logical replication launcher "
             ├─61148 "postgres: TimescaleDB Background Worker Scheduler "
             ├─61283 "postgres: TimescaleDB Background Worker Scheduler "
             └─66529 "postgres: grafana_backend grafana 10.11.12.13(50788) idle"
```

Enabling embedded reporting in Kubernetes and Red Hat OpenShift Container Platform

To enable embedded reporting in a native Kubernetes cluster, such as Azure Kubernetes Service, Amazon Elastic Kubernetes Service, or Red Hat OpenShift, update your Turbonomic `charts_v1alpha1_xl_cr.yaml` file to create a containerized `timescaledb` pod that runs the PostgreSQL database.

Note:

In Red Hat OpenShift, Grafana does not accept the global security context. Instead, set the Red Hat OpenShift security context (`scc`) at the group level. Assuming `turbonomic` is the project where Turbonomic is deployed, run the following command from the command line in the Red Hat OpenShift cluster:

```
oc adm policy add-scc-to-group anyuid system:serviceaccounts:turbonomic
```

If Turbonomic is deployed in a different project or namespace, replace `turbonomic` with your project or namespace. Although setting `scc` at the group level is not considered a best practice, this is the only option to enable Grafana in Red Hat OpenShift.

1. First, enable only the Timescaledb pod to ensure that the reporting database comes up first before you enable any of the other reporting components.

   Add the following lines to your Turbonomic `charts_v1alpha1_xl_cr.yaml` file:

   ```
   timescaledb:
     enabled: true
   postgresql:
     postgresqlPassword: <your_password>
     persistence:
       size: 500Gi
   ```

   a. Optionally, to set resource requests and limits, add lines similar to the following example to the `postgresql` section:

   ```
   postgresql:
     persistence:
       size: 500Gi
   ### Optional to set resource requests and limits
       resources:
         requests:
           memory: 1Gi
   ```

   b. **Red Hat OpenShift only:** Add the following lines to enable the security context.

   ```
       volumePermissions:
         enabled: true
   ```

2. Apply the updated `charts_v1alpha1_xl_cr.yaml` file and wait for the `timescaledb` pod to be ready.

   a. Confirm that the `timescaledb` pod is ready in the user interface (UI) for your container platform or from the command line.

   b. Confirm that the PostreSQL database is ready by checking that the logs for the `timescaledb` pod include a `database system is ready to accept connections` message.

3. Next, enable the extractor properties, reporting, and Grafana reporting components.

   Add the following lines to your `charts_v1alpha1_xl_cr.yaml` file:

   ```
   properties:
     global:
       dbs.postgresDefault:
         rootPassword: <same password as postgresqlPassword field.>
     extractor:
       grafanaAdminPassword: [your_private_password]
   grafana:
     adminPassword: [your_private_password]
     enabled: true
     grafana.ini:
       database:
         password: [your_private_password]
         type: postgres
   reporting:
     enabled: true
   ```

4. Apply the updated `charts_v1alpha1_xl_cr.yaml` file.

   The following example shows a sample configuration for the Turbonomic `charts_v1alpha1_xl_cr.yaml` with all the reporting components enabled and the Red Hat OpenShift security context requirements configured. The optional lines to set resource requests and limits are specified in the `postgresql` section:

   ```
   #default for OCP apiVersion: charts.helm.k8s.io/v1 kind: Xl metadata:  name: xl-release spec:
   #Red Hat OpenShift only
   #obtain uid from project, using annotation openshift.io/sa.scc.uid-range or scc.supplemental-
   groups
   ```

```
#for embedded reporting use this uid as the value for securityContext: in the following
sections
securityContext:
  fsGroup: 1000760000
### Red Hat OpenShift only - if needed
nginx:
  nginxIsPrimaryIngress: false
  httpsRedirect: false
### End of Red Hat OpenShift only section ##
properties:
  global:
    dbs.postgresDefault:
      rootPassword: <same password as postgresqlPassword field.>
  extractor:
    grafanaAdminPassword: [your_private_password]
grafana:
  adminPassword: [your_private_password]
  enabled: true
  grafana.ini:
    database:
      password: [your_private_password]
      type: postgres
reporting:
  enabled: true
timescaledb:
  enabled: true
postgresql:
  postgresqlPassword: <your_password>
  persistence:
    size: 500Gi
### Optional to set resource requests and limits
    resources:
      requests:
        memory: 1Gi
### End of optional section
### Red Hat OpenShift only
  securityContext:
    runAsUser: 1000760000
  volumePermissions:
    enabled: true
    securityContext:
      runAsUser: 1000760000
```

5. To enable the **Embedded Reporting** link in the Turbonomic UI, delete the API pod.

   From your container platform UI or the command line, find the pod that begins with `api-`, for example `api-7887c66f4b-shndq`, and delete it.

   Run the following command to get the full pod names, run the command,

   ```
   kubectl get pods -n turbonomic
   ```

   Find the pod that begins with `api`. For example, assume that the entry is:

   ```
   ...
   api-7887c66f4b-shndq                          1/1     Running   0
   ...
   ```

   Run the following command:

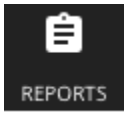
   ```
   kubectl delete pod -n turbonomic api-7887c66f4b-shndq
   ```

6. In the Turbonomic UI, click **Reports** in the main navigation menu.

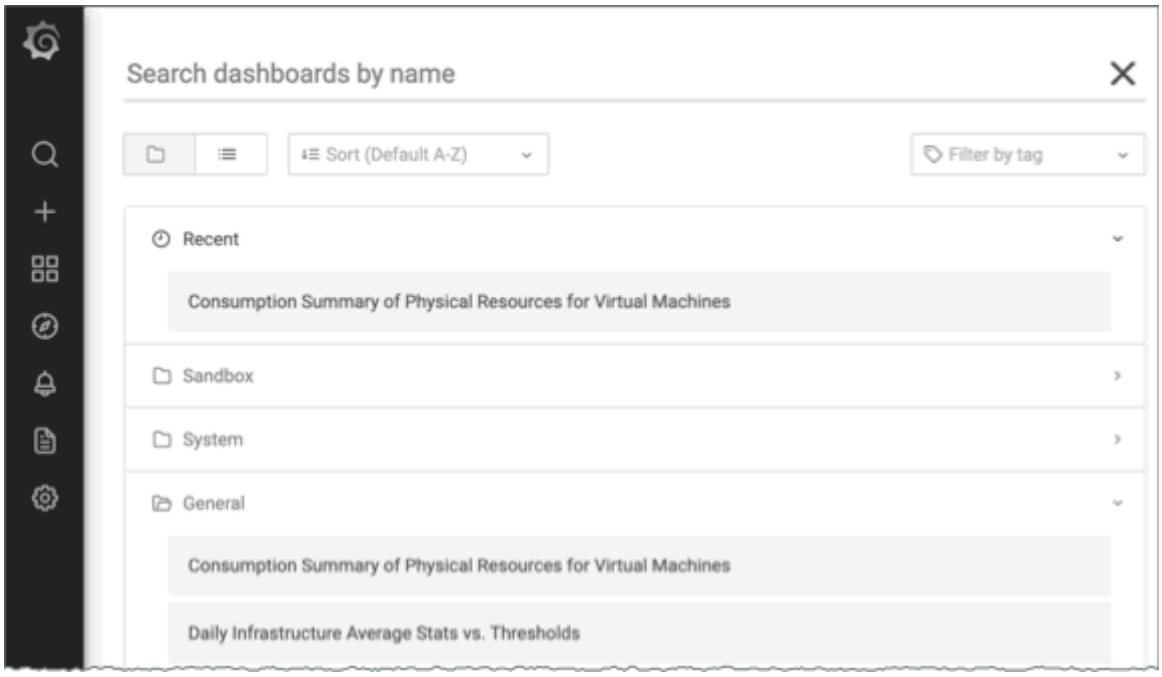
   You are redirected to the Grafana reports UI.

Navigating to the Embedded Reports page

After you enable embedded reports, open the Turbonomic user interface and click **Reports** in the navigation bar.



Dashboards and charts open in a new browser tab.



You can search for specific dashboards or browse folders to find the dashboards you want. You can also create custom reports.

Dashboards and charts are powered by the Grafana® observability platform. With Grafana, you can navigate the existing dashboards, and to make your own charts and dashboards with no coding required. If you are new to Grafana and need help with getting started, read the documentation available at:

https://grafana.com/docs/grafana/latest/

Note:

Turbonomic no longer has an agreement to use the Enterprise license of Grafana, and includes the open source Community license instead. Be aware that the Community license does not support PDF reports. Contact your Turbonomic representative for guidance on PDF reports.

Creating custom reports

To create custom reports, you must run SQL queries against the embedded reports database (`Turbo Timescale`).



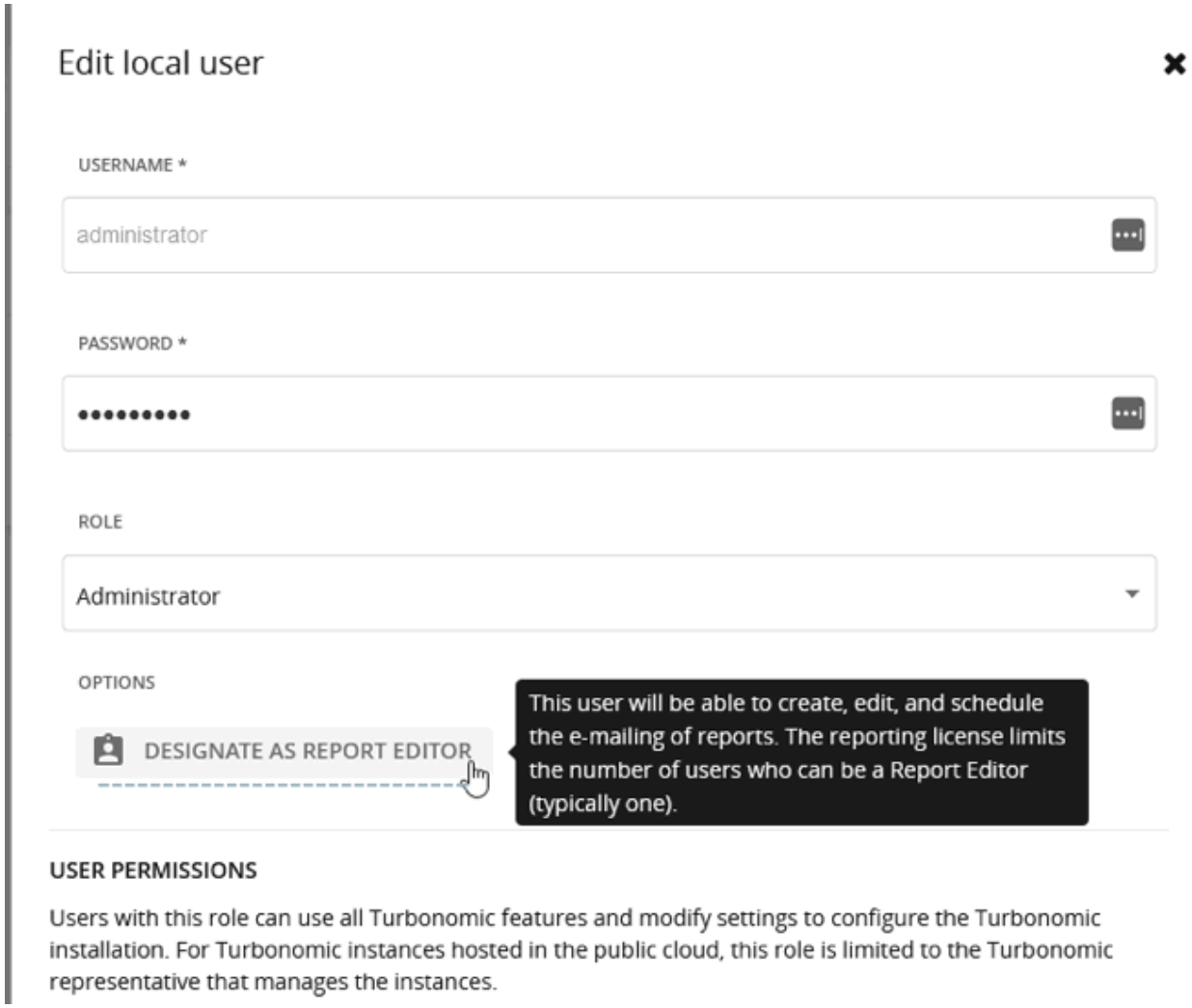
The database schema includes certain tables against which you can run queries. For more information about the schema, see Integration - Embedded reporting.

(Optional) Report editing

To create and edit reports, a user must have Report Editor privileges. This user can have any role, as long as it is not a *shared* or *scoped* role. Only one user per Turbonomic instance is allowed to have Report Editor privileges. You can grant these privileges to another user.

To create a user account with Report Editor privileges:

1. In the Turbonomic user interface, navigate to **Settings / User Management**.
2. Choose the user account that you want to configure as a Report Editor. You can either edit an existing account or create a new one.
3. Choose a role for the user. The user can have any role, as long as it is not a *shared* or *scoped* role.
4. In the **Options** section, choose **DESIGNATE AS REPORT EDITOR**.



5. Set any other properties for the user account that you want, and save the user account.

   Note:

   It can take up to 30 minutes before the Reports page shows the Report Editor username. This usually occurs if you have changed the Report Editor multiple times.

   To access reports from the user interface, a user must have the Administrator or Site Administrator role, or a non-administrator role without a defined scope. For example, a user with the Observer role but without a scope can access reports.

   The default Shared Observer and Shared Adviser roles require scopes, so users with these roles cannot access reports.

Embedded reports storage requirement estimates

The embedded reports feature uses a TimescaleDB server to manage the chart data, which is a PostgreSQL server that runs with the TimescaleDB extension. Configure the datastore for your Turbonomic instance so it has enough space to support the TimescaleDB requirements.

When you initially enable embedded reports, estimate the storage that you need and configure the platform storage. If you already enabled embedded reports, check your current storage configuration and decide whether it meets your needs now and into the future.

The storage that your TimescaleDB requires depends on:

- Data retention period

How long to store the TimescaleDB data.

- The size of your environment

  The count of entities Turbonomic manages in your environment. This count changes over time. Think of it as the average number of entities in your environment over the data retention period.

Also, increased entity count increases the data requirement, as does other activity. Storage requirements can increase over time for reasons such as the following examples:

- You add entities such as workloads, application components, storage, or hosts to your environment.
- You configure new targets.

Storage estimates lookup tables

Turbonomic investigates the TimescaleDB storage requirements for different topologies and retention periods. The following table lists the estimates that were calculated. Your environment might have different requirements.

	Number of entities						
Retention period	**10k**	**25k**	**50k**	**100k**	**250k**	**500k**	**1000k**
6 months	36 GB	91 GB	182 GB	364 GB	910 GB	1.8 TB	3.6 TB
1 year	72 GB	181 GB	361 GB	723 GB	1.8 TB	3.5 TB	7 TB
2 years	144 GB	361 GB	721 GB	1.4 TB	3.5 TB	7.2 TB	14 TB

The default installation grants a disk quota of 200 GB to the TimescaleDB. For the default installation, the database can support the following entity counts:

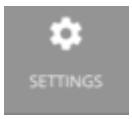
Retention period	**Entity count**
6 months	55k
1 year	27k
2 years	14k

Setting the data retention period

By default, Turbonomic sets the retention period for embedded reports to be 365 days. You can see the currently set retention period, and change it in the Turbonomic user interface.

To execute these actions, navigate to the Maintenance Options page:

1. Navigate to the Settings Page.

   Click to navigate to the Settings Page.
2. Choose Maintenance Options.

   Maintenance Options
3. Set the data retention period for embedded reports.

In the **Data Retention** group of controls, find the field for **Saved Reporting Data**, which displays the current data retention period for embedded reports, in days. The default is 365 days.

To change the retention period, enter a different number of days, and then click **Apply Settings**.

Increasing storage capacity for TimescaleDB

If you estimate the storage requirements for embedded reports after you install Turbonomic, you might learn that you need to increase the storage capacity that is available to the TimescaleDB.

The platform uses Logical Volume Management (LVM) to manage the VM disks. To increase database storage, add a disk to the VM, and then use it to extend the LVM logical volume, `/dev/turbo/var_lib_mysql`. This logical volume serves both the historical database and the embedded reports database. When you do that, you increase the quota for the TimescaleDB.

For more information, see Increasing Available Disk Space

Estimating entity count

To get a sense of entity count in your environment, search the Turbonomic log files for an INFO message that lists the number of entities for each discovery cycle. You can search for the string `INFO [Stages$BroadcastStage]`. The INFO string is similar to the following example:

```
topology-processor-6f6486df64-zf 2021-09-27 20:51:33,724 INFO [Stages$BroadcastStage] :
Successfully sent 1505 entities within topology...
```

This example shows that you have 1505 entities in the topology. Consider how your inventory changes over time. For example, you can check the entity count over time to see whether it increases regularly.

Alternatively, you can search the logs with the following command to see the counts and sizes of the entities in the full supply chain. This example assumes that Turbonomic is deployed in the `turbonomic` namespace. If your deployment is in a different namespace, edit the command to specify the correct namespace after the `-n` flag.

```
kubectl logs $(kubectl get pods -n turbonomic | grep 'rsyslog' | awk '{print $1}') | grep
broadcast | grep -i "successfully sent"
```

In Red Hat OpenShift environments, the corresponding command is similar to the following example.

```
oc logs $(oc get pods -n turbonomic | grep 'rsyslog' | awk '{print $1}') | grep broadcast |
grep -i "successfully sent"
```

Custom reports

Built-in and custom reports are displayed on the Grafana dashboard page. You can import reports from the Turbonomic open source site, or you can create your own using SQL Server queries into the data sets.

Importing reports

You can find a set of custom reports to import on the Turbonomic Open Source site at https://github.com/turbonomic/visualization/tree/main/embedded-visualization. This page lists different categories of reports that you can use to track your environment.

To import a custom report:

1. In the embedded reports page, create a folder to store your imported JSON files.

   In embedded reports page, navigate to **Dashboards / Browse**. on that page, click **New Folder** to create a folder for you imported JSON files.

   For example, create a folder named `{{ site.data.vars.Product_Short }}_Github_Reports`.

   After you create the folder, navigate back to the **Dashboards / Browse** page.

2. Navigate to the report you want to import.

   Go to https://github.com/turbonomic/visualization/tree/main/embedded-visualization and browse to find the report you want.

3. Copy the JSON file to the clipboard.

   In the report entry, navigate to the JSON file. In GitHub, display the file as **Raw JSON**, then select the JSON and copy it to the clipboard.

4. Import the JSON file into your Reports.

   In the embedded reports page, click **Import**, and then paste the JSON you copied into the **Import Via Panel JSON** field. Then click **Load**.

   To complete the import:

   - Give the name you want for the report.
   - Choose the folder that you just created to store the report.
   - If you want to share this report with other Grafana installations, make a note of the UID, or change it to a value you can remember.
   - Click **Import**.

The embedded reports page now displays your imported report.

Creating reports

To compose custom reports, your Turbonomic user account must have **Report Editor** permission. When you navigate from Turbonomic to the Grafana Reports page, in addition to browsing dashboards, you can create new dashboards. To create a custom report:

1. Create a dashboard.

   Navigate to the Grafana Dashboards page, then click the **+** icon in the menu bar to create a new dashboard.

2. Click to create a new panel.

   The page for the new panel includes a tabbed section with a **Query** tab.

3. In the **Query** tab, set the **Data Source** field to *Turbo Timescale*.

   This value sets up the panel to display data from the {{ site.data.vars.Product_Short }} Timescale database.

4. Edit the panel's query.

   The panel displays a default query. You can edit this query to change the data that the panel displays. At the end of the default query, click **Edit SQL**. This displays a field where you can specify the SQL that you want.

   For information about experimenting with different queries, see Exploring SQL Queries

   To help you build these queries, this document describes the schema tables, and provides some sample queries against them.

Exploring SQL queries

To work with SQL queries, your {{ site.data.vars.Product_Short }} user account must have **Report Editor** permission. To explore queries:

1. Navigate to the Explore page.

   The page displays with a sample query already in place

2. Choose to explore Turbo Timescale.

   Next to the page title, **Explore**, you can choose which database to work with. Choose the database `Turbo Timescale`.

If you installed an external TimescaleDB, be sure to choose th name that you gave to that database.

3. Edit the **Explore** query.

   The page displays a default query. You can edit this query to change the data that the page displays. At the end of the default query, click **Edit SQL**. This displays a field where you can specify the SQL that you want.

   To test your query, click **Run Query**. If your query includes data points over time, you can choose **Format As: Time Series**. For tabular data, choose **Format As: Table**.

   To help you build these queries, this document describes the schema tables, and provides some sample queries against them.

Schema tables for Embedded Reporting

The embedded reports data set includes the following Schema Tables to represent Turbonomic data.

action_group

Data Object	Type	Nullable	Primary	Description	Reference
id	int	False	False		
type	enum	False	False	The specific action type - Provision, Reconfigure, Scale, etc.	action_type
category	enum	False	False	The category of the given action.	action_category
state	enum	False	False	That action type.	action_state
mode	enum	False	False	The action mode - automated, manual, etc.	action_mode
risks	text[]	True	False	An array of strings to describe the risks that drive the action.	

cloud_service_cost

Data Object	Type	Nullable	Primary	Description	Reference
time	timestamp with time zone	False	True		
account_oid	bigint	False	True	The ID of the cloud account or subscription this service is associated with.	
cloud_service_oid	bigint	False	True	The ID of the cloud service.	
cost	real	False	False	The cost of the service.	

completed_action

Data Object	Type	Nullable	Primary	Description	Reference
recommendation_time	timestamp with time zone	False	False	When analysis recommended the action.	
acceptance_time	timestamp with time zone	False	False	For manual actions, when the action was accepted for execution.	

Data Object	Type	Nullable	Primary	Description	Reference
completion_time	timestamp with time zone	False	False	When the action completed. Can be for success or failure.	
action_oid	bigint	False	False	The ID of the action. Use this to get action details.	
type	enum	False	False	The action type.	action_type
severity	enum	False	False	The action severity.	severity
category	enum	False	False	The action category.	action_category
target_entity_id	bigint	False	False	The ID of the entity that the action affects. For example, for a VMem resize, the VM that is resized.	
involved_entities	bigint[]	True	False	An array of IDs for entities involved in the action. This includes the target entity.	
description	text	False	False	The text description of the action.	
savings	double precision	False	False	The cost savings this action results in. A negative savings indicates an investment.	
final_state	enum	False	False	Whether the action succeded or failed.	terminal_state
final_message	text	False	False	A message for the final state. If the action failed, this should contain an error message.	
attrs	jsonb	True	False	Additional (often type-specific) info for the action. For example, in resizes this will contain the commodity type and from/to amounts.	action.attrs

entity

Data Object	Type	Nullable	Primary	Description	Reference
oid	bigint	False	True	The entity ID.	
type	enum	False	False	The entity type.	entity_type
name	text	False	False	The current display name for the entity. While display names may change over time, this is always the most recent display name.	

Data Object	Type	Nullable	Primary	Description	Reference
environment	enum	True	False	Whether the entity is in an on-prem, cloud, or hybrid environment.	environment_type
attrs	jsonb	True	False	Additional data for the entity.	entity.attrs group.attrs
first_seen	timestamp with time zone	False	False	When the entity was first discovered in the topology.	
last_seen	timestamp with time zone	False	False	When the entity last existed in the topology. For an entity that still exists, the value is 9999-12-31T23:59:59.	

entity_cost

Data Object	Type	Nullable	Primary	Description	Reference
time	timestamp with time zone	False	False		
entity_oid	bigint	False	False	The entity ID.	
category	enum	False	False	The category of service cost - IP service, storage service, on-demand or RI compute, etc.	cost_category
source	enum	False	False	The type of billing data that provides the cost data.	cost_source
cost	real	False	False	The calculated cost.	

entity_savings

Data Object	Type	Nullable	Primary	Description	Reference
time	timestamp with time zone	False	True	For a realized saving or investment, when the action was executed. For a missed saving or investment, when the action was last recommended.	
entity_oid	bigint	False	True	The object ID for the entity that has seen a saving or investment.	
savings_type	enum	False	True	Whether the action resulted in an investment or saving, and whether the action executed or not.	savings_type
stats_value	real	False	False	The value of the investment or saving for the related action.	

entity_types_with_cost

Data Object	Type	Nullable	Primary	Description	Reference
type	enum	True	False	The entity type.	entity_type

file

Data Object	Type	Nullable	Primary	Description	Reference
volume_oid	bigint	False	True	If present, the ID of the volume that stores this file.	
path	text	False	True	The path to the file.	
type	enum	False	False	The file type - ISO, log file, snapshot, etc.	file_type
file_size_kb	bigint	True	False	The file size in kb.	
modification_time	timestamp with time zone	True	False	When the file was last modified.	
storage_oid	bigint	False	False	If present, the ID of the disk that stores this file.	
is_attached	boolean	False	False	Whether the file is attached or unattached.	
hash	bigint	False	False		

historical_entity_attrs

Data Object	Type	Nullable	Primary	Description	Reference
time	timestamp with time zone	False	False	When this attribute was recorded.	
entity_oid	bigint	False	False	The ID of the entity this attribute pertains to.	
type	enum	False	False	The attribute type.	attr_type
bool_value	boolean	True	False	If the attribute is boolean, the boolean value.	
int_value	int	True	False	If the attribute is an integer, the integer value. If the attribute is an enum, this is the OID of the enum in the pg_enum table (to guard against name or order changes). The int_value should be castable to the enum via a custom cast (e.g. int_value::entity_state = 'POWERED_ON').	
long_value	bigint	True	False	If the attribute is a long, the long value.	
double_value	double precision	True	False	If the attribute is a double, the double value.	
string_value	text	True	False	If the attribute is a string, the string value.	
int_arr_value	int[]	True	False	If the attribute is a list of ints, the list values. If the attribute is a list of enums, these are the OIDs of the	

Data Object	Type	Nullable	Primary	Description	Reference
				enums in the pg_enum table (to guard against name or order changes).	
long_arr_value	bigint[]	True	False	If the attribute is a list of longs, the long values.	
string_arr_value	text[]	True	False	If the attribute is a list of strings, the string values.	
double_arr_value	float8[]	True	False	If the attribute is a list of doubles, the double values.	
json_value	jsonb	True	False	If the attribute is a JSON object, the given object.	

metric

Data Object	Type	Nullable	Primary	Description	Reference
time	timestamp with time zone	False	False	When analysis discovered thIS metric. For headroom metrics, when the headroom plan was created.	
entity_oid	bigint	False	False	The ID of the entity this metric pertains to.	
type	enum	False	False	The metric type.	metric_type
provider_oid	bigint	True	False	The ID of the entity that provides this metric. For example, for VMem, the ID of the host that provides the memory.	
key	text	True	False	The commodity key. This value can be null if no key is present.	
current	double precision	True	False	Current utilization of commodity in provider entity.	
capacity	double precision	True	False	The capacity of the commodity in the provider entity.	
utilization	double precision	True	False	Overall utilization of the commodity in the provider entity.	
consumed	double precision	True	False	Amount of commodity currently used by consumer entity.	
peak_current	double precision	True	False	Peak amount of current used commodity in the provider entity.	
peak_consumed	double precision	True	False	Peak amount of commodity currently used by the consumer entity.	
entity_type	enum	False	False	The type of the consumer entity.	entity_type

pending_action

Data Object	Type	Nullable	Primary	Description	Reference
recommendation_time	timestamp with time zone	False	False	When analysis recommended this action.	
action_oid	bigint	False	False	The action ID. Use this to get more action details.	
type	enum	False	False	the action type.	action_type
severity	enum	False	False	The action severity	severity
category	enum	False	False	The action category	action_category
target_entity_id	bigint	False	False	The ID of the entity this action affects.	
involved_entities	bigint[]	True	False	An array of IDs for all the entities involved in the action. Includes the target entity.	
description	text	False	False	The text description of the action.	
savings	double precision	False	False	The cost savings this action results in. A negative savings indicates an investment.	
attrs	jsonb	True	False	Additional (often type-specific) info for the action. For example, in resizes this will contain the commodity type and from/to amounts.	action.attrs

pending_action_stats

Data Object	Type	Nullable	Primary	Description	Reference
time	timestamp with time zone	False	False		
scope_oid	bigint	False	False		
entity_type	enum	False	False		entity_type
environment_type	enum	False	False		environment_type
action_group	int	False	False		
prior_action_count	int	False	False		
cleared_action_count	int	False	False		
new_action_count	int	False	False		
involved_entity_count	int	False	False		
savings	real	False	False		
investments	real	False	False		

run_health_checks

Data Object	Type	Nullable	Primary	Description	Reference
check_name	text	True	False	The display name of the health check.	
passed	boolean	True	False	Whether the health check passed or failed.	
messages	text[]	True	False	The text description of the health check results.	

scope

Data Object	Type	Nullable	Primary	Description	Reference
seed_oid	bigint	False	True	The ID of the entity whose scope is the subject of the given record. For example, for a cluster-to-host relationship, the seed can be the cluster, and the scoped entity would be the host.	
scoped_oid	bigint	False	True	ID of the entity that is part of the seed scope.	
scoped_type	enum	False	False	Entity type for the scoped entity.	entity_type
start	timestamp with time zone	False	True	When this relationship first appeared in the topology. This covers the initial appearance, or an appearance after a gap (the relationship was temporarily broken).	
finish	timestamp with time zone	False	False	When this relationship last appeared in the topology. A value of 9999-12-31T23:59:59 means the relationship is not yet known to have terminated.	

scope_to

Data Object	Type	Nullable	Primary	Description	Reference
oid	bigint	True	False		
from_time	timestamp with time zone	True	False		
to_time	timestamp with time zone	True	False		
seed	bigint	True	False		

scope_to_types

Data Object	Type	Nullable	Primary	Description	Reference
oid	bigint	True	False		
from_time	timestamp with time zone	True	False		
to_time	timestamp with time zone	True	False		
seed	bigint	True	False		
e_type	enum	True	False		entity_type

topology_stats

Data Object	Type	Nullable	Primary	Description	Reference
time	timestamp with time zone	False	False		
attrs	jsonb	True	False		

wasted_file

Data Object	Type	Nullable	Primary	Description	Reference
path	text	True	False	The file path.	
file_size_kb	bigint	True	False	The file size.	
modification_time	timestamp with time zone	True	False	When the file was last modified.	
storage_oid	bigint	True	False	The ID of the storage that contains the file.	
storage_name	text	True	False	The name of the storage (denormalized for reporting).	

JSON data objects

The following JSON objects represent data about specific entities in your environment.

ActionCommodity

Data Object	Description	Type
capacity	The available capacity of the given commodity.	float
percentileUtilization	For entities and resources that support it, the calculated percentile of utilization.	Double
used	The amount of the resource currently in use.	float
utilization	The percentage of capacity in use.	float

ActionEntity

Data Object	Description	Type	Reference	Repeated
oid	The ID of the entity.	Long		
name	The entity name. This corresponds to the name that displays in the user interface.	String		
type	The entity type.	/enums/entity_type	entity_type	
attrs		JSON	entity.attrs	
newMetric		JSON	Commodity	True
newRelated		JSON	RelatedEntity	True

ActionImpactedCost

Data Object	Description	Type
onDemandCost	For on-demand billing, the entity cost.	Float
onDemandRate	For on-demand billing, the rate of cost.	Float
riCoveragePercentage	For RIs, the percentage of RI coverage.	Float

ActionImpactedCosts

Data Object	Description	Type	Reference
afterActions	A description of what the cost will be after the action executes.	/shared/ ActionImpactedCost	ActionImpactedCost
beforeActions	A description of the current cost, before the action executes.	/shared/ ActionImpactedCost	ActionImpactedCost

ActionImpactedEntity

Data Object	Description	Type	MapKeyType	Reference
affectedCosts	A description of the before and after costs for this entity.	/shared/ ActionImpactedCosts		ActionImpactedCosts
affectedMetrics	A description of the before and after metrics for this entity.	/shared/ ImpactedMetric	/enums/ metric_type	ImpactedMetric
name	The entity name.	String		
oid	The entity ID.	Long		
settings	For entities that support percentile calculations, the aggressiveness and observation period for this entity.	/shared/EntitySettings		EntitySettings
type	The entity type.	/enums/entity_type		entity_type
attrs		JSON		entity.attrs
newMetric		JSON	Commodity	True
newRelated		JSON	RelatedEntity	True

BuyRiInfo

Data Object	Description	Type	Reference
computeTier	The compute tier, or workload family for the action. The ActionEntity object will be for an entity of type COMPUTE_TIER.	/shared/ActionEntity	ActionEntity
count		Integer	

Data Object	Description	Type	Reference
masterAccount	The account that manages the RI billing.	/shared/ActionEntity	ActionEntity
region	The region that will host the RIs.	/shared/ActionEntity	ActionEntity
target	The cloud account or subscription that will manage the RIs.	/shared/ActionEntity	ActionEntity

Commodity

Data Object	Description	Type
capacity		Double
consumed		Double
current		Double
peakConsumed		Double
type		String
utilization		Double

CommodityChange

Data Object	Description	Type	Reference
attribute		String	
commodityType	The type of commodity the action will change - MEM, IO_THROUGHPUT, etc.	/enums/metric_type	metric_type
from	The commodity value before the action executes.	Float	
percentileChange	For entities and commodities that support percentile calculations, the before and after values for percentile utilization and settings.	/shared/ CommodityPercentileChange	CommodityPercentileChange
target	A description of the entity this action will change.	/shared/ActionEntity	ActionEntity
to	The commodity value after the action executes.	Float	
unit	The disp;layed units of measurement.	String	

CommodityPercentileChange

Data Object	Description	Type
after	The percentile utilization after the action executes	Double
aggressiveness	The aggressiveness setting used to calculate the percentile.	Integer
before	The percentile utilization before the action executes.	Double
observationPeriodDays	The observation period used to calculate the percentile.	Integer

DeleteInfo

Data Object	Description	Type	Format
filePath	The path to the file.	String	
fileSize	The size of the file.	Double	
lastModifiedTimestamp	When the file was last modified.	String	yyyy-MM-dd'T'HH:mm:ss'Z'
unattachedDays	How many days the file has been unattached.	Integer	
unit	The displayed unit of measure for the file size.	String	

entity.attrs

Data Object	Description	Type	SupportedEntityTypes	Repeated	Reference
attachment_state	Whether a given volume is attached or unattached.	/enums/AttachmentState	VIRTUAL_VOLUME	False	AttachmentState
connected_networks	A list of network names that a given VM is connected to.	MULTI_TEXT	VIRTUAL_MACHINE	True	
cpu_model	For a physical host, the model of CPU on that host.	TEXT	PHYSICAL_MACHINE	False	
guest_os_name	The name of the OS running on a given VM.	TEXT	VIRTUAL_MACHINE	False	
guest_os_type	The specific type for the OS running on a given VM.	/enums/OSType	VIRTUAL_MACHINE	False	OSType
is_local	Whether a storage is local to the machine, or networked.	BOOLEAN	STORAGE	False	

Data Object	Description	Type	SupportedEntityTypes	Repeated	Reference
model	The specific model of the given physical host.	TEXT	PHYSICAL_MACHINE	False	
num_cpus	The number of CPUs on the given physical host.	NUMBER	PHYSICAL_MACHINE	False	
tags	A list of tags that are assigned to the given entity.	TEXT_MULTIMAP (Embedded Reporting) or List (Data Exporter)	ALL	False	
targets	The target that manages the given entity.	/shared/Target	ALL	True	Target
timezone	The timezone that is set for the clock on the physical machine.	TEXT	PHYSICAL_MACHINE	False	
vendor_id		TEXT	VIRTUAL_VOLUME,BUSINESS_ACCOUNT		

EntitySettings

Data Object	Description	Type
percentileAggressiveness	The current aggressiveness setting.	Integer
percentileObservationPeriodDays	The current observation period.	Integer

group.attrs

Data Object	Description	Type	Repeated	Reference	SupportedEntityTypes	SupportedGroupTypes
dynamic	Whether this group is static or dynamic. true for dynamic.	BOOLEAN	False		STORAGE_CLUSTER COMPUTE_CLUSTER GROUP K8S_CLUSTER BILLING_FAMILY RESOURCE_GROUP	RESOURCE_GROUP COMPUTE_CLUSTER BILLING_FAMILY GROUP K8S_CLUSTER STORAGE_CLUSTER

Data Object	Description	Type	Repeated	Reference	SupportedEntityTypes	SupportedGroupTypes
member_types		MULTI_TEXT	True		STORAGE_CLUSTER COMPUTE_CLUSTER GROUP K8S_CLUSTER BILLING_FAMILY RESOURCE_GROUP	RESOURCE_GROUP COMPUTE_CLUSTER BILLING_FAMILY GROUP K8S_CLUSTER STORAGE_CLUSTER
origin		TEXT	False		STORAGE_CLUSTER COMPUTE_CLUSTER GROUP K8S_CLUSTER BILLING_FAMILY RESOURCE_GROUP	RESOURCE_GROUP COMPUTE_CLUSTER BILLING_FAMILY GROUP K8S_CLUSTER STORAGE_CLUSTER
tags		TEXT_MULTIMAP (Embedded Reporting) or List (Data Exporter)	False		GROUP RESOURCE_GROUP COMPUTE_CLUSTER K8S_CLUSTER STORAGE_CLUSTER BILLING_FAMILY	ALL

Data Object	Description	Type	Repeated	Reference	SupportedEntityTypes	SupportedGroupTypes
targets		/shared/Target	False	Target	GROUP RESOURCE_GROUP COMPUTE_CLUSTER K8S_CLUSTER STORAGE_CLUSTER BILLING_FAMILY	ALL

ImpactedMetric

Data Object	Description	Type	Reference
afterActions	A description of what the metric utilization will be after the action executes.	/shared/ActionCommodity	ActionCommodity
beforeActions	A description of the current utilization, before the action executes.	/shared/ActionCommodity	ActionCommodity

MoveChange

Data Object	Description	Type	Repeated	Reference
from	Where the entity will move from.	/shared/ActionImpactedEntity	False	ActionImpactedEntity
resource		/shared/ActionEntity	True	ActionEntity
to	Where the entity will move to.	/shared/ActionImpactedEntity	False	ActionImpactedEntity

RelatedEntity

Data Object	Description	Type
name		String
oid		Long
type		String
relatedEntityIdentifier		String

Target

Data Object	Description	Type
category		String

Data Object	Description	Type
entityVendorId		String
name		String
oid		Long
type		String

action.attrs

JSON Object	Description	Type	MapKeyType	Reference
buyRiInfo	For an action to buy RIs, additional information about the RIs.	/shared/BuyRiInfo		BuyRiInfo
deleteInfo	For an action to delete a file, a description of that file.	/shared/DeleteInfo		DeleteInfo
moveInfo	For a move action, a description of from and to for the move.	/shared/MoveChange	/enums/ entity_type	MoveChange
resizeInfo	For a resize action, a description of the commodity changes.	/shared/ CommodityChange	/enums/ metric_type	CommodityChange
scaleInfo	For scaling a cloud VM, a description of the from and to for the VM move.	/shared/MoveChange	/enums/ entity_type	MoveChange
target	Extra information about the entity this action impacts.	/shared/ ActionImpactedEntity		ActionImpactedEntity

Enums

These enums specify values for certain states or types.

AttachmentState

- ATTACHED:
- UNATTACHED:

OSType

- LINUX:
- LINUX_WITH_SQL_ENTERPRISE:
- LINUX_WITH_SQL_STANDARD:
- LINUX_WITH_SQL_WEB:
- RHEL:
- SUSE:

- UBUNTU_PRO:
- UNKNOWN_OS:
- WINDOWS:
- WINDOWS_BYOL:
- WINDOWS_WITH_SQL_ENTERPRISE:
- WINDOWS_WITH_SQL_STANDARD:
- WINDOWS_WITH_SQL_WEB:

action_category

- COMPLIANCE:
- EFFICIENCY_IMPROVEMENT:
- PERFORMANCE_ASSURANCE:
- PREVENTION:
- UNKNOWN:
- SAVING:

action_mode

- AUTOMATIC:
- DISABLED:
- EXTERNAL_APPROVAL:
- MANUAL:
- RECOMMEND:

action_state

- ACCEPTED:
- CLEARED:
- FAILED:
- FAILING:
- IN_PROGRESS:
- POST_IN_PROGRESS:
- PRE_IN_PROGRESS:
- QUEUED:
- READY:
- REJECTED:
- SUCCEEDED:
- BEFORE_EXEC:
- IN_PROGRESS_NON_CRITICAL:

action_type

- ACTIVATE:
- ALLOCATE:
- BUY_RI:
- DEACTIVATE:
- DELETE:

- MOVE:
- NONE:
- PROVISION:
- RECONFIGURE:
- RESIZE:
- SCALE:
- START:
- SUSPEND:

attr_type

- ENTITY_STATE:
- NUM_VCPU:
- VOLUME_ATTACHED:
- IDLE_POWER:

cost_category

- IP:
- ON_DEMAND_COMPUTE:
- ON_DEMAND_LICENSE:
- RESERVED_LICENSE:
- RI_COMPUTE:
- SPOT:
- STORAGE:
- TOTAL:
- CLOUD_COMMITMENT_USAGE:

cost_source

- BUY_RI_DISCOUNT:
- ENTITY_UPTIME_DISCOUNT:
- ON_DEMAND_RATE:
- RI_INVENTORY_DISCOUNT:
- TOTAL:
- UNCLASSIFIED:
- CLOUD_COMMITMENT_DISCOUNT:

entity_state

- FAILOVER:
- MAINTENANCE:
- POWERED_OFF:
- POWERED_ON:
- SUSPENDED:
- UNKNOWN:

entity_type

- ACTION_MANAGER:
- APPLICATION:
- APPLICATION_COMPONENT:
- APPLICATION_COMPONENT_SPEC:
- APPLICATION_SERVER:
- AVAILABILITY_ZONE:
- BILLING_FAMILY:
- BUSINESS:
- BUSINESS_ACCOUNT:
- BUSINESS_APPLICATION:
- BUSINESS_ENTITY:
- BUSINESS_TRANSACTION:
- BUSINESS_USER:
- CHASSIS:
- CLOUD_COMMITMENT:
- CLOUD_SERVICE:
- COMPUTE_CLUSTER:
- COMPUTE_RESOURCE:
- COMPUTE_TIER:
- CONTAINER:
- CONTAINER_PLATFORM_CLUSTER:
- CONTAINER_POD:
- CONTAINER_SPEC:
- DATABASE:
- DATABASE_SERVER:
- DATABASE_SERVER_TIER:
- DATABASE_TIER:
- DATACENTER:
- DESIRED_RESERVED_INSTANCE:
- DESKTOP_POOL:
- DISK_ARRAY:
- DISTRIBUTED_VIRTUAL_PORTGROUP:
- DOCUMENT_COLLECTION:
- DPOD:
- GROUP:
- HCI_PHYSICAL_MACHINE:
- HYPERVISOR_SERVER:
- INFRASTRUCTURE:
- INTERNET:
- IO_MODULE:
- IP:
- K8S_CLUSTER:

- LICENSING_SERVICE:
- LOAD_BALANCER:
- LOGICAL_POOL:
- MAC:
- MOVER:
- NAMESPACE:
- NETWORK:
- NETWORKING_ENDPOINT:
- NODE_POOL:
- OPERATOR:
- PHYSICAL_MACHINE:
- PORT:
- PROCESSOR_POOL:
- REGION:
- RESERVED_INSTANCE:
- RESERVED_INSTANCE_SPECIFICATION:
- RESOURCE_GROUP:
- RIGHT_SIZER:
- SAVINGS:
- SERVICE:
- SERVICE_ENTITY_TEMPLATE:
- SERVICE_PROVIDER:
- STORAGE:
- STORAGE_CLUSTER:
- STORAGE_CONTROLLER:
- STORAGE_TIER:
- STORAGE_VOLUME:
- SWITCH:
- TARGET:
- THIS_ENTITY:
- THREE_TIER_APPLICATION:
- VIEW_POD:
- VIRTUAL_APPLICATION:
- VIRTUAL_DATACENTER:
- VIRTUAL_MACHINE:
- VIRTUAL_MACHINE_SPEC:
- VIRTUAL_VOLUME:
- VLAN:
- VM_SPEC:
- VPOD:
- WEB_SERVER:
- WORKLOAD_CONTROLLER:
- _NONE_:

- DATA_WAREHOUSE:
- DATA_WAREHOUSE_TIER:
- NODE_GROUP:
- POWERVM_PHYSICAL_MACHINE:

environment_type

- CLOUD:
- HYBRID:
- ON_PREM:
- UNKNOWN_ENV:

file_type

- CONFIGURATION:
- DISK:
- ESXCONSOLE:
- ISO:
- LOG:
- MEMORY:
- SNAPSHOT:
- SWAP:

metric_type

- ACTIVE_SESSIONS:

  Count of the number of actively logged in sessions Count
- BALLOONING:

  Number of kb the host is dedicated for ballooning kilobytes
- BUFFER_COMMODITY:

  Utilization of a port buffer (fabric, UCS specific) megabytes
- CARBON:

  Carbon dioxide equivalent (CO2e) emissions for a given entity grams
- CONNECTION:

  Number of connections onto the application Count
- COOLING:

  acceptable temperature range that is utilized by an entity percentage
- CPU:

  CPU used by host (current column) or host CPU consumed by VMs (consumed column). MHz
- CPU_ALLOCATION:

  For VMs (consumed column) - VM buys from VDC or from desktop pool (VDI). For hosts - PMs sells to VDCs. MHz
- CPU_HEADROOM:

  Number of VMs that can fit in all of the available CPU (on a cluster) Count
- CPU_PROVISIONED:

Capacity is based on the over-provisioned value in policy MHz

- CPU_READY:

  Ready queue ms
- DB_CACHE_HIT_RATE:

  The percentage of accesses that result in cache hits. Percentage
- DB_MEM:

  DB mem utilization or DM capacity kilobytes
- DTU:

  Capacity defined in the cloud Count
- ENERGY:

  Electricity consumed by a given entity over a period of time watt-hours
- EXTENT:

  Internal NA
- FLOW:

  Amount of network traffic per second between two nodes kilobytes/sec
- FLOW_ALLOCATION:

  bytes
- HEAP:

  Amount of heap configured for an application kilobytes
- IMAGE_CPU:

  For VDI, usage of the business user MHz
- IMAGE_MEM:

  For VDI, usage of the business user kilobytes
- IMAGE_STORAGE:

  For VDI, usage of the business user megabytes
- IO_THROUGHPUT:

  Amount of traffic through the IO adapter kilobytes/sec
- MEM:

  Physical machine - usage of memory. VM - consumed memory. kilobytes
- MEM_ALLOCATION:

  Physical machine - what the host provides to VDCs. VM (consumed column) - what the VM buys from VDC or desktop pool. kilobytes
- MEM_HEADROOM:

  Number of VMs that can fit in the available memory on the cluster Count
- MEM_PROVISIONED:

  Capacity, based on the over-provisioned value in policy kilobytes
- NET_THROUGHPUT:

  Amount of traffic through the Network adapter kilobytes/sec
- POOL_CPU:

  For VDI, usage of the desktop pool MHz
- POOL_MEM:

For VDI, usage of the desktop pool kilobytes

- POOL_STORAGE:

  For VDI, usage of the desktop pool megabytes

- PORT_CHANEL:

  Traffic on ports for UCS Mgr/HPE1 kilobytes/sec

- POWER:

  Electricity consumed by a given entity watts

- PROCESSING_UNITS:

  unit of measure for shared processing power across one or more virtual processors PU

- Q16_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q1_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q2_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q32_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q3_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q4_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q5_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q64_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q6_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q7_VCPU:

VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- Q8_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- QN_VCPU:

  VM - Sum of the Ready Queue usage of all of the CPU on the VM. To normalize the value, divide it by the number of vCPU. Host - Average of the underlying Ready Queue values from all of the corresponding VMs with the same number of vCPUs. ms

- REMAINING_GC_CAPACITY:

  Percentage of time not spent on garbage collection Percentage

- RESPONSE_TIME:

  Application response ms

- SLA_COMMODITY:

  User-defined value Number

- STORAGE_ACCESS:

  Number of I/O (read + write) within two discovery cycles count

- STORAGE_ALLOCATION:

  Allocated storage megabytes

- STORAGE_AMOUNT:

  Used storage megabytes

- STORAGE_HEADROOM:

  Count of VM that can fit for the storage in a cluster in terms of storage Count

- STORAGE_LATENCY:

  Acceptable latency. User can set capacity for most storage, is discovered for NetApp. ms

- STORAGE_PROVISIONED:

  Capacity, based on the over-provisioned value in policy megabytes

- SWAPPING:

  Memory swapping bytes/sec

- THREADS:

  Threads in use Count

- TOTAL_HEADROOM:

  Min value of CPU, Mem, IO, of the headrooms Count

- TOTAL_SESSIONS:

  Total number of possible sessions Count

- TRANSACTION:

  Transactions for the application transactions/sec

- TRANSACTION_LOG:

  Size of the transaction log megabytes

- VCPU:

  CPU in use MHz

- VCPU_LIMIT_QUOTA:

  Amount of virtual CPU limit a pod has allocated against the namespace quota millicores
- VCPU_REQUEST:

  Virtual CPU request allocated by a pod against the node allocatable capacity millicores
- VCPU_REQUEST_QUOTA:

  Amount of virtual CPU request a pod has allocated against the namespace quota millicores
- VCPU_THROTTLING:

  Throttling of container virtual CPU that could impact response time percentage
- VIRTUAL_PROCESSORS:

  Guest OS representation of the capacity of the VM VP
- VMEM:

  Memory sold by a VM. Memory sold and bought for a container kilobytes
- VMEM_LIMIT_QUOTA:

  Total amount of virtual memory limit for all pods allocated to the namespace against the namespace quota kilobytes
- VMEM_REQUEST:

  Virtual memory request allocated by a pod against the node allocatable capacity kilobytes
- VMEM_REQUEST_QUOTA:

  Amount of virtual memory request a pod has allocated against the namespace quota kilobytes
- VSTORAGE:

  Storage utilized by an entity megabytes
- CONCURRENT_QUERIES:
- GPU:
- GPU_MEM:
- GPU_REQUEST:
- GPU_REQUEST_QUOTA:
- LLM_CACHE:
- QUEUING_TIME:
- SERVICE_TIME:

savings_type

- MISSED_INVESTMENTS:

  Actions you could have taken that would inscrease your investment to provide more resources to your applications.
- MISSED_SAVINGS:

  Actions you could have taken that could have reduced your costs.
- REALIZED_INVESTMENTS:

  Actions you have taken that inscrease your investment to provide more resources to your applications.
- REALIZED_SAVINGS:

  Actions you have taken that have reduced your costs.

severity

- `CRITICAL`:
- `MAJOR`:
- `MINOR`:
- `NORMAL`:

terminal_state

- `FAILED`:
- `SUCCEEDED`:

Configuring an external TimescaleDB

You can configure External Reports to use an external deployment of TimescaleDB. You can use this as an alternative to the default deployment.

To enable an external deployment, you must complete the following tasks:

- Deploy or access an external TimescaleDB instance. This instance is a PostgreSQL server with the TimescaleDB extension installed. The DB server must be deployed in a way that is accessible to your installation of Turbonomic. It can be deployed on the cloud or in an on-prem VM. As of Turbonomic 8.14.2, the embedded reports feature currently supports PostgreSQL 15.x and TimescaleDB 2.17.1. Prior to Turbonomic 8.14.2, the embedded reports feature supported PostgreSQL 12.x and TimescaleDB 2.10.2. For more information, see the following resources:
  - Creating an External TimescaleDB
  - Install TimescaleDB
- Provision users, databases, and schemas on the TimescaleDB instance. To provision these database objects, you can manually provision or enable Turbonomic to automatically provision. For more information, see:
  - Manual provisioning of users and databases
  - Automatic provisioning of users and databases
- Edit properties in the Turbonomic `charts_v1alpha1_xl_cr.yaml` file. The location of the `charts_v1alpha1_xl_cr.yaml` file depends on whether you are configuring an OVA installation or a Kubernetes Node installation of Turbonomic.
- Optionally, enable secure connections between Turbonomic and the TimescaleDB instance. For information about enabling TLS/SSL for database connections, see the PostgreSQL documentation Secure TCP/IP Connections with SSL.

  This link provides steps to create a self-signed certificate that works with embedded reporting. However, self-signed certificates are not recommended for production instances.

Creating an external TimescaleDB

You can create an external TimescaleDB service that you host in your environment. Alternatively, you can use an existing TimescaleDB service that is deployed on the cloud or in your enterprise by the DB Management team.

The TimescaleDB service that Turbonomic uses must satisfy these conditions:

- **Deployment**: The TimescaleDB is a PostgreSQL server with the TimescaleDB extension. It must be deployed in a way that is accessible to your installation of Turbonomic. It can be deployed on the cloud or in an on-prem VM.

  For information about installing Timescale, see Install TimescaleDB.

- **Supported version**: As of Turbonomic 8.14.2, the embedded reports feature currently supports PostgreSQL 15.x and TimescaleDB 2.17.1. Prior to Turbonomic 8.14.2, the embedded reports feature supported PostgreSQL 12.x and TimescaleDB 2.10.2.
- An account with global R/W privileges on the TimescaleDB instanceTurbonomicThe embedded reports feature requires a specific set of databases to be created on the TimescaleDB instance. It is typical practice for an administrator to manually create the necessary databases. Alternatively, you can create a global R/W user account for the Turbonomic platform so it can create the databases automatically.
- **Entry point**: The TimescaleDB must provide an entry point through a DNS Name or IP address that you can access from Turbonomic. You configure this access in the Turbonomic `charts_v1alpha1_xl_cr.yaml` file.

To create the TimescaleDB, complete the following tasks:

1. Install PostgreSQL by running the following command:

   ```
   sh -c 'echo "deb http://apt.postgresql.org/pub/repos/apt $(lsb_release -cs)-pgdg
   main" > /etc/apt/sources.list.d/pgdg.list' wget -qO- https://www.postgresql.org/media/keys/
   ACCC4CF8.asc | sudo tee /etc/apt/trusted.gpg.d/pgdg.asc &>/dev/null apt update export
   DEBIAN_FRONTEND=noninteractive ; export NEEDRESTART_MODE=a ; apt install postgresql
   postgresql-client -y
   ```

2. Install TimescaleDB by running the following command:

   ```
   curl -s https://packagecloud.io/install/repositories/timescale/timescaledb/script.deb.sh |
   sudo bash export DEBIAN_FRONTEND=noninteractive ; export NEEDRESTART_MODE=a ; apt install
   timescaledb-2-postgresql-15 timescaledb-2-loader-postgresql-15 -y
   ```

3. Restart PostgreSQL:

   ```
   systemctl restart postgresql
   ```

4. Create a global R/W user account that can be used to create databases on the DB service.

   For information about automatically creating databases and users through R/W access, see Automatic Provisioning of Users and Databases.

5. Optional: Manually create the users and databases that embedded reports use.

   If you cannot grant Turbonomic global R/W access to the DB, you must manually create the databases and users for embedded reports.

   For information about manually creating databases and users, see Manual Provisioning of Users and Databases.

Manual Provisioning of users and databases

For Turbonomic to automatically create the required databases for embedded reports, the Turbonomic `charts_v1alpha1_xl_cr.yaml` file must specify credentials for a user account that has global R/W access to the DB service. To ensure this access, you can store the credentials in cleartext in the `charts_v1alpha1_xl_cr.yaml` file.

If Automatic Provisioning is appropriate for your environment, then you can go directly to Automatic Provisioning of Users and Databases. That topic is where you specify the global R/W account that embedded reports can use to access your TimescaleDB.

If Automatic Provisioning is not appropriate for your environment, you must access the DB service with an R/W account and manually add the databases and user accounts that embedded reports can use. Then, after you add the databases to your TimescaleDB, you will edit the Turbonomic `charts_v1alpha1_xl_cr.yaml` file to use these databases.

Updating PostgreSQL configuration

Before you proceed with provisioning the databases and users, update the PostgreSQL configuration file. Use the following steps:

1. Navigate to `/etc/postgresql/15/main/postgresql.conf`.
2. Use following property values in `postgresql.conf` file:
   a. Under `Connection Settings`:

   ```
   listen_addresses = '*'
   ```

   b. Under `Shared Library Preloading`:

   ```
   shared_preload_libraries = 'timescaledb'
   ```

3. Restart PostgreSQL:

   ```
   systemctl restart postgresql
   ```

Manually provisioning databases and users on the TimescaleDB

The following steps provision the database objects that embedded reporting requires. These examples use default names for illustration. You can substitute your own names for databases and users. Remember the names that you use because you must match them as you edit the Turbonomic `charts_v1alpha1_xl_cr.yaml` file.

To provision the databases and users, open a command session on the TimescaleDB, and run the following commands:

- Log in to PostgreSQL:

  ```
  su - postgres -c 'psql -U postgres -p 5432'
  ```

- Create two databases - one for `extractor` data, one for `grafana` data: The extractor database manages the Turbonomic data stream, and the Grafana database manages data for reporting within Grafana.

  Run the following commands, where you can specify your own database names instead of `extractor` and `grafana`:

  ```
  CREATE DATABASE extractor;
  CREATE DATABASE grafana;
  ```

- Create database users: For the extractor database, you create an R/W user, a read-only group, and a read-only user. Also create an R/Wn user for the Grafana database.

  Run the following commands, where you can specify your own names for instances of USER or ROLE:

  ```
  -\- main read/write user for extractor data
  CREATE USER extractor PASSWORD '<password>';
  -\- group for users with read-only access to extractor data
  CREATE ROLE readers_extractor_extractor;
  -\- read-only user for extractor data as a member of that group
  CREATE USER query PASSWORD '<password>';
  GRANT CONNECT ON DATABASE extractor TO readers_extractor_extractor;
  GRANT readers_extractor_extractor TO query;
  -\- read-write user for grafana data
  CREATE USER grafana_backend PASSWORD '<password>';
  ```

- Create and prepare the schema for extractor data. Connect to the extractor database by using `\c <database_name>` command. Run the following commands, where you can specify your own name for the SCHEMA, and you grant privileges to the users and roles you previously created:

  ```
  CREATE SCHEMA extractor;
  -\- read/write user has full access
  GRANT ALL PRIVILEGES ON SCHEMA extractor TO extractor;
  -\- all users in readers group have read-only access
  GRANT USAGE on SCHEMA extractor TO readers_extractor_extractor;
  GRANT SELECT ON ALL TABLES IN SCHEMA extractor TO readers_extractor_extractor;
  -\- make sure readers get access to any tables added in the future
  ALTER DEFAULT PRIVILEGES IN SCHEMA extractor GRANT SELECT ON TABLES TO
  readers_extractor_extractor;
  ```

```
-\- make the extractor and query users use the extractor schema by default
ALTER ROLE extractor SET search_path TO `extractor`;
ALTER ROLE query SET search_path TO `extractor`;
-\- install the timescaledb plugin into the extractor database using the extractor schema
CREATE EXTENSION timescaledb SCHEMA extractor;
```

- Create and prepare the schema for Grafana data. Connect to the Grafana database by using `\c <database_name>` command. Run the following commands, where you can specify your own name for the SCHEMA, and you grant privileges to the users and roles you previously created:

```
CREATE SCHEMA grafana;
-\- read/write user has full access
GRANT ALL PRIVILEGES ON SCHEMA grafana TO grafana_backend;
-\- make sure the grafana user uses the grafana schema by default
ALTER ROLE grafana_backend SET search_path TO 'grafana';
```

- Exit from PostgreSQL using `\q`.
- Set the environment variable:

```
PG_HBA_CONF="/etc/postgresql/15/main/pg_hba.conf"
```

- Copy the content to a .bak extension file:

```
cp ${PG_HBA_CONF} "${PG_HBA_CONF}.bak"
```

- Run the following 6 commands in the listed order:

```
echo "# enable remote access from other hosts (all ipv4 and ipv6 addresses)" | sudo tee -a
${PG_HBA_CONF} > /dev/null

echo "host  all       all       0.0.0.0/0        md5" | sudo tee -a ${PG_HBA_CONF} > /dev/null

echo "host  all       all       {Your_IP}/32     md5" | sudo tee -a ${PG_HBA_CONF} > /dev/null

echo "host  all       all       ::/0          md5" | sudo tee -a ${PG_HBA_CONF} > /dev/null

sed -i 's/host all all 127.0.0.1\/32/#host all all 127.0.0.1\/32/' ${PG_HBA_CONF}

sed -i 's/host all all ::1\/128 md5/#host all all ::1\/128 md5/' ${PG_HBA_CONF}
```

- Restart PostgreSQL:

```
systemctl restart postgresql
```

The previous commands provision the required databases and users. Your provisioning must be complete and correct for embedded reports to properly collect data and display it in Grafana reports and dashboards.

Editing the CR file for manually created databases

After you provision the required database objects, edit the Turbonomic `charts_v1alpha1_xl_cr.yaml` file to use these databases.

To edit the `charts_v1alpha1_xl_cr.yaml` file:

1. Open the `charts_v1alpha1_xl_cr.yaml` file for editing.

   The location of the file depends on the type of Turbonomic installation that you are configuring:

   - For an OVA installation of Turbonomic:

     Open an SSH terminal session on your Turbonomic instance.

     Log in with the System Administrator that you set up when you installed Turbonomic:

     – Username: `turbo`

     – Password: `<your_private_password>`

     Then edit the file:

/opt/Turbonomic/kubernetes/operator/deploy/crds/charts_v1alpha1_xl_cr.yaml

- Turbonomic on a Kubernetes node or node cluster:

  Open the following file for editing:

  deploy/crds/charts_v1alpha1_xl_cr.yaml

2. Specify the endpoint for connecting to the external database.

   You can use the database service DNS, or you can use an IP address.

   Add the endpoint as the externalTimescaleDBIP property in the spec: global: section of the .cr file:

```
spec:
  global:
    externalTimescaleDBIP: <host-or-IP>
```

3. Specify Turbonomic access to the databases that you provisioned.

   As you specify users and databases, be sure to match the names you used to provision users, databases, and schema. These instructions use the default naming.

   You specify:

   - Global read-only access to the extractor database /spec/properties/global/dbs This grants the extractor and the api components read-only access to the extractor database that you previously provisioned. Those components use the query user account.
   - Read/write access to the extractor and grafana databases /spec/properties/extractor/dbs This grants the extractor component read/write access to the extractor and grafana databases that you previously provisioned.
   - Read/write access for Grafana to access the grafana database /spec/grafana/grafana.ini/database

   Edit the charts_v1alpha1_xl_cr.yaml file to add the following entries:

```
spec:
  properties:
    global:
      dbs:
        extractor:
          databaseName: extractor
          schemaName: extractor
          query:
            userName: query
            password: <defaault-password>
    extractor:
      dbs:
        extractor:
          userName: extractor
          password: <default-password>
        grafana:
          schemaName: grafana
  grafana:
    grafana.ini:
      database:
        name: grafana
        user: grafana_backend
        password: <default-password>
```

4. If not already enabled, then enable Reporting, TimescaleDB, and Grafana. The following need to exist under the spec section in charts_v1alpha1_xl_cr.yaml:

```
reporting:
  enabled: true

timescaledb:
  enabled: true

grafana
   enabled: true
```

5. Run the following 3 commands to apply the changes:

```
systemctl restart postgresql

kubectl delete pod <extractor-xxx>

kubectl delete pod <grafana-xxx>
```

Automatic provisioning of users and databases

For Turbonomic to automatically create the required databases for embedded reports, the Turbonomic `charts_v1alpha1_xl_cr.yaml` file must specify credentials for a user account that has global R/W access to the DB service. To enable this access, store the credentials in cleartext in the `charts_v1alpha1_xl_cr.yaml` file.

If Automatic Provisioning is not appropriate for your environment, you must access the DB service with an R/W account and manually add the databases and user accounts that embedded reports can use. See Manual Provisioning of Users and Databases.

To enable an external TimescaleDB for embedded reports, you must edit the Turbonomic `charts_v1alpha1_xl_cr.yaml` file.

1. Open the `charts_v1alpha1_xl_cr.yaml` file for editing.

   The location of the file depends on the type of Turbonomic installation that you are configuring:

   - For an OVA installation of Turbonomic:

     Open an SSH terminal session on your Turbonomic instance.

     Log in with the System Administrator that you set up when you installed Turbonomic:

     – Username: `turbo`

     – Password: `<your_private_password>`

     Then edit the file:

     `/opt/Turbonomic/kubernetes/operator/deploy/crds/charts_v1alpha1_xl_cr.yaml`

   - Turbonomic on a Kubernetes node or node cluster:

     Open the following file for editing:

     `deploy/crds/charts_v1alpha1_xl_cr.yaml`

2. Specify the endpoint for connecting to the external database.

   For the connection endpoint, specify either the fully qualified domain name of the external server or its IP address.

   Add the endpoint to the `spec/global/externalTimescaleDBIP` property in the `charts_v1alpha1_xl_cr.yaml` file:

```
spec:
...
  global:
    externalTimescaleDBIP: <host-or-IP>
```

3. Specify the global R/W account that Turbonomic uses to log in to the TimescaleDB service.

   This enables Turbonomic to automatically create the databases, schemas, and users that embedded reports requires.

   Note:

   Only complete this step if you want to enable global RW access to the TimescaleDB service in the `charts_v1alpha1_xl_cr.yaml` file. You specify the account credentials in clear text.

   If you choose not to specify the global account here, then you must manually provision the database objects that embedded reports requires. See Manual Provisioning of Users and Databases.

To grant the access Turbonomic needs, specify the username and password for an account that has global R/W privileges on the TimescaleDB instance. Add these credentials in the `/spec/properties/global/dbs/postgresDefault` block of the cr.yaml file, as follows:

```
spec:
  properties:
    global:
      dbs:
        postgresDefault:
          rootUserName: <root-user>
          rootPassword: <root-password>
```

Reference: Working with YAML files

YAML is the primary file format to create and configure resources on kubernetes, including everything to do with the Turbonomic platform. The Custom Resource YAML provides a convenient, single place to define the majority of configuration details for Turbonomic. When editing YAML files, you must be careful to respect the file syntax, including indents in the file. General rules for edits include:

- Always uses spaces, not tabs, for all indentation. Since spacing and indentations matter, and can yield an invalid result or a parameter completely skipped over, you should work with YAML files with an editor that supports using vertical lines associated with indentations to visually spot a misalignment. If your editor of choice makes this difficult, you can use the linux `expand` utility when you're done, to convert tabs to equivalent spaces.
- Indention uses two spaces per level.
- Be careful to keep the same indentation for all properties in a given section.
- Never use the same property name twice in the same section. Doing this will render the YAML file invalid, though you will likely not see any notification of a problem. Rather, all but one of the property definitions will be silently ignored.

Spacing matters

For the Turbonomic Custom Resource, indentation defines where parameters are applied (globally or to specific components), so ensure you line up the text appropriately. The following example shows examples of specifications that are applied at a global level (`{“spec”:{“global”:[{“tag”:”8.16.4”}]}}`) which sets the container image tag for all instances. Then indented you see properties that are global for the remote database (`{“spec”:{“properties”:{"global":[{“dbPort”:”6033”}]}}}`) describes a property of `dbPort` that would be set for remote DB connections.

Each line is indented two spaces from the higher level.

```
spec:
  global:
    repository: turbonomic
    tag: 8.16.4
  properties:
    global:
      dbPort: 6033
  kubeturbo:
    enabled: true
  aws:
    enabled: true
```

Combine properties correctly

A YAML file is read beginning to end, and if there are different parameters that apply to the same component, they need to be combined. The following example shows a YAML where properties for the `ui` component of image tag *and* memory limit resources.

```
spec:
  global:
    repository: turbonomic
```

```
    tag: 8.16.4
  ui:
    image:
      tag: 8.16.4
  properties:
    global:
      dbPort: 6033
  kubeturbo:
    enabled: true
  aws:
    enabled: true
  ui:
    resources:
      limits:
        memory: 4Gi
```

This YAML will not set both image tag and memory limit resources for the `ui` component because they are set in two different sections. The memory limits would be applied since that is the last section to be read, and it would overwrite the first set of properties with the image tag. The following YAML will set both image tag and memory limit resources for the `ui` component:

```
spec:
  global:
    repository: turbonomic
    tag: 8.16.4
  ui:
    image:
      tag: 8.16.4
    resources:
      limits:
        memory: 4Gi
  properties:
    global:
      dbPort: 6033
  kubeturbo:
    enabled: true
  aws:
    enabled: true
```

To avoid having the same property name twice in the same section, search for an existing instance of the key you want to add. With the complex embedded structure in many k8s resources, this can be quite challenging; however, you can use an online validator that does not accept duplicate keys to check your entire CR file (such as YAMLint - The YAML Validator).

Finding a property

This documentation refers to specific properties in the CR file using a "path" expression. For example, assume the path `/spec/global/repository` to designate a `repository` property in the file. You can find the property in the file as follows:

1. Find a line that says `spec:` with no indentation at all.
2. Between that line and the next unindented line (not counting comments, which start with #), find a line that says `global:` and is at the next level of indentation.
3. Between that line and the next line with the same indentation, find a line that starts with `repository:`. That line is where the addressed property is defined.

In this example, find the `repository` property specified by `/spec/global/repository`:

```
apiVersion: charts.helm.k8s.io/v1alpha1
kind: Xl
metadata:
  name: xl-release
spec:
  properties:
    global:
      repository:        # This is NOT the correct property
        ...

  # Global settings
  global:
    repository:          # This is the one you want
```

The first `repository` property is not at `/spec/global/repository`, but at `/spec/properties/global/repository`.

Chapter 16. Compliance

Turbonomic adheres to a number of standard certifications. Turbonomic is part of the IBM Cloud Services ISMS ISO 27001:2013, 27017:2015, 27018:2014, and 27701:2019 certification program. The ISO standards specify the requirements for establishing, implementing, maintaining, and continually improving an information security management system. They also include requirements for the assessment and treatment of information security risks. See IBM Cloud compliance programs for more information. The IBM certificates are available online.

- ISO 27001 is a widely-adopted global security standard outlining the requirements for information-security management systems and provides a systematic approach to managing company and customer information based on periodic risk assessments.
- ISO 27017 gives guidelines for information-security controls applicable to the provisioning and use of cloud services, as well as implementation guidance for both cloud service providers and cloud service customers.
- ISO 27018 establishes commonly accepted control objectives, controls, and guidelines for implementing measures to protect Personally Identifiable Information (PII) in accordance with the privacy principles in ISO 29100 for the public cloud computing environment.
- ISO 27701 provides guidance for establishing, implementing, maintaining and continually improving a Privacy Information Management System.

Turbonomic team reviews security and privacy-related activities for compliance with IBM requirements.

Assessments and audits are completed by the Turbonomic team to confirm compliance with its information security policies.

Workforce security education and awareness training is completed by the Turbonomic team on an annual basis. IBM personnel are reminded of their job objectives, and their responsibility to meet ethical business conduct and Turbonomic security obligations.

Chapter 17. Glossary

This glossary provides terms and definitions for the Turbonomic software and products.

The following cross-references are used in this glossary:

- *See* refers you from a nonpreferred term to the preferred term or from an abbreviation to the spelled-out form.
- *See also* refers you to a related or contrasting term.

A

action execution schedule
An automation policy setting in Turbonomic that determines when specific generated actions are executed.

action mode
A policy setting in Turbonomic that specifies the degree of automation for generated actions. An action mode can direct Turbonomic to execute actions automatically, or post actions for review.

Active Directory (AD)
A hierarchical directory service that enables centralized, secure management of an entire network, which is a central component of the Microsoft Windows platform.

AD
See Active Directory.

affinity rule
A VMware rule that specifies that the members of a selected virtual machine Distributed Resource Scheduler (DRS) group can or must run on the members of a specific host DRS group.

agent
A process that performs an action on behalf of a user or other program without user intervention or on a regular schedule, and reports the results back to the user or program.

anti-affinity rule
A VMware rule that specifies that the members of a selected virtual machine Distributed Resource Scheduler (DRS) group cannot run on the members of a specific host DRS group.

API
See application programming interface.

application
One or more computer programs or software components that provide a function in direct support of a specific business process or processes.

application component
A software component, application code, or a unit of processing that consumes resources, such as memory or CPU, to enable it to perform its function.

application programming interface (API)
An interface that allows an application program that is written in a high-level language to use specific data or functions of the operating system or another program.

application resource management (ARM)
A top-down, application-driven approach that continuously analyzes applications' resource needs and generates fully automatable actions to ensure applications always get what they need to perform. ARM is the Turbonomic mission to simultaneously optimize performance, compliance, and cost of applications and infrastructure in real time.

ARM
See application resource management.

availability set
A logical grouping in Microsoft Azure that isolates VM resources from each other to limit the impact of failure.

B

ballooning
A technique for hypervisors to reclaim memory from virtual machines that they host by claiming physical pages on a guest OS and allocating the pages to other consumers.

bring your own license (BYOL)
A licensing model in which corporate customers use their existing licenses flexibly, on-premise or in the cloud, without additional fees. Under this model, the customer remains responsible for managing these licenses, including renewals.

business application
a logical grouping of application entities and nodes that work together to compose a complete application as end users would view it. Turbonomic users can monitor overall performance, make resourcing decisions, and set policies in the context of their business applications.

business transaction
A logical grouping of underlying elements in the supply chain of the Application View in Turbonomic. These elements, which represent all elements that fulfill a user-initiated request, could include services, applications, components, database servers, containers, and virtual machines.

buyer
An entity in the Turbonomic environment that consumes resources from another entity, such as a virtual machine that buys memory resources from a host.

BYOL
See bring your own license.

C

central processing unit (CPU)
A measurable unit of processing capacity on a host. CPU is typically expressed on the host as the number of cores, with a given processing speed.

cloud-native (cloud native)
Pertaining to a model for building and running applications or services that takes advantage of cloud computing principles such as containers, microservices, and dynamic orchestration.

cloud native
See cloud-native.

cluster
A collection of complete systems that work together to provide a single, unified computing capability.

container
A standard unit of software that packages code and dependencies so that applications runs quickly and reliably from one computer environment to another.

controller
In Kubernetes, a control loop that watches the state of a cluster, then makes or requests changes where needed to move the current cluster state closer to the required state.

CPU
See central processing unit.

D

database memory (DBMem)
A monitored resource in Turbonomic that is the measurement of memory utilized by a database server. Turbonomic uses this resource to drive actions for databases, instead of the Vmem on the hosting VM.

data transfer object (DTO)
An object that carries data between processes to reduce the number of method calls.

DBMem
See database memory.

desired state
The state of an operating environment that assures application performance while achieving efficient use of resources and complying with business rules and constraints.

DTO
See data transfer object.

dynamic group
A group whose membership can change according to specified criteria such as entity names, resource capacity, or tags.

F

fabric target
A service that unites compute, network and storage access into a cohesive system.

G

ghost volume
See wasted storage.

Grafana
An open source analytics and visualization platform to monitor, search, analyze, and visualize metrics.

group
A named collection of like entities that can be used in Turbonomic to set scope to policies, plans, charts, and so forth.

H

heap
The portion of a VM or container's memory allocated to individual applications.

hot-add
The addition of CPU and memory, or both, without requiring rebooting.

hypervisor
A service that creates and runs virtual machines (VMs), providing them compute and storage resources. The hypervisor treats on-prem resources such as CPU, memory, and storage as a pool that can be easily reallocated between existing guests or to new virtual machines.

I

IAM
See identity and access management.

identity and access management (IAM)
The process of controlling access of authorized users to data and applications, while helping companies comply with various regulatory requirements.

identity provider (IdP)
A provider who offers user authentication as a service. Authentication of a user's sign in details is performed by the identity provider who creates, manages, and maintains the user's identity information. Identity providers enable users in an organization to single sign-on (SSO) to one or more systems.

IdP
See identity provider.

Information Technology Infrastructure Library (ITIL)
A best-practice framework for facilitating the delivery and management of IT services.

information technology service management (ITSM)
A set of workflows and tools to develop, deliver, and manage IT services. ITSM is used to handle incidents, service requests, problems, and changes, all of which are typically linked through an ITSM platform, such as ServiceNow.

input/output operations per second (IOPS)
A standard computing benchmark used to determine the best configuration settings for servers.

IOPS
See input/output operations per second.

ISO image
An electronic file that contains the identical content of an optical disc, including the folder and file hierarchy. Using an .ISO file extension, ISO images are created to distribute the disc's data over a network in order to burn a CD or DVD at the destination computer.

ITIL
See Information Technology Infrastructure Library.

ITSM
See information technology service management.

L

latency
The time delay between the moment an operation is initiated, and the moment it begins to take effect.

license
A legal agreement that authorizes the use of proprietary information including, but not limited to, copyrighted or patented information.

lift and shift
A strategy for moving an application or operation from one environment to another without redesigning the workflow for the app or operation.

M

market
In Turbonomic, a model of the data center environment that represents each entity, such as physical machines, virtual machines, storage device, volume, or application component, as a buyer or seller in a market. Turbonomic uses virtual currency to give a budget to buyers and assign cost to resources so that users can compare the cost of application transactions with the cost of space on a disk or physical space in a data center.

microservice
A set of small, independent architectural components, each with a single purpose, that communicate over a common lightweight API.

N

namespace

1. A virtual cluster within a Kubernetes cluster that can be used to organize and divide resources across multiple users.
2. A logical container in which all the names are unique. The unique identifier for an artifact is composed of the namespace and the local name of the artifact.

network interface card (NIC)
A physical hardware component that attaches computers to a computer network.

NFU
See normalized factor unit.

NIC
See network interface card.

non-disruptive mode
A policy setting for VMs that prevents actions that require a reboot or introduce disruptions from executing automatically.

normalized factor unit (NFU)
For a reserved instance (RI) in AWS environments, a measure of RI capacity that is used to compare or combine that capacity for different template families.

O

OAS
See OpenAPI Specification.

observation period
For Turbonomic analysis, the time span of sampling for the data that is used to calculate utilization percentiles.

OpenAPI Specification (OAS)
A standard, programming language-agnostic interface description for HTTP APIs, which allows both humans and computers to discover and understand the capabilities of a service without requiring access to source code, additional documentation, or inspection of network traffic.

Open Virtual Appliance (OVA)
A package that is deployed in a virtual environment to create virtual machines and virtual appliances. The OVA package is an archive file that contains the OVF directory.

Open Virtualization Format
A packaging standard designed to address the portability and deployment of virtualization appliances.

orchestration
The automated configuration, management, and coordination of computer systems, applications, and services.

OVA
See Open Virtual Appliance.

OVF file
A package consisting of many files that is deployed in a virtual environment to create virtual appliances.

P

plan market
A snapshot of the real-time market that is created in Turbonomic to simulate what-if scenarios against an environment. The plan market starts with the same entities, utilization levels, placement constraints and analysis settings as the real-time market has.

policy schedule
A schedule that determines that an automation policy is active only at designated times. For example, a policy schedule can set a policy to automatically resize or start VMs for the entire month of December, in anticipation of an increase in demand.

R

ready queue
A queue for VMs that are waiting for available CPU.

regex
See regular expression.

regular expression (regex)
A set of characters, meta characters, and operators that define a string or group of strings in a search pattern.

remaining garbage collection capacity (remaining GC capacity)
The percentage of CPU time not spent on garbage collection (GC). Turbonomic uses this metric in conjunction with Heap Utilization when making scaling decisions for application component entities.

remaining GC capacity
See remaining garbage collection capacity.

Representational State Transfer (REST)
A software architectural style that guides the design and development of the architecture for the web. REST defines a set of constraints for the architecture of Internet-scale distributed hypermedia systems, such as the web.

reserved instance (RI)
A reservation of cloud resources and capacity that is purchased at a set price for a given time period. Unlike on-demand pricing, the purchaser commits to paying for all of the hours of the contract's term. As a benefit, the hourly rate is lowered significantly.

response time
The elapsed time between entering an inquiry or request and receiving a response.

REST
See Representational State Transfer.

return on investment (ROI)
The calculation an entity makes in each Turbonomic market cycle to decide what action, if any, it wants to take. ROI equals revenue divided by expenses.

RI
See reserved instance.

ROI
See return on investment.

S

SaaS
See software as a service.

SAML
See Security Assertion Markup Language.

Secure Shell (SSH)
A network protocol for secure data exchange between two networked devices. The client can use public-key and private-key authentication, or password authentication, to access the remote server.

Security Assertion Markup Language (SAML)
An XML framework for exchanging authentication and authorization information.

seller
An entity in the Turbonomic environment that provides resources to another entity, such as a host that sells CPU resources to a virtual machine.

service
A measurable function that is part of an internal or user-initiated request, such as a request to update inventory.

service provider (SP)
Any company that provides services for a fee to its customers, such as telecommunication companies, application service providers, enterprise IT, and Internet service providers.

Simple Network Management Protocol (SNMP)
A set of protocols for monitoring systems and devices in complex networks. Information about managed devices is defined and stored in a Management Information Base (MIB).

SNMP
See Simple Network Management Protocol.

software as a service (SaaS)
A model of software deployment whereby software including business processes, enterprise applications, and collaboration tools, are provided as a service to customers through the cloud.

SP
See service provider.

SQL
See Structured Query Language.

SSH
See Secure Shell.

static group
A group with a fixed list of members.

Structured Query Language (SQL)
A standardized language for defining and manipulating data in a relational database.

supercluster
Two or more clusters that are merged into a single group of providers. This enables Turbonomic to move workload from a host in one of the clusters to a host in any of the merged clusters.

T

target
A resource or workload management service in a virtual environment that is connected to Turbonomic

tenant
An entity that accesses services through a subscription that is associated with an account or namespace.

V

VCPU
See virtual CPU.

VDC
See virtual datacenter.

VDI
See virtual desktop infrastructure.

virtual CPU (VCPU)
A measure of the CPU resources on a VM. VCPU is typically expressed in a VM as the number of cores.

virtual datacenter (VDC)
A collection or pool of resources that groups the resources around specific requirements or business needs. vDCs can implement boundaries for a private cloud infrastructure, and then can establish tenant groups on that infrastructure.

virtual desktop infrastructure (VDI)
An infrastructure that consists of desktop operating systems hosted within virtual machines on a centralized server.

virtual memory (VMem)
The allocated memory capacity in a virtual machine.

virtual storage (VStorage)
The allocated virtual storage capacity.

VMem
See virtual memory.

VStorage
See virtual storage.

W

wasted storage
Storage volumes that are not actively used by any workload in the system. Typically, volumes that are wasted storage are paid for but unused.

workload controller
A Kubernetes controller that watches the state of a user's pods and then requests changes where needed.

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